Alcatel 9400 UX

User Manual

3CC08991ATAA TQBJA 01

Status Released

Change Note

Short Title Alcatel 9400UX

All rights reserved. Passing on and copying of this document, use and communication of its contents not permitted without written authorization from Alcatel.

13, 15, 18, 23, 25 or 38 GHz Small and Medium Capacity Digital Microwave Link





with supervision
946LUX11/946LUX12 (RQ2)
946LUX40 (SNMP)



PAGE INTENTIONALLY LEFT BLANK

Customer Service Support:

a Team tuned to your needs for your entire satisfaction



Congratulations on having bought your equipment from Alcatel.

We hope that it will give you full satisfaction.

For any additional information, about your Alcatel Welcome Center (for Technical Support or for repair process), please contact your Accatel Contract Manager.

PAGE INTENTIONALLY LEFT BLANK

Table of contents

1 – Foreword 15

1.1 – Structure of the manual 15

1.2 – Using the manual 15

1.3 – Safety instructions 16

1.3.1 – General rules 16

1.3.2 – Symbols 16

1.3.3 – Symbols used in the document 17

1.3.4 – Certificate of compliance with European safety standards 18

1.3.5 – Eco–Declaration 19

1.3.6 – Public exposition to electromagnetic fields 20

2 – Description of the equipment 21

2.1 – Alcatel 9400 Family 21

2.1.1 – Simplified description 21

2.1.2 – Configuration 22

2.1.3 – ALCATEL 9400 UX features 24

2.1.4 – Capacity configurations 29

2.1.5 – Typical hardware configurations 29

2.2 – Operation 30

2.2.1 – 1+0 configuration 30

2.2.2 – Basic 1+1 configuration 34

2.2.3 – 1+1 configuration with multiplexer/demultiplexer protection 36

2.3 – Engineering service channels 38

2.3.1 – ESC number 1 39

2.3.2 – ESC number 2 40

2.3.3 – ESC numbers 3 and 4 40

2.3.4 – ESC number 5 41

2.4 – Alarms, telesignalling and remote controls (TS/TC) 42

2.5 – Loopback options 42

2.6 – Equipment management 44

2.7 – IDU controls, indications and connectors 45

2.7.1 – Classic main IDU 45

2.7.2 – Light IDU 47

2.7.3 – Extension IDU 48

2.7.4 – Access IDU 49

2.8 – Technical characteristics 51

2.9 – Frequency agility bands 58

3 – Installation 61

3.1 – Delivery 61

3.1.1 – Checking the configuration 61

3.2 – Labels on the equipment 62

3.2.1 – Typical installation dimensions (1+0 configuration with integrated 1 ft) 66

3.3 – Installing the equipment 68
3.3.1 – Information required for installation 68
3.3.2 – Precautions concerning electromagnetic compatibility (EMC) and safety 68
3.3.3 – Tools required 68
3.3.4 – Fasten torques for the screws and connectors 69
3.4 – Installing 19” IDUs 69
3.4.1 – Installation on feet or on a wall mounting 70
3.4.2 – Laborack or 9U subrack installation 70
3.5 – Outdoor part installation 73
3.5.1 – General information 73
3.5.2 – Choosing antenna polarization 74
3.5.3 – Installing the configuration with pole mounting 1 + 0 (9400UXI102) 75
3.5.4 – Installing the configuration with pole mounting 1 + 1 (9400UXI101) 81
3.5.5 – Installation with non-integrated antenna 87
3.6 – Wiring 92
3.6.1 – Wiring the power supply to the 19” indoor units 92
3.6.2 – Equipment earthing 93
3.6.3 – IDU/ODU wiring 95
3.6.4 – Wiring the 34 Mbit/s tributary and auxiliary 2 Mbit/s stream 96
3.6.5 – Wiring n x 2 Mbit/s tributaries 96
3.6.6 – Engineering service channel wiring 98
3.6.7 – Alarm and remote control wiring (TS/TC) 98
3.6.8 – Engineering service channel ESC2 telephone handset 99
3.7 – Wiring between a terminal’s units 99
3.7.1 – Main IDU - extension IDU wiring 99
3.7.2 – Wiring the IDUs in 1+1 with MUX protection 99
3.8 – Wiring between terminals of a station 101
3.8.1 – Tributary wiring 101
3.8.2 – ESC wiring 101

4 – Configuration and operation software 103
4.1 – Installing the software 103
4.1.1 – Warning note 103
4.1.2 – Installing the software 105
4.2 – 9400 UX applications 110
4.2.1 – Introduction 110
4.2.2 – Menu bar 110
4.2.3 – Global synthesis line 112
4.2.4 – How to open a session 113
4.2.5 – List of applications 114
4.3 – “Administrative functions” application 115
4.3.1 – Starting in Default configuration 115
4.3.2 – “File” and “Receive” menus 117
4.3.3 – “Operator” menu 117
4.4 – “Installation parameters” application 120
4.4.1 – “Tables” menu 120
4.4.2 – “File” menu 123

4.5 – “Operation parameters” application 124
4.5.1 – “Terminal” menu 125
4.5.2 – “Network” menu with 946LUX11 or 946LUX12 software (RQ2) 128
4.5.3 – “Network” menu with 946LUX40 software (SNMP) 132
4.5.4 – “Thresholds” menu 135
4.5.5 – “Alarms” menu 138
4.5.6 – “Summary” menu (946LUX12 and 946LUX40 version of the software) 140
4.5.7 – “File” menu 140
4.6 – “Software downloading”application 143
4.7 – “Alarms, Status and Controls” application 146
4.7.1 – Opening the application 146
4.7.2 – Main screen 146
4.7.3 – Viewing alarm and synthesis information 147
4.7.4 – Accessing secondary NE screens 148
4.7.5 – Exiting secondary screens 149
4.7.6 – Alarm synthesis line 150
4.7.7 – “View” menu 151
4.7.8 – “History” menu 151
4.7.9 – “Remote_Control” menu 153
4.8 – “Performance Monitoring (G821)” application 155
4.8.1 – “Display” menu 156
4.8.2 – “Save” menu 157
4.9 – “Radio Transmission Parameters” application 158
4.10 – “Remote inventory” application 161
4.10.1 – “File” menu 161
4.10.2 – “Receive” menu 162
4.11 – “Maintenance Memory” application 164
4.11.1 – “File” menu7 165
4.11.2 – “Receive” menu 166
4.11.3 – “Clear” menu 166
4.12 – “G784 Performance Monitoring” application 167
4.12.1 – “File” menu 168
4.12.2 – “View” menu 169
4.12.3 – “Mode” menu 169
4.12.4 – “Configure” menu 175
4.13 – “Analogue measurements” application 177
4.13.1 – “Mode” menu 177
4.14 – Local supervision 178
4.14.1 – Station A of network with supervision/station of network without supervision . . . 178
4.14.2 – Station B in a network with supervision 179

5 – Commissioning 181
5.1 – Order of commissioning for a link 181
5.2 – Installing and commissioning station A (phase 1) 182
5.2.1 – Preliminary operations 182
5.2.2 – Setting up the installation parameters 183
5.2.3 – Setting up the operation parameters 183
5.2.4 – Checking out operation 184

5.3 – Installing and commissioning station B 186
5.3.1 – Preliminary operations 186
5.3.2 – Pointing the antenna with pole mounting 1+1 (9400UXI101) 186
5.3.3 – Pointing the antenna with pole mounting 1+0 (9400UXI102) 187
5.3.4 – Checking out operation 187
5.4 – Commissioning the link from station A (phase 2) 188
5.4.1 – Pointing the antenna 188
5.4.2 – Checking the radio transmission parameters 188
5.4.3 – Remote tributary loop 189
5.4.4 – Checking the transmit/receive switching function 189
5.4.5 – Checking quality 190
5.4.6 – End of commissioning 191
5.5 – Reversing the order of link installation and commissioning 191

6 – Operation, servicing and maintenance 193
6.1 – Network supervision 193
6.1.1 – Network supervised with the Alcatel network management system 193
6.1.2 – Network without the Alcatel network management system 193
6.2 – Preventive maintenance 193
6.3 – Corrective maintenance 193
6.3.1 – Methodology 193
6.3.2 – Analyzing IDU alarm indicators 193
6.3.3 – Using the operating programs to trace alarms 194
6.3.4 – Corrective servicing 195
6.4 – Screens displayed on opening the AS&C application 196
6.5 – Analyzing the “zoom” displays of the functional screens 198
6.5.1 – 1+0 configurations 198
6.5.2 – 1+1 configurations without MUX protection 199
6.5.3 – 1+1 configurations with MUX protection 200
6.6 – Analyzing units and modules 203
6.6.1 – Main, extension or Light IDU 203
6.6.2 – Main and standby outdoor units (ODU) 204
6.6.3 – LIU module or function 207
6.6.4 – Power supply module (PSU) 207
6.6.5 – ESC 3 and 4 module 208
6.6.6 – ENVT module 208
6.6.7 – MCU boards and SCU board in 1+1 configuration with Mux protection 210
6.6.8 – SCU boards 213
6.7 – Changing IDU 214
6.7.1 – Replacing like for like a Light or classic IDU (1+0 or basic 1+1) 214
6.7.2 – Replacing a Light IDU with a classic IDU (1+0 only) and vice versa 214
6.7.3 – Replacing like for like a main IDU with MUX protection 214
6.7.4 – Replacing like for like an extension IDU 215
6.7.5 – Note on IDU compatibility 215
6.8 – Changing ODU 216

7 – Changing configurations 217
7.1 – Changing frequency 218
7.1.1 – Changing frequency within the same sub-band 218
7.1.2 – Changing frequency within another sub-band or a different band 219
7.2 – Changing tributaries 220
7.2.1 – Enabling inactive tributaries 220
7.2.2 – Changing the tributary impedance (classic IDU) 220
7.2.3 – Changing the tributary impedance (Light IDU versions only) 228
7.2.4 – Changing bit rate by software 229
7.2.5 – Changing bit rate by changing an LAU and LIU tributary boards 230
7.3 – Changing software key 232
7.4 – Changing a NE or IP physical address 233
7.4.1 – From the station configured as a network element 233
7.4.2 – From a station with mediation function (master key or local supervision) 233
7.5 – Upgrading software (from 946LUXn to 946LUXn+1) 234
7.5.1 – Modifying IDU 234
7.5.2 – Modifying PC software 234
7.5.3 – Downloading NE software 234
7.5.4 – Migration from 946LUX1x to 946LUX40 and from 946LUX40 to 946LUX1x 234
7.6 – Opening an IDU 235
7.6.1 – “Indoor” unit configuration 236
7.7 – Telephone service channel 238
7.7.1 – Deleting, inserting or changing telephone numbers 238
7.7.2 – Configuring the ESC 2 238
7.8 – Hardware configuration of service channels 3, 4 and 5 (classic and extension IDU) 238
7.8.1 – Configuring ESC 3, 4 and 5 in the main IDU without ESC extension 238
7.8.2 – Configuring ESC 3, 4 and 5 in the ESC extension unit 241
7.8.3 – Adding an ESC extension IDU 249
7.9 – Configuring ESC3 and the NMS ports (Light IDU versions) 250
7.9.1 – Switch positions 250
7.9.2 – Methodology 250
7.10 – Configuring remote indication loops 251
7.10.1 – Classic (main) IDU remote indication 251
7.10.2 – Light IDU versions remote indication 252
7.11 – Changing redundancy configurations 253
7.11.1 – Changing from 1+0 to 1+1 without ESC extension or MUX protection 253
7.11.2 – Upgrading from 1+0 to 1+1 without multiplexer protection 254
7.11.3 – Upgrading from 1+0 to 1+1 with multiplexer protection and access idu 254
7.11.4 – Adding MUX protection with access IDU to a 1+1 configuration 255
7.11.5 – Upgrading from 1+0 to 1+1 with multiplexer protection without access idu 255
7.11.6 – Adding MUX protection without access IDU to a 1+1 configuration 256
7.12 – Changing power level 257
7.12.1 – All frequencies bands except 9418UXR201 and A9418UXR202 257
7.12.2 – Frequencies bands without RTPC 257
7.13 – Changing 1+1 HSB coupler polarization 259
7.14 – Changing SIMM memories board 260

7.15 – Replacing consumable items 262
7.15.1 – MCU board battery 262
7.15.2 – Changing fans 263
7.15.3 – Changing a fuse of an IDU 265

Appendix 1 – Human exposure to electromagnetic fields 267
A.1.1 – Introduction 267
A.1.2 – Standards and regulations 267
A.1.2.1 – ICNIRP guidelines 267
A.1.2.2 – European regulation 267
A.1.3 – Evaluation of safety perimeters 268
A.1.4 – Safety perimeters 268
A.1.5 – Implementation of protection rules 270
A.1.6 – Conclusion 270

Appendix 2 – Installation information 271

Appendix 3 – Set of consumables 273

Appendix 4 – Service kit 274
A.4.1 – Service kit 9400UXT102 for ODU 94xxUXR201 to 94xxUXR203 274
A.4.2 – Short service kit 9400UXT113 for A9400UX flat ODU 275

Appendix 5 – Assembling “N” type coaxial connectors 276
A.5.1 – Type 1 fitting on cable 1AC001100022 276
A.5.2 – Type 2 mounting on cable 1AC001100022 277

Appendix 6 – Pin out of user ports 278
A.6.1 – Tributaries 278
A.6.1.1 – 2 Mbit/s tributaries 278
A.6.1.2 – 34 Mbit/s tributaries 279
A.6.2 – Engineering service channels 279
A.6.2.1 – Supervision bus interface 279
A.6.2.2 – Telephone ESC with selective calling 280
A.6.2.3 – Engineering service channels ESC 3, ESC 4 and ESC 5 281
A.6.3 – Alarms and TS/TC 284
A.6.3.1 – “ESC3 / ALARM I/O” connector on the Light IDU versions 284
A.6.3.2 – “ALARMS I/O” connector on the main IDU 284
A.6.4 – “F” connector 284
A.6.4.1 – IDU “F” connector 284

Appendix 7 – List of equipment alarms 285

Appendix 8 – List of remote controls 290

Appendix 9 – List of RTP application alarms 292

Appendix 10 – Creating user profiles 294

Appendix 11 – NECTAS configuration 295
A.11.1 – Nectas 3.79 295
A.11.1.1 – Updating the date and time on an unsupervised network 295
A.11.1.2 – Time synchronization 296
A.11.1.3 – Editing the RTP polling period (POLRTP variable) 296
A.11.1.4 – Modification of the polling pediod of analogue measurements (POLMEA variable) 297
A.11.2 – Nectas 4.1x 298
A.11.2.1 – Time and date updating of a network without supervision 298
A.11.2.2 – Editing the RTP polling period (POLRTP variable) 298
A.11.2.3 – Modification of the polling pediod of analogue measurements (POLMEA variable) 299

Appendix 12 – List of models and commercial items 300
A.12.1 – Hardware 300
A.12.2 – Software 302
A.12.3 – Commercial items 305

Appendix 13 – List of abbreviations 308

INDEX 311

PAGE INTENTIONALLY LEFT BLANK

1 – Foreword

1.1 – Structure of the manual

This manual is for users with a sound knowledge of how to operate and install microwave links and how to use a PC-based craft terminal running the Windows t operating system. With it, you should quickly be able to operate the equipment. It is not intended to replace the training services that we can provide for your particular needs.

The manual is divided into seven sections followed by appendixes and an Index.

- **Foreword**
- **Description of the equipment**
- **Installation**
- **Configuration and operation software**
- **Commissioning**
- **Operation, Servicing and Maintenance**
- **Changing configurations**
- **Appendixes**
- **Index**

1.2 – Using the manual

With this manual, you should be able to commission and operate the described equipment to a basic level.

You should always read this manual in conjunction with the attached “Update” document (if provided) so that you are aware of the latest equipment upgrades.

Manual updates

This edition of the manual describes hardware and software releases of the revision indexes (and above), detailed in Appendix 12.

In cases where an equipment upgrade affects the content of the manual, the relevant modification should be inserted in the “**Update**” document, with the same reference number, but with code type VE (instead of TQ).

When the number or extent of the changes justifies it, they should be incorporated in the body of the manual and the manual’s revision index should be incremented.

Previous versions of this issue

Previous versions of this documentation are available on request

Note: MS-DOS, MICROSOFT and WINDOWS are registered trademarks of Microsoft Corporation.

1.3 - Safety instructions

1.3.1 - General rules

The following general safety precautions must be observed by the installer and the operator. ALCATEL assumes no liability for the customer's failure to comply with these requirements.

Ground the equipment: For Safety Class 1 equipment, always connect the earth conductor of the power cable to an appropriate earthing device.

DO NOT operate the product in an explosive atmosphere or in presence of flammable gases or fumes.

For protection against fire: replace the line fuse(s) only with fuse(s) of the same voltage and current rating and type.

Dangerous voltages: Users must not remove equipment covers or shields. The installation and maintenance procedures described in this manual are for use by service-trained personnel only.

DO NOT operate equipment which may be damaged: Whenever it is possible that the safety protection features built into this equipment have been impaired, ISOLATE FROM THE POWER SUPPLY and do not use the equipment until safe operation can be verified by service-trained personnel. If necessary, return the equipment to Alcatel After Sales for service and repair.

DO NOT substitute parts or modify equipment: Return the product to Alcatel Customer Service for servicing and repair.

Each main IDU contains a lithium battery to back up data. This battery should be changed every 7 to 10 years, when inventory memory becomes empty following a power outage (see § 7.15.1).

CAUTION
There is a risk of explosion if the incorrect battery is used.
Only replace with a battery of the same type:
M4T28-BR12SH1, Supplier: STMicroélectronics
Observe all applicable national standards regarding the disposal of used batteries.
Do not burn and do not recharge.

1.3.2 - Symbols

1.3.2.1 - Danger symbols

When subsystems and modules have warning labels, it is extremely important to follow their instructions.

These labels are designed to indicate dangerous situations; they may contain any standard symbol or any text considered necessary to protect users and employees.

The most frequent danger situations and symbols are:

Danger or general warning



Prompts the user to refer to the manual.

Dangerous electrical voltages



Close to dangerous voltages (>42.4 V AC peak, 60 V DC; power level ≥240 VA) you will find this warning label



Presence of heat-radiating mechanical parts

1.3.2.2 - Earth symbols



Terminal for connecting the protective earth conductor in power supply wiring



Other earth terminal

1.3.2.3 - Other symbols



Indicates compliance with European standards



Emissions frequency: check that this complies with the standards in use in the country.



Radiation alert

1.3.3 - Symbols used in the document

These symbols alert the reader the possible risks. They indicate:

- the cause and type of danger,
- the possible consequences,
- the preventive action.

1.3.3.1 - Warning

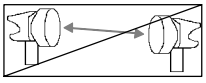


- **protection of personnel,**
- warning of a **possible dangerous situation,**
- danger of fatal or serious injury.

1.3.3.2 - Precautions



- **protection of equipment,**
- warning of a procedure, practice or condition that **could be dangerous** to equipment or its environment,
- danger of damage to the equipment or its environment; permanent loss of data possible.



- This symbol, introducing the description of a procedure, indicates that it will cause the link to be temporarily disconnected.



- This symbol, introducing the description of a procedure, indicates that it cannot be continued without a full knowledge of the data contained in the procedure sheet concerning the stations concerned.

1.3.4 – Certificate of compliance with European safety standards

DECLARATION OF CONFORMITY

We, ALCATEL CIT
5 rue Noël Pons
92734 Nanterre Cedex
France

Declare, under our responsibility, that the products

Terminal Station ALCATEL 9400 LX, LX/UX, et UX

- Frequency ranges	1.35GHz – 1.53GHz 1.7GHz – 2.7GHz 7.1GHz – 8.5GHz 12.75GHz – 13.25GHz 14.4GHz – 15.35GHz 17.7GHz – 19.7GHz 21.2GHz – 23.6GHz 24.5GHz – 26.5 GHz 37GHz – 39.5 GHz
- Interfaces	E1, E3
- Power supply	18 –60 VDC
- Modulation	4 QAM and 16 QAM

to which this certificate relates are in conformity, provided that they are installed and maintained in accordance with :

- the "state of the art",
- manufacturer's instructions,

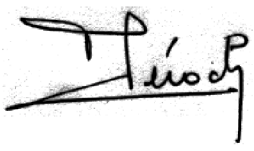
and used under normal conditions,
with the requirements of the following European Directive :

1999/5/CE of 09 March 1999, annex III, (Directive on radio equipement and telecommunications terminal equipement : R&TTE Directive).

- EMC : ETSI EN 301 489 1 & 4 (version 1.2.1)
- SAFETY : EN 60950 (1992 + A1 + A2 + A3 + A4)
- RADIO : EN 301 751 V1.1.1 (2000-12)
  ETSI standards : EN 300 197, EN 300 198, EN 300 431, EN 300 630, EN 300 633, EN 301 216, EN 301 128

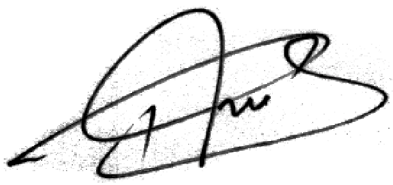
The tests having made it possible to establish this certificate were carried out by:

LCIE	CETECOM	Alcatel-Italia
33 av du Général Leclerc	ICT Services GmbH	Via Trento 30
Fontenay-aux-Roses	Saarbrücken	Vimercate
FRANCE	GERMANY	ITALY



M. PEROCHE
ALCATEL-CIT Quality Department

6th February 2003

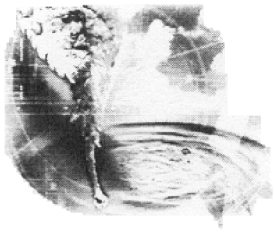


P. FOUET
WTD/F Quality Department

1.3.5 - Eco-Declaration

Product environmental attributes declaration form according to ECMA-TR/70

Small and Medium Capacity Digital Microwave Link 9400 UX



Supplier's Name : **ALCATEL CIT**

Supplier's Address : 10 rue Latécoère
78141 VELIZY CEDEX

Department : Wireless Transmission Division
5, rue Noël PONS
92737 NANTERRE Cedex (FRANCE)

Declares (based on product specification or tests results obtained from sample testing that the product :

Product Name : 9400 UX

Models	Indoor	Outdoor 13 ,15 , 18 G Hz	Outdoor 23 ,25, 38 G Hz
Weight (in kg , non packed)	3	10	8
Dimension (in mm , non packed)	44 x 450 x 235	250x250x250	220x220x250

Conforms to the environmental specifications , as listed in the Eco-declaration below .

Name , function & signature
of Authorized person

Vélizy, 2003/01/20

M. PEROCHE
ALCATEL CIT Quality Director

1.3.6 - Public exposition to electromagnetic fields

The public protection towards electromagnetic fields emitted by the antenna of the 9400UX is one of main requirements of the R&TTE Directive. An evaluation of the security measures to be implemented is presented in Appendix 1.

2 – Description of the equipment

2.1 – ALCATEL 9400 Family

The Alcatel 9400 is a family of digital point–to–point microwave radio systems, designed to satisfy the various digital transmission needs of public and private networks, for a large range of applications.

The Alcatel 9400 UX family covers the frequency range from 13 to 38 GHz, necessary to satisfy the largest range of propagation conditions as well as network configurations that gives also the ability to have a highest spectrum efficiency from 13 up to 38 GHz band with a 16QAM version.

It covers the following capacities:

- 2x2, 4x2, 8x2, 16x2 Mbit/s, 34+2 Mbit/s with the 4QAM modulation and 8x2, 16x2 Mbit/s; 34+2 Mbit/s with 16QAM modulation.

The Alcatel 9400 UX family is today used worldwide in:

- cellular networks,
- fixed access networks for Public Telecom or CLECs (Competitive Local Exchange Carrier), including:
  - infrastructure of Wireless IP networks,
  - direct complementary point–to–point access in LMDS (Local Multipoint Distribution Services) networks,
- private networks,
- also in: Utility networks, Security/Defense networks…

Typically the distances achieved are:

- from 5 to 30 km for the microwave frequencies (13 to 18 GHz),
- from a few hundred meters up to 10 km for the millimeter wave frequencies (23 to 38 GHz).

2.1.1 – Simplified description

2.1.1.1 – General

The A9400 UX is fully compliant with the international requirements in the 13, 15, 18, 23, 25 and 38 GHz bands with the main Tx/Rx separations available depending on the national regulations.

Alcatel 9400 UX – 2x2 to 16x2 Mbit/s or 34 + 2 Mbit/s					
Designation	**Band (GHz)**	**Frequency (GHz)**	**ETSI Standard**	**Frequency plans**	**Duplex difference (MHz)**
9413 UX 4QAM	13	12.75–13.25	EN 301 128	ERC 12–02 ITU–R Rec F497–6	266
9413 UX 16QAM	13	12.75–13.25	EN 301 128	ERC 12–02 ITU–R Rec F497–6	266
9415 UX 4QAM	15	14.4–15.35	EN 301 128	ERC 12–07 ITU–R Rec F636–3	728, 315, 420, 490
9415 UX 16QAM	15	14,4–15,35	EN 301 128	ERC 12–07 ITU–R Rec F636–3	728, 315,420, 490
9418 UX 4QAM	18	17.7–19.7	EN 301 128	ERC 12–03 ITU–R Rec F595–3	1010, 1008,340, 1560
9418 UX 16QAM	18	17,7–19,7	EN 301 128	ERC 12–03 ITU–R Rec F595–3	1010, 1008, 340, 1560

Designation	Band (GHz)	Frequency (GHz)	ETSI Standard	Frequency plans	Duplex difference (MHz)
9423 UX 4QAM	23	21.2-23.6	EN 300 198	ERC 13-0 2 annex A ITU-R Rec F637-3	1008, 1197, 1200, 1232
9423 UX 16QAM	23	21.2-23.6	EN 300 198	ERC 13-0 2 annex A ITU-R Rec F637-3	1008, 1197, 1200, 1232
9425 UX 4QAM	25	24.5-26.52	EN 300 431 Class 1	ERC 13-02 annex B ITU-R Rec F748-3	1008
9425 UX 16QAM	25	24.5-26.52	EN 300 431 Class 4	ERC 13-02 annex B ITU-R Rec F748-3	1008
9438 UX 4QAM	38	37-39.5	EN 300 197	ERC 12-01 ITU-R Rec F749-1	1260
9438 UX 16QAM	38	37-39.5	EN 300 197	ERC 12-01 ITU-R Rec F749-1	1260

(*) Please consult factory if other Tx/Rx separations are requested.

The Alcatel 9400 UX system features high spectrum efficiency (minimum use of RF bandwidth for a given capacity). The spectrum efficiency is optimized with the use of a **4 QAM or 16 QAM modulation** (4 or 16 states Quadrature Amplitude Modulation) with digital filtering, providing compliance with the relevant ETSI spectrum masks and spurious emissions requirements (ERC Rec 74-01).

Capacity	(Mbit/s)	2x2	4x2	8x2	16x2 34+2
RF channeling (4QAM modulation)	(MHz)	3.5	7	14	28
RF channeling (16QAM modulation)	(MHz)	-	-	7	14

2.1.2 - Configuration

2.1.2.1 - Unprotected configuration

An Alcatel 9400 UX unprotected (1+0) radio terminal is made up of the following elements:

- an **Outdoor Unit (ODU)**: incorporating the complete RF transceiver (modem, RF units, branching filter) and an integrated or separated antenna.The ODU is capacity independent,
- an **Indoor Unit (IDU)**: incorporating the baseband processing and offering tributaries interfaces as well as service channel and supervision.

The IDU is frequency independent (same unit from 13 up to 38 GHz).

Different IDU configurations are available depending on system configuration:

- a light version available in 1+0 configuration and up to 4x2 Mbit/s,
- a classic version allowing all possible configurations (1+0, 1+1, capacity up to 16x2 or 34+2 Mbit/s).

These two IDU versions are compatible with a same ODU and can be mixed in the same network.

- a **single coaxial cable** of a length which can be up to 300 m.

Main IDU and Equipment Craft Terminal



1+0 9400 UX Outdoor Unit on its pole mounting, with 30 cm integrated antenna



New generation unit 9400UX flat ODU

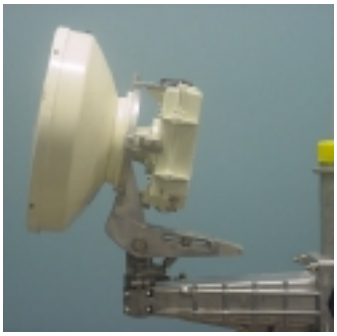


Figure 1 - Alcatel 9400 UX 1+0 configuration

2.1.2.2 – Protected configuration

Two types of problem can impact the availability of a radio link: equipment failures and propagation problems.

1+1 protected configurations are used to increase the availability of the link compared to 1+0 unprotected configuration, by adding redundant functions to the equipment on the Outdoor and Indoor parts:

- to secure the link in case of temporary (transient alarm) or definitive (equipment failure) malfunction,
- to improve the performances of the equipment towards propagation problem (in some outdoor units configurations only).

The (1+1) configurations include an automatic errorless Rx switching.

If the link is protected, then wayside traffic and auxiliary channels are automatically protected.

Various protected configurations are available with the following applications:

ODU CONFIGURATION		**APPLICATIONS**	
	9400 UX		
1+0	1x ODU 1x antenna single polarization	Non–protected	
1+1 HSB	2x ODU at same frequency 1x coupler 1x antenna single polarization	Protection against failures	
1+1 HSB SD	2x ODU at same frequency 2x antennas single polarization	Protection against failures	Improvement of performances versus multipath propagation
1+1 FD	2x ODU at different frequencies 2x antennas single polarization (or 1 antenna dual polarization)	Protection against failures	Improvement of performances versus multipath propagation

SD : Space Diversity, FD: frequency diversity

The above mentioned outdoor units configurations must be associated with one of the following indoor units configurations:

IDU CONFIGURATION		APPLICATIONS	
1+0	classic IDU Light IDU	Non-protected	
1+1 BASIC PROTECTION	MAIN IDU (classic) EXTENSION IDU (without MUX-DEMUX option)	Automatic switching: selection of the best receive path and operational transmitter (in case of HSB)	
1+1 WITH MUX-DEMUX PROTECTION	MAIN IDU EXTENSION IDU (with MUX-DEMUX option) ACCESS IDU	Automatic switching: selection of the best receive path and operational transmitter (in case of HSB)	Protection of the tributaries access including redundant mux-demux

2.1.3 - ALCATEL 9400 UX features

2.1.3.1 - Equipment flexibility

Software controlled frequency:

The in-field agility of the 9400 UX is provided on a quarter frequency plan (synthesizer step 250 kHz). Only four different ODUs maximum are needed to cover the full frequency band.

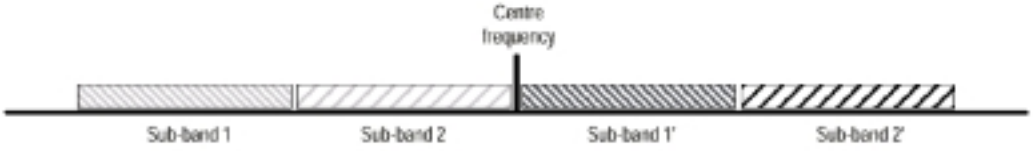


Figure 2 - Sub-band breakdown

The synthesized , software controlled, RF local oscillator allows an easy frequency setting.

Software controlled capacity:

The Alcatel 9400 UX system provides unique capacity agility features, controlled by software up to a maximum bit rate. This maximum can be adjusted from 2x2 Mbit/s to 16x2 Mbit/s using the appropriate software key.

The system capacity can be configured, by software, between 2x2 Mbit/s and the maximum authorized by the software key.

Moreover, capacity upgrade can be made easily changing only the software key without hardware change.

Transmit power control:

In certain cases, the output power of the equipment has to be decreased, to reduce interference towards other links in case of very short hops. This feature improves the flexibility of the network design and furtherly more increases the frequency re-use.

Different implementations of transmit power control are available depending on the frequency bands:

- At 13 GHz (3CC09297Axxx),15 GHz (3CC08880Axxx and 3CC08978Axxx) and 18 GHz, insertion of fixed attenuators (of 3, 6 or 10 dB) on transmission side in the IDU unit.
- for the 13 GHz 4QAM (3CC11737Axxx) and the 16QAM (3CC 11736Axxx), a software transmit power control of 20 dB in 1 dB step can be provided (as a software key option with classic IDU, embedded within the Light IDU)
- for the 15 GHz 4QAM (3CC12979Axxx), a software transmit power control of 30 dB in 1 dB step can be provided (as a software key option with classic IDU, embedded within the Light IDU)
- in the millimetric frequency bands (23 to 38 GHz), a software transmit power control of 30 dB in 1 dB step for the 4QAM and of 20 dB in a 1 dB step for the 16 QAM, can be provided (as a software key option with classic IDU, embedded within the Light IDU).

- for all new generation ODU (ref. A9400 UX flat ODU) frequency bands for the 4 or 16QAM, the software power transmit control of 30 dB in 1db step can be provided as software key.

Software key parameters:



Figure 3 - Software key

The software key, used with the classic IDU, provides flexibility on the configuration parameters of the equipment, as well as its options:

- maximum configuration definition: 1+0 or 1+1,
- maximum capacity definition: 2x2 Mbit/s up to 16x2 Mbit/s,
- set-up of transmit frequency range (7 to 40 GHz by default but configurable upon customer specific request),
- authorization for Transmit Power Control option (all bands except 18 GHz),
- authorization for Embedded Mediation option (master terminal or slave terminal),
- authorization for Remote Craft Terminal functionality option.
- authorization for 16QAM modulation switching (A9400 UX flat ODU)

The Light IDU features a fixed embedded configuration (no removable software key):

- maximum configuration: 1+0,
- capacity: 2x2 Mbit/s up to 4x2 Mbit/s,
- Transmit Power Control option enabled (all bands except 18 GHz),
- Embedded Mediation option with the supervision of only one network element,
- Remote Craft Terminal functionality option enabled.

Link Identity Coding:

The amount of microwave links, especially in urban areas puts the problem of possible interferers during installation and turn-on phase.

The digital frame incorporates Link Identity coding capabilities to prevent the capture of an unwanted signal.

2.1.3.2 - Ease of installation

The Alcatel 9400 UX systems have been designed to allow quick installation and turn-on.

Compact Outdoor Unit :

The Outdoor unit (ODU) is housed in a very compact and robust weatherproof (IP 55) container; it is delivered with a solar shield which provides a protection to the outdoor electronics in hot and very sunny climatic zones.

The Outdoor Unit (1 or 2 depending on the configuration 1+0 or 1+1) can be rapidly installed on suitable poles with their own fastening system.

The specific pole mounting is equipped with an accurate alignment system - presetting and fine “elevation” and “bearing” tuning - to permit easy pointing with the integrated antenna. A very compact pole mounting has been optimized for 1+0 configuration, it is common to 9400UX – from 13 to 38 GHz.

The 9400 UX outdoor unit can be used with an integrated antenna 300 or 600 mm diameter or separate antennas if a larger diameter is needed.

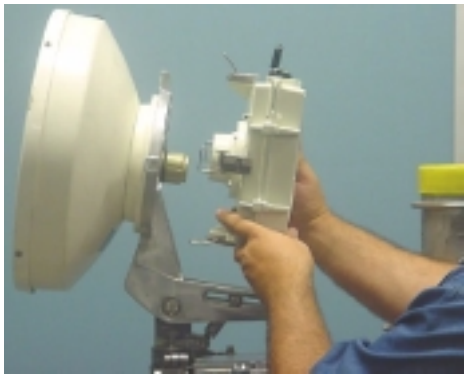
The ODU is fixed by means of quick latches. This system allows to change the ODU without altering the antenna alignment.

Two ODU generations are available. The new generation ODU, A9400 UX flat ODU, is lighter than the previous range from 94xxUXR201 to 94xxUXR203*. It is 4 and 16QAM compatible and software configurable only.

* xx is for the band frequency of the Outdoor unit. For example, a unit referenced 9413UXR204 depends on the 13GHz frequency band



ODU 94xxUXR201, 94xxUXR202 or 94xxUXR203



ODU A9400 UX flat ODU

Figure 4 - Installation of the ODU

The alignment of the antenna can be easily performed with the use of a dedicated service kit. This kit , when connected to the ODU, is able to provide the receive signal level, and the connection to the other end of the link through a telephone service channel (when applicable).

Different IDU mechanical versions:

The small dimensions of the **19" IDU** (classic or light) – 1U high for all capacities, depth 235mm – allow for easy installation in standard 19" racks and saves a lot of space where there are tight constraints like shelters or base stations in cellular or LMDS networks.

The 19" IDU can also be installed as a desktop unit.



Figure 5 - Light 19" IDU

Single coaxial cable:

A single coaxial cable of up to 300 m achieves the connection between the ODU and the IDU, so that a high flexibility is left for locating the radio and antenna.

Simple commissioning:

All the standard Site Acceptance Tests can be done with a single PC to commission the link, making extensive use of built-in supervision features:

- continuous performance check and alarm status overview,
- numerous baseband loopback facilities,
- built-in PRBS generator and error counter.

The transmit side incorporates a built–in PRBS (Pseudo Random Bit Sequence) generator able to insert a standard test signal for the link Bit Error Rate measurements, in order to facilitate initial turn–on. The bit error rate measurement is made through an optional application software available on the PC. The PRBS can be switched on remotely.

2.1.3.3 – Equipment software features

The Alcatel 9400 UX terminal provides a F interface to a Craft Terminal (PC), which enables alarm monitoring, quality measurement, as well as configuration of the equipment.

The Alcatel 9400 UX Network Element (NE) supports a basic group of applications, listed hereafter, which provide simplified testing, operation and maintenance of the equipment:

- "Administrative Functions":
  - Depending on the operator needs, different user profiles can be created with an associated password to access or not to the applications (in read–only or read/write mode),
- "Installation parameters" and "Operation parameters":
  - Declaration of the equipment hardware settings for ODU and IDU: type of configuration protected or unprotected, frequency band, number of tributaries, presence of options…
  - Configuration of: capacity, Link Identity Code, Tx & Rx frequencies, output power (if available through software), 2 Mbit/s tributaries cross–connect…,
- "Alarm, Status and Control" (AS&C):
  - Possibility to activate or inhibit alarm indications,
  - Possibility to configure alarms as urgent or non–urgent,
  - The software displays a synthesis of the monitored Urgent, Non Urgent & Attended alarms (which are also available via dry loops),
  - Possibility to set the threshold values at which the alarms are triggered,
  - Possibility to set telecommands such as : loopbacks, manual switch, transmit power mute…
- "Performance Monitoring G821" (except for LUX40):
  - Link quality monitoring : Errored Seconds, Severely Errored Seconds, Unavailable Seconds.
- "Radio Transmission Parameters" (RTP)
- Tx and Rx power, BER, initial diagnostic on Tx and Rx alarms.

As an extension, the Network element can support a group of advanced applications which provide the operator with an enhanced level of equipment management:

- "Remote inventory":
  - The remote inventory is used to read the information allowing a complete identification of the selected Network Element (Indoor Unit and Outdoor Unit): serial number, date of manufacture, reference and revision index of the hardware and software releases.
- "Performance Monitoring G784", including G826:
  - This application allows the performance analysis of the Terminal Points supported by the NE in accordance with ITU–T Rec G784.
- "Maintenance Memory":
  - The maintenance memory application allows to display all dated events stored in the equipment when the Craft Terminal is not connected.
- "Analog measurements":
  - This application is used to take measurements on various measurements points and displays them on the screen: min Rx power, current Rx power, max BER, current BER, status of secondary power supply voltages..

2.1.3.4 - Network Management

The network management solutions proposed are based on an advanced concept offering the following advantages :

Global synthesis view:

The management system offers a global synthesis view, displaying in real time the synthesis of the alarms of the complete network, as well as an alarm synthesis line for each Network Element of the network. This feature allows a very quick and efficient trouble shooting.

Network Element view:

Through the alarm synthesis line of a Network Element, it is possible to zoom into the selected terminal in order to get an advanced trouble shooting using the Alarm Status & Control application.

Remote Craft Terminal management (RCT):

A remote mediation function can be activated on any Alcatel 9400 network element (slave configuration). From the CT connected to a remote network element, the operator gets the complete view of the network and is able to perform the same operations of supervision allowed from the craft terminal connected to the embedded mediation equipment (alarms, configurations…).

The CT connected to the mediation and the RCT are able to run in the same time. Conflict accesses are managed at the Mediation level.

The Remote Mediation function is available with a dedicated software option (offered as a standard solution with Light IDU).

Up to two (2) RCTs are allowed to run in the same time into the network.

For networks up to 64 Network Elements (called a cluster), a stand-alone solution called **Embedded Mediation Function** is proposed.

It allows to display the global synthesis view as well as the NE view of each equipment of the cluster (as described here-above).

To perform this function, the following equipment is required:

- a 9400 terminal with a classic IDU equipped with a master software key (Embedded mediation function),
- different 9400 Terminals (with classic or Light IDU versions).

Interface to a higher NMS level, can also be proposed, using:

- the Alcatel Telecommunication Management Network (TMN), based on Alcatel RQ2 protocol, able to manage Alcatel plesiochronous and synchronous transmission equipment, providing advanced network management features.
- or using a hierarchical management structure based on the new Simple Network Management Protocol (SNMP), which can manage the 9400 UX equipment, including also 9470 LX/UX.

2.1.4 – Capacity configurations

2.1.4.1 – Classic IDU version

The capacity configurations depend:

- on the type of line interface units (LAU and LIU) installed in the IDU(s),
- the software key used.

The table below summarizes the capabilities according to the bit rates supported by the IDU of the equipment and the type of software key used.

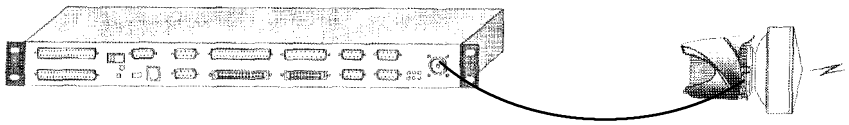
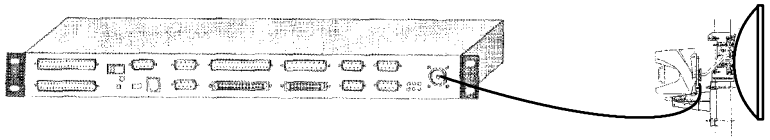
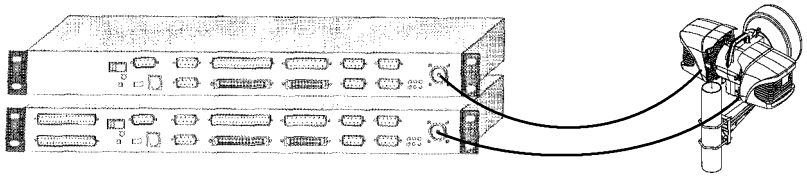
IDU TYPE	BIT RATES SUPPORTED BY SOFTWARE KEY			
	2x2	4x2	8x2	16x2
4 x 2	2x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s
8 x 2	2x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s 8x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s 8x2 Mbit/s
16 x 2	2x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s 8x2 Mbit/s	2x2 Mbit/s 4x2 Mbit/s 8x2 Mbit/s 16x2 Mbit/s
34	–	–	–	34 Mbit/s +2 Mbit/s auxiliary

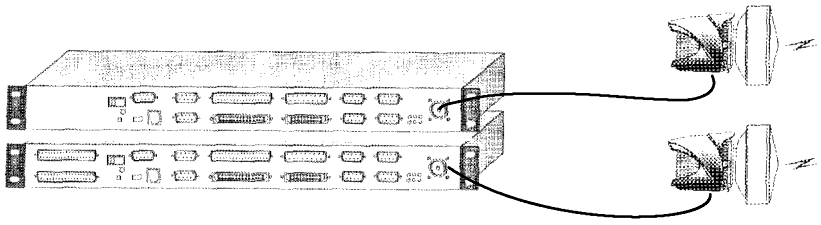
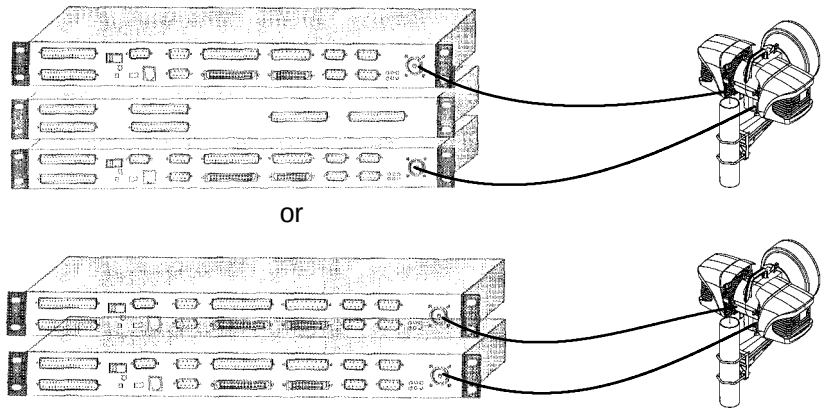
2.1.4.2 – Light IDU versions

The Light IDU versions cover the following bit rates (software configured):

- 2x2 Mbit/s,
- 4x2 Mbit/s.

2.1.5 – Typical hardware configurations

1+0 configuration integrated antenna	
1+0 configuration with separate antenna	
1+1 HSB configuration using an HSB coupler and integrated antenna	

1+1 HSB or frequency diversity configuration with two antennas	
1+1 HSB configurations with multiplexer protection	

The block diagrams representing the main configurations appear in Figures 6 to 9 on the following pages.

2.2 – Operation

2.2.1 - 1+0 configuration

A block diagram of this configuration is given:

- in figure 6 for the classic IDU version,
- in figure 7 for the Light IDU versions.

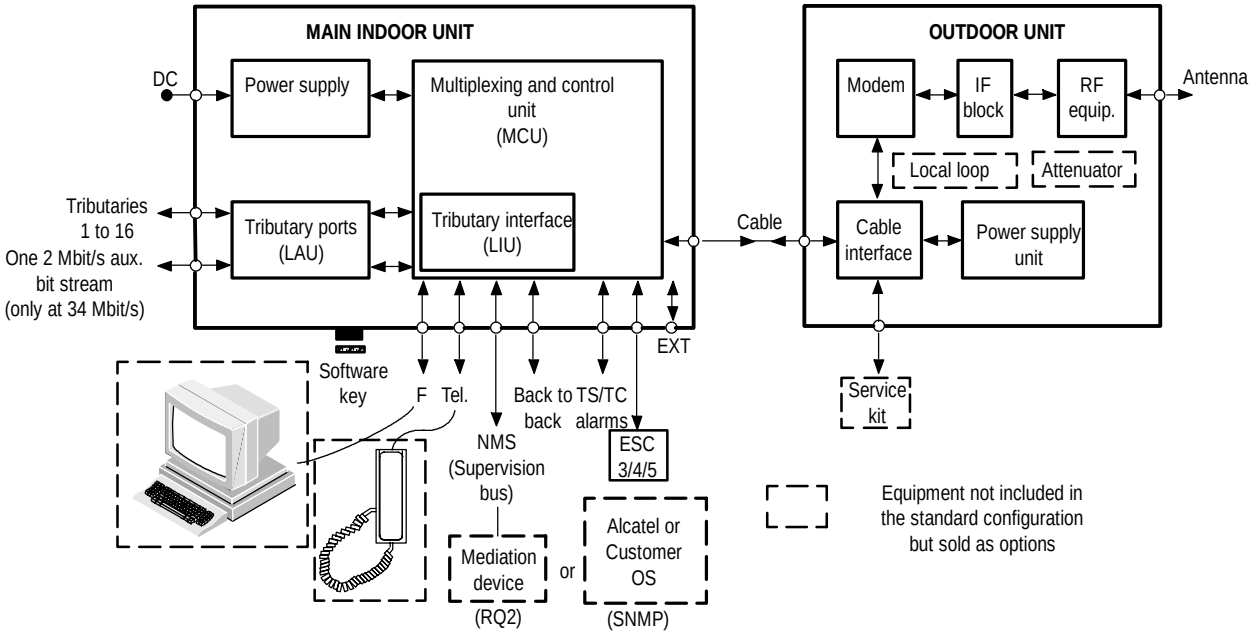


Figure 6 - Block diagram of a 1+ 0 station (classic IDU version)

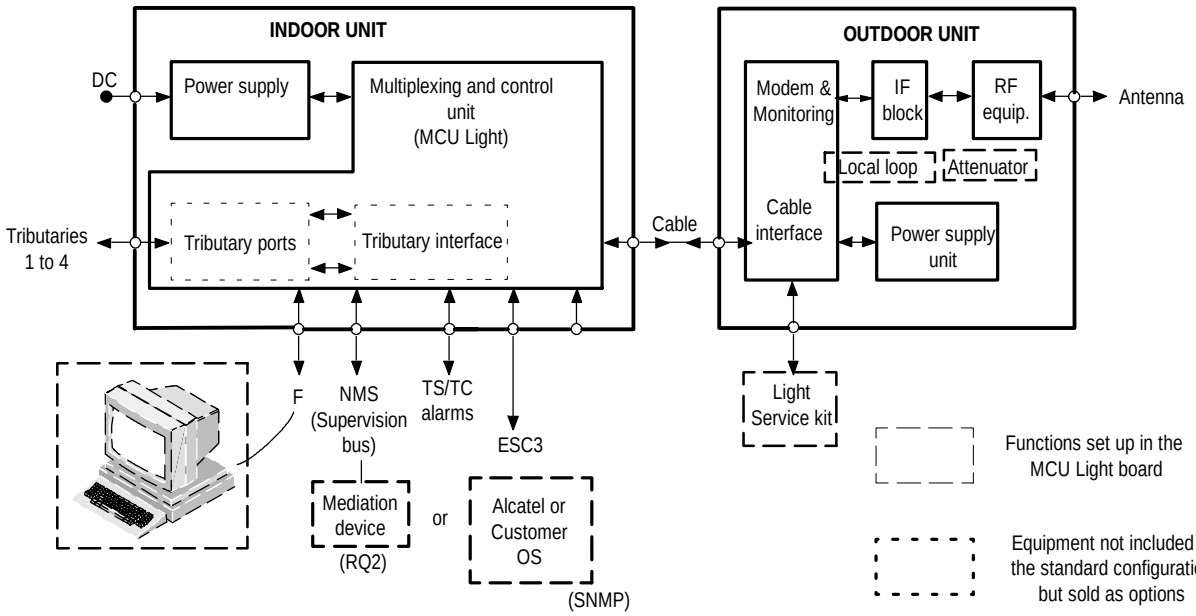


Figure 7 - Block diagram of a 1+ 0 station (Light IDU versions)

2.2.1.1 – Operation of the main IDU (InDoor Unit) (classic IDU version)

The **Line Access Units (LAU)** offer all the passive functions required to provide transmit and receive G.703 interfaces for the tributaries and 2 Mbit/s auxiliary bit stream (when there is one). This mainly entails impedance matching. Two types of board can be used:

- LAU 3CC06059Axxx: eight ports for 2 Mbit/s tributaries (two boards of this type are used for the 16 x 2 Mbit/s configuration),
- LAU 3CC06061Axxx: one port for 34 Mbit/s + one 2 Mbit/s auxiliary bit stream port.

The **LIU (Line Interface Unit)** boards provide all the active functions required for transmit and receive mode G.703 interfacing of tributaries and the 2 Mbit/s auxiliary bit stream (mainly HDB3 { HCMOS code conversion). The LIU boards are daughter boards that plug into the MCU board. Three types of board can be used:

- LIU 3CC05818Axxx: four ports for 2 Mbit/s tributaries,
- LIU 3CC06026Axxx: eight ports for 2 Mbit/s tributaries,
- LIU 3CC06119Axxx: eight port extension unit, used in conjunction with the eight port LIU board, 3CC06026Axxx, for the 16 x 2 Mbit/s configuration,
- LIU 3CC06118AAxx: one port for 34 Mbit/s + one 2 Mbit/s auxiliary bit stream port.

The **Multiplexing and Control Unit (MCU)** handles:

- multiplexing/demultiplexing functions, to assemble the aggregate for transmission and restore the tributaries from the received aggregate,
- cross-connecting of 2 Mbit/s tributaries,
- dropping/insertion of engineering service channels (up to 5) and the 2 Mbit/s auxiliary bit stream (when there is one):
  - S ESC N_1: reserved for the supervision channel,
  - S ESC N_2: telephone channel with selective calling,
  - S ESC N_3: V.11 or V.28 64 kbit/s interface,
  - S ESC N_4 and N_5: V.11 64 kbit/s interfaces (not available for the 2 x 2 Mbit/s rate),

- connection of local supervision system, for connecting a supervision PC; depending on the software versions, two types of SIMM memory are used:
  - a basic version for 946LUX11 and 946LUX12 operating system versions (RQ 2)
  - an extended version (option sold by reference 9400UXB267) for 946LUX40 software (SNMP)
- processing of external commands: telesignalling and remote controls (TS/TC),
- control of the outdoor unit,
- interfacing with the outdoor unit, for:
  - power feed to the ODU,
  - transmitting and receiving the aggregate,
  - transmitting and receiving ODU supervision function transmission and reception,
  - telephone ESC transmission and reception between the IDU and the ODU,
- storage of events in battery backed maintenance memory.

The **Power Supply Unit (PSU)** is available in two versions, 24 V or 48 V DC floating, for primary power supplies. It provides DC/DC conversion to generate the necessaries voltages required to operate the indoor unit and the primary power supply voltage of the outdoor unit.

2.2.1.2 – Operation of the Light IDU versions

The **Light MCU (Light Multiplexing and Control Unit)** handles:

- all passive functions (mainly impedance matching) and active functions (mainly HDB3 { CMOS code conversion) needed for transmit and receive mode G.703 interfacing of the four tributaries,
- multiplexing/demultiplexing functions, to create the aggregate to be transmitted and recreate the tributaries from the received aggregate,
- cross connecting of 2 Mbit/s tributaries,
- engineering service channel drop and insert functions:
  - ESC N_1: reserved for the supervision channel,
  - ESC N_3: V.11 or V.28 64 kbit/s interface,
- connection of local supervision system, for connecting a supervision PC. The memory required for the 946LUX12 system software (RQ 2) is installed on the Light MCU board. To operate the LUX40 software (SNMP), additional SIMM memory must be added (option sold by reference 9400UXB267).
- processing of external controls: telesignalling and remote controls (TS/TC),
- control of the Outdoor unit,
- interfacing with the Outdoor unit for:
  - transmission and reception of the aggregate,
  - transmission and reception of ODU supervision channels,
  - ODU power feed.

For the **Light IDU,** the types of power supply available are:

- 19" Light IDU: ¦ 24 V DC or ¦ 48 V DC floating (This version is a low power consumption version, allowing the operation of the Light IDU without fans).

2.2.1.3 – IDU/ODU cable

A coaxial cable, only available in 50 ohm impedance version, with a maximum length of 300 meters (when of standard type), connects the IDU to the ODU. This carries:

- the digital data streams between the IDU and the ODU, comprising:
  - the aggregate obtained from tributary multiplexing,
  - the ODU control signals,
  - in the case of a classic IDU, the IDU/ODU telephone channel (connected to ESC N_2).
- the ODU's primary DC power supply voltage.

2.2.1.4 – OutDoor Unit (ODU) operation

The reference of the outdoor unit depends on the frequency band.

The **cable interface** handles:

- reception and transmission of the data streams,
- separation of the signals and the power supply voltage to feed the unit's subsystems,
- the interface for the aggregate, for transmission or reception, with the modem.

The **Modem** (modulator/demodulator) function receives the aggregate from the cable interface and processes the baseband signal for microwave transmission by 4QAM or 16QAM modulation, as follows:

- in transmission:
  - digital filtering and error correction coding (FEC),
  - digital/analogue conversion of bit streams I and Q from the encoding function;
- in reception:
  - a part of the automatic gain control (AGC) function,
  - analogue/digital conversion of the I and Q bit streams from the demodulation function.

It interfaces with the IF unit for the transmit and receive the I and Q bit streams.

The **IF block** handles modulation and demodulation of the aggregate, using the following functions:

- in transmission:
  - analogue filtering of the I and Q bit streams (two filters that can be switched according to capacity),
  - 4QAM or 16QAM modulation of an IF carrier, according to the frequency plan adopted, generated by a local internal oscillator,
  - level amplification and tuning (AGC), based on measurement of the rms power output from the block;
- in reception:
  - amplification, with AGC, of the received signal, transposed via the receive system according to the frequency plan adopted,
  - demodulation of the signal, so that the I and Q bit streams can be recovered,
  - amplification and analogue filtering of the bit streams (two filters that can be switched according to capacity),
  - amplification, with AGC, before transmission of the I and Q bit streams to the Modem board.

The **transmit mixer** transposes the intermediate frequency to a microwave frequency. Frequency agility is obtained by a synthesized local oscillator (LO).

The transmit RF signal is then transmitted to the **power amplifier**, monitored by an output power slaving device and a circuit for cutting off carrier transmission if a fault is detected in the transmit circuits.

For the 13 GHz and the 23 to 38 GHz frequency bands, the transmitted power can be adjusted in 1 dB steps by software configuration (software key option needed), within a range of:

- –20 dB from nominal output power for 13 GHz (94xxUXR202 and 94xxUXR203),
- –30 dB from nominal output power for 15, 23, 25, 38 GHz (4QAM RTPC),
- –20 dB from nominal output power for 23, 25, 38 GHz (94xxUXR203 16QAM).
- –30 dB from nominal output power for all frequency bands of the ODU 9400UX flat ODU.

For the 13, 15 and 18 GHz frequency bands (without RTPC option), attenuators can be inserted on the transmit output (optional).

The receive RF signal from the branching filter, is applied to the **RF receive circuit**, comprising a cascaded RF low noise amplifier (except at 38 GHz), a mixer and an IF amplifier. The mixer, which receives the amplified RF frequency, transposes the RF signal to a receive IF signal using the same LO as the transmit mixer.

When amplified, the signal is sent to the IF block's receive circuits for demodulation.

The **branching filter** comprises a duplexed filter for separating the transmit and receive RF signals present at the antenna port. It can accommodate the **RF loopback** (94xxUXR201, 94xxUXR202 and 94xxUXR203, and included into A9400UX flat ODU), which uses a conversion stage for converting transmit and receive frequencies for test purposes.

The **PSU** receives the primary voltage from the IDU via the link cable and, from it, generates the voltages required to operate the subsystems of the Outdoor unit.

2.2.1.5 - ESC extension configuration

This extension, **available only in the classic IDU version**, can be used to customize ESC N_3 and N_4 and 5 as detailed in § 2.3. It requires the presence of an extension IDU with a dedicated **SCU (**Service Channel Unit**)** board.

2.2.2 - Basic 1+1 configuration

This configuration is available only in the classic IDU version.

It comprises (Figure 8) a main IDU (identical to the one in the 1+0 configuration), an extension IDU and two ODUs.

The extension unit contains a **Service Channel Unit (SCU)** which handles the following functions:

- provision of the user interfaces for ESCs 3 to 5,
- control of the second ODU,
- interfacing of the second ODU's link cable.

The MCU of the main unit sends the SCU the aggregate to be transmitted and receives the receive aggregate from it. The MCU is responsible, in reception, for switching between the normal and standby channels.

There are three configurations:

- Hot Standby (HSB), one antenna port
- Hot Standby, two antenna ports
- Frequency diversity.

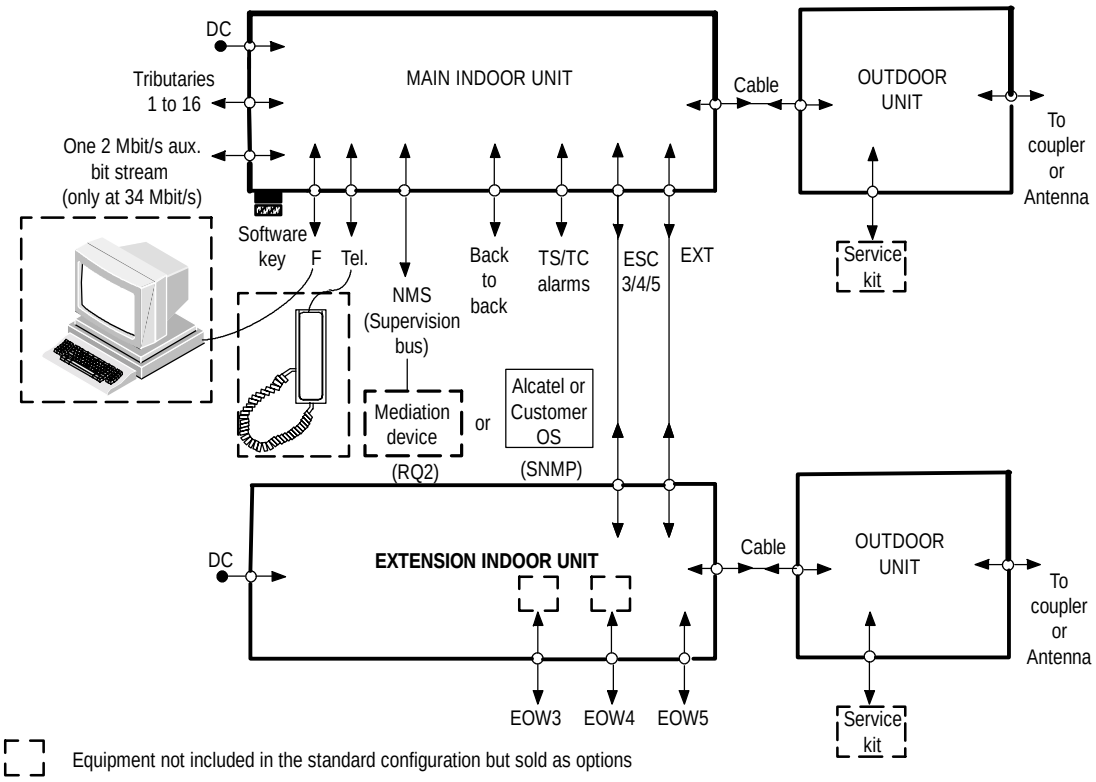


Figure 8 - Block diagram of 1+1 station

2.2.2.1 – HSB configuration with one antenna port

The two ODUs are connected by an **HSB coupler** which provides a 10 dB attenuation on the standby channel. In addition, by design, the standby amplifier's power is:

- reduced by 15 dB for the ODU which does not permit an output power control,
- cut off for the ODU which permits the output power control.

The HSB switching of a transmitter is based on detection of a fault which can originate from a failure:

- of the cable interface,
- of the ODU power supply,
- of the transmit IF synthesizer,
- of the RF synthesizer,
- of the amplifier.

The system switches back to the normal channel automatically as soon as it is disconnected and reconnected again without alarm (because of imbalance in the system), in transmission and in reception.

Switching can also be remotely controlled, by manually switching to the standby channel. The remote control for cutting off transmit power does not cause a switchover.

2.2.2.2 – HSB configuration with two antenna ports

This configuration uses either a double polarization antenna or two antennas (in this case, it also adds space diversity). One system sends (the standby transmitter is cut off). Operation is the same as in the previous configuration, apart from the switching, in which reversibility is not required because the link budgets are identical for both channels.

2.2.2.3 – Frequency diversity configuration

The two ODUs send the aggregate simultaneously at two different frequencies. The MCU chooses the best of the two signals on reception. This configuration also uses two antenna ports.

2.2.3 – 1+1 configuration with multiplexer/demultiplexer protection

This configuration is available only in the classic IDU version.

It can be used (see Figure 9) to protect the link from failures of the MCU board in the main IDU. It comprises:

- a main IDU, as in the 1+0 configuration but whose tributaries are configured in high impedance.
- an extension IDU.
- an access IDU.
- two ODUs.

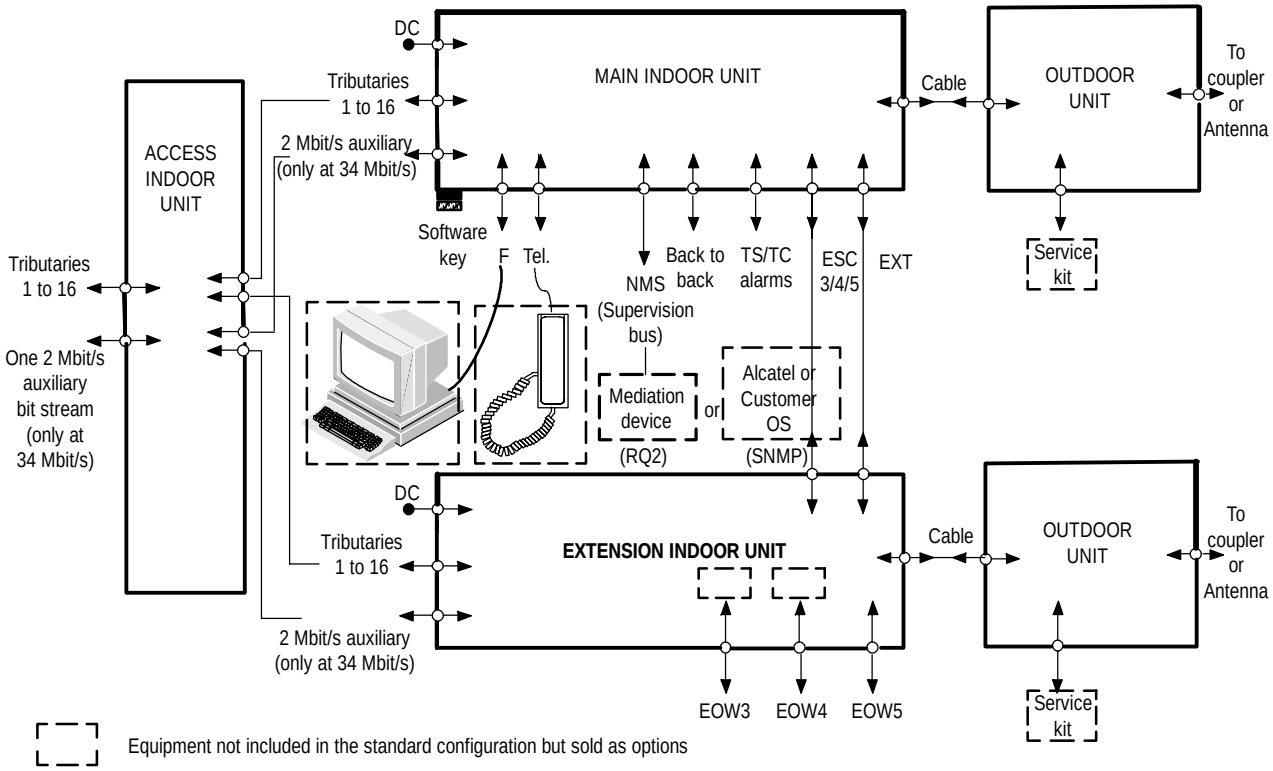


Figure 9 - Block diagram of a 1+1 station with multiplexer protection

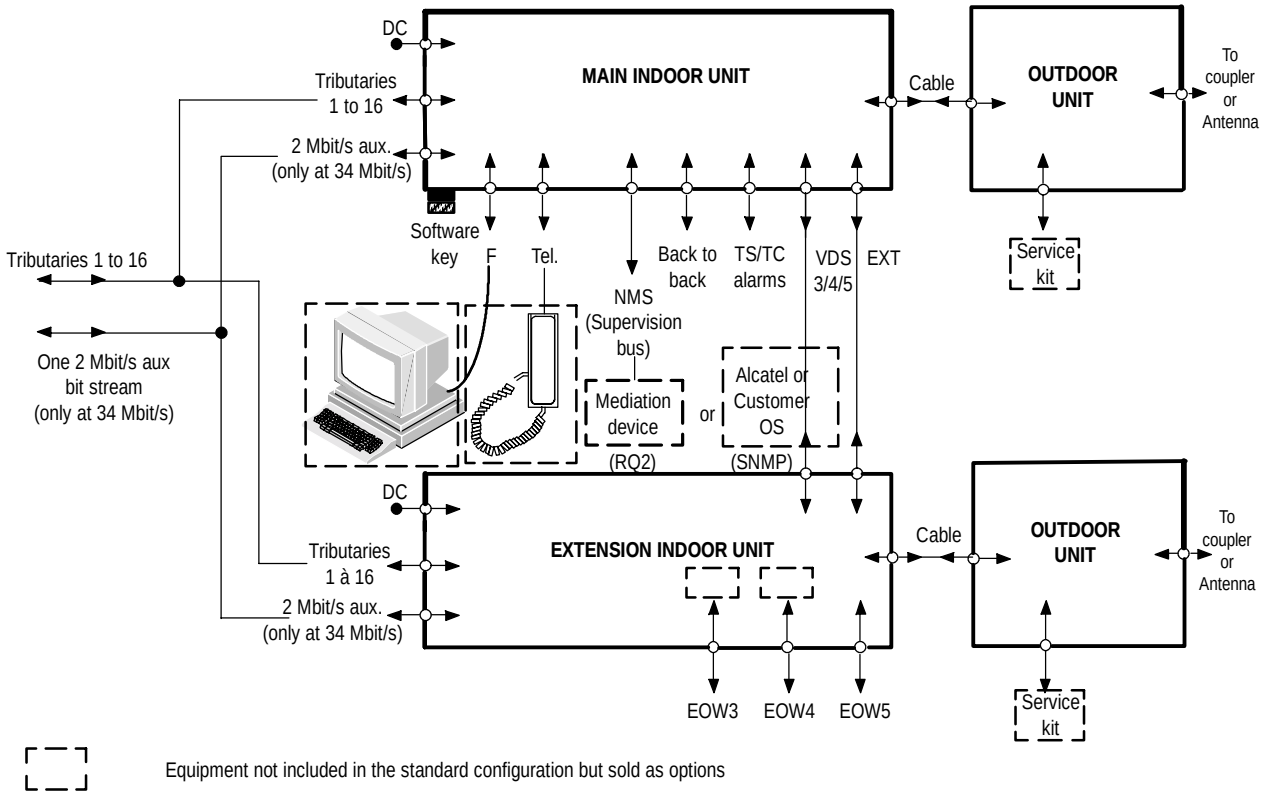


Figure 10 - Block diagram of a 1+1 station with multiplexer protection without access unit

The **extension IDU** contains the same LAU and LIU boards as the main IDU, configured in high impedance, plus an SCU board to handle the following functions:

- hitless switching on reception,
- user interfaces for ESCs 3 to 5,
- customization of ESCs 3 and 4 by the addition of daughter boards (see § 2.3),
- control of the second ODU,
- interfacing for the second ODU's link cable.

The MCU boards of the main IDU and the SCU board of the extension IDU exchange the aggregates sent and received.

The **access IDU** contains a dedicated LAU combination board which handles:

- in transmission, distribution of the tributary signals to the main and extension IDUs,
- in reception, transmission of the tributary signals from the active IDU.

All the basic 1+1 configurations are also available in 1+1 configurations with multiplexer protection (refer to § 2.2.2.1 to 2.2.2.3).

2.3 – Engineering service channels

The equipment comprises:

- **in the classic IDU configuration**: five engineering service channels (ESC N_1 to ESC N_5), except in the 2 x 2 Mbit/s configuration (only three ESC).
- **in the Light IDU configuration**: two engineering service channels (ESC N_1 and ESC N_3).

In the main classic IDU and in the Light IDU, the characteristics of these engineering service channels are predefined.

The extension IDU (classic IDU version) can be used to customize ESC N_3, N_4 and N_5. This unit has a SCU board installed with daughter boards to customize ESC N_3 and N_4.

The table below summarizes the ESC options available according to whether the configuration includes a main IDU (classic, Light) or a classic main IDU with an extension IDU.

ESC	1+0 CONFIGURATION WITH Light IDU	1+0 BASIC CONFIGU-RATION WITH classic IDU	OTHER CLASSIC IDU CONFIGURATIONS
ESC N_1	Supervision	Supervision	Supervision (1)
ESC N_2		Telephone with selective calling	Telephone with selective calling (1)
ESC N_3	V11 or V28 64 kbit/s	V11 or V28 64 kbit/s (2)	Depending on daughter board used: (3) S **64 kbit/s point-to-point, codirectional G.703 interface,** ref.: 3CC05710AAxx S **Four-wire analogue, 600 ohms**, ref.: 3CC05711AAxx S **64 kbit/s point-to-point, V11 or V28** ref.: 3CC05712AAxx S **Digital asynchronous (1 x 9600 bit/s or 2 x 4800 bit/s), V11 point-to-point or point-to-multipoint** (three-way coupler), ref.: 3CC05713AAxx S **Digital asynchronous (1 x 9600 bit/s max. or 2 x 4800 bit/s max.), V28 point-to-point or point-to-multipoint** (three-way coupler), ref.: 3CC06482AAxx
ESC N_4 (Bit rate > 2x2 Mbit/s)		64 kbit/s V11 (2)	As ESC N_ 3 (3)
ESC N_5 (Bit rate > 2x2 Mbit/s)		64 kbit/s V11 (2)	**64 kbit/s V11 or V28** (3)

(1) Access on main IDU.

(2) Access on main IDU if there is no extension IDU.

(3) Access on extension IDU.

2.3.1 – ESC number 1

ESC N_1 supports the bus reserved for supervision. The bus is implemented between the stations by a radio link (channel 1 and channel X in a 1+1 configuration) and is available in the stations, at connectors NMS1 and NMS2, for setting up point-to-multipoint links (Figure 11).

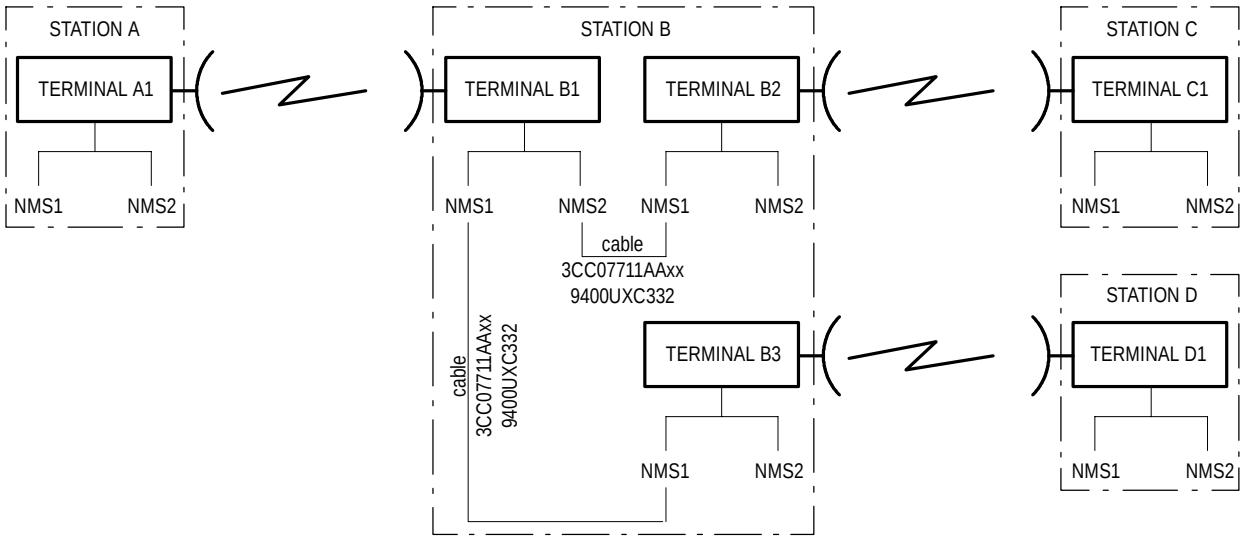


Figure 11 - Supervision links

With 946LUX11 and 946LUX12 software (RQ2 interface), each terminal is identified by a physical address from 0 to 255, with address 0 reserved for the supervisory station and address 1 to the maintenance. Two terminals in the same network should not have the same physical address.

With the 946LUX40 software (SNMP interface), 254 up to 65534 addresses are available when the sub–network is choosen in class B. Two terminals in the same network should not have the same IP address.

As many configurations are possible, it is recommended to contact Alcatel for the network configuration.

With integrated mediation (**classic IDU configuration option**), one of the stations in the network can be designated as a “master” station. It then supervises a network of up to 63 Alcatel 9400LX or UX network elements.

Supervision bus continuity is provided:

With 946LUX11 and 946LUX12 software (RQ2 interface) via a switching matrix defined in the system software (“Operation Parameters” application, see § 4.5.2). This matrix will interconnect radio (MSU1), cable (NMS1, NMS2) and equipment (SCC1) input/output ports via the bus (Figure 12A). The six interconnections are normally used (“Standard configuration” option in the 946LUX11 software, or all the boxes checked in the 946LUX12 version).

With the 946LUX40 software (SNMP interface) each port is enabled or disabled in the system software (“Operation Parameters” application, see § 4.5.3): MSU1 radio inputs/outputs, NMS1 and NMS2 cables.

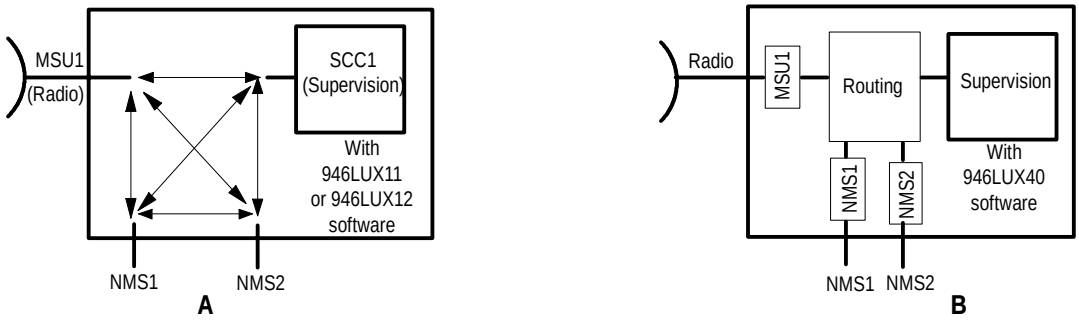


Figure 12 - Supervision bus continuity

2.3.2 - ESC number 2

Engineering service channel N_2 is a telephone channel with selective calling **available only in the classic IDU version**. The handset is connected to the  connector on the main IDU. Each terminal is assigned a call number between 011 and 999 (not including X00) on installation. The number 00 is for general calling. Two terminals in the network should not have the same telephone number. ESC N_2 is available (Figure 13):

- in digital form, at the “BACK TO BACK” connector of the main IDU, for digital transfers in stations with two terminals back-to-back.
- in analogue form, at the “AUDIO 1” and “AUDIO 2” connectors of the main IDU, for setting up a point-to-multipoint link,

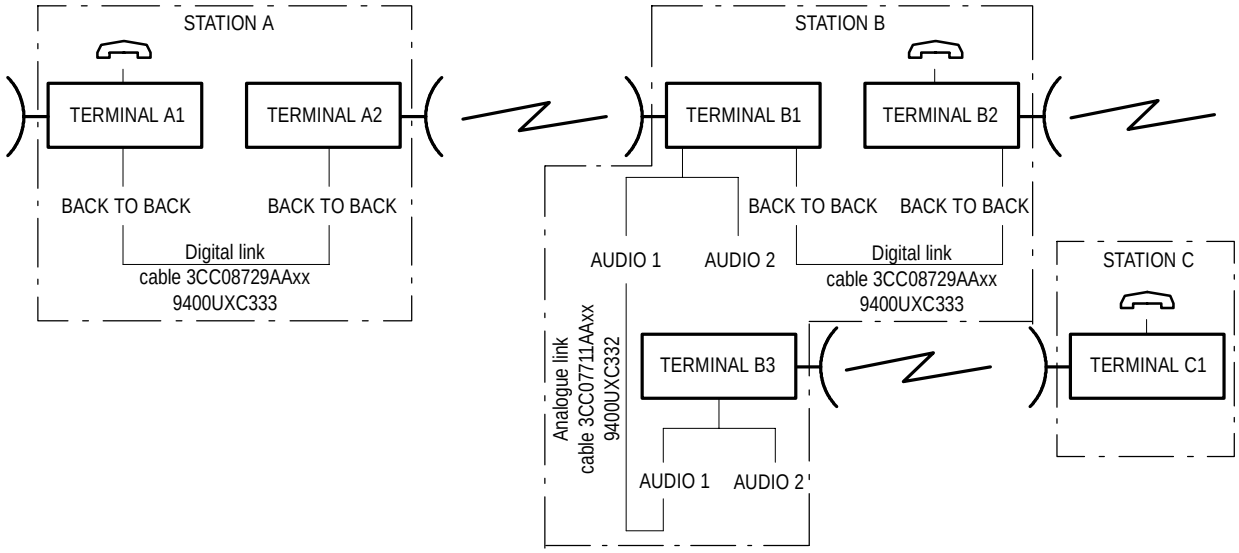


Figure 13 - Telephone ESC

2.3.3 - ESC numbers 3 and 4

2.3.3.1 - Without extension IDU

If there is no extension IDU:

- ESC N_3 is a V11 or V28 point-to-point 64 kbit/s channel,
- ESC N_4 is a V11 point-to-point 64 kbit/s channel (not applicable to the Light IDU versions).

The V11 or V28 equipment connects to the “ESC. EXT.” connector of the main IDU.

2.3.3.2 - With extension IDU (not applicable to the Light IDU versions)

With an extension IDU, the ESCs can be digital (G.703 or V11/V28 64 kbit/s, V11 or V28 9.6 or 4.8 kbit/s) or analogue (four-wire telephone channel + TTL interface of RON/TRON type), point-to-point or point-to-multipoint. The ESCs are customized by daughter boards described in § 7.9.

POINT-TO-POINT ESC

Connection for the ESC equipment:

- ESC N_3: to the “ESC 3-1” connector of the extension IDU.
- ESC N_4: to the “ESC 4-1” connector of the extension IDU.

Connection of two terminals back-to-back: each of the terminals must have an extension IDU with an ESC board of the same type. The link is set up by a cable linking ports “ESC 3-1” (ESC N_3) and “ESC4-1” (ESC N_4) on the extension IDUs of the two terminals.

POINT-TO-MULTIPOINT ESC

This type of ESC can be used only for asynchronous digital links.

Connection for the ESC equipment:

- ESC N_3: to connectors “ESC 3-1” and “ESC3-2” of the extension IDU,
- ESC N_4: to connectors “ESC 4-1” and “ESC4-2” of the extension IDU.

The connection of the terminals for ESC N_3 is illustrated in Figure 14. The connection principles are the same for ESC N_4, replacing “ESC 3-1” with “ESC 4-1” and “ESC 3-2” with “ESC 4-2”.

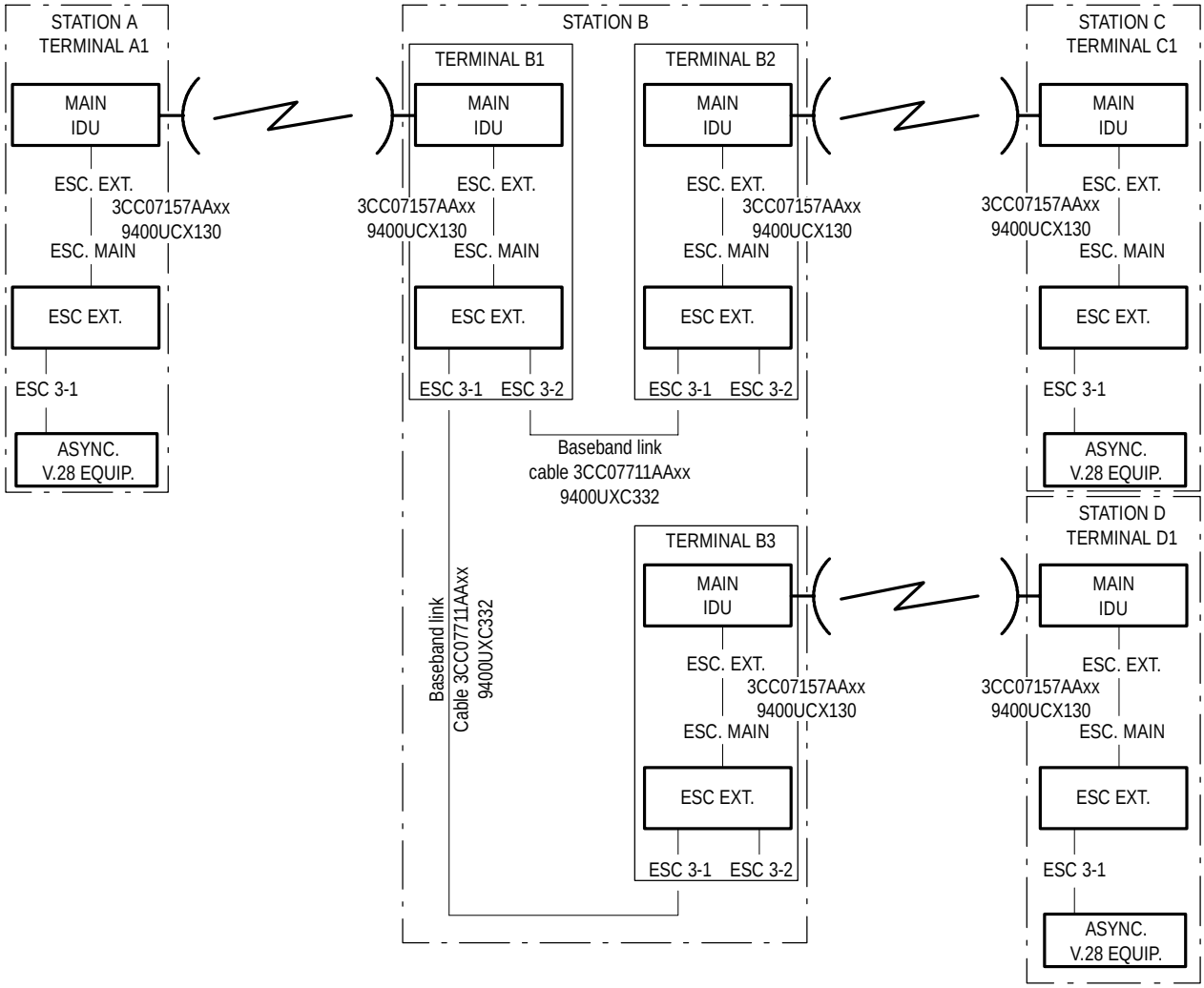


Figure 14 - Point-to-multipoint engineering service channel ESC3

2.3.4 - ESC number 5

Channel not available in the Light IDU versions.

2.3.4.1 - Without extension IDU

As ESC N_4.

2.3.4.2 - With extension IDU

ESC N_5 can only be a V11 or V28 point-to-point link.

Connection of the ESC equipment: to connector ESC 5 of the extension IDU.

Connection of two terminals back-to-back to form a repeater station: each of the terminals must have an extension IDU with an ESC board of the same type. The link is implemented by a cable connecting the “ESC 5” ports of the extension IDUs of the two terminals.

2.4 – Alarms, telesignalling and remote controls (TS/TC)

The equipment has:

- **Three loops, respectively preassigned** for “Urgent alarm” (URG), “Non-urgent alarm” (NURG) and “Attended” (ATT) states. Each alarm or event generated by the equipment can be classified as Urgent, Not Urgent, Inhibited or Status by the supervisory software (§ 4.5.5).
- **A remote control loop**, available to the user.
  The three contacts of this loop (common, N/C and N/O) are accessible:
  - S on the “ALARM. I/O” connector of the main classic IDU,
  - S on the “ESC 3/ALARM I/O” connector of the Light IDU,

  The loop formed by the common and normally closed contacts is normally closed (in the absence of an alarm condition or remote control).
- **eight housekeeping alarms** (remote indications via dry loops), available to the user. The remote indication loops are configured as normally open and can be configured as normally closed, depending on the setting of the swaps on the MCU board, as described in § 7.10.1. A name can be assigned to each loop, in the software, on installation. These loops are available:
  - S on the “ALARM. I/O” connector of the main classic IDU,
  - S on the “ESC 3/ALARM I/O” connector of the Light IDU unit,

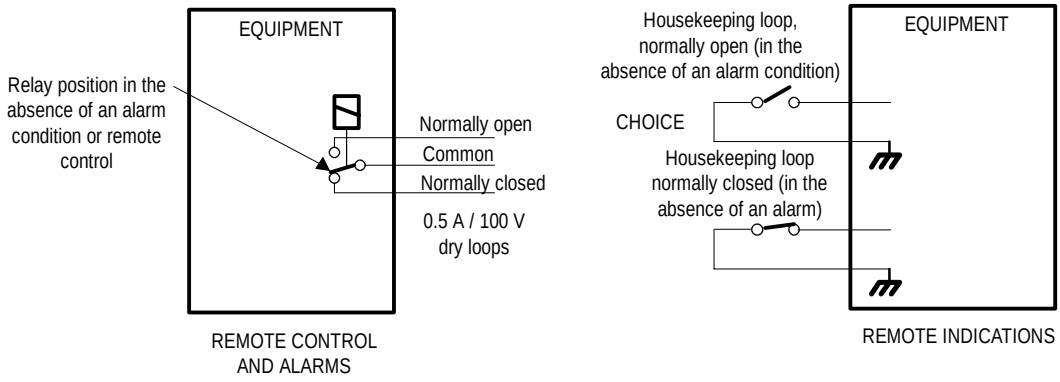


Figure 15 - Remote control, alarms and remote indications

2.5 – Loopback options

The loopback options are provided on the active channel to facilitate commissioning and maintenance operations. Figure 16 shows the positioning of the loops according to the configuration.

Each **local loop** loops the aggregate:

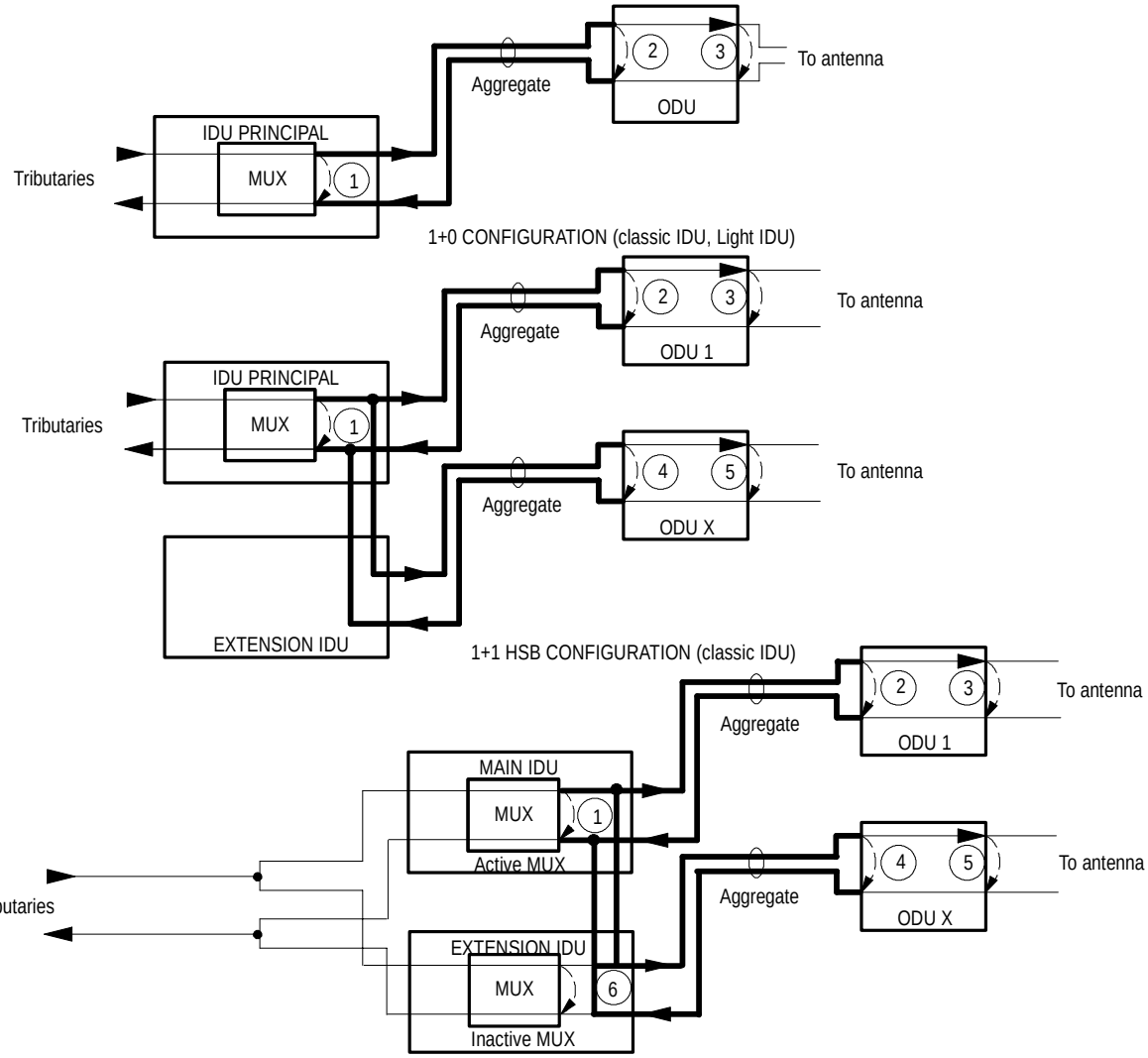
- at the output of the main IDU ①, and the extension IDU ⑥,
- at the input of the main ODU ② and the extension ODU ④, to check the cable link between the IDU and the ODU (this command cuts off output power),
- at the output of the main ODU ③ and the extension ODU ⑤ if the RF loopback option has been selected (the RF loop cuts transmission).

CAUTION:

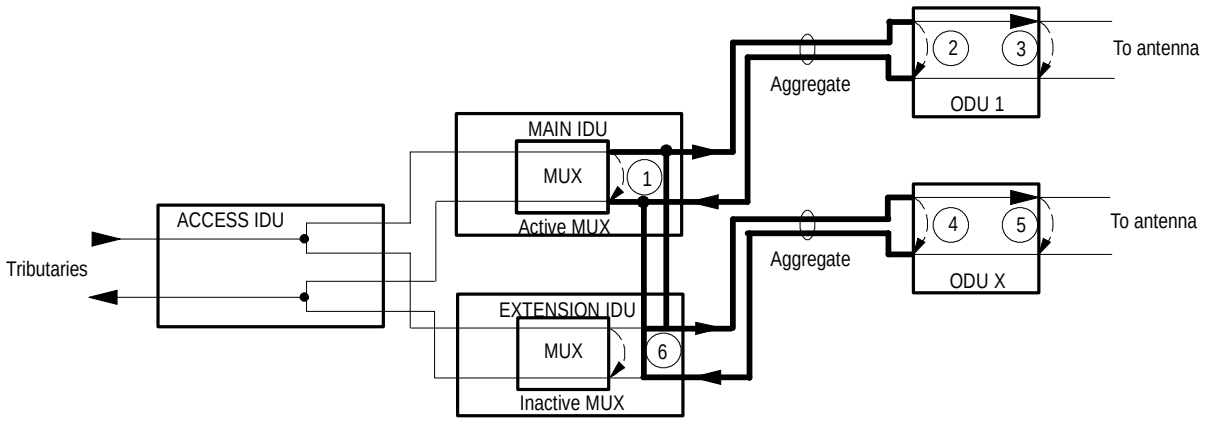
S **A local loop disconnects the link and isolates it from the network.**

S **The RF loopback configuration requires the remote transmitter to be cut off by remote control.**

*



BASIC HSB 1+1 CONFIGURATION WITH MULTIPLEXER PROTECTION (classic IDU) without access unit



BASIC 1+1 HSB CONFIGURATION WITH MULTIPLEXER PROTECTION (classic IDU)

Figure 16 - Locations of local loops

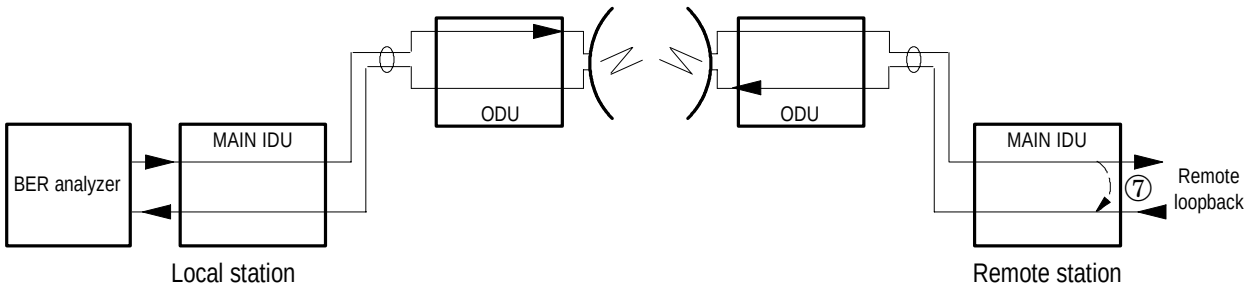


Figure 17 - Checking a hop using the remote loop option

The remote loop remote control function must be executed on the remote station.

Remote loopback ⑦ can be used in a station's IDU to loop the receive output of a tributary to its transmit input. This way, you can use a BER analyzer connected to the local station to check out a hop (Figure 17).

- **Before setting up a loop, you must assign identical link identity codes for both transmit and receive directions (see § 4.5.1) to avoid generating an alarm condition.**
- **For RQ2 supervision networks, disconnect the terminal from the network management system before setting up a loop.**

2.6 – Equipment management

When installing and commissioning the equipment, the "Installation parameters" (see § 4.4) and "Operation parameters" (see § 4.5) applications of the supervision function are used to load the link parameters (configuration, bit rate, frequency, etc) from the craft terminal.

The configuration is saved in two memories, one located in the IDU, the other in the ODU.

On power up, the microprocessor responsible for administering the equipment compares the content of the two memories:

- if the contents are identical, it loads their content into the various modules of the IDU and the ODU, and switches the equipment to operational mode,
- if the contents are different:
  - On the IDU, the three "Urgent alarm", "Non urgent alarm" and "Alarm attended" LEDs (see § 2.7) on the front panel of the main IDU flash simultaneously. The equipment waits for the "IDU-ACO ODU" button on the main IDU to be pressed to select a configuration to download.

Nota : A configuration is valid if one of the two IDU or ODU leds is lighted.

The default configuration is obtained by attempting to load the configuration of equipment that is not included (for example, disconnect the ODU, power up the IDU and choose to download the ODU configuration when the LEDs are flashing at the same time). You must then reload the entire equipment configuration from the CT.

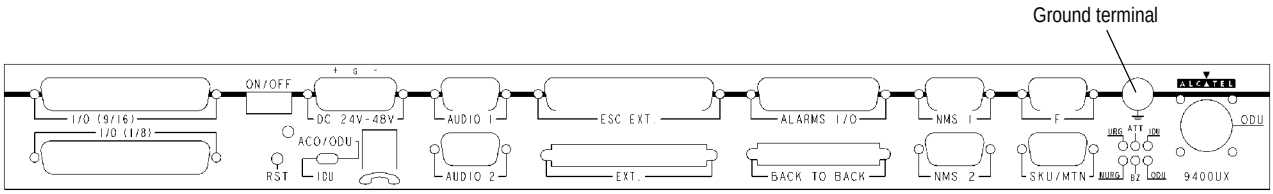
Nota : When the IDU and ODU leds are off, then the equipment starts automatically in default configuration.

2.7 – IDU controls, indications and connectors

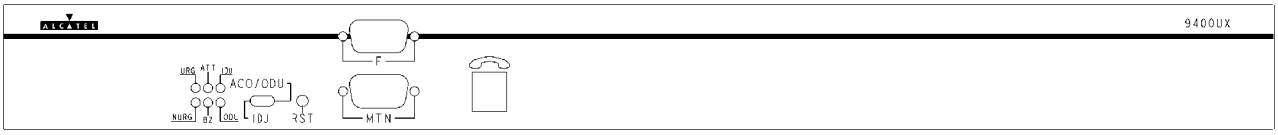
The indoor units have a “Connector” panel and an “Operation” panel (Figure 18). The operation panel carries controls and indicators which duplicate those on the “Connector” side. The tables in the sections that follow describe these components.

In some configurations, some of the items described below may be omitted.

2.7.1 - Classic main IDU



Connector panel of the main IDU



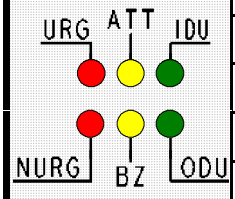
Front panel of the main IDU

Figure 18 - Controls, indicators and connectors on the main IDU

The connector panel of the main IDU is fitted with:

- indicators showing the operational status of the equipment at a glance,
- connectors.

The components are listed in the table below.

ITEM	TYPE	FUNCTION
	Items shown on a shaded background appear on both panels.	
	Red LED	URG: Urgent alarm present.
	Red LED	NURG: Non-urgent alarm present.
	Yellow LED	ATT: Alarm attended, acknowledged by the operator.
	Yellow LED	BZ: Busy (telephone ESC busy).
	Green LED	IDU: No IDU alarm or IDU configuration is valid.
	Green LED	ODU: No ODU alarm or ODU configuration is valid.
ON/OFF	Two-position switch	For powering the unit up/down. The green LED beneath this switch indicates when the unit is powered up.
RST	Pushbutton	Resets the unit’s software.
IDU-ACO/ODU	Momentary action two-position switch	S On startup: selects the IDU or ODU configuration if the configuration data is lost S In operation: ACknOwledge (acknowledgement of an urgent alarm)/test the LEDs.

ITEM	TYPE	FUNCTION
I/O (1/8)* I/O (9-16)*	Female 37-way sub-D connectors	**2 Mbit/s port version:** Tributaries 1 to 16. **34 Mbit/s port version:** 2 Mbit/s auxiliary bit stream (connector on the LAU board). 34 Mbit/s ports on 1.6/5.6 coaxial sockets mounted on a plate installed in place of the top “I/O (9/16)” connector.
DC 24V-48V *	Male 3-way sub-D connector	Primary power supply input (24 V or 48 V DC).
	RJ11 connector	ESC telephone handset port.
AUDIO 1 * AUDIO 2 *	Female 9-way sub-D connector	Interface for ESC N°2 (4-wire telephone channel + E/M signalling). A buzzer alerts the operator if a call is received over the telephone channel.
ESC. EXT. *	Female 37-way sub-D connector	S Link with “ESC MAIN” connector on the extension IDU for customizing ESCs 3, 4 and 5. S Interface for ESCs 3, 4 and 5 if there is no extension unit.
ALARM. I/O *	Female 25-way sub-D connector	Remote controls and telesignalling (TS/TC).
NMS 1 * NMS 2 *	Female 9-way sub-D connector	Local supervision network interfaces.
F *	Female 9-way sub-D connector	Supervisory PC interface.
SKU/MTN	Female 9-way sub-D connector	S Connector for the software key according to the configuration, enabling the equipment to operate (must be fitted). S Also used for maintenance (factory tests).
EXT	Female 68-way mini-D connector	Link with the extension unit’s “MAIN” connector.
BACK TO BACK	Female 50-way mini-D connector	Link with the corresponding connector on the other direction’s main IDU.
ODU	Female N connector	Link with the ODU. This connector includes a protective spark gap.

* Connectors with pinout detailed in section 3 and in Appendix 6.

2.7.2 – Light IDU

2.7.2.1 – Light IDU (19” version)

The Light IDU version has only a “Connector” panel (Figure 19) with:

- switches and a software reset button,
- display components for ascertaining the operational state of the equipment at a glance,
- connector ports.

It is equipped with the items listed in the table below.

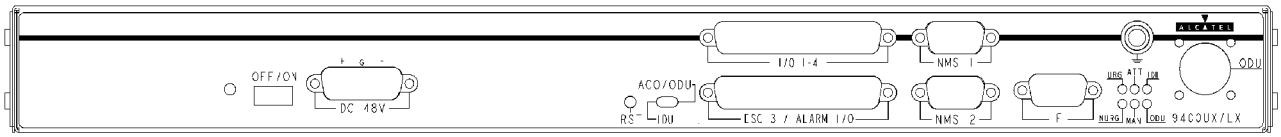
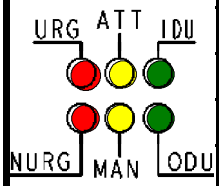
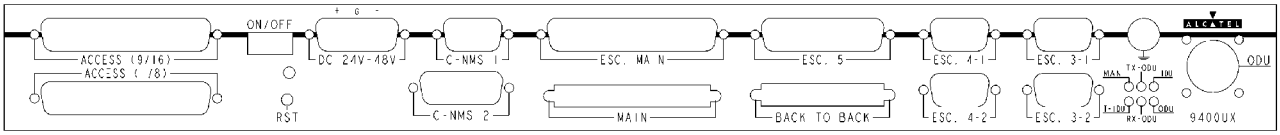


Figure 19 - Controls, indicators and connectors on the Light IDU

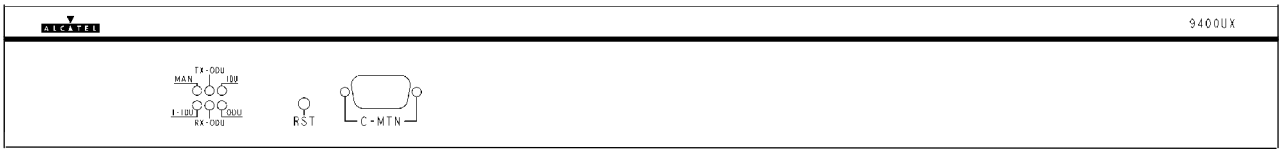
ITEM	TYPE	FUNCTION
ODU	Female N connector	Link with the ODU. This connector includes a protective surge arrestor.
	Red LED	URG: Urgent alarm present
	Red LED	NURG: Non-urgent alarm present
	Yellow LED	ATT: Alarm attended, acknowledged by the operator
	Yellow LED	MAN: Remote control indicator
	Green LED	IDU: No IDU alarm or IDU configuration is valid
	Green LED	ODU: No ODU alarm or ODU configuration is valid
F *	Female 9-way Sub-D connector	Interface with supervisory PC
NMS 1 * NMS 2 *	Female 9-way Sub-D connectors	Supervision network bus interfaces.
I/O 1-4 *	Female 37-way Sub-D connector	Tributaries 1 to 4
ESC 3 / ALARM I/O *	Female 37-way Sub-D connector	S ESC Nº 3 interface S Remote controls and telesignalling (TS/TC)
IDU-ACO/ODU	Momentary action two-position switch	S On startup: select the IDU or ODU configuration if the configuration is lost S In operation: ACknOwledge (Acknowledgement of urgent alarms/LED test).
RST	Pushbutton	Software reset.
DC 48V or 24V *	Male 3-way Sub-D connector	Primary power supply input (24 V or 48 V DC).
ON/OFF	Stable two-position switch	For powering the unit up/down The green LED to the left of this switch indicates that the unit is powered up.

* Connectors with pin-out details in section 3 and in Appendix 6.

2.7.3 – Extension IDU



Extension IDU connector panel



Extension IDU operation panel

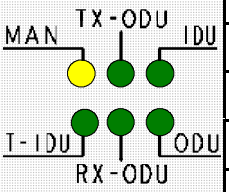
Figure 20 - Extension IDU controls, indicators and connectors

The connector panel of the extension IDU is fitted with:

- indicators showing the operational status of the equipment at a glance,
- connectors.

In the first three columns of the table below, a cross indicates the presence of the components according to the type of extension unit:

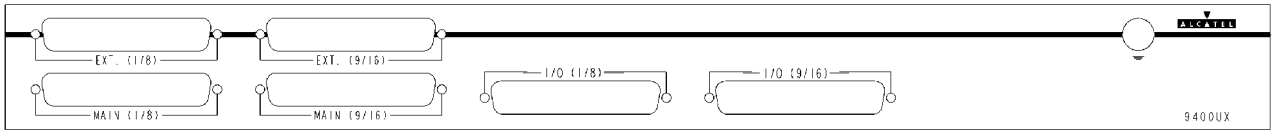
- Type A : ESC extension unit.
- Type B : 1+1 extension unit with no multiplexer protection.
- Type C : 1+1 extension unit with multiplexer protection.

A	B	C	ITEM	TYPE	FUNCTION
				Items shown on a shaded background appear on both panels.	
	X	X		Yellow LED	MAN: Indicates that a remote control has been sent to the equipment from a supervisory PC. The LED remains on until the remote control disappears.
				Green LED	TX-ODU: Transmit traffic on standby ODU.
				Green LED	IDU: No alarms on extension IDU.
				Green LED	T-IDU: Traffic on extension multiplexer/demultiplexer.
				Green LED	RX-ODU: Receive traffic on standby ODU.
				Green LED	ODU: No alarms on standby ODU.
X	X	X	ON/OFF	Two-position switch	For powering the unit up/down. The green LED beneath this switch indicates when the unit is powered up.
	X	X	RST	Pushbutton	Resets the unit's software.
		X	ACCESS (1-8) ACCESS (9-16)	Female 37-way sub-D connector	Link with access unit's "EXT. (1/8)" and "EXT. (9/16)" connectors.
X	X	X	DC 24V-48V *	Male 3-way sub-D connector	Primary power supply input (24 V or 48 V).
X	X	X	ESC. MAIN	Female 37-way sub-D connector	Link with the main IDU's "ESC EXT" connector for ESC extension.

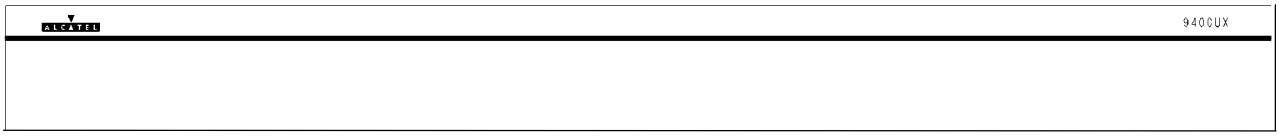
A	B	C	ITEM	TYPE	FUNCTION
X	X	X	MAIN	Female 68-way mini-D connector	Link with the main IDU's "EXT" connector.
X	X	X	ESC. 5	Female 25-way sub-D connector	Port for ESC N°5.
X	X	X	BACK TO BACK	Female 50-way mini-D connector	Not used.
X	X	X	ESC. 4-1 * ESC. 4-2 *	Female 9-way sub-D connectors	Ports 1 and 2, ESC N°4.
X	X	X	ESC. 3-1 * ESC. 3-2 *	Female 9-way sub-D connector	Ports 1 and 2, ESC N°3.
	X	X	ODU	Female N connector	Link with the ODU. This connector includes a protective spark-gap.
Options			C-NMS1 C-NMS2	Female 9-way and 15-way sub-D connectors	Not used.

* Connectors with pinout detailed in Appendix 6.

2.7.4 - Access IDU



Access IDU connector panel



Access IDU front panel

Figure 21 - Access IDU connectors

The connector panel of the access indoor unit is fitted with the components described in the table below:

ITEM	TYPE	FUNCTION
I/O (1/8)* I/O (9/16)*	Female 37-way sub-D connector	**2 Mbit/s access version:** Tributaries 1 to 16. **34 Mbit/s access version:** 2 Mbit/s auxiliary bit stream port. 34 Mbit/s ports on 1.6/5.6 coaxial sockets on a panel mounted in place of the top “I/O (9/16)” connector.
MAIN (1/8) MAIN (9/16)	Female 37-way sub-D connectors	**2 Mbit/s access version:** Tributaries links 1 to 16 to “I/O (1/8)” and “I/O (9/16)” connectors on the main IDU. **34 Mbit/s access version:** Tributaries 34 Mbit/s and 2Mbit/s to “I/O (1/8)” and “I/O (9/16)” connectors on the main IDU.
EXT. (1/8) EXT. (9/16)	Female 37-way sub-D connectors	**2 Mbit/s access version:** Tributaries links 1 to 16 to “I/O (1/8)” and “I/O (9/16)” connectors on the main IDU. **34 Mbit/s access version:** Tributaries links 34 Mbit/s and 2 Mbit/s to “ACCESS (1/8)” and “ACCESS (9/16)” connectors.

* “Customer” connectors, with pinout detailed in section 3.

2.8 – Technical characteristics

The values indicated hereafter are typical values.

REFERENCE STANDARDS AND FREQUENCY PLANS					
Equipment	**Reference Standard**	**Band (GHz)**	**Frequency (GHz)**	**Channeling**	**Standard * Tx/Rx separation (MHz)**
9413 UX 4QAM	EN 301 128	13 GHz	12.75–13.25	ERC 12–02 ITU–R Rec F497–6	266
9413 UX 16QAM	EN 301 128	13 GHz	12.75–13.25	ERC 12–02 ITU–R Rec F497–6	266
9415 UX 4QAM/16QAM	EN 301 128	15 GHz	14.4–15.35	ERC 12–07 ITU–R Rec F636–3	728, 315, 420, 490
9418 UX 4QAM/16QAM	EN 301 128	18 GHz	17.7–19.7	ERC 12–03 ITU–R Rec F595–3	1008, 1010, 340, 1560, 1092.5
9423 UX 4QAM	EN 300 198	23 GHz	21.2–23.6	ERC 13–02 annex A ITU–R Rec F637–3	1008, 1197, 1200, 1232
9423 UX 16QAM	EN 300 198	23 GHz	21.2–23.6	ERC 13–02 annex A ITU–R Rec F637–3	1008, 1197, 1200, 1232
9425 UX 4QAM	EN 300 431	25 GHz	24.5–26.5	ERC 13–02 annex B ITU–R Rec F748–3	1008
9425 UX 16QAM	EN 300 431	25 GHz	24.5–26.5	ERC 13–02 annex B ITU–R Rec F748–3	1008
9438 UX 4QAM	EN 300 197	38 GHz	37–39.5	ERC 12–01 ITU–R Rec F749–1	1260
9438 UX 16QAM	EN 300 197	38 GHz	37–39.5	ERC 12–01 ITU–R Rec F749–1	1260

(*) consult Alcatel if non–standard Tx/Rx separations are requested.

(*) consult Alcatel for availability of these options

RF CHANNELING				
Capacity (Mbit/s)	2x2	4x2	8x2	16x2 / 34+2
RF Channeling (MHz) **(4QAM modulation)**	3.5	7	14	28
RF Channeling (MHz) **(16QAM modulation)**	–	–	7	14

IN-FIELD TUNABILITY RANGE FOR 94XXUXR201, 94XXUXR202* OR 94XXUXR203*			
Equipment	**Max frequency agility band depending on duplex difference (MHz)**	**Output Power (dBm)**	**Output Power Setting option (dB)**
9413 UX 4QAM	116	25	** Fixed attenuator: 3, 6, or 10 dB for 3CC08879AAxx
9413 UX 4QAM RTPC	116	25	+25 dBm to 5 dBm with 1 dB step for 3CC12690AAxx
9413 UX 16QAM	116	21	+21 dBm to 1 dBm with1 dB step for 3CC12683AAxx
9415 UX 4QAM	220	21	Fixed attenuator: 3, 6, or 10 dB for 3CC08880AAxx
9415 UX 4QAM RTPC	220	24	+24 dBm to –6 dBm with 1 dB step for 3CC12979AAxx
9418 UX 4QAM	480	16.5 24.5*	Fixed attenuator: 3, 6, or 10 dB
9423 UX 4QAM RTPC	500	19	+19 to –11 dBm with 1 dB step
9423 UX 16QAM	500	17	+17 to –3 dBm with 1 dB step
9425 UX 4QAM RTPC	450	18	+18 to –12 dBm with 1 dB step
9425 UX 16QAM	450	16	+16 to –14 dBm with 1 dB step
9438 UX 4QAM RTPC	560	16	+16 to –14 dBm with 1 dB step
9438 UX 16QAM	560	14	+14 to –6 dBm with 1 dB step

(*) xx is for the Outdoor IDU frequency band. For example, an IDU referenced 9413UX202 depends on the 13Ghz frequency band.

(**) consult Alcatel for availability of these options

TUNABILITY RANGE FOR NEW GENERATION ODU A9400UX flat ODU					
Equipment		**Duplex difference (MHZ)**	**Frequency agility band (MHz)**	**Output power (dBm)**	**Output power setting (dB)**
9413 UX	**4 QAM**	266	116	+ 24	+24 to –6 dBm with 1 dB step
	16 QAM			20	+20 to –10 dBm with 1 dB step
9415 UX	**4 QAM**	315, 420, 490, 728	220	+ 24	+24 to –6 dBm with 1 dB step
	16 QAM			20	+20 to –10 dBm with 1 dB step
9418 UX	**4 QAM**	340, 1008,1010, 1560	480	+ 22	+22 to –8 dBm with 1 dB step
	16 QAM			19	+19 to –11 dBm with 1 dB step
9423 UX	**4 QAM**	1008, 1197, 1200, 1232	500	+ 19	+19 to –11 dBm with 1 dB step
	16 QAM			+ 16	+16 to –14 dBm with 1 dB step
9425 UX	**4 QAM**	1008	450	+ 17	+17 to –13 dBm with 1 dB step
	16 QAM			+ 14	+14 to –16 dBm with 1 dB step
9438 UX	**4 QAM**	1260	560	+ 16	+16 to –14 dBm with 1 dB step
	16 QAM			+ 13	+13 to –17 dBm with 1 dB step

TYPICAL BER THRESHOLDS FOR 94xxUXR201, 94xxUXR202 OR 94xxUXR203* in dBm								
Equipment	**2x2**		**4x2**		**8x2**		**16x2 / 34+2**	
	10^{-3}	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**
9413 UX 4QAM	–95	–92	–92	–89	–89	–86	–86	–83
9413 UX 16QAM	–	–	–	–	–83	–80	–80	–77
9415 UX 4QAM	–94	–91	–91	–88	–88	–85	–85	–82
9418 UX 4QAM	–94	–91	–91	–88	–88	–85	–85	–82
9423 UX 4QAM	–92	–89	–89	–86	–86	–83	–83	–80
9423 UX 16QAM	–	–	–	–	–82	–79	–79	–76
9425 UX 4QAM	–91	–88	–88	–85	–85	–82	–82	–79
9425 UX 16QAM	–	–	–	–	–81	–78	–78	–75
9438 UX 4QAM	–87	–84	–84	–81	–81	–78	–78	–75
9438 UX 16QAM	–	–	–	–	–77	–74	–74	–71

*: Typical values at antenna port

BER THRESHOLDS FOR NEW GENERATION ODU A9400UX flat ODU*									
Equipment		**2x2**		**4x2**		**8x2**		**16x2 / 34+2**	
		10^{-3}	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**
9413 UX	**4 QAM**	–95	–92	–92	–89	–89	–86	–86	–83
	16 QAM	–	–	–	–	–84	–81	–81	–78
9415 UX	**4 QAM**	–95	–92	–92	–89	–89	–86	–86	–83
	16 QAM	–	–	–	–	–84	–81	–81	–78
9418 UX	**4 QAM**	–93	–90	–90	–87	–87	–84	–84	–81
	16 QAM	–	–	–	–	–82	–79	–79	–76
9423 UX	**4 QAM**	–93	–90	–90	–87	–87	–84	–84	–81
	16 QAM	–	–	–	–	–82	–79	–79	–76
9425 UX	**4 QAM**	–92	–89	–89	–86	–86	–83	–83	–80
	16 QAM	–	–	–	–	–81	–78	–78	–75
9438 UX	**4 QAM**	–91	–88	–88	–85	–85	–82	–82	–79
	16 QAM	–	–	–	–	–79	–76	–77	–74

*: Typical values at antenna port

CONFIGURATION		
Additional branching losses per terminal *	**Tx**	**Rx**
1+0 or1+1 FD (Main & Stand–By channels)	0	0
1+1 HSB 1 antenna		
• Main Channel	1	1
• Stand–By channel	11	11

* : Typical values ODU unit inputs, antenna access

1+0/1+1 FD SYSTEM GAIN – AT ANTENNA PORT (INCLUDING DUPLEXER LOSS) FOR ODU 94xxUXR201, 94xxUXR202 OR 94xxUXR203								
Equipment	**2x2**		**4x2**		**8x2**		**16x2 / 34+2**	
	10^{-3}	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**
9413 UX 4QAM	120	117	117	114	114	111	111	108
9413 UX 16QAM	–	–	–	–	104	101	101	98
9415 UX 4QAM	115	112	112	109	109	106	106	103
9415 UX 4 QAM RTPC	118	115	115	112	112	109	109	106
9418 UX 4QAM	111	108	108	105	105	102	102	99
9423 UX 4QAM	111	108	108	105	105	102	102	99

1+0/1+1 FD SYSTEM GAIN – AT ANTENNA PORT (INCLUDING DUPLEXER LOSS) FOR ODU 94xxUXR201, 94xxUXR202 OR 94xxUXR203								
Equipment	**2x2**		**4x2**		**8x2**		**16x2 / 34+2**	
9423 UX 16QAM	–	–	–	–	99	96	96	93
9425 UX 4QAM	109	106	106	103	103	100	100	97
9425 UX 16QAM	–	–	–	–	97	94	94	91
9438 UX 4QAM	103	100	100	97	97	94	94	91
9438 UX 16QAM	–	–	–	–	91	88	88	85

1+0/1+1 System gain – At antenna port (including duplexer loss) for A9400UX flat ODU									
Equipment		**2x2**		**4x2**		**8x2**		**16x2 / 34+2**	
		10^{-3}	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**	**10^{-3}**	**10^{-6}**
9413 UX	**4 QAM**	119	116	116	113	113	110	110	107
	16 QAM					104	101	101	98
9415 UX	**4 QAM**	119	116	116	113	113	110	110	107
	16 QAM					104	101	101	98
9418 UX	**4 QAM**	115	112	112	109	109	106	106	103
	16 QAM					101	98	98	95
9423 UX	**4 QAM**	112	109	109	106	106	103	103	100
	16 QAM					98	95	95	92
9425 UX	**4 QAM**	109	106	106	103	103	100	100	97
	16 QAM					95	92	92	89
9438 UX	**4 QAM**	107	104	104	101	101	98	98	95
	16 QAM					92	89	90	87

ANTENNA GAIN (IN dB , typ GAIN)							
	Integrated		**Non–integrated**				
	30 cm	**60 cm**	**Flextwist loss ***	**60 cm**	**90 cm**	**120 cm**	**180 cm**
13 GHz		>35	0.35		>38.3	>41	>44.6
15 GHz	>30.5	>36	0.5		>39.5	>42	>45.5
18 GHz	>32.7	>38	0.7		>41.6	>44	>47.3
23 GHz	>34.4	>39.6	0.7		>43.1	>45.6	>48.9
25 GHz	>35.6	>40.7	0.7		>44.2	>46.8	
38 GHz	>39.3	>44.2					

* For those referenced in Appendix 12.3 Commercial items

ALCATEL 9400 UX – ALL FREQUENCY BANDS	
Synthesizer step	250 kHz
RF stability	+/– 10 ppm
Spectrum masks	Compliant with the relevant ETSI standard
Spurious emissions	Compliant with the ETSI standard Compliant with ERC Rec 74–01
Equaliser Type	7 coefficients
Link Identity Code	32 codes

TRIBUTARY INTERFACES**	
type	2 Mbit/s ITU–T G.703 120 ohms or 75 Ohms 34 Mbit/s ITU–T G.703 75 Ohms
Jitter	ITU–T G. 823/ G. 921
Man–Machine Interface	
visual	7 LEDs
other	F interface for Personal Computer (Windows™) Interface for Telephone Service Channel (ESC n°2 on classic IDU only)
Alarms	
Equipment alarms	3 outputs
Environmental events	8 inputs / 1output

In the case of a 75 Ohms connection, the respect of the G703 norms and CEM norms is only guaranteed by ALCATEL if “spider” adaptaters are used 3CC07885Axxx or 3CC07759Axxx made for this purpose and with the IDU configuration.

POWER SUPPLY		
DC nominal (primary)	**24 VDC**	+/– 20 to +/– 40 VDC
	48 VDC	+/– 39 to +/– 60 VDC
Polarity		+ or – or floating (isolated ground)
Power consumption for ODU 94xx UXR201, 94xx UXR202 OR 94xx UXR203		1+0 : 60W with classic IDU 1+0 : 50 W with Light IDU 1+1 : 120W
Power consumption for A9400UX flat ODU		1+1 : 30 W

ELECTROMAGNETIC COMPATIBILITY / SAFETY	
EMC	EN 301 489 1 & 4 (version 1.2.1)
Safety	EN 60 950
Power supply	EN 300 132
Lightning protection	Symmetric protection implemented in ODU & IDU +/– 1kV on cable ground according to IEC 61000–4–5 with 2 wave types: 1.2/50 µs and 10/700 µs

IDU–ODU CABLE	
Type	1AC001100022
Fire resistance	IEC 332–1
Max distance	300 m

ENVIRONMENTAL CONDITIONS		
Storage		EN 300 019 class 1.3
Transport		EN 300 019 class 2.3
Operation	**IDU**	EN 300 019 class 3.2 Temperature range : –10 to +55°C
	ODU	EN 300 019 class 4.1, Protection factor IP 55 (IEC529 standard) Temperature range : –33 to +55°C

MECHANICAL		
Classic Main IDU, Extension IDU, Access IDU		44x450x235 mm / 3 kg (19” / 1U)
Light IDU		44x450x235 mm / 3 kg (19” / 1U)
ODU 94xxUXR201 94xxUXR202 94xxUXR203	**13 to 18 GHz**	250x250x250 mm / 10kg
	23 to 38 GHz	220x220x250 mm / 8 kg
ODU A9400UX flat ODU	**13 to 38 GHz**	235x235x72 mm / 4 kg

For the typical installation dimensions, see Chapter 3

2.9 – Frequency agility bands

ALL VALUES EXPRESSED IN MHz FOR A 3.5 MHz CHANNEL.

Equipment	Duplex difference	Tx IF	Tx Band	F min	F max
9413 UX	266	1106	1	12751.75	12864.25
			2	12861.75	12978.25
			1P	13017.75	13130.25
			2P	13127.75	13244.25
9415 UX	420	1260	1	14502.75	14723.50
			2	14719.75	14940.25
			1P	14922.75	15143.25
			2P	15139.75	15360.25
	490	1330	1	14404.75	14632.25
			2	14628.75	14856.25
			1P	14894.75	15122.25
			2P	15118.75	15346.25
	728	1568	1	14502.75	14618.25
			1P	15230.75	15346.25
	315	1155	1	14635.75	14765.25
			2	14761.75	14898.25
			1P	14950.75	15080.25
			2P	15076.75	15213.25

Equipment	Duplex difference	IF	Tx Band	F min	F max
9418 UX	340	1180	1	18581.75	18698.25
			2	18701.75	18818.25
			1P	18921.75	19038.25
			2P	19041.75	19158.25
	1560	2235	1	17701.75	18138.25
			1P	19261.75	19698.25
	1010	1850	1	17701.75	18199.25
			2	18180.75	18678.25
			1P	18711.75	19210.25
			2P	19190.75	19698.25
	1008	1848	1	17703.75	18199.25
			2	18182.75	18678.25
			1P	18711.75	19207.25
			2P	19190.75	19686.25
9423 UX	1008	1848	1	22001.75	22490.25
			2	22101.75	22590.25
			1P	23009.75	23498.25
			2P	23109.75	23598.25
	1232	1907	1	21201.75	21786.25
			2	21781.75	22366.25
			1P	22433.75	23018.25
			2P	23013.75	23598.25
	1200	1875	1	21201.75	21818.25
			2	21781.75	22398.25
			1P	22401.75	23018.25
			2P	22981.75	23598.25
	1197	1872	1	21204.75	21818.25
			2	21784.75	22398.25
			1P	22401.75	23015.25
			2P	22981.75	23595.25

Equipment	Duplex difference	IF	Tx Band	F min	F max
9425 UX	1008	1848	1	24549.75	24996.25
			2	24997.75	25444.25
			1P	25557.75	26004.25
			2P	26005.75	26452.25
9838 UX	1260	2100	1	37059.75	37616.25
			2	37619.75	38176.25
			1P	38319.75	38876.25
			2P	38879.75	39436.25

3 - Installation

3.1 - Delivery

When you receive the equipment in its packaging:

- Check the condition of the packaging.
- If damaged, make your reservations known to the carrier as quickly as possible.

3.1.1 - Checking the configuration

The configuration is normally supplied in:

- one package containing the IDUs and ODUs according to the configuration, with connector manual and test results book,
- one or more packages containing the antennas, HSB coupler and installation accessories, service telephone, cables, etc.

3.1.1.1 - Content of the ODU/IDU boxes

Light IDU **CONFIG.**	**Nr. of boxes**	**CONTENT**
1+0	1 or 2	1 Light IDU (19") + 1 ODU + 1 connector kit + documentation

Classic IDU **CONFIG.**	**Nr. of boxes**	**CONTENT**
1+0	1	1 main IDU + 1 ODU + 1 software key + 1 connector kit + documentation + eventually 1 extension IDU.
1+1	1 or 2	1 main IDU + 1 ODU + 1software key + 1 connector kit + documentation.
		1 extension IDU + 1 ODU + 1 connector kit.
1+1 with MUX protection	1 or 2	1 main IDU and if necessary 1 access IDU + 1 ODU + 1 software key + 1 connector kit + documentation.
		1 extension IDU + 1 ODU + 1 connector kit.

3.1.1.2 - Unpacking

- Unpack the equipment according to the instructions on the packaging.
- Take an inventory and identity any missing items. If the delivery does not match the delivery advice note, notify ALCATEL **within 48 hours of receipt of the equipment**.
- Leave the equipment readily accessible during the work.

	**IMPORTANT NOTE: NEVER REMOVE THE ROUND YELLOW PADS USED TO SEAL THE OUTDOOR SYSTEM.**

3.1.1.3 - Storage

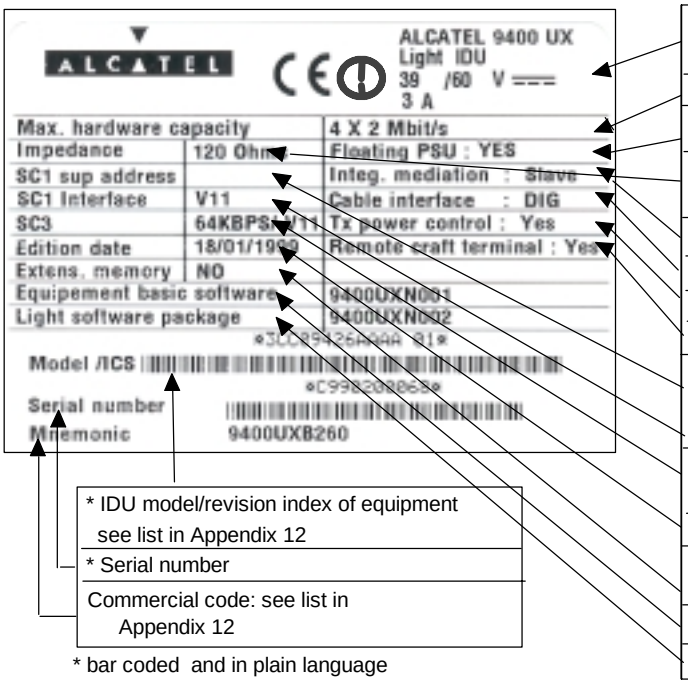
If installation is to be deferred, the type of packaging will define the equipment storage conditions:

- cardboard boxes should be warehoused indoors, in a dry, well aired room.
- wooden or plywood boxes may be stored outdoors, provided that they are protected from rain and the sun's rays.

3.2 – Labels on the equipment

The labels below are also affixed to the boxes to indicate their contents.

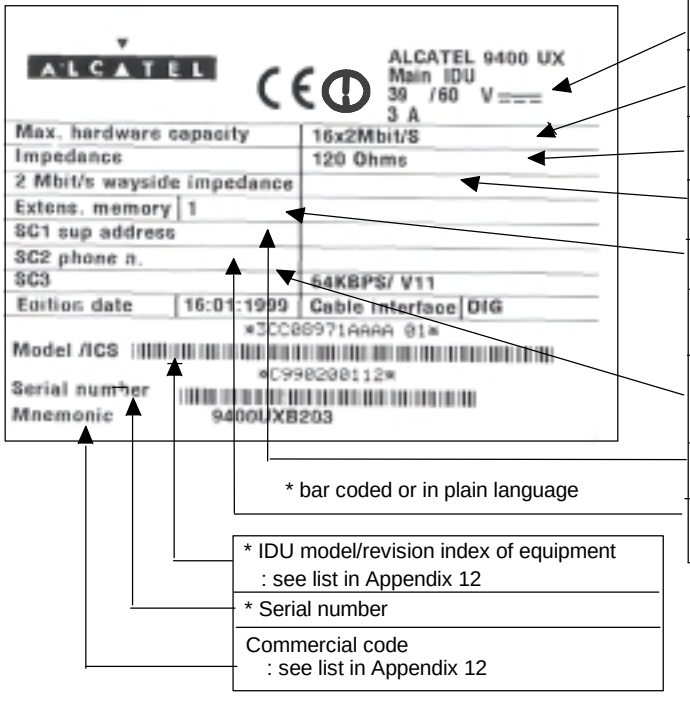
LIGHT IDU LABEL INFORMATION



Power supply voltage value	20 to 40 V= 39 to 60 V =
Maximum IDU capacity	4x2 Mbit/s
Floating power supply	YES
2 Mbit/s tributary port impedance	75 ohms or 120 ohms
Integrated mediation function	Slave
Cable interface type	DIG: Digital version
Transmitted power control	YES
Remote craft terminal	YES
Address of unit for supervision bus	Not completed in the factory.
Management interface type	V.11 or V.28
Service channel 3 interface type	V.11 or V.28
Edition date	DD / MM / YYYY
Memory extension	YES + Commercial code or NO
UX basic software	9400UXN001
Light UX software	9400UXN002

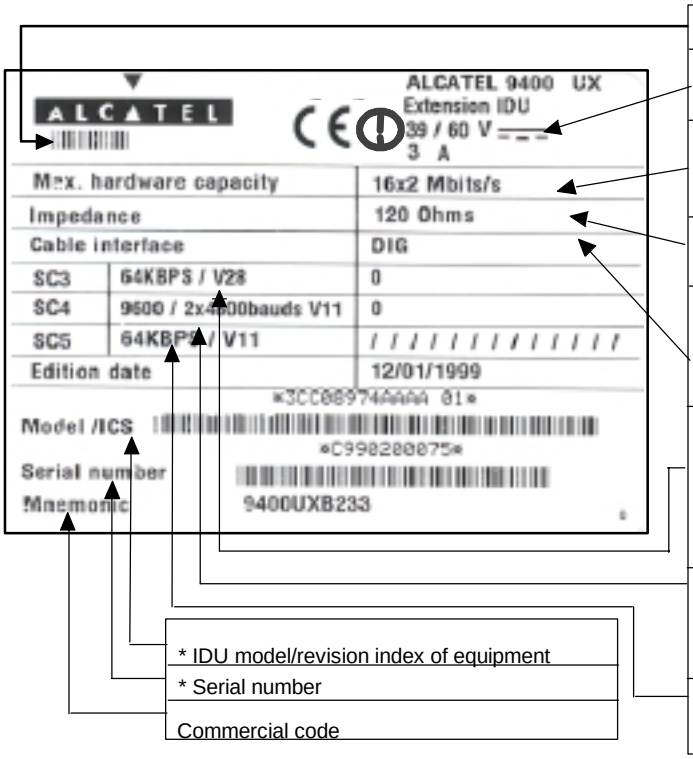
The configuration depends on the factory delivery.

CLASSIC MAIN IDU LABEL INFORMATION



Power supply voltage value	20 to 40 V = 39 to 60 V =	
Maximum IDU capacity (according to equipment)	4x2 Mbit/s ; 8x2 Mbit/s 16x2 Mbit/s ; 34+2 Mbit/s	
2 Mbit/s tributary port impedance	75, 120 ohms, high impedance (mux protection) or automatic	
2 Mbit/s aux bitstream port impedance	if any, 75 ohms or 120 ohms or high impedance	
Memory extension	YES + Commercial code or NO	
Cable interface type	ANA : Analogue version DIG : Digital version	
Service channel 3 port type	V.11 or V.28	(Meaningless if an extension IDU for dedicated SCs is used)
IDU address for supervision bus	not entered in the factory	
Audio service channel phone number	not entered in the factory.	

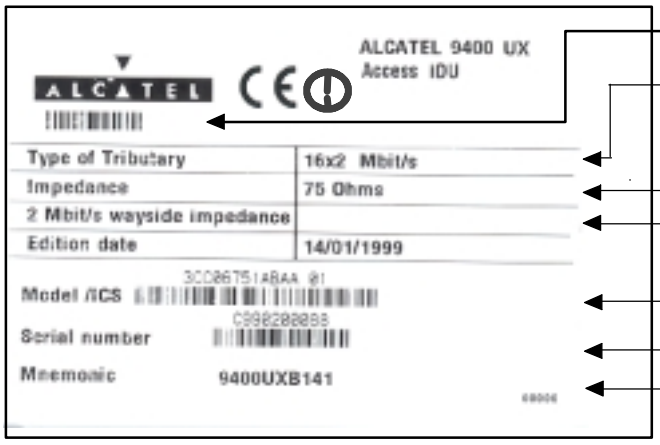
TYPICAL LABEL ON THE EXTENSION IDU



* Reserved for Alcatel	
Power supply voltage value	20 to 40 V = 39 to 60 V =
Maximum IDU capacity (according to equipment)	4x2 Mbit/s ; 8x2 Mbit/s 16x2 Mbit/s ; 34+2 Mbit/s
2 Mbit/s tributary port impedance	75, 120 ohms, high impedance (mux protection) or automatic
Cable interface type	ANA : Analogue version DIG : Digital version
Service channel 3 port type + corresponding commercial code	64 kbit/s G.703 codirectional 64 kbit/s V.11 or V.28 300-3400 Hz 4 wire 9600 / 2 X 4800 bauds, V.11 9600 / 2 X 4800 bauds, V.28
SC4 port type + corresponding commercial code	Options as for SC3
SC5 board type	V.11 V.28

* bar coded or in plain language

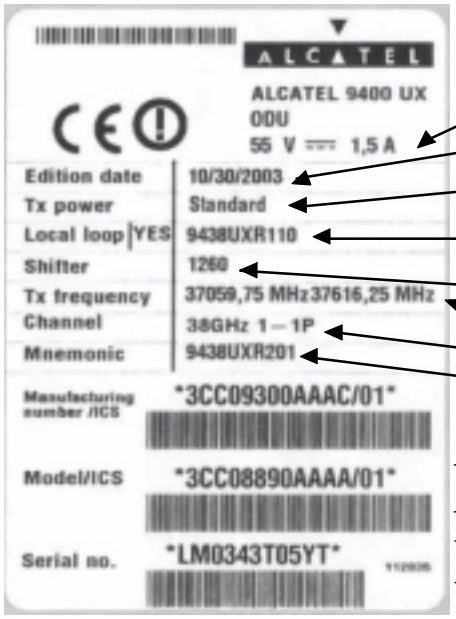
TYPICAL LABEL ON THE ACCESS IDU



* Reserved for Alcatel	
Tributary type	4x2 Mbit/s ; 8x2 Mbit/s 16x2 Mbit/s ; 34+2 Mbit/s
Tributary port impedance	75 ohms 120 ohms
2 Mbit/s auxiliary stream port impedance	75 ohms or 120 ohms or not applicable (if not provided)
* IDU model	
* Serial number	
Commercial code	

* bar coded or in plain language

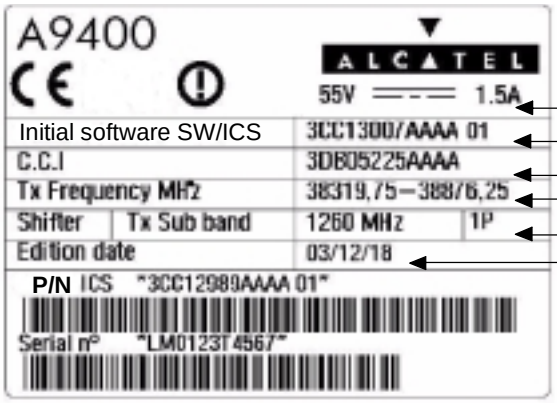
TYPICAL LABEL ON THE ODU 94xxUXR201 94xxUXR202 94xxUXR203



Power supply voltage value	
Edition date	
Transmit power for 15 and 18 GHz	Std = Standard High= High power
RF local loopback	YES + commerc. code; NO
Duplex difference	
Min and maxi operating frequencies	for transmission
Canal	
Commercial code	
*Industrial code / revision index	
* ODU model/revision index	
* Serial number	

* bar coded or in plain language

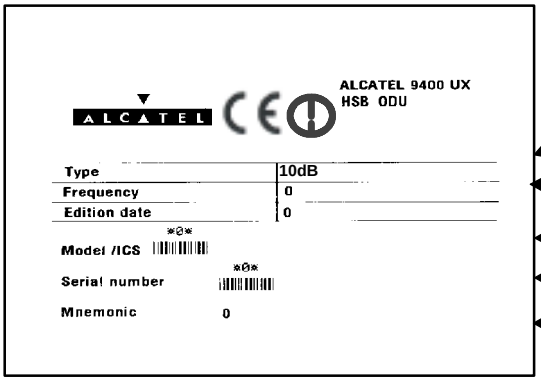
TYPICAL LABEL ON THE ODU A9400UX FLAT ODU



Power supply voltage value	
ODU Software (when learning factory)	
Commercial code	
Min and max operation frequencies	for transmission
Duplex difference	
Edition date	
*Industrial code / revision index	
* Serial number	

* bar coded or in plain language

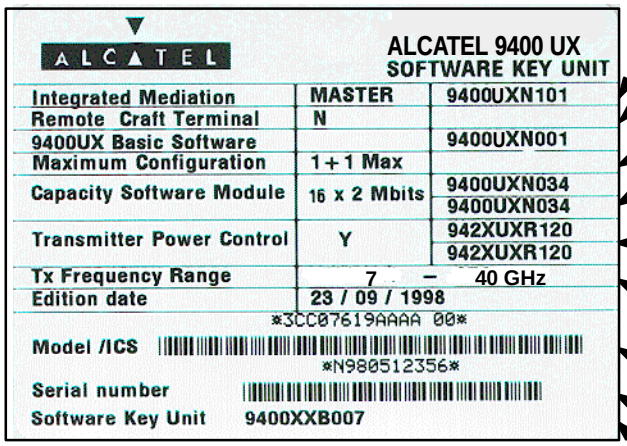
TYPICAL LABEL ON THE HSB (HOT STANDBY) COUPLER



Coupling value	10 dB
Operating frequency band	13 ; 15 ; 18 ; 23 ; 25 ; 38 GHz
* Unit mode	
* Serial number	
Commercial code	

* bar coded or in plain language

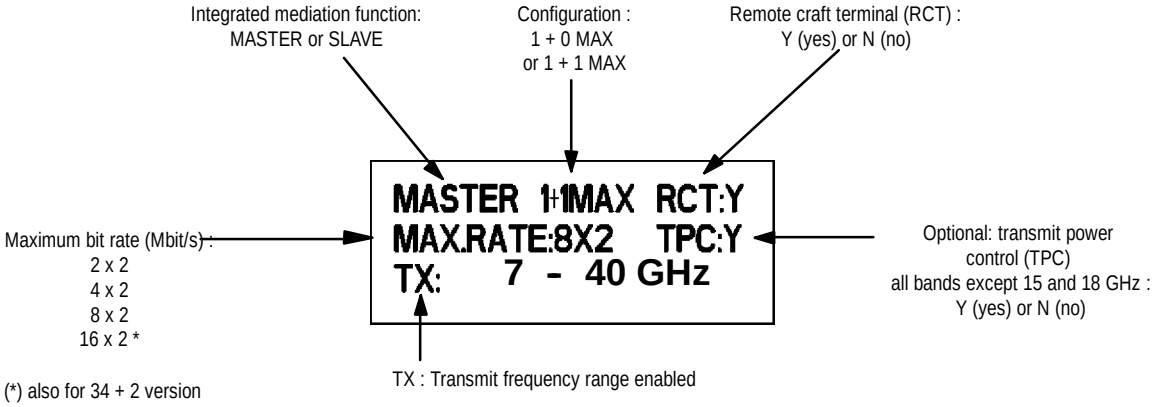
SOFTWARE KEY LABEL ON THE BOX



Integrated mediation function: + commercial code	Master or Slave
RCT (remote craft terminal)	Y (yes) + commercial code or N (no)
Max. configuration	1+0 max. or 1+1 max
Max bit rate + commercial code of each software module	2 x2 ; 4 x2 ; 8 x2 ; 16 x 2 ; 34 Mbit/s
TPC: only for 13, 23 to 38 GHz bands	Y (yes) + commercial code or N (no)
Transmit frequency range	
* Model of key / Revision index	
* Serial number of key	
Commercial code	

* bar coded or in plain language

SOFTWARE KEY LABELS



Typical label affixed to the software key

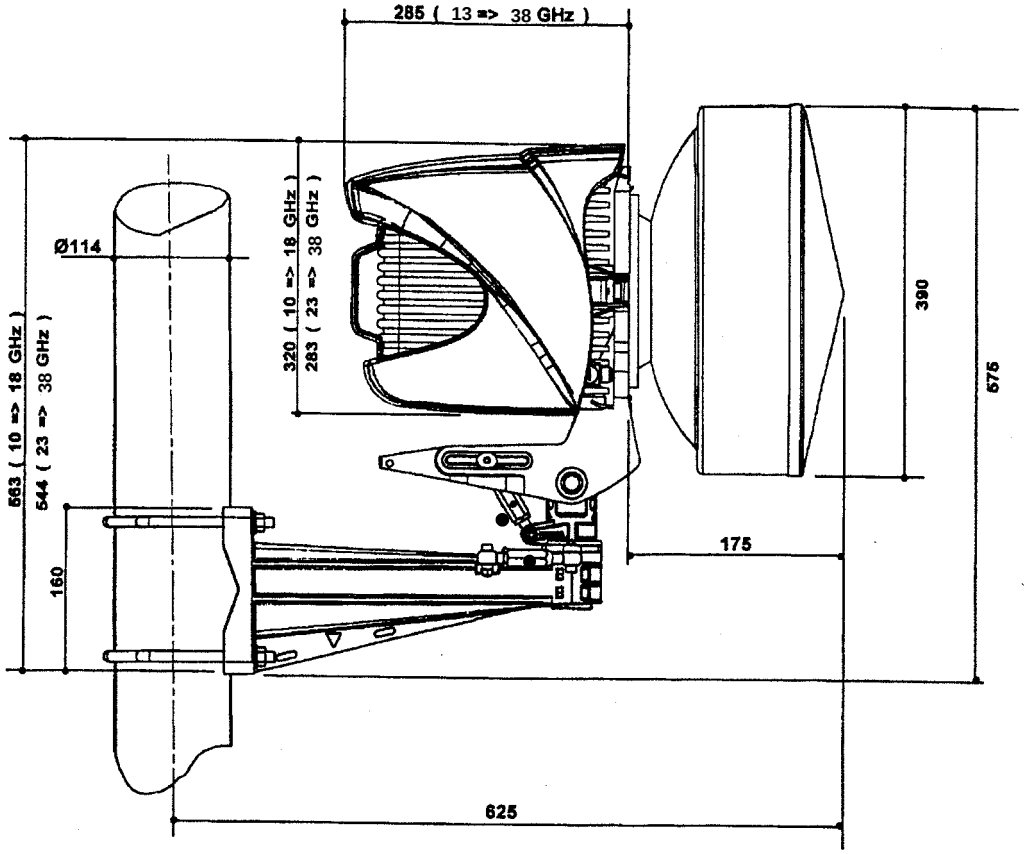
Item label
(on other side)



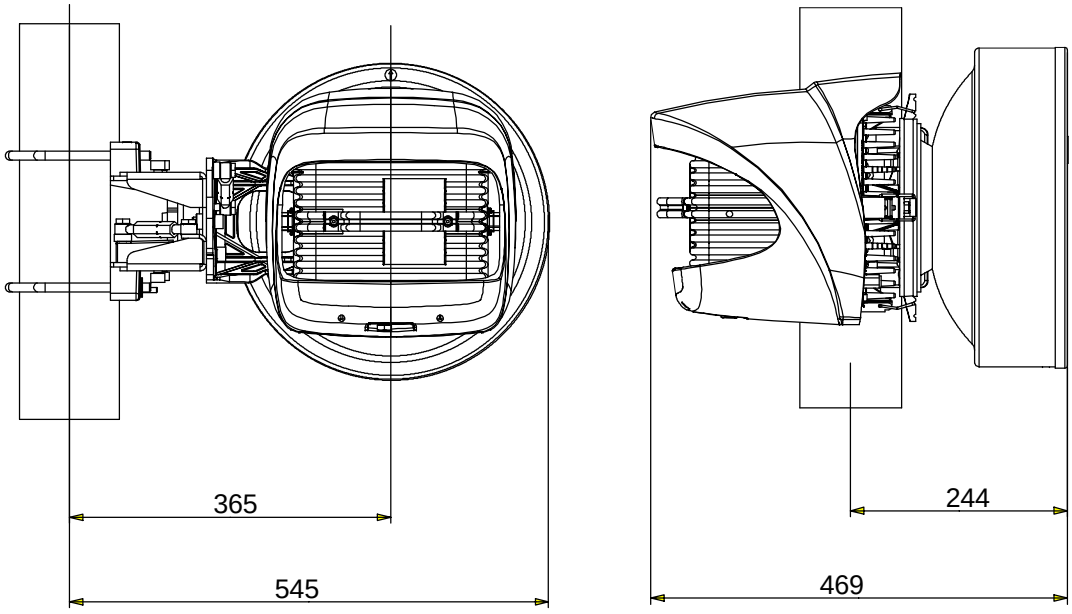
3.2.1 – Typical installation dimensions (1+0 configuration with integrated 1 ft)

3.2.1.1 – ODU 94xxUXR201 A 94xxUXR203

Pole mounting 1+1

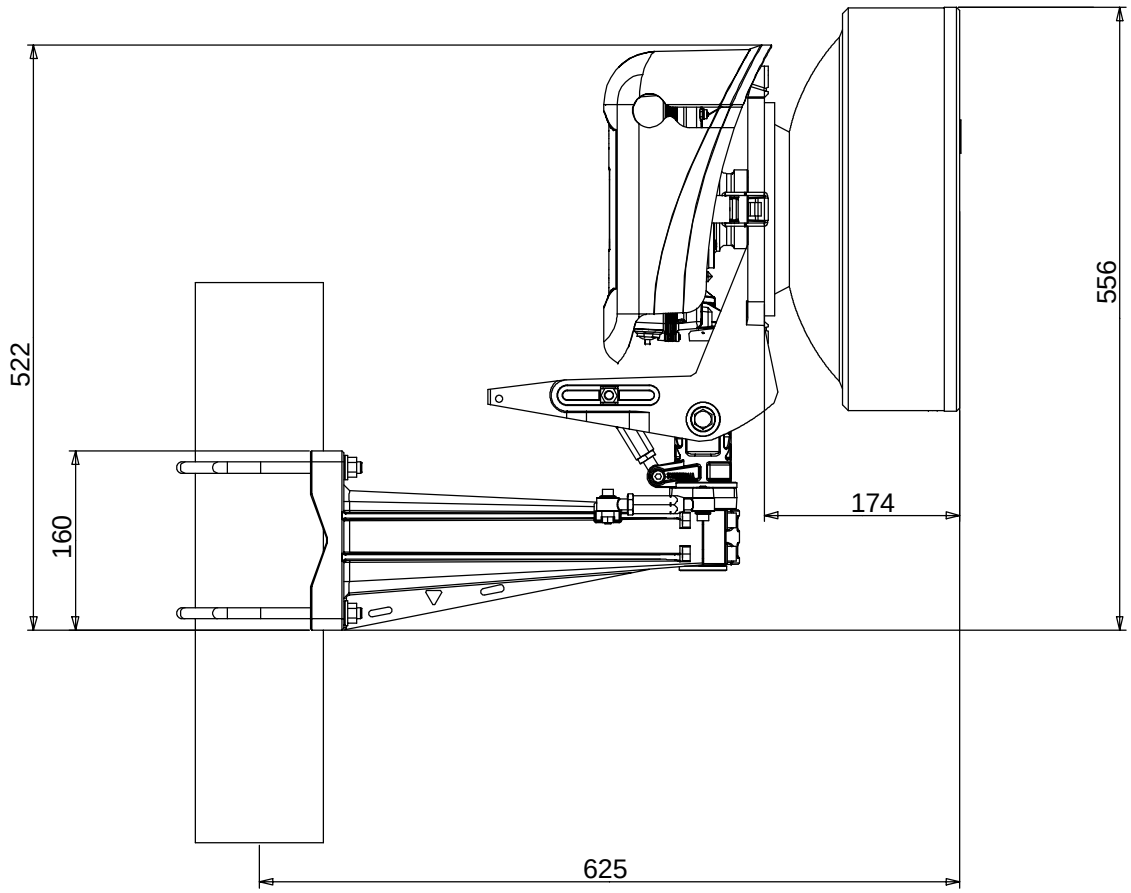


Pole mounting 1+0

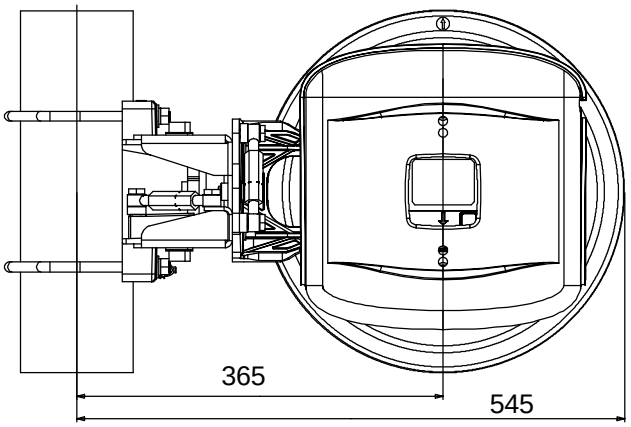


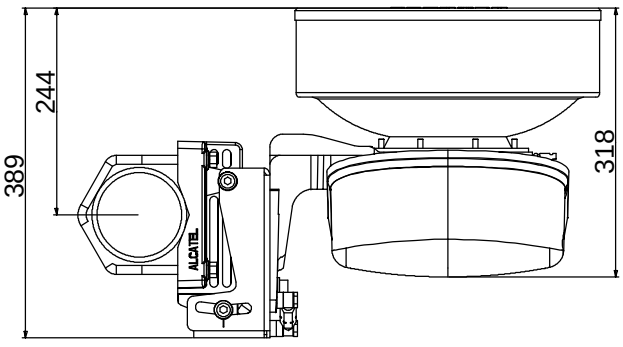
3.2.1.2 – ODU A9400 UX flat ODU

Pole mounting 1+1



Pole mounting 1+0





3.3 – Installing the equipment

It is assumed that the installer has already installed the other peripheral equipment (cable trays, multiplexes, etc).

3.3.1 - Information required for installation

Appendix 2 contains a sheet for you to complete to collate all the general information needed for the installation procedure.

3.3.2 - Precautions concerning electromagnetic compatibility (EMC) and safety

1) Installation is designed to meet all new requirements concerning electromagnetic compatibility and safety.

2) The EMC performance of the equipment depends largely on installation practices (cable installation, earthing connections, etc) which should be based on best trade practices.

3.3.3 - Tools required

The list of the tools required to install the microwave links of the 9400 UX family is given below (applies to all frequency bands).

Tool		Usage examples
2.5 mm Allen key (for M3 screw)	x	For adjusting the polarization of the antenna and the HSB coupler
3 mm Allen key (for M4 screw)	x	For adjusting the polarization of the antenna and the HSB coupler
5 mm Allen key (for M6 screw)	x	For mounting the antenna, the hot standby coupler
6 mm Allen key (for M8 screw)	x	For pointing the antenna
8 mm Allen key (for M10 screw)	x	For adjusting the polarization of the HSB coupler
“Pipe” wrench and **13** mm flat wrench		For mounting the pole mounting for a separate antenna
16/17 mm box wrench and flat wrench	x	For fixing the pole mounting and fine tuning the antenna and various fastenings
16/17 mm torque wrench		For fixing the pole mounting and various fastening with the correct torque
5 mm flat wrench		For dismantling/reassembling the cover on classic main, extension and access IDUs
8 mm flat torque wrench, Radiall, code: R282320		For fitting integrated attenuators in the ODU
20 mm flat torque wrench	x	For fitting the “N” coaxial plugs
3.5 x 100 mm screwdriver (for slotted head screws)	x	Indoor installations: connectors, software key, handles, or for changing the polarization of the antenna and for “non integrated” assembly
10 x 100 mm screwdriver (for slotted head screws)		For mounting the “pole mounting” in the case of a separate antenna.
Manually or with a screwdriver (for slotted head screws)		For mounting the solar shield of ODU A9400 flat ODU

The above tools marked with an “**x**” are included in a “station toolkit”, ref: **9400UXT103.**

Depending on installation, additional equipment may be useful:

- “Mars Actel” OSA3 Compax insertion and extraction tool, for wiring compax terminal blocks (mars actel IDC) COMPAX; and Sub-D plug crimping tool, ref: **608868-1 (AMP)**.

Of course, normal installer’s tools are essential: drill, drill bits, soldering iron, cable tie pliers, terminal pliers, etc.

A set of consumables, ref: **9400UXT002** for “one link” is available as an option.

For commissioning, an optional “service kit” (commercial code **9400UXT102**) can be used: this can be used to measure the AGC voltage (relative to the received field) using a DC voltmeter, and includes a headset/microphone assembly for connecting to the telephone service channel*.

* It is also possible to order just the “service kit” cable, under reference **9400UXT112.**
A compass and a pair of binoculars (not supplied) are useful for roughly pointing the antenna.

3.3.4 - Fasten torques for the screws and connectors

The table below shows the maximum fasten torques requested.

Screw caliber	Maximum fasten torques
	Ncm
Screw M3	61
Screw M4	150
Screw M6	490
Screw M8	1190
Screw M10	2380
Connector N (nut fastened onto the cable)	200

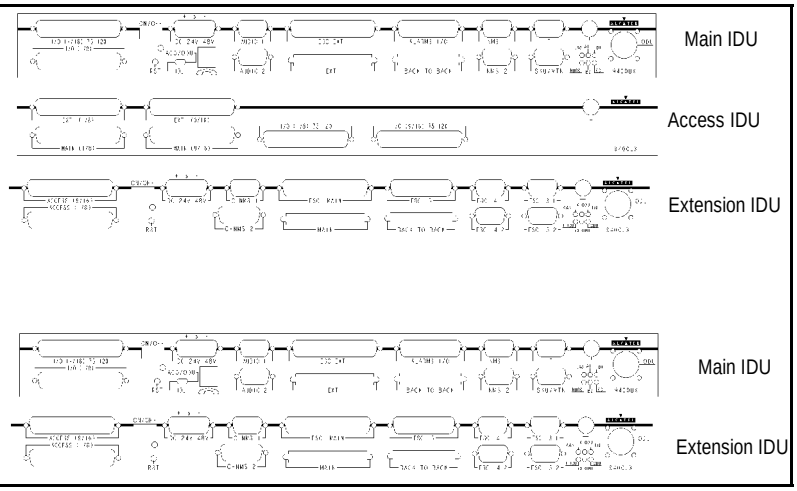
3.4 - Installing 19” IDUs

IDUs can be installed:

- front connector (Light IDU has only one connector panel),
- with the operation panel to the front.

They can be installed:

- on a desktop or on a wall mounting,
- in a laborack or 9U subrack.

Caution: when several IDUs are used, they must be stacked in the order shown opposite	

Before installation, check the labels for IDU configuration information (bit rates and port impedances), type of remote indication loops, EOW configuration.

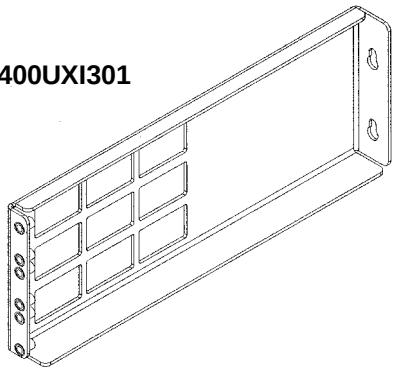
	CAUTION: do not place anything on the indoor units

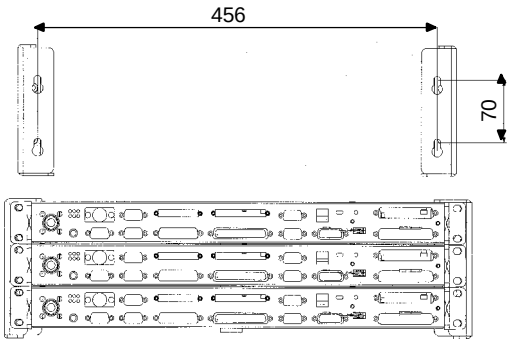
3.4.1 - Installation on feet or on a wall mounting

For desktop installation, fit each unit with its four feet (kit ref: **9400UXI303**), the feet clip onto the bottom of the unit.

To install units on a wall mounting, use kit ref.: **9400UXI301** which can support a maximum of three units. For this wall mounting, the feet of the units must not be fitted. Kit **9400UXI301** includes: two brackets, four fixings for the brackets (M6 x 50 socket cap screws, onduflex washers and 4620 expansion bolts), and fixings by the front handles for three units (washers 19621 and screws EX 10804).

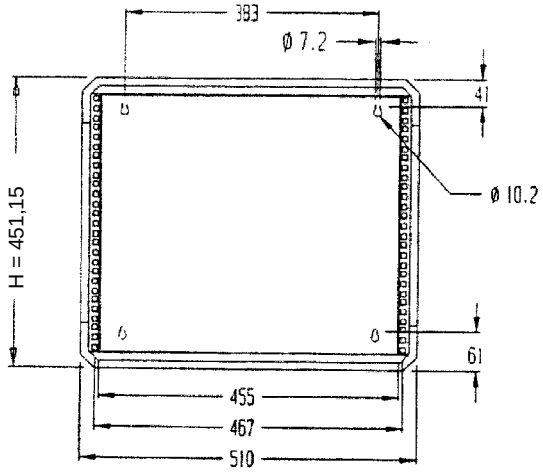
kit : **9400UXI301**

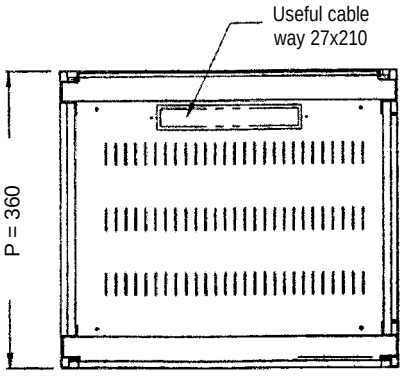




3.4.2 - Laborack or 9U subrack installation

3.4.2.1 - Installing the 9U unit (Ref. 9400UXI302)





3.4.2.2 - Alcatel recommendations for IDU installation in 9U rack

The recommendations depend on IDU configurations (1+ 0, 1+1 basic or 1+1 mux protected)

- 1+1 mux protected configuration : each group of 2 or 3 IDU shall be separated by 1U.
- 1+0 or 1+1 basic configuration : each IDU shall be separated by 1U.

This configuration supposes that room temperature should not exceed +45°C (and that the temperature at the IDU level never exceeds 55°C) .

These recommendations allow good installation conditions (place for cable interconnections) and good operation conditions (cooling of the IDU).

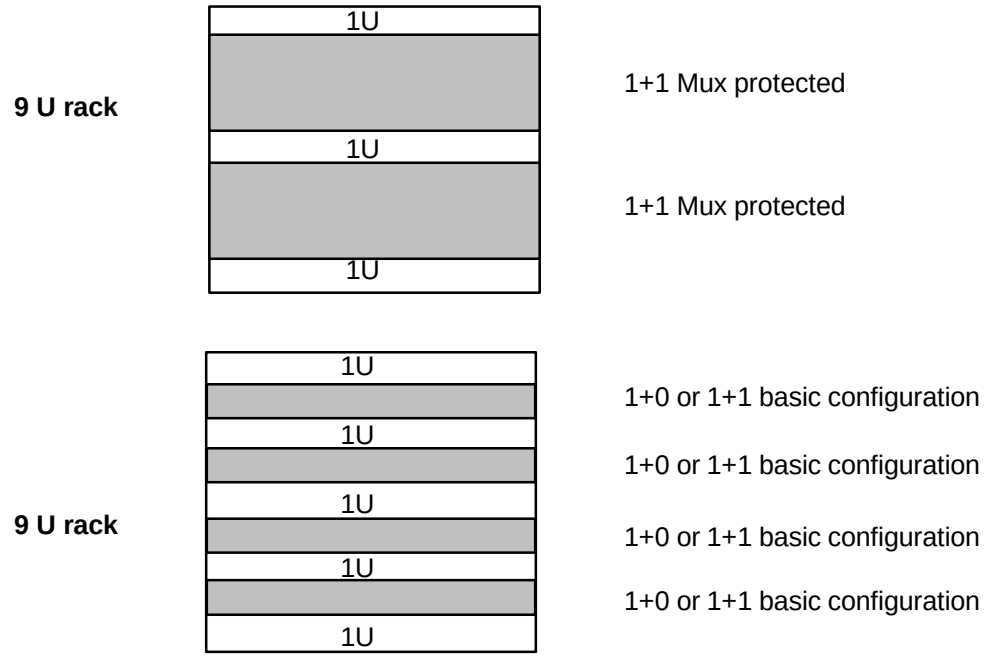
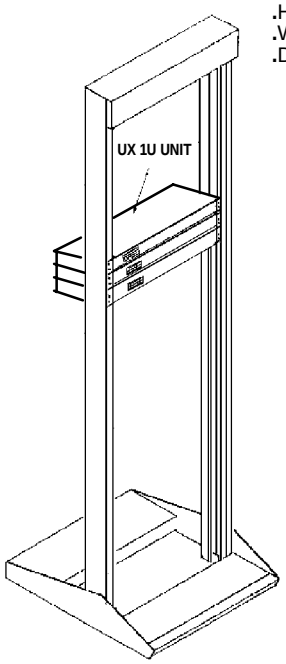


Figure 22 - Example of installation in 9U Rack

3.4.2.3 – Laborack installation

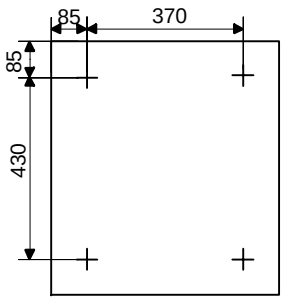


Laborack dimensions :

- Height = 2050 mm
- Width = 540 mm
- Depth = 600 mm

Ref. : **9400XXI300**

Drilling template



Front

- As standard, the laborack will be secured to the floor by four M6 socket cap screws supplied with the equipment.
- In the planned position, drill the floor in accordance with the template:
  drill holes: 12 mm ∅
  depth: 60 mm
- Fit the plugs and position the laborack
- Fix the laborack using the screws

When you have correctly positioned the fixing brackets on the 19" unit, the equipment will be held by the front panel by four screws fitted into the laborack's cage nuts.

3.4.2.4 - Alcatel recommendations for IDU installation in rack

The recommendations depend on IDU configurations (1+ 0, 1+1 basic or 1+1 mux protected)

- 1+1 mux protected configuration : each group of 2 or 3 IDU shall be separated by 1U.

If a group of two IDU is used, aeration can be improved leaving 1U between the two IDU.

- 1+0 or 1+1 basic configuration : each IDU shall be separated by 1U.

This configuration supposes that room temperature at IDU level should not exceed +55°C

Note : Be sure the cable path lets dismantling IDUs in 1+1

3.5 – Outdoor part installation

3.5.1 – General information

3.5.1.1 - Pole mounting

The mechanical assembly is supplied complete, mechanically assembled, with screw fastenings kit and ground terminals included in a plastic bag inside the casting.

The standard mechanical system is mounted on a pipe with a diameter of:

- **90 to 114 mm for 300 mm diameter antennas,**
- **114 mm for 600 mm diameter integrated antennas.**

	**CAUTION**: **If the pipe diameter is smaller than the one mentioned above, the fixing brackets of the pole mounting must be replaced by the kit 9400UXI103 that improves the wind resistance of the whole equipment. This kit is especially recommended for the 600 mm antennas on 76 to 90 mm tubes.**

There are also supports for fixing the pole to the wall: please consult us.

Various antenna types and sizes can be used, together with radio systems operating at different frequencies and in a variety of configurations.

	**CAUTION**: **Take care not to damage the O-Ring on the nose of the antenna connection.** **If the ODU is not to be mounted immediately after the pole mounting of the antenna, protect the connection nose (from water, dirt and impact).**

For installation with non-integrated antenna, please refer to the antenna manufacturer recommendations.

3.5.1.2 - ODU

9400 UX Outdoor units are designed for assembly:

- either with an integrated antenna connected directly to the Outdoor equipment, see § 3.5.4.1 and 3.5.4.5,
- or with one or more non-integrated antennas, see § 3.5.5., mounted with flextwist guides.

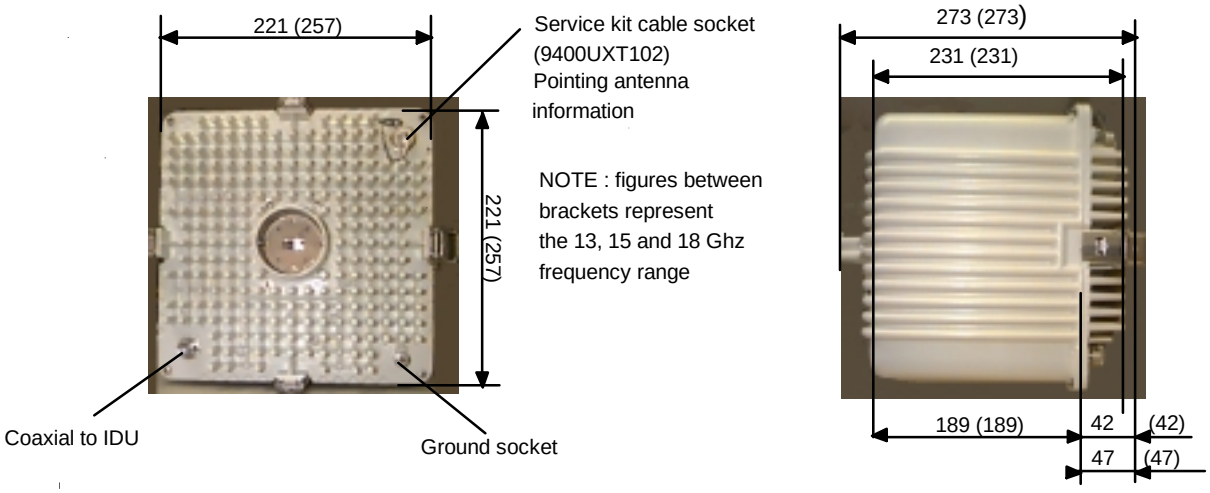


Figure 23 - ODU 94xxUXR201 94xxUXR202 94xxUXR203

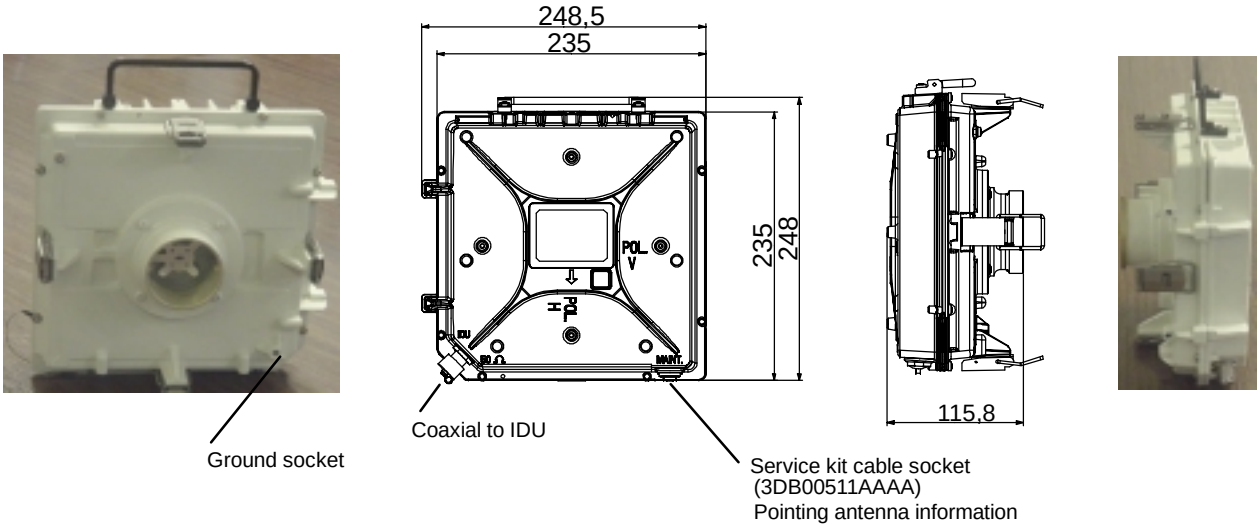
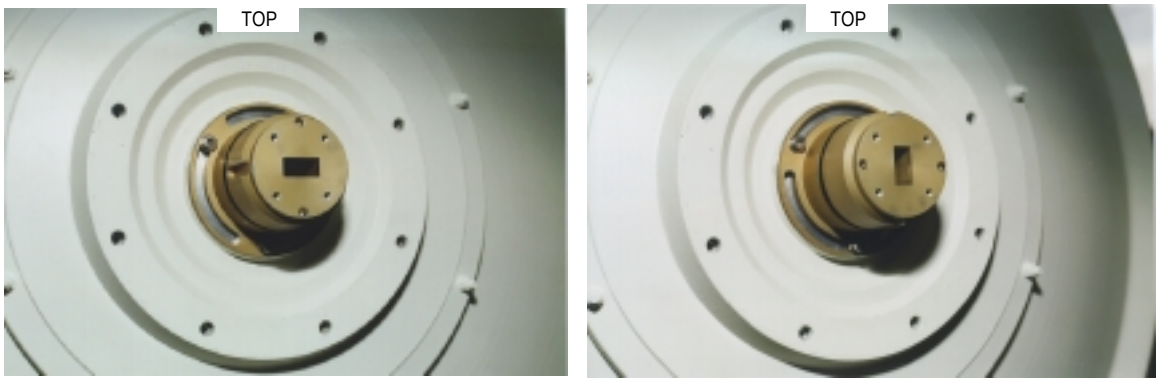


Figure 24 - ODU A9400UX FLAT ODU

	CAUTION: **Never remote the round yellow pads that are used to seal the system on the ODU.** **Do not stand on the ODUs, as they are not designed to support the weight of a man.**

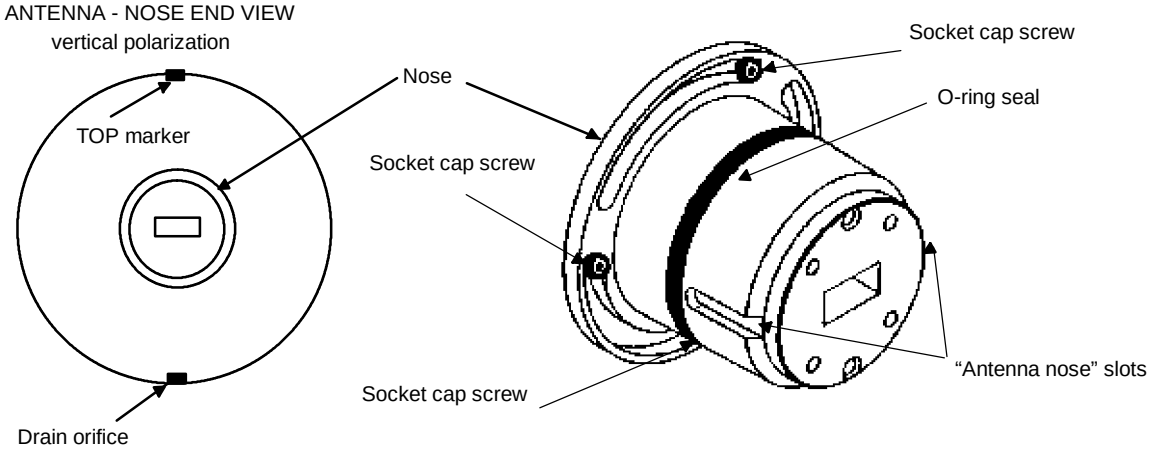
3.5.2 - Choosing antenna polarization

The antennas are normally supplied with vertical polarization.



Vertical polarization

Horizontal polarization



To change the polarization: undo the three socket cap screws and turn the nose through 90 degrees then tighten the screws again.

⚠	**CAUTION**: **The seal should be coated with silicone grease before fitting (the grease is provided with the antenna).**

3.5.3 - Installing the configuration with pole mounting 1 + 0 (9400UXI102)

The antenna is screwed on the pole mounting **9400UXI102**. The ODU is mounted with quick latches.

Overall antenna steer (with turnbuckles set to the "maximum") is:

- Azimuth: 360 degrees for a 300 mm and 600 mm diameter antenna, around the pipe.

The steer obtained by the turnbuckles is:

- Elevation: ± 25 degrees,
- Azimuth: ± 10 degrees.

To avoid obstacles (wall too close, etc), you can fix the pole mounting on any side of the pipe. The dimensions of the outdoor equipment are given in § 3.2.1**.**

NOTE: **The pole mounting 9400UXI102 is not designed to be fixed directly to the wall**

Procedure :

- 1) Position the antenna vertically (nose horizontal), with the drain hole in the bottom part and free of obstacles.
- 2) Insert the 7 M6 screws ① with their "onduflex" springy crinkle washers ② , tighten and secure the screws (these screws can be found in a plastic bag located inside the casting.

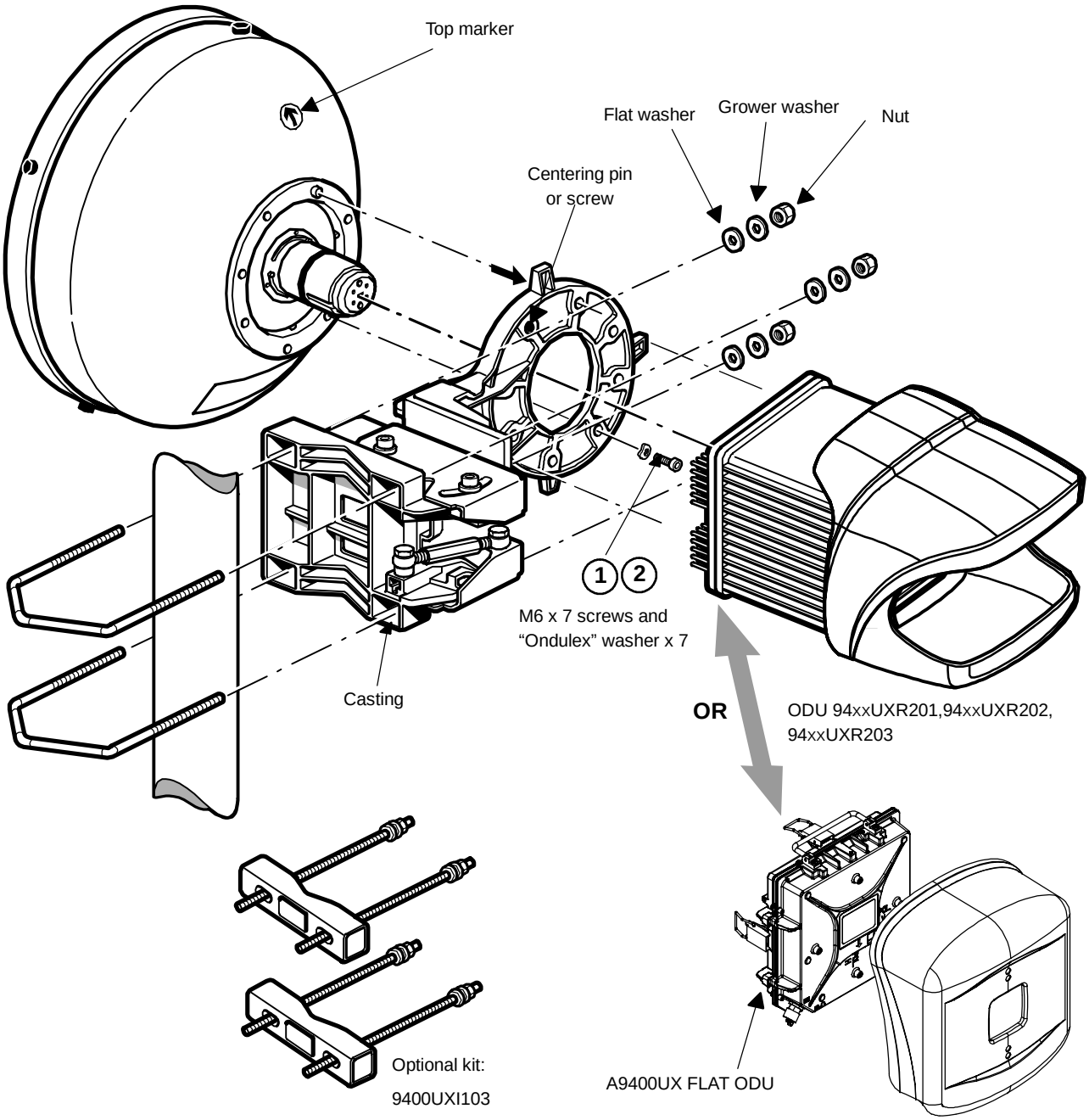
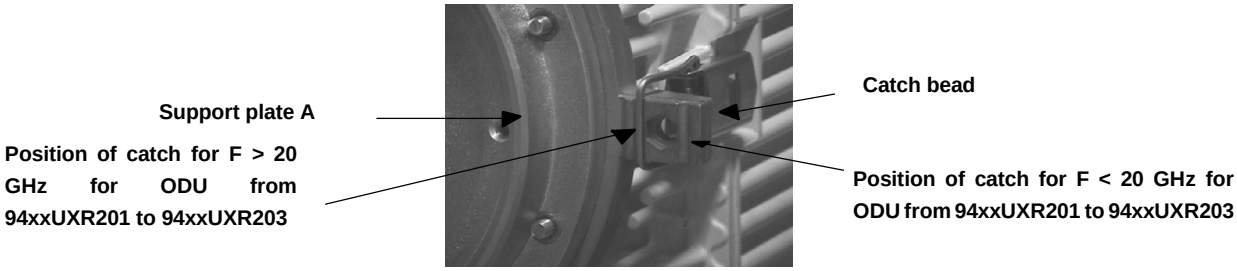


Figure 25 - Pole mounting 1+0 configuration (9400UXI102)

	**CAUTION: For the fasten torques of the screws, refer to chapter 3.3.4.**

3.5.3.1 - Installing the ODU 94xxUXR201, 94xxUXR202, 94xxUXR203

- Remove the solar shield by undoing the M6 fixing screw with a 16 mm flat wrench.
- Take the ODU by the handle, offer up the two protrusion bosses inside the “nose” of the unit to the two grooves in the “nose” of the antenna (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the top catch on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch,** and finish with the **top catch**.



- Reposition the solar shield via the back of the ODU from 94xxUXR201 to 94xxUXR203, **centre the bottom ribs of the solar shield** in the "grooves" of the ODU. Slide fully home, and **moderately** tighten by hand the screw (M6) in the recess on the underside of the ODU. Check the assembly by trying to extract the solar shield from its recess.

Note: The locking screw is not a fixing, it is simply intended to safeguard against inadvertent dismantling; do not overtighten.

	**CAUTION**: **If the screw cannot be reached by hand, use a 16 mm flat wrench (as an indication, fasten to a torque of 1mN, that is 0.1mdaN).**

REMINDER:

- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male "noses".
- When the ODU is fitted, the polarization of the antenna can be ascertained from the position of the ODU handle:

  handle arranged vertically = horizontal polarization, and vice versa.

Two choices per polarization are possible:

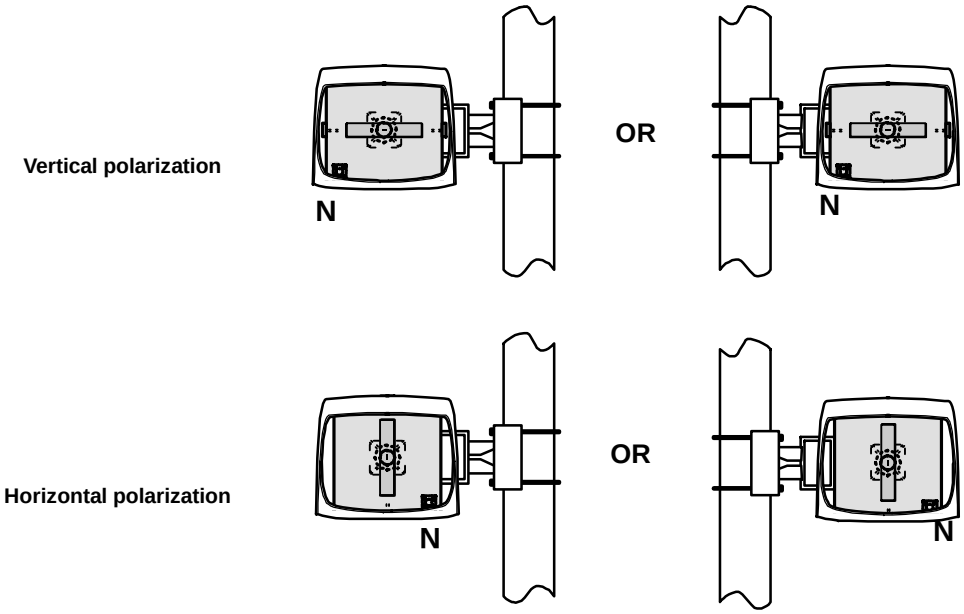
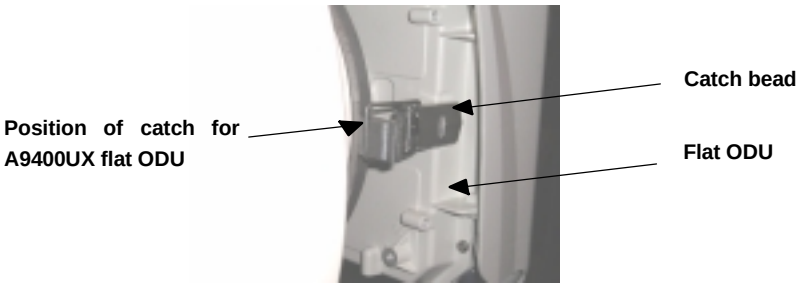


Figure 26 - Installing the ODU 94xxUXR201 to 94xxUXR203

3.5.3.2 - Installing the flat ODU

- Remove the solar shield by undoing the M6 fixing screw with a 16 mm flat wrench.
- Take the ODU by the handle, offer up the two protrusion bosses inside the "nose" of the unit to the two grooves in the "nose" of the antenna (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the top catch on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch,** and finish with the **top catch**.



- Flange the ODU handle.
- Reposition the solar shield via the back of the flat ODU. Slide fully home, and **moderately** tighten by hand the screw (M4) on the ODU.

CAUTION : When demounting, if the screws cannot be reached by hand, use a screw-driver.

- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male "noses".
- When the ODU is mounted, the 50 Ω coaxial jack is always facing down, the natural reading of the POL H or POL V etching shows the antenna polarization (see Figure 27).

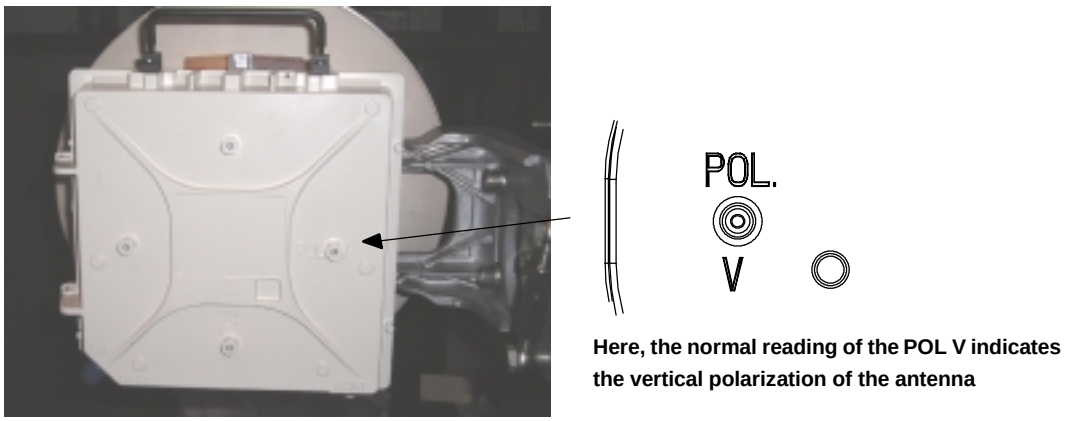


Figure 27 - Installing the ODU A9400UX flat ODU

- In vertical or horizontal polarization, the solar shield remains fixed in the same position and the ODU is the only part which can rotate 90 degrees when changing the polarization. (See Figure 28).

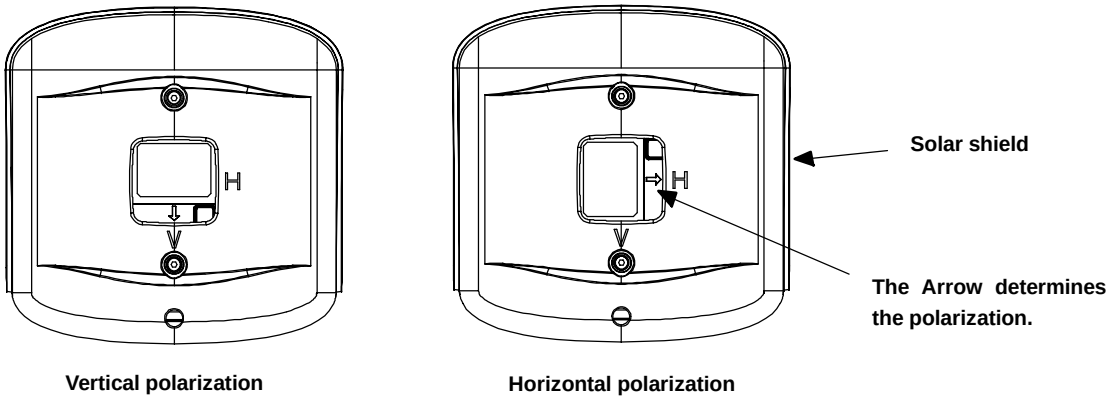


Figure 28 - Solar shield installation

3.5.3.3 - Installing on the pipe.

CAUTION: **Do not pick up the ODU/antenna assembly by the ODU handle.**

Note: Be aware that azimutal coarse pointing of antenna is done at this phase.

3.5.3.4 – Coarse pointing the antenna

Coarse pointing of the antenna is carried out as follows:

- **Roughly adjust the elevation orientation** of the “pole mounting” in the direction of the remote station. Use the antenna top marker (see figures in § 3.5.2), a compass and binoculars, then secure the brackets.
- **Azimutal coarse pointing** has to be carried out when installing the pole mounting.

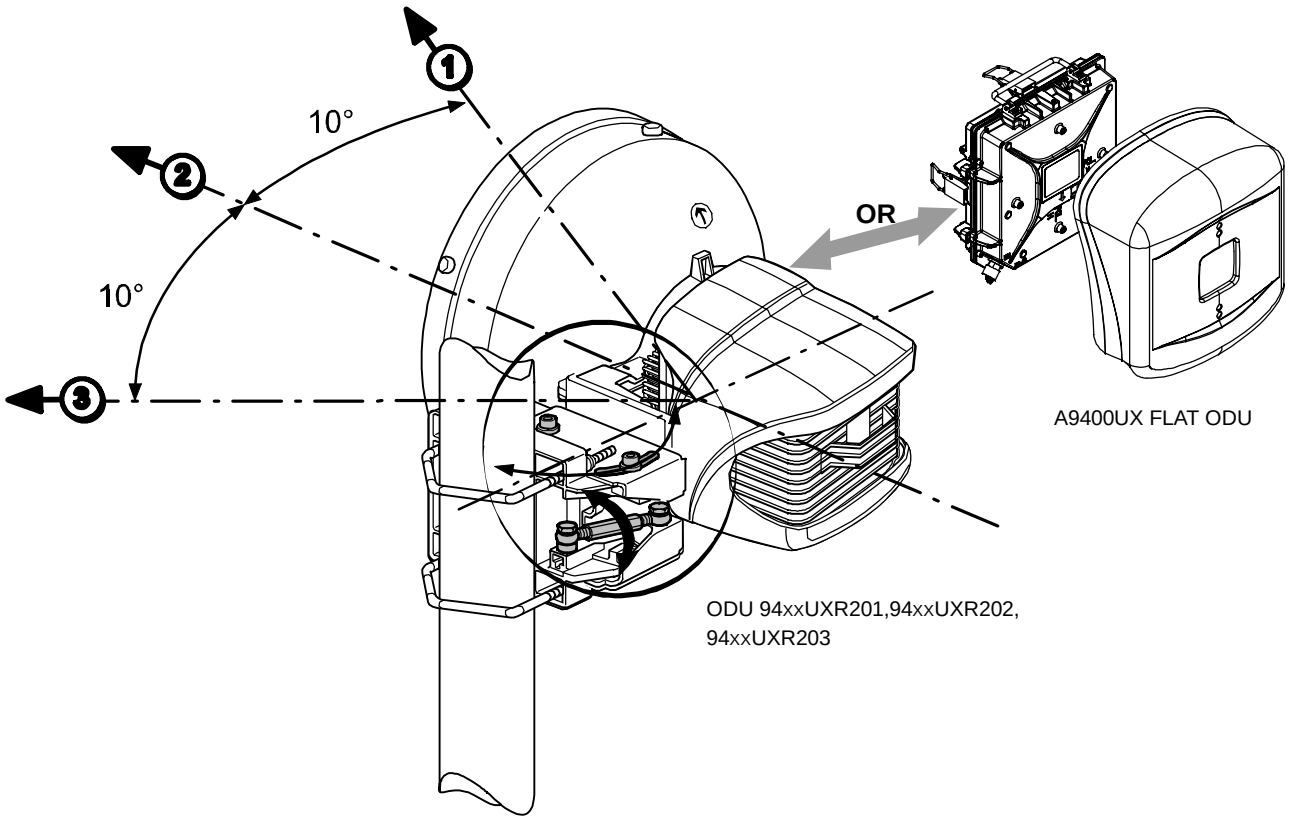


Figure 29 - Azimutal pointing of antenna

- **Elevation alignment** is performed only with the elevation turnbuckles**.** If the elevation angle is higher that (+ 5°) or lower than (– 5°), it is necessary to:
  - remove completely one of the 2 screws fixing the antenna support on the azimutal support and,
  - insert it in the third hole accessible.

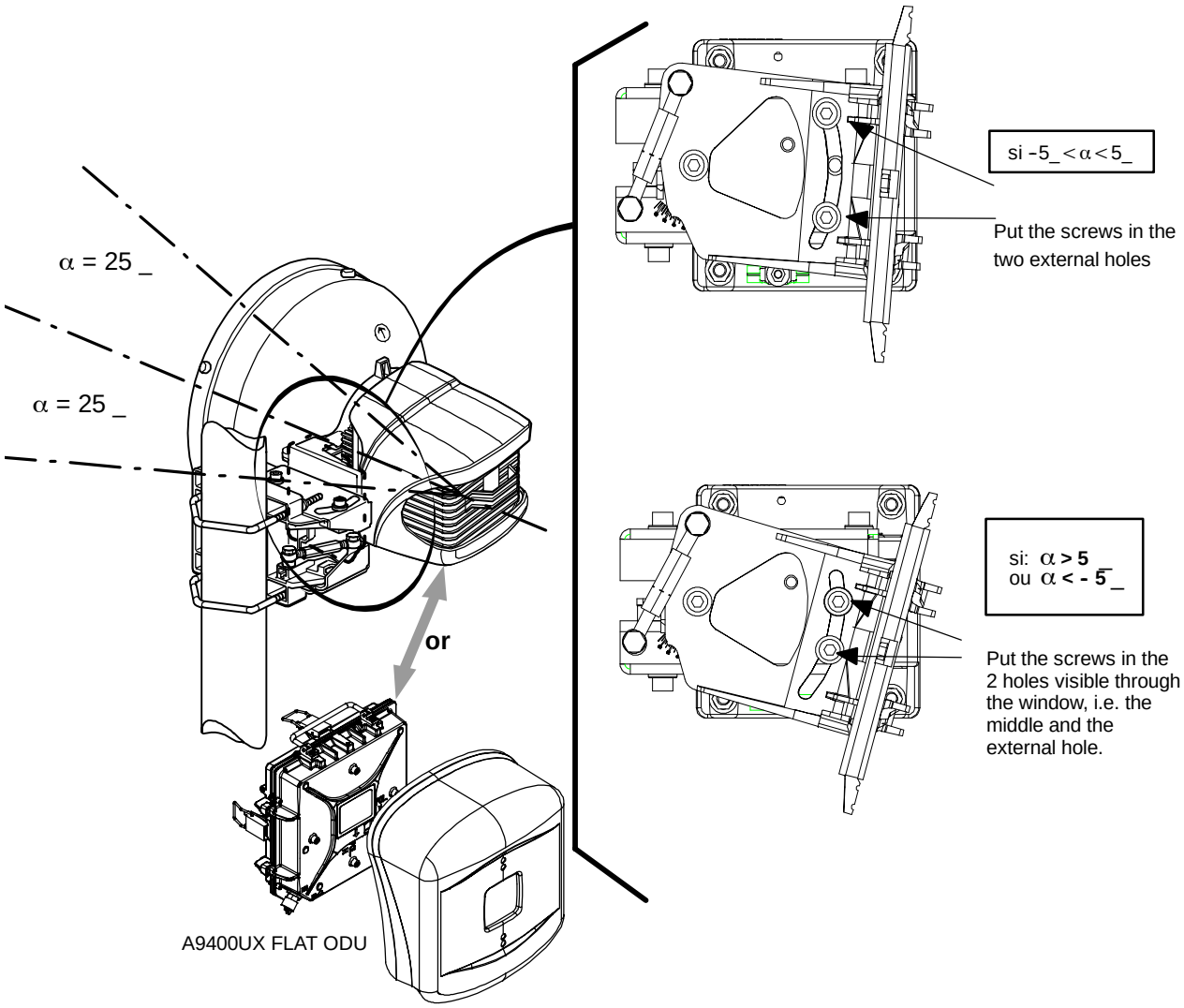
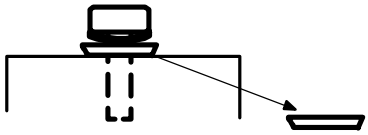


Figure 30 - Elevation coarse pointing of antenna

	**CAUTION:** **In either case, the support must be fixed with 3 screws, 2 put in the window and 1 corresponding to its axis of rotation.** **All screws must be tightened and secured only after fine pointing of the antenna**

	**CAUTION:** **Take care to put the 2 screws in the window with the flat washers positioned as shown is the figure below.**



3.5.4 - Installing the configuration with pole mounting 1 + 1 (9400UXI101)

This assembly is valid only for antenna diameters of up to 600 mm (two feet). The antenna is screwed to the pole mounting 9400UXI101. The ODU is mounted with quick latches.

Overall antenna steer (with turnbuckles set to the “maximum”) is:

- azimutal: 360 degrees for a 300 mm diameter antenna and 180 degrees for a 600 mm diameter antenna,
- elevation: ± 25 degrees.

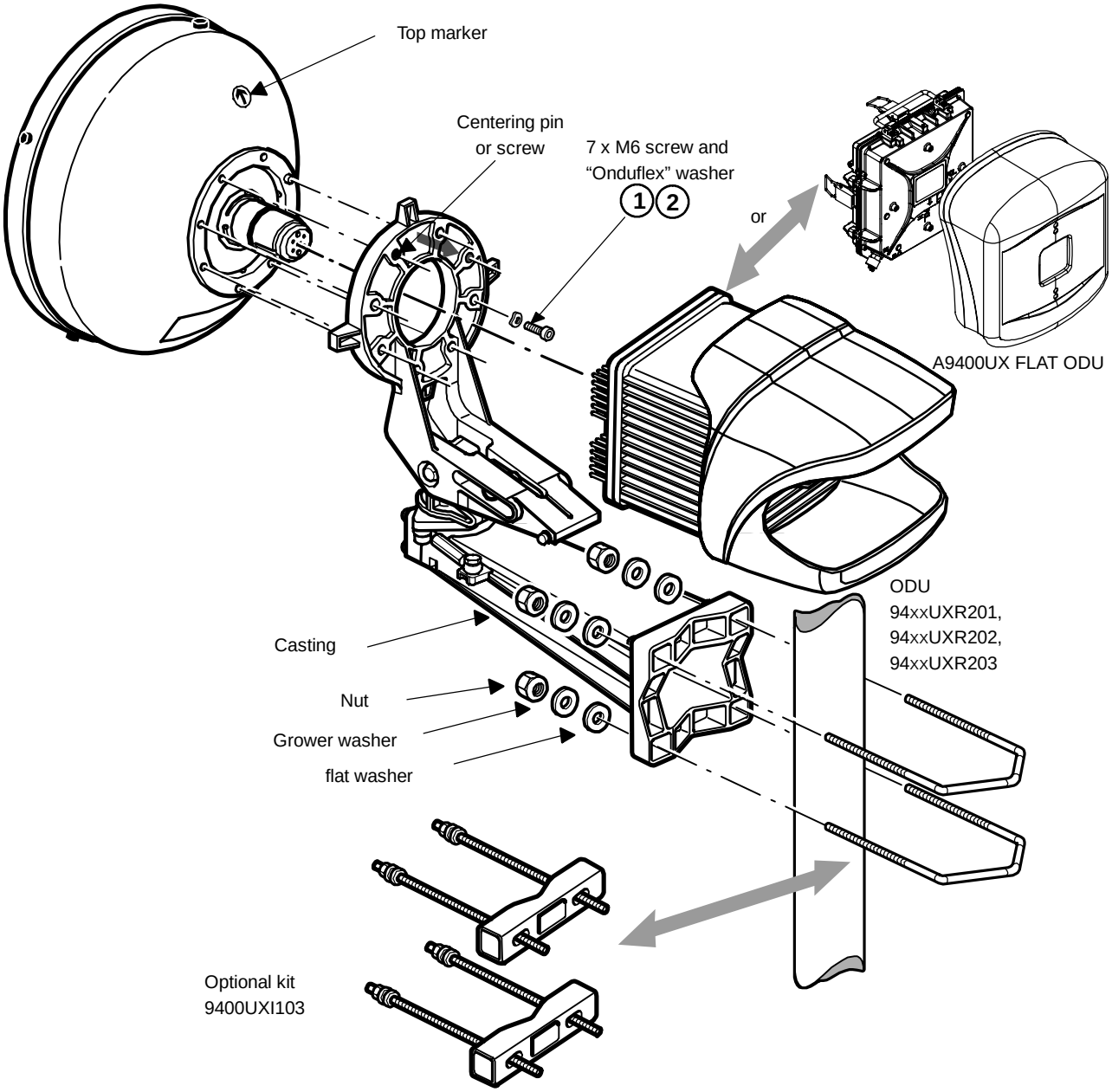
The steer obtained by the turnbuckles is:

- azimutal: ± 9 degrees,
- elevation: ± 9 degrees.

If you have physical difficulties assembling the mounting (wall too close, obstacle, etc), the “azimutal” turnbuckle can be moved and mounted on the other side of the pole mounting arm.

Procedure :

- 1) Position the antenna vertically (nose horizontal), with the drain hole in the bottom part and free of obstacles.
- 2) Insert the 7 M6 screws ① with their “onduflex” springy crinkle washers ② , tighten and secure the screws (these screws can be found in a plastic bag located inside the casting).

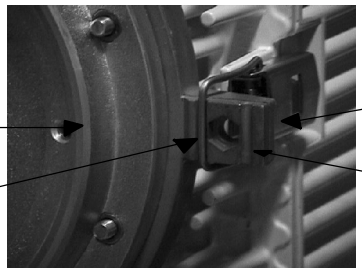


	CAUTION: For the fasten torques of the screws, refer to chapter 3.3.4.

Figure 31 - 9400UXI101 configuration

3.5.4.1 – Installing the ODU 94xxUXR201, 94xxUXR202, 94xxUXR203

- Remove the solar shield by undoing the M6 fixing screw with a 16 mm flat wrench.
- Take the ODU by the handle, offer up the two protrusion bosses located inside the “nose” of the unit and facing the two grooves in the “nose” of the antenna (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the **top catch** on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch,** to complete with the **top catch**.



- Swivel the antenna support through 90° on its azimuthal axis to release it from the pole. Immobilize the vertical axis by the five screws (see Figure 32, item ③).
- Reposition the solar shield via the back of the ODU, **centre the bottom ribs of the solar shield** in the “grooves” of the ODU. Slide fully home, and **moderately** tighten by hand the screw (M6) in the recess on the underside of the ODU. Check the assembly by trying to extract the solar shield from its recess.

Note: The locking screw is not a fixing, it is simply intended to safeguard against inadvertent dismantling; do not overtighten.

	**CAUTION**: **If the screw cannot be reached by hand, use a 16 mm flat wrench (as an indication, fasten to a torque of 1mN, that is 0.1mdaN).**

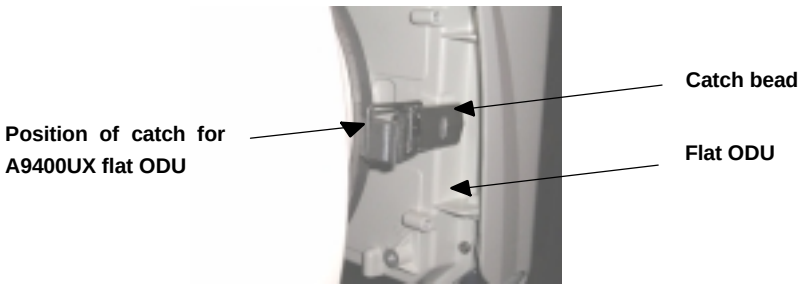
REMINDER:

- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male “noses”.
- For a 1+0 configuration, when the ODU is fitted, the polarization of the antenna can be ascertained from the position of the ODU handle:

  handle arranged vertically = horizontal polarization, and vice versa.

3.5.4.2 – Installing the ODU 9400UX flat ODU

- Remove the solar shield by undoing the M4 fixing screw manually or with a screw driver.
- Take the ODU by the handle, offer up the two bosses located inside the “nose” of the unit and facing the two grooves in the “nose” of the antenna (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the **top catch** on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch,** to complete with the **top catch**.



- Swivel the antenna support through 90° on its azimutal axis to release it from the pole. Immobilize the vertical axis by the five screws (see Figure 32, item ③).

- Flange the ODU handle.
- Reposition the solar shield via the back of the flat ODU. Slide fully home, and **moderately** tighten by hand the screw (M4) on the ODU.

	**CAUTION : When unscrewing, if the screws cannot be reached by hand, use a screw-driver.**

- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male "noses".
- When the ODU is mounted, the 50-ohm coaxial jack is always facing down, the natural reading of the POL H or POL V etching shows the antenna polarization (see Figure 27).

3.5.4.3 - Installing on the pipe.

	**CAUTION**: **Do not pick up the ODU/antenna assembly by the ODU handle.**

3.5.4.4 - Pointing the antenna

- **1) Roughly adjust the elevation orientation** of the "pole mounting" in the direction of the remote station. The elevation turnbuckle should remain in its middle position. Use the antenna top marker (see figures in § 3.5.2), a compass and binoculars, then secure the brackets ② in optional Kit 9400UXI103.

	**CAUTION**: **The bracket nuts should be fastened to a torque of 3 m.daN +/- 20%. The brackets must be clean and without grease except on the threads.**

- **2) Roughly point the antenna** support towards the required azimuthal by undoing the "azimuthal" locking screw. The azimuthal turnbuckle should remain in its middle position (approximately 85 mm centres); swivel the antenna support vertically for adjustment purposes, then retighten the "azimuthal" locking screw.

	**CAUTION**: **The azimuthal turnbuckle should not exceed 94 mm centre distance when fine tuning the adjustment.**

	**CAUTION**: **Never mount integrated antennas and ODUs under the pole mounting support.** **Never mount the elevation axis horizontally:** **ODUs must be above the pole mounting and not laterally offset.**

- **3)** On the "pole mounting", undo the 4 + 1 M8 screws (item ③ - see figure below) so that you will be able to fit the ODU later.

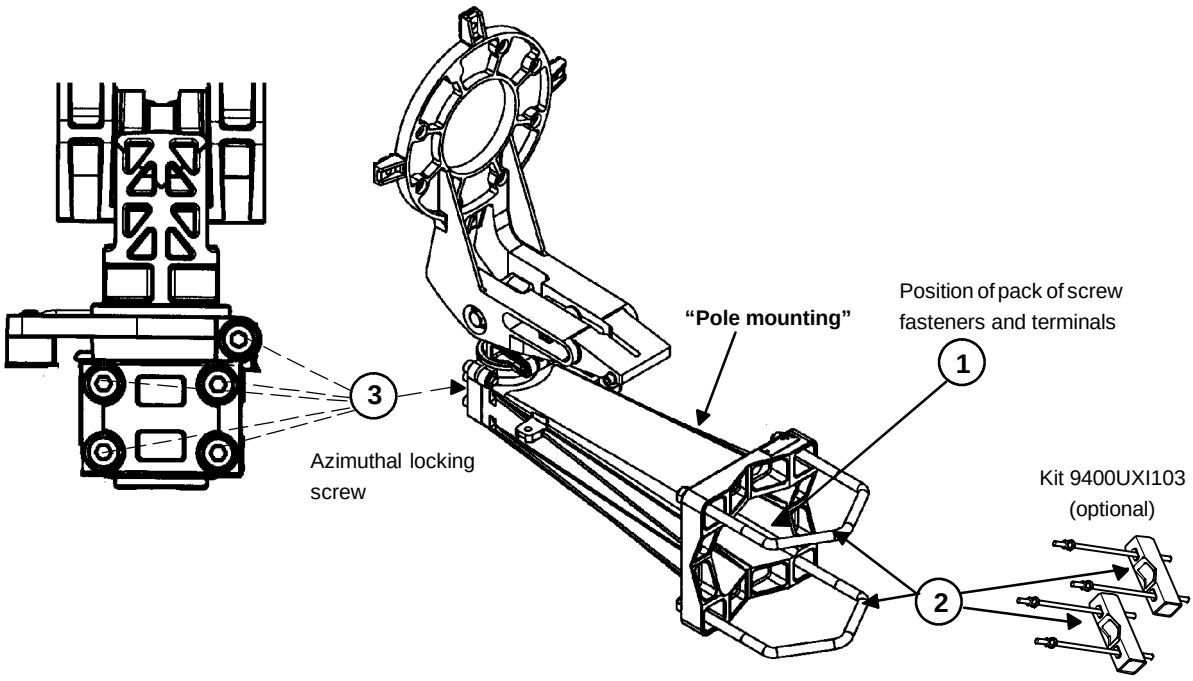


Figure 32 - Pointing the antenna

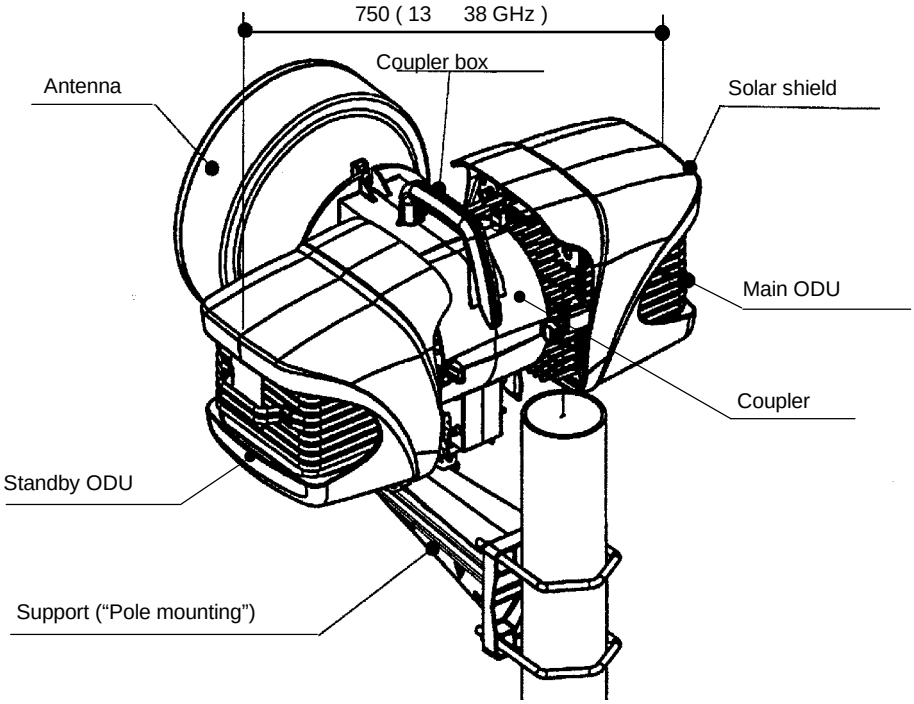
- **4)** Position the U bolts ② on the pole, fit the flat washers, "grower" washers and nuts.

⚠	**CAUTION: For the fasten torques of the screws, refer to chapter 3.3.4.**

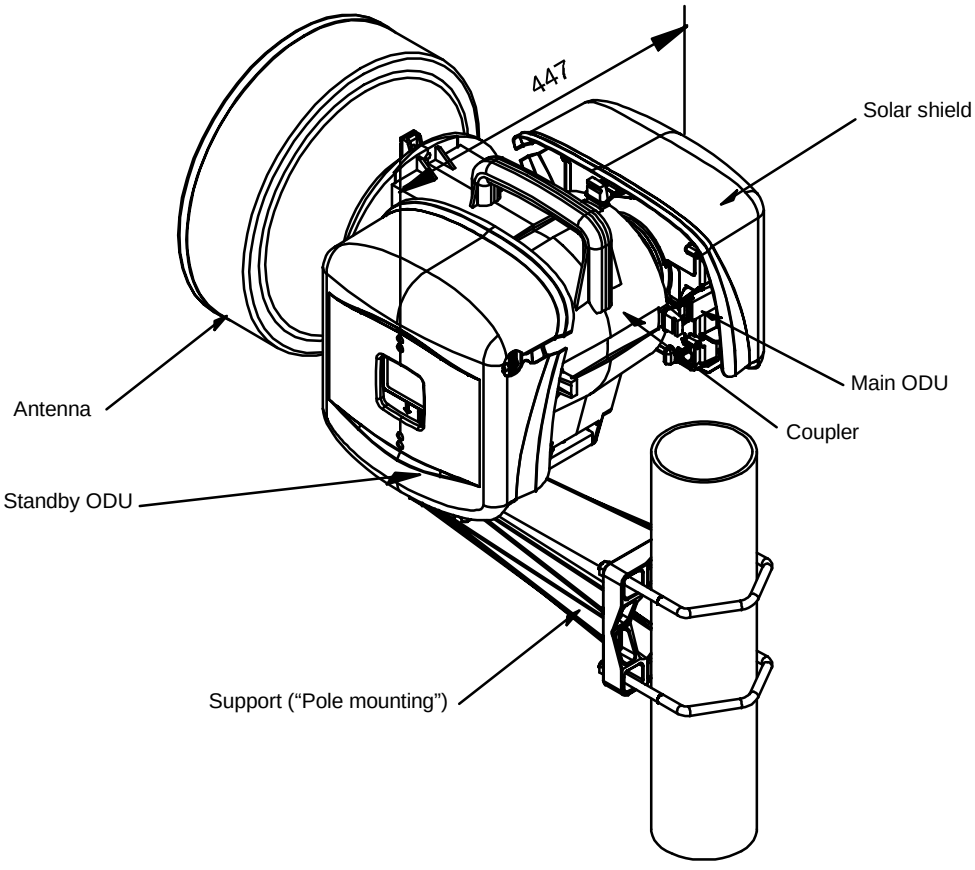
3.5.4.5 - Installing the (1+1) HSB configuration with integrated antenna

TYPICAL 1 + 1 HSB CONFIGURATION WITH COUPLER

- ODU 94xxUXR201 to ODU 94xxUXR203



- ODU 9400UX flat ODU



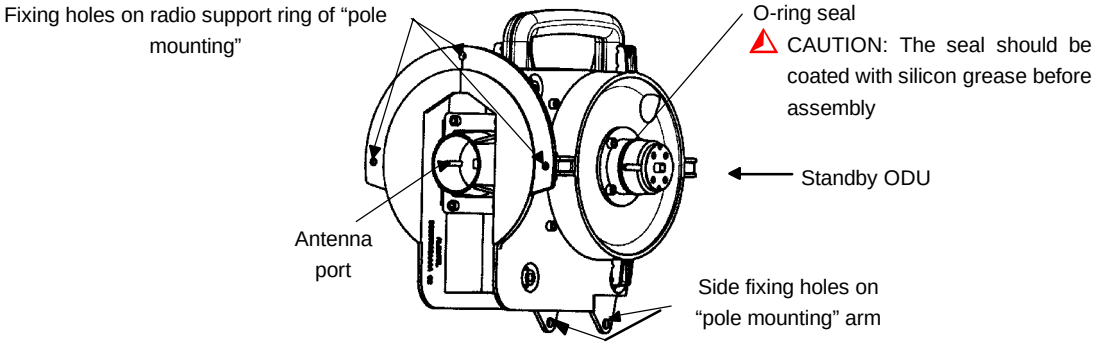
INSTALLING THE HSB COUPLER

IMPORTANT NOTE: NEVER REMOVE THE ROUND YELLOW PADS THAT ARE USED TO SEAL THE OUTDOOR SYSTEM.

The coupler is supplied with vertical polarization (see § 3.5.2). The procedure for changing the polarization of the HSB coupler is described in that section (see **§ 7.13**). This must be done before installation.

To install the HSB coupler assembly:

- take it by the handle, offer up the two bosses located in the “nose” of the coupler on the side of the antenna port, to the grooves in the nose of the antenna,
- ***Do not remove the sealing window*** located on the antenna port. The ODU/coupler/antenna assembly requires **no additional seal on the flanges**; the flanges are smooth and sealing is provided by the O−ring seal located around the male “noses”.
- fix the coupler using three M6 x 35 mm screws and nuts in the holes in the top “lugs” of the pole mounting.
- secure the coupler by the two side screws (M6 x 30) on the pole mounting.



CAUTION: **The seal should be coated with silicon grease before assembly (the grease is provided with the HSB).**

INSTALLING ODU

The positions of the main and standby ODUs are shown in the previous figures; for installing the ODUs, see § 3.5.4.1.

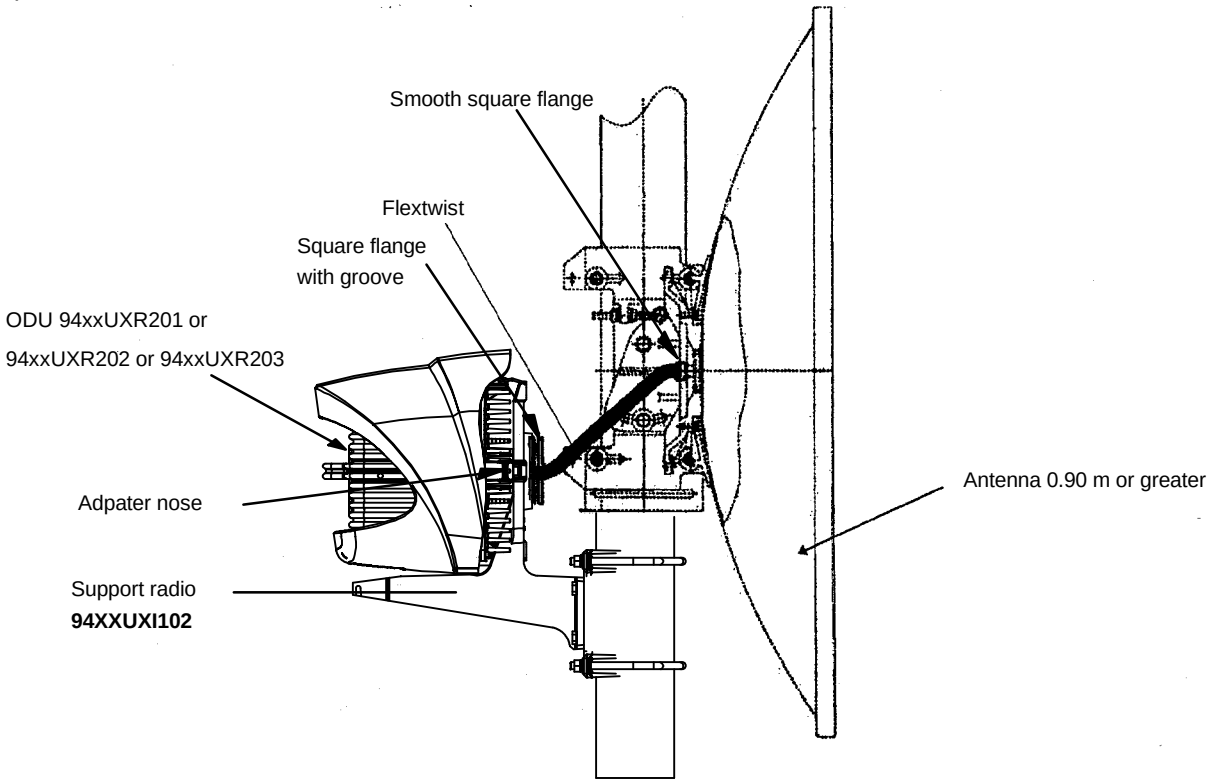
3.5.5 - Installation with non-integrated antenna

3.5.5.1 - Method

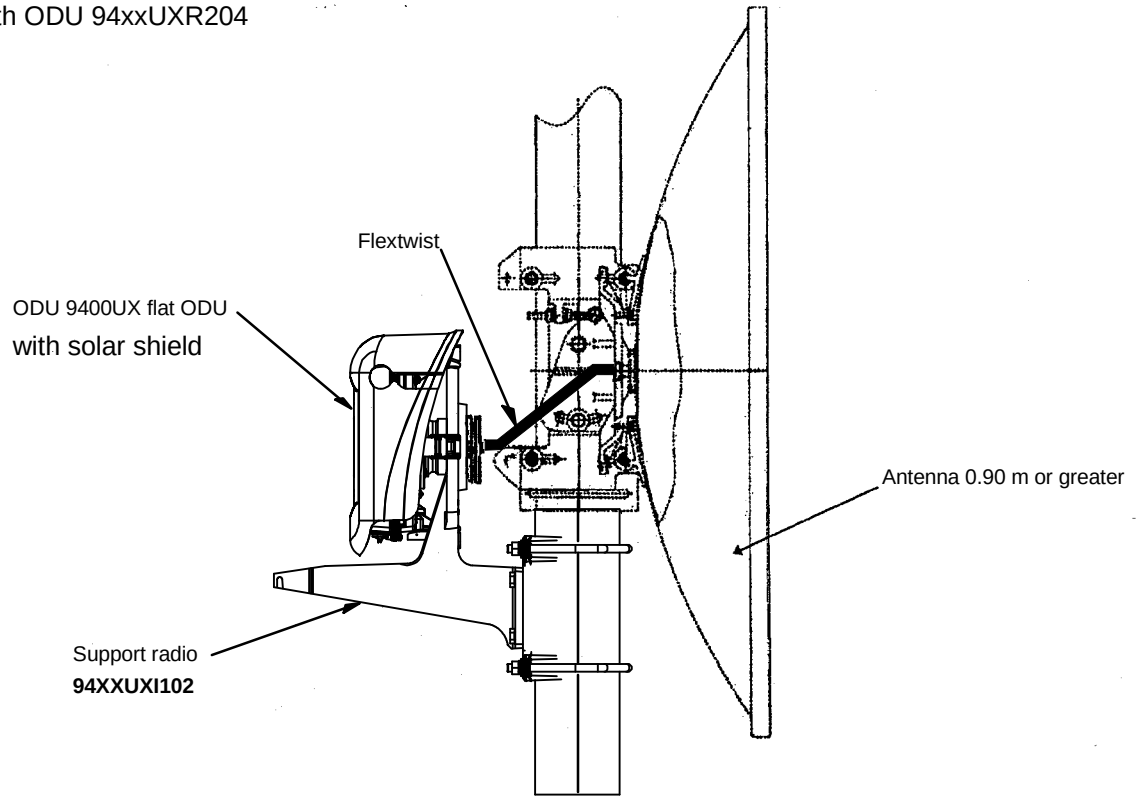
- Install the antenna on the pipe, according to the instructions given by the manufacturer and supplied with the antenna, and prepoint the antenna towards the remote station using a compass and a pair of binoculars,
- Install the pole mounting on the pipe, see § 3.5.4.3,
- Install the HSB coupler on the pole mounting (only for 1+1 configuration), see § 3.5.4.5,
- Install the ODU, see § 3.5.4.1,
- Install the flextwist (suited to the frequency) between the pole mounting and the antenna, see § 3.5.5.6.

3.5.5.2 – Typical installation

a) with ODU 94xxUXR201 or 94xxUXR202 or 94xxUXR203



b) with ODU 94xxUXR204



3.5.5.3 – Assembling and installing radio support

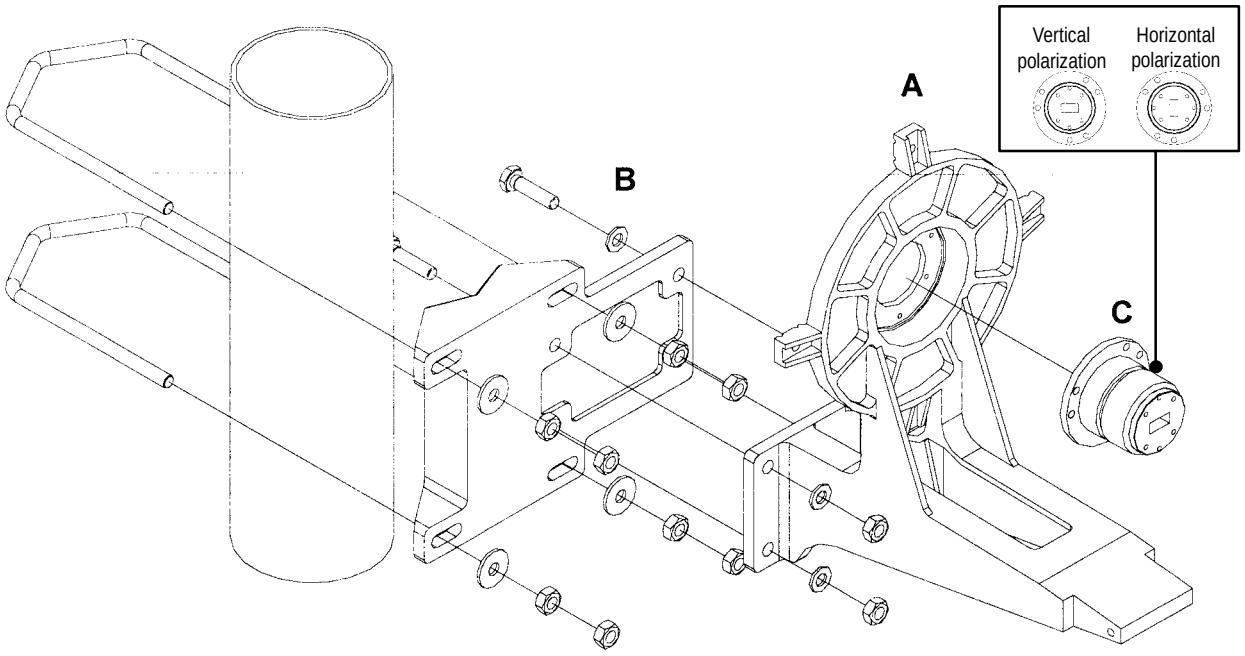
For installation, you will need the following:

- The ODU support, ref: **1AB128510002 is** regardless of frequency band.
- a microwave nose,
- a 600 mm long “flextwist” waveguide.

Note : As our ODU supports are chromate plated or tinned on the “flextwist”, be careful with the materials used facing the unit to prevent any galvanic couple.

Frequency range	Antenna side flange	Flextwist flanges	Radio side flange	Flextwist reference
13 GHz	PBR 120	UBR 120/PBR 120	UBR 120	9413UXI103
15 GHz	PBR 140	UBR 140/PBR 140	UBR 140	9415UXI103
18 GHz	PBR 220	UBR 220/PBR 220	UBR 220	94XXUXI103
23/25 GHz				
38 GHz	Only integrated antennas			

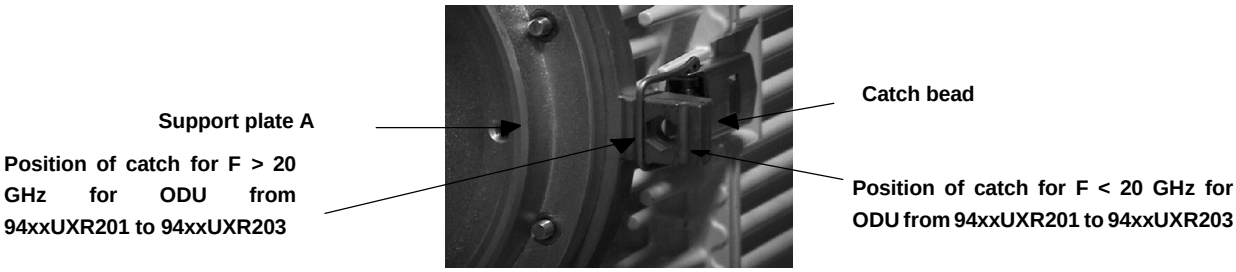
The pole mounting kit comprises the parts detailed below.



- Choose the horizontal or vertical polarization to match that of the ODU before fixing nose **C** (type dependent on the frequency) on plate **A** with the three M4 x 16 socket cap screws (3 mm Allen key).
- Choose the **direction of assembly** of plate **B** (on the right or on the left of the pipe); this should match that of the antenna, then bolt the plate **A** onto plate **B** with four M10 x 35 hex nuts and four washers.
- Offer up the assembly to the antenna support pipe, then secure it with the M10 U bolts, nuts, lock nuts and washers.
- Check that all the screw fastenings are correctly tightened.

3.5.5.4 - Installing the ODU 94xxUXR201, 94xxUXR202, 94xxUXR203

- **1)** Remove the solar shield by undoing the M6 fixing screw with a 16 mm flat wrench.
- **2)** Take the ODU by the handle, offer up the two bosses inside the “nose” of the unit to the two grooves in the “adapter nose” (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the top catch on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch** and finish with the **top catch**.

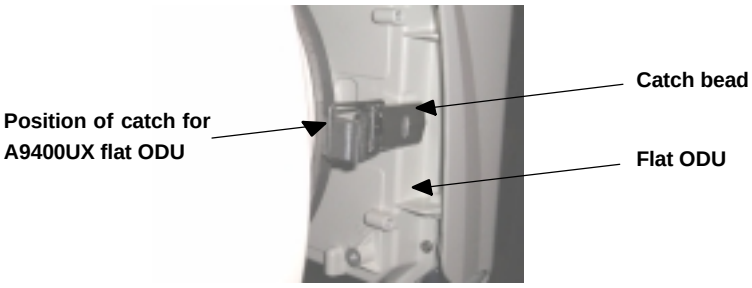


- 3) Reposition the solar shield via the back of the ODU, **centre the bottom ribs of the solar shield** in the “grooves” of the ODU. Slide fully home and, **by hand,** tighten the screw (M6) in the recess on the underside of the ODU. Check the assembly by trying to extract the solar shield from its recess

*(torque, by way of indication: 1 mN = 0.1 mdaN)

3.5.5.5 - Installing the flat ODU

- Remove the solar shield by undoing the M6 fixing screw with a 16 mm flat wrench.
- Take the ODU by the handle, offer up the two bosses located inside the “nose” of the unit and facing the two grooves in the “nose” of the antenna (the ODU-IDU connecting cable entry should be located in the bottom part of the ODU). Still holding the ODU, engage without fastening the **top catch** on the support, begin by fastening the **two side catches** on the support, then fasten the **bottom catch,** to complete with the **top catch**.



- Flange the ODU handle.
- Reposition the solar shield via the back of the flat ODU. Slide fully home, and **moderately** tighten by hand the screw (M4) on the ODU.

CAUTION : When unscrewing, if the screws cannot be reached by hand, use a screw-driver.

- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male “noses”.
- The ODU/antenna assembly requires **no additional seal on the flanges**; the two ends are smooth. Sealing is provided by the O-ring seal around the male “noses”.
- When the ODU is mounted, the 50-ohm coaxial jack is always facing down, the natural reading of the POL H or POL V etching shows the antenna polarization (see Figure 27).

3.5.5.6 – Installing the “Flextwist” waveguide

Refer to the example given in § 3.5.5.2 and the table in § 3.5.5.3.

This 600 mm long twistable flexible waveguide is supplied complete with gaskets and fasteners. At one end, it has a smooth square flange (to be mounted on the antenna) and at the other end, a grooved square flange designed to accommodate an O-ring seal (mounted at the ODU end). The kit comprises:

- one 600 mm “flextwist” waveguide,
- one O-ring seal for square flange,
- six M4 x 20 or M3 x 16 stainless steel socket cap screws (13/15 GHz or 18/23/25 GHz versions respectively),
- six M4 x 12 (13/15 GHz) or M3 x 12 (18/23/25 GHz) stainless steel socket cap screws,
- 12 18/10 stainless Z4 (13/15 GHz) or Z3 (18/23/25 GHz) flat washers,
- 12 18/10 stainless B4 (13/15 GHz) or B3 (18/23/25 GHz) “Onduflex” springy crinkle washers ,
- six 18/10 stainless HM4 (13/15 GHz) or HM3 (18/23/25 GHz) hex nuts,
- one record of measurements taken on the waveguide.



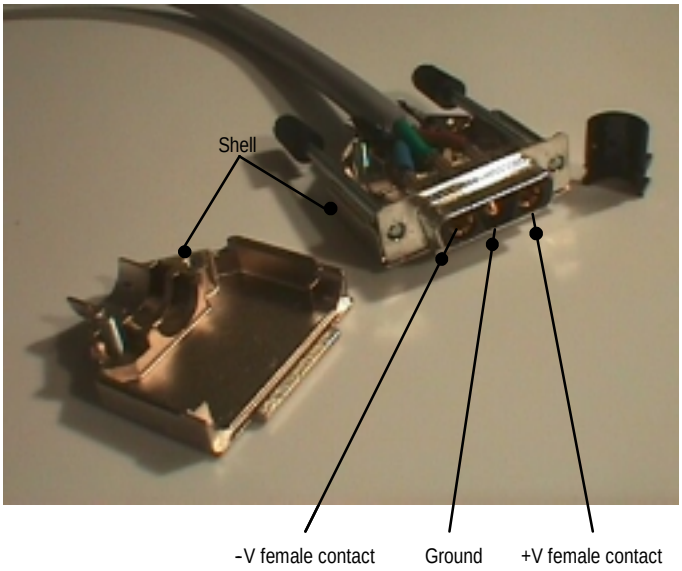
REMEMBER TO FIT THE GASKETS AND TIGHTEN THE SCREWS.

3.6 – Wiring

3.6.1 - Wiring the power supply to the 19” indoor units

3.6.1.1 - Direct connection with the power supply

The 48 V DC or 24 V DC connection kit is for setting up the power connection between the main or extension IDU and the distribution panel (one for each terminal). On the user side, **the plus or minus pole can be grounded** in the current version of the equipment.



Plug + shell reference: 3CC08048AAxx or ABxx
(supplied with the connector kit)

For 48 V supply, 3 x 1.5 mm^2 connecting cable (minimum), flexible, unshielded

For 24 V supply, 3 x 2.5 mm^2 connecting cable (minimum) flexible, unshielded

Preparing the connecting cable

3.6.1.2 - Power panel and IDU

19”, 3U internal power panel (ref: 9400XXC511)

This provides six ways protected by 6A fuses, providing remote connection to the “operator’s” power input.



Wall mounted power unit (ref: 9400XXC512)

This provides six ways protected by 6A fuses, for remote connection to the “operator’s” power input.



For both these products, the fuses used are **8.5 x 31.5 6A 400V gG** domestic type cartridges.

The cables supplied to connect the IDUs are:

- 48 V: **3CC08165AAxx**(3 x 1.5 mm^2),
- 24 V: **3CC08209AAxx** (3 x 2.5 mm^2).

To connect the “operator’s” input, the recommended cable is:

- 48 V : **3CC08211AAxx** (3 x 4 mm^2),
- 24 V : **3CC08212AAxx** (3 x 10 mm^2).

3.6.2 - Equipment earthing

Earthing connections are made on three separate parts:

- the indoor units,
- the outdoor units,
- the pole mounting mechanical assembly.

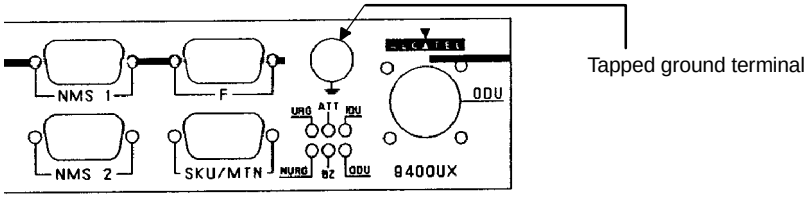
Earthing is bonded using green/yellow insulated cable with a minimum cross sectional area of 1x16 mm^2 towards the grounding vertical run. Wiring should be kept as short as possible. The overall length for all of the connections must not exceed 20 metres.

In addition, the IDU/ODU cable may be grounded using kit: **9400UXI204**

	**The ground connection for the ODU part should not be done through the lightning conductor.**

3.6.2.1 - On 19” indoor units (IDU)

The ground terminal comprises a tapped terminal (connector side of the unit, alongside the IDU/ODU coaxial socket), to which is fastened (by its M6 screw) a crimp terminal.



3.6.2.2 - On outdoor units (ODU)

The mechanical ground terminal comprises a tapped terminal located on the front of the ODU (see § 3.5). Earthing connections are by crimp terminal (this is supplied with the pole mounting’s kit of screw fasteners).

3.6.2.3 - On the pole mounting assembly

The ground terminal comprises a tapped hole located at the back of the cast part which supports the radio unit in the 1+0 configuration and the HSB coupler in a 1+1 configuration. The earth/ground terminal is based on a crimp terminal (this terminal is supplied with the pole mounting’s kit of screw fasteners).

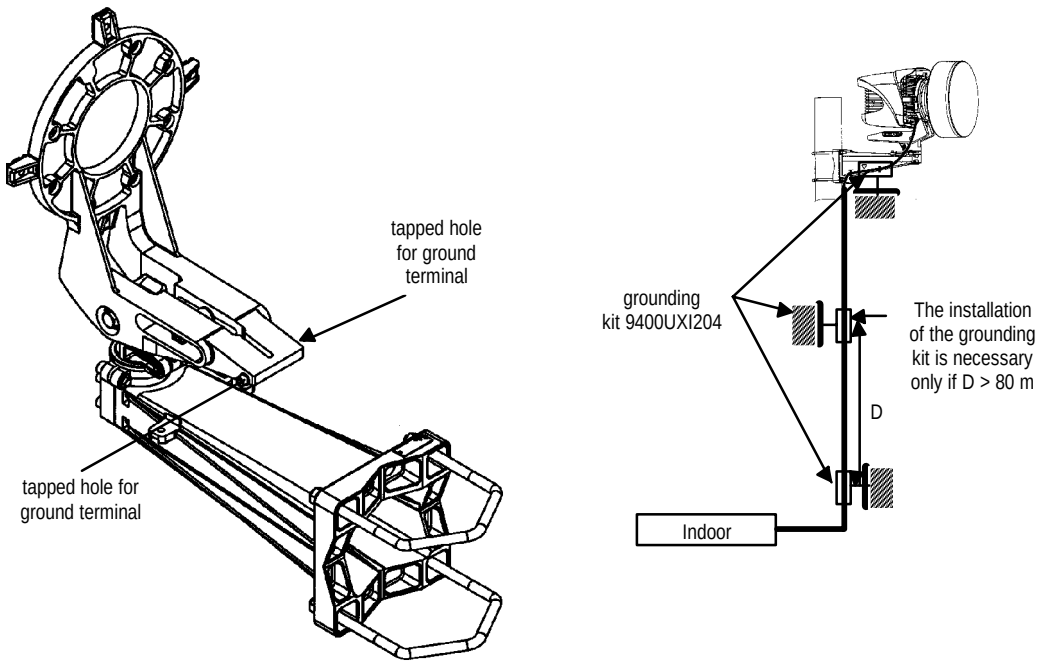


Figure 33 - Grounding the "pole mounting" and the cable

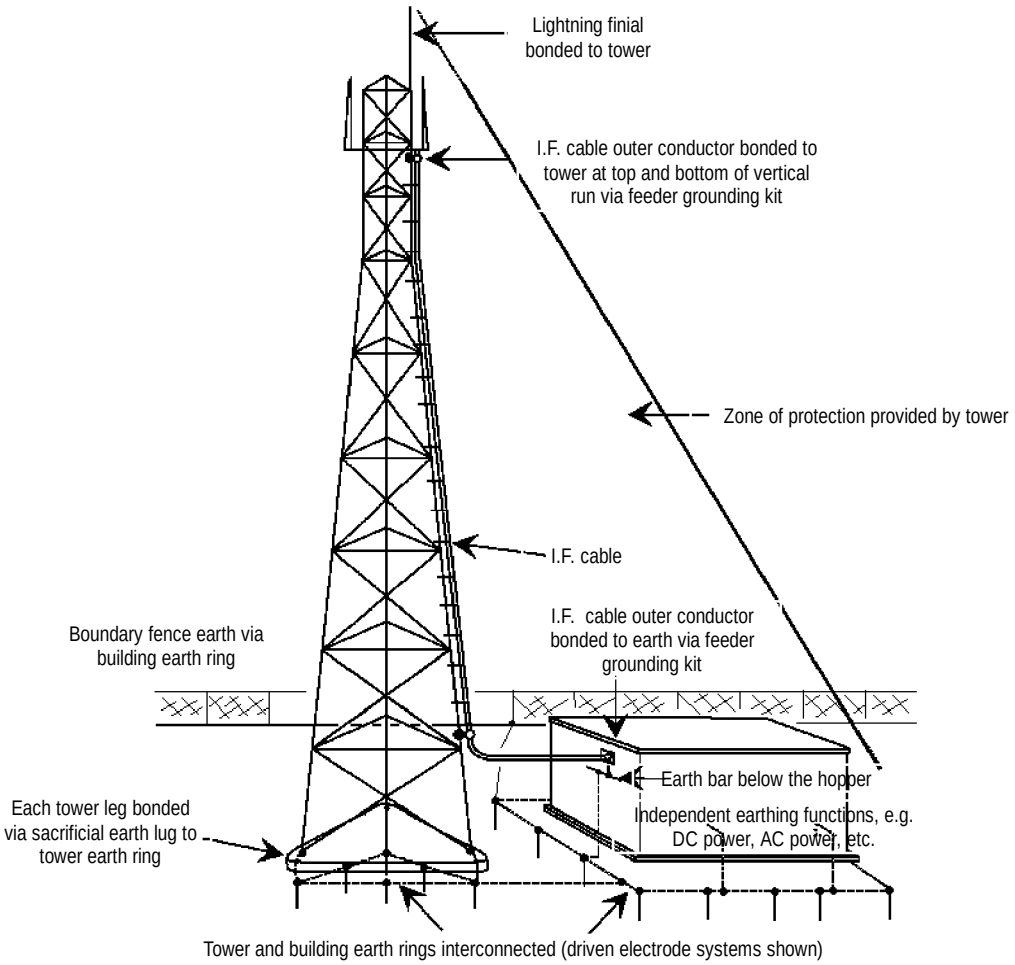


Figure 34 - Typical example of good earthing practice [ETSI document Draft EG 200 053 V1.4.3 (2002-02) (Page 91/145), informative]

3.6.3 – IDU/ODU wiring

The link is provided by a single 50 ohm coaxial cable (for each IDU/ODU link). For protection against interference and to ensure that it does not radiate interference (EMC requirements), the cable used is double-shielded, 11 mm in diameter, with an outer PU sheathing (for UV protection) with a maximum length of **300 meters** approved today under the reference **1AC001100022**.

The outer sheathing of the cable is marked at approximately 60 cm intervals with the cable reference **1AC001100022**. The cable is terminated by male, elbow "N" type connectors, ref: **9400UXI203**. The connectors are watertight, but, for added protection, when connected, preferably wind self–amalgamating tape supplied with the consumables kit (optional) around the connector.

Instructions for fitting these connectors are given in Appendix 5.

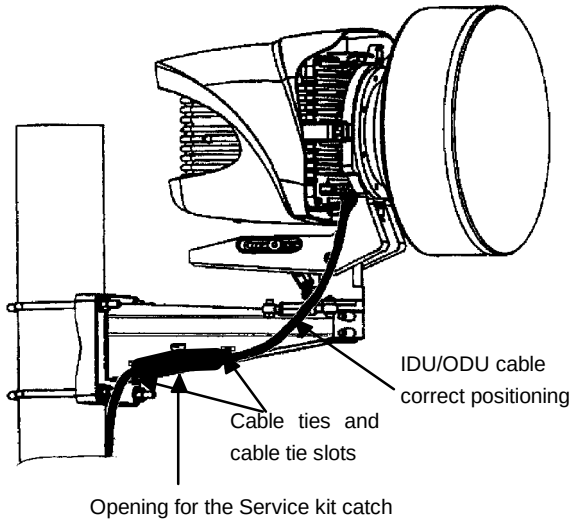
The greatest possible attention should be paid to the wiring of the coaxial plugs and to compliance with the cable installation rules, such as: minimum bending radius of 80 mm, watertightness, etc. If ordered by the customer, a cable earthing kit may be supplied; this is for sites where radio frequency interference is high, for example: radio stations, TV transmitters, etc. Kit ref: **9400UXI204** (see figure 33).

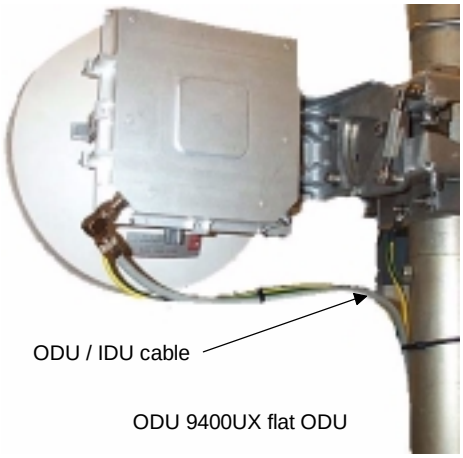
Instructions for the use of the cable earthing kit are supplied with the kit.

In a 1 + 1 configuration, the difference in length between IDU / ODU cables must not exceed 20 metres.

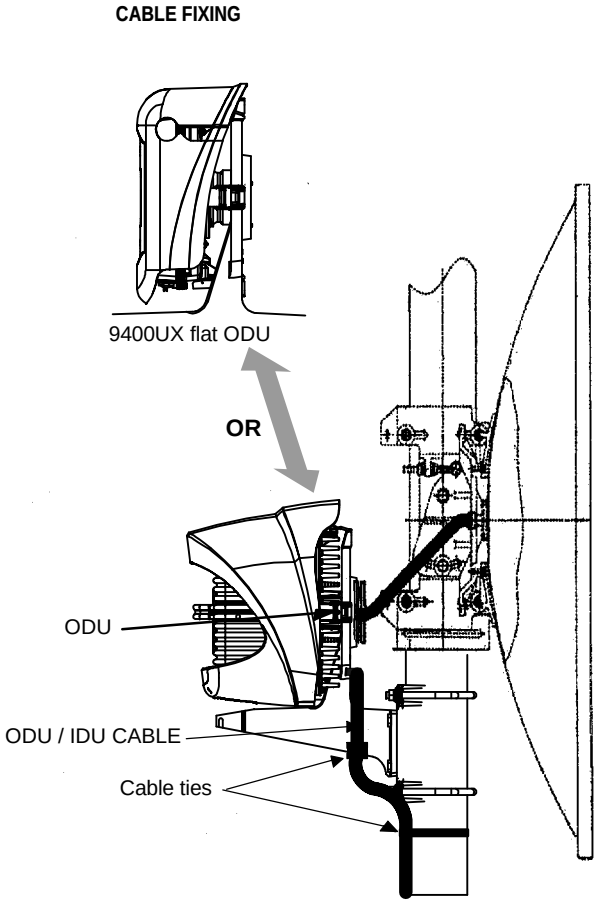
3.6.3.1 – Fixing the IDU/ODU cable

The coaxial cable is fixed to the pole mounting as rigidly as possible to avoid repetitive movement caused by mechanical or atmospheric vibrations that might in turn damage the cable or connector.





ODU 9400UX flat ODU



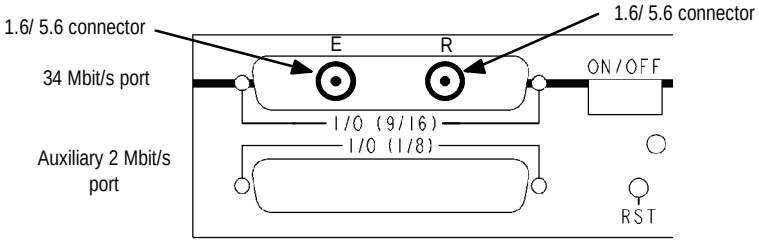
The cable is fixed along the supporting pipe or pole by the normal methods (Colson cable ties are supplied in the optional consumables kit)

3.6.4 - Wiring the 34 Mbit/s tributary and auxiliary 2 Mbit/s stream

The 34 Mbit/s port is implemented by 1.6/5.6 connectors located on a plate installed at:

- "**I/O (9/16)**" on the main IDU for 1+0 and 1+1 configurations,
- "**I/O (9/16)**" on the access IDU for 1+1 configurations with MUX protection.

If a BNC connector is required for the 34 Mbit/s stream, use a BNC/1.6/5.6 adapter kit, ref: **9400XXI405.**



The auxiliary 2 Mbit/s port uses female 37-pin Sub-D connectors:

- "**I/O (1/8)**" on the main IDU,
- "**I/O (1/8)**" on the access IDU.

SIGNAL	PIN				LAU : 3CC06061AAxx LIU : 3CC06118AAxx
	In+	In-	Out+	Out-	
Auxiliary 2 Mbit/s	9	28	19	37	X

In the 75 Ω version, "In+" and "Out+" are the live input and output points and "In-" and "Out-" are the ground connectors for the signals.

3.6.5 - Wiring n x 2 Mbit/s tributaries

3.6.5.1 - Connections

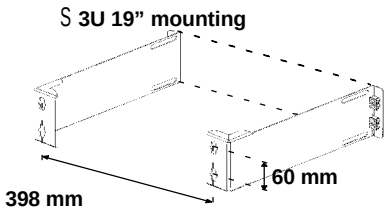
The user ports will be via one of the following ports described in **Appendix 6.1.1**:

- 37-way "**I/O (1/8)**" and "**I/O (9/16)**" connectors on the main IDU,
- 37-way "**I/O (1 - 4)**" connectors on the Light IDU version,
- 37-way "**I/O (1/8)**" and "**I/O (9/16)**" connectors on the access IDU (1+1 version with MUX protection), when the access IDU exists.
- on 3CC13659AAxx cables (1+1 with MUX protection without access IDU)

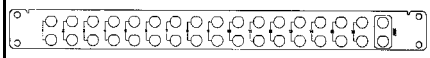
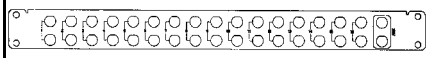
In 120 ohms, if the non-distributor option is selected, wire **the braces supplied with the IDU-ODU connector kit (ref: 3CC07899AAxx)** using a crimping tool **(608868-1, from AMP).**

3.6.5.2 - Wall mounting (9400UXI301) indoor units

S The distributors described below can be fixed to a wall using this kit comprising two brackets + fixing screws.

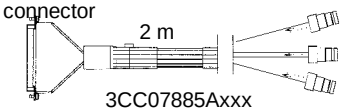
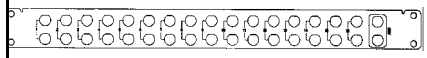
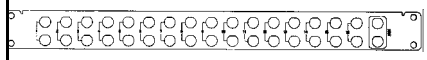
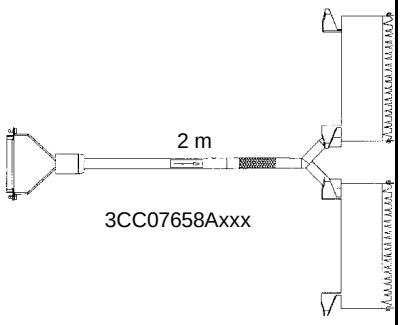
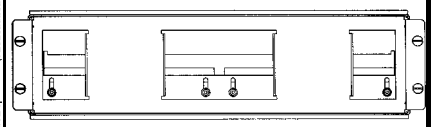
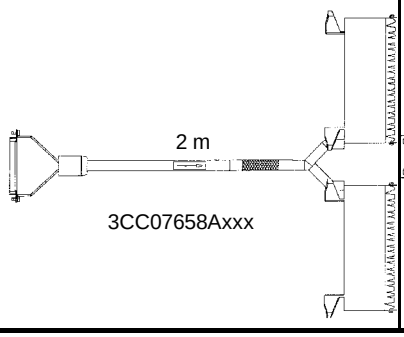
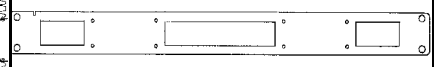


3.6.5.3 – 4 x 2 Mbit/s wiring with distributor

COMMERCIAL CODE OF DISTRIBUTOR ASSEMBLY	OUTPUT TYPE	CABLE	DISTRIBUTOR
9400XXI404	75 Ω 1.6/5.6	37-pin connector 2 m  3CC07885Axxx	1 Unit  3CC08061AAxx 1 Unit = 44.45 mm
1 x 9400XXI404 9400XX405	75 Ω BNC	37-pin connector 2 m  3CC07759Axxx	1U  3CC08061ABxx

For 120 Ω wiring, use the cables provided for the 8 x 2 Mbit/s version.

3.6.5.4 – 8 x 2 Mbit/s wiring with distributor

COMMERCIAL CODE OF DISTRIBUTOR ASSEMBLY	OUTPUT TYPE	CABLE	DISTRIBUTOR
9400XXI408	75 Ω 1.6/5.6	37-pin connector 2 m  3CC07885Axxx	1U  3CC08061AAxx
2 x 9400XXI408 9400XXI405	75 Ω BNC	37-pin connector 2 m  3CC07759Axxx	1U  3CC08061ABxx
9400XXI408	120 Ω	2 m  3CC07658Axxx	3U EMC  3CC07810AAxx
	120 Ω	2 m  3CC07658Axxx	1U  3CC08062AAxx non EMC

3.6.5.5 – 16 x 2 Mbit/s wiring with distributor

COMMERCIAL CODE OF DISTRIBUTOR ASSEMBLY	OUTPUT TYPE	CABLE	DISTRIBUTOR
9400XXI416	75 Ω 1.6/5.6	37-pin connector 2 m **2** x 3CC07885Axxx	1U 3CC08061AAxx
9400XXI416 4 x 9400XXI405	75 Ω BNC	37-pin connector 2 m **2** x 3CC07759Axxx	1U 3CC08061ABxx
9400XXI416	120 Ω	2 m **2** x 3CC07658Axxx	3U EMC 3CC07810AAxx
	120 Ω	2 m **2** x 3CC07658Axxx	1U 3CC08062AAxx non EMC

3.6.6 – Engineering service channel wiring

This wiring is for connecting the user's terminal equipment to the ESC connector. For the link cable, use the connectors supplied in the connector kit shipped with the IDU-ODU assembly. The pinout of the connectors is given in **Appendix 6.2**.

3.6.7 – Alarm and remote control wiring (TS/TC)

The TS/TC interface on:

- the "**ALARMS I/O**" connector of the **main IDU**,
- the "**ESC3 / ALARM I/O"** connector of the **Light IDU versions.**

is used to:

- supply three signals (outputs) to the equipment (dry loops, 0.5 A/100 V).
- supply (output) one remote control (dry loop, 0.5 A/100 V).
- connect eight housekeeping alarm inputs: nº1 to nº8 alarms. These alarms are formed by normally opened or normally closed loops, see § 7.10.

For the connecting cable, use the connector provided in the connector kits supplied with the IDU–ODU assembly. The connector pinout is given in the **Appendix 6.3**.

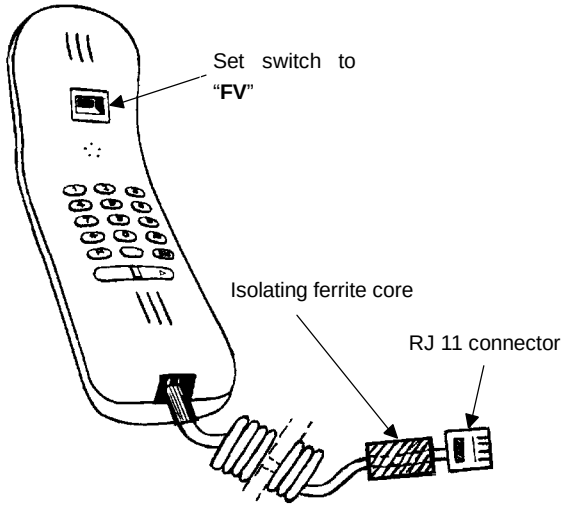
3.6.8 – Engineering service channel ESC2 telephone handset

The service channel telephone handset (classic IDU only, ref: **9400XXB000,** optional kit) is as shown below. It connects to the IDU, via the RJ11 connector with the handset symbol; it uses DTMF (tone) dialling.

	**To safeguard against EMC problems, remember to clip the isolating ferrite core on the cable. This is supplied in the main IDU connector kit.**

The handset can be connected to the main IDU, via the front panel or connector panel.

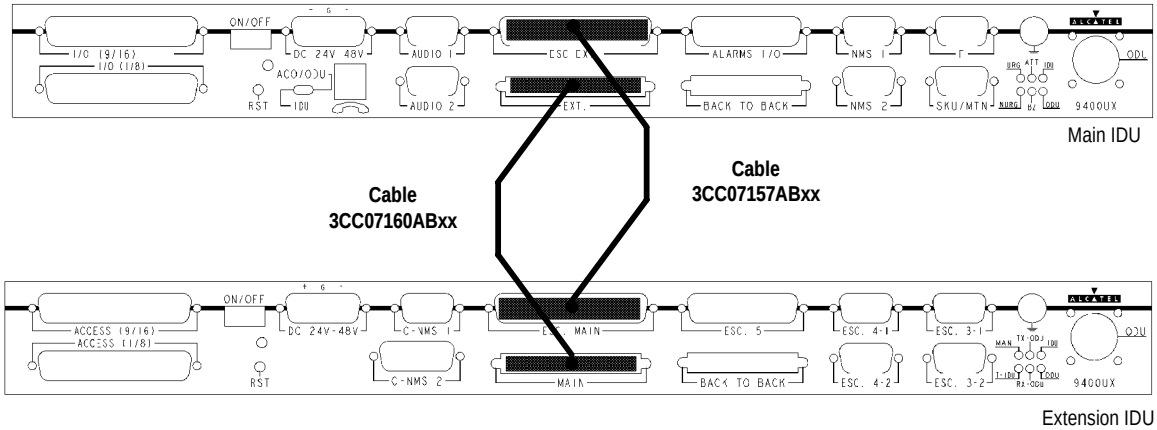
Caution: to operate, set the handset switch to “FV”.



3.7 – Wiring between a terminal’s units

3.7.1 – Main IDU - extension IDU wiring

This is for 1+1 configurations of all types and 1+0 configurations with extension.

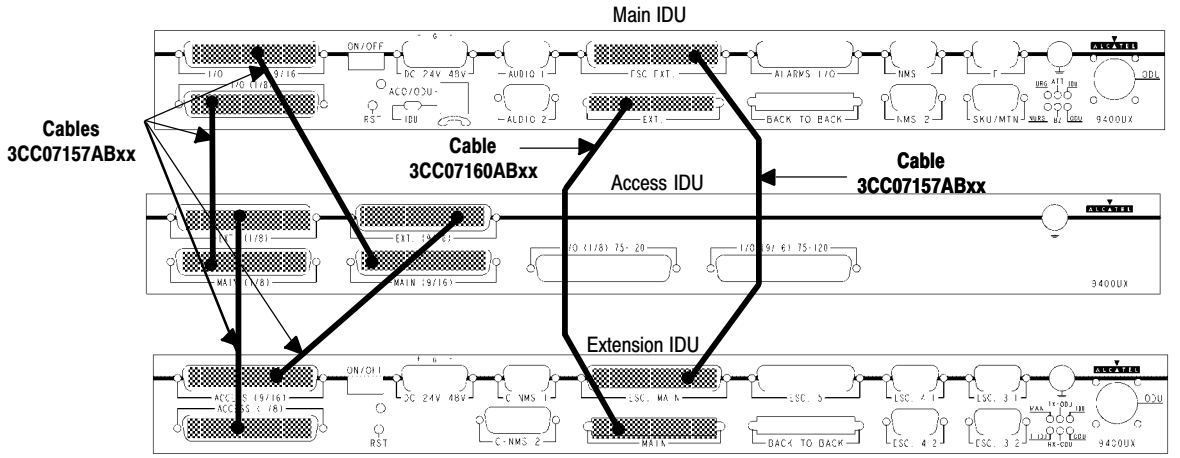


Connect the **ESC EXT** connector on the main IDU to the **ESC MAIN** connector on the extension IDU using a **cable ref: 9400UXCI130 (3CC07157AAxx), with male 37-way connectors (L = 300 mm).**

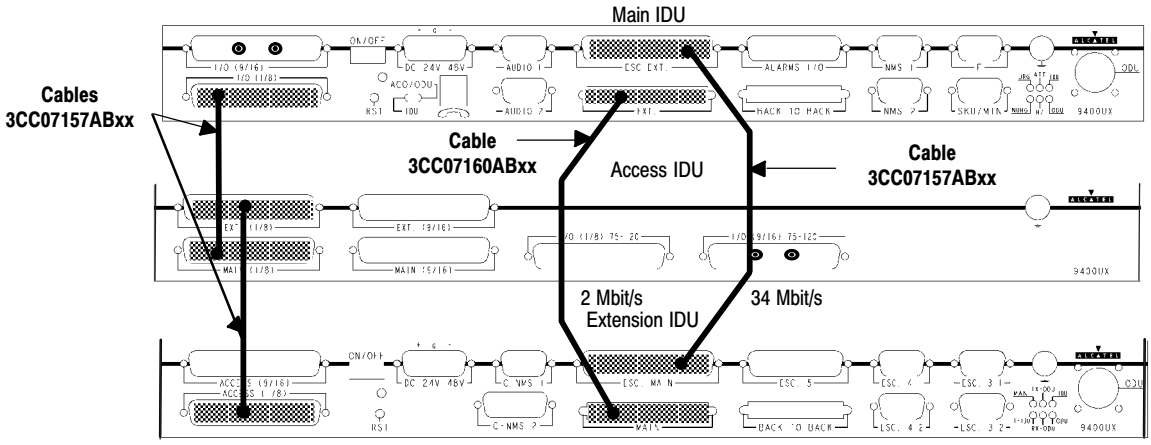
Connect the **EXT** connector on the main IDU to the **MAIN** connector on the extension IDU via a **cable ref: 9400UXB221 (3CC07160AAxx), with 68-way rectangular plugs (L = 300 mm).**

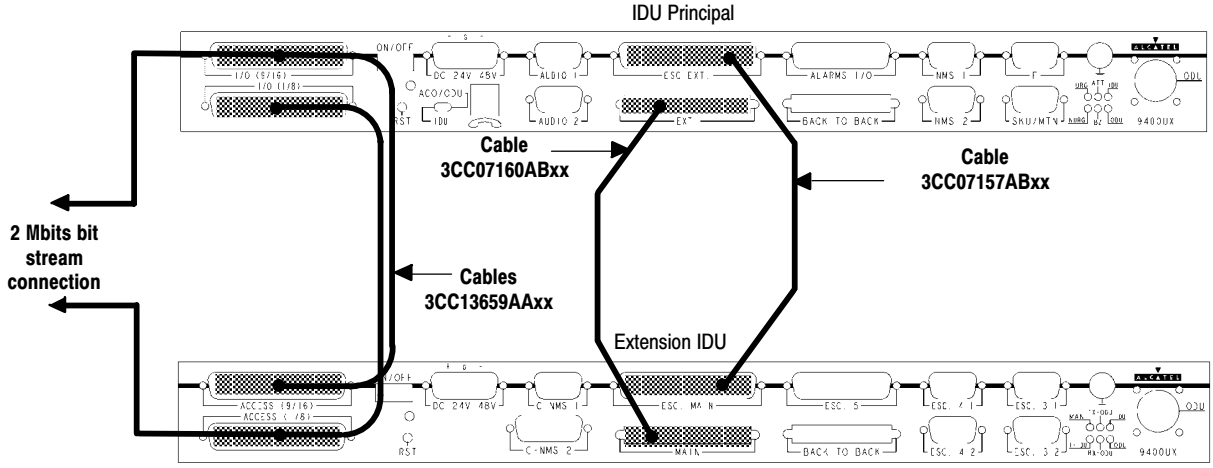
3.7.2 – Wiring the IDUs in 1+1 with MUX protection

This applies to 1+1 configurations with multiplex protection.



At N x 2 Mbit/s





Configuration with MUX protection without access IDU

Four identical cables (two if the number of tributaries is ⩽ 8) **ref: 3CC07157ABxx, 37-pin male plugs, L = 300 mm** are used to connect to the following connectors on the access IDU:

- **MAIN (1/8)** to the **I/O (1/8)** connector of the main IDU.
- **EXT (1/8)** to the **ACCESS (1/8)** connector of the extension IDU.
- **MAIN (9/16)** to the **I/O (9/16 75-120)** connector of the main IDU.
- **EXT (9/16)** to the **ACCESS (9/16)** connector of the extension IDU.

NOTE: When the 1+1 configuration with multiplexer protection is selected, the units must be arranged ***physically*** from top to bottom, as follows: main unit, access unit, extension unit.

3.8 – Wiring between terminals of a station

The wiring in this case is to provide continuity of tributaries and service channels between the various terminals of the particular station.

3.8.1 - Tributary wiring

Set up the wiring to interconnect the various tributaries from the N x 2 Mbit/s distributors or the 34 Mbit/s connectors of each terminal **according to the wiring diagram specified by the network administrator.**

3.8.2 - ESC wiring

3.8.2.1 - ESC1 (supervision channel)

STATION WITH TWO 2-WAY TERMINALS

Connect the **NMS1** connector on the first unit to the **NMS1** connector on the second unit using cable ref: **9400UXC332** (**3CC07711AAxx)**, 2 m long.

STATION WITH THREE 3-WAY TERMINALS

Connect the **NMS1** connector on the first unit to the **NMS1** connector on the second unit using cable ref: **9400UXC332** (**3CC07711AAxx)** 2 m long**.**

Connect the **NMS2** connector on the second unit to the **NMS1** connector on the third unit using cable ref: **9400UXC332 (3CC07711AAxx)**, 2 m long**.**

NODE EXCHANGE

With LUX11 and LUX12 software (bus RQ2)

The supervision bus can be used to extend cabling to n units by linking the **NMS2** connector of one to the **NMS1** connector of the next, as many times as is needed using cable ref: **9400UXC332 (3CC07711AAxx),** 2 m long.

	**Caution**: You must never create rings in the RQ2 bus.

With LUX40 software (SNMP bus)

The supervision bus is used to extend the wiring to N items of equipment by linking the **NMS2** connector of one to the **NMS1** connector of the next, as many times as is needed, using cable ref: **9400UXC332 (3CC07711AAxx),** 2 m long.

SNMP bus protection: the SNMP bus supports meshed connections, for example, by linking the **NMS2** connector of the nth terminal to the **NMS1** connector of the first terminal, and so on.

3.8.2.2 - ESC2 wiring (analogue channel)

This channel is not provided in the Light IDU versions.

STATION WITH TWO 2-WAY TERMINALS

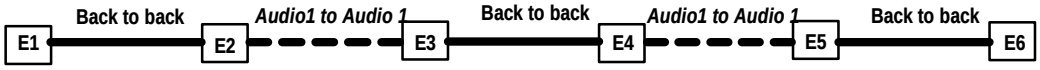
Connect the **BACK TO BACK** connectors of the main IDUs via cable ref: **9400UXC333 (3CC08729AAxx),** 2 m long

STATION WITH THREE 3-WAY TERMINALS

- Connect the **BACK TO BACK** ports of the first unit and second unit via cable ref: **9400UXC333 (3CC06688AAxx),** 2 m long,
- and connect the **AUDIO1** ports of the second unit to the **AUDIO1** ports of the third unit using cable ref: **9400UXC332 (3CC07711AAxx),** 2 m long.

The connections can be extended to n terminals according to the figure below:

- Connect the **BACK TO BACK** ports of the terminals (1 and 2), (3 and 4), (5 and 6), etc, via cable ref: **9400UXC333 (3CC08729AAxx),** 2 m long,
- and connect the **AUDIO1** ports of terminals (2 and 3), (4 and 5), (6 and 7), etc, via cable ref: **9400UXC332 (3CC07711AAxx),** 2 m long .



3.8.2.3 - ESC 3, 4, 5 wiring

CLASSIC IDUS

Set up specific wiring using the connectors supplied with the IDU connector kit

- Connector pinout information is given in Appendix 5
  - 9-way connector **3CC07920AAxx**
  - 25-way connector **3CC08221AAxx**
  - 37-way connector **3CC07899AAxx**

LIGHT IDU

Only service channel 3 is available.

- Prepare specific wiring with 37-way connector **3CC09311AAxx** supplied with the connector kit.

4 – Configuration and operation software

Systems are commissioned, operated and maintained via a portable PC with the “9400 LX/UX application” supervision software installed which runs in Windows(TM) environment. The computer connects directly to the main IDU (F connector, item ① in the figure below), via a 9-way/9-way cable (1AB 054120027) supplied with the set of software installation diskettes or CD–ROM.

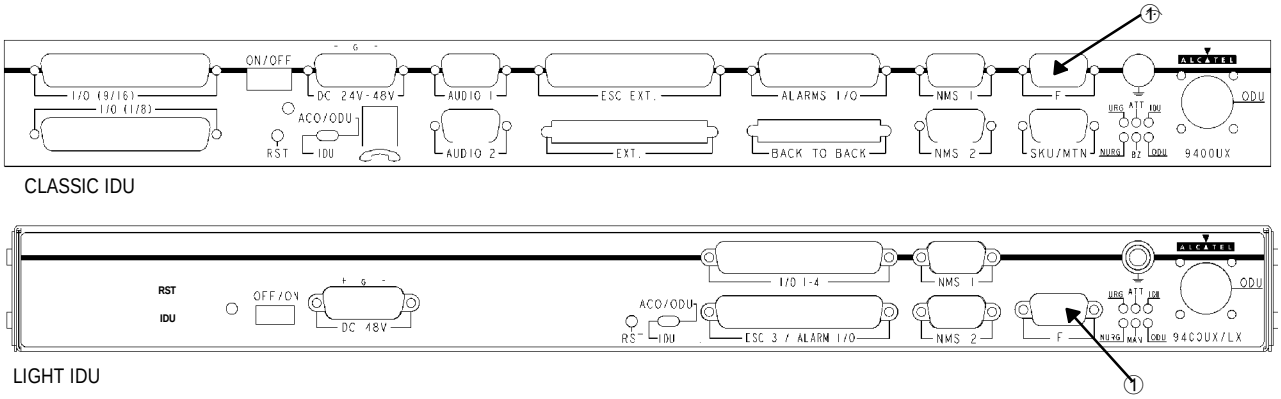


Figure 35 - Controls, indicators and connectors on the IDU

This manual applies to the 946LUX11, 946LUX12, and 946LUX40 versions of the software installed on the PC. Applicability of the various versions is given in the table below:

CONFIGURATION	946LUX11	946LUX12	946LUX40
Classic IDU	X	X	X
Light IDU		X	X

For the terminal to which it is connected and for the remote terminal (when the link is set up), this software can be used to:

- display and administer all alarms, status, remote controls,
- read and modify the configuration (bit rate, frequencies, etc),
- display transmission parameters (received power, bit error ratio, etc),
- monitor performance (G.821),
- download software.

Optionally, the extended version of the 946LUX11 and 946LUX12 software or the standard version of 946LUX40 software, supports four additional applications:

- Remote inventory,
- Event storage,
- Performance monitoring (G.784),
- Analogue measurements.

4.1 – Installing the software

4.1.1 - Warning note

You are assumed to be familiar with:

- use of the PC installed and its devices,
- operation of the PC in MS-DOS and WINDOWS 95, 98, NT or 2000 *.

* MS-DOS, MICROSOFT and WINDOWS are registered trademarks of MICROSOFT Corporation

The PC must have the following **minimum** specifications:

- Pentium or compatible processor operating at 700 MHz or above,
- 128 Mbytes of RAM,
- serial asynchronous communication port,
- 4 Gbytes hard disk,
- CD–ROM or DVD–ROM drive,
- 3.5” high density floppy drive (1.44 Mbytes),

The operating system validated are the following:

- Windows 95, 98, NT or 2000* installed on the hard disk.

Before installing the system, check the following points:

- if there is a virtual disk, uninstall it,
- delete the following Windows programs: manufacturer’s overlay (Dashboard, Tabworks), programs in the startup group,
- check that the system date and time are correct and if not, adjust them,
- disable the screen saver,
- make sure that no other software is running at the same time as NECTAS.

* MS-DOS, MICROSOFT and WINDOWS are registered trademarks of MICROSOFT Corporation

4.1.2 - Installing the software

4.1.2.1 - NECTAS

Nota: NOCTAS is equivalent to NECTAS for a 1322NX-SRL mediation and NOCTAS can be installed in place of NECTAS for IDUs.

NECTAS VERSION 3.79 ON DISKETTES

SOFTWARE	No. of diskettes	Diskette labels	Diskette Reference	DESCRIPTION
NECTAS 3.79	3	WGA 736	3CC08597AAxx	System software, version 3.79 for Windows 3.11, Windows 95 or 98.

NECTAS VERSION 4.1X ON CD-ROM

SOFTWARE	CD-ROM Reference	DESCRIPTION
NECTAS 4.1x	3AL78606AAxx	System software, version 4.11 or above for Windows NT or 2000 with set up procedure.

4.1.2.2 - 946LUX11

SOFTWARE	No. of diskettes	Diskette labels	Diskette Reference	DESCRIPTION
RTP	1	BQA 751	3CC04457AAxx	Radio transmission parameters
946LUX11	2	BQA 758	3CC07386AAxx	9400 UX equipment software
946LUX11 software extension	2	WGA 740	3CC07386ABxx	946LUX11 add-on applications (Diskettes supplied only if this option has been requested)

4.1.2.3 - 946LUX12

946LUX12 ON DISKETTES

SOFTWARE	No. of diskettes	Diskette labels	Diskette Reference	DESCRIPTION
RTP	1	BQA 751	3CC04457AAxx	Radio transmission parameters
946LUX12	2	WGA 737	3CC08734AAxx	9400 UX equipment software
946LUX12 software extension	2	WGA 741	3CC08734ABxx	946LUX12 add-on applications (Diskettes supplied only if this option has been requested)

946LUX12 ON CD-ROM

SOFTWARE	CD-ROM Reference	DESCRIPTION
RTP	3CC13244AAxx	Radio transmission parameters
946LUX12	3CC13241AAxx	9400 UX equipment software
946LUX12 software extension	3CC013243AAxx	946LUX12 add-on applications

⚠	**CAUTION: For 16QAM ODU please use the software release noted in Appendix 12.2 or above.**

4.1.2.4 - 946LUX40

946LUX40 ON DISKETTES

SOFTWARE	No. of diskettes	Diskette labels	Diskette Reference	DESCRIPTION
RTP	1	BQA 751	3CC04457AAxx	Radio transmission parameters
946LUX40	2	WGA743	3CC01117AAxx	9400 UX equipment software

946LUX40 ON CD-ROM

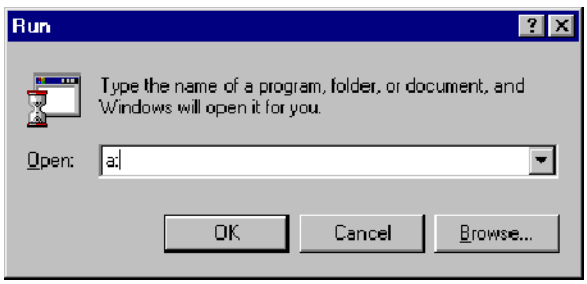
SOFTWARE	CD-ROM Reference	DESCRIPTION
RTP	3CC13244AAxx	Radio transmission parameters
946LUX40	3CC13242AAxx	9400 UX equipment software

⚠	**CAUTION: For 16QAM ODU please use the software release noted in Appendix 12.2 or above.**

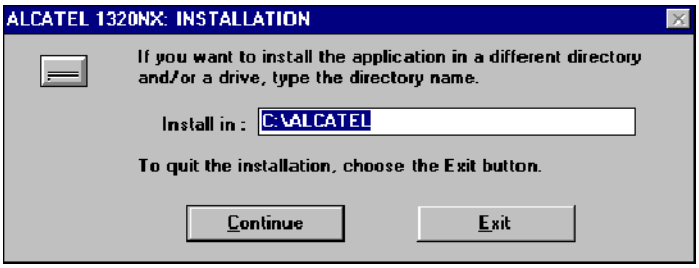
SOFTWARE INSTALLATION FROM DISKETTES

The system files are installed on the hard disk, in the basic C:\ALCATEL directory.
Procedure:

- Insert the system diskette (WGA 736, diskette 1/3) in the drive.
- Click “Start” in the WINDOWS taskbar and select “Run”.

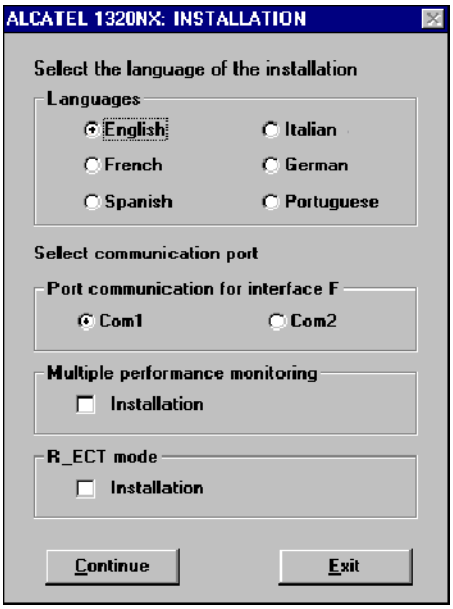


- Type a: in the “Open” text box then click “Browse”. The files on the diskette are then listed in a window. From these, select “Install.exe” or “Setup.exe” depending on software release. Click “Open” then “OK”. The following window appears:



Do not alter the path specified in this dialogue box

- Click “Continue” then follow the instructions displayed by the installation program.



Do not check the “Installation” boxes in the “Multiple performance monitoring” and “R_ECT mode” fields.

- Choose the same language for all the software installed to maintain consistency in the system software.

Proceed in the same way to install each of the application programs. These should be installed in the following order:

- NECTAS,
- RTP,
- 946LUXxx (diskette 2.2 of the software is not used for French language installation),
- 946LUX11 extension or LUX12 extension (diskette 2/2 of the software is not used for French language installations) if this option has been ordered.

Note:

The 946LUXxx files are automatically installed in the directory C:\ALCATEL\946LUXxx.

When you have finished installing the link, reboot the PC by “Start”, “Shut Down”. Check the “Restart” option and click “OK”. Wait for the Windows desktop to reappear and run the NECTAS software. The “**Alarm Synthesis**” application is run automatically.

Accessing the applications entails entering the user name and password. Each user is assigned a profile defining the applications to which he or she is allowed access.

The network administrator is responsible for defining the profiles and assigning them to the various users, as described in § 4.3.

On leaving the factory, all equipment recognizes two predefined “name/password” pairs (all in uppercase):

- ADMIN/ADMIN: Administrator profile (defined in the equipment and not modifiable),
- USER/USER: User profile “write mode operator” profile, see § 4.3.3 in appendix 10).

The administrator is advised to:

- change the password of the ADMIN user,
- delete the user name USER as soon as some operators have been profiled.

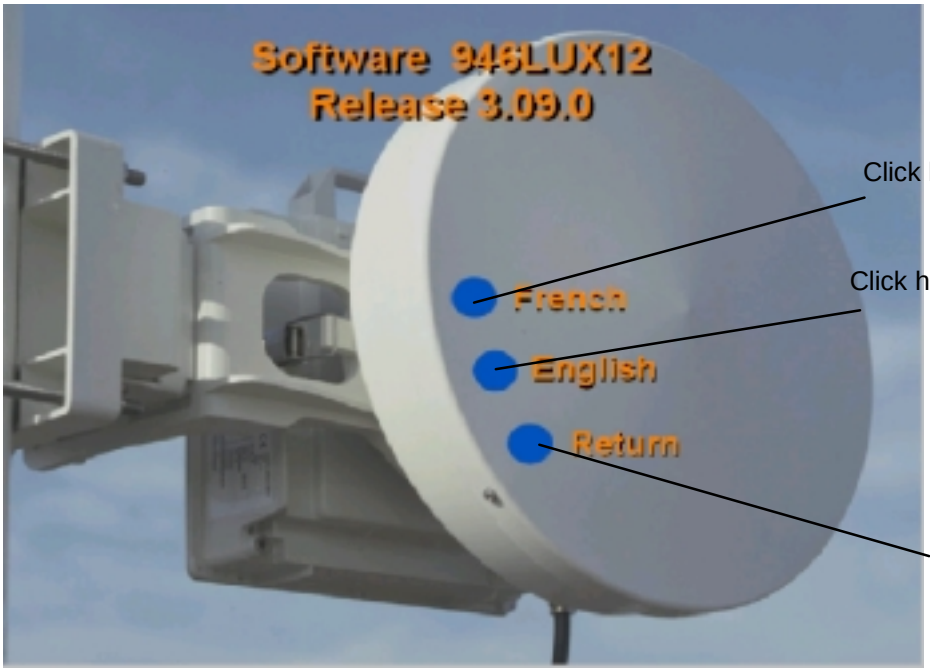
INSTALLATION OF 946LUX12, 946LUX40, RTP OR 946LUX12 EXTENSION SOFTWARES FROM THE CD–ROM

- The PC must have the minimum specifications listed in chapter 4.1.1
- Insert the CD that contains one of the software to be installed on the CT.
  - S The following screen appears automatically, without the intervention of the operator, on the PC explorer in English if the local parameters of the PC are configured in another language than French. The screen is displayed automatically in French if the local parameters are configured in French.



Click here to install the software

  - S Click on the “Software” button
  - S The software installation screen appears as follows:



Click here for a French installation

Click here for an English installation

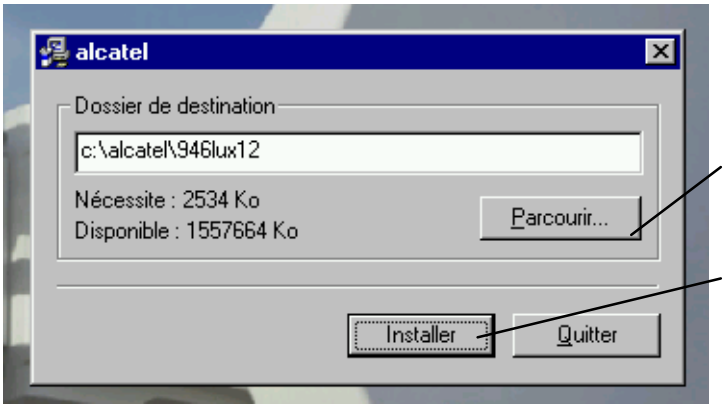
Back to the previous screen

  - S When you choose English, the software installation is in English. The files are automatically installed in the default directory C:/ALCATEL/946LUXxx for the 946LUXxx software and in C:/ALCATEL for the RTP or in any other directory you choose by clicking on “Browse” (see next screen).



S If you choose “French”, a warning window appears. The files are installed automatically in the default directory C:/ALCATEL/946LUXxx and C:/ALCATEL for the RTP or in any other directory you choose by clicking “OK” (see the next screens).





Click here to choose another directory than the one proposed

Click here to start the installation in French

4.2 – 9400 UX applications

4.2.1 - Introduction

This manual describes how to use the craft terminal **only for the applications associated with the 9400 UX equipment**.

These applications can be accessed from the "**Alarm Synthesis**" (Alarm summary) main menu on the craft terminal (figure 36).

The operator is assumed to be familiar with the WINDOWS environment and the various resident applications.

The screens provided in this manual are examples only. The screens you see on site will depend on the final configuration and may differ from these. In particular, they may include the mention 946LUX11, 946LUX12 or 946LUX40 depending on the software version installed.



Figure 36 - Alarm summary screen

The alarm summary screen comprises:

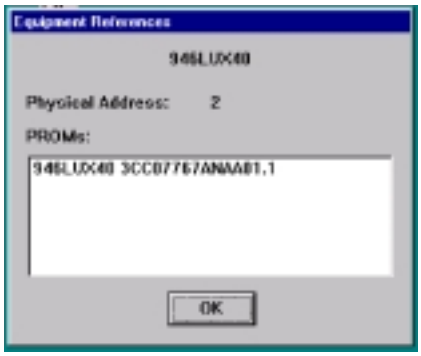
- a menu bar providing access to the applications,
- a global summary, or "synthesis", line, showing a summation of all the alarms on all the network elements.

4.2.2 - Menu bar

4.2.2.1 - "View" menu

Classic IDU with "Slave" key or Light IDU configured as Network Element.





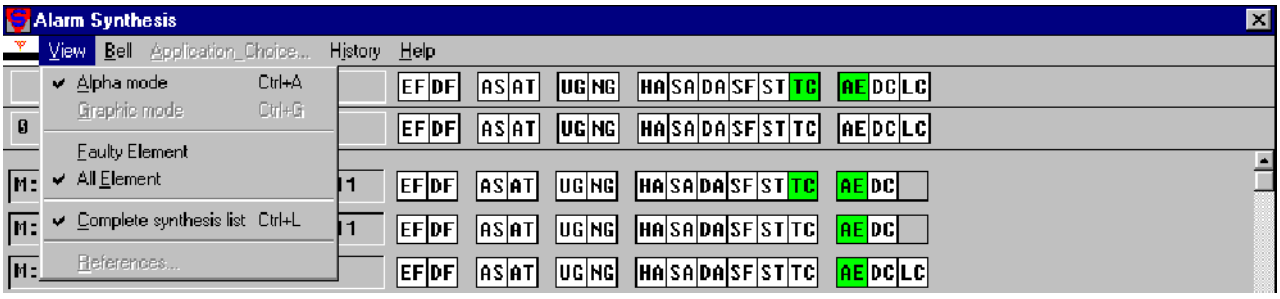
Reference

Displays information concerning the selected NE (reference of onboard software with revision index).

The onboard software (946LUXxx) must be compatible with that loaded in the PC.

The reference of the onboard software should be entered in the sheet in Appendix 2.

Classic IDU with "Master" key configured as mediation function.



Refer to § 4.2.4 for information on accessing the applications from these screens.

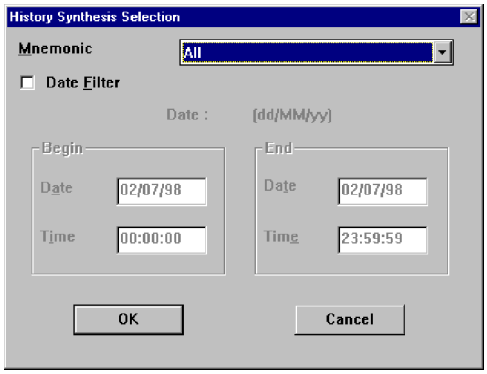
4.2.2.2 – "Bell" menu



- **Disable**: Locally and permanently disables (until the software is closed) the audible alarm on the PC. If the alarm bell is enabled, you will see a " p " symbol alongside the command in the pulldown menu. Simply click the command to toggle it on or off.
- **PC Cut Off:** Cancels the alarm on the PC until a new alarm appears.
- **Alarm cut-off**: Broadcasts an acknowledgement of urgent and non urgent alarms:
  - in the equipment, if under local supervision,
  - in all the network elements appearing in the alarm summary, if operating in master supervision mode.

4.2.2.3 – "History" menu



	**"Display" and "Print" commands** Select the events to be viewed or printed. **Mnemonic** Selects all events for a single mnemonic ("All" option: Selects all the mnemonics). **Date Filter** Check this box to select all events between a start date and time and an end date and time that you specify in the appropriate text boxes.
	"File" command Backs up the history log as ASCII text in the Sy_memo.0 file in the C:\Alcatel\ directory. This file can be transferred to diskette using the standard WINDOWS commands and viewed on another workstation. History can be initialized by suppressing the Sy_memo1.0, Sy_memo2.0, Err_cod1.log and Err_cod2.log files in C:\Alcatel\ directory.

The history log display screen contains the following information:

- the date and time when the event started,
- the name of the NE concerned,
- the type of NE,

- the type of alarm or event,
- event start or end indication (appearance/disappearance).

Two log files, each holding up to 1000 events, are used: the second of these is filled after the first. When the second is full, new events overwrite the contents of the first.

The history log is updated in real time even when left open.

4.2.2.4 - "Help" or "?" menu

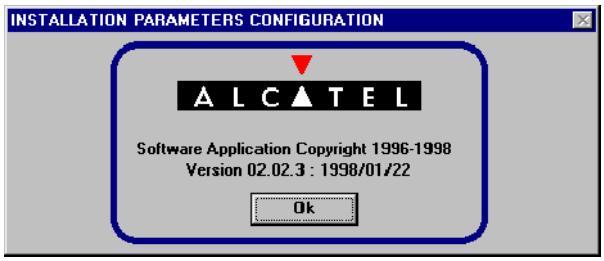
This menu often comprises two submenus. It may also appear in the form of a question mark ("?"):

HELP

This menu, just like this manual, is designed to assist the user.

ABOUT

This option calls up a panel on screen showing the **software release and copyright information:**



Simply click "**Ok**" to quit this screen.

4.2.3 - Global synthesis line

The global synthesis line comprises 15 information boxes, explained below:

MNEMO	COLOUR	MEANING	COMMENT
EF	Red	Equipment Failure	The Network Element is no longer responding or has failed
DF	Red	Dialogue Failure	S Command not understood S Transmission error S Protocol error
AS	Green	Alarm Stop	Switches off the audible alarm (not used in the 9400 UX)
AT	Magenta	Alarm Attended	Indication that an alarm is present and acknowledged
UG	Red	Urgent Alarm	Indication requiring prompt attention
NG	Yellow	Non urGent Alarm	Indication not requiring prompt attention
HA	Magenta	Housekeeping Alarm	Environment, or housekeeping, alarm, (for example, in a microwave station, door open, generator set fault, abnormal temperature rise, etc)
SA	Magenta	Service Affected	Transmission problem requiring prompt attention
DA	Magenta	Distant Alarm	Alarm detected that is not local to the NE
SF	Magenta	Signal Failure	Signal failure at the equipment's customer input
ST	Green	STatus	The equipment has received a remote control
TC	Green	Terminal Connected	A CT is connected to equipment

MNEMO	COLOUR	MEANING	COMMENT
AE	Green	Access Enabled	Access to the remote controls (in the “AS&C” Alarms, Status and Controls application) and to the configuration (“Equipment configuration” and “Operational configuration” applications) is enabled by a command from a hierarchical level above the CT
DC	Black	Default Configuration	The equipment is in a default configuration. It needs a complete configuration to be downloaded for the first commissioning.
LC	–	Loss of Configuration	Not used in 9400 XX.

4.2.4 - How to open a session

When you run NECTAS, the alarm synthesis screen appears with the “**Application_Choice...**” menu:

- If the “**Application_Choice...**” menu is grayed out (Inaccessible), the key on the equipment is a “Master” key. Select “View/All Element” menu to view all the network elements declared in the network, then click “M:0–S:0” to select it. The “Application_Choice...” menu is then accessible.
- Click the “**Application_Choice...**” menu. The screen for you to enter the password then appears.



- Give your name and password and click “**OK**”. (For more details on user profiles and passwords, refer to § 4.3).

If it is not in the default configuration, a list of the applications (Figure 37) that you can access, according to your profile, is then displayed.

Each application has its own help function, which can be accessed via a pull-down menu.

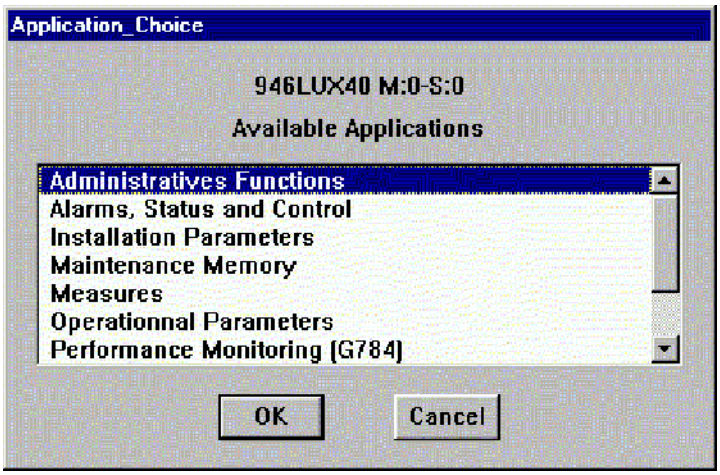


Figure 37 - List of applications specific to the 94xx UX range

Otherwise, before you can access the application, the software will ask you the necessary parameters to start your session (see § 4.3).

4.2.5 - List of applications

The functions performed by the various applications, and their uses, are summarized in the table below.

APPLICATION	TYPE	FUNCTION	USE
Administrative functions	Administration	Managing passwords and access to the applications	Installation, operation and maintenance
Installation parameters	Commissioning	Managing a network element's hardware configuration	Commissioning Changing the hardware configuration
Operational parameters		Managing a network element's operational configuration	Commissioning Changing the NE or network configuration
Software downloading		Updating network element software	Updating network element software
Alarms, Status and (remote) Controls	Operation	Displaying alarms and status information Selecting and sending remote controls	Routine operation
Performance monitoring (G.821)		Viewing information contained in the various performance counters for the different entities (links, sections, etc)	Preventive maintenance Checks after commissioning or changing the network configuration
Radio Transmission Parameters		Monitoring the main transmission parameters (power levels, bit error ratios) Providing an initial diagnostic on transmit and receive alarms	Routine operation Preventive maintenance Checks after: S commissioning or changing the network configuration S restarting after corrective maintenance work
Remote inventory	Extended 946LUX11 or 946LUX12 applications for operation (standard version of and 946LUX40)	Reading inventory information from network elements	Network management
Event logging		Reading events logged in NEs	Operation Corrective maintenance
Performance monitoring (G.784)		Analyzing the performance levels of terminal points (TP) supported by an NE in accordance with ITU-T Recommendation G.784	Preventive maintenance Checks after commissioning or changing the network configuration
Analogue measurements		Viewing measurement results performed on the various points of an NE	Operation Preventive maintenance Corrective maintenance

4.3 – "Administrative functions" application

The "Administrative functions" application lets:

- the administrator manages operators' rights of access to the various applications via predefined "profiles",
- each operator changes his or her password as stored in the equipment.

The way the various profiles are created is explained in Appendix 10.

Path: Alarm synthesis → Application_Choice... → Administrative functions

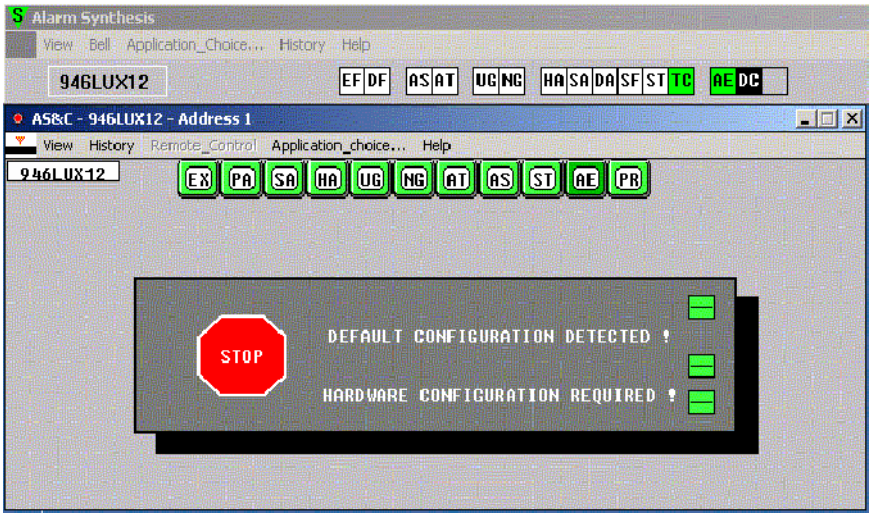


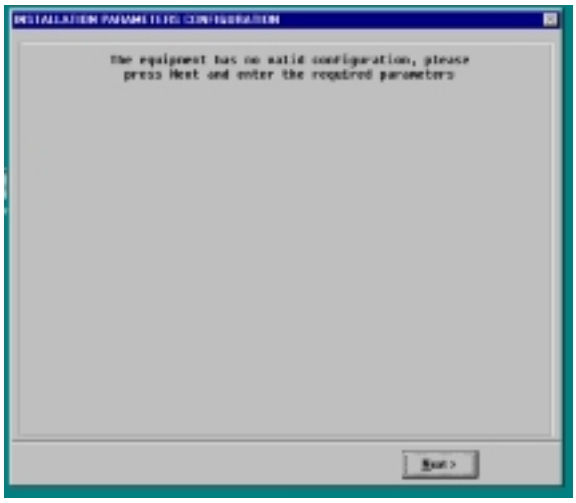
4.3.1 – Starting in Default configuration

This default starting application only exists for the LUX12 (ICS08 version and later) and LUX40 (ICS05 version and later) applications.

The default configuration is used by IDU when the IDU/ODU assembly is programmed for the first time or when the ODU configuration is selected on the IDU using the switch (see § 5.2.1) while the ODU is disconnected.

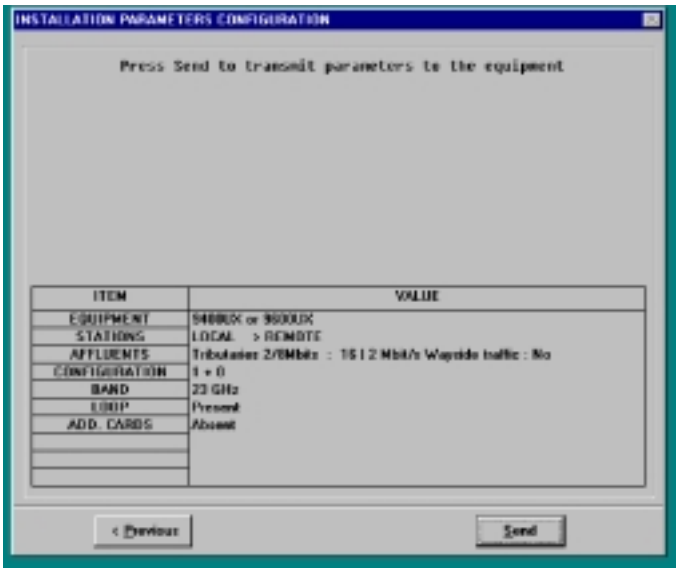
Path: Alarm synthesis → Application_Choice... → Installation parameters





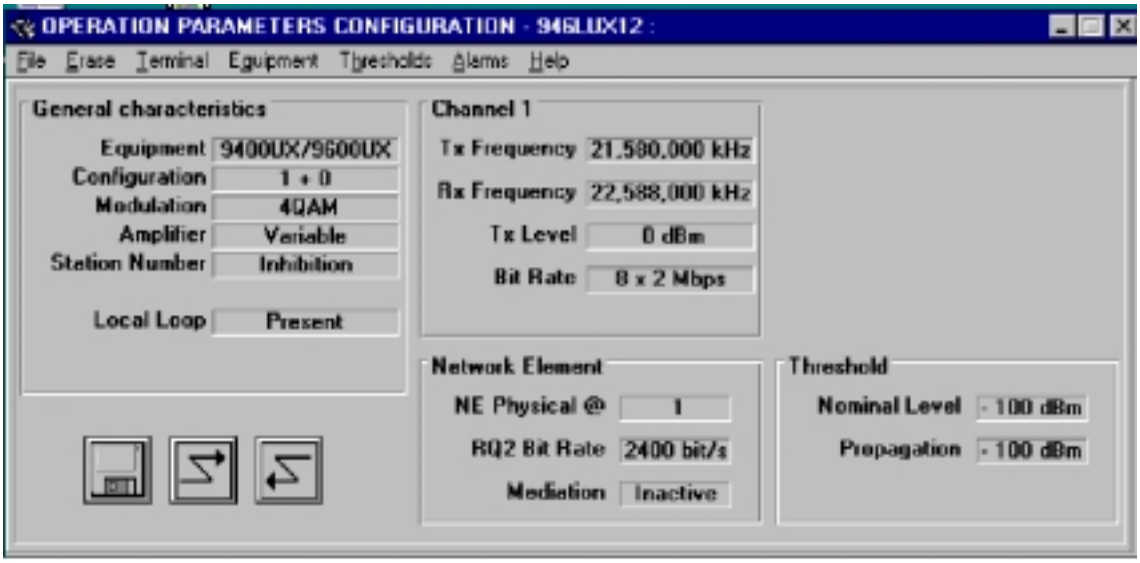
Click ‘Next’. Different sub–menus appear in which you can declare the hardware parameters of the equipment.

Follow the sequence as described in § 4.4. Information displays in the following screen:



Click ‘Send’ to validate the equipment configuration.

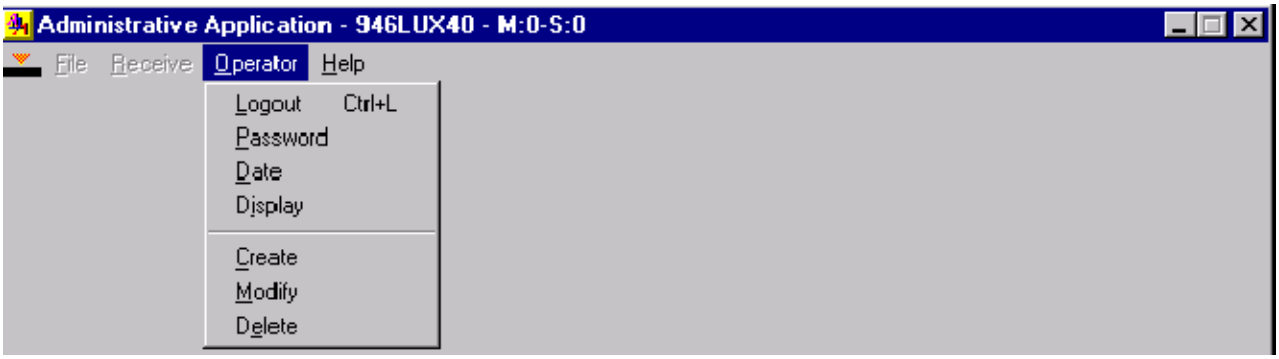
Hardware parameters are applied to the equipment and the configuration application of the operational parameters is automatically launched in order to end the system configuration of the bit rate, frequencies and tributary mixer parameters (except for 34 Mbits/sec equipment).

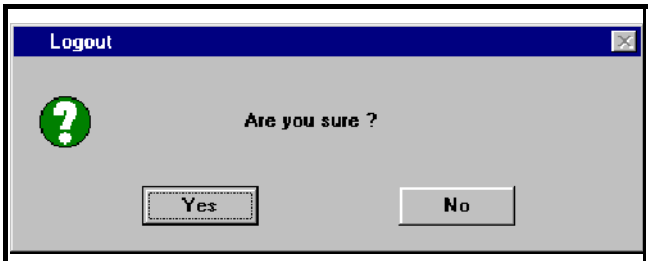


4.3.2 – “File” and “Receive” menus

The “File” and “Receive” menus cannot be accessed with the 946LUX11, 946LUX12, or 946LUX40 application from 9400 UX equipment.

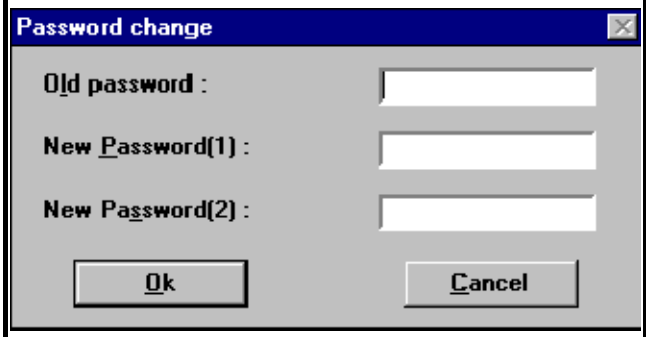
4.3.3 – “Operator” menu





Logout

Command for disconnecting from the equipment. A dialogue box appears for you to confirm the operation.



Password

Command for you to modify your password in the NE for which the session is opened.

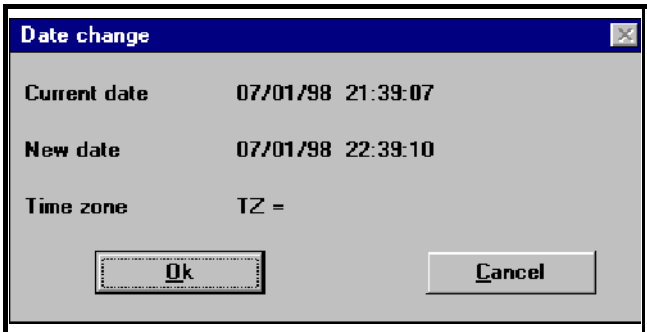
Enter the current password in the “Old password” field, making sure that you enter the correct sequence of uppercase and lowercase characters (only the letters A to Z and digits from 0 to 9 are allowed).

Enter your new password in the next two fields then click “**Ok**”.

If you do not enter the two new passwords identically, a dialogue box appears prompting you to repeat the operation.

If you enter the old password incorrectly, a warning message is displayed.

NOTE: After three unsuccessful attempts to change your current password in a particular session, you are automatically disconnected from the application.



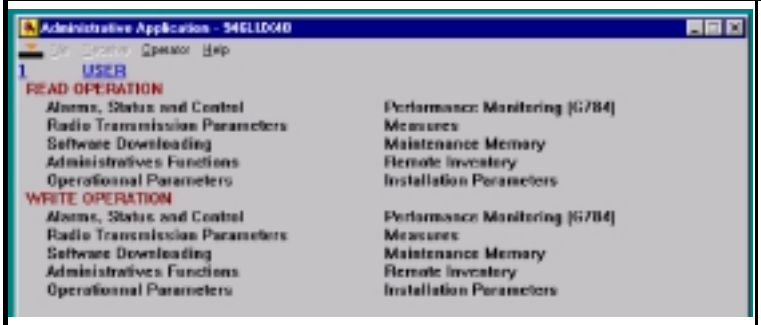
Date

Shows the date and time:

- Current date: date and time stored in the equipment.
- New date: PC date and time.

The date and time cannot be modified from 9400 UX equipment.

The date and time can be modified only from a mediation CT (see Appendix 11).

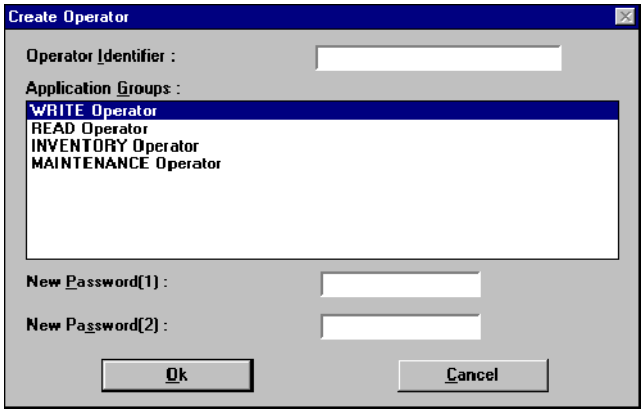


Display

Command enabling:

- you to view your profile, but not your password.
- the administrator to display the profiles of all operators, except their passwords.

The screen lists the read and write operations you can perform.



Create

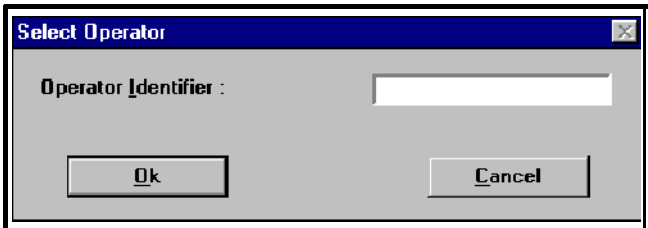
Accessible only to an operator with the "Administrator" profile

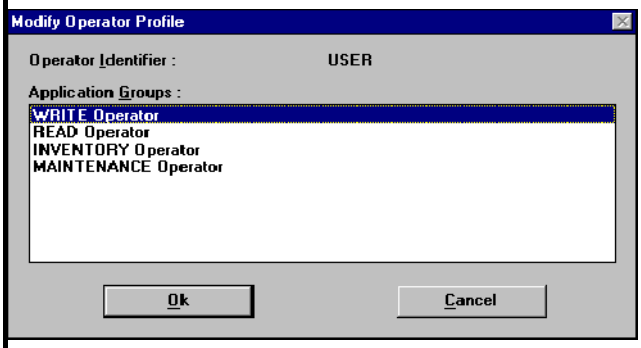
Command used by the administrator to assign a profile to a new operator in the NE for which the session is opened.

The maximum number of operators is 10.

- Enter the operator name in the "Operator Identifier" field.
- Select one of the profiles from the "Application Groups" list box.
- Enter the temporary password (at least four characters) in each of the next two fields.
- Click "**Ok**".

If the operator name already exists, a warning message is displayed.



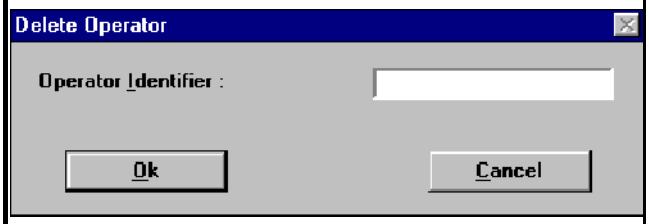


Modify

Accessible only to an operator with the "Administrator" profile

Command used by the administrator to modify an operator's profile, in the NE for which the session is opened. The operator's password cannot be changed. A dialogue box appears in which to enter the name of the operator. When confirmed, a new dialogue box appears from which to choose the new profile from a list.

If the operator is not known by the system, a warning message is displayed.



Delete

Accessible only to an operator with the "Administrator" profile

Command used by the administrator to remove an operator from the equipment for which the session is opened. The dialogue appears in which to enter the name of the operator. If the request is accepted, a prompt for confirmation is displayed.

If the operator is not known, a warning message is displayed.

4.4 – “Installation parameters” application

Before altering the installation parameters, check that there are no remote controls active.

The “Installation parameters” application is used to declare the equipment hardware parameters to the management software.

Path: Alarm Synthesis → Application_Choice... → Installation Parameters

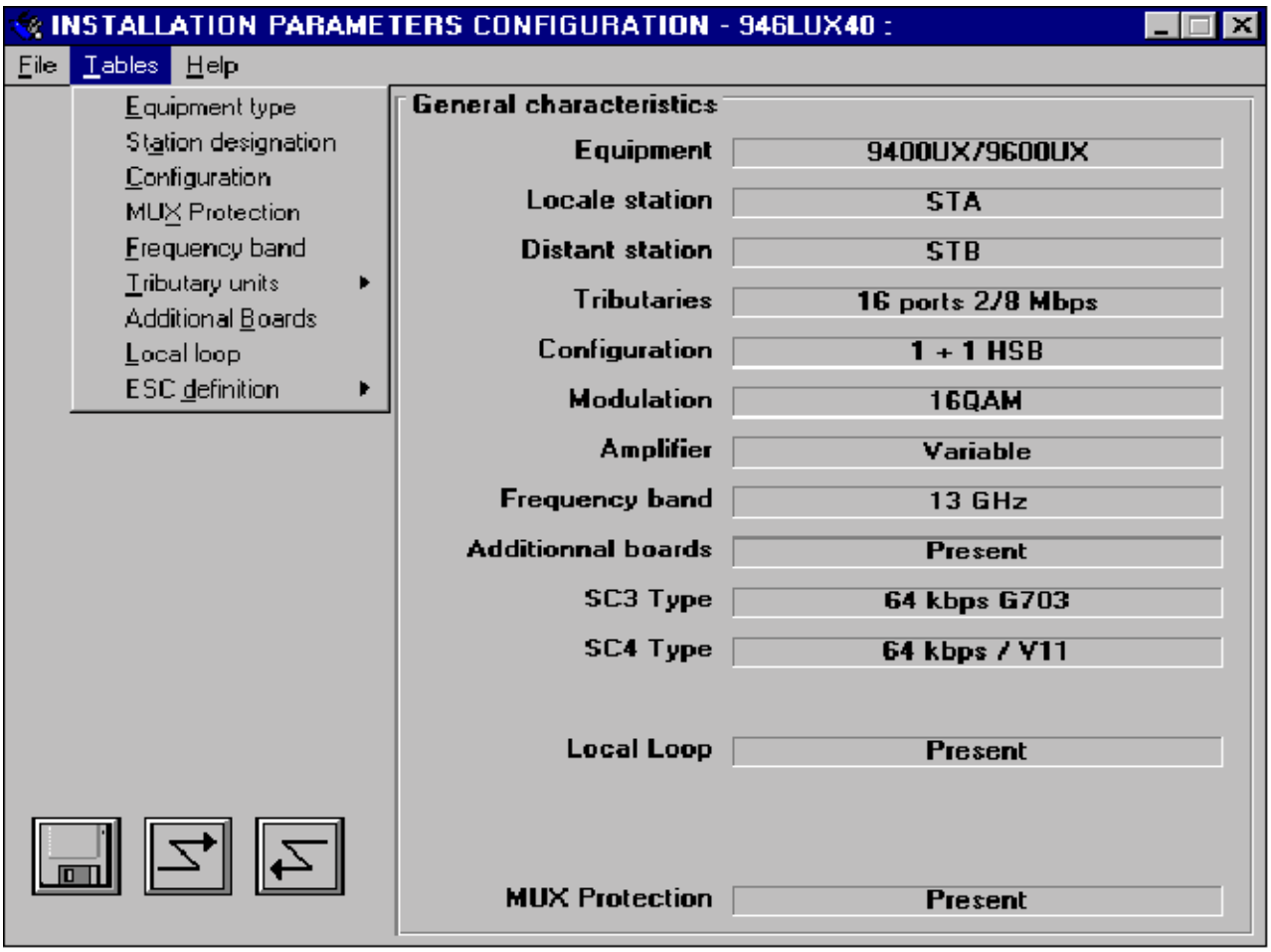
When the application is started up, the current settings are read from the equipment and uploaded to the craft terminal. A progress indicator bar shows the progress of the upload.

When the settings have been uploaded, the **“INSTALLATION PARAMETERS CONFIGURATION”** screen for the connected NE is displayed.

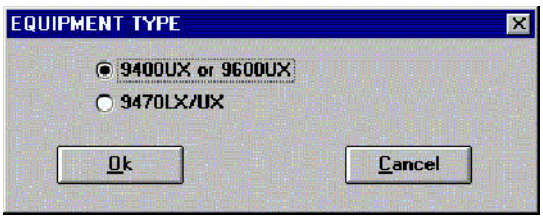
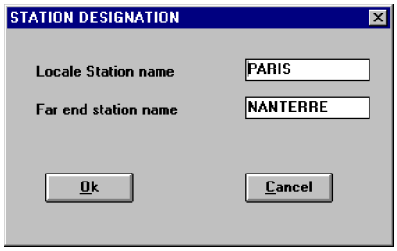
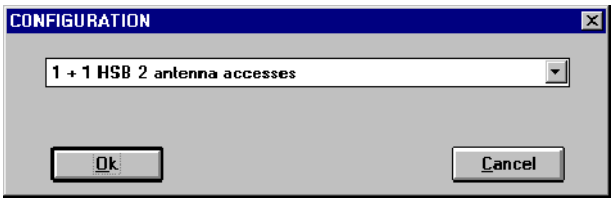
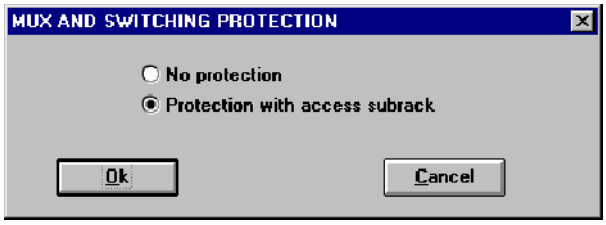
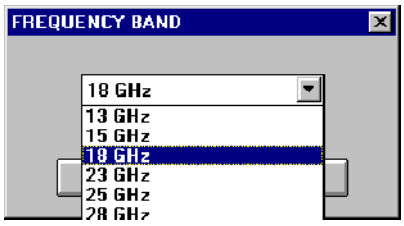
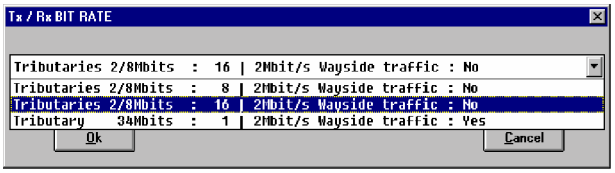
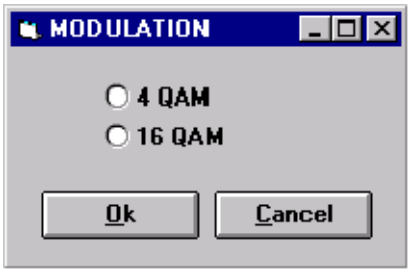
For a classic IDU, no configuration can be stored if the “Absent key” alarm is generated.

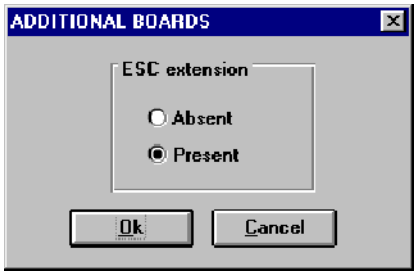
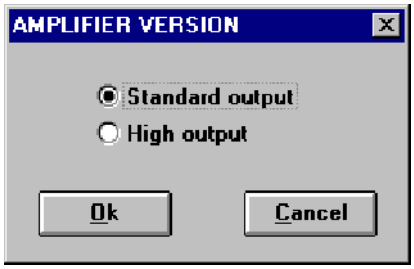
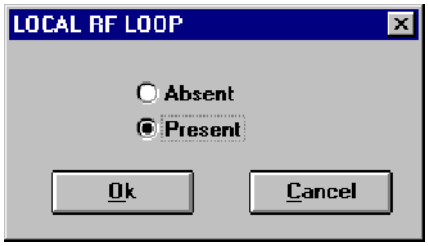
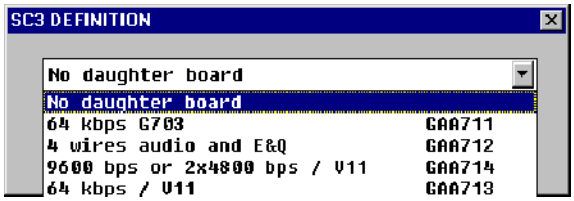
4.4.1 – “Tables” menu

The number of submenus in this menu depends on the type of equipment connected.

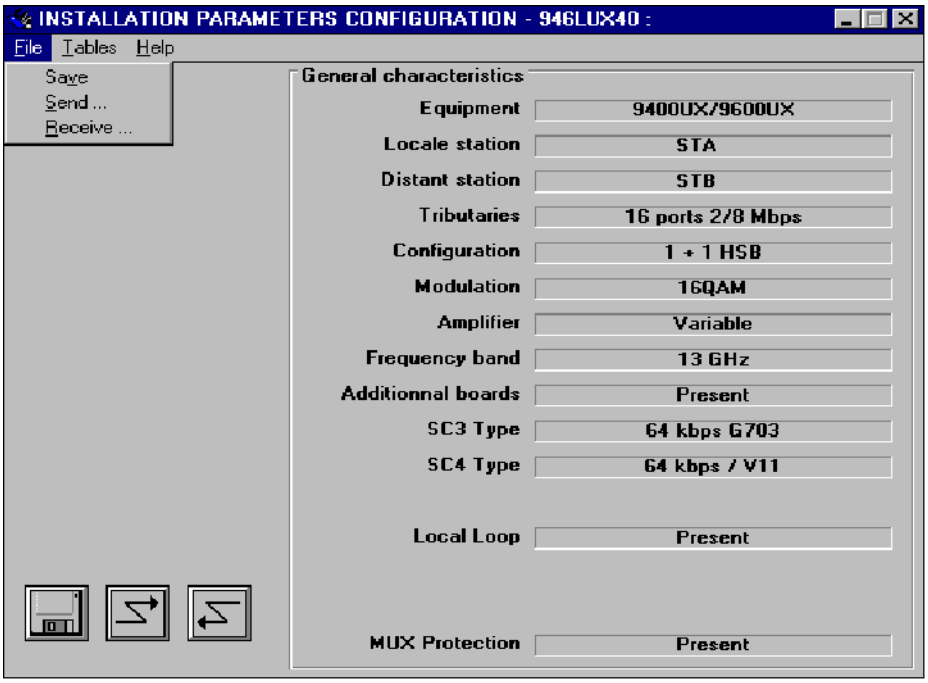


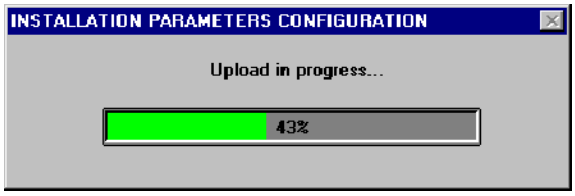
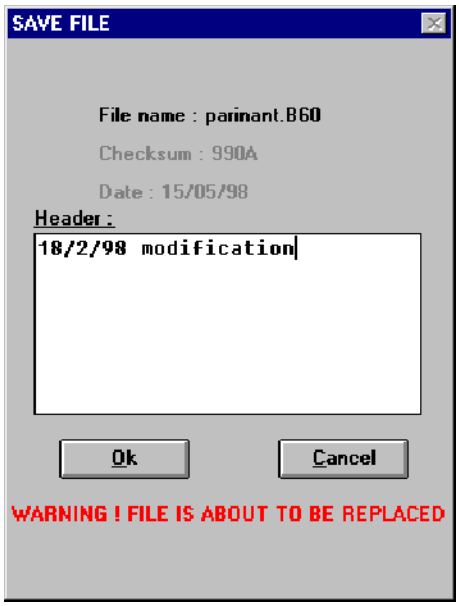
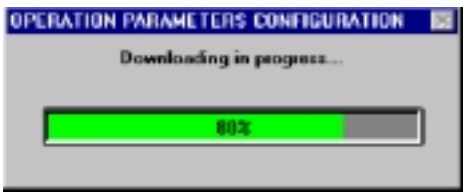
The submenus are for declaring the equipment's hardware settings.

	**Equipment type** Select 9400 UX
	**Station designation** Gives the names of the local and far end stations (names must not exceed eight characters, or include characters prohibited by MS-DOS)
	**Configuration** This is for selecting the correct option for the equipment configuration from the pull-down list
	*1+1 configuration option* **MUX protection** Select one of the two buttons depending on the absence or presence of an "access" IDU.
	**Frequency band** Select the frequency band for the equipment from the pull-down list
	**Tributary units** Select the maximum number of tributaries allowed by the LAU/LIU boards installed in the equipment
	**Selection of the modulation** To be configured depending on the delivered hardware and the capacities of the software key for the ODU A9400UX flat ODU.

	*1+0 configuration option* **Additional Boards** Click the correct button for the configuration (depending on whether an “ESC extension” IDU is absent or present)
	*Type of amplifier for the versions without RTCP* **Amplifier** Select the correct button for the ODU configuration (standard output power or high output power)
	*Option* **Local (RF) loop** Select the correct button for the local RF loopback hardware configuration. For the 9400UX flat ODU, select “Present”
	*“ESC extension” IDU option or 1+1 configuration* **ESC 3 or 4 definition** For each of engineering service channels ESC 3 and ESC 4, select the type of daughter board inserted in the “extension” IDU. If the unit has no daughter board to customize ESC 3 (or 4), choose the “no daughter board” option.

4.4.2 – “File” menu



	**Receive** Uploads the current NE configuration and updates the configuration displayed on the CT.
 	**Save** Saves the current file. The file is saved by the same name, so a message warning you that the existing file is about to be deleted appears. You can enter a comment in the “Header” field. **Send** Sends the NE the new configuration. When you confirm the request to save the file, the screen displays a progress indicator bar (bottom screen). In classic IDU, if the software key is incompatible with the new configuration, the new configuration is disregarded. Progress indicator bar displayed while the configuration is being downloaded

4.5 – “Operation parameters” application

Before altering the installation parameters, check that there are no remote controls active.

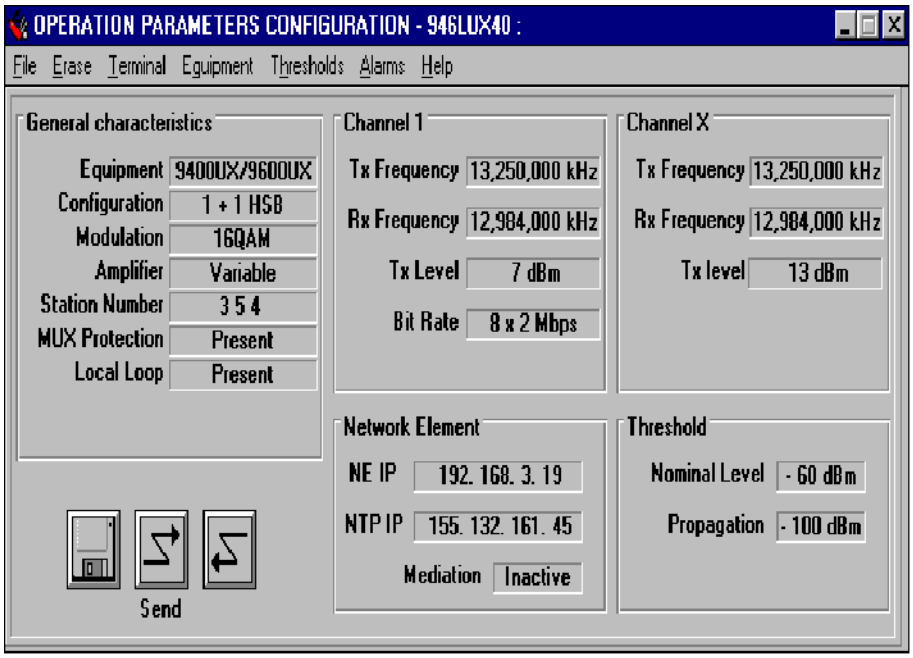
- **This application should be run only after closing the “Installation parameters” application.**
- **For a classic IDU, no configuration can be stored if the “key absent” alarm is present.**

The “Operation parameters” application is for declaring the operating parameters of the equipment for which the hardware parameters have already been defined and transmitted to equipment via the “Installation parameters” application.

Path: Alarm Synthesis → Application_Choice... → Operational Parameters

When you run the application, the current parameters are read from the equipment and uploaded to the CT. A progress indicator bar is displayed showing the progress of the upload.

The following main screen appears when uploading has finished. If the configuration includes inconsistencies (undefined frequencies, bit rates inconsistent with the hardware configuration, etc), intermediate screens appear prompting you to correct the inconsistent settings.



946LUX12, or 946LUX40

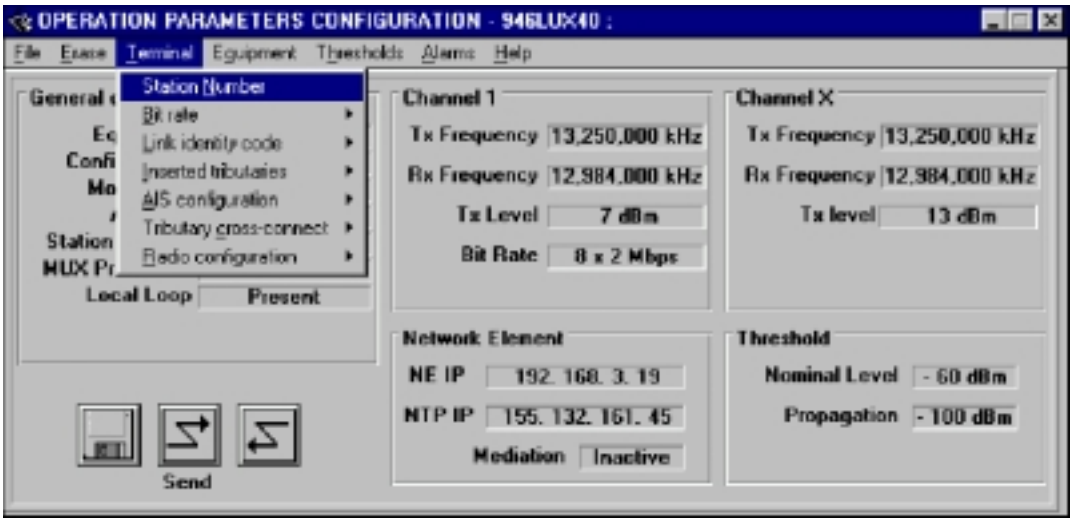
(Summary not available for LUX11)

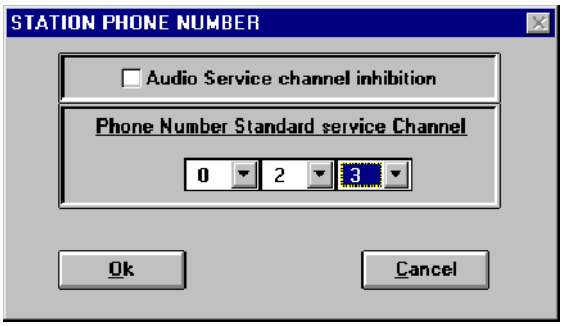
Note:

- when the application is initiated, the configuration parameters are automatically uploaded from the NE.
- the CT of the station and the master CT must not modify the installation and operation parameters at the same time.

The screens for the “Terminal”, “Equipment”, “Thresholds” and “Alarms” menus are for declaring the network element’s operation parameters.

4.5.1 – “Terminal” menu



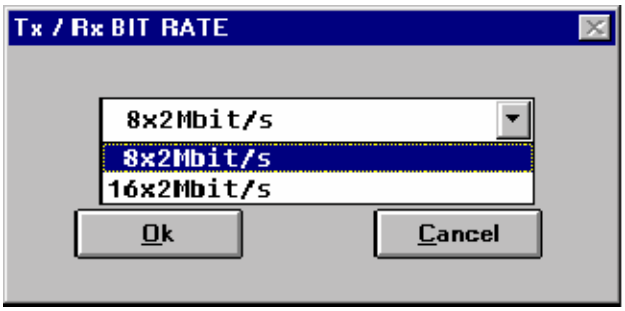


Station Number (Not displayed with Light IDU versions.

This is for selecting the telephone number of the station for ESC 2.

Possible values: 011 to 999, but there must not be more than one zero in the number.

If the “Audio service channel inhibition” box is checked: the terminal has no telephone.

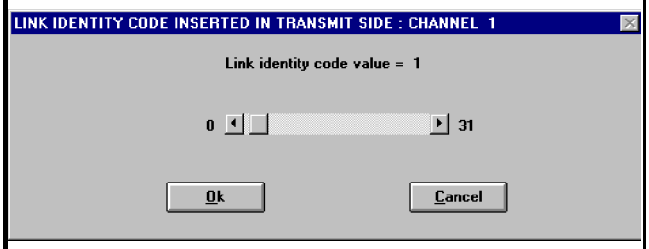


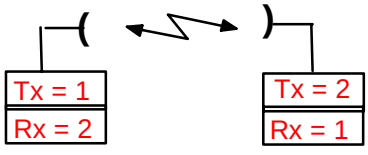
Bit rate

This is for choosing the operational bit rate.

The bit rate is limited by the maximum bit rate allowed by the software key and/or the maximum bit rate allowed by the hardware configuration.

For a Light IDU, only the 2x2 and 4x2 Mbit/s rates are displayed.





Link identity code

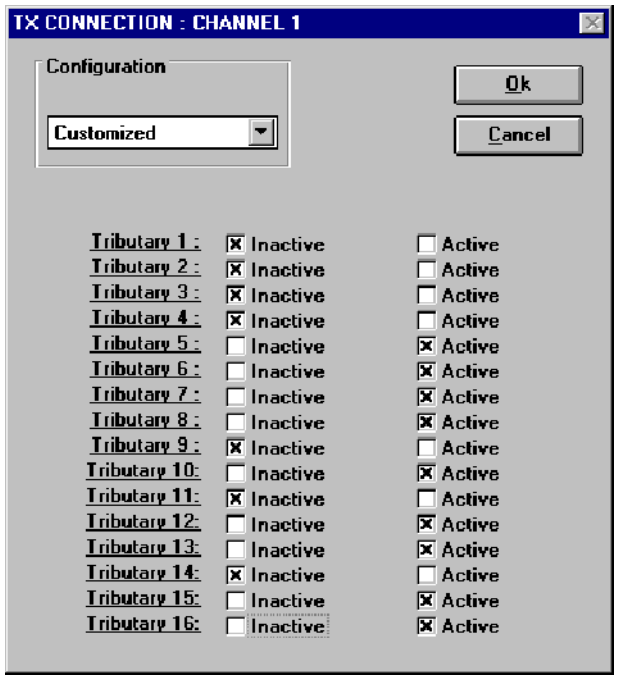
Two dialogue boxes (transmit and receive).

This is for protecting transmission by introduction of a code on transmission (possible values: 0 to 31). This same code must be inserted on reception in the far end equipment so that the received code can be compared with the code awaited.

Similarly, on reception, the programmed code must be the same as the one programmed for transmission in the far end station.

Different codes are recommended for the two directions.

Note: When using a loop, reset identical codes to avoid the occurrence of an alarm and restore the initial configurations when the loop is cleared (see example).



Inserted tributaries

Dialogue box displayed for transmission and reception.

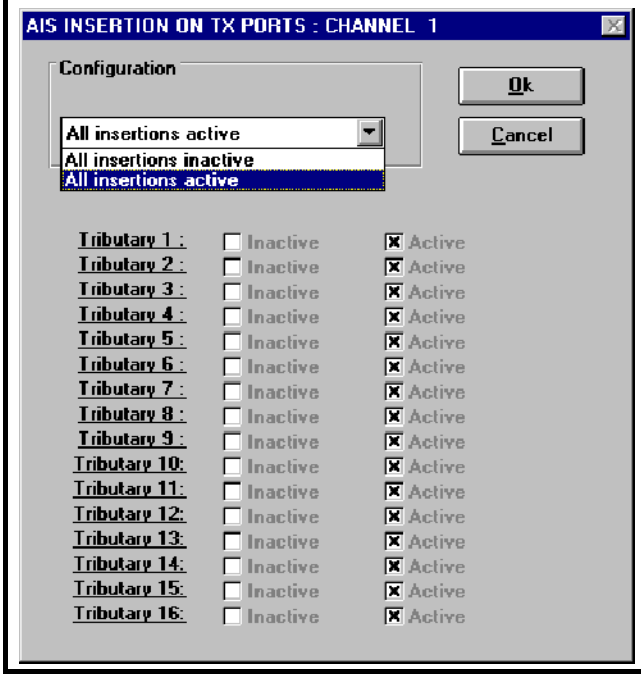
Enables or disables each of the tributaries. A tributary that is not loaded and active generates an alarm.

The “Configuration” list is for activating or deactivating all tributaries simultaneously.

The “Customized” option is for activating or deactivating tributaries individually by checking the appropriate boxes.

The number of tributaries displayed depends on the option selected in the “bit rate” screen.

In Light IDU versions, two or four tributaries are displayed.



AIS configuration

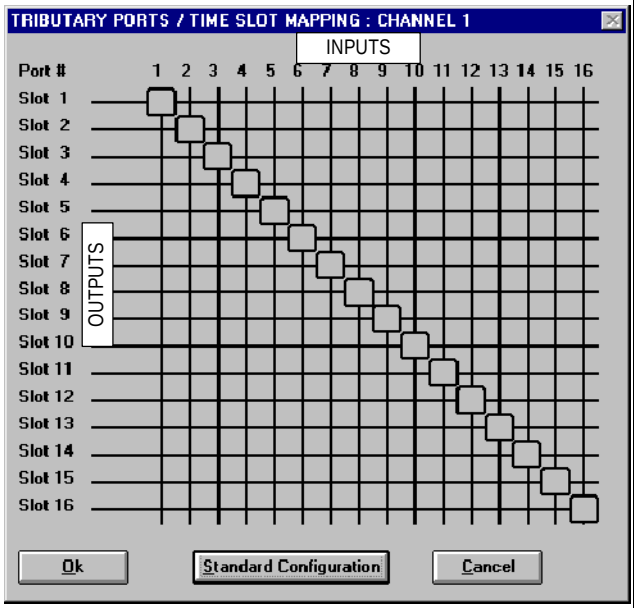
Dialogue box displayed on transmission and reception.

For specifying whether AIS insertion is active or inactive for the tributaries.

The “Configuration” list box lets you activate or deactivate all insertions simultaneously.

The number of tributaries displayed depends on the option selected in the “bit rates” screen.

In Light IDU versions, two or four tributaries are displayed.



Tributary ports/Time slot mapping

This is for modifying the order in which tributaries are transmitted.

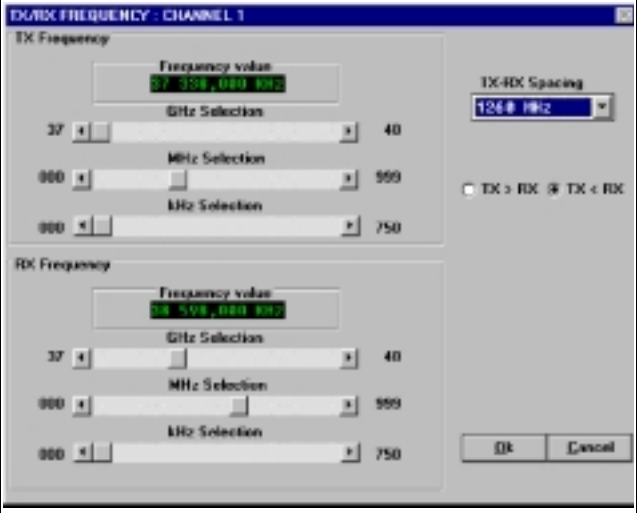
When you click the point of intersection of column i and row j, incoming tributary N_ i in the local terminal will leave at port N_ j of the far end terminal.

The “Standard configuration” button reinstates the configuration shown opposite.

The columns represent the inputs and the rows represent the outputs.

The number of tributaries displayed depends on the option selected in the “bit rates” screen.

In Light IDU versions, two or four rows and columns are displayed.



Radio configuration/Tx frequency
Radio configuration/Rx frequency

(example given for LUX40)

Dialogue box displayed for transmission and reception.

Three horizontal bars are provided for adjusting frequencies in:

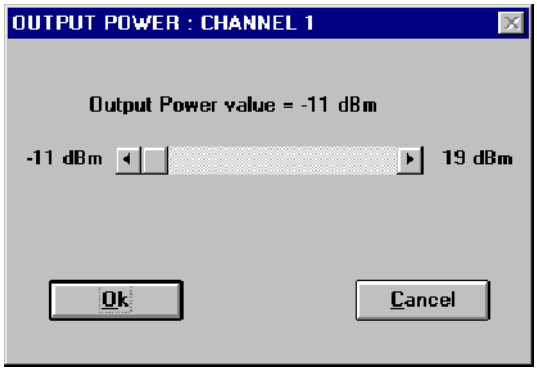
- GHz (step = 1 GHz),
- MHz (step = 1 MHz),
- kHz (step = 250 kHz).

The values that can be programmed are limited by the frequency band of the equipment and, in the case of the classic IDU, by software key.

Check that the duplex separation ($FX_{transmit}$ – $RX_{receive}$), anterior to the 06 issue, is in accordance with the values entered.

If the separation between the transmit frequency and the receive frequency is slightly different from the duplex separation of the equipment, the “configuration error” alarm does not appear. However, it will appear if an RF loopback configuration is set up.

In 1+1 HSB configurations, the transmit and receive frequencies of the standby channel, although identical to those of the normal channel, must be programmed by the operator.



13, 15, 18, 23, 25 and 38 GHz in RTCP configuration

Radio configuration/Output power

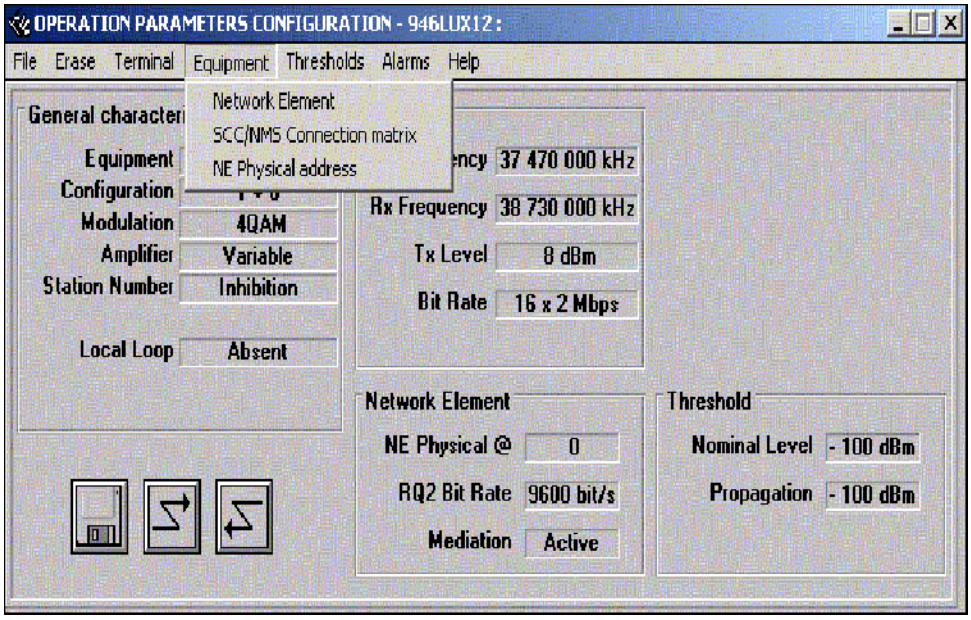
Adjustment of the output power of the ODU. For possible minimum and maximum values, please see § 2.8.

With the classic IDU, these values may be limited by the software key.

These values may also be limited by the ODU.

4.5.2 – “Network” menu with 946LUX11 or 946LUX12 software (RQ2)

This menu manages the parameters used to insert the NE in the network management system, in other words, **it is used to manage the NE from the Mediation Function of the Master Station**.



946LUX11 or 946LUX12

The “**Network element**” submenu provides access to a screen in which you can select the equipment as:

- network element,
- mediation function.

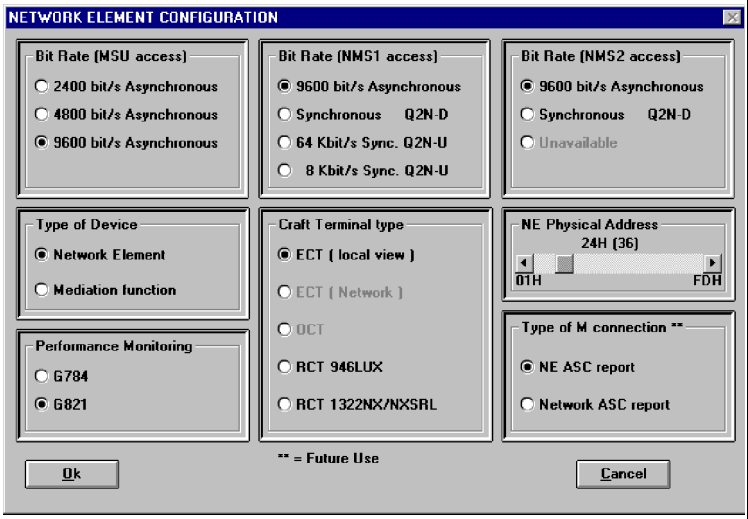
Secondary network element addressing is available only for mediation functions.

These two cases are explained below.

When the PC is connected to a terminal with a software key incorporating the RCT option (always included in the Light IDU versions), the “RCT946LUX” option can be used to obtain a copy of the network supervision PC’s screen (1322 NX or 946LUXxx craft terminal).

In local supervision mode, you can access the local equipment and remote equipment.

4.5.2.1 – Network element configuration



Network Element

Bit Rate: The bit rate selected must match that of the mediation function's RQ2 bus. The bit rate selected in the "Bit rate (MSU access)" box is automatically copied into the other boxes.

Type of device: The unit is declared as a network element or mediation function (see § 4.14).

Craft Terminal type:

The terminal type "RCT 946LUX" appears only if the unit is an NE with mediation and fitted with an RCT (Remote Craft Terminal) type software key.

The RCT options are available only if the software key allows this function.

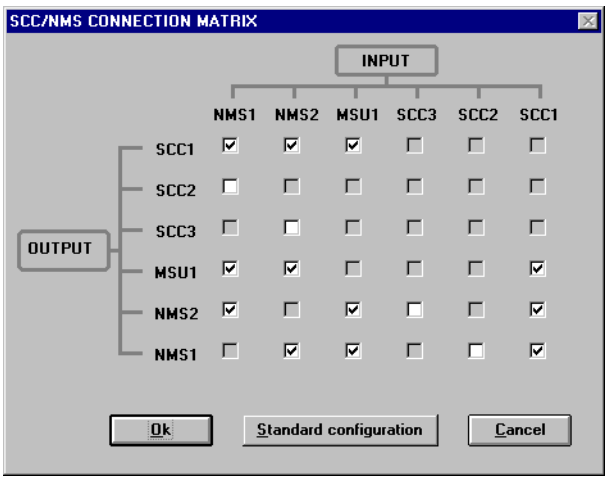
NE Physical Address: This address (1 to 253 or 01H to FDH in hexadecimal) is defined by operating the scrollbar.

Performance Monitoring: Selects the type of performance measurement. The G784 type is available only if the extended 946LUX11 or 946LUX12 software has been installed.

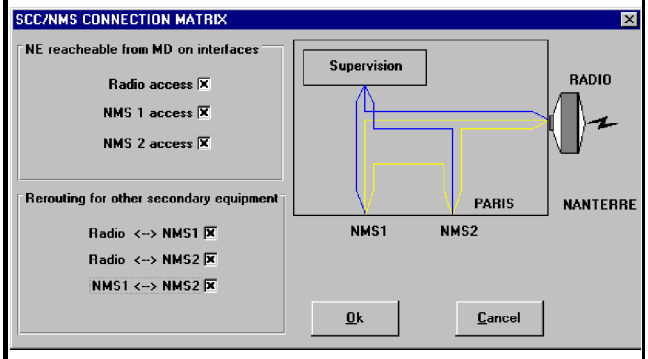
Note: When the new configuration is sent, the NE supervision is reset. This action, which takes a few minutes, occurs when:

- the physical address of the NE is changed
- switching is done between G784 ⇆ G821 mode
- equipment type is modified

4.5.2.2 - SCC/NMS Connection matrix



946LUX11



946LUX12

Screen for setting up, for a particular NE, the connections used to transmit information from the network management system to the selected NEs (see § 2.3.1).

946LUX11

Normally, the standard configuration will be selected, unless otherwise indicated by the network administrator.

946LUX12

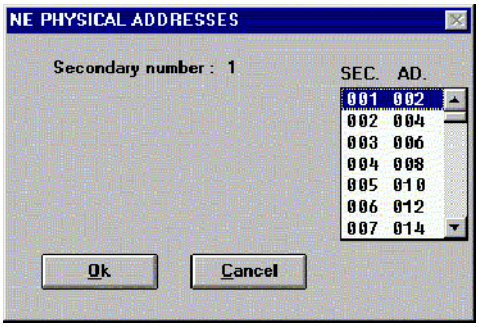
The links are set up:

- in the lefthand window, by clicking the appropriate box,
- in the righthand window, by clicking the name of the starting point of the link then, holding down the left mouse button, moving the cursor to the name of the other end of the link.

Links are deleted in the same way.

4.5.2.3 - Secondary NE addressing

These menus appear only if the NE is configured as a mediation function (see § 4.5.2.1).



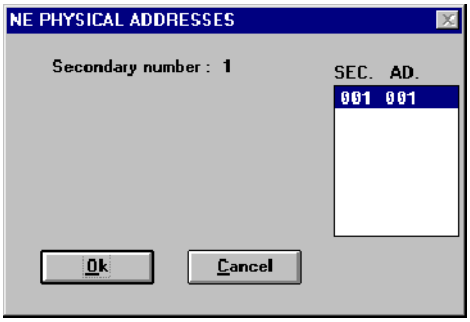
Secondary NEs - Master key

Screen not displayed in the Light IDU versions, since only one secondary network element with a predefined address is accessible.

Option displayed when the element is declared as Mediation Function (master key).

Gives a list of the physical addresses of each secondary network element (slave station) managed by the Mediation Function.

If you double click on a network element or click the "**Ok**" button after selecting a network element, a dialogue box opens for you to select the physical address below.



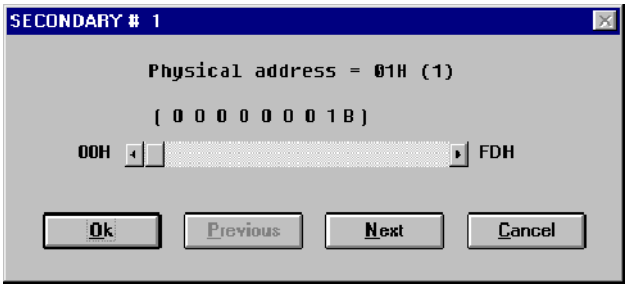
Secondary NEs – Slave key or Light IDU

The local supervision function can be used to view another network element in addition to the local equipment.

Option displayed when the element is declared as Mediation Function to enable local supervision (slave key) or in the case of the Light IDU versions.

The NE can supervise just one far-end network element, so only one physical NE address is displayed.

If you double click on the NE or click the "**Ok**" button after selecting a NE, the following dialogue box appears:

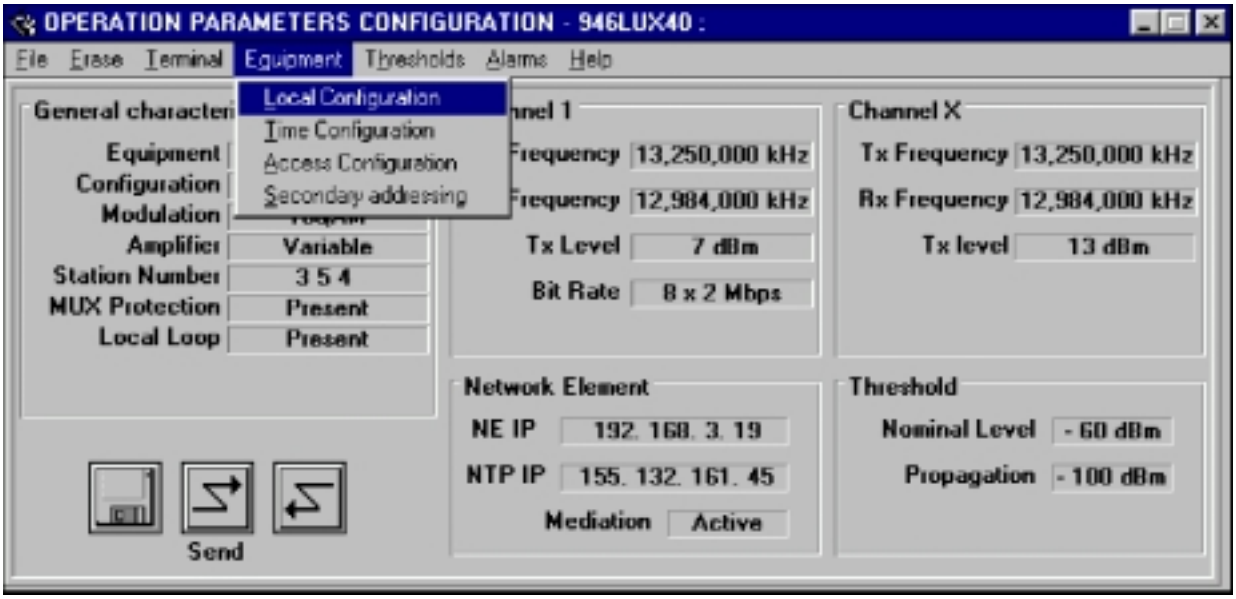


Dialogue box for modifying the physical address of the remote equipment to be viewed.

Address 00: No NE connected.

4.5.3 – "Network" menu with 946LUX40 software (SNMP)

This menu manages the parameters used to insert the NE in the network management system, in other words, **it is used to manage the NE from the Mediation Function of the Master Station**.



The "**Local configuration**" submenu provides access to a screen in which you can select the equipment as:

- network element,
- mediation function

The "Secondary Addressing" submenu is available only for equipment declared as mediation.

When the PC is connected to a terminal with a key incorporating the RCT option, the "PED946LUX" option can be used to obtain a copy of the network supervision PC's screen (946xx craft terminal).

In local supervision mode, you can access the local equipment and a remote network element.

4.5.3.1 – Example of network managed via an SNMP bus

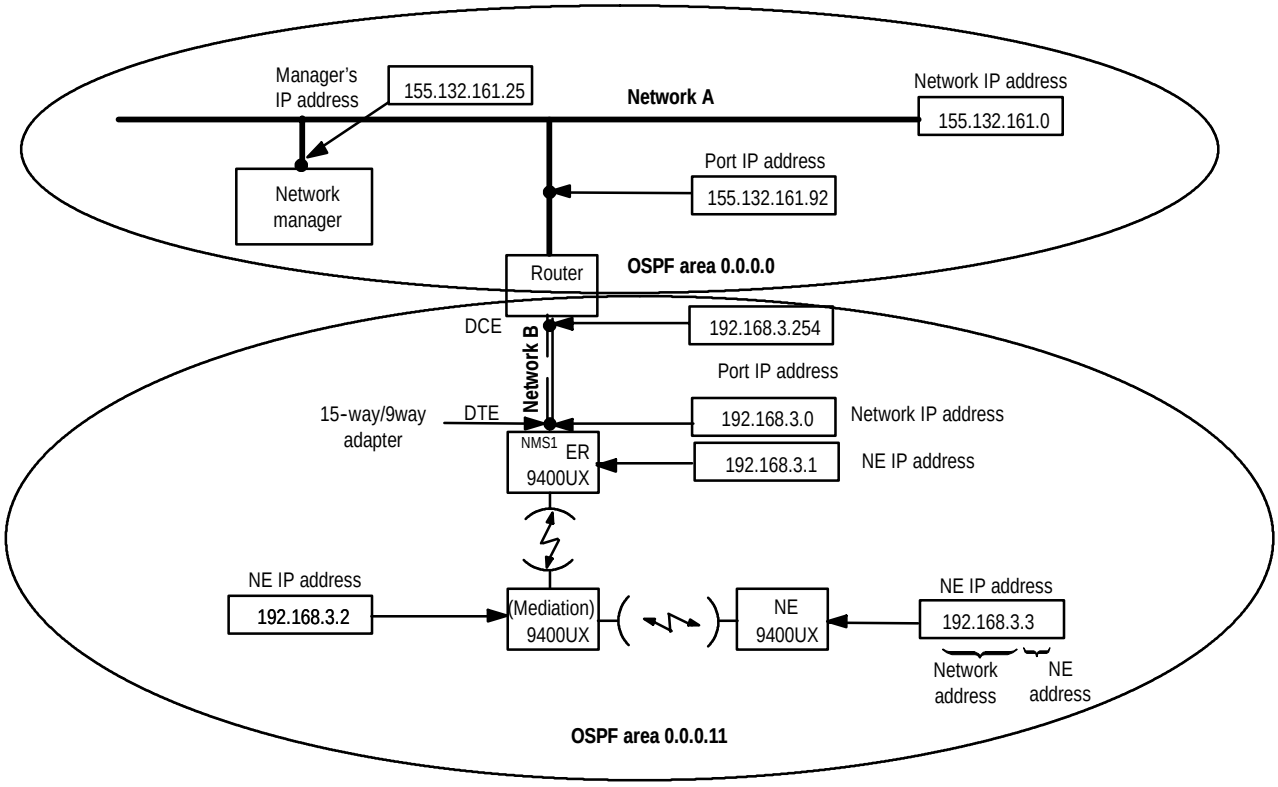


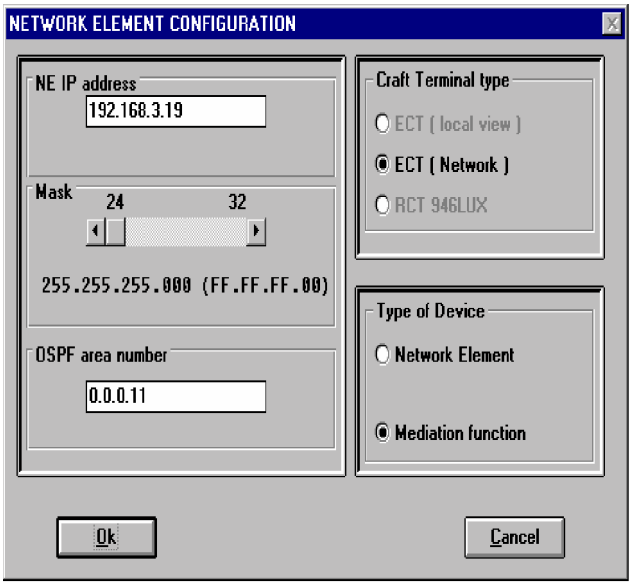
Figure 38 - Example of network with SNMP interface

In this type of network management, the 9400 network can be managed directly via the network manager or via a mediation function in the 9400 network. **The SNMP bus operates at 64 kbit/s, synchronous.**

The connection with a router is made with referenced cables 3CC10970AAxx for an IDU DCE with a female 15 pin connector and 3CC10969AAxx for an IDU DTE with a male 15 pin connector.

4.5.3.2 - N.E. configuration

⚠	**All the NE (Network Element) of a same OSPF area must be configured with the same subnetwork mask.**



Type of Device: The unit is declared as a Network Element or mediation function.

Craft Terminal type: The type of terminal **"RCT 946LUX"** does not appear unless the unit is an NE with mediation fitted with a type RCT software key. (RCT = Remote Craft Terminal).

No possible options.

NE IP address :
This IP (Internet Protocol) address is coded on 4 bytes A, B, C, D.

Class B or C addresses are attributed to the NE and the router according to the size of the network.

It is possible to address sub-networks.

OSPF area: The routing protocol used is OSPF (Open Shortest Path First). Routing information is exchanged between NEs in the same area. When different areas are used, they must be connected to area "0" (backbone) using routers (Area border routers).

Never configure a NE as being in OSPF area 0.0.0.0.

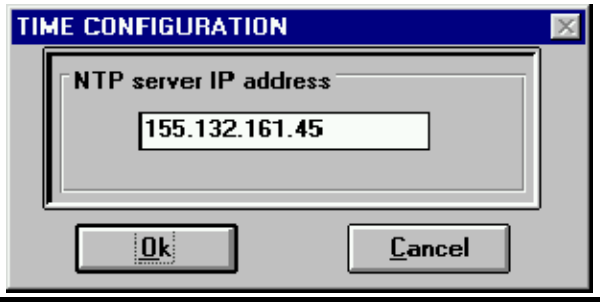
No more than 63 NE in the same OSPF area.

It is possible to address sub-networks.

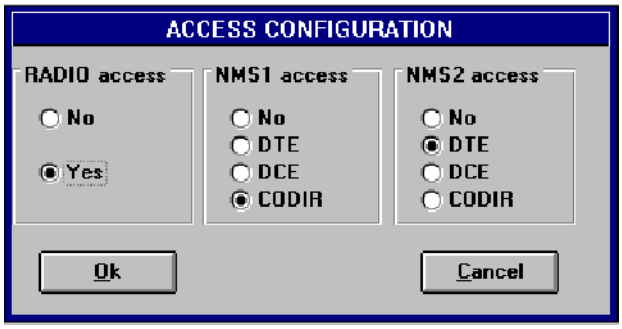
Mask: The mask must be adjusted according to the adressing plan of the network.

Note: A reinitializing of the NE requires a few minutes and is done in the following cases:

- Modifying the NE IP address,
- Modifying the type of device,
- Modifying the number of the OSPF area,
- Modifying the subnetwork mask.



Type in the NTP server IP address. Alternatively, leave 0.0.0.0. In this case, the NE will adopt the NE master time.

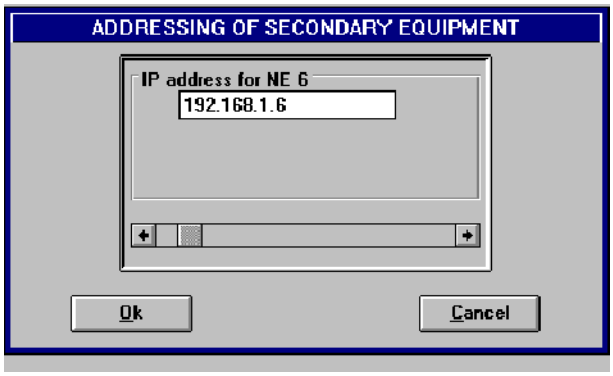


Access configuration

Screen used for a given network element to set up connections for transmitting information from the network management system to the selected elements (see § 2.3.1).

- NE ports NMS1 and NMS2 in a 9400 network should be set to "**CODIR**".
- NMS ports connected to a router should be set to "**DTE**", with the corresponding port on the router set to DCE.

Links are eliminated by clicking "**No**".



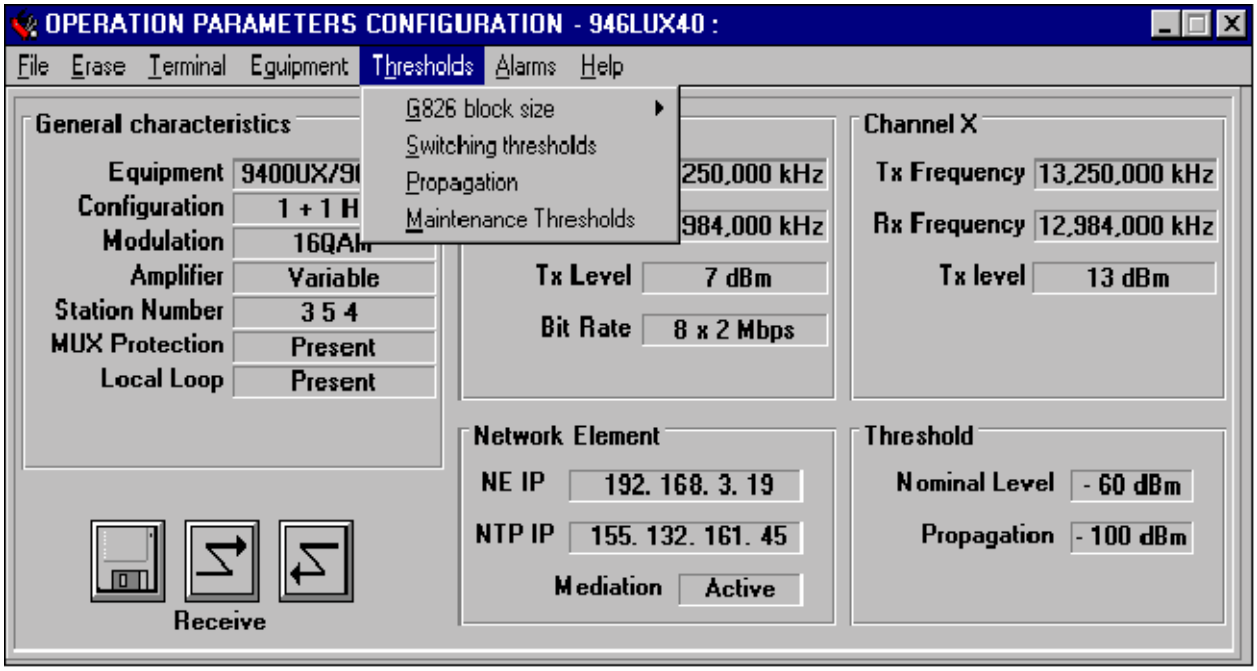
Addressing of secondary equipment

Option displayed **when the network element is declared as a mediation** (master key).

This function is used to enable all the NEs that the mediation device is required to monitor.

- use the moving scale to select the number of the NE (above the window), enter its IP address in the window and confirm by clicking "**Ok**".

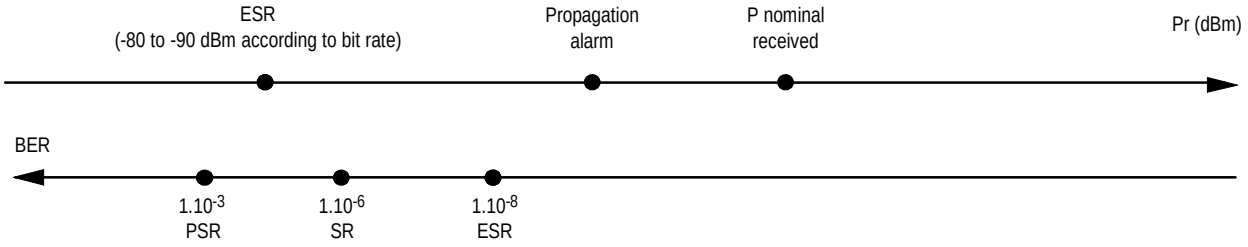
4.5.4 - "Thresholds" menu

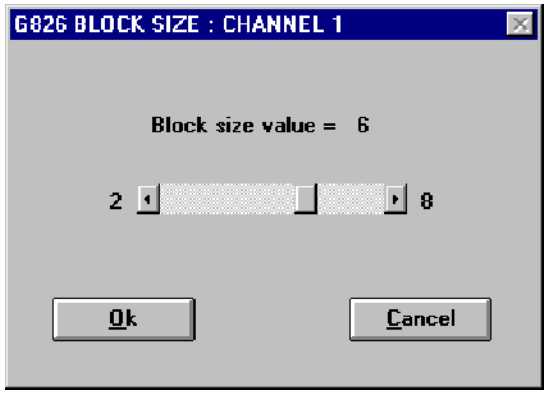


The “Propagation” and “Alarm thresholds” menus are used to define:

- the threshold value at which the propagation alarm is triggered, compared to the nominal receive power value (value obtained by link circulation),
- the alarm activation thresholds associated with switching requests in a 1+1 configuration. Two thresholds can trigger the early switching request: received power level or error ratio.

These thresholds should be applied in the order indicated by the figure below:





This screen appears only if the G.784 option has been selected (network screen / network element)and for bit rates between 2x2 Mbps and 8x2 Mbp.

G826 block size

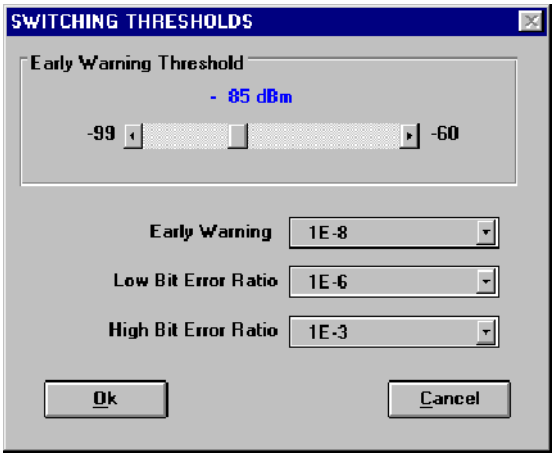
Set the block size to:

1 at 2x2 Mbps,

2 at 4x2 Mbps,

4 at 8x2 Mbps,

8 at 16x2 Mbps.



Switching thresholds/Alarm triggering thresholds (1+1 configuration)

Early warning alarm. Early Switching Request **(ESR):**

- Horizontal bar: Selects the absolute received power level which the *Early Warning* alarm is triggered.
- Pull-down list: Selects the BER threshold at which the *early warning* alarm is triggered.

Low Bit Error Ratio. .Switching Request **(SR):**

Selects the BER threshold of which the *low Bit Error Ratio* switching request alarm is triggered.

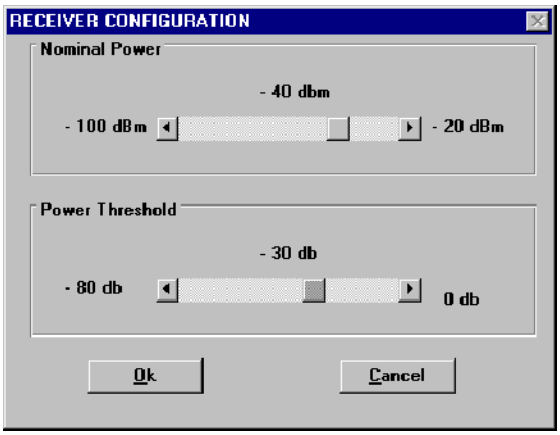
High Bit Error Ratio. Priority Switching Request **(PSR):**

Selects the BER threshold at which the *High Bit Error Ratio* switching request alarm is triggered.

If the AIS is active, the selected error rate determines the AIS engaging.

The threshold settings should be such that:

ESR < SR < PSR

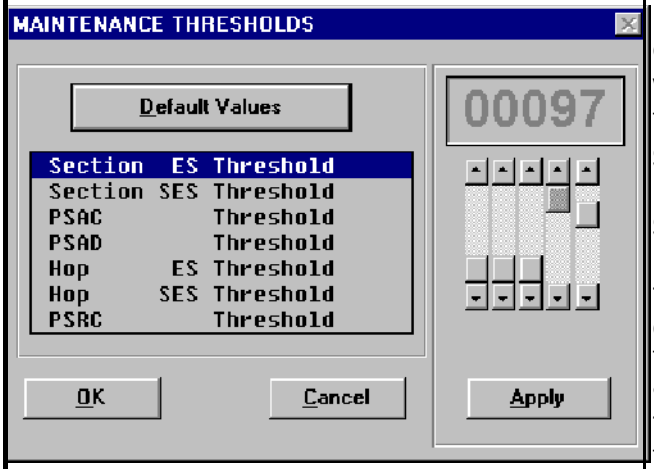


Propagation

Nominal Power: Indicates the nominal received power (derived from link circulation).

Power Threshold: Selects the attenuation value compared to nominal receive power at which the propagation alarm is triggered.

In this example, the alarm is triggered when the received power falls below -70dBm (-40dBm - 30dB).



Maintenance Thresholds

Defines the thresholds at which maintenance alarms are triggered.

The equipment analyzes and logs ES, SES, PSAC, PSAD and PSRC, etc, using counters that are reset every 24 hours.

When one of the counters exceeds the programmed threshold, it triggers:

- a prompt maintenance alarm for the first four thresholds (this alarm appears immediately),
- a deferred action maintenance alarm for the last three thresholds (this alarm appears immediately).

These alarms are normally disabled in the initial NE configuration when shipped. They are enabled via the “Alarms” menu of the “Operation parameters” application (§ 4.5.5).

These alarms are reset by the remote control function “Remote_Control/All/MAINTENANCE ALARMS : Reset” in the “Alarms, States and Remote controls” application.

The threshold values programmed in the factory are given in the table below.

The acronyms used are as follows:

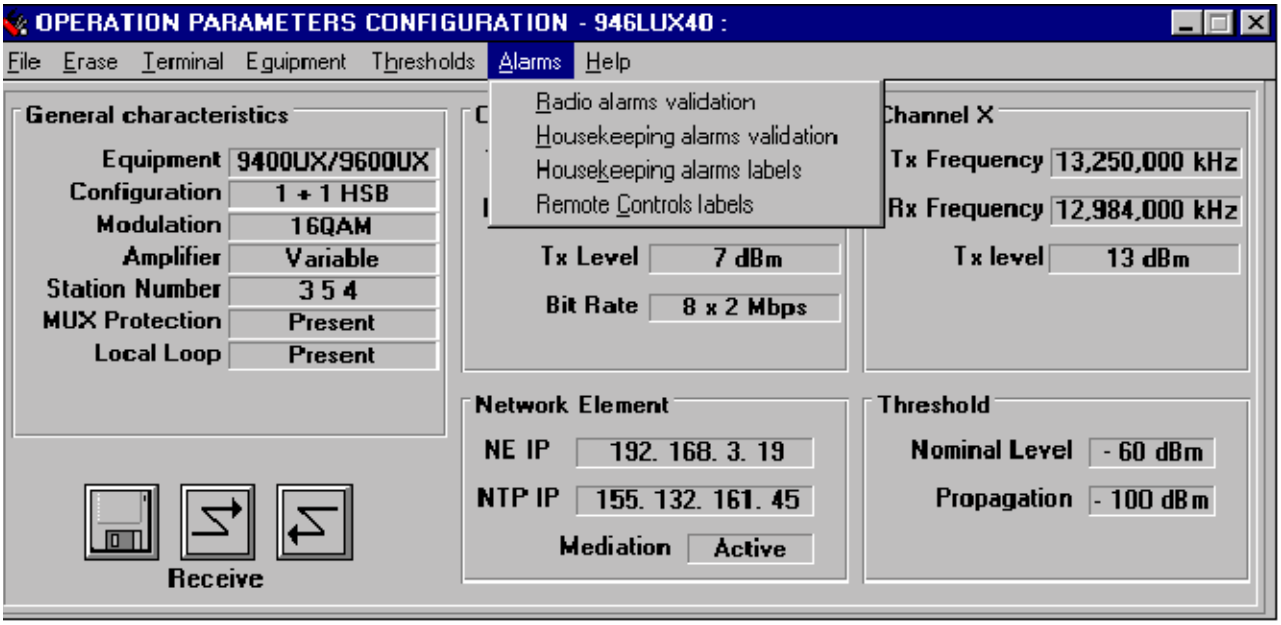
ACRONYM	NORMAL CHANNEL	STANDBY CHANNEL*
PSAC	Number of switchovers from active channel to standby channel	Number of switchovers from any active channel to the standby channel
PSAD	Number of one-second periods during which the active channel is backed up	Number of one-second periods during which any active channel is backed up
PSRC	Number of automatic switching requests observed on the normal channel	Number of automatic switching requests observed on the normal channel

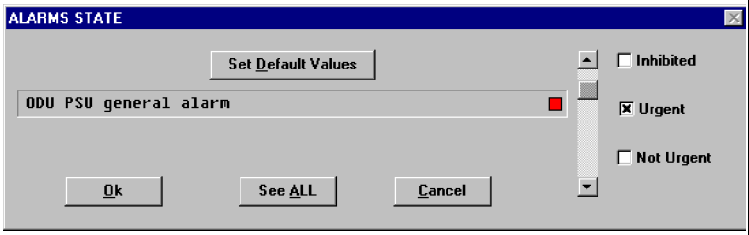
* Not applicable in 1+0 configurations.

Factory-set threshold values

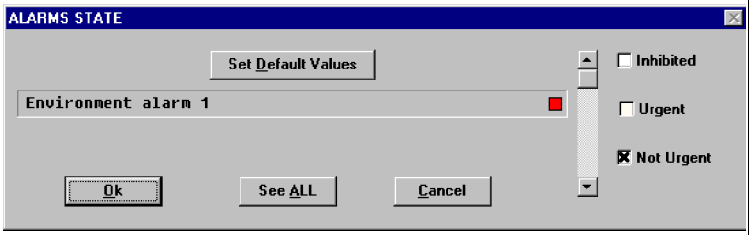
THRESHOLD	VALUE
Section ES Threshold	97
Section SES Threshold	2
PSAC Threshold	100
PSAD Threshold (Meaningless)	86400
Hop ES Threshold	97
Hop SES Threshold	2
PSRC Threshold	100

4.5.5 - "Alarms" menu





Radio alarms



Housekeeping alarms

Radio alarms or Housekeeping alarms validation

Displaying the list of alarms:

- Use the scroll bar in the righthand part of the dialogue box.
- Click the "See ALL" button and a list of alarms (see table in Appendix 7) appears. You can double click an alarm to return to the dialogue box with the alarm selected.

Changing the alarm criterion:

Assign the selected alarm criterion by checking one of the boxes. A coloured box appears alongside the alarm name, depending on the criterion you select: "Inhibited" (Grey), "Urgent" (Red) or "Not Urgent" (Yellow).

If you do not check any of the boxes, the criterion is set to "State", coloured magenta.

Changing the criterion for all alarms:

Click the "Set default values" button to set the criteria for all the alarms managed by the current configuration in a single operation:

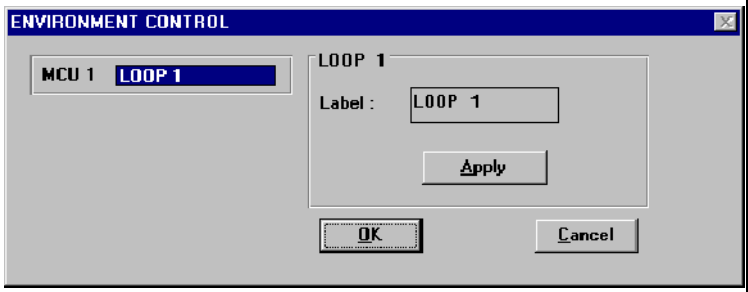
- The alarms are declared as *Urgent alarms*
- Remote controls and indications are declared as *Not urgent alarms*



Housekeeping alarms labels

Select one of the loops available from the lefthand column and, after double clicking the "Label" box, assign a name (up to 12 ASCII characters).

Click "Apply" then "OK" in that order to exit.



Remote Controls (loops) labels

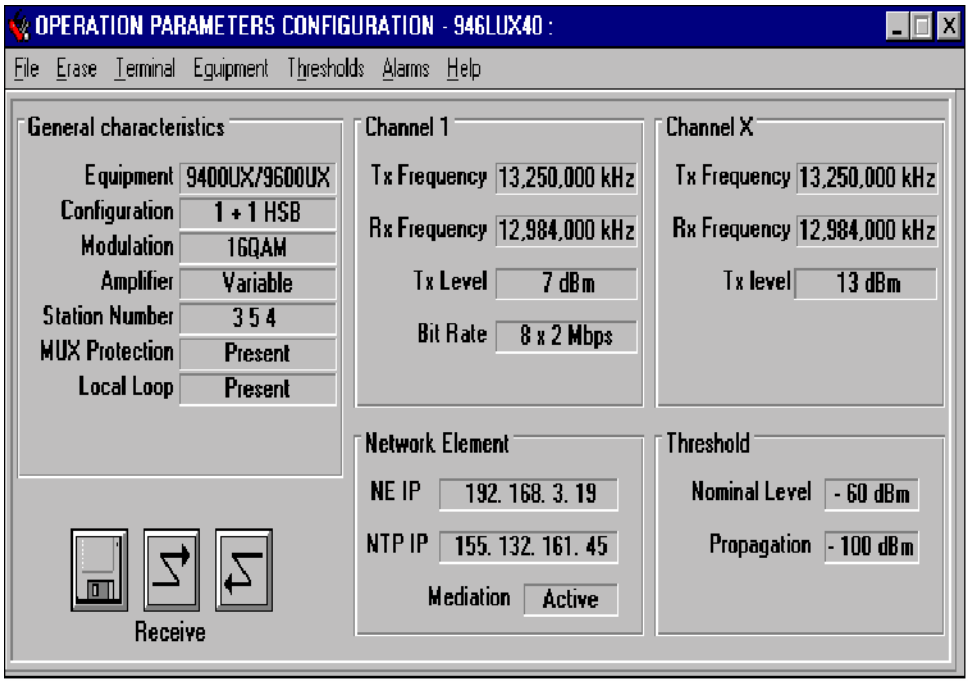
The principle here is the same as for the environment loops.

4.5.6 – “Summary” menu (946LUX12 and 946LUX40 version of the software)

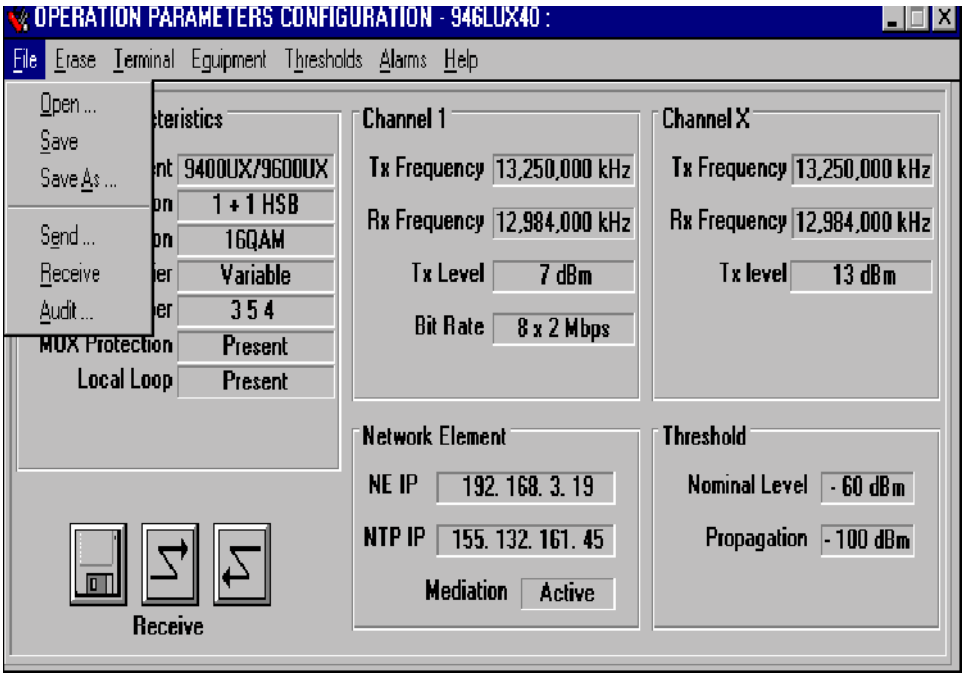
This menu calls up the window below, summarizing the equipment settings. You exit the screen by clicking “Hide summary” in the menu bar (LUX12 versions lower than index 6).

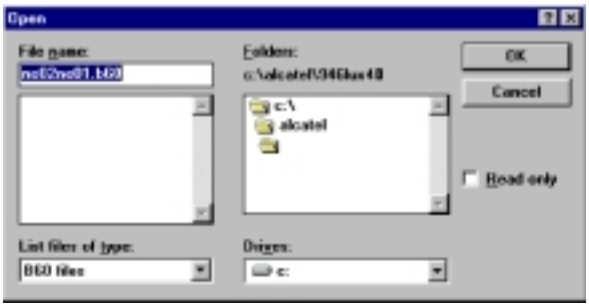
You can edit and modify these settings only by a double click on the parameter you want to modify.

“Config summary” screen in the 946LUX40 software



4.5.7 – “File” menu





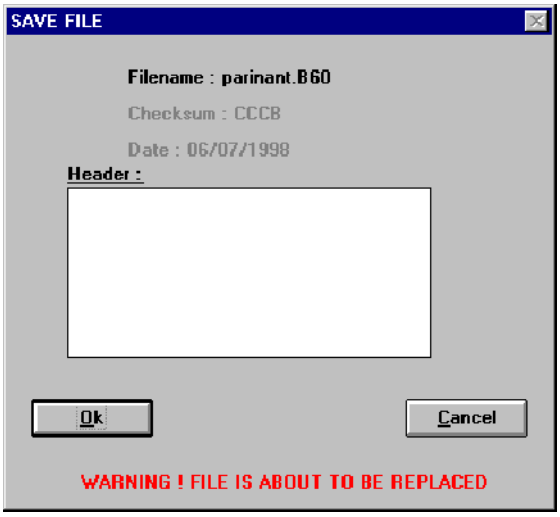
Open ..

Opens a “Configuration table” file saved previously. The directory containing open files is:
c:\alcatel\946luxxx.

The command uses the variables of the station name to name the files. When the local station name is PARI(S) and that of the far-end station is NANT(ERRE), the file name may be:

parinant.b60 (for the file used), parinant.old or parinant.p01 to parinant.p09 (for the files saved).

Note: Before you open a configuration file, the current configuration is saved as parinant.old.



Save

Saves the configuration tables in the file. The directory for open files is:
c:\alcatel\946luxxx.

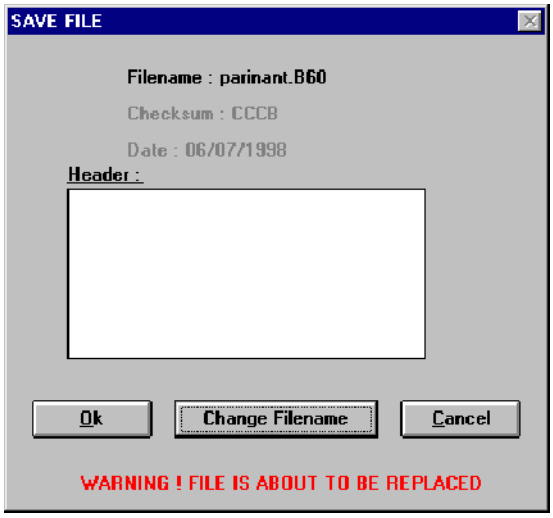
The command uses the variables of the station name to name the files. When the name of the local station is PARI(S) and that of the far-end station is NANT(ERRE), the file name is:

parinant.B60

Note: A warning message is displayed.

△

The configuration is not sent to the equipment.



Save As ...

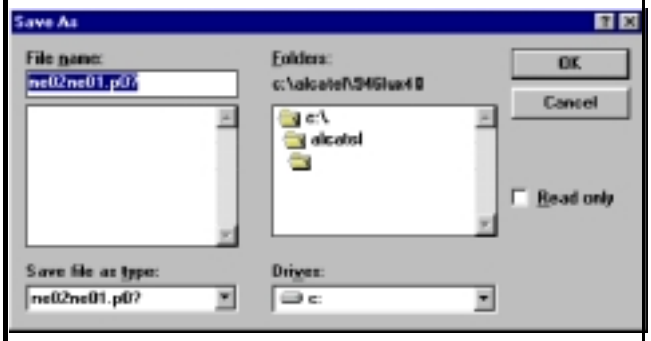
The “Save as” command opens the same window as the “Save” command, but with an extra “Change Filename” button. If you click this button, the following window opens.

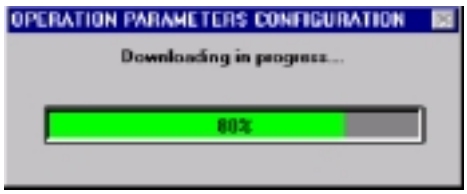
Saves the configuration tables in a file. The directory for open files is:
c:\alcatel\946luxxx.

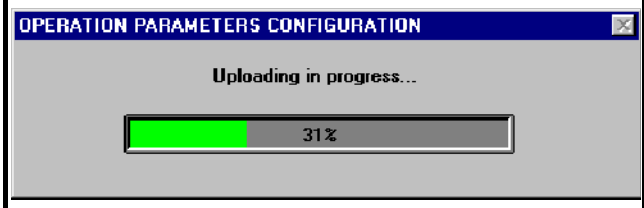
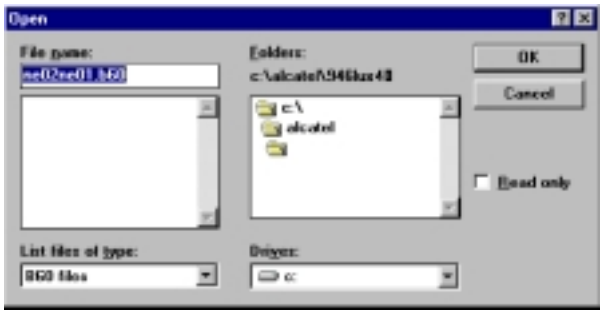
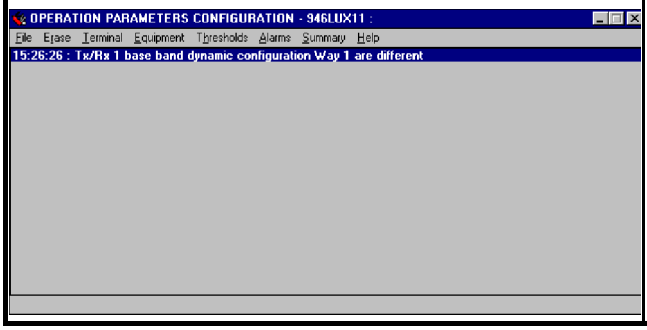
The command uses the variables of the station name to name the files. When the name of the local station is PARI(S) and that of the far-end station is NANT(ERRE), the file name is:

parinant.p01 to parinant.p09 (Replace the “?” with a value from 1 to 9 as required).

Note: A confirmation prompt window is displayed.

<table>
<tr>
<td> 


</td>
<td>Send ...
Sends the current configuration tables to the NE.
These tables are saved first.
All the configuration parameters are then down-loaded to the NE.
Note:
S In the case of a classic IDU, the configuration tables may be rejected by the NE when a software key problem arises.
S Wait for downloading to finish before carrying out any other operations. Download progress is indicated by a progress indicator bar (see screen below).

Configuration download progress indicator bar.</td>
</tr>
<tr>
<td> 
</td>
<td>Receive
Command for:
S Uploading configuration tables from the NE.
S Matching the current CT configuration to the NE configuration.
Note: When the application is started up, the configuration tables are automatically uploaded from the NE.</td>
</tr>
<tr>
<td> 
</td>
<td>Audit
Compares the current configuration on the CT with the configuration files previously saved.
This menu is used in particular to check changes made to the configuration between opening the program and the time of the audit.
Files are select from the screen displayed (Open).</td>
</tr>
<tr>
<td> 
</td>
<td>The result appears on the background of the application's main window. Only configuration tables that do not match appear.
The "Erase" menu lets you clear the result of the audit from the window.</td>
</tr>
</table>

4.6 – “Software downloading”application

This application can be used to download the onboard version of the Network Element software.

The download can be executed from:

- an **ECT** (Equipment Craft Terminal) connected to the NE,
- an **OCT** (Office Craft Terminal) connected to the mediation device,
- an **LCT** (Local Craft Terminal) at the **CS** (Central Station),

to all the stations of a network (to update MCU board software in 9400 UX equipment).

The download operation is in three stages:

- **installation of the new software release** on the craft terminal’s hard disk, from the diskette supplied by Alcatel,
- **downloading of the software** to the RAM of the NEs that need to receive it,
- **activation of the software** in each NE.

NOTES

For a given NE, the application must not be run simultaneously from more than one craft terminal.

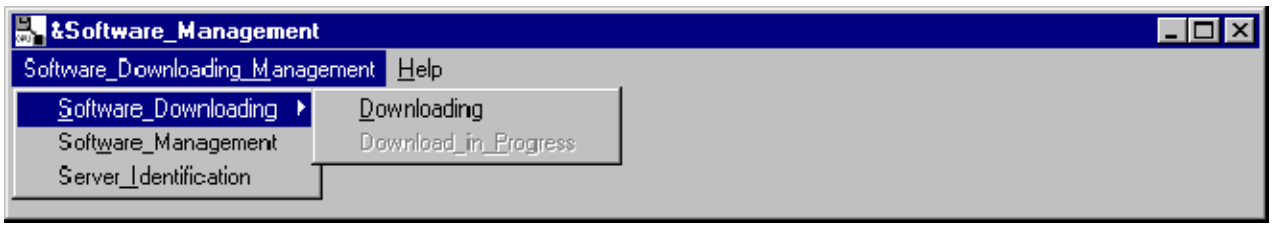
For a given type of NE and craft terminal, the operation for installing the software is performed just once, on receipt of the diskette supplied by Alcatel, containing the software release. Only the downloading and activation stages need to be repeated for all the NEs of the network supervised by the craft terminal.

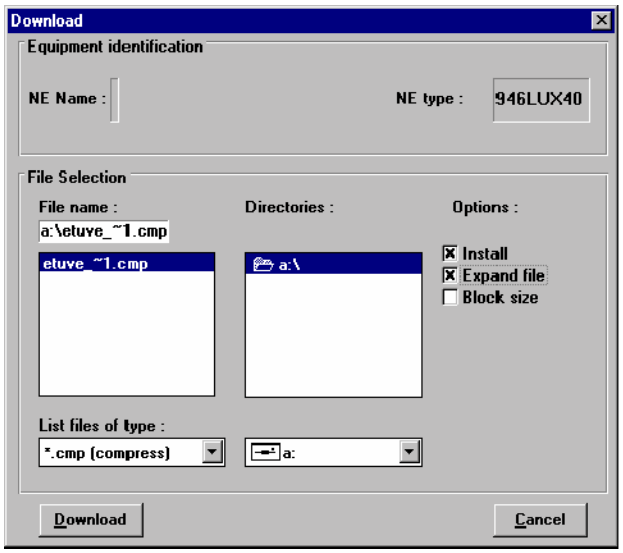
When you have opened the application:

- **if a download had been begun previously** with an unfinished transfer of data to RAM, a window appears prompting you to finish the transfer;
- **if a software release had been previously transferred** to NE RAM but not activated, a window appears prompting you to activate the software.

Path: Alarms Synthesis → Application_Choice... → Software downloading

“SOFTWARE_DOWNLOADING_MANAGEMENT” MENU





Software Downloading

Selects the source file for the download.

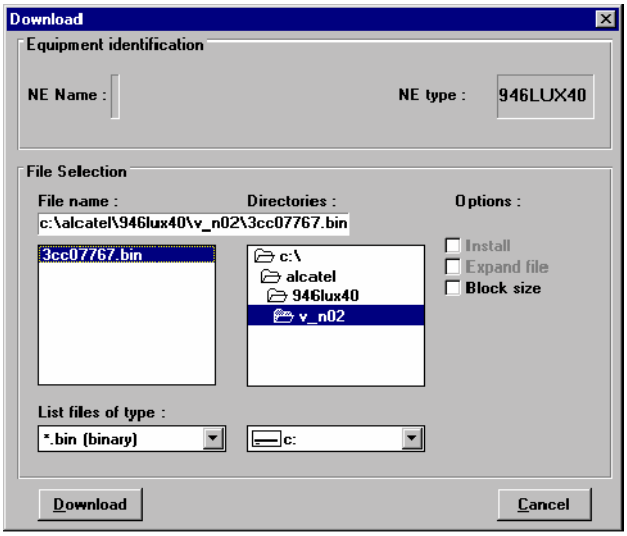
With installation:

Downloading following delivery of a new software release by ALCATEL.

The “Install” and “Expand file” boxes should both be checked (.cmp, type compressed files supplied on diskette for installation from drive a:\).

Block size (247 bytes by default) may be reduced if a high BER reading is obtained on the link, by checking the appropriate box (see later in this table).

Click the “Download” button to initiate the download. Progress indicator bars are displayed for you to monitor installation and downloading progress.



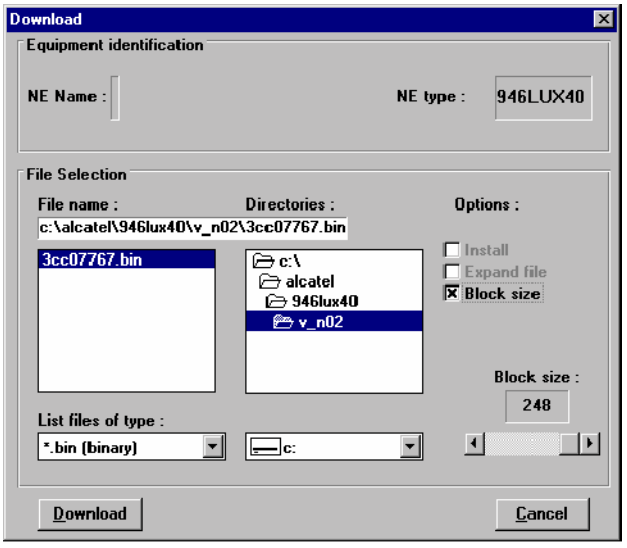
Without installation:

To be used if an installation had been completed previously. The “Install” and “Expand file” boxes can no longer be accessed. The .bin type files to be installed must be selected from the hard disk C:\. The block size can be adjusted.

Click the “Download” button to initiate the download. A progress indicator bar is displayed for you to monitor the progress of the download.

Note:

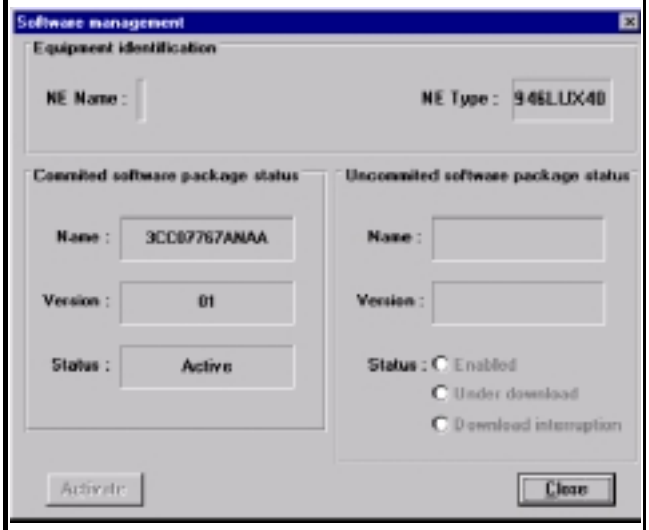
- Installation and expansion of the .cmp file supplied on diskette, which includes transferring of the file to the craft terminal’s hard disk followed by its de-compression to obtain a .bin binary file, takes about 45 minutes.
- The process of transferring the .bin file to the NE takes at least half an hour.



Adjusting the block size

Use the cursor that appears when the corresponding box is checked.

To be used only if a high bit error ratio is obtained.



Software management

For identifying the MCU's software release.

Equipment identification:

- NE name: name of network element
- NE type: type of network element

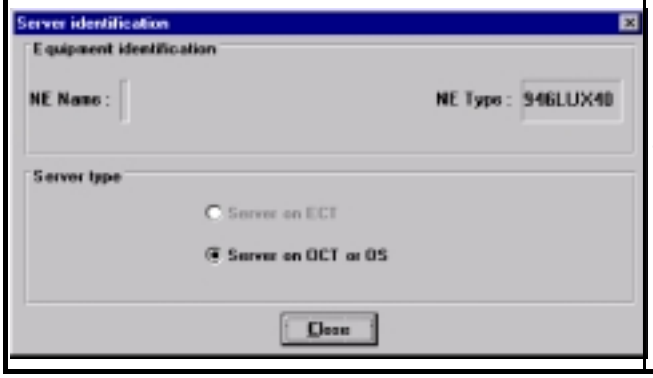
Committed software package status

- Name: the identity of the software (ALCATEL reference)
- Version: the software revision index
- Status: the system software activated is the software downloaded into the NE's flash memory

Uncommitted software package status

- Name: software identity (ALCATEL reference)
- Version: software revision index
- Status:
  Enabled: Software transferred into RAM and available for activation (by clicking the "Activation" (Activate) button to transfer from RAM to flash memory)
- *Under download:* Status that applies during installation and data transfer
- *Download interruption:* Status that applies after a command to abort the download

When you click the "Activate" button, the uncommitted software becomes committed.



Server Identification

Identification of the craft terminal from which downloading is performed.

The server is the PC performing the download

Read only information screen

NE Name: identifies the name of the NE

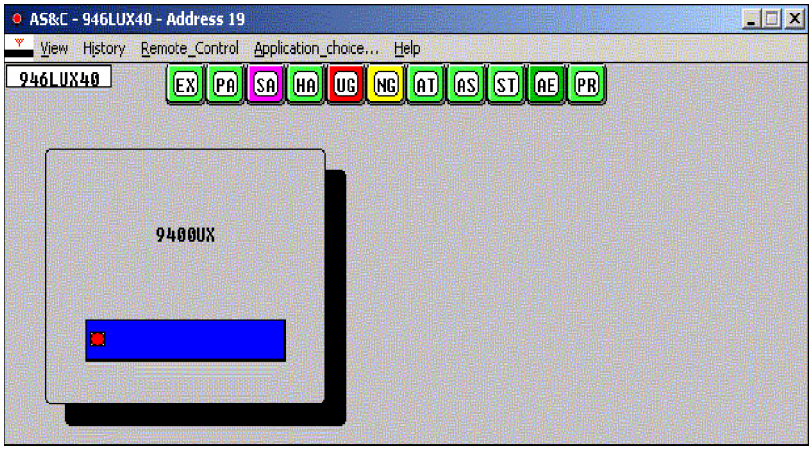
NE Type: identifies the type of NE

Server type: indicated by the button that is checked

4.7 – “Alarms, Status and Controls” application

4.7.1 - Opening the application

Path: Alarm synthesis → Application_Choice... → Alarms, Status & Controls



When you click the blue rectangle, the main screen is displayed.

4.7.2 - Main screen

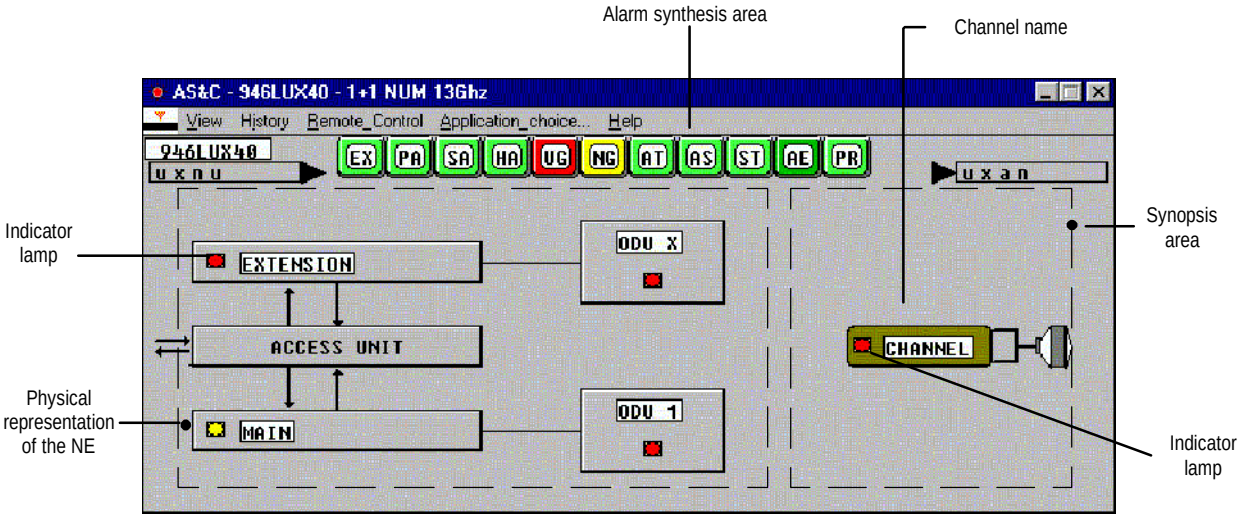


Figure 39 - Alarms, Status and Controls application screen for the 9400 UX (1+1 configuration with protection switching)

This screen contains all the dynamic information concerning the connected NE. It normally comprises three information areas:

- **a synthesis part**, in the top portion of the screen, containing:
  - information and alarm boxes supported by the NE,
  - the type of NE connected, the name of the station and of the far end station;
- **a functional part** displaying a synopsis representation of the equipment (in this case, on the right of the screen);
- **a physical representation** of the equipment in which you can locate “board” alarms or internal functions (in this case, on the left of the screen).

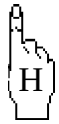
From this screen, you can obtain and view alarm status information on the various components of the connected NE.

In the case of an internal fault on a board or NE, the indicator lamp changes colour in the physical representation of the NE concerned. Presence of this symbol on a background other than green indicates that an alarm has been detected; the colour will depend on the severity of the alarm:

- **red background**: **Urgent** alarm,
- **yellow background**: **Non Urgent** alarm,
- **magenta background**: Status: Alarm that is **neither Urgent nor Non Urgent**,
- **green background**: **No alarm**, normal state or alarm disabled.

4.7.3 – Viewing alarm and synthesis information

To obtain the meaning of an alarm or status indication, simply position the cursor (appears with several different shapes) on the graphic representation of the alarm or state and click.

	General shape of the selection pointer. This cursor is moved by the mouse to the various graphic objects displayed on screen. Depending on the object to which it points, it can assume the shape of a hand or magnifying glass, as shown below. It can also be moved from one graphic object to another by the cursor control keys on the keyboard.
	The hand calls up an information bubble concerning the element to which it points, such as an alarm or configuration fault message. When the cursor is moved by the keyboard, the “Return” key toggles the information bubble on and off.
	The magnifying glass is used to activate the “Zoom” function on network element synopsis, to provide access to additional NE information. When the cursor is moved via the keyboard, the “Return” key activates the “Zoom” function on the equipment to which the cursor points.

4.7.4 – Accessing secondary NE screens

See screens in Figures 40 and 41.

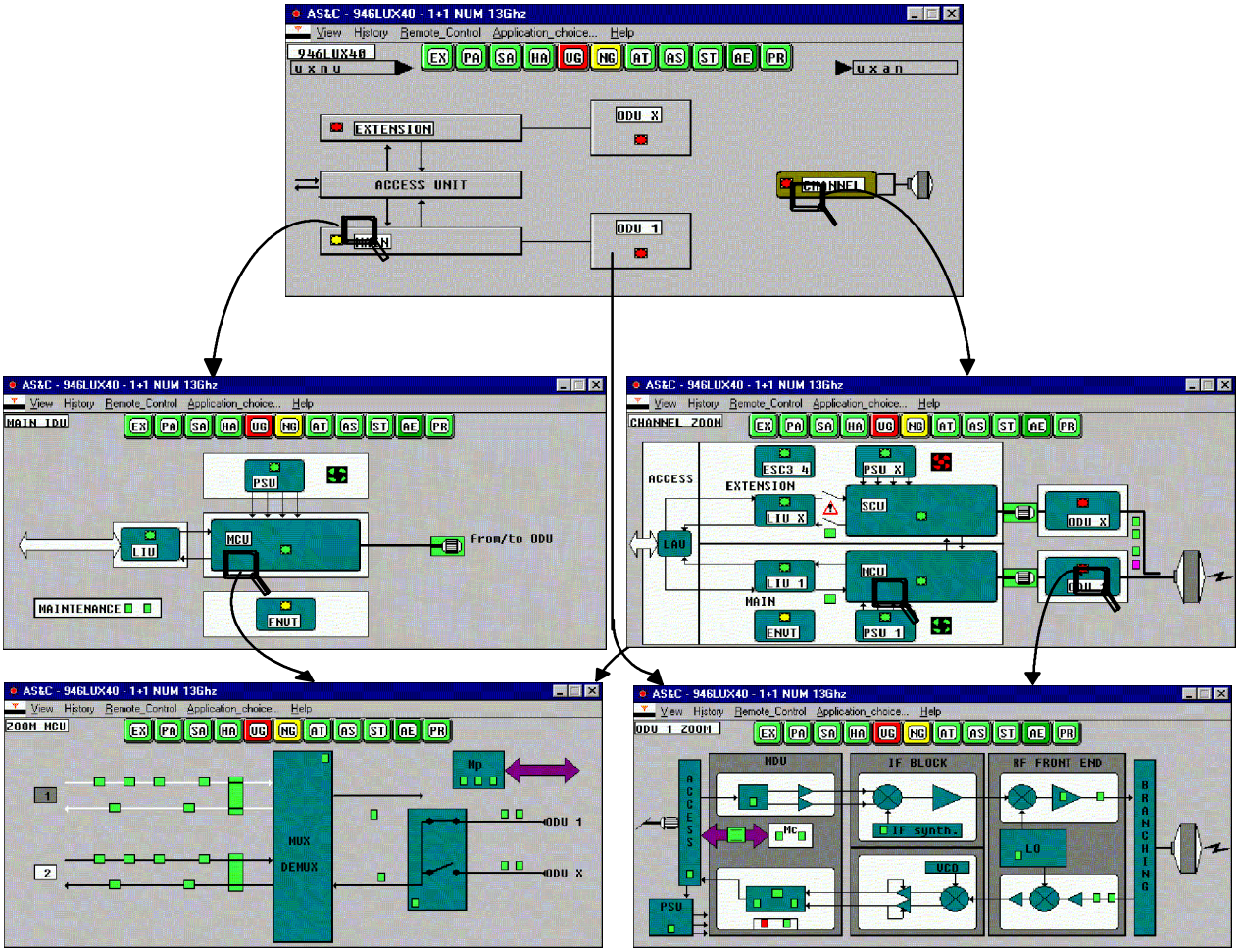


Figure 40 - Paths to the secondary screens (1+1 configuration with protection switching and 946LUX40)

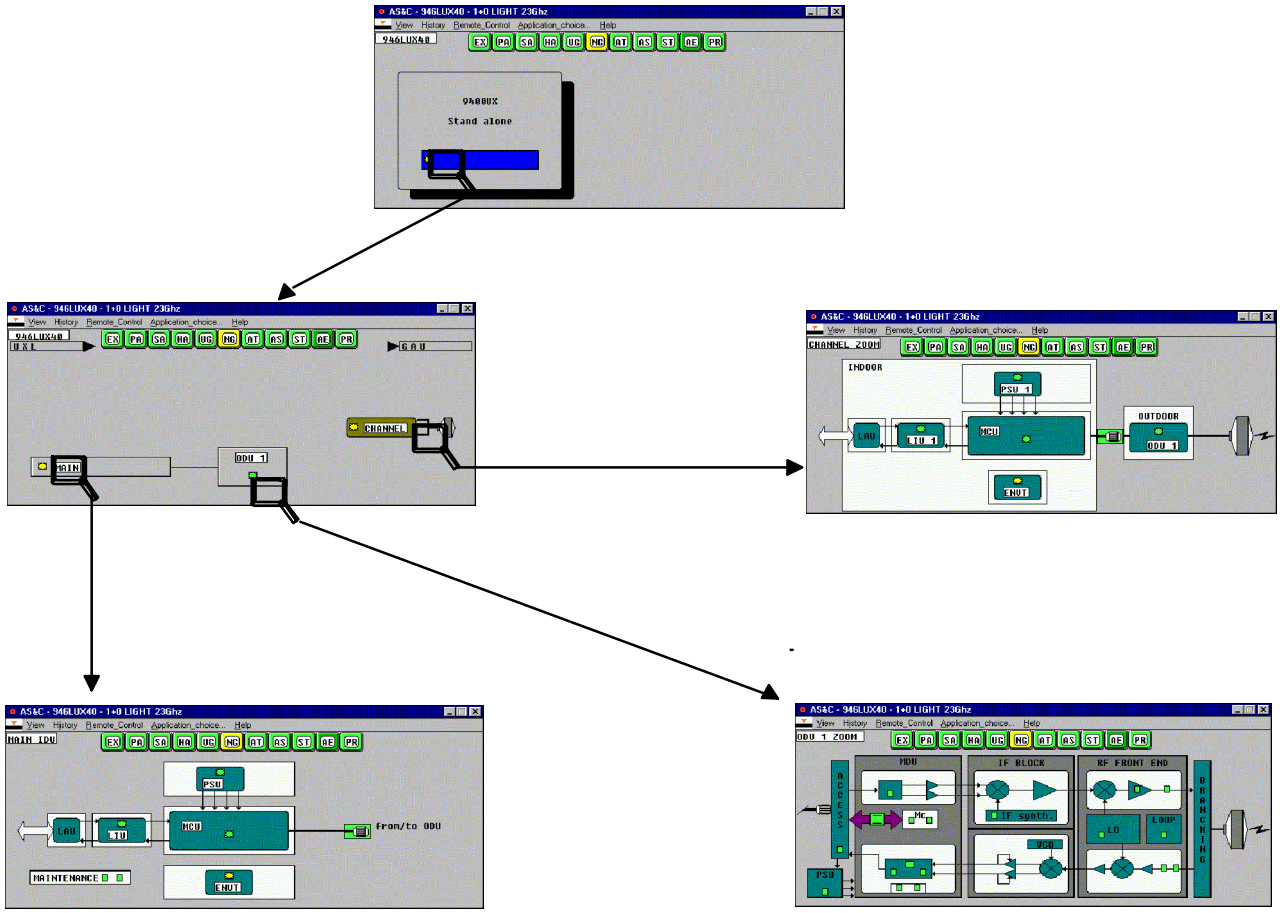


Figure 41 - Paths to the secondary screens (1+0 configuration, Light IDU versions and 946LUX40)

Example of access to the secondary screens by zooming in on the synopsis diagram:

In the functional part of the main screen, you can access various secondary screens detailing the components of the NE.

You zoom in on the components by moving the cursor **to the name of the element concerned** (white background). When the cursor changes from an arrow to a magnifying glass, click to obtain the secondary screen (in this case, in the top screen of Figure 41 or in the second screen of Figure 41, click "**CHANNEL**").

The secondary "**CHANNEL**" screen (middle right screen) shows the synopsis diagram of the channel and provides access to the names of its component subsystems and their associated alarms.

If you click the magnifying glass on the representation of the MCU, the bottom left screen appears, detailing the MCU.

Example of access to the secondary screens from the physical representation:

In the main screen, on the physical representation, click MAIN when the arrow changes to a magnifying glass. The synopsis diagram of the IDU then appears. In the same way, if you click the MCU, you obtain details of the MCU.

The "**View**" menu (see § 4.7.7) provides an alternative way of accessing the secondary screens.

4.7.5 - Exiting secondary screens

- Click "**View**" and "**First**" to return to the main screen,
- Click "**Previous**" or the right mouse button, or press "Escape" to return to the previous screen,
- Click the  button in the top right corner to close the application immediately.

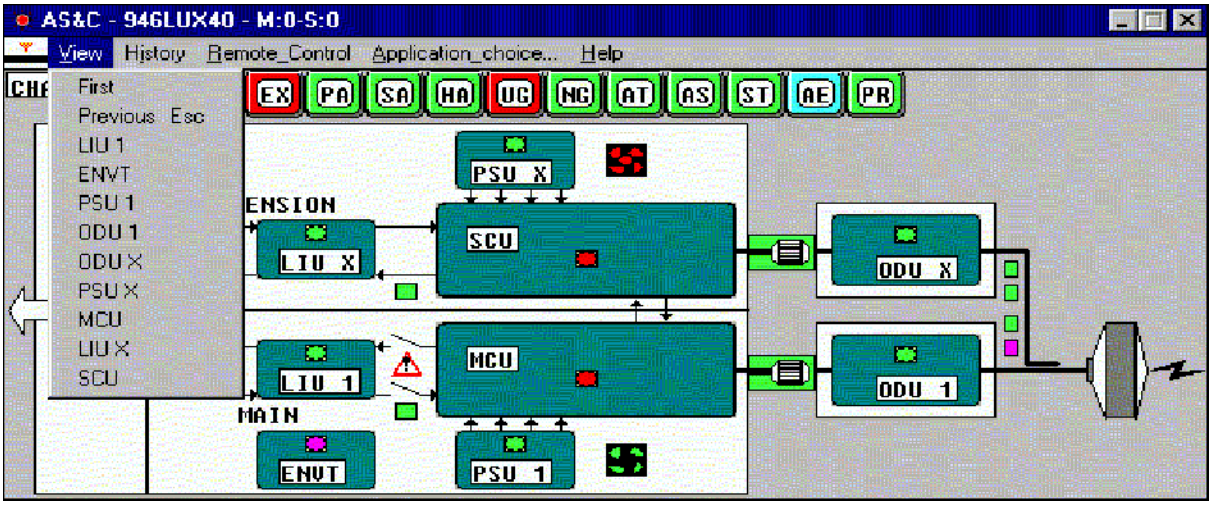
4.7.6 – Alarm synthesis line

The various acronyms in the information boxes that make up the alarm synthesis line are explained below:

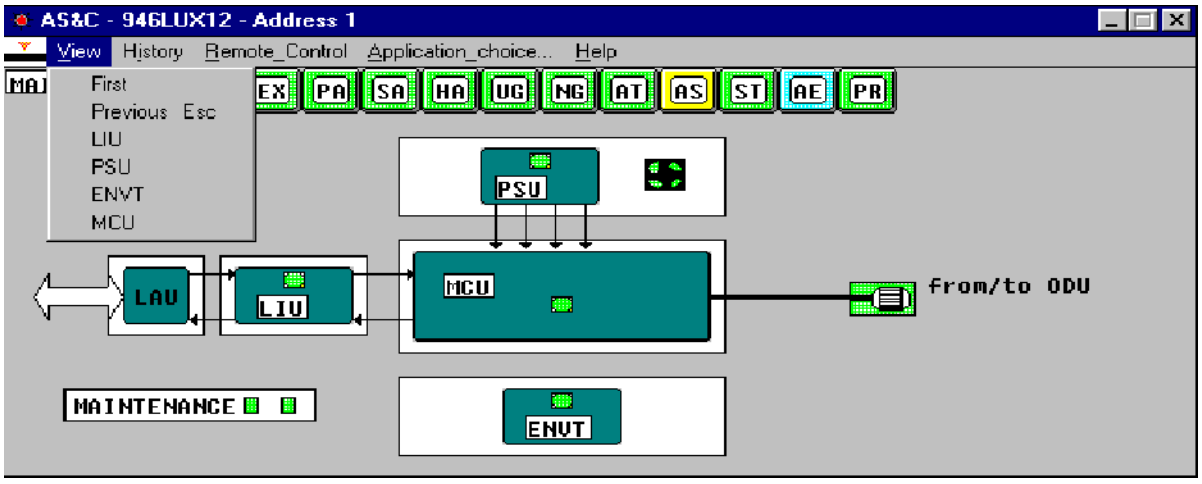
EX	PA	SA	HA	UG	NG	AT	AS	ST	AE	PR

MNEMO	COLOUR	MEANING	COMMENT
EX	Red	External Alarm	Synthesis of alarms associated with a signal failure or a signal fault on the tributaries or auxiliary channels
PA	Magenta	Propagation Activity	Received field below a threshold defined on configuration
SA	Magenta	Service Affected	Transmission problem requiring prompt attention
HA	Magenta	Housekeeping Alarm	Housekeeping, or environment alarm (one of the remote indication loops wired to the equipment is generating an alarm)
UG	Red	UrGent alarm	Alarm indication requiring prompt attention
NG	Yellow	Non urGent alarm	Alarm indication not requiring immediate attention
AT	Magenta	Alarm Attended	Alarm indication that has been acknowledged
AS	–	–	Not used
ST	Yellow	STatus	The equipment has received a remote control
AE	Green White if access not enabled	Access Enabled	Access to the remote controls (in AS&C) and configuration (CFG) applications is enabled by a command from a hierarchical level above the CT (OS)
PR	Red	Provisioning Refused	Appears only in the 946LUX12 and 946LUX40 versions: The software key is incompatible with the equipment. Check that the equipment configuration is consistent with those permitted by the key. If not, change the IDU (see § 6.7). Meaningless for the Light IDU versions. 946LUX11 version: See § 6.6.6.

4.7.7 – “View” menu



1+1 configuration with MUX protection and 946LUX40



1+0 configuration or Light IDU version and 946LUX12

The “**View**” menu calls up a list of the functions described in the synopsis diagram and containing additional information, as well as a means of accessing the first screen of the menu or the previous screen.

This information can also be obtained by moving and clicking the magnifying glass cursor, on the name of the block concerned.

4.7.8 – “History” menu

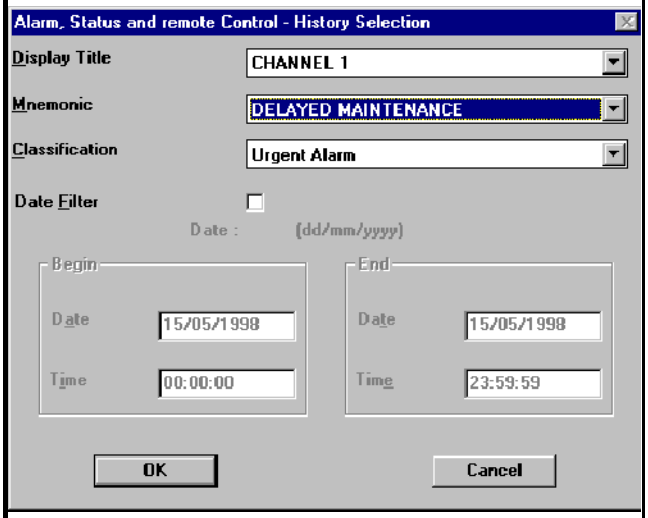
The “History”, or log, provides information on all the events that have occurred on the NE. It cannot be viewed until an event occurs. As long as no event has been recorded, the option remains greyed out and inaccessible.

The history comes into operation only when the application is opened. When the application is opened, the “**History**” menu shows the physical address of the NE and all alarms active on opening the application. It then displays events as they occur, with their date and timestamps.

When the application is closed, events occurring since the last time the log was consulted (in other words, the last time the application was closed for the NE) are logged. They will then be displayed when the application is reopened, with the date and time of reopening (an * character indicates that the date and time are those at the time of reopening and not the actual time the event occurred).

Up to 1000 events can be logged. Beyond this number, the most recent event overwrites the oldest event.





"Display..." and "Print..." commands

Select alarms to be displayed or printed out.

Display Title

Selects alarms for a single item. For example, alarms concerning just one function ("All" option: Selects all items).

Mnemonic

Selects alarms concerning a single mnemonic across all items ("All" option: Selects all mnemonics).

Classification

Selects alarms assigned the same criterion: Urgent, Non Urgent ("All the classifications" option: Selects without criteria).

Date Filter

When you check this box, selects alarms between a start date and time and an end date and time, to be entered in the appropriate boxes.

"Save..." command

Saves events affecting the NE, in chronological order. No selection options.

The events are logged, in text file form, in the al_memo.SN file, where SN is the logical number of the NE in the list. In the case of a slave key, the local equipment is number 0 and the remote equipment is number 1.

The file can be read using any Windows text editor such as NOTEPAD.

To reset the history, remove the al_memo1.SN and al_memo2.SN files from the c:\alcatel\l\luxxx directory

The "History" screen shown in Figure 42 contains the following information:

- the date when the application was opened,
- the events, arranged from most recent to oldest,
- for each event:
  - the time the event started,
  - the name of the item,
  - the alarm mnemonic,
  - urgency criterion (Urg, Nurg, ...),
  - start or end of alarm.

The scroll bar can be used to view alarms that have occurred but could not be fitted on the screen.

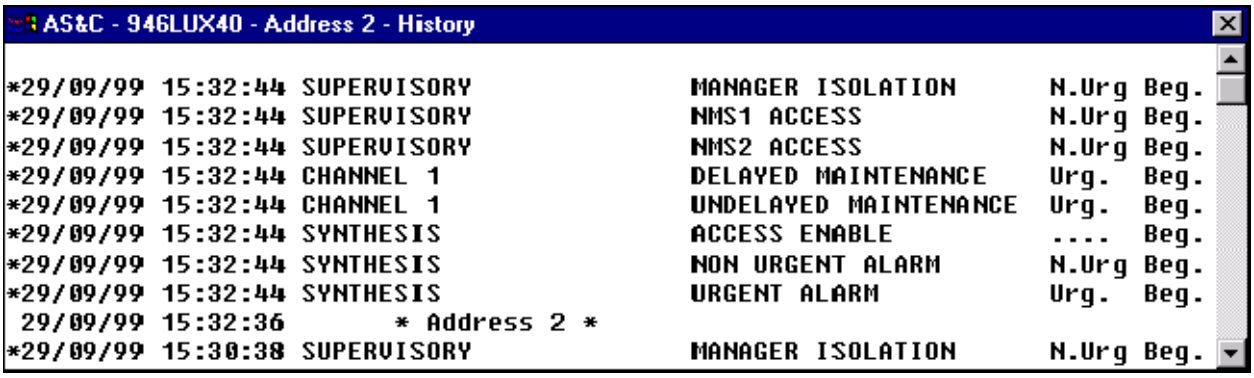
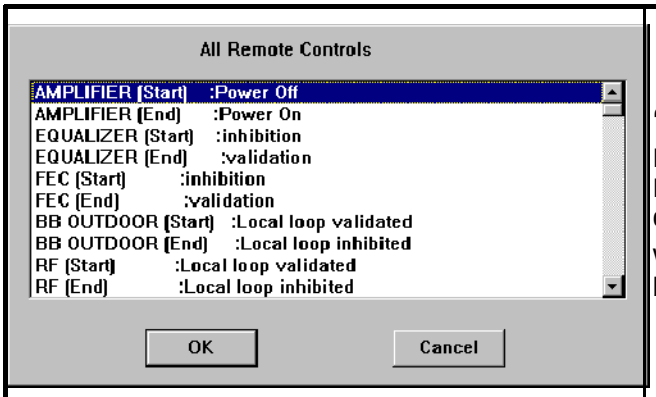


Figure 42 - View of the "History" log

4.7.9 – "Remote_Control" menu

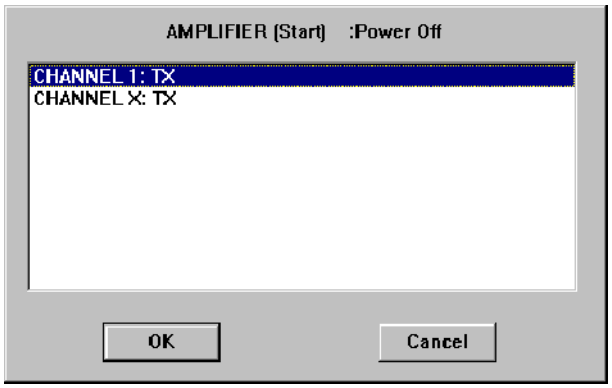




"All" command

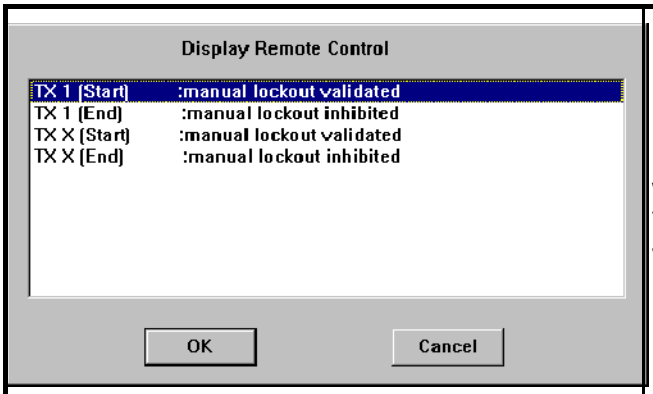
Lists all the remote controls that can be sent to the NE (the list of remote controls is given in Appendix 8).

When you select a remote control, the dialogue box below opens.



List of all items that can be affected by the remote control selected above.

When you confirm the remote control, an acknowledgement message is displayed.



“Current Display” command

List all the remote controls that can be associated with one (or more) items graphically represented on the current screen.

When you confirm the remote control, an acknowledgement message is displayed.



Remote control acknowledgement message.

Transmission of a remote control from the **CT** will be allowed only if the **AE** (Access Enable) mnemonic appears on a green background in the alarm synthesis line. Permission to send the remote control is subject to authorization from the central station (if there is one), or entry of a password, if one was introduced when the software was installed. Otherwise, the message “Remote control not allowed” appears.

Warning: In a 1+1 HSB configuration, a transmit power alarm appears when the remote control “RF local loop validated” is sent on the transmitter that is not active since this remote control is inhibited.

That alarm disappears when disconnecting the cable between the IDU and the ODU.

Warning: When a local loop of the equipment is done, with an interface with analogic cables, the supervision link of the network does not work.

4.8 – "Performance Monitoring (G821)" application

Not applicable for 946LUX40.

The "**Performance monitoring (G821)**" application lets you view information contained in the performance counters. These counters record:

- Errored seconds (ES): one-second intervals that include at least one error,
- Severely errored seconds (SES): one-second intervals with a bit error ratio greater than 1.10^{-3}.
- US (unavailable seconds): period beginning with the first ten consecutive SES and ending at the start of ten periods with no consecutive SESs.

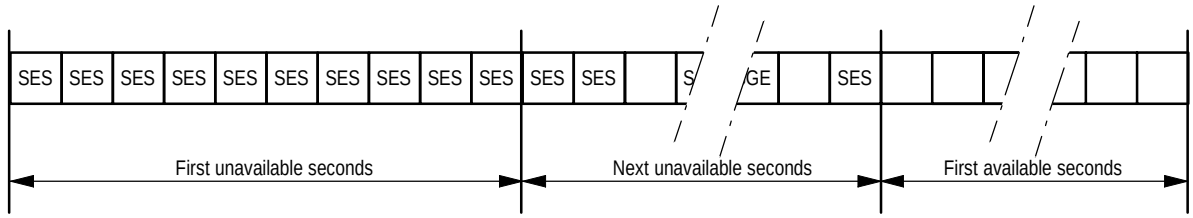
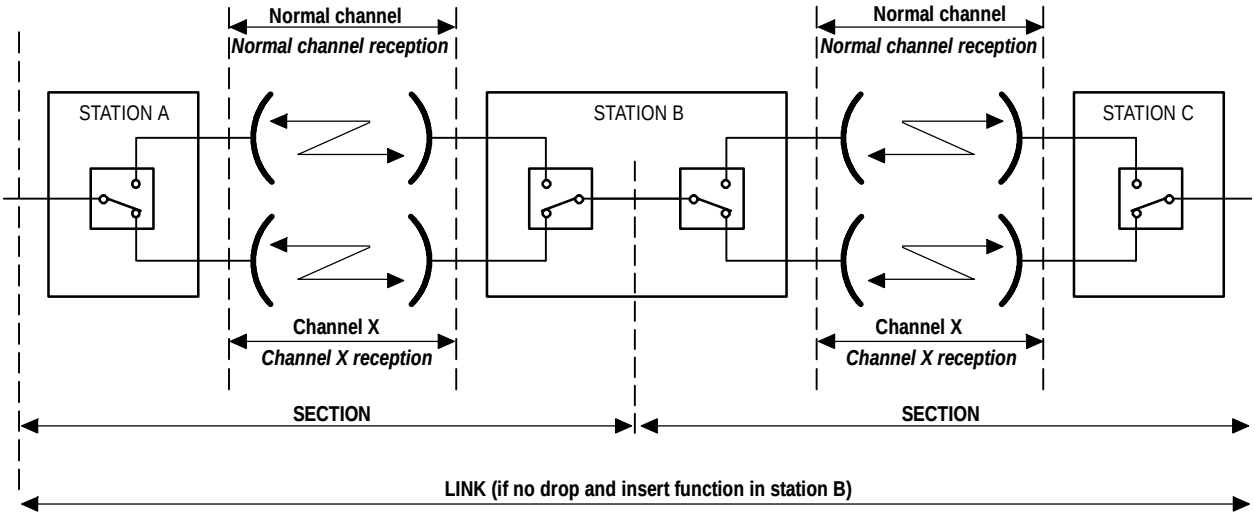


Figure 43 - Example of unavailable periods

These values are defined for a link, a section, a channel, the G.821 definition of which is given in Figure 44.



Note: In 9400 LUX 1+1 (classic IDU), the repeater stations do not exist: LINK= SECTION
In 9400 LUX 1+0 (classic IDU and Light IDU) the repeater stations and "standby" channel do not exist, so LINK = SECTION = CHANNEL

Figure 44 - Elements of a link (in G.821)

When the application is opened, the date of opening and the address of the connected network element are displayed, with the values accumulated in the counters since the application was last closed. These counters are reset when the application is next closed.

This information is generated and sent by the network elements, and can be displayed in G821 format.

Path: Alarm Synthesis → Application_Choice... → Performance Monitoring G821

When the application is started up, the first screen gives all the monitoring information on the connected network element. By default, all the measures are displayed every 60 seconds as shown in Figure 45. The title of the screen identifies the NE concerned.

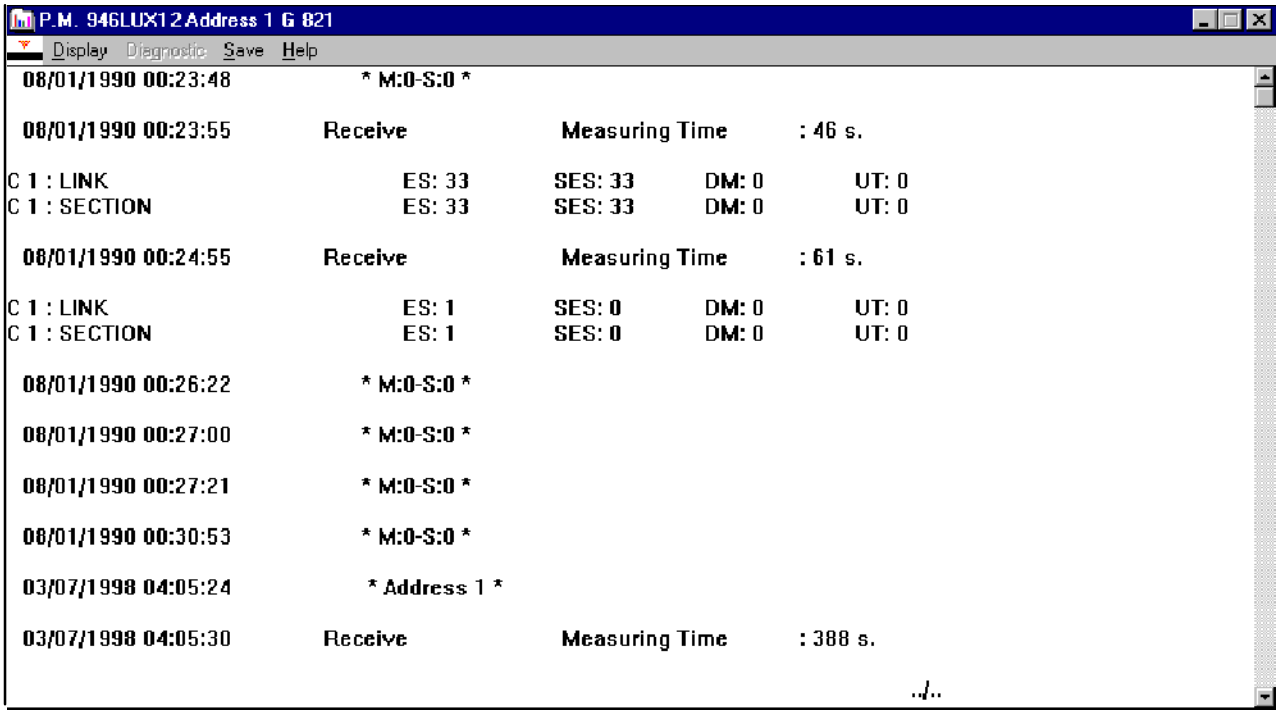


Figure 45 - First performance monitoring (G821) screen

In case there is no event error, no measurement is displayed.

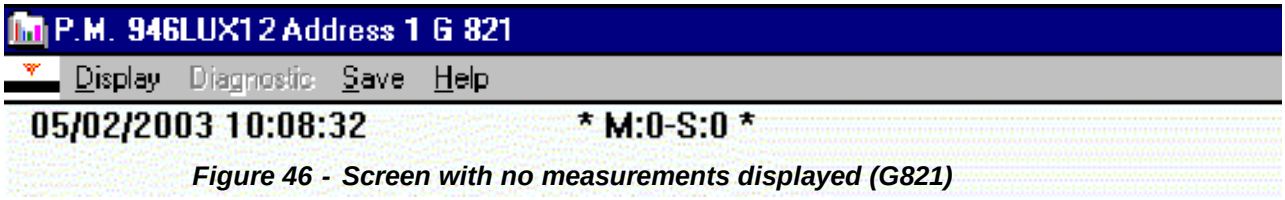


Figure 46 - Screen with no measurements displayed (G821)

4.8.1 – "Display" menu

The "Display" menu lets you view the performance measurements recorded by the equipment.

This information is displayed in chronological order, as shown in Figure 45.

A scroll bar lets you view all the events logged.

VERTICAL SCROLLING

Vertical scrolling can be controlled by the mouse or by the keyboard (using the up and down arrows). To scroll through one event at a time, click the arrow boxes on the scroll bar.

The symbol ../.. in the bottom right corner of the page indicates that the display of the latest information received is incomplete; scroll the table down to view the next block of information.

G821 PERFORMANCE MEASUREMENTS

the information displayed includes:

- the date and time of start of measurement,
- the type of counter: Receive,
- the measurement duration in seconds,

- for each entity (section, link, channel):
  - the identity of the entity,
  - an indication of loss of data if necessary,
  - the readings of the counters attached to the entity.

If there are no errors, no corresponding display appears on screen.

The display is refreshed at 60-second intervals, and updated to include all new errors that might have occurred.

4.8.2 - "Save" menu

The **Save** menu lets you save performance monitoring information to disk (**File**) or to printer (**Print**).

"FILE" option

This option lets you save the performance monitoring information supplied by the network element to disk, in the network element directory and in the file named PM_MEMO.sn (sn=secondary number). This ASCII file can be viewed by the operator and analyzed outside of the application using a package like EXCEL, for instance. It can contain a maximum of 2000 events. To reset the performance monitoring information, remove the pm_memo1.sn and pm_diag2.ads from the c:\alcatel\luxxx

"PRINT" option

With this option, all the performance monitoring information can be printed out in chronological order.

4.9 – “Radio Transmission Parameters” application

The **Radio Transmission Parameters** (**RTP**) application (Figure 47), lets you:

- monitor the trends of the main transmission parameters (transmitted power, received power, bit error ratio),
- run an initial diagnostic on transmit and receive alarms.

No more than two RTP applications can be opened simultaneously, on two different stations.

Path: Alarm Synthesis → Application_Choice... → Radio Transmission Parameters

The polling interval of the RTP application can be set via the POLRTP variable in the Win.ini file, as explained in Appendix 11.

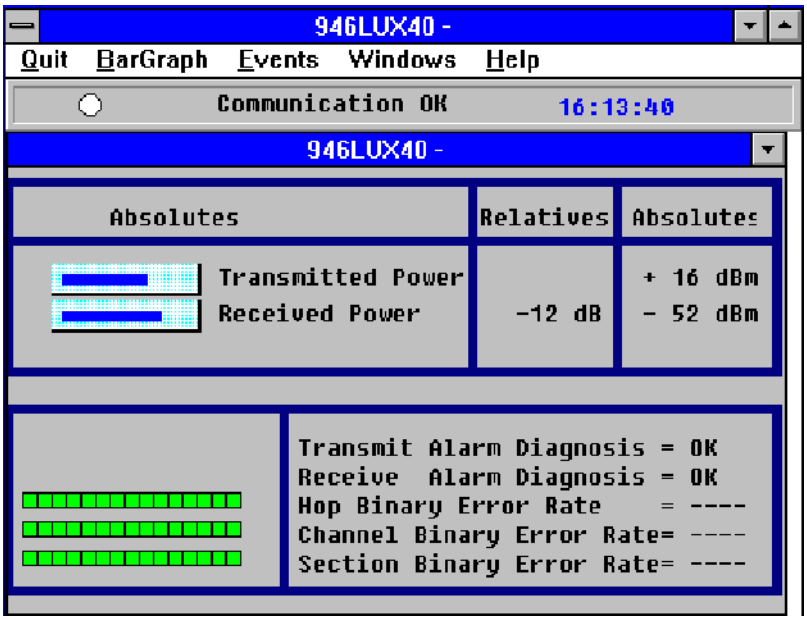


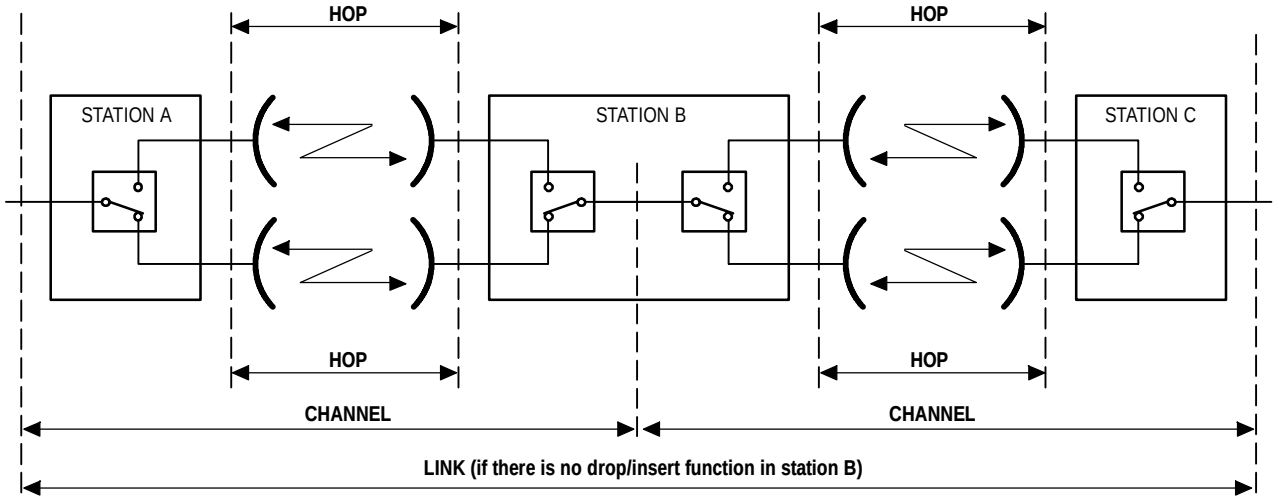
Figure 47 - Radio Transmission Parameters (RTP) screen

FIELD	FUNCTION
Communication	Indicates the state of communication with the remote NE: – “**Communication OK**”, the remote NE is contacted at intervals of less than three seconds, – “**No response**”, communication with the remote NE has been lost for more than three seconds; a string of “????” characters appears in place of the data. – Flashing green signal if the remote NE is contacted. – White signal if communication with the remote NE is lost. The right end of this information line shows the current time.
Transmitted Power	Shows the transmitted power level as an absolute value. In an HSB configuration, the value displayed is the transmitted power set point value for the standby transmitter.
Received Power	Shows the received power level, as an absolute value (dBm) and relative to the nominal threshold defined in the configuration.

FIELD	FUNCTION
Alarm Diagnosis	Shows the alarm furthest back up the transmission chain for transmission, reception or both directions in the case of a repeater station. **Click once** on the alarm mnemonic to see its meaning, **Click twice** and the list of mnemonics with their meanings is displayed.
Binary Error Rate	Shows the bit error ratio (BER): **Hop, Channel and Section BER** information. Reading the bargraph: Each division of the bargraph that becomes red represents a BER value as shown below. 5.10^{-8} | 1.10^{-7} | 2.10^{-7} | 5.10^{-7} | 1.10^{-6} | 2.10^{-6} | 5.10^{-6} | 1.10^{-5} | 2.10^{-5} | 5.10^{-5} | 1.10^{-4} | 2.10^{-4} | 5.10^{-4} | 1.10^{-3} | cut-off

Nota : The names of Hop, Channel and Link are different from the ones used in G821 et G784.

Figure 48 is a visual representation of the concepts of channel, hop and section in RTP.



Note: In the 9400 UX, the repeater stations do not exist so SECTION = CHANNEL

Figure 48 - Elements that make up a link (in RTP)

COMMANDS IN THE RTP SCREEN

The commands in the top part of the **RTP** screen are accessed by clicking once. The functions of these commands are described in the table below.

COMMAND	FUNCTION
Quit	Closes the RTP application
BarGraph	Activates or deactivates graphic display: – of the bit error ratio, – of transmitted power, – of received power. A display "without bargraph" can be selected to view two RTP windows representing different network elements side by side.
Events	Used, when the application is open, to display a list of the latest events (80 events for a terminal), as explained in the rest of this table.
Windows	Three RTP application representation and layout options: – icon form, – screens cascaded from the top left corner of the screen, – different coloured screens. **Caution**: the RTP window is positioned in the foreground of the NECTAS window (network display) when using the NECTAS window. To access the RTP screens: – use the ALT TAB key combination, or – double click the background to call up the "task list" box and then double click RTP.

When you select the "**Events**" command in the RTP screen, the "**Events List 946LUXxx**" screen shown below appears.

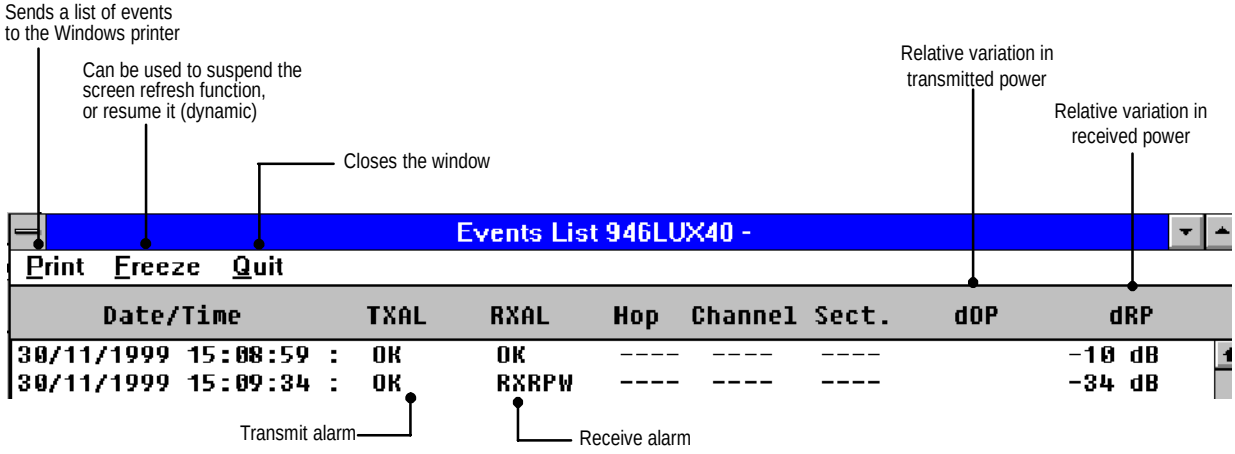


Figure 49 - List of most recent events

The meanings of the alarm messages are given in Appendix 9.

The relative power levels represent variations relative to values defined on configuration.

The menu bar contains the following menus:

- "Freeze" when the display refresh function is enabled,
- "Dynamic" when the display refresh function is fixed.

Application available only with the extended 946LUX11/946LUX12 software and the standard /946LUX40 software version.

4.10 – "Remote inventory" application

This application can be used by an operator or NMC manager to access inventory information concerning the selected network element. The information is designed to assist in the administration of the network elements as performed by the customer's maintenance and repair centres.

The inventory data can be accessed from a craft terminal or from a remote management device. The data is displayed at the request of the operator and can be printed out.

All the data contained in backup **.rur** files provides for complete identification of a replaceable unit (generic name RU), and can be used to facilitate management of the equipment by installation and maintenance teams.

The various data fields shown by the application contain the following information:

- identity of Alcatel CIT or subsidiaries (ACIT, ASEL, ASES, ATEL),
- the mnemonic representing the type of network element (see list in Appendix 12),
- the reference of the replaceable unit (for example 3CC05714AAAA) and its revision index,
- reference of the resident software installed (for example 3CC08538AAAA) and its revision index,
- the CLEI code (Bellcore specification), on up to ten characters,
- the place of manufacture (four characters) (reserved for ALCATEL),
- serial number,
- date of manufacture (YYMMDD format).

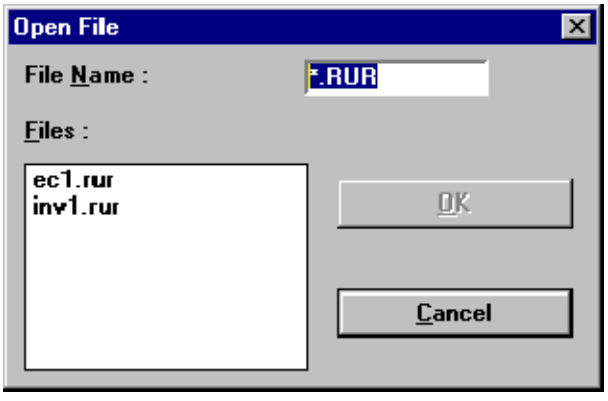
The information supplied can be used by the operator to identify a unit remotely for management and administration purposes.

All the inventory data is loaded before delivery to the customer, during the factory tests and/or inspections. This cannot be modified by the operator.

Path: Alarm Synthesis → Application_Choice... → Remote Inventory

4.10.1 - "File" menu



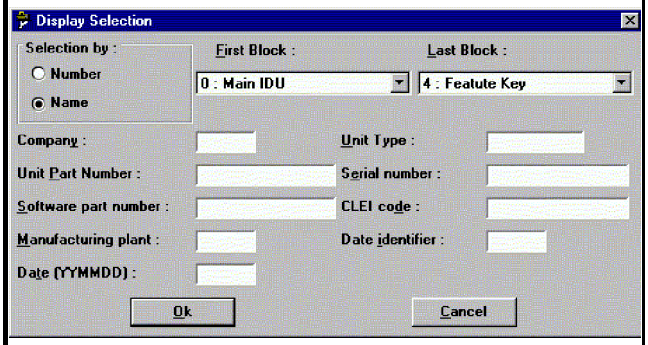
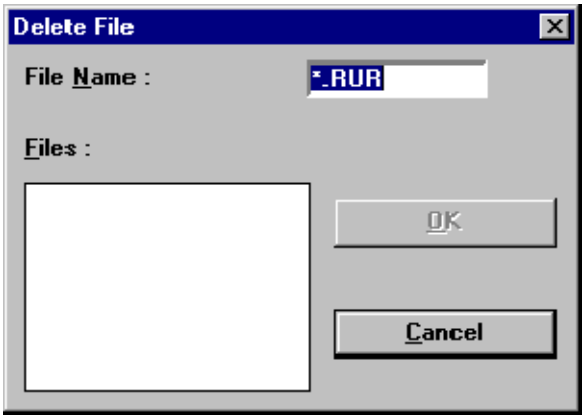


Open...

For reading and displaying, with a selection filter, the NE data previously backed up on disk (file name with the .rur extension). The selection filter (see below) appears when you confirm the selected file.

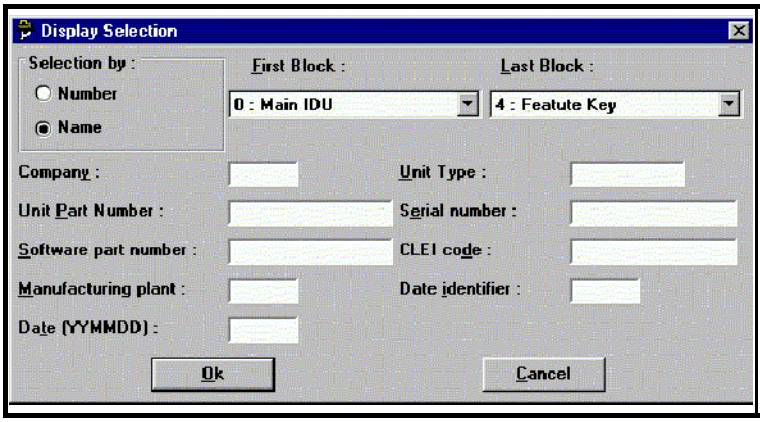
The events previously displayed are lost.

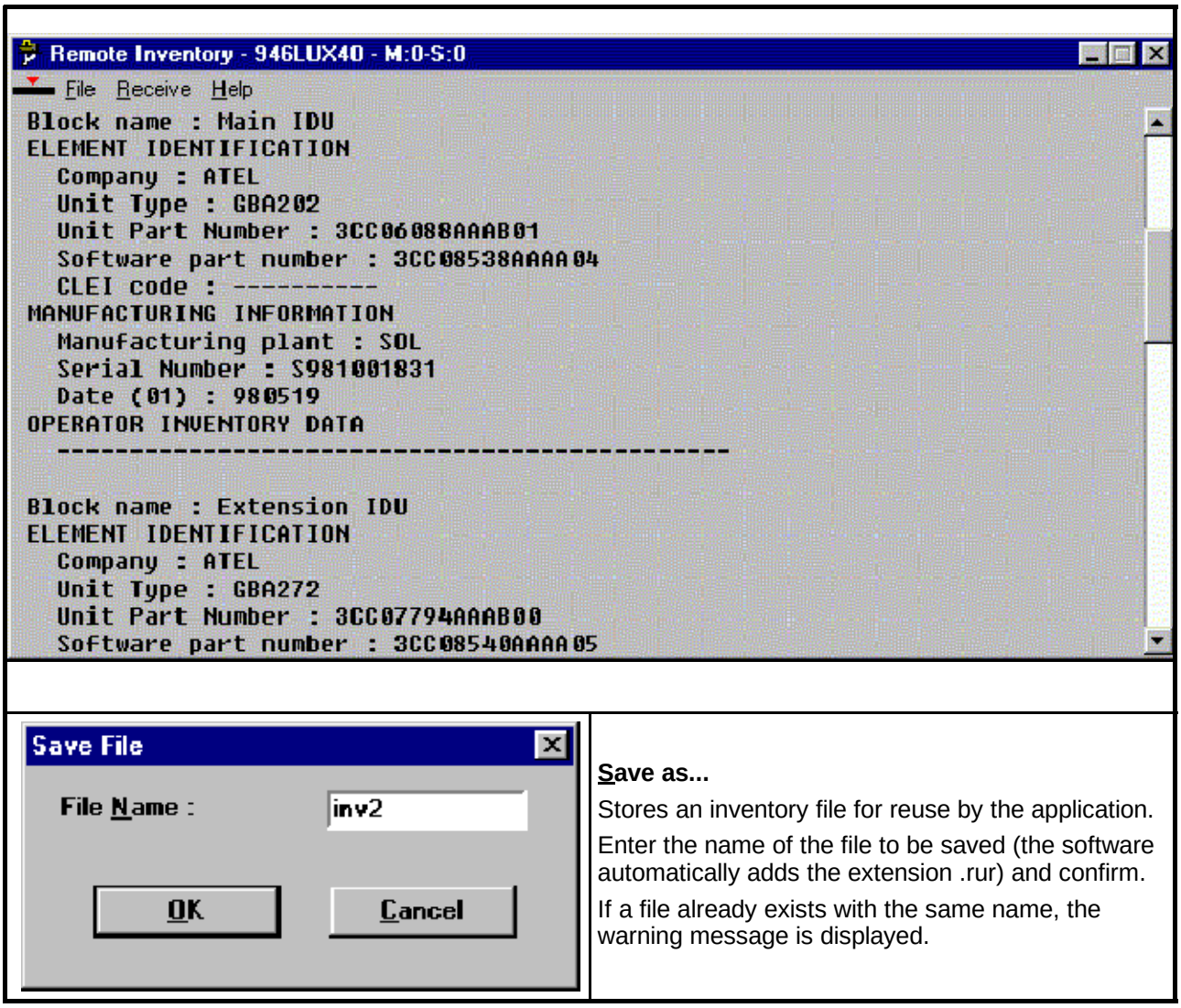
The display takes the form shown in § 4.10.2.

	Filter for selecting data to be displayed.
	**Delete...** Deletes a backup file (with the extension .rur) chosen from the list. A prompt for confirmation is displayed.
	Print Prints out the data in the .rur file selected from the list.

4.10.2 - "Receive" menu



	**Display...** To read and display, according to the selection filter opposite, data contained in the NE. When you confirm, the screen below appears (in this case, no selection filter was specified).



Save as...

Stores an inventory file for reuse by the application.

Enter the name of the file to be saved (the software automatically adds the extension .rur) and confirm.

If a file already exists with the same name, the warning message is displayed.

Application available only with the extended 946LUX11/946LUX12 software and the standard /946LUX40 software version.

4.11 – "Maintenance Memory" application

The "Maintenance Memory" application can be used to store the appearances and disappearances of events occurring when the link between the operating system and the NE is cut, or when the CT is not connected.

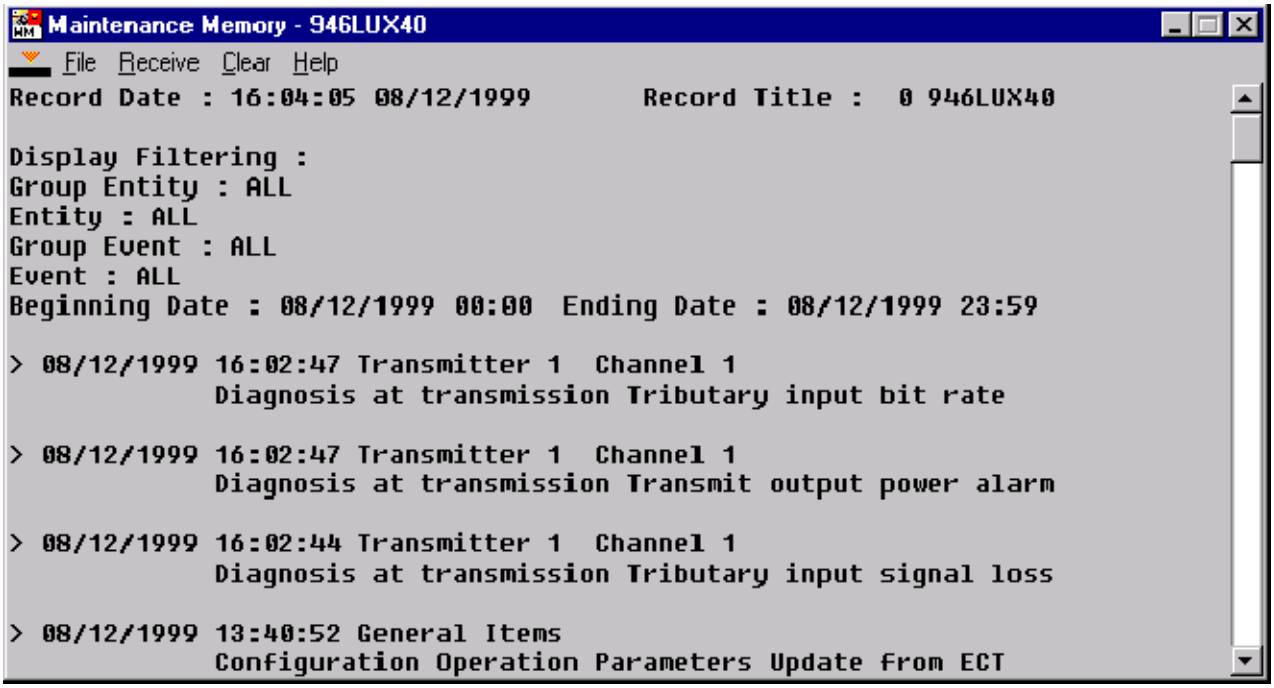
The NE supports an event memory for each physical or logical entity. This memory will store 256 radio events or 40 miscellaneous management–related events.

Reading the memory does not delete it, but each NE can reset the content of the memory. Access is protected by the access rights mechanism.

The maintenance memory is read at the request of the operator.

You can delete the content of the maintenance memory by a specific command.

Path: Alarm Synthesis → Application_Choice... → Event Memory



To be pertinent, before the saving, the events list requires a putting on time command if it is not automatic.

Meaning of the signs:

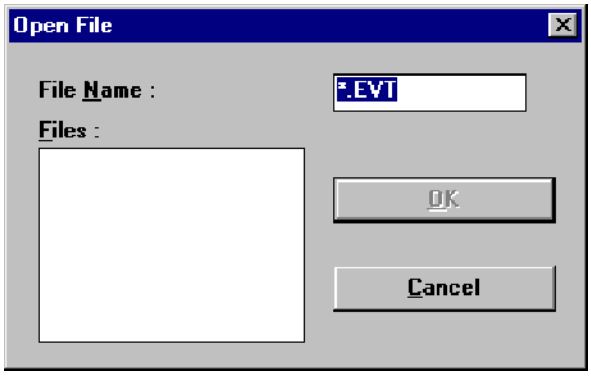
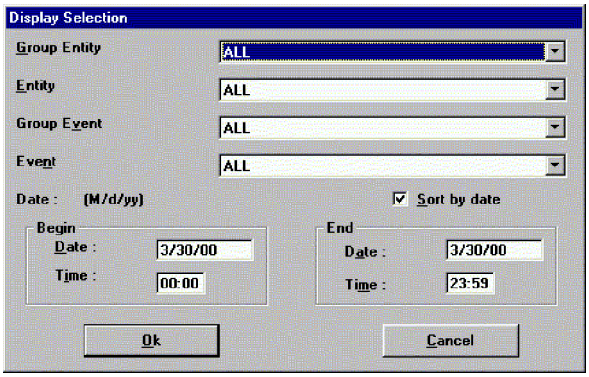
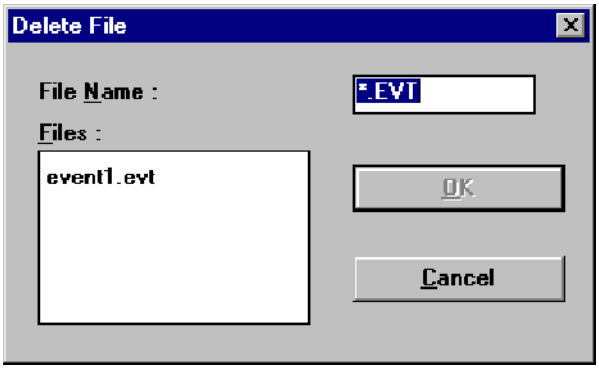
> beginning of an event

< end of an event

? non significant date

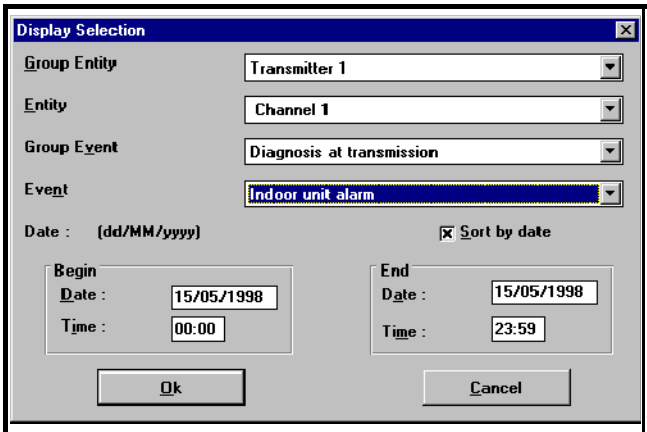
4.11.1 – "File" menu7



	**Open...** To read and display, with a selection filter, network element data previously saved on the disk (file with the extension .evt). The selection filter (see below) is displayed when you confirm the file selected. The data previously displayed is lost.
	**Display...** Filter for selecting the events to be displayed.
	**Delete...** To delete a backup file (with the extension .evt), chosen from the list. A prompt for confirmation is displayed.
	Print...To print out data from the .evt file selected from the open list.

4.11.2 – "Receive" menu



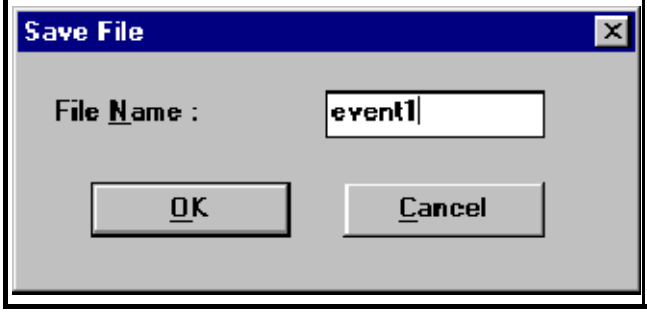


Display...

To read and display, according to the selection filter opposite, events contained in maintenance memory.

Sort by date:

- checked: events are displayed in chronological order
- box not checked: events are displayed by groups of entities.



Save as ...

To save an event file that can be reused by the application.

Enter the name of the file to be saved (with the extension .evt) and confirm.

If a file already exists with the same name, a warning message is displayed.

4.11.3 – "Clear" menu



Clears the maintenance memory. A message is displayed prompting you to confirm the operation.

Application available only with the extended 946LUX11/946LUX12 software and the standard 946LUX40 software version.

4.12 – "G784 Performance Monitoring" application

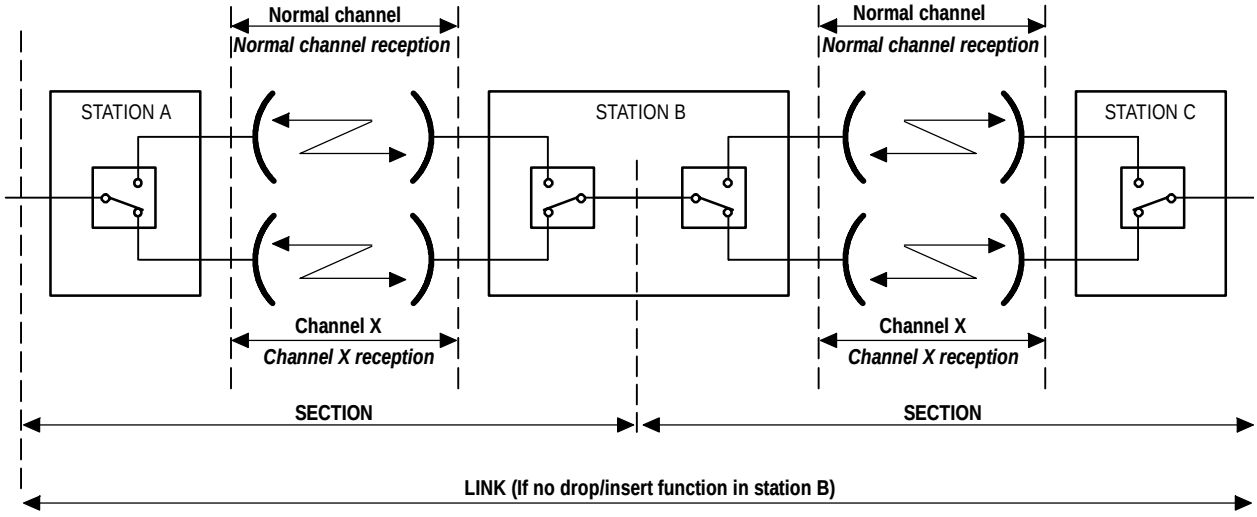
This application can be used to analyze the performance of the Terminal Points (TP) supported by a network element in accordance with ITU-T Recommendation G784 concerning the SDH equipment supervision.

Nota 1: The events analyzed by the G.784 application are composed of:
error rate measurements
number of alarm threshold overtaking
number of radio circuits switching
In the document, the word "error"covers also the "event" notion.

Nota 2 : The G.784 recommendation analyzes the performances of the circuit established between two Terminal Points at the traffic transportation level (digital trains).
The application of this concept is extended to the radio circuits level to enable the user to control the operational running of the link.
Consequently, some counters indicate events that do not concern the G826 analysis that takes care of the supervision of the traffic quality such as the switching number at reception (the traffic is protected by the hitless switching function).

The TPs are divided into various classes. Each TP class requires an appropriate error count format.

Certain TPs correspond to the sections and links defined, in G.784 terms, as shown in Figure 50.



Note: In the 9400 UX 1+1 (classic IDU), the repeater stations do not exist so: LINK = SECTION
In the 9400 UX 1+0 configuration (classic, Light IDU), the repeater stations and "Standby" channel do not exist, so LINK = SECTION = CHANNEL

Figure 50 - Items that make up a link (in G.784 terms)

Users of the G784 application can transfer events detected (all kinds or those selected by sort criteria) either continuously (AUTOMATICALLY), or upon request (MANUALLY).

In automatic mode, the measurement results are counted over two accumulation periods, every quarter hour or once a day; in manual mode, the measurements are run at the request of the operator.

Path: Alarm Synthesis → Application_Choice... → Performance Monitoring (G784)

When you start up the application, the G784 window (Figure 51) opens. Display defaults to the "**manual**" and "**day**" modes.

The selections made in the "**View**" and "**Mode**" menus are summarized in the status bar along the bottom of the application's window.

All the events display boxes assigned to a TP and a counting period (24 hours or 15 minutes) appear pale blue.

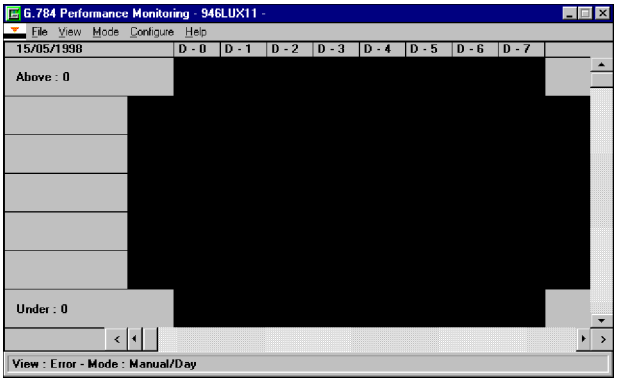
The application remains in manual mode until a display request is sent.

The “**Above**” field:

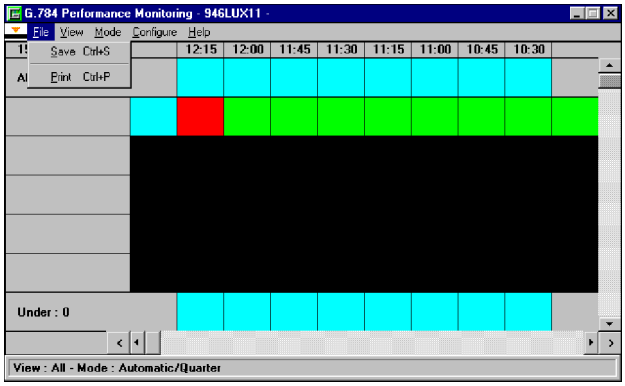
- shows the number of TPs not shown in the top part of the window.

The “**Under**” field:

- gives the number of TPs not shown in the bottom part of the window.



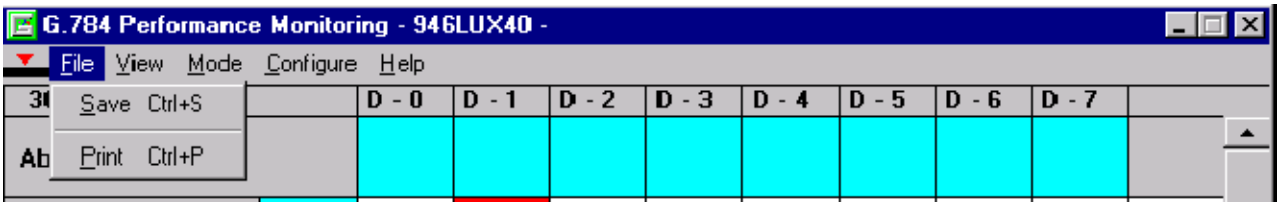
“Day” mode/No TP started up



“Quarter hour” mode/TPs started up

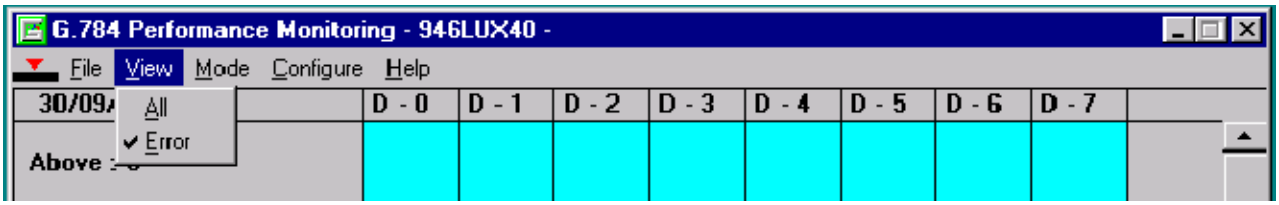
Figure 51 - G784 window

4.12.1 - “File” menu



	**Save** Saves the content of the error counters, in ASCII format, in the NE’s directory. Enter the name of the file to be saved (with the extension .cnt) and confirm. This file can then be used only in a word processing or spreadsheet application, and not by the application itself. This file can be imported then in a spreadsheet (like Excel) but it cannot be managed by the application itself.
	Print Prints out the state of the counters.

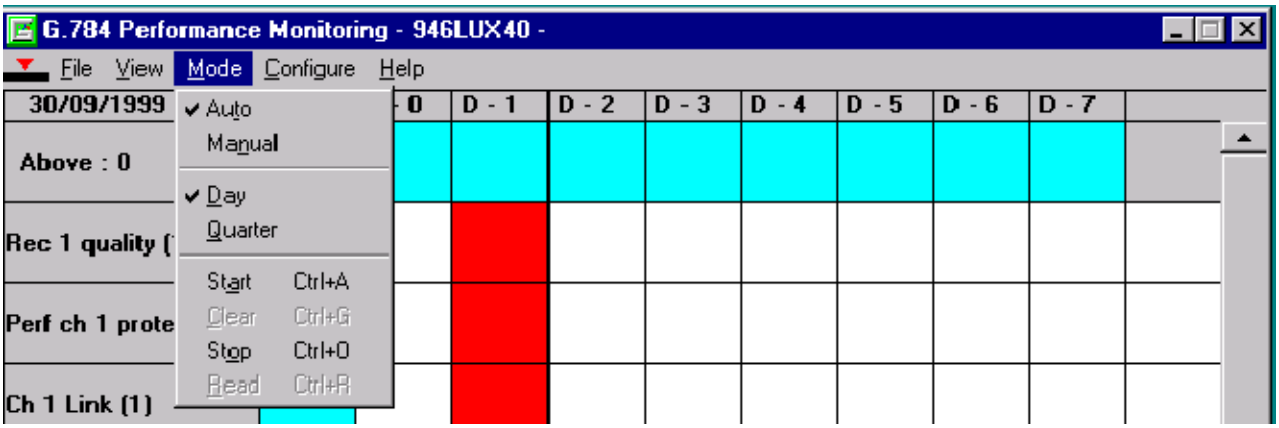
4.12.2 – "View" menu



Selects the terminal points (TP) being monitored for display on screen (all or just those affected by errors).

The display is refreshed automatically at the end of each accumulation period.

4.12.3 – "Mode" menu



MODE	MEANING
Auto	Automatically transfers the content of the TP error counters at the start of each accumulation period. In this mode, the "Clear" and "Read" commands are inaccessible.
Manual	Default mode when the application is started up. The error counters are transferred at the request of the operator using the "Read" command.
Day	For selecting accumulation periods: S Day: the error count begins each day at midnight. S Quarter: the error count begins at the 0, 15, 30 and 45 minute points of each hour. This option adjusts the time scale in the application's main screen (one column for each day or for each quarter hour).
Quarter	The content of the counters in the time periods leading up the opening of the application can be viewed (19 quarter hour counters and 9 day counters at most) by clicking "Read". The list of TPs started up in quarter hour mode may differ from that of TPs started up in day mode.

After a manual or automatic read, an error indication is displayed in a box representing the "**Terminal point - Time**" pairs of values, for each terminal point monitored (TP) and for each accumulation period.

Each individual box of the "**Terminal point - Time**" pair is assigned a colour representing the value of the error. The following table explains the colour code used.

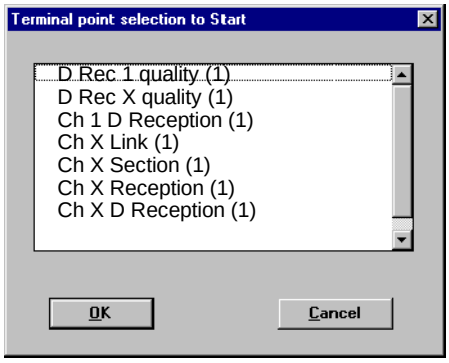
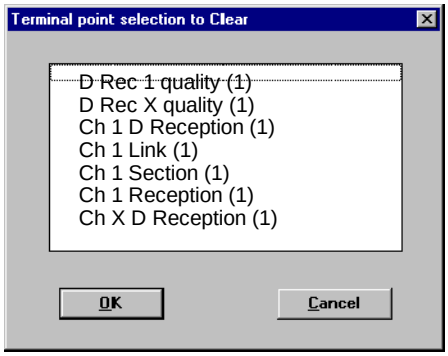
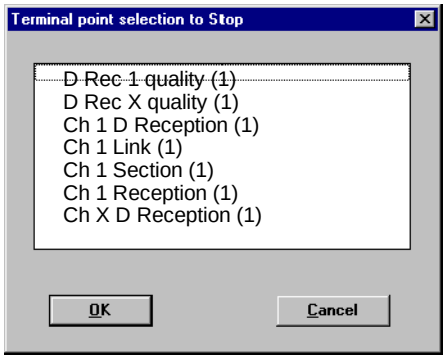
PRIORITY	MEASUREMENT RESULT	COLOUR
Low	No measurement	Pale blue
	No error	Green
	Loss of data	White
	Value below lower criterion threshold	Red
	Value below higher criterion threshold	Yellow
High	Value above higher criterion threshold	Red

For each TP displayed in the window, an error summary appears from left to right in the corresponding line.

You can browse vertically (to select TPs) and horizontally (through days or quarter hours) using the scroll bars.

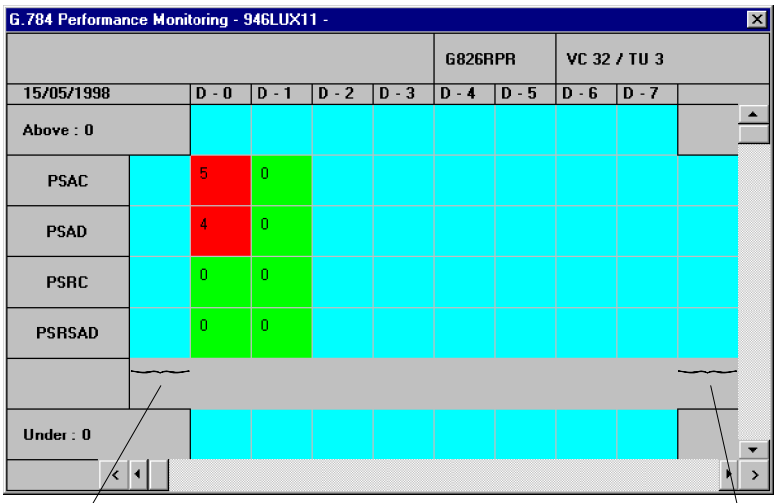
Notes

- For all the TP not displayed in the window, an error summary at the top and bottom of the area shown on screen appears on the corresponding line.
- The colour coding of the events counter summaries are the same as those in the table above.
- There is also a summary on the left and on the right of the area displayed.

	**Start** Starts display of the error counters for the TP. Select the TP or TPs to be started from the list (TPs not yet started) and confirm.
	**Clear** *Accessible only in manual mode* For the TPs selected, clears all the error counts stored since the application was started up from NE memory. A message prompting for confirmation is displayed.
	**Stop** Selects the TPs for which the operator wants to stop viewing the error counters. The list of TPs displayed contains only those currently started. **Read** *Accessible only in manual mode* Collects the latest totals recorded by the NE, and updates the data displayed (in automatic mode, the data is uploaded at the end of each accumulation period).

Details of TP counters

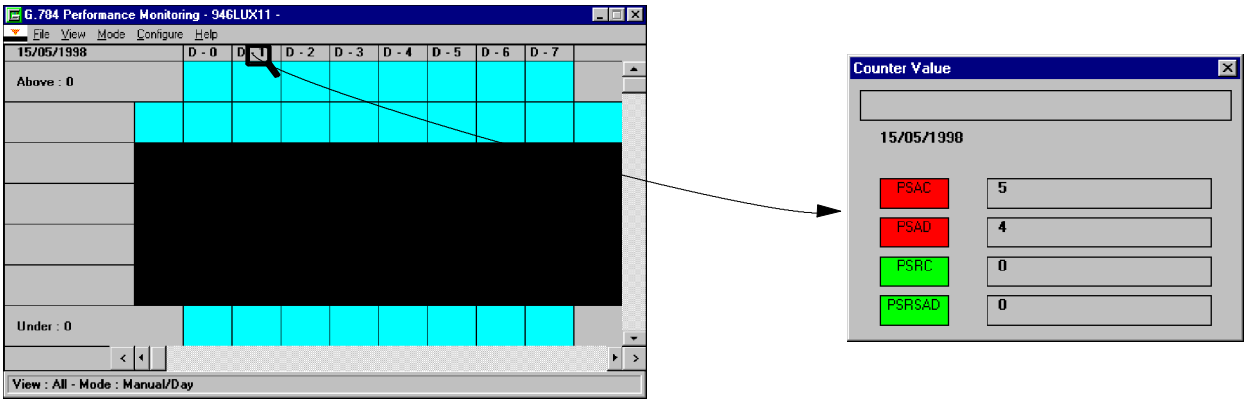
When you click a TP title in the lefthand column, the screen zooms in on detail of the events counters for that TP. The readings of the counters appear in the boxes representing each date:



Column summarizing the most serious faults after the time period displayed

Column summarizing the most serious faults prior to the time period displayed

When you position the cursor on a date in the window, the magnifying glass appears so that you can zoom in on the counter readings for that date:



The counter readings appear as follows:

- values above 1 000 000 in the form of N 10^m,
- colours according to the results of comparison with the thresholds.

The error counters displayed depend on the TP selected.

“Receive quality” TP

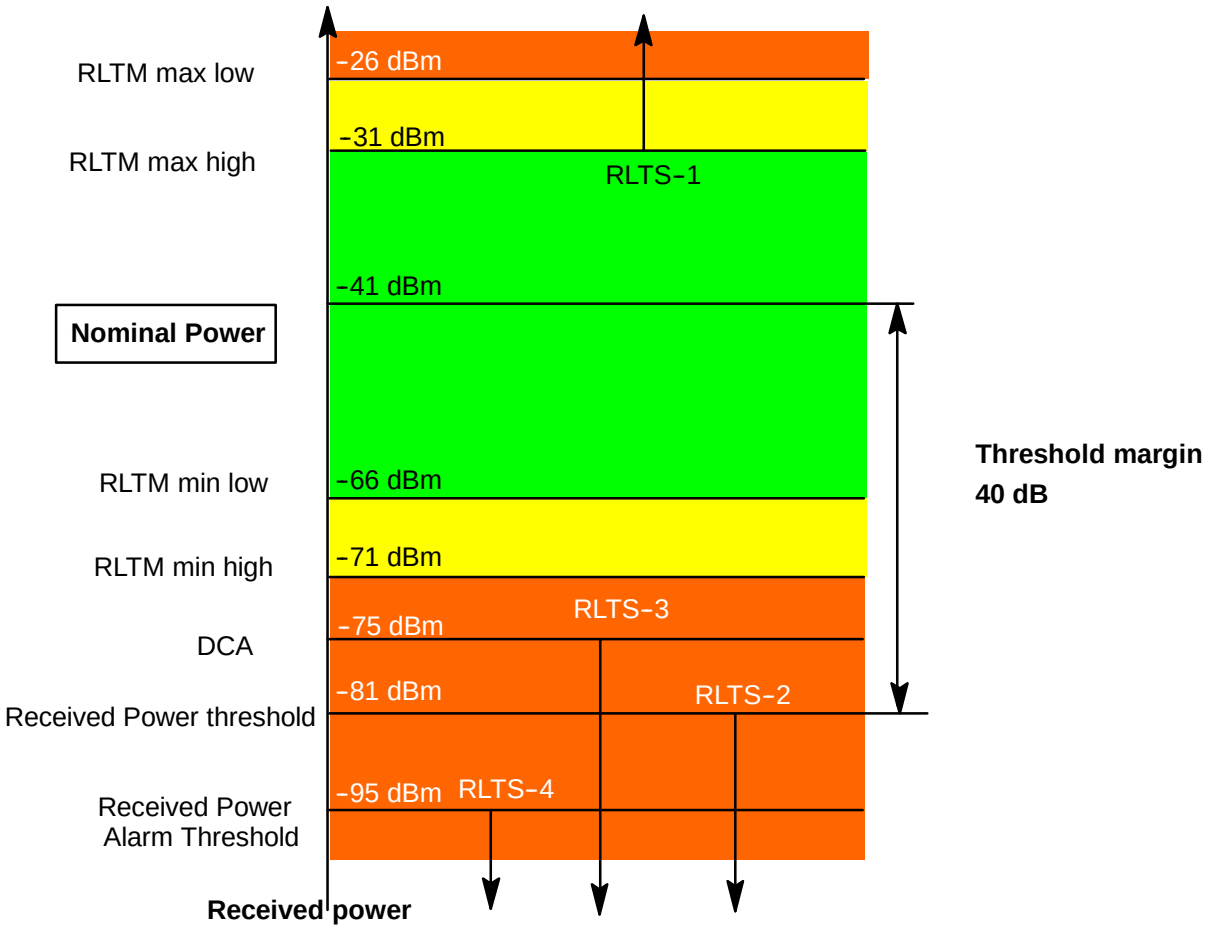
The six error counters for which readings are displayed are:

ACRONYM	MEANING
RLTS-1	Number of one-second periods during which the received power has been more than 10 dB above nominal power.
RLTS-2	Number of one-second periods during which received power has exceeded the Alarm propagation threshold (nominal power – threshold margin).
RLTS-3	Number of one-second periods during which received power has exceeded the early switching request threshold (Receiver).
RLTS-4	Number of one-second periods during which received power has exceeded the received power alarm threshold.
RLTMMax	Maximum estimated received power value during the observation period.
RLTMMin	Minimum estimated received power value during the observation period.

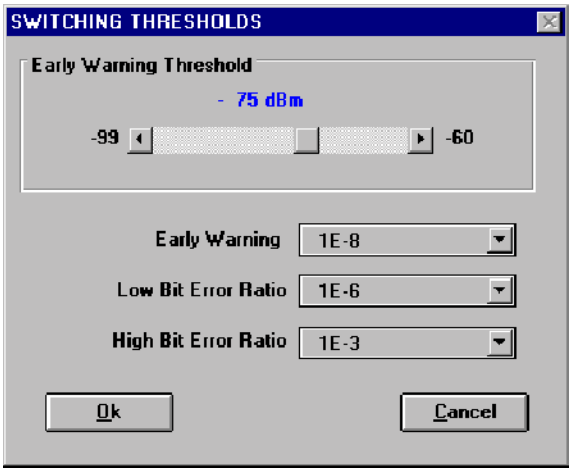
Note:

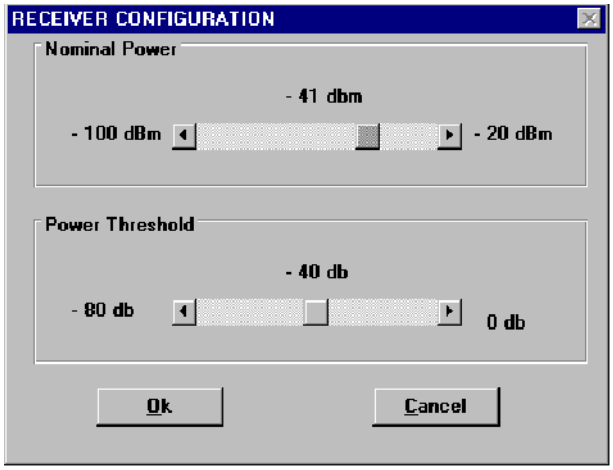
- RLTS-N: **R**eceived **L**evel **T**hreshold **S**econd
- RLTMMin **R**eceived **L**evel **T**ide **M**ark **M**in
- RLTMMax **R**eceived **L**evel Tide **M**ark **M**ax

Example of quality measurement at reception :



The following example is in accordance with the thresholds defined in the operational configuration below and in the G826 thresholds configuration (see chapter 4.12.4)





“Switching performance” TP

The four error counters for which readings are displayed are:

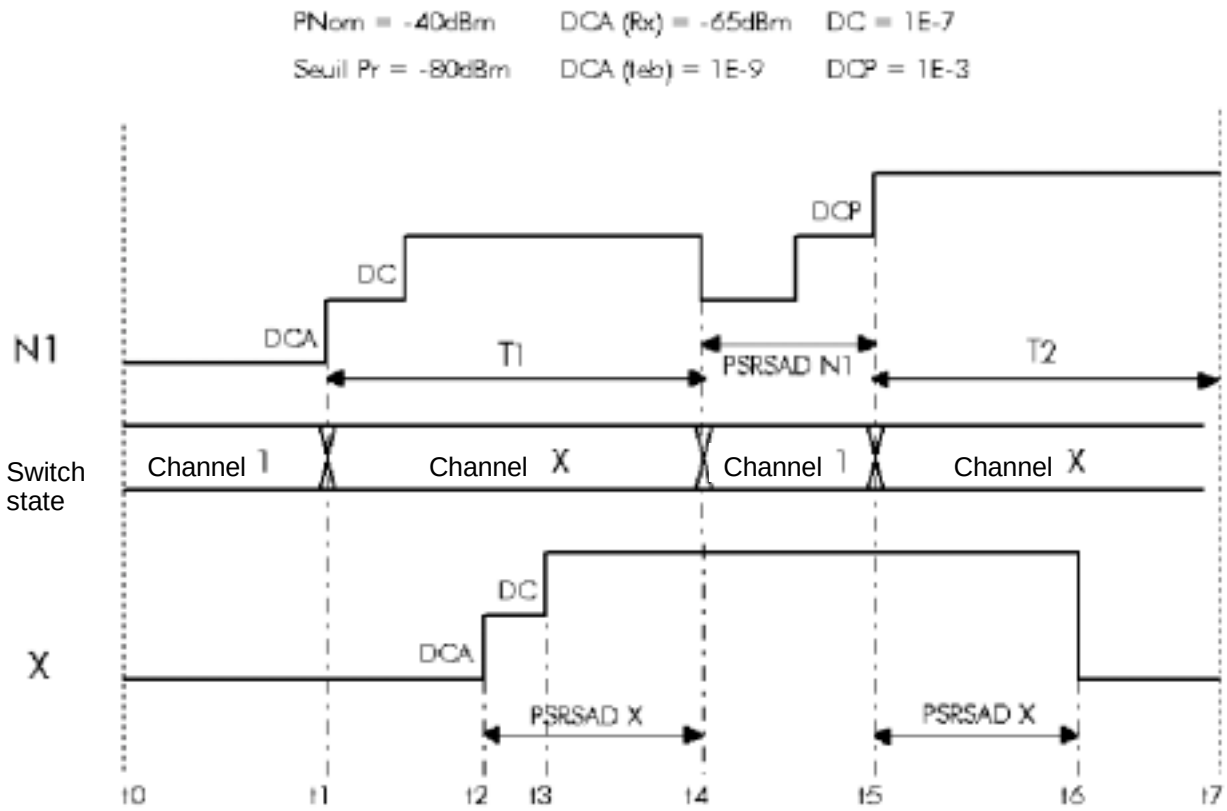
ACRONYM	NORMAL CHANNEL	STANDBY CHANNEL*
PSAC	Number of switchovers from active channel to standby channel	Number of switchovers from any active channel to the standby channel
PSAD	Number of one-second periods during which the active channel has been backed up	Number of one-second periods during which any active channel has been backed up
PSRC	Number of automatic switching requests observed on the normal channel	Number of automatic switching requests observed on the normal channel
PSRSAD	Number of one-second periods during which it has been impossible to meet a channel switching request	Number of one-second periods during which the standby channel has received automatic switching requests and has been unable to satisfy them because it is already backing up another channel (meaningless for UX systems).

Note:

- PSAC: **P**rotection **S**witch **A**ctual **C**ount
- PSAD: **P**rotection **S**witch **A**ctual **D**uration
- PSRC: **P**rotection **S**witch **R**equest **C**ount
- PSRSAD: **P**rotection **S**witch **R**equest **S**ervice **A**ffecting **D**uration

* Not applicable in 1+0 configurations.

Example of switching performance measurement



N1	Considering	G784 counter	Calculated
PSAC	2	2	2
PSAD	T1+T2	664	666
PSRC	1	1	1
PSRSAD	PSRSAD N1	88	88

X	Considering	G784 counter	Calculated
PSAC	2	2	2
PSAD	T1+T2	664	666
PSRC	1	1	1
PSRSAD	PSRSAD X	373	373

Checking by calculation of the G784 counters values:

t0 = 9:30:00
t1 = 9:32:26
t2 = 9:36:49
t3 = 9:38:03
t4 = 9:40:15
t5 = 9:41:43
t6 = 9:44:30
t7 = 9:45:00

T1 = t4 - t1 = 469
T2 = t7 - t5 = 197

PSAD = T1+T2 = 666

PSRSAD N1 = t5 - t4 = 88
PSRSAD X = (t4 - t2) + (t6 - t5) = 373

Nota: the timing is in the record of the AET application.

“Link”, “Section” and “Receive” TP

The three error counters for which readings are displayed are:

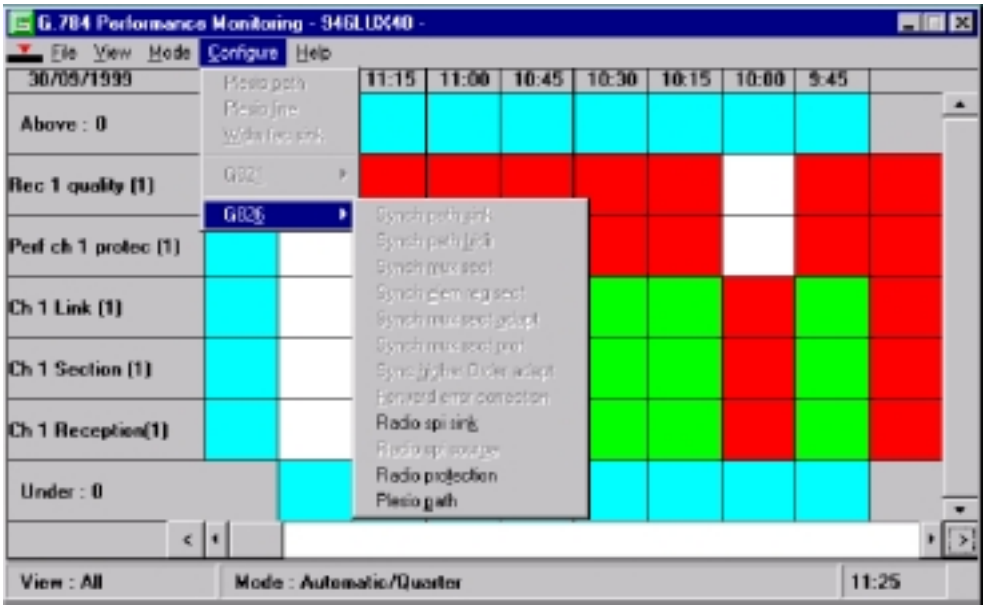
ACRONYM	MEANING
BBE	Number of errored blocks that are not part of an SES.
ES	Number of errored seconds.
SES	Number of severely errored seconds.

Note:

- BBE: **B**ackground **B**lock **E**rror
- ES: **E**rrored **S**econd
- SES: **S**everely **E**rrored **S**econd

All isolated BBE must manage an ES.

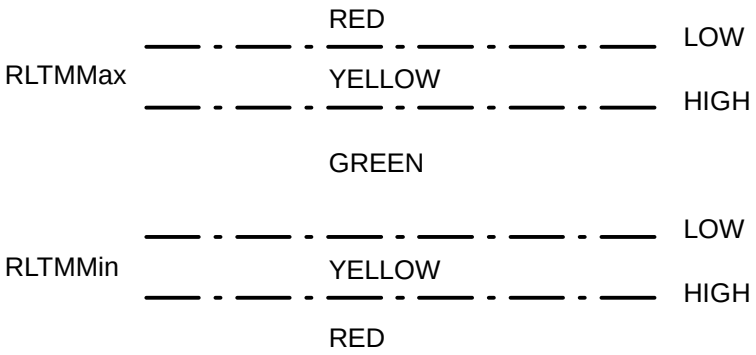
4.12.4 – "Configure" menu



Setting menu for the HIGH and LOW thresholds for G826.

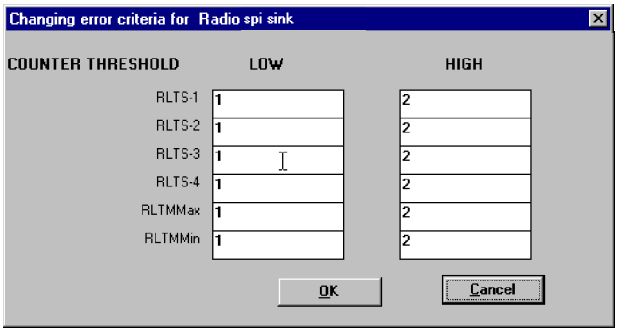
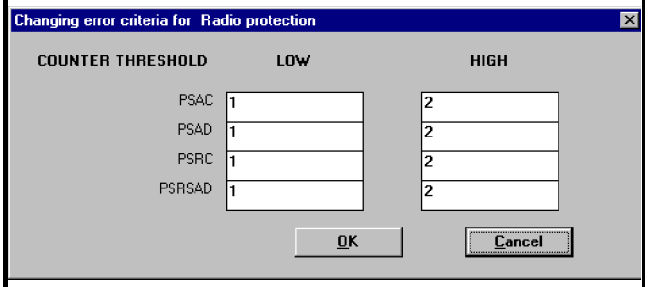
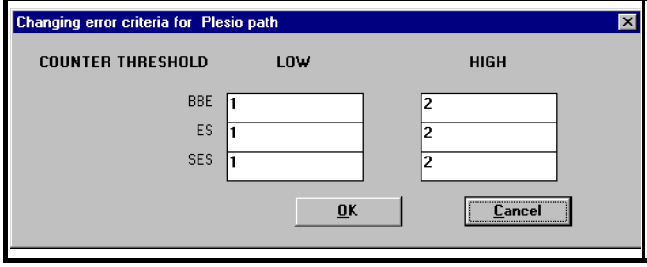
The setting of the HIGH and LOW thresholds allows the user to control the displaying criteria for representing levels of degradation independently for each counter and for each type of TP.

Example: For the RLTM thresholds of the received quality, the displayed colors will be in accordance with the following figure:



The meaning of the other colors (pale blue, green and white) is not modifiable by the user (see chapter 4.12.3).

The change of these thresholds remains permanent for a given equipment, even after the closing of the application. They can be modified any time by the user and be effective immediately.

	**G826/Radio spi sink** Thresholds for “Receive quality”
	**G826/Radio protection** Thresholds for “Switching performance”
	**G826/Plesio path** Thresholds for: S “Link” S “Section” S “Reception”

Some counters of the “Radio Protection” group, indicate events that do not concern the G.826 analysis that takes care of the supervision of the traffic quality such as:

- PSAC,
- PSAD,
- PSRC.

These indications give information on the efficiency of the radio diversity.

Application available only with the extended 946LUX11/946LUX12 software and the standard 946LUX40 software version.

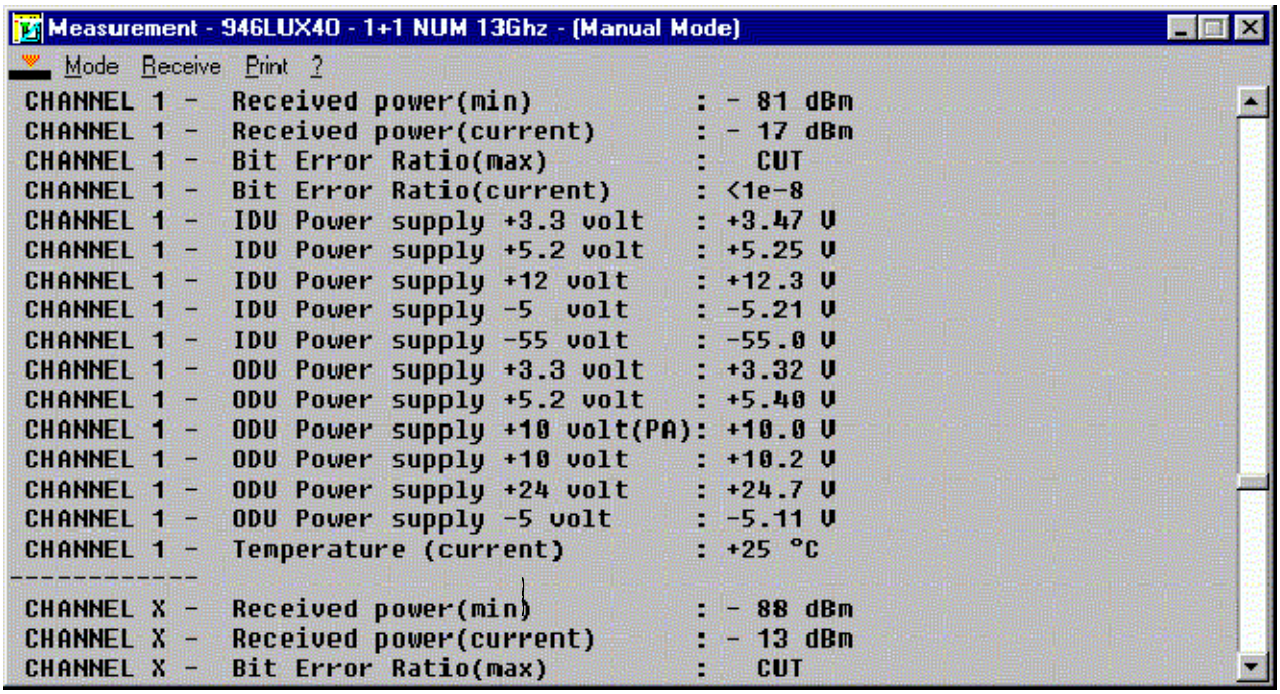
4.13 – "Analogue measurements" application

This application is used to take measurements on various points of an NE and display them on screen.

Path: Alarm Synthesis → Application_Choice... → Analogue Measurements

When the application is opened, a window appears containing the readings of the last measurements performed. The date and time of the measurement appear at the top of the window.

The polling interval for the analogue measurement refresh function can be adjusted via the POLMEA variable in the 946LUXxx.ini file, as described in Appendix 11.



4.13.1 – "Mode" menu



MODE	FUNCTION
Auto	All of the measurements defined for each network element are performed every 60 seconds (period can be configured in the WIN.INI file, as specified in appendix 11). The content of the measurement display window is refreshed automatically at the end of each period. The "Receive" menu is disabled in this mode.
Manual	The network element performs its measurements without displaying the readings. The display is obtained at the request of the operator (by clicking "Receive").

4.14 – Local supervision

The 9400 UX supervision system can be used to supervise the local station and any remote station in the network. The physical address of the remote equipment must be known to be able to select local supervision mode.

If there is a network supervision function, the supervision bus must be disconnected before setting the station to local supervision mode. There are two possibilities depending on whether you are located at station A or station B, relative to the network supervision point.

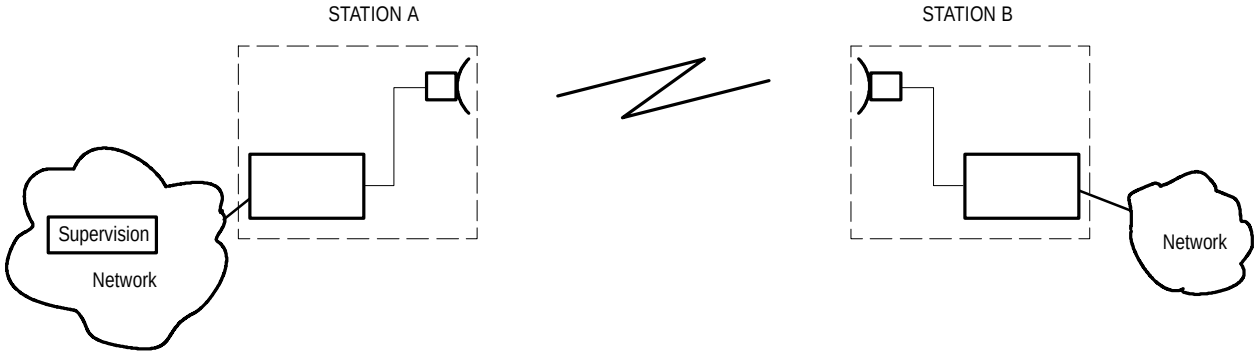


Figure 52 - Relative positions of stations A and B

4.14.1 – Station A of network with supervision/station of network without supervision

4.14.1.1 – Switching to local supervision mode

CAUTION: Before modifying the parameters, make a note of the current configuration (Physical address and SCC NMS connection matrix with the 946LUX11 and 946LUX12 software, or IP address, number of OSPF area, mask of subnetwork and Access configuration with the 946LUX40 software)

- Run the “Operation parameters” application.
- In the “Equipment” menu, make the configuration changes summarized in the table below.

SUBMENU	CONFIGURATION
Network Element	Type of Device: Check the “Mediation function” button.
SCC/NMS Connection matrix (for 946LUX11 and 946LUX12 software) or Access configuration (for the 946LUX40 software)	Disconnects the supervisory network to isolate the branch containing stations A and B (example in Figure 52). For this: S with 946LUX11 leave only the MSU1→SCC1 and SCC1→MSU1 connections. S with 946LUX12, and 946LUX40 check only the “radio access” button.
Secondary equipment	With 946LUX11/12 software. click “001” in the “SEC” column and enter the physical address of the station B equipment. With the 946LUX40 software, click the first network element and enter the IP address of the station B equipment.

- Send the file.
- Wait about two minutes for the supervision system to be re-initialized and reconfigured.
- Check that the screen shows both NEs (M:0–S:0 for station A and M:0–S:1 for station B) using: “View/All Element”.

4.14.1.2 - Exiting local supervision

You exit the local supervision function as follows:

- Go to station A (M:0-S:0).
- Run the “Operation parameters” application.
- In the “Equipment” menu, set up the configuration summarized in the table below:

SUBMENU	CONFIGURATION
Network Element	Type of Device: Check the “Network Element” button. Restore the initial equipment address.
SCC/NMS Connection matrix (946LUX11/946LUX12) or Access configuration (946LUX40)	Restore the previous configuration or the standard configuration.

- In the “File” menu, click “Send” to download the new configuration to the equipment.
- Wait for the equipment to reset (about two minutes), then return to the “ALARM SYNTHESIS” function.
- Check that there are no alarms by opening the AS&C application.

4.14.2 - Station B in a network with supervision

4.14.2.1 - Switching to local supervision mode

- Connect the network supervisor and ask him to modify the SCC-NMS connection matrix to isolate the branch containing stations A and B (example in Figure 52). For this:
  - in 946LUX11, leave only the MSU1→SCC1 and SCC1→MSU1 connections
  - in 946LUX12 and 946LUX40, check only the “Radio access” button.
- In the “Equipment” menu of the “Operation parameters” application, set up the configurations summarized in the table below.

SUBMENU	CONFIGURATION
Network element	Equipment type: Check the “Mediation” button.
Secondary equipment	With the 946LUX11/12 software click “001” in the “SEC” column and enter the physical address (or the IP address in the case of the 946LUX40 software) of the station A equipment.

- Send the file.
- Wait about two minutes for the supervision system to be reset and reconfigured.
- Check that the screen shows the two items (M:0-S:0 for station B and M:0-S:1 for station A) by: “View/All elements”.

4.14.2.2 - Exiting local supervision

You exit the local supervision function as follows:

- Station A:
  - Go to station A (M:0-S:1)
  - Run the “Operation parameters” application.
  - In the “Equipment/SCC/NMS connection matrix” menu (or Network/Access configuration for 946LUX40 software), restore the standard configuration.
  - Send the configuration.

- Station B:
  - Go to station B (M:0–S:0):
  - Run the “Operation parameters” application.
  - In the “Equipment/Network element” menu, check the “Network element” button under “Equipment type”, and retransmit the equipment’s initial physical address (or IP address for LUX40).
  - Send the configuration.
- Wait for the equipment to reset (about two minutes) then return to “ALARM SYNTHESIS”.
- Check that there are no alarms by opening the AS&C application.

5 - Commissioning

A very precise knowledge of the link configuration is absolutely vital before you begin commissioning. For this, complete the sheets for stations A and B according to the model supplied in Appendix 2.

5.1 - Order of commissioning for a link

The commissioning operations described below are for a link comprising a station A and a station B.

If the network includes supervision, station A is the one located between the supervisory station and station B (Figure 53). Installation and commissioning starts with station A.

If station B has to be commissioned first, apply the procedure in § 5.5.

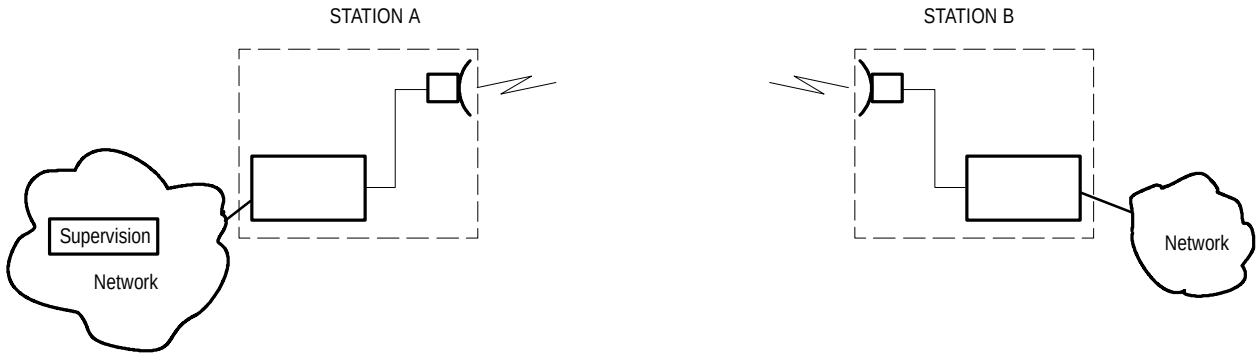


Figure 53 - Relative positions of stations A and B

The stages are:

- prepare the PC (see § 4.1),
- install station A and roughly point the antenna towards station B,
- commission station A,
- install station B and roughly point the antenna towards station A,
- commission station B and point the antenna, connect the tributaries and the service channels,
- return to station A to:
  - fine tune the pointing of the antenna,
  - carry out all the commissioning checks (switching to local supervision mode),
  - check out the link,
  - cancel the local supervision function,
  - connect the tributaries and engineering service channels.

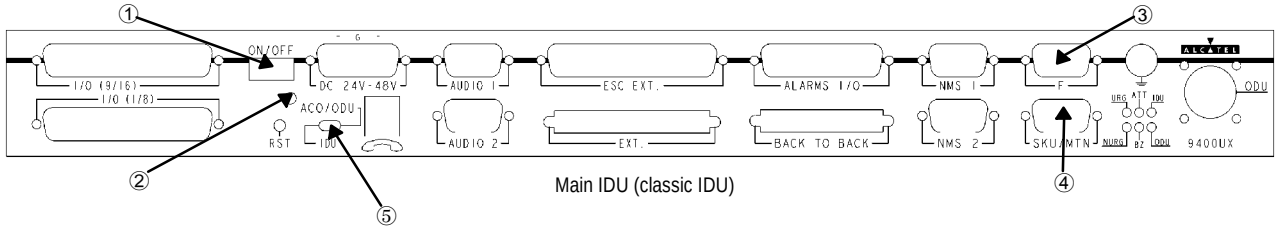
EQUIPMENT REQUIRED

The following equipment is required for commissioning:

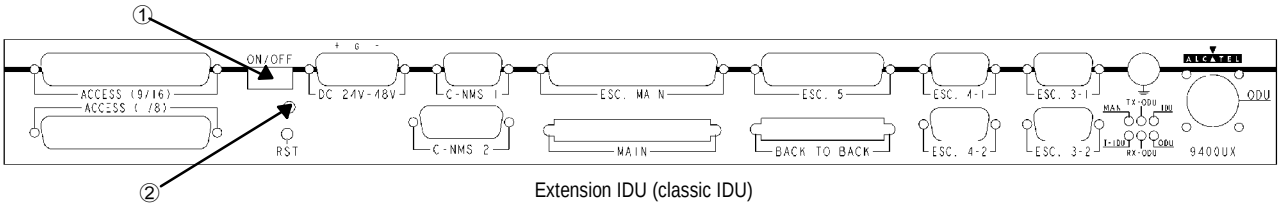
- Multimeter.
- Portable PC running the supervisory software (see § 4.1).
- PC-NE link cable, supplied with the system diskettes.
- A “service kit” (for measuring AGC and the telephone link between the IDU and the ODU).
- A 2 Mbit/s or 34 Mbit/s BER analyzer, with an impedance of 75 or 120 Ω, depending on the configuration of the link to be installed.
- A compass and a pair of binoculars.
- Tools as defined in § 3.3.3.

5.2 – Installing and commissioning station A (phase 1)

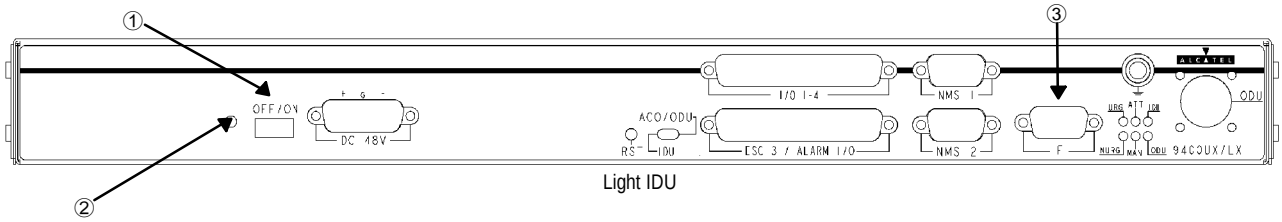
5.2.1 - Preliminary operations



Main IDU (classic IDU)



Extension IDU (classic IDU)



Light IDU

- Install station A according to the instructions in section 3, and in particular those in § 3.5.3.4 or 3.5.4.4 describing how to roughly point the antenna of station A towards station B.
- Check that the correct power supply voltage is present (24 or 48 V), with the correct polarity, on the power supply outlet.
- Check that there are no cables connected to an NMS port; any that are connected must be disconnected.
- **Connect the IDU(s)/ODU(s) link cable(s)** (at the IDU).
- **In the case of a configuration with classic IDU, check that a software key enabling the configuration** is connected to the SKU/MTN connector ④ (refer to § 3.2, Comments on the information marked on the software key label).

- Power up the IDU(s) (via switch ① on the connector panel):

  § The green LED ② should come on.

- Connect the PC to the equipment using the interface cable supplied (serial port – connector marked “F” on the IDU ③).
- Wait for the ATT, URG and NURG LEDs to start flashing out of sync on the main IDU. The LED flashing occurs if IDU and ODU have never been configured together (otherwise, refer to § 4). Download the IDU configuration, by pushing switch ⑤ to the IDU setting (IDU configuration taken as a reference). After two minutes at most (for the configuration to be registered), the LEDs should stop flashing simultaneously.
- Run “NECTAS” software, the “Alarm synthesis” application opens. Start the “Installation parameters” application (refer to § 4.4). If the LUX12 versions you run is under index 08 and LUX40 versions under index 05, go direct to § 5.2.2. For all other LUX12 versions above index 08 and LUX40 versions above index 05, the default application opens. Just complete the hardware parameters successively in the different menus.
- In the case of a light IDU configuration, go direct to § 5.2.2. In all other cases, proceed as follows.

  Run the “Alarms, Status and Controls” application,

  zoom in on “Channel 1” or the “main” IDU,

  zoom in on ENVT,

  check that there is no “software key” alarm.
- If there is a “software key” alarm, carry out the same checks that you would carry out after a change of key (see § 7.3).

5.2.2 - Setting up the installation parameters

- Run the “Installation parameters” application. Open in turn each of the items in the “Table” menu, modifying the values as required according to the station’s configuration (refer to the station’s sheet).
- Upload the configuration by “File/send” (see § 4.4.2, “File” menu).
- **Close the application.**

5.2.3 - Setting up the operation parameters

- Run the “Operation parameters” application. In turn, open each of the items in each of the menus, modifying the values as required according to the station’s configuration (refer to the station’s configuration sheet).



§ **Check that the duplex separation between transmit and receive frequencies is correct.**

§ **For a 1+1 HSB configuration, check that the transmit and receive frequencies of channels 1 and X are the same.**

- Upload the configuration by “File/Send” (see § 4.5.7, “File” menu).
- **Close the application.**

5.2.4 - Checking out operation

Checking operation entails setting up local and remote loopback configurations as defined in § 2.5.

5.2.4.1 - Checking operation in a 1+0 configuration

1 Connect the 2 or 34 Mbit/s BER analyzer (according to the configuration) to an active tributary.

2 Open the “Alarms, Status and Controls” application and set up an IDU baseband loop (remote control: “MCU>BB IDU [Start] :local loop validated”).

3 Check the bit error ratio. If zero, continue checking; otherwise, check the wiring, that the LAU and LIU boards of the unit match (and their impedances) and the configuration of the equipment. If everything is OK, change the IDU.

4 Cancel the IDU baseband loop (remote control: “MCU>BB IDU [End] :local loop inhibited”).

5 Set up the baseband radio loop (remote control: “BB OUTDOOR [Start] :Local loop validated”).

6 Check the bit error ratio. If zero, continue checking; otherwise, check the state of the IDU/ODU link coaxial cable.

7 Cancel the radio baseband loop (remote control: “BB OUTDOOR [End] :Local loop inhibited”.

8 Set up the RF loop (if this option is available in the equipment) via the remote control: “RF [Start] : Local loop validated”.

9 Check the bit error ratio. If zero, continue checking via operation 10; otherwise, check the radio configurations and correct the configuration errors.

10 Cancel the RF loop (remote control: “RF [End]: Local loop inhibited”).

11 Check that there are no remote controls still active (indicated by a yellow coloured non-urgent alarm and a green state).

12 Close the “Alarms, Status and Controls” application.

13 Run the “Radio transmission parameters” application.

14 Check that the “Transmit alarm diagnostic” is “OK” and close the application.

15 If an alarm is detected or an incorrect value is found, refer to section 6 and correct as required.

16 Disconnect the BER analyzer.

5.2.4.2 – Checking out operation in a 1+1 configuration

1 Connect the 2 or 34 Mbit/s BER analyzer (according to the configuration) to an active tributary.

2 Open the “Alarms, Status and Controls” application and force operation to channel 1 via the remote controls:

- “TX 1 [Start] :manual lockout validated”,
- “MCU>RX 1 [Start] :manual lockout validated”.

3 On channel 1, carry out operations 2 to 11 in § 5.2.4.1.

4 Force operation to channel X via the remote controls:

- “TX X [Start] :manual lockout validated”,
- “MCU>RX X [Start] : manual switch. validated”.

5 On channel X, carry out:

- operations in 5 to 11 in § 5.2.4.1 for a basic 1+1 HSB configuration,
- operations 2 to 11 in § 5.2.4.1 for a 1+1 HSB configuration with multiplexer protection.

6 Cancel forcing on channel X via the remote controls:

- “TX X [End] :manual lockout inhibited”,
- “MCU>RX X [End] :manual switch. inhibited”.

7 In a 1+1 configuration with multiplexer protection, perform operations 8 to 12. Otherwise, go directly to operation 13.

8 Force operation to the standby multiplexer/demultiplexer via the remote control:

- “MUX/DEMUX [Start] :standby switching validated”.

9 Perform operations 2 to 4 in § 5.2.4.1.

10 Force operation to the normal multiplexer/demultiplexer via the remote control:

- “MUX/DEMUX [Start] :normal lockout validated”.

11 Perform operations 2 to 4 in § 5.2.4.1.

12 Cancel forced operation on the normal multiplexer/demultiplexer via the remote control:

- “MUX/DEMUX [End] :normal lockout inhibited”.

13 Check that there are no remote controls still active (indicated by a yellow coloured non-urgent alarm and a green state).

14 Close the “Alarms, Status and Controls” application.

15 Run the “Radio Transmission Parameters” application.

16 Check that the “Transmit alarm diagnostic” is “OK” and close the application.

17 If an alarm is detected or an incorrect value found, refer to section 6 and correct as required.

18 Disconnect the BER analyzer.

5.3 - Installing and commissioning station B

5.3.1 - Preliminary operations

- Commission station B by carrying out the same operations as those described for station A in § 5.2.1 to 5.2.4, apart from the RF loop operations.

5.3.2 - Pointing the antenna with pole mounting 1+1 (9400UXI101)

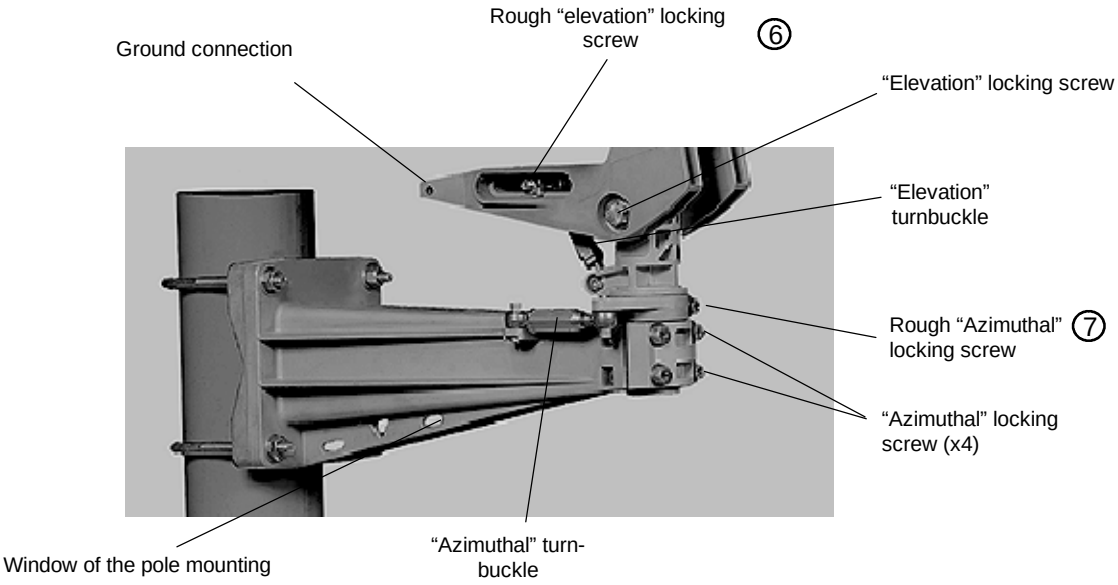


Figure 54 - Pole mounting 1+1 adjustments (9400UXI101)

- Attach the hook of the service kit (ref. **9400UXT102**) in the window of the “Pole mounting”.
- Connect connector P01 (see Appendix 4) to the “MAINT” port on the front panel of the ODU.
- Check the tightness of the two “rough” pointing screws ⑥ and ⑦, tighten with a 5 mm Allen key.
- Connect the voltmeter, set to “DC” range, to P02/P03 of the service kit 9400UXT102 or to P02/P03 of the service kit 9400UXT113. With the two turnbuckles, fine tune the azimuthal and elevation settings (Figure 54) to obtain a maximum voltage on the voltmeter (AGC DC voltage of between 0 and 5 V).
- Retighten the two azimuthal locking screws using the “box” wrench.
- Tighten the four front elevation screws using the 5 mm Allen key (tighten in criss-cross fashion).
- Finish the operation by tightening the azimuthal and elevation turnbuckle lock nuts with the 16 mm flat wrench.

	**CAUTION: For the fasten torques of the screws, refer to chapter 3.3.4.**

5.3.3 – Pointing the antenna with pole mounting 1+0 (9400UXI102)

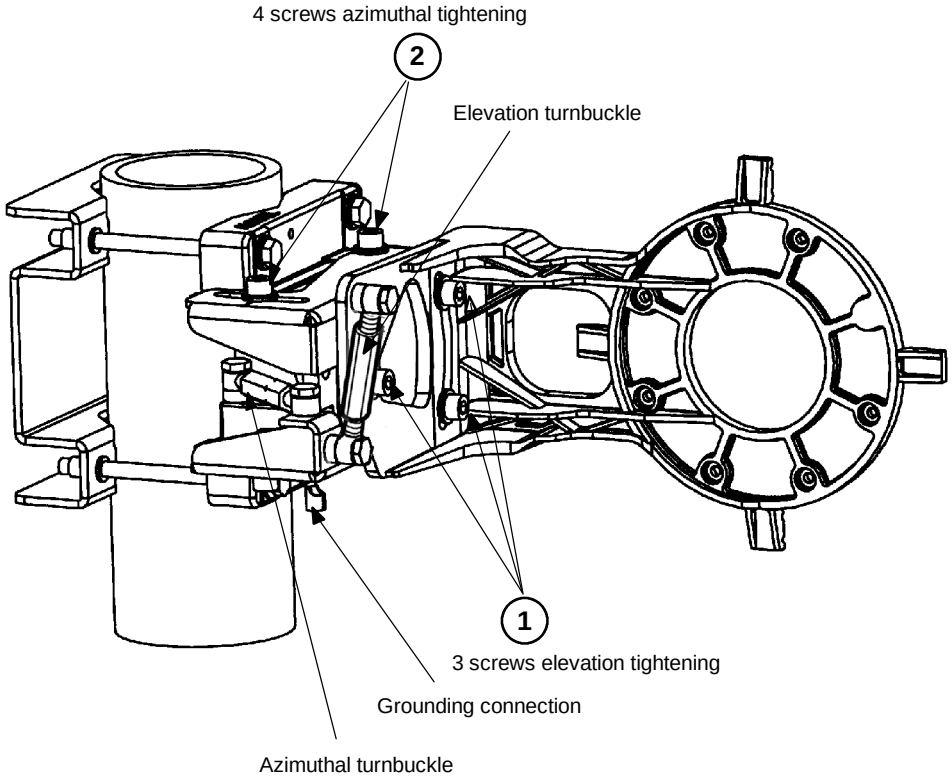


Figure 55 - Pole mounting 1+0 (9400UXI102)

- Connect connector P01 of the service kit (see **Appendix 4)** to the “MAINT” port on the front panel of the ODU.
- Connect the voltmeter, set to “DC Volts”, to P02/P03 of the service kit 9400UXT102 or to P02/P03 of the service kit 9400UXT113.
- Be sure that the three screws ① and the four screws ② are a bit slackened, just enough to allow movement of the different parts of the pole mounting. These screws must not be too much slackened, otherwise, the fine–tuning of alignment will be imperfect. Use the 8 mm Allen key.
- With the two turnbuckles and the 16 mm flat wrench, fine tune the elevation and azimuthal settings until you obtain the maximum voltage reading in the voltmeter.
- Tighten the three screw ① and the four screws ② with torque 3 mdaN.

Note: Fine pointing of the antenna is described in § 5.3.2.

5.3.4 – Checking out operation

- Using the local loop configurations (apart from the RF loop), repeat the operations carried out on station A (see § 5.2).
- Run the “Radio transmission parameters” application.
- Check that the received power level is greater than –90 dBm. This check must be carried out on the normal channel and, if appropriate, the standby channel.

  If the received power is less than this value, the reason is probably an antenna coarse pointing problem or a transmission problem in station A.
- Connect the tributary, supervision and ESC cables previously disconnected.
- If an operator is available in another station of the network and there is a telephone in station B, test the telephone ESC (general call using number 00).
- Test the ESC 3, ESC 4 and ESC 5 connections, if applicable, and if the corresponding equipment is installed in the station.

5.4 – Commissioning the link from station A (phase 2)

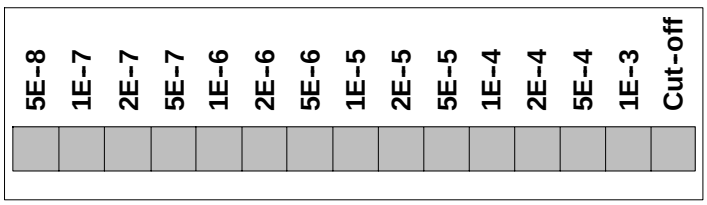
5.4.1 – Pointing the antenna

- Adjust the orientation of the antenna in the same way as for station B (see § 5.3.2).

5.4.2 – Checking the radio transmission parameters

- Switch station A to local supervision mode (see § 4.14).

  Display the equipment and check that there are no EF (equipment fail) alarms.
- Select M:0–S:0 (Station A).
- Run the “Radio transmission parameters” and “Alarms, Status and Controls” applications.
- Check that the “Communication OK” indicator flashes and there are no transmit and receive alarms. If there are, use the “Alarms, Status and Controls” application to analyze and, if necessary (if the fault is in the configuration), correct the cause of the alarm.
- Note down the received power level. This must be measured on each channel in a 1+1 configuration, by forcing a switchover on reception to one or the other channel using the remote controls:
  - “MCU>RX 1 [Start] :manual lockout validated” to force to channel 1,
  - “MCU>RX X [Start] :manual switching validated” to force to channel X.
- Disable the forcing remote controls.
- Select M:0–S:1 (Station B).
- Run the “Radio Transmission Parameters” and “Alarms, Status and Controls” applications.
- Check that the “Communication OK” indicator flashes and there are no transmit and receive alarms. If there are, use the “Alarms, Status and Controls” application to analyze and, if necessary (if the fault is in the configuration), correct the cause of the alarm.
- Note down the received power level. This must be measured on each channel in a 1+1 configuration, by forcing a switchover on reception to one or the other channel using the remote controls:
  - “MCU>RX 1 [Start] :manual lockout validated” to force to channel 1,
  - “MCU>RX X [Start] :manual switch. validated” to force to channel X.
- The received power levels should match the nominal received power indicated on the station configuration sheet to within ±6 dB. If the difference is greater:
  - Check that there are no abnormal propagation conditions (storm or heavy rainfall, for example).
  - In the case of a 1+1 HSB configuration with integrated coupler, use the “Alarms, Status and Controls/channel” application on each station to check that the standby channel is not being used for reception and/or transmission.
  - Check the pointing of the antennas (if necessary, pointing to a secondary lobe).
- If the problem persists, contact the network administrator or “Alcatel customer service”.
- If the level is correct, check the reading on the BER bargraph (all the LEDs should be on green).



If alarms are indicated by the RTP application, proceed as follows for the stations with an alarm condition:

- Select the M:0–S:0 or M:0–S:1 station.
- Run the “Alarms, Status and Controls” application, run a diagnostic and correct as necessary according to the instructions in § 6.3.
- For a configuration error (tributaries wrongly declared inactive or active, for example), run the “Installation parameters” and “Operation parameters” applications for the station concerned and make the necessary changes. The installation and operation parameters of station B can be modified remotely.



Be careful not to disconnect the link by setting the parameters wrongly!

- Close the “Alarms, Status and Controls” application.

5.4.3 - Remote tributary loop

A remote loop on an ***active*** tributary is needed for the checks described in § 5.4.4 and 5.4.5.

- Run the “Alarms, Status and Controls” application on station B (M:0–S:1).
- Set up the remote loop configuration as follows:
  - In the “Remote controls” menu, choose “All”.
  - Select the remote control “[MCU>TRIB. x [Start] : distant loop validated”. (x : the number of the tributary on which the loop is set up).
  - Click “OK”.
- In station A, connect the BER analyzer to the port of tributary x and check that there are no errors.

5.4.4 - Checking the transmit/receive switching function

These checks must be carried out only in frequency diversity 1+1 and 1+1 HSB configurations.

- Set up the remote loop on an active tributary.
- Open the “Alarms, Status and Controls” applications for the M:0–S:0 (Station A) and M:0–S:1 (Station B).
- In both stations, initiate a switchover to the standby channel as described below.

5.4.4.1 - 1+1 configuration

MENU	Submenu	1+1 FRQ DIV CONFIGURATION	1+1 HSB CONFIGURATION
Remote controls	All		Click “TX X [Start] :manual lock-out validated” and confirm.
Remote controls	All	Click “MCU>RX X [Start] :manual switch. validated” and confirm.	Click “MCU>RX X [Start] :manual switch. validated” and confirm.

- Check that the switchover is completed (see the graphic representation of the MCUs in stations A and B using the “Alarms, Status and Controls” applications).

NOTE: Receive mode switchovers must be executed without errors.

- In frequency diversity 1+1 configuration, there is no transmit switching.
- In 1+1 HSB configurations, transmit switching causes transient dropouts on the link and a loss of received data indicated by a burst of errors.

- In both stations, initiate a switchover to the normal channel as described below:

MENU	Submenu	1+1 DIV. CONFIGURATION	1+1 HSB CONFIGURATION
Remote control	All		Click “TX 1 [Start] :manual lock-out validated” and confirm.
Remote control	All	Click “MCU>RX 1 [Start] :manual lockout validated” and confirm.	Click “MCU>RX 1 [Start] :manual lockout validated” and confirm.

- Check that the switchover is completed (see the graphic representation of the MCUs in stations A and B using the “Alarms, Status and Controls” application). The above note still applies.
- Reinstate automatic switching mode (cancel any yellow coloured non-urgent alarms and green states) and cancel the remote controls previously activated as described below:

MENU	Submenu	1+1 DIV. CONFIGURATION	1+1 HSB CONFIGURATION
Remote control	All		Click “TX 1 [End] :manual lockout inhibited” and confirm.
Remote control	All	Click “MCU>RX 1 [End] :manual lockout inhibited” and confirm.	Click “MCU>RX 1 [End] :manual lockout inhibited” and confirm.

5.4.4.2 - 1+1 configurations with multiplexer protection

- In both stations, force operation to the normal multiplexer/demultiplexer as described below:

MENU	Submenu	ACTION
Remote controls	All	Click “MUX/DEMUX [Start] :standby switching validated” and confirm.

- Check that the switchover is completed (see the graphic representation of the MCUs in stations A and B using the “Alarms, Status and Controls” application).
- MUX switchovers cause a microinterruption of the link which results in the appearance of error bursts.
- Check that there are no errors (after traffic has stabilized on the normal MUX).
- In both stations, initiate a switchover to the standby multiplexer as described below:

MENU	Submenu	ACTION
Remote controls	All	Click “MUX/DEMUX [Start] :standby switching validated” and confirm.

- Check that the switchover is completed (see the graphic representation of the MCUs in sections A and B using the “Alarms, Status and Controls” application).
- Check that there are no errors.
- Reinstate the automatic switching mode (cancel any yellow-coloured non-urgent alarms and green states) by sending the remote control:

MENU	Submenu	ACTION
Remote controls	All	Click “MUX/DEMUX [End] :standby switching inhibited” and confirm.

5.4.5 - Checking quality

- Run the “Performance monitoring (G821)” or “G784” application, “Display” menu (see § 4.8). This application keeps a count (errored seconds, severely errored seconds, degraded minutes and unavailable seconds) on the link (results given for each link, channel, etc).

Duration and results:

Application running time: 2 hours.
Tolerance: the readings must be less than or equal to those given in the table below:

EVENT	AGGREGATE BIT RATE			
	4 Mbit/s	8 Mbit/s	16 Mbit/s	34 Mbit/s
ES	2	3	4	4
SES	0	0	0	0

EVENT	4 Mbit/s	8 Mbit/s	16 Mbit/s	34 Mbit/s
DM	Meaningless			
US	0	0	0	0

NOTE:

If fading is observed over a period of less than 15 minutes, the fading period is not taken into account.
If fading is observed over a period of longer than 15 minutes, the test will be extended as required.

For the 946LUX40 only the “G784 Quality Monitoring” application is available.

- When you have finished the checks, close the “Performance monitoring” application.

5.4.6 - End of commissioning

To terminate the commissioning process:

- Cancel the remote loop as follows:
  - In the “Remote controls” menu, choose “All”.
  - Select the remote control “MCU>TRIF. x [End] :distant loop inhibited”. (x is the number of the tributary that was looped).
  - Click “OK”.



- **Clear the maintenance memory (see § 4.11.3) and the maintenance alarms using the remote control “Maintenance Alarms: reset”.**
- **Set different link identity codes for the transmit and receive directions, see § 4.5.1**

- Reconnect the tributary, monitoring and ESC cables previously disconnected.
- Check that there are no remote controls still active (indicated by a yellow coloured non-urgent alarm).
- Exit the local supervision function as in § 4.14.2.2.
- If the network is under supervision, incorporate the link in the supervision network:
  - call the supervision centre and ask the operator to declare the link (giving the addresses of the two terminals).
  - Wait for the network to be cut over.



Never leave configured equipment in the local supervision mode when commissioning is finished.

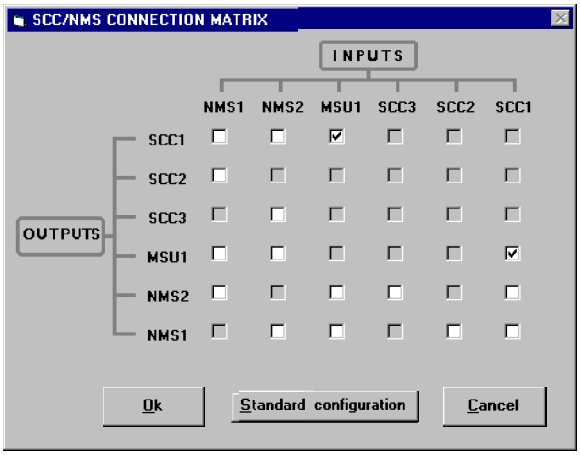
5.5 - Reversing the order of link installation and commissioning



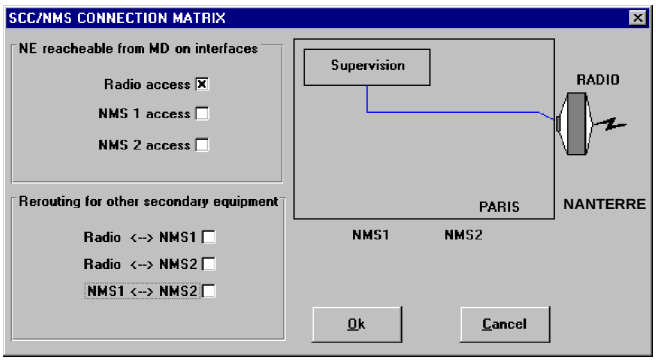
Note: This concerns only equipment supervised via the RQ2 protocol (946LUX11 and 946LUX12).

If the order of installation described in § 5.1 is not possible for a link comprising a station A and a station B, beginning with station A, proceed as follows:

- install station B and roughly point the antenna towards station A,
- commission station B,
- install station A and roughly point the antenna towards station B,
- commission station A and point the antenna, then connect all the engineering service channels,
- using the “Operation parameters / equipment / SCC/NMS connection matrix” menu (see § 4.5.2), configure the matrix as shown in the screen below:



946LUX11



946LUX12

- return to station B and carry out the following operations:
  - fine tune the pointing of the antenna,
  - carry out all the commissioning checks (switching to local supervision mode) as described in § 5.4,
  - check out the link,
  - from station B, restore the SCC/NMS connection matrix to the standard configuration in station A by opening the “Equipment/SCC/NMS connection matrix” screen and proceeding as follows:
    946LUX11 software: click the “Standard configuration” button,
    946LUX12 software: check all six boxes in the screen.
  - cancel the local supervision function.

6 - Operation, servicing and maintenance

6.1 - Network supervision

6.1.1 - Network supervised with the Alcatel network management system

The network supervision station receives all alarms generated by the network elements. With the supervision applications, the operator analyzes the alarms and initiates appropriate corrective maintenance operations (see the procedures later in this document).

If the results of the performance monitoring programs are unsatisfactory, analysis functions can be used to locate the hops involved, and perform the necessary maintenance operations to restore original performance levels on the link.

6.1.2 - Network without the Alcatel network management system

On detecting a failure, the operator must go to one of the stations on the failed hop, equipped with a portable PC (craft terminal, CT), connect it to the F interface of the main IDU and run the AS&C, RTP and G.821 or G.784 (extended 946LUX11/946LUX12, or standard 946LUX40) applications to analyze the fault and determine which element has failed.

6.2 - Preventive maintenance

This kind of maintenance is performed when carrying out a corrective maintenance procedure, or on a routine visit to all the equipment of the station. It involves inspecting the units and their interconnections (connectors, cables, sockets, etc), as well as all the equipment involved in the operation of the station (power supply, batteries, solar panels, earthing, etc) and, in case of doubt, checking the suspect parts, taking all the necessary precautions to avoid disrupting data transmission over the link (using the 1+1 switching capability if available, etc). Preventive maintenance is recommended for the following units:

- MCU board battery:
  - every 7 to 10 years (see § 7.15.1).
- Fan unit:
  - every 2 to 3 years or in case of alarm.

    The system uses 2 fans. An alarm occurs when one fan fails or when the IDU exceeds the normal operating temperature. Both fans should be changed at the same time (see § 7.15.2).

If a main IDU 3CC0897xxxxB and an extension 3CC08920xxxB are used in accordance with chapter 3.4 recommendations (1U separation) this preventive maintenance could be forgot.

6.3 - Corrective maintenance

6.3.1 - Methodology

- Check the condition of the cables, connectors and connections, and of all the components that are involved in the operation at the station (power supply, battery, earthing, etc),
- Analyze the IDU alarm indicators,
- If alarms are indicated, locate them using the configuration and operation programs installed on a CT and analyze them and carry out the necessary corrective procedures (change of unit, module, cable, etc).

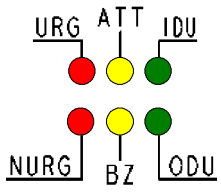
6.3.2 - Analyzing IDU alarm indicators

The alarm indicators on the IDUs provide an indication of the network elements that are affected by alarms:

Besides the information given below (classic IDU and Light IDU indications), the URG, NURG and ATT LEDs can flash in a particular way to indicate equipment status information, as explained below:

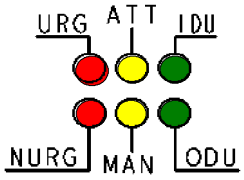
- URG, NURG and ATT LEDs flashing in sequence: change of configuration or software startup phase (lasts about two minutes),
- URG, NURG and ATT LEDs flashing simultaneously: the configurations of the IDU and ODU do not match. The equipment is waiting for the operator to operate the IDU/ODU-ACO switch.

Meaning of indications on the main IDU (classic):



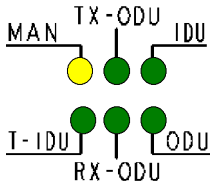
Red LED	⇒ URG	ON :	Urgent alarm indicated
Red LED	⇒ NURG	ON :	Non-urgent alarm indicated
Yellow LED	⇒ ATT	ON :	Alarm attended, acknowledged by operator
Yellow LED	⇒ BZ	ON :	Busy (telephone ESC busy)
Green LED	⇒ IDU	ON :	No alarm on main IDU or valid configuration
Green LED	⇒ ODU	ON :	No alarm on main outdoor unit or valid configuration

Meaning of indications on the Light IDU:



Red LED	⇒ URG	ON :	Urgent alarm indicated
Red LED	⇒ NURG	ON :	Non-urgent alarm indicated
Yellow LED	⇒ ATT	ON :	Alarm attended, acknowledged by operator
Yellow LED	⇒ MAN	ON :	Active remote control signal present
Green LED	⇒ IDU	ON :	No alarm on IDU equipment or valid configuration
Green LED	⇒ ODU	ON :	No alarm on ODU equipment or valid configuration

Meaning of indications on the extension IDU (if installed):



Yellow LED	⇒ MAN	ON :	Remote control from the supervision function
Green LED	⇒ T-IDU	ON :	Traffic on extension multiplexer/demultiplexer
Green LED	⇒ T-IDU	OFF :	Traffic on main multiplexer/demultiplexer
Green LED	⇒ TX-ODU	ON :	Transmit traffic on standby ODU
Green LED	⇒ TX-ODU	OFF :	Transmit traffic on main ODU
Green LED	⇒ RX-ODU	ON :	Receive traffic on standby ODU
Green LED	⇒ RX-ODU	OFF :	Receive traffic on main ODU
Green LED	⇒ IDU	ON :	No alarm on extension IDU
Green LED	⇒ ODU	ON :	No alarm on standby ODU

Note: One green LED ⇒ IDU / ODU OFF indicates an alarm on the equipment.

6.3.3 - Using the operating programs to trace alarms

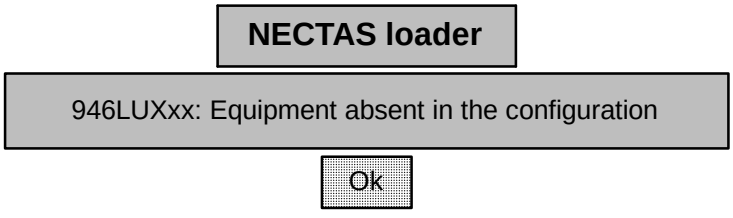
- Connect the CT to port F via a cable ref: 1AB054120027,
- Open the NECTAS application;

If the following message appears on screen:



1) Check that the IDU is indeed powered up and that the initialization phase is finished,
2) Check that the cable from port F is connected to serial port A on the computer,
3) Check the cable (pin-to-pin links: 1 ↔ 1, 2 ↔ 2, ...9 ↔ 9),

If the following message appears on screen:



4) Click "**Ok**" and download the 946LUXxx system software,

5) Otherwise, change the IDU.

- Open “Applications...” and enter the correct name and password (§ 4.2.4),
- Open the “Radio Transmission Parameters” application,

  1) Analyze the received power level or levels,

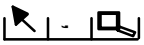
  2) Analyze the transmit/receive alarm or alarms,

  3) Analyze the BER readings.

  This will enable you to identify the most probable causes of the alarms in the equipment or in the far end equipment.
- Open the “AS&C” program.

  Refer to § 4.7 for general information concerning the AS&C application and how to use it.

  1) Reminders:

  - Cursor shapes:  when the cursor changes to a hand, clicking the left mouse button calls up the alarm message, and when it changes to a magnifying glass, the same action can be used to zoom in on the corresponding subsystem.
  - Alarm criteria:

  Caution: the alarm criteria can be modified on configuration (§ 4.5.5); the colouring of the LEDs encodes the criteria as follows:

  red: urgent alarm,

  yellow: non-urgent alarm,

  green: no alarm or alarm disabled,

  magenta: status.

  2) Method:

  - Analyze the alarm synthesis line (see § 4.2.3), the TC square should be green; if the CD square is in reverse video, the configuration has been lost; enter all of the station configuration (see § 5.2.1 to 5.2.4),
  - “Zoom” in on the subsystem(s) with the alarm conditions.

Each time the “Alarm synthesis” message appears on screen, use the “Zoom” function to obtain more details on the alarm type or types on the module affected. This alarm will not be documented in the rest of this document.

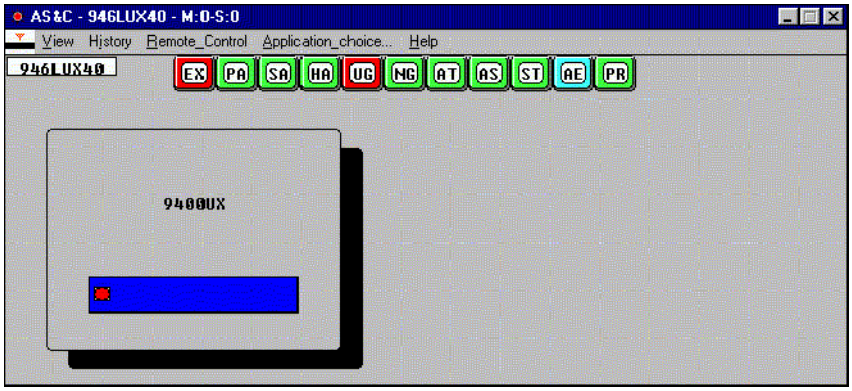
6.3.4 - Corrective servicing

In a 1+1 configuration with a MUX/DEMUX protection, it is the spare MUX/DEMUX that is chosen by default at the starting even if an alarm is present on this equipment. Then, you must force the main MUX/DEMUX with the remote control and then change the spare unit and go back to automatic mode with the remote control.

With the analog cable interface, the G826 counters are not correctly evaluated. The G821 counters must be used (except LUX40).

6.4 – Screens displayed on opening the AS&C application

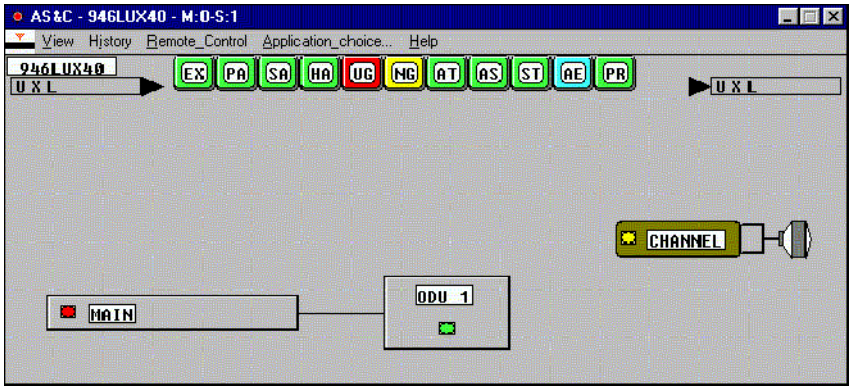
9400 UX with 946LUX12, and 946LUX40 software only



946LUX40 opening screen

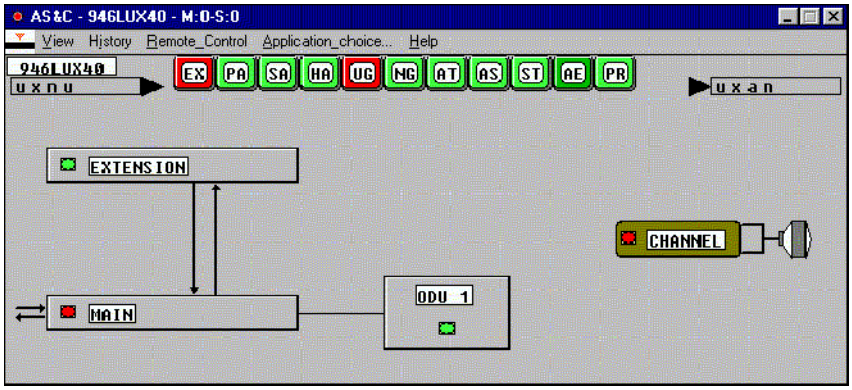
9400 UX

Look through the following screens to find yours and refer to the appropriate section by “zooming” in on the channel. **Exception:** If the alarm appears only in the “main” IDU, “zoom” in on this unit and refer to § 6.6.1.



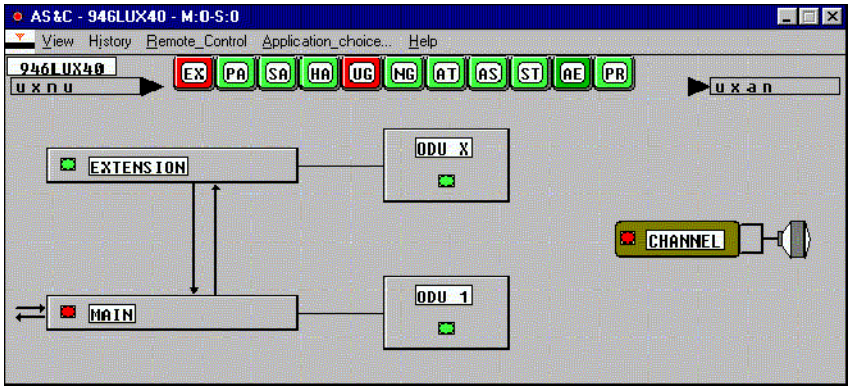
1+0 configuration

See § 6.5.1



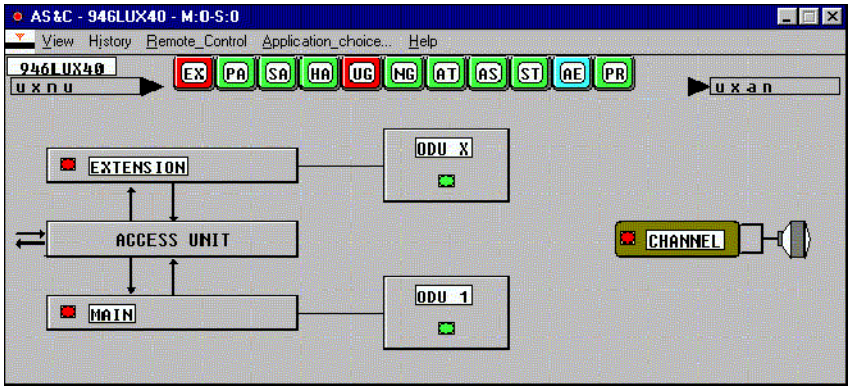
1+0 configuration with ESC extension

See § 6.5.1



1+1 configuration without Mux protection

See § 6.5.2



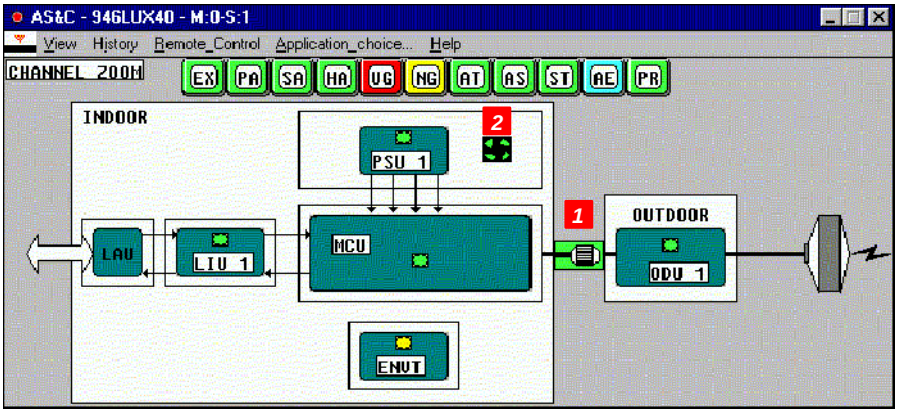
1+1 configuration with Mux protection

See § 6.5.3

6.5 – Analyzing the “zoom” displays of the functional screens

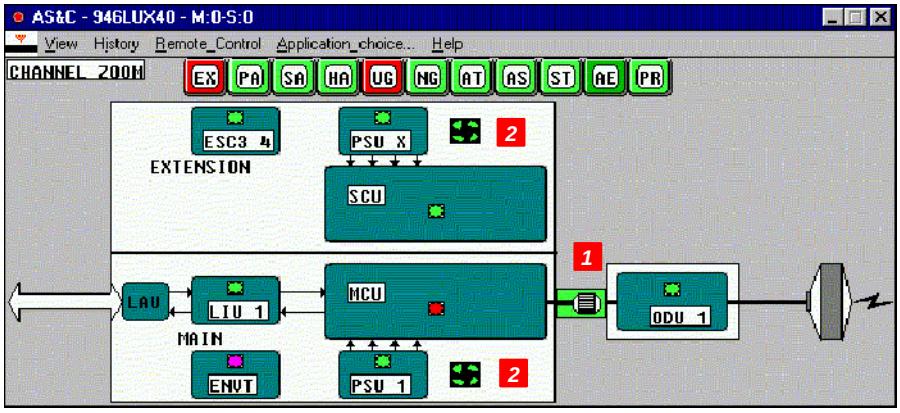
6.5.1 - 1+0 configurations

9400 UX channel configuration with 946LUX12, and 946LUX40 software



Channel “Zoom” in 1+0 classic or light configuration

ODU ⇒ § 6.6.2
LIU function ⇒ § 6.6.3
PSU module ⇒ § 6.6.4
ENVT module ⇒ § 6.6.6
MCU board ⇒ § 6.6.7

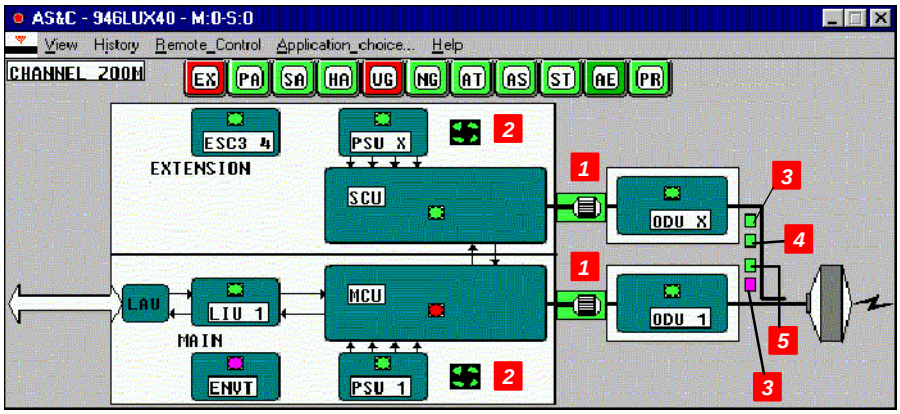


1+0 configuration with ESC extension

ODU ⇒ § 6.6.2
LIU module ⇒ § 6.6.3
PSU module ⇒ § 6.6.4
ESC 3 4 module ⇒ § 6.6.5
ENVT module ⇒ § 6.6.6
MCU board ⇒ § 6.6.7
SCU board ⇒ § 6.6.8

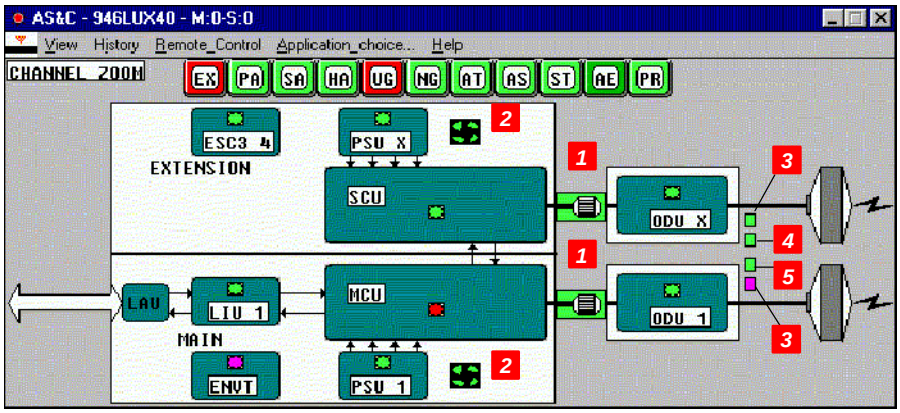
Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
1	CABLE	URG alarm	*Alarm.* Cable problem: disconnected, incorrectly fastened, cut, short-circuited or bad contact inside a connector. If the cable is short-circuited, the IDU PSU is probably generating an alarm and crackling. Check that the power supply alarm is cleared when you disconnect the cable. Repair the cable, check its connectors, change the defective items or the cable.
2	AIR DISPLACE-MENT UNIT	NURG alarm	*Air displacement unit* (fan) *or temperature alarm.* Due to fan failure or too high temperature reach. If they are rotating well, it means that the IDU is installed in a too high temperature environment. One or two fans failed; change the fans or the IDU (see § 7.15.2).

6.5.2 – 1+1 configurations without MUX protection



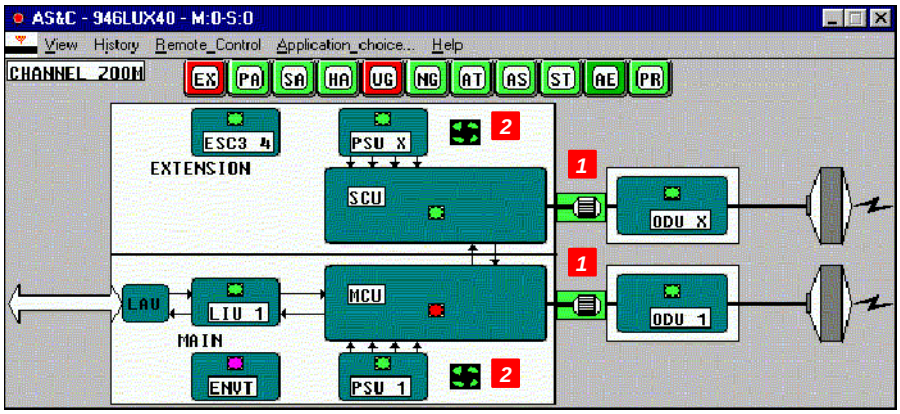
1+1 HSB configuration
One antenna

ODU ⇒ § 6.6.2
LIU module ⇒ § 6.6.3
PSU module ⇒ § 6.6.4
ESC 3 4 module ⇒ § 6.6.5
ENVT module ⇒ § 6.6.6
MCU board ⇒ § 6.6.7
SCU board ⇒ § 6.6.8



1+1 HSB configuration
Two antennas

ODU ⇒ § 6.6.2
LIU module ⇒ § 6.6.3
PSU module ⇒ § 6.6.4
ESC 3 4 module ⇒ § 6.6.5
ENVT module ⇒ § 6.6.6
MCU board ⇒ § 6.6.7
SCU board ⇒ § 6.6.8



1+1 configuration with
frequency diversity

ODU ⇒ § 6.6.2
LIU module ⇒ § 6.6.3
PSU module ⇒ § 6.6.4
ESC 3 4 module ⇒ § 6.6.5
ENVT module ⇒ § 6.6.6
MCU board ⇒ § 6.6.7
SCU board ⇒ § 6.6.8

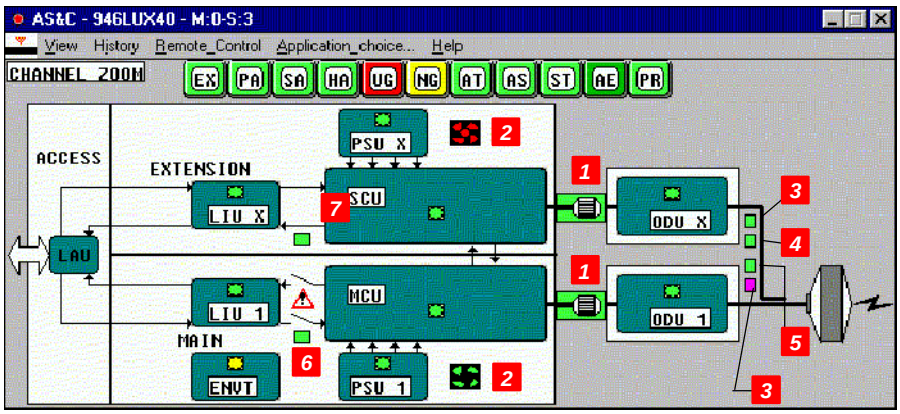
N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
1	CABLE	URG alarm	*Alarm.* Cable problem: disconnected, incorrectly fastened, cut, short-circuited or bad contact inside a connector. If the cable is short-circuited, the IDU PSU is probably generating an alarm and crackling. Check that the power supply alarm is cleared when you disconnect the cable. Repair the cable, check its connectors, change the defective items or the cable.

Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
2	AIR DISPLACE-MENT UNIT OR TEMPERATURE	NURG alarm	*Air displacement unit* (fan) *or temperature alarm.* One or two fans failed; change the fans of the IDU (see § 7.15.2). Make sure the room temperature does exceed 55 degrees Celsius for the IDU.
3	TX X TX 1	Status	*Transmitter switched indication.* Indicates the transmitter working in HSB mode.
4	COMMON LOSS	URG alarm	*Common loss alarm.* In HSB mode, indicates a switchover on alarm if the switchover on alarm option has been enabled, see § 4.5.1. **CAUTION**, this alarm can be cleared only by shutting down and starting up the equipment. NOTA: This alarm is inhibited.
5	MANUAL MODE	NURG state	*Forced switching mode at transmission.* Indicates a switchover forced by remote control on the normal or stand-by transmitter.

REMOTE CONTROLS

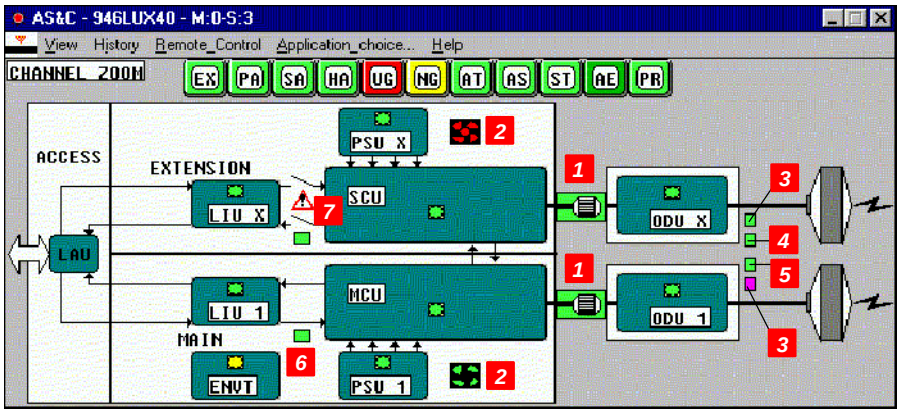
Remote control activation/deactivation label	Effect of activation
TX 1 [Start]/[End]: manual lockout validated/inhibited	Manual lockout on channel 1 in transmission (in 1+1 HSB conf.). **Remote station**: Automatic disabling after five minutes
TX X [Start]/[End]: manual lockout validated/inhibited	Manual lockout on channel X in transmission (in 1+1 HSB conf.). **Remote station**: Automatic disabling after five minutes

6.5.3 - 1+1 configurations with MUX protection



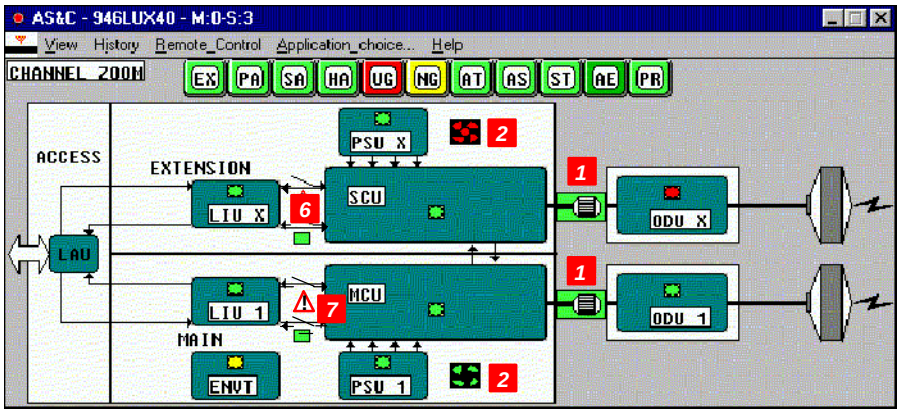
1+1 HSB configuration
One antenna with
Mux protected

Multiplexing and demultiplexing by the main IDU's MCU board



1+1 HSB configuration Two antennas with Mux protected

Multiplexing and demultiplexing by the main IDU's MCU board



1+1 frequency diversity configuration with Mux protected

Multiplexing and demultiplexing by the extension IDU's SCU board

ODU ⇒ § 6.6.2	ESC 3 4 module ⇒ § 6.6.5
LIU module ⇒ § 6.6.3	ENVT module ⇒ § 6.6.6
PSU module ⇒ § 6.6.4	MCU and SCU boards ⇒ § 6.6.7

Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
1	CABLE	URG alarm	*Alarm.* Cable problem: disconnected, incorrectly fastened, cut, short-circuited or bad contact inside a connector. If the cable is short-circuited, the IDU PSU is probably generating an alarm and crackling. Check that the power supply alarm is cleared when you disconnect the cable. Repair the cable, check its connectors, change the defective items or the cable.
2	AIR DISPLACE-MENT UNIT OR TEMPERATURE	NURG alarm	*Air displacement unit* (fan) *or temperature alarm.* One or two fans failed; change the fans or the IDU (see § 7.15.2). Make sure the room temperature does exceed 55 degrees Celsius for the IDU.
3	TX X TX 1	Status	*Transmitter switched indication.* Indicates the transmitter working in HSB mode.

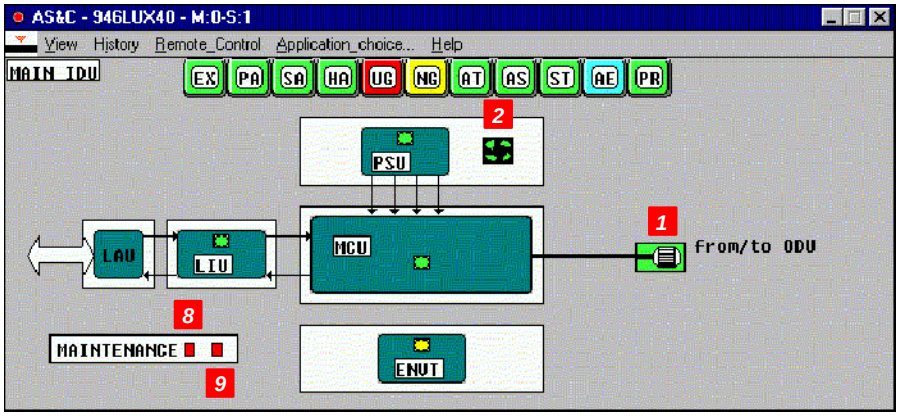
Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
4	COMMON LOSS	URG alarm	*Common loss alarm.* See table in § 6.5.2 **CAUTION**, this alarm can be cleared only by shutting down and starting up the equipment. NOTA: This alarm is inhibited.
5	MANUAL MODE	NURG state	*Forced switching mode at transmission.* See § 6.5.2
6	SWITCHING MODE	NURG state	*Switching mode manual.* Manually selected by remote control on the multiplexing/demultiplexing equipment (MCU of the main IDU or SCU of the extension IDU).
7	SCU SWITCH-ING MCU SWITCH-ING	NURG state	Indicates multiplexing/demultiplexing equipment not used.

REMOTE CONTROLS

Remote control activation/deactivation label	Effect of activation
TX 1 [Start]/[End]: manual lockout validated/inhibited	Manual override on channel 1 in transmission (in 1+1 HSB conf.). **Remote station**: Automatic disabling after five minutes
TX X [Start]/[End]: manual lockout validated/inhibited	Manual override on channel X in transmission (in 1+1 HSB conf.). **Remote station**: Automatic disabling after five minutes
MUX/DEMUX [Start]/[End]: normal lockout validated/inhibited	“Normal” multiplexer/demultiplexer override (1+1 configuration with MUX protection). **Remote station**: Automatic disabling after five minutes
MUX/DEMUX [Start]/[End]: standby switching validated/inhibited	”Standby” multiplexer/demultiplexer override (1+1 configuration with MUX protection). **Remote station**: Automatic disabling after five minutes

6.6 – Analyzing units and modules

6.6.1 – Main, extension or Light IDU



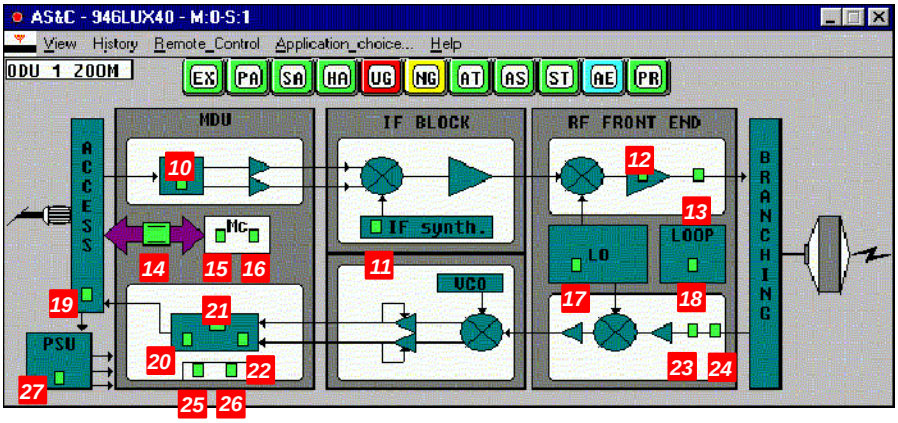
Main, extension or Light IDU

N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
1	CABLE	URG alarm	See § 6.5.1
2	AIR DISPLACE-MENT UNIT	NURG alarm	See § 6.5.1
8	UNDELAYED MAINTENANCE	URG alarm	*Undelayed* (prompt) *maintenance alarm.* See definition in § 4.5.4; use the History log programs to detail the alarms. Having traced and identified the fault, repair as appropriate. **Stop alarm**: “Remote_Controls/All/Maintenance alarms: Reset”.
9	DELAYED MAIN-TENANCE	URG alarm	*Delayed* (deferred) *maintenance alarm.* See definition in § 4.5.4; use the History log programs to detail the alarms. Having traced and identified the fault, repair as appropriate. **Stop alarm**: “Remote_Controls/All/Maintenance alarms: Reset”.

REMOTE COMMANDS

Remote control activation/deactivation label	Effect of activation
MAINTENANCE ALARMS: reset	Reset of maintenance alarm counters and cancellation of URG(ent) alarms maintenance alarms

6.6.2 - Main and standby outdoor units (ODU)



Main outdoor unit (ODU)

On installation, any inconsistency between the presence of the RF loop and its declaration will cause numerous alarms to be generated, and a transmit or receive frequency value error.

Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
10	SYNCHRONIZING	URG alarm	*Block synchronization alarm.* Bit rates not supported by the modulator. Check the configuration of the installation parameters (§ 4.4.1) and operation parameters (§ 4.5.1). Check the number of 2 or 34 Mbit/s inputs against the installation sheet. Otherwise, replace the ODU (see § 6.8).
11	IF SYNTHETIZER	URG alarm	*Tx synthesizer alarm.* Transmit IF synthesizer unlocked. On installation: check the transmit frequency, duplex separation and the compatibility of the frequency with the ODU's RF band. Otherwise replace the ODU (see § 6.8).
12	AMPLIFIER	R_C State	*Amplifier mode.* Power transmitted by channel 1 or X amplifier cut-off.
13	OUTPUT POWER	URG alarm	*Output power alarm.* Fall of output power from the channel 1 or X transmit amplifier; check that there is no transmitted power cut-off remote control; otherwise, replace the ODU (see § 6.8).
14	UNIT MISSING	URG alarm	*Unit missing* or no dialogue. Absence of ODU or dialogue with the modem's microcontroller. Check that the link's coaxial cable is correctly connected and not damaged; check all of the configuration, otherwise replace the ODU (see § 6.8).
15	GENERAL ALARM	URG alarm	Cannot access a hardware interface (component or bus). Inconsistency or loss of inventory data. If, after new power on, the alarm persists, replace the ODU (see § 6.8).
16	CONFIGURATION ERROR	URG alarm	*Configuration error.* Check the configuration, in particular the declaration of the RF loop. In the 15 and 18 GHz frequency bands, incoherence between transmit amplifier and transmitted power declaration. No respect of duplex separation. Frequency band problem. Reversal of Tx and Rx frequencies. Otherwise replace the ODU (see § 6.8).

Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
17	RF SYNTHETIZER	URG alarm	*RF synthesizer alarm.* Incompatibility between the transmit and receive frequency values requested and those supported by the channel 1 or X synthesized RF LO; check the transmit and receive frequencies and the duplex separation. Otherwise replace the ODU (see § 6.8).
18	RF LOOPBACK	R_C State	*Local loop*(back). RF local loop on normal channel 1 or standby channel X (option).
19	BB LOOP	R_C State	*Local loop*(back). Local baseband loopback at the channel 1 or standby channel X ODU input.
20	FEC INHIBITION	R_C State	*Forward Error Correction inhibition.* Forward error corrector code disabled on normal channel 1 or standby channel X receive.
21	CARRIER UNLOCKED	URG alarm	*Carrier unlocked or demodulator loss of signal.* No signal at the demodulator input. Check the frequency of the channel's transmitter and the transmit and receive bit rates; see below for information on analyzing propagation problems.
22	EQUALIZER INHIBITION	R_C State	*Receive equalizer inhibition.* Receive equalizer disabled on normal channel 1 or standby channel X.
23	PROPAGATION	URG alarm	*Propagation alarm.* Propagation alarm on channel 1/X: Depending on the values selected in "Operation parameters/Thresholds/Propagation"; check below for information on analyzing propagation problems.
24	RECEIVE POWER	URG alarm	*Receive power alarm.* The field received by normal channel 1 or standby channel X is too weak: check the frequency and transmitted power of the channel's far-end transmitter; see below for information on analyzing propagation problems.
25	EARLY WARNING	URG alarm	*Early switching request on received field or bit error ratio .* on the RF channel: Depending on the values set in "Operation parameters/thresholds/switching thresholds" (switching request inoperative in 1+0 version; see below, analysis of propagation problems)
26	LOW BER	URG alarm	*Performance switching request on low error ratio.* on RF channel: According to values set in "Operation parameters/thresholds/switching thresholds" (switching request inoperative in 1+0 version; see below, analysis of propagation problems).
27	PSU ALARM	URG alarm	*Power supply alarm.* At least one of the secondary power supply voltages on ODU 1 or standby X is faulty. Replace the ODU (see § 6.8).

PROPAGATION ALARM

This alarm appears when the receive power is weaker than the threshold defined in "Operation parameters/Thresholds/Propagation".

1) Open the "Radio Transmission Parameters" application and read the received power value.

2) If the power value is compatible with the link budget, check that the nominal power and threshold declared in the "Thresholds/Propagation" menu of the "Operation Parameters" application are correct, correct if necessary and check that the alarm disappears.

3) Otherwise, switch to local supervision mode to check far-end station B, if possible:

- Check the received power level at B.
- If this differs significantly from that obtaining at A, first of all check that stations A and B have not switched to the HSB standby channel (10 dB loss each time), check that there is no manual switching to the standby channels and that the power level at B has not been reduced (23–38 GHz ranges if available through software, Operation Parameters/output power), otherwise the far-end ODU has probably failed. The same symptom may be caused by accidentally cutting off the amplifiers of station B by remote control.
- If the power levels are not very different, it may be that:

  – One of the antennas is not accurately pointed (check the AGC voltages): repoint and check that the alarm disappears.

  – Abnormal conditions are hampering propagation: presence of a sheet of water or waterlogged ground along the path of the link, reflections from one or more layers of the troposphere caused by abnormal stratification of the atmosphere, a fairly frequent condition in tropical climates, natural screen or construction strongly masking the first “Fresnel ellipsoid”, rain storms, etc.

4) Remember to cancel the local supervision mode.

EARLY WARNING ALARM

If just the EARLY WARNING alarm is lit, check the dBm level of the alarm threshold; it should be less than the propagation threshold; correct if necessary via “Operation Parameters/Thresholds/Switching thresholds”.

EARLY WARNING, LOW BER, DCP, CARRIER UNLOCKED AND PROPAGATION ALARMS

When propagation and performance alarms are generated at the same time, it may be that:

- There is a receive frequency problem in station A, or a transmit frequency problem in far-end station B, or a bandwidth problem (bit rate not the same in both stations). Check and correct as necessary in station A.

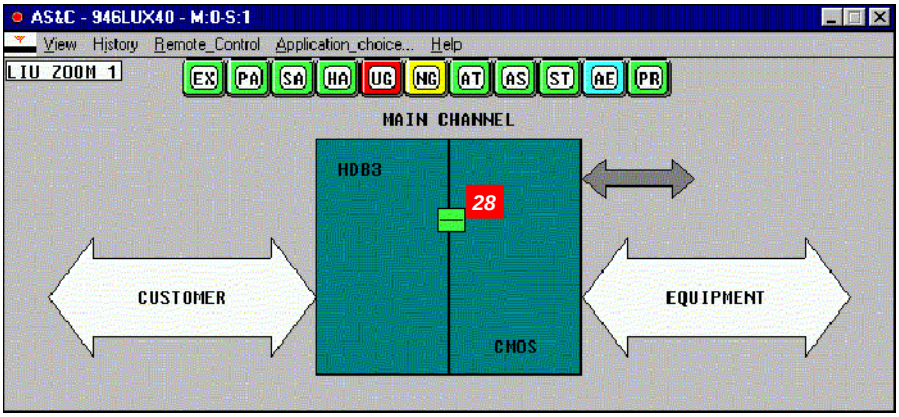
  Check the equivalent components of the far–end equipment by switching to local supervision mode when possible (there may be a EF: Equipment Fault in the far-end equipment preventing the far-end station from being viewed).
- There are abnormally bad propagation conditions.
- The antennas need accurately pointing.
- The far-end ODU has failed.

It will probably be necessary to go to the far-end equipment to complete the diagnostic.

REMOTE CONTROLS

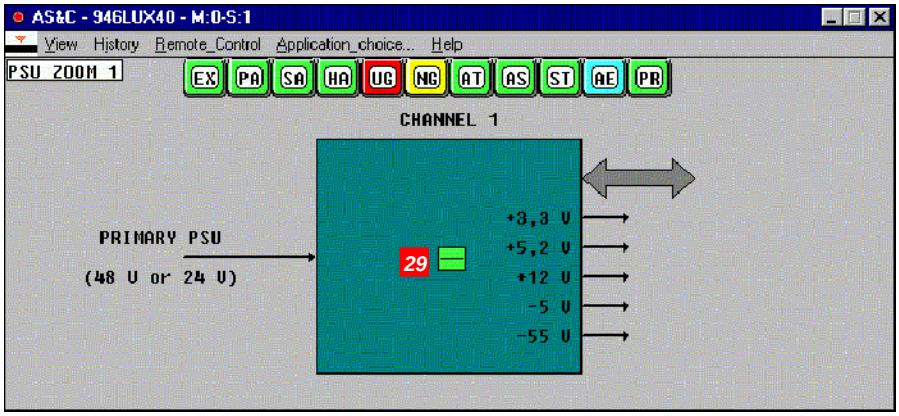
Remote control activation/deactivation label	**Effect of activation**
AMPLIFIER [Start]/[End]: Power Off/On	Power amplifier for each ODU. **Remote station**: Automatic disabling after five minutes
BB OUTDOOR [Start]/[End]: local loop validated/inhibited	Local baseband loopback at the input of each ODU. **Remote station**: Remote control rejected
EQUALIZER [Start]/[End]: inhibition/validation	Equalizer for each ODU
FEC [Start]/[End]: inhibition/validation	Error correcting code for each ODU
RF [Start]/[End]: local loop validated/inhibited	Local RF loopback for each ODU **Remote station**: Remote control rejected

6.6.3 – LIU module or function



N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
28	CONFIGURATION ERROR	URG alarm	Classic IDU: Check that the LAU and LIU boards are correct for the configuration (number of inputs, bit rates), and modify if necessary. Light IDU: Check the configuration of the light MCU board. Modify if necessary. Otherwise, change the classic or Light IDU (see § 6.7). **Caution!** Inconsistencies between the 2 Mbit/s tributary impedance hardware configuration and the software declaration of these tributaries will not cause alarms in classic or light 9400 UX equipment.

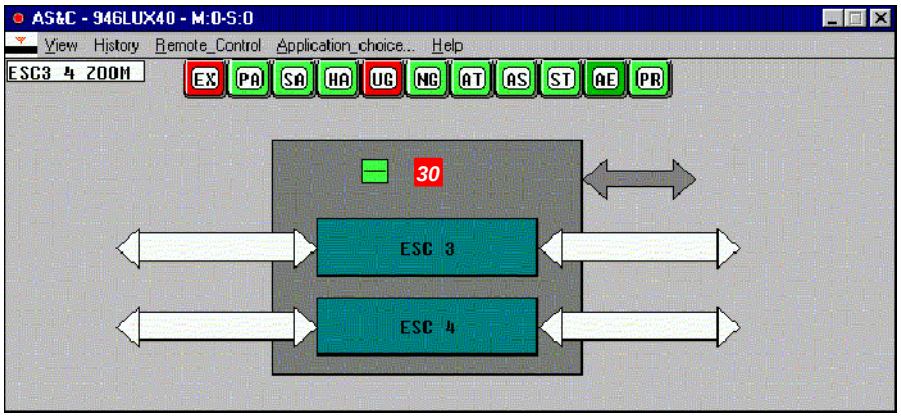
6.6.4 – Power supply module (PSU)



N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
29	GENERAL ALARM	URG alarm	At least one of the secondary power supply voltages of the main or extension IDU is faulty. A short circuit may have occurred in the IDU/ODU link cable; disconnect the cable and check whether the alarm disappears or remains. If the alarm disappears, check the cable and repair it if necessary. Otherwise, use the “Analogue measurements” application (extended 946LUX11/12) to check the state of the voltages. If the alarm persists, replace the IDU (see § 6.7)

6.6.5 - ESC 3 and 4 module

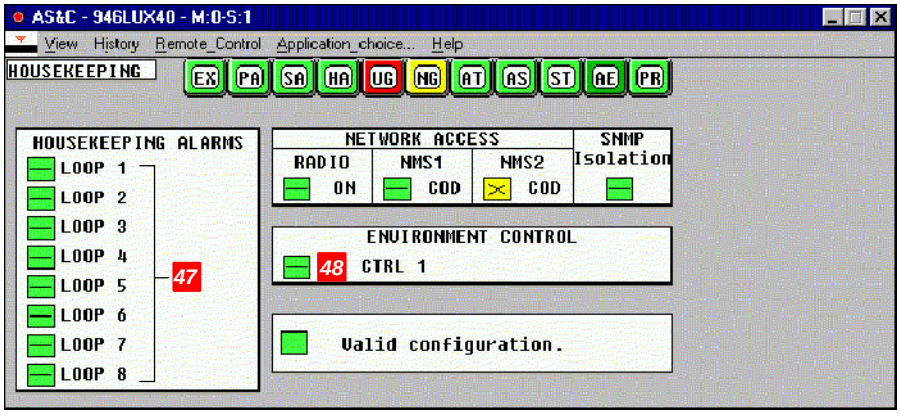
Module omitted from light 9400 UX and 9400 UX 1+0 equipment with no ESC extension



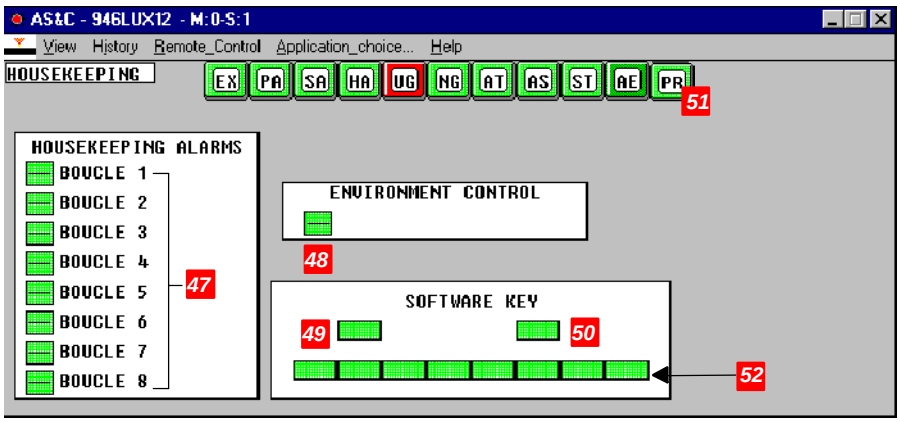
N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
30	CONFIGURATION ERROR	URG alarm	Inconsistency between the daughter board(s) installed and the configuration declared on installation and the definition of ESCs 3 and 4. Check for the physical presence and installation conditions of the ESC daughter board(s) (see § 7.8). If necessary, change the daughter board(s). Otherwise change the IDU (see § 6.7).

6.6.6 - ENVT module

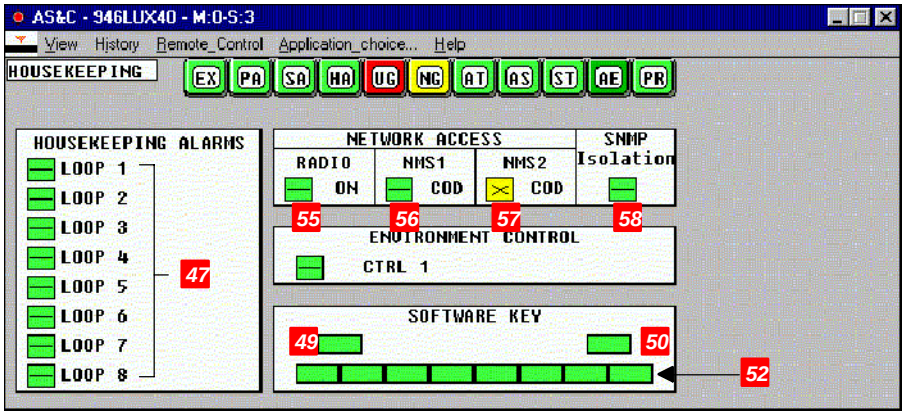
Light 9400 UX under 946LUX40



Classic 9400 UX under 946LUX12



Classic 9400 UX under 946LUX40



Any key absent alarm will prevent you modifying the configuration.

N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
47	HOUSEKEEPING 1 to 8	NURG state	Station environment dry loop alarms: depending on how each loop is assigned (see § 7.10).
48	ENVIRONMENT CTRL 1	R_C State	Environment remote control number 1 active.
49	SOFTWARE KEY MISSING	URG alarm	Check that the software key is correctly positioned on the SKU/MTN connector. If not, replace it and if the alarm persists, replace the IDU (see § 6.7).
50	SOFTWARE KEY ALARM	URG alarm	Alarm that appears when you change key, when the new key does not have the same characteristics as the old key. Confirm the new key by opening the “Operation parameters” application and down-loading the configuration (File/Send). The key alarm should disap-pear.
51	PROVISIONING REFUSED	URG alarm	Software key is incompatible with equipment. Verify coherence between equipment configuration and those permitted by the key. Otherwise replace the IDU (see § 6.7). This alarm can also appear if changes are made to the “Installation parameters” while alarm *49* or *50* is present. Reset the supervision function.
52	TERM. SHUTDOWN IN 24, 21, ..., 3H	URG alarm	Time remaining in multiples of three hours before the link will be cut off because of a key alarm. Replace a key.
55	RADIO PORT	URG alarm	Radio link cut off. To determine the cause of the fault, examine the other equipment alarms, starting from the highest alarm grouping level.
56	NMS1 PORT	URG alarm	Cable on NMS1 port not connected or defective or bad router configuration.
57	NMS2 PORT	URG alarm	Cable on NMS2 port not connected or defective.
58	MANAGER ISOLATED	URG alarm	The manager is not polling the network elements after being declared (contact the manager for confirmation).

REMOTE CONTROLS

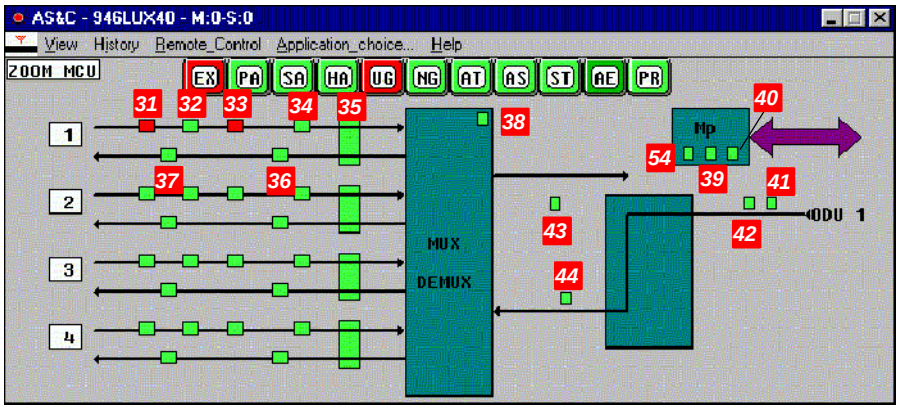
Remote control activation/deactivation label	Effect of activation
EQT RESTART: equipment restart	Restarts the equipment
RC 1 [Start]/[End]: Housekeeping 1 RC enabled/disabled	Housekeeping remote control

The way the ENVIRONMENT (housekeeping) alarms and ENVIRONMENT (housekeeping) remote controls are assigned depend on the station-specific wiring operations performed on installation (see § 3.6.7 in the Installation section 3).

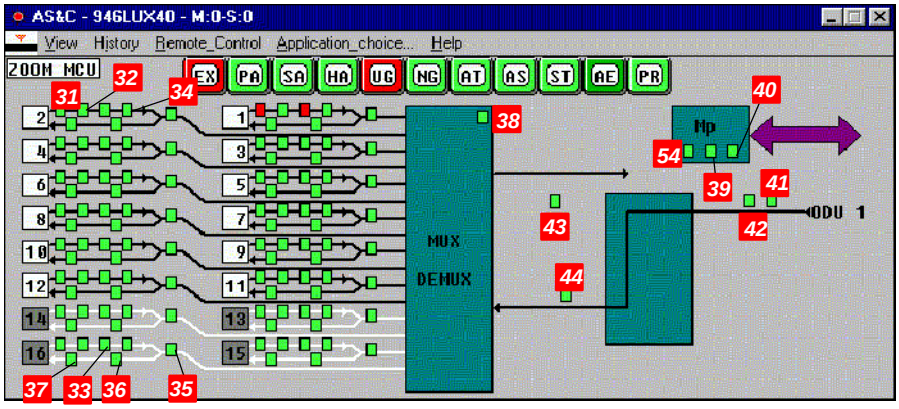
6.6.7 - MCU boards and SCU board in 1+1 configuration with Mux protection

The MCU board screens below are given by way of example:

- Tributaries active: tributary number on white background, lines black.
- Tributaries inactive: tributary number on grey background, lines white.

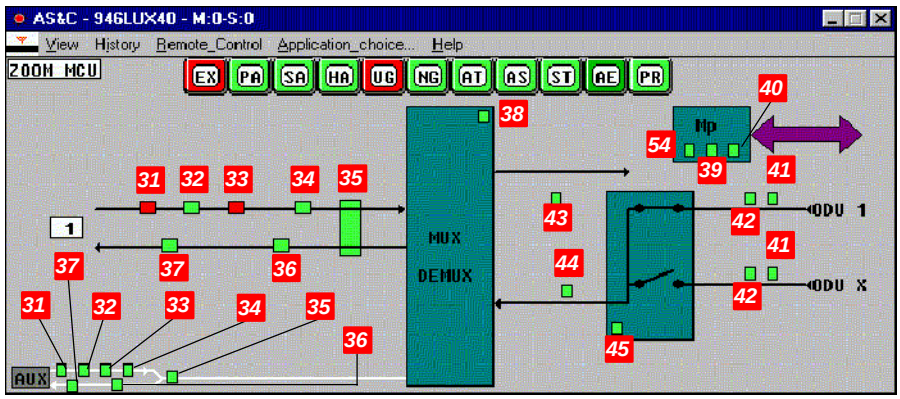


MCU board
1+0 configuration
(Light and classic IDU
4 x 2 Mbit/s
Tributaries1 2, 3, 4 active



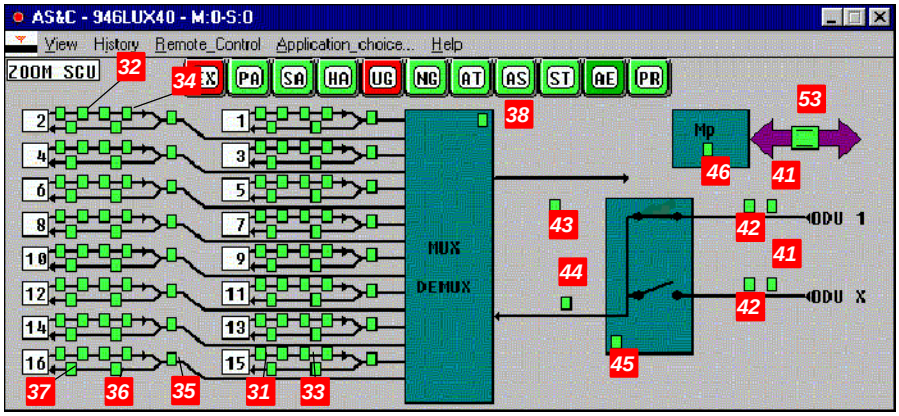
MCU board
1+0 configuration
(Classic IDU)
16 x 2 Mbit/s

Tributaries 1 to 12 active
Tributaries 13 to 16 inactive



MCU board
1+1 configuration
(Classic IDU)
34 + 2 Mbit/s

34 Mbit/s tributary active
auxiliary 2 Mbit/s tributary inactive



SCU board

1+1 configuration with Mux protection

16 x 2 Mbit/s

16 x 2 Mbit/s tributaries active

N°	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
31	TRIB.i or AUX 2MB: SIGNAL LOSS	URG alarm	No input signal for tributary n_i or the auxiliary 2 Mbit/s stream. Check the tributary wiring and for presence of the signal
32	TRIB.i or AUX 2MB: UN-EXPECTED	URG alarm	The signal received at the input of tributary n_i or of the auxiliary 2 Mbit/s bit stream does not match the expected signal (bit rate, code, etc). Check the signal received. **Caution!** Never loop the ports of unused tributaries at the ends of a link, disable them by the software.
33	TRIB.i or AUX 2MB: TX AIS	URG alarm	Insertion of an AIS on transmission over tributary n_i or the auxiliary 2 Mbit/s bit stream if the transmit AIS insertion option is enabled (“Operation parameters/Terminal/AIS/send/*active*”)
34	TRIB.i RC or AUX 2 MB RC: SEQ.INS.	R_C State	Pseudo-random pattern inserted on tributary n_i or the auxiliary 2 Mbit/s stream (only one insertion possible)
35	TRIB.i RC or AUX 2 MB RC: DIST. LOOP	R_C State	Remote baseband loopback (ODU port) on tributary n_i or auxiliary 2 Mbit/s stream
36	TRIB.i RC or AUX 2 MB RC: BER MEAS.	R_C State	Bit error ratio measurement on tributary n_i or auxiliary 2 Mbit/s stream
37	TRIB.i or AUX 2MB: RX AIS	URG alarm	Generation of an AIS in reception on tributary n_i or the auxiliary 2 Mbit/s bitstream if the receive AIS insertion option is active (“Operation parameters/Terminal/AIS/reception/*active*”)
38	MUX_DEMUX ALARM MUX_DEMUX GEN. ALARM	URG alarm	Multiplexer of main IDU MCU board or extension IDU SCU board failed in 1+1 configuration with MUX protection. Replace the failed IDU (see § 6.7)
39	MAIN UNIT	URG alarm	General main IDU failed alarm; check the state of inventory memory (see § 6.7) ; replace the IDU (see § 6.7)
40	SUPERVISION UNIT	URG alarm	MCU board of main IDU failed; replace the IDU (see § 6.7)
41	RX 1 HIGH BER RX X HIGH BER	URG alarm	Priority switching request on high bit error ratio or loss of receive mode frame sync on channel 1 or X. Check the ODU alarms
42	RX 1 LOSS OF FRAME RX X LOSS OF FRAME	URG alarm	Loss of receive frame sync on channel 1 or X caused by a receive problem resulting in the transmission of an AIS. Check the received field level and the frequency configuration of the RF channel. Check the ODU alarms

Nº	MESSAGE	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
43	LOCAL LOOPBACK	R_C State	Local baseband loopback on main IDU or extension IDU
44	LINK IDENTITY CODE	URG alarm	Bad frame identification code received on switched channel. Check the consistency of the transmit and receive code configurations. If necessary, check the received field level (for receipt of a disturbing signal)
45	FORCED SWITCHING MODE	R_C State	Receive channel selected by MCU by manual forcing
53	EXTENSION MISSING	URG alarm	Extension IDU absent or no dialogue with MCU. Check that the link wiring is correctly connected and undamaged; otherwise replace the extension IDU (see § 6.7)
54	CONFIGURATION ERROR	URG alarm	Inconsistency between the bit rates requested by the MCU board and those supported by the LIU board or boards. Check the installation configuration. Not used in the Light 9400 UX version.

TRIBUTARY PROBLEM

- First check the condition and positions of the tributary cables and connectors.
- Check the tributary cross-connect matrices.
- Connect a BER analyzer to one of the active tributaries; send a baseband (BB) local loop remote control via the MCU; check that looped transmission is completed correctly.
- If the BER measurement result in BB loopback configuration is correct, the fault is on the equipment upline from the IDU/ODU system.
- If not:
  - for a 1+0 or 1+1 configuration without Mux protection, change the main IDU (see § 6.7), and check that the fault disappears;
  - for a 1+1 configuration with Mux protection, switch over to the alternative equipment;
  - if the fault disappears, change the first IDU;
  - if the fault persists, change one IDU, switch over to that IDU and check that the fault disappears. If not, check the measuring instruments.

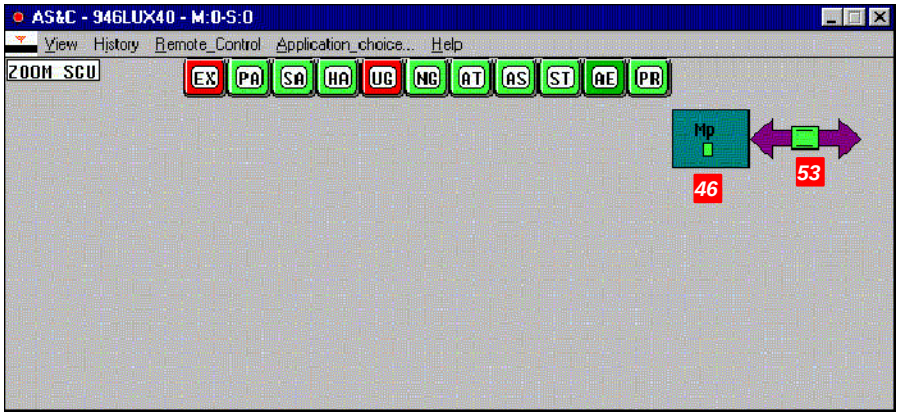
REMOTE CONTROLS

Remote control activation.deactivation label	Effect of activation
MCU>TRIB. x [Start]/[End]: pseudo rand. seq. ins. valid./inhib.	Insertion of a pseudo random pattern on tributary N_ x (for tests on an active tributary)
MCU>AUX. [Start]/[End]: pseudo rand. seq. ins. valid./inhib.	Insertion of a pseudo-random pattern on the 2 Mbit/s auxiliary bit stream (with the 34 Mbit/s rate)
MCU>TRIB. x [Start]/[End]: BER measure validated/inhibited	Measurement of bit error ratio on tributary N_ x (using the pseudo-random bit pattern above on an active tributary)
MCU>AUX. [Start]/[End]: BER measure validated/inhibited	Measurement of bit error ratio on the 2 Mbit/s auxiliary bit stream (using the above pseudo-random bit pattern) (with the 34 Mbit/s rate)
MCU>TRIB. x [Start]/[End]: distant loop validated/inhibited	Remote loopback on tributary N_ x
MCU>AUX. [Start]/[End]: distant loop validated/inhibited	Remote loopback on the 2 Mbit/s auxiliary bit stream (with 34 Mbit/s rate)

Remote control activation.deactivation label	Effect of activation
MCU>BB IDU [Start]/[End]: local loop validated/inhibited	Local baseband loopback at the output of the IDU for main IDU and extension IDU in the case of a 1+1 type configuration with MUX protect **Remote station**: Remote control rejected
MCU>RX 1 [Start]/[End]: manual lockout validated/inhibited	Manual switching override on main receiver (in 1+1 configuration). **Remote station**: Automatic disabling after five minutes
MCU>RX X [Start]/[End]: manual lockout validated/inhibited	Manual switching override on standby receiver (in 1+1 configuration). **Remote station**: Automatic disabling after five minutes

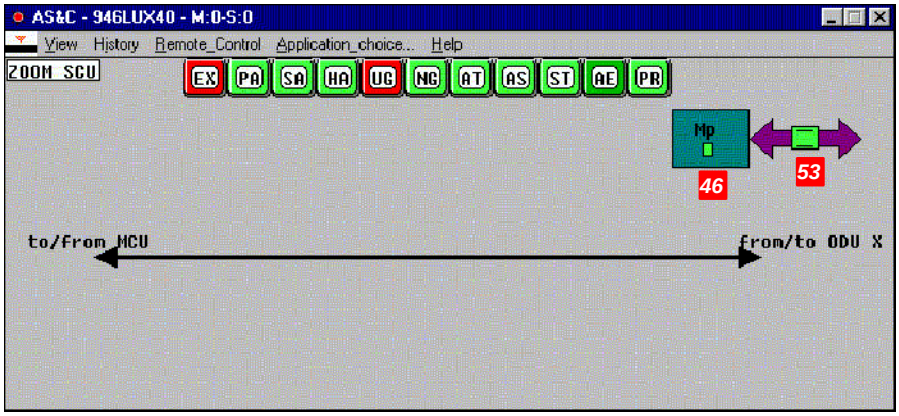
6.6.8 – SCU boards

The two screens below represent the SCU board in a 1+0 configuration with ESC extension and a 1+1 configuration without Mux protection.



SCU board

1+0 configuration with ESC extension



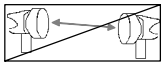
SCU board

1+1 configuration without Mux protection

Nº	ALARM	TYPE	MOST PROBABLE CAUSE AND CORRECTIVE ACTION
46	EXTENSION UNIT	URG alarm	General extension IDU alarm. The extension IDU or the SCU board has failed; replace the extension IDU (see § 6.7)
53	EXTENSION MISSING	URG alarm	No extension unit or absence of dialogue with the MCU. Check that the link wiring is correctly connected and undamaged; otherwise replace the extension IDU (see § 6.7)

6.7 – Changing IDU

6.7.1 - Replacing like for like a Light or classic IDU (1+0 or basic 1+1)



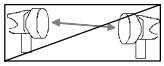
- Power down the Light IDU / main IDU. Disconnect it from the power source.
- Disconnect all the cables connected to the unit you are changing (remember to remove the software key).
- Change the Light IDU / main IDU after checking that the characteristics match (number of inputs, impedance, etc). The impedances of the IDU can be reconfigured in any case (see § 7.2.2) and configuration is not needed with IDU which reference ends with the C letter (3CC.....C) provided cables referenced by Alcatel are used.
- Reconnect all the cables to the unit and the software key.
- Power up the IDU.

  Note: For the Plug-in IDU, simply insert it in the BTS, it will power on automatically.
- Wait for the URG, NUR and ATT LEDs to start flashing simultaneously.
- Configure the IDU from the ODU by setting the IDU/ODU switch to the "ACO/ODU" position. Wait about two minutes for the terminal to be reconfigured. The alarm LEDs should go off and the equipment LEDs should turn green.
- Check that there are no alarms in the AS&C application.
- Update the station's installation form (see model in Appendix 2).

In the event of an error, particularly if the pushbutton is set to the wrong side, re-enter all of the station configuration (see section 5, Commissioning).

Note: *In the case of a basic 1+1 configuration, power* ***down*** *the* ***extension IDU****.*

6.7.2 - Replacing a Light IDU with a classic IDU (1+0 only) and vice versa



The procedure for changing the equipment is the same as for like for like replacement, but, in some cases, the ODU configuration may not be taken into account. Disconnect the IDU-ODU connecting cable and using the information in the installation form (see model in Appendix 2), download the 1+0 configuration from the CT (see section 5 – Commissioning).

- Update the station's installation form (see model in Appendix 2).

6.7.3 - Replacing like for like a main IDU with MUX protection

- Force operation using the remote control "MUX/DEMUX [Start]: forced standby enabled", and switchover to the multiplexer/demultiplexer on the extension IDU's SCU board.
- Force transmission and reception to the standby ODU.
- Power down the main IDU. Disconnect from the power source.
- Disconnect all the cables connected to the unit you are changing (remember to remove the software key from the main IDU). For IDU configured with a MUX protection without Access IDU, only disconnect the part of the cables connected directly to the IDU.
- Change the main IDU after checking that the specifications are the same (number of inputs, impedance, etc).
- Reconnect all the cables to the unit starting with the lower MAIN/EXT cable (60way connector) taking care not to damage the contacts and also the software key from the removed unit.

- Power up the main IDU.
- Wait for the URG, NUR and ATT LEDs to start flashing simultaneously.
- Send the ODUs configuration to the IDU by pushing immediately on the IDU/ODU switch to the “ACO/ODU” position (the lower green LED must be lighted).



Warning: An error of handling at this moment, will result in a connection break with a loss of traffic.

The flashing of the leds is limited to 1 minute. The switch has to be activated as soon as the led starts flashing otherwise the IDU sends its configuration to the ODUs and breaks the connection.

- The alarm leds should turn off and the 2 green leds should light on the main IDU (this stage can take around one minute according to the ODU type).
- Check that there are no alarms in the AS&C application and that the yellow led of the extension IDU is turned off, otherwise cut off the forcing on the receiver of the spare ODU.
- Update the station’s installation form (see model in Appendix 2).

In the event of an error, particularly if the pushbutton is set to the wrong side, re-enter all of the station configuration (see section 5, Commissioning).

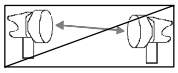
6.7.4 - Replacing like for like an extension IDU

- Check that the transmit and receive multiplexers are switched to the main channel.
- Power down the extension IDU and disconnect it from the power source.
- Disconnect all the cables connected to the unit you are changing.
- Change the unit after checking that the specifications are all the same (number of inputs, impedance, etc).
- Reconnect all the cables to the unit.
- Power up the unit.
- Check that the alarm LEDs are off.
- Check that there are no alarms in the AS&C application.
- Update the station’s installation form (see model in Appendix 2).

6.7.5 - Note on IDU compatibility

It is possible for the two IDUs to have different but compatible configurations for use in the station. In such cases, you must alter the station’s configuration parameters through the “Installation Parameters” and/or “Operation Parameters” applications.

6.8 – Changing ODU



Only in the case of a 1+0 configuration.

- Switch–off the IDU connected to the ODU to be changed.
- Disconnect the IDU–ODU cable.
- Change the ODU after checking that the characteristics of the two ODUs are the same.
- Reconnect the IDU–ODU cable.
- Switch–on the IDU.
- In the case of a standby ODU, wait two minutes. The reconfiguration takes place automatically.
- For a main ODU, wait for the URG, NUR and ATT LEDs to start flashing simultaneously. Configure the ODU from the IDU by pushing the IDU/ODU switch to the “IDU” position. Wait about two minutes for the terminal to be reconfigured. The alarm LED should go off and the equipment LEDs should turn green.
- Check that there are no alarms in the AS&C application.
- Update the station’s installation form (see model in Appendix 2).

In the event of an error, particularly if the pushbutton is set to the wrong side, re-enter all the station configurations (see section 5**,** Commissioning).

- If you are replacing an ODU 16 QAM UX or an ODU 13 or 15 GHz with RTPC by a A9400UX flat ODU, you must systematically upgrade the IDU software and the software key. This software upgrade is simply recommended,for the other ODU even when you replace them with a flat ODU.

7 - Changing configurations

Changes to the transmission network may entail upgrading the equipment configurations to satisfy new requirements. The 9400 UX system is designed to support these changes simply by modifying the software configuration, or by physically modifying the equipment and its configuration.



Before any change of configuration, disable all remote controls.

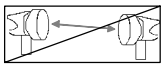
For any work on the boards (dismantling/reassembly, change of configuration), the operator must wear a grounded wrist strap (for example, the 3M "Disposable Wrist Strap", reference 2209).THESE OPERATIONS MUST BE CARRIED OUT ONLY BY TECHNICIANS, AUTHORIZED BY ALCATEL.

Update the reference documents and the labels in accordance with the new configuration.

Some of these operations may entail **opening the IDU to change the hardware configuration (§ 7.6)**. These changes may lead to:

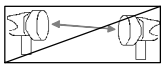
- **changing the IDU** (§ 6.7):
  - replacing like for like a light or classic IDU (1+0 or basic 1+1) (§ 6.7.1),
  - replacing a Light IDU with a classic IDU (1+0 only) or vice versa (§ 6.7.2),
  - replacing like for like a main IDU with multiplexer protection (§ 6.7.3),
  - replacing like for like an extension IDU (§ 6.7.4),
- **changing the ODU** (§ 6.8),
- **changing from a 4QAM to a 16QAM**:
  - upgrade IDU and PEX to last software version (Appendix 12.2), if necessary, refer to § 4.1.2 and 4.6 for an upgrade,
  - replace the 4QAM ODU with the 16QAM ODU (§ 6.8),
- **changing the channel frequency** (§ 7.1) :
  - changing frequency within the same sub-band (§ 7.1.1),
  - changing frequency within another sub-band or a different band (§ 7.1.2),
- **changing the tributaries** (§ 7.2) :
  - enabling inactive tributaries (§ 7.2.1),
  - changing the impedance of a tributary (classic IDU) § 7.2.2,
  - changing the impedance of a tributary (Light IDU) § 7.2.3,
  - changing the bit rate by software (§ 7.2.4),
  - changing the bit rate by changing the LAU and LIU tributary boards (§ 7.2.5),
- **changing the software key** (§ 7.3),
- **supervision**
  - using local supervision (§ 4.14),
  - changing the physical address of the NE (§ 7.4),
  - upgrading the software (§ 7.5),
- **telephone service channel ESC2** (if any) § 7.7
  - inserting, deleting or changing the telephone number in a station (§ 7.7.1),
  - hardware configuring service channel 2 (§ 7.7.2),

- **service channels ESC 3, 4 and 5** (§ 7.8)
  - configuring service channels 3, 4 and 5 in the main IDU without ESC extension (§ 7.8.1),
  - configuring service channels 3, 4 and 5 in the ESC extension unit (§ 7.8.2),
  - adding an ESC extension IDU (§ 7.8.3)
  - configuring the service channel and network ports of a Light IDU (§ 7.9),
- **configuring remote indication loops** (§ 7.10),
  - main classic IDU (§ 7.10.1),
  - light IDU (§ 7.10.2),
- **changing redundancy configuration** (§ 7.11)
  - upgrading from 1+0 to 1+1 without ESC extension or MUX protection (§ 7.11.1),
  - upgrading from 1+0 to 1+1 without MUX protection (§ 7.11.2),
  - upgrading from 1+0 to 1+1 with multiplexer protection (§ 7.11.3),
  - adding multiplexer protection to a 1+1 configuration (§ 7.11.4),
- **changing transmitted power level** (§ 7.12),
- **changing the HSB coupler polarization** (§ 7.13).
- **changing SIMM memories board** (§ 7.14),
- **replacing the following** (§ 7.15),
  - MCU board battery (§ 7.15.1),
  - IDU fan (§ 7.15.2),
  - IDU fuses (§ 7.15.3).

The presence of the  symbol at the start of a procedure indicates that it will involve temporarily disconnecting the link.

7.1 - Changing frequency

7.1.1 - Changing frequency within the same sub-band



A change of frequency within the same sub-band does not entail moving to each station of the link. The change of frequency on remote station (B) can be applied from local station (A) using the local supervision function, or from the general supervision terminal.

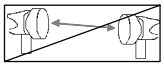


The frequency of the remote station must always be modified before that of the local station.

- Update the information sheet needed to install the station (Appendix 2).
- Switch to local supervision mode in station A (§ 4.14).
- Select “M:0-S:1” (Station B).
- Open the “Operation parameters” application (§ 4.5.1).
- For each channel, modify the transmit and receive frequencies in the “Terminal/Radio configuration/Tx frequency” and “Terminal/Radio configuration/Rx frequency” menus.
- Check that the duplex separation between the transmit and receive frequencies is correct.
- In a 1+1 HSB configuration, check that the transmit and receive frequencies of the two channels are the same.

- Download the new configuration.
- Close the application.
- Select “M:0–S:0” (Station A).
- Open the “Operation parameters” application (§ 4.5.1).
- For each channel, modify the transmit and receive frequencies in the “Terminal/Radio configuration/Tx frequency” and “Terminal/Radio configuration/Rx frequency” menus.
- Check that the duplex separation between the transmit and receive frequencies is correct.
- Download the new configuration.
- Close the application.
- Check that there are no alarms in the AS&C application.
- Exit the local supervision function (§ 4.14.2.2).

7.1.2 – Changing frequency within another sub-band or a different band



Changing frequency in a different sub-band entails changing the ODU (one or two units according to the configuration).

Changing frequency in another frequency band also entails changing the antennas and, if necessary, the HSB coupler.

This means that, in both cases, changing the frequency entails going to each station. The configuration is changed locally in each station by software.

The following procedures must be carried out **on each station**:

- Update the information sheet needed to install the station (Appendix 2).
- Power down the IDUs.
- Adjust the outdoor part according to the new configuration (exchanging ODUs and, if necessary, antennas and the HSB coupler), referring to section 3, “Installation”.
- Repeat the commissioning procedures as described in section 5, in particular:
  - If there is a change of frequency band, adjust the installation parameters (“Installation parameters/Tables/Frequency band”).
  - In all cases, adjust the operational parameters (“Operation Parameters/Terminal/Radio configuration/Tx frequency” and “Operation Parameters/Terminal/Radio configuration/Rx frequency”).
- Check that the duplex separation between the transmit and receive frequencies is correct.
- In a 1+1 HSB configuration, check that the transmit and receive frequencies of the two channels are the same.
- Download the new configuration.
- Close the application.
- Check that there are no alarms.

7.2 - Changing tributaries

7.2.1 - Enabling inactive tributaries

- Update the information sheet need to install the station (Appendix 2).
- Open the “Operation parameters” application.
- Open the “Terminal/Inserted tributaries” menu (see § 4.5.1).
- Activate the required tributaries on each channel, for transmission and reception.
- Send the new configuration.
- Close the “Operation parameters” application.
- Open the AS&C application and check that the new active tributaries are shown with black lines and a number on a white background.

7.2.2 - Changing the tributary impedance (classic IDU)

7.2.2.1 - Impedances according to the configuration

The tributaries ports are compatible with the G703 UIT-T standard.

For the n x 2 Mbits tributaries type, the impedance can vary from 75 to 120 Ω and reciprocally only by configuration.

These different configurations can be carried out by:

- configuration modification of the L.A.U. boards alone
- connection of the cable 75 Ω referenced 3CC07885ADAA for the 4 x 2 Mbits with 1,6 / 5,6 connectors or referenced 3CC07885AEAA for the 8 x 2 Mbits with 1,6 / 5,6 connectors or 3CC07759ADAA for the 4 x 2 Mbits with BNC connectors or 3CC07759AEAA for the 8 x 2 Mbits with BNC connectors whereas the IDU is forced in 120 Ω.
- configuration of the L.A.U. boards and the IDU port
- automatic configuration when connecting the cables in 1+0 or in 1+1 provided that the codified Alcatel cables are used or that the cabling is respected as defined in Appendix 6.1.1.1.

These configuration possibilities are listed in the table in the chapter 7.2.2.2.

7.2.2.2 - Configuration of the impedances according to the configuration

During the lifetime of a product, various versions of the material manufacturing are made and to ease the explanation of configuration modifications of these various versions, we use the following acronyms (**IDU ❶**, **IDU ❷**,**CABLE ❶**, **CABLE ❷**).

DEFINITION

IDU ❶		
L.A.U boards	**References of the main IDU**	**References of the extension IDU**
LAU 8x2 Mbits/s: 3CC06059AAxx	16X2 NUM 120Ω: 3CC08971AAAx* 16X2 NUM 75Ω: 3CC08971ABAx* 16X2 H.I 1+1 Full: 3CC08971ACAx* 8X2 NUM: 3CC08972AAAx*	EXT FULL 16X2: 3CC08974AAxx* EXT 8X2: 3CC08975AAAx*
LAU 4x2 Mbits/s: 3CC06058AAxx	4X2 NUM: 3CC08973AAAx*	EXT 4X2: 3CC08976AAAx*

* x = A or B

IDU ❷		
L.A.U boards	**References of the main IDU**	**References of the extension IDU**
LAU 8x2 Mbits/s: 3CC06059ACxx	16X2 NUM: 3CC08971AAAC 8X2 NUM: 3CC08972AAAC	16X2 NUM: 3CC08974AAAC 8X2: 3CC08975AAAC
LAU 4x2 Mbits/s: 3CC06058AAxx	4X2 NUM: 3CC08973AAAC	EXT 4X2: 3CC08976AAAC

CABLE ❶ (see details § 3.6.5)			
Impedance	**Reference**	**Connector**	**Mbits/s**
75Ω	3CC07885AAxx	1,6/5,6	4X2
	3CC07885ABxx	1,6/5,6	8X2
	3CC07759AAxx	BNC	4X2
	3CC07759ABxx	BNC	8X2
120Ω	3CC07658AAxx	distribution bar	8X2

CABLE ❷		
Reference	**Connector**	**Mbits/s**
3CC07885ADxx	1,6/5,6	4X2
3CC07885AExx	1,6/5,6	8X2
3CC07759ADxx	BNC	4X2
3CC07759AExx	BNC	8X2

GENERAL PRINCIPLE

In 1+0 or 1+1 configuration, the impedance modifications from 120 { 75 Ω are made on the L.A.U. boards.

In 1+1 configuration with the multiplexer protection, the main IDU and the extension IDU are positioned in high impedance and it is the IDU port that is configured in 75 or 120 Ω.

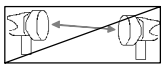
In 34 Mbits (with only the **IDU ❶**), the impedance modifications are made on the 3CC06061AAxx board as indicated in chapter 7.2.2.4.

In n X 2 Mbits/s with **IDU ❶**, to modify the impedance from 120 { 75 Ω or to get high impedance (HZ) the straps have to be positioned on the 3CC06059AAxx or 3CC06058AAxx as indicated in chapter 7.2.2.4.

In n X 2 Mbits/s with **IDU ❷**, the impedance modifications are made automatically when connecting the cables in 1+0 or in 1+1 provided that the codified Alcatel cables are used or that the cabling is respected as defined in Appendix 6.1.1.1. It is only when using the 3CC06059ACxx or 3CC06058ABxx board has to be configured as indicated in chapter 7.2.2.4.

If the IDU ❷ is used, it is strongly recommended to change or to remain in automatic position and it is only when using the cable ❷ that it is mandatory to change in manual configuration.

7.2.2.3 - Method



- Update the information sheet needed to install the station (Appendix 2).
- Power down the main IDU and, if necessary, the extension IDU. Disconnect them from the power source.
- Disconnect the tributary ports and engineering service channels between IDUs.
- Open the IDUs concerned (§ 7.6).
- Set the strap on the LAU board or boards to the 75 ohms, 120 ohms or protection setting as required.
- Close the units.
- Reconnect the connectors of the tributaries and service channels to the IDUs.
- Connect the IDUs to the power source. Power them up again.
- Check that there are no alarms.

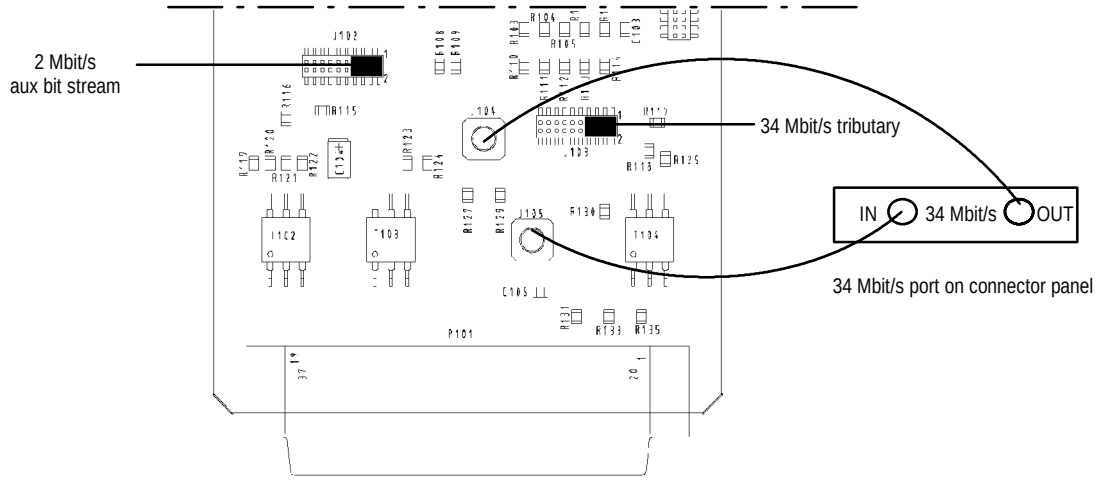
7.2.2.4 – LAU boards for main and extension IDUs

CONFIGURING THE 34 MBIT/S LAU BOARD

Port impedance levels:

- 34 Mbit/s tributaries must be set to the 75 ohms or protection setting, as indicated in figure 56,
- 2 Mbit/s auxiliary bit streams must be set to the 75 ohms, 120 ohms or protection setting, as indicated in figure 57.

S **Tributary positions**



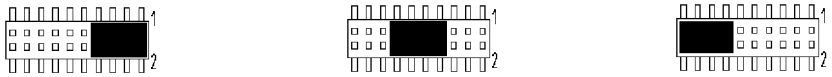
***Positions of impedance straps and ports on the 34 Mbit/s LAU board,* 3CC06061AAxx**

S **Impedance strap positions**



34 Mbit/s in 75 ohms **34 Mbit/s in protection mode**

Figure 56 - Position of the straps in 34 Mbits/s



2 Mbit/s in 75 ohms **symmetrical 2 Mbit/s in 120 ohms** **2 Mbit/s in protection mode**

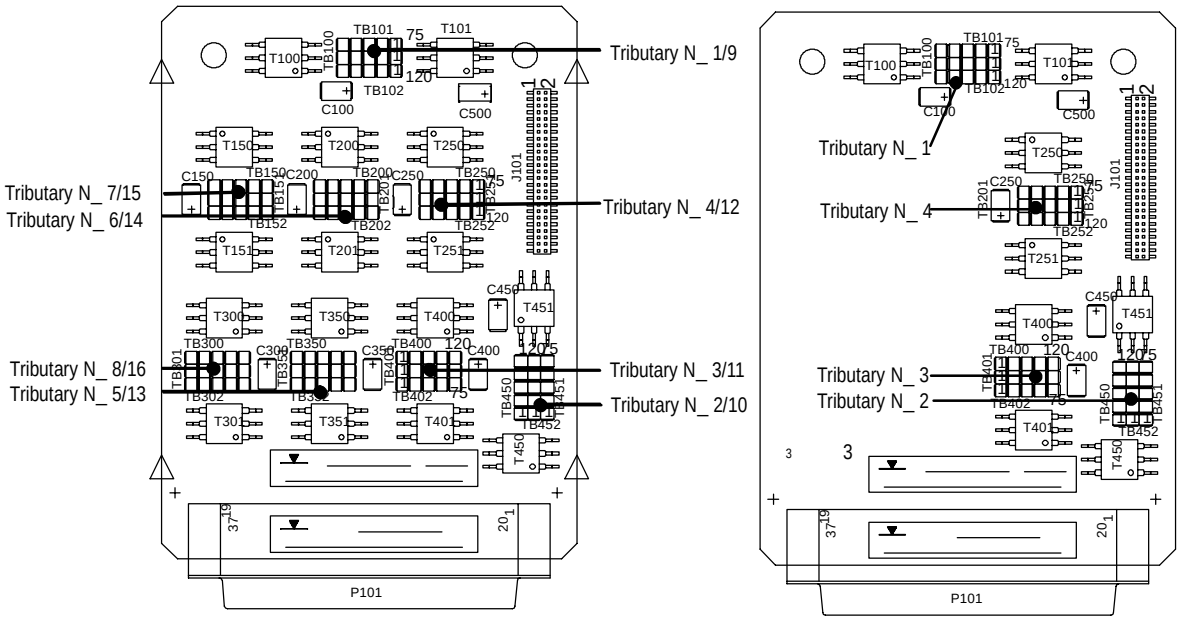
Figure 57 - Position of the straps in 2 Mbits/s

CONFIGURING THE TRIBUTARIES OF THE N X 2 MBIT/S LAU BOARDS

- 3CC06059AAxx or 3CC06058AAxx

The tributary port impedances must be set to the 75 ohms, 120 ohms or protection settings, as indicated below:

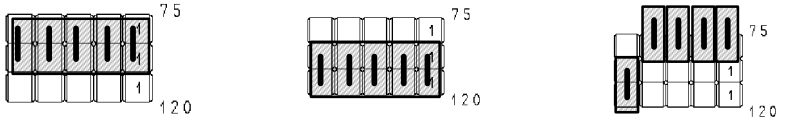
- **Tributary positions**



3CC06059AAxx **3CC06058AAxx**

Position of the tributaries on the boards

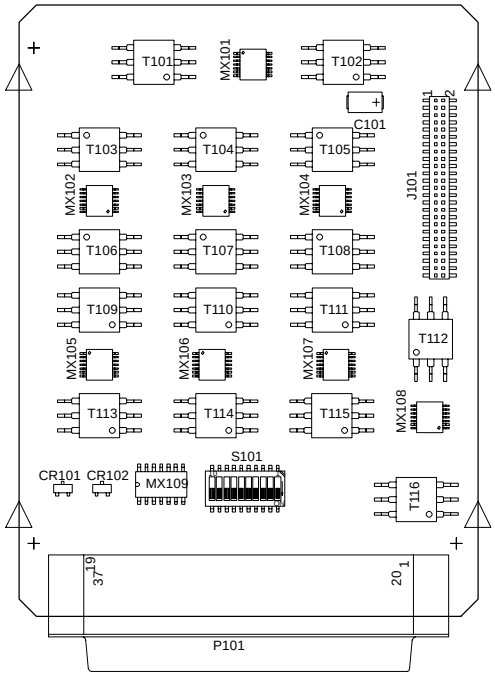
- **Strap positions**



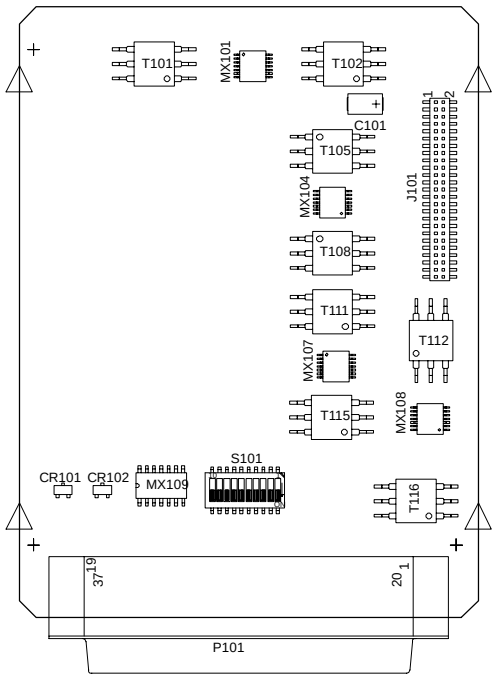
75 ohms **120 ohms** **Protection with port**

The 75 ohm and 120 ohm settings are marked on the PCB

- 3CC06059ACxx or 3CC06058ABxx

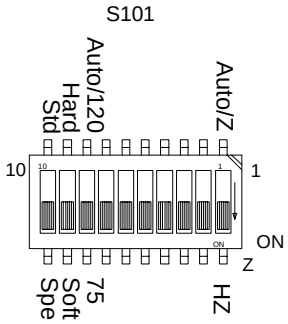


3CC06059ACxx

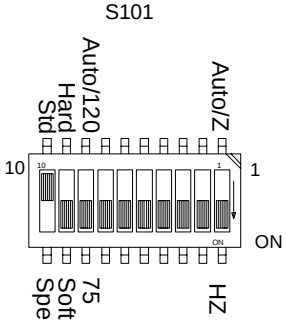


3CC06058ABxx

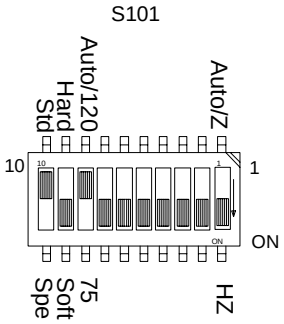
S **Switches position**



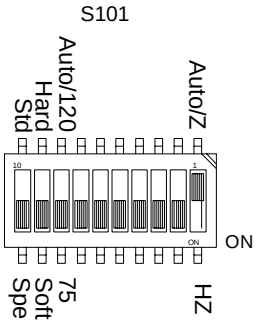
Automatic position (recommended), standard factory configuration with IDU ❷



Manual forcing 120 ohms, use of cable ❷



Manual forcing 75 ohms, (for information)



Manual forcing in protection mode with port (for information)

The white part of the above drawing indicates the switches position.

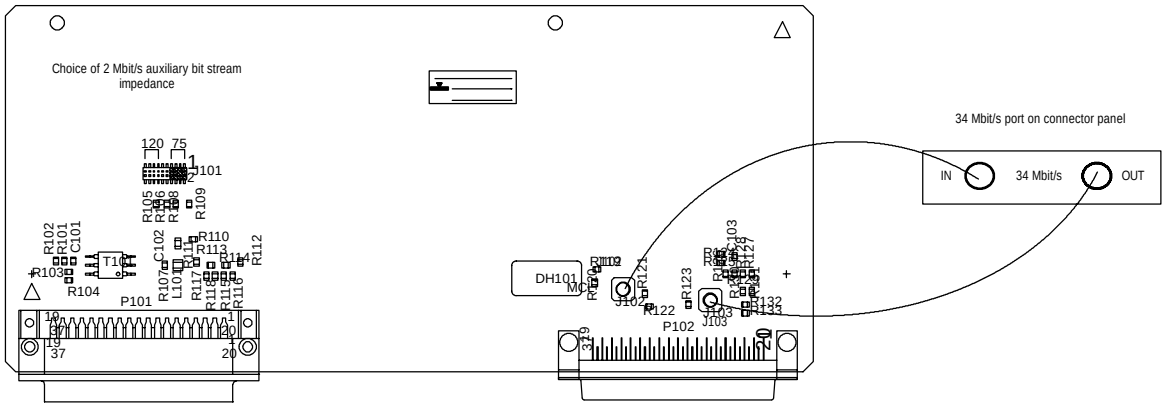
The automatic mode position covers all the cases of use of cable ❶ (or those respecting the cabling given in appendix 6.1.1.1) and the 1+1 protected.

7.2.2.5 – LAU combination boards in access IDUs

CONFIGURING THE 34 MBIT/S LAU COMBINATION BOARD IN ACCESS IDUS

- **Set the straps for the auxiliary 2 Mbit/s portion of the 34 Mbit/s bit rate on board 3CC07443AAxx**
  - S auxiliary 2 Mbit/s tributary port impedance must be set to 75 ohms or 120 ohms.

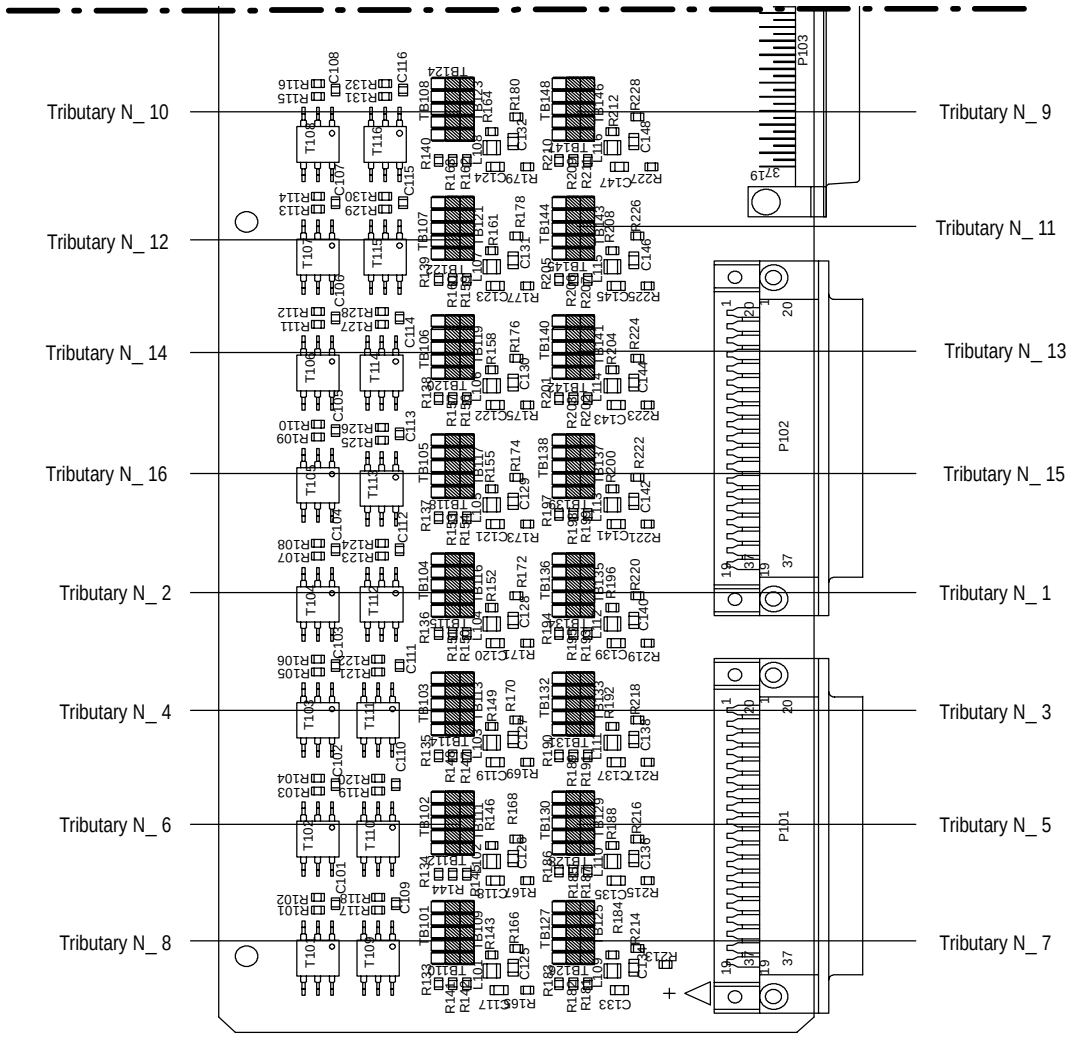
Note: The impedance of the 34 Mbit/s tributary port must not be modified, it is set to 75 ohms.



Position of 2 Mbit/s impedance strap and "tributary" ports on board **3CC07443AAxx**

CONFIGURING THE TRIBUTARIES ON THE N X 2 MBIT/S LAU COMBINATION BOARD IN ACCESS IDUS

- **positions of N x 2 Mbit/s tributaries on board 3CC06765AAxx**



Board* 3CC06765AAxx *: tributary positions

- **Positions of straps for N x 2 Mbit/s on board 3CC06765AAxx**



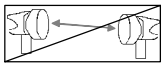
75 ohms **120 ohms**

The 75 ohms and 120 ohms settings are marked on the PCB

On board **3CC06765AAxx**, the impedances of the tributary ports must be set to 75 ohms or 120 ohms.

7.2.3 - Changing the tributary impedance (Light IDU versions only)

7.2.3.1 - Methodology



- Update the information sheet needed to install the station (Appendix 2).
- Power down the Light IDU.

  Disconnect it from the power source.

  Note: For the Plug-in IDU, extract it from the BTS.
- Disconnect the tributary ports and engineering service channel from the IDU.
- Open the IDU.
- Set the switches to the 75 ohms or 120 ohms position as required.
- Close the unit.
- Reconnect the connectors of the tributaries and service channel to the IDU.
- Connect the IDU to the power source.

  Power up again.

  Note: For the Plug-in IDU, simply insert it in the BTS, it will power on automatically.
- Check that there are no alarms.

7.2.3.2 - Configuring the n x 2 Mbit/s tributaries

The port impedance values of tributaries 1 to 4 can be set to **75 ohms** or **120 ohms** using switch modules**:**

- **S105 for channel 1**
- **S104 for channel 2,**
- **S103 for channel 3,**
- **S102 for channel 4,**

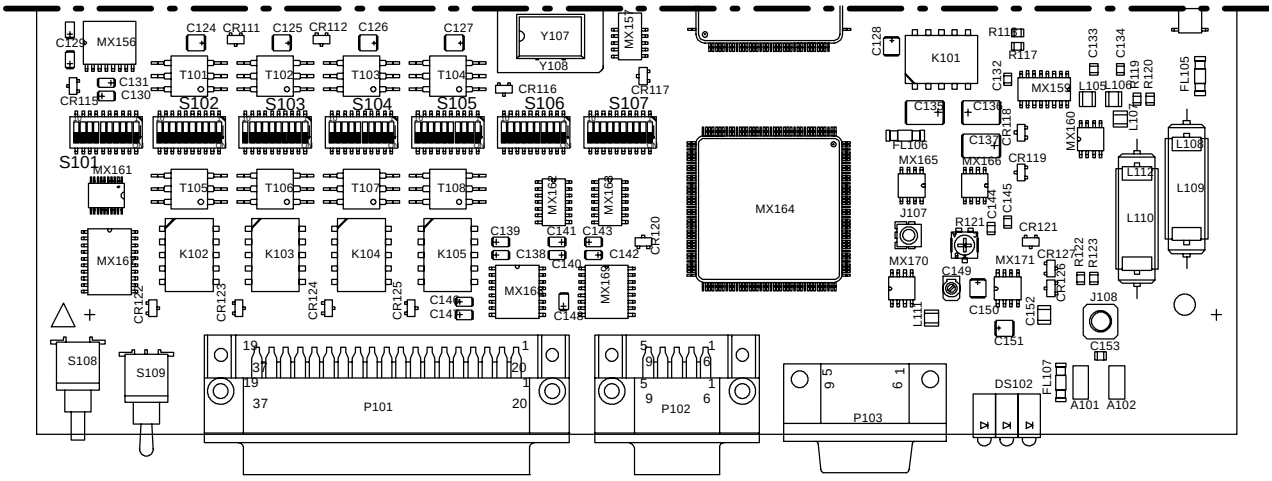
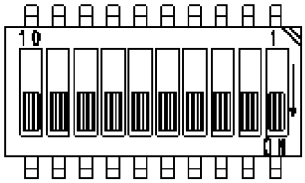


Figure 58 - Positions of switches on the MCU board

Each switch module has ten switches, as shown below.



- **Switches 5 to 10** are for choosing the 75 or 120 ohms impedance for the channel concerned.
- **Switches 2, 3, 4** are for choosing the type of ground in 75 ohms configuration.
- **The combination of settings of switches 1** on **S104** and **S105** can be used to identify the type of engineering practice containing the SCU board: switch 1 of S102 and S103 is not used.

Figure 59 - Locating and using the switches on the switch module

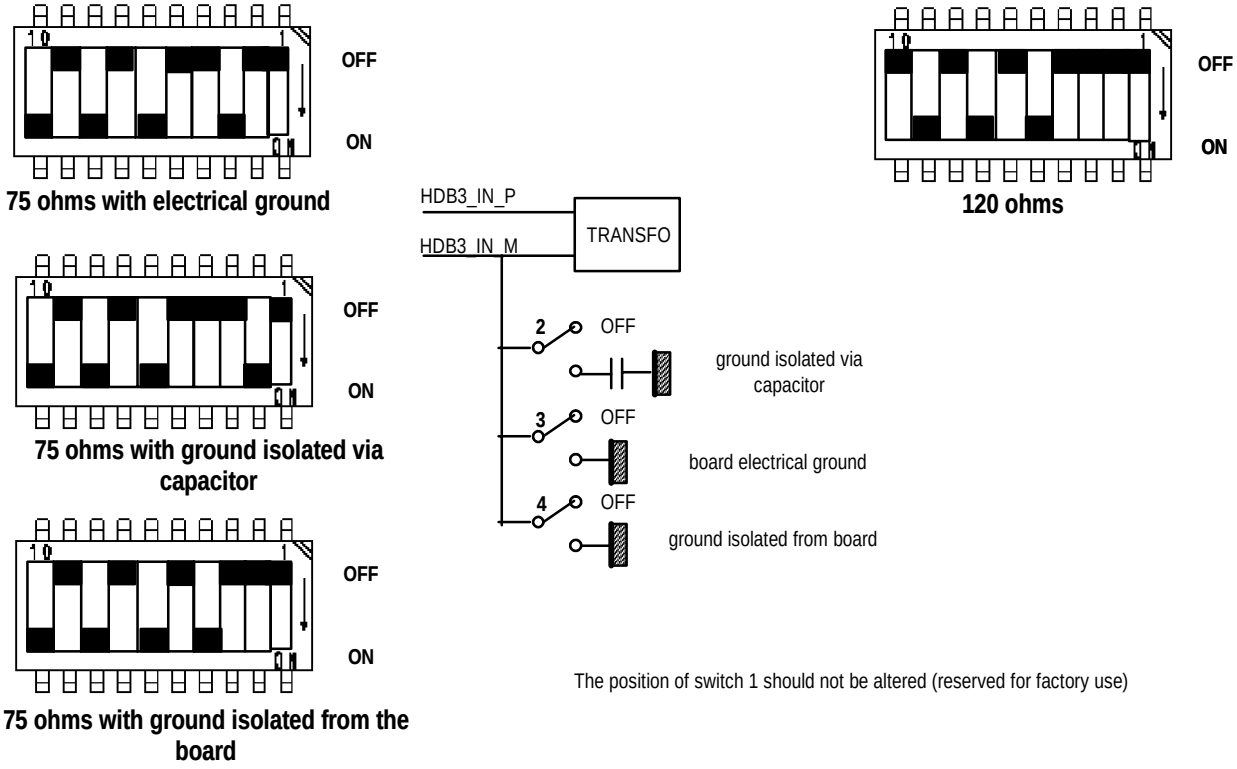
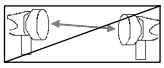


Figure 60 - Configuring the impedance of a tributary

7.2.3.3 – Identifying the type of engineering practice

The combination of **switches 1 on S104 and S105** can be used to identify the type of engineering practice containing the MCU board. This setting is reserved for Alcatel's use.

7.2.4 – Changing bit rate by software



In the classic IDU version, choose, from the list of available bit rates, another bit rate offered by the software key–LAU and LIU board combination (for example, change from a 4x2 Mbit/s configuration to an 8x2 Mbit/s configuration).

In the Light IDU versions, you can change from a 2x2 Mbit/s configuration to a 4x2 Mbit/s configuration (or vice versa).

This operation can be performed only if the **number of tributaries used is less than the number of tributaries available on the equipment**. If this condition is satisfied, you must also have **a software key that is compatible with the new bit rate**. If not, you must first change the key as described previously in § 7.3.



The bit rate of the remote station must always be modified before that of the local station.

- Update the information sheet needed to install the station (Appendix 2).
- Open the “Operation parameters” application.
- Open the “Terminal/Bit rate” menu (see § 4.5.1). Specify the new bit rate.
- Open the “Terminal/Inserted tributaries” menu (see § 4.5.1). Activate the required tributaries on each channel, for transmission and reception.
- Send the new configuration.
- Close the “Operation parameters” application.
- Open the AS&C application and check that the new active tributaries are shown with black lines and a number on a white background.

7.2.5 - Changing bit rate by changing an LAU and LIU tributary boards

These operations concern only classic IDUs.

7.2.5.1 - Procedure

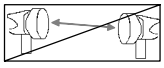
The case in which a change of bit rate does not involve changing the LAU and LIU boards is described in § 7.2.

The table below summarizes the procedures according to the various changes of configuration.

UPGRADE	1+0 or 1+1 CONFIGURATIONS	1+1 CONFIGURATIONS WITH MULTIPLEXER PROTECTION	
	MAIN IDU	MAIN/ EXT. IDU	ACCESS IDU (if present)
4 inputs # 8 inputs	Change boards: LAU: 3CC06058xxx ! 3CC06059xxx LIU: 3CC05818AAxx ! 3CC06026AAxx and configuration of main IDU: 2 Mbit/s tributaries: 75/120 ohms	Change boards: LAU: 3CC06058xxx ! 3CC06059xxx LIU: 3CC05818AAxx ! 3CC06026AAxx with configuration of main and extension IDUs: 2 Mbit/s tributaries: protection	Check the configuration of the new port impedances (75 or 120 ohms)
8 inputs # 16 inputs	Add boards: LAU: 3CC06059Axxx LIU: 3CC06119AAxx and main IDU configuration: 2 Mbit/s tributaries: 75/120 ohms	Add boards: LAU: 3CC06059Axxx LIU: 3CC06119AAxx with configuration of main and extension IDUs: 2 Mbit/s tributaries: protection	Check the configuration of the new port impedances (75 or 120 ohms) Add link cables for tributaries 1 to 16 between the access IDU and the main and extension IDUs
n x 2 Mbit/s # 34 Mbit/s	Change boards: LAU: ! 3CC06061Axxx LIU: ! 3CC06118AAxx and main IDU configuration: 34 Mbit/s: 75 ohms 2 Mbit/s aux. stream: 75/120 ohms	Change boards: LAU: ! 3CC06061Axxx LIU: ! 3CC06118AAxx and configuration of main and extension IDUs: 34 Mbit/s: protection 2 Mbit/s aux. stream: protection	Change the LAU combination board: ! 3CC07443Axxx and configuration of access IDU: 34 Mbit/s: 75 ohms 2 Mbit/s aux. stream: 75/120 ohms
34 Mbit/s # n x 2 Mbit/s (n ≤ 4)	Change boards: LAU: 3CC06061Axxx ! 3CC06058Axxx LIU: 3CC06118AAxx ! 3CC05818AAxx and configuration of main IDU: 2 Mbit/s tributaries: 75/120 ohms	Change boards: LAU: 3CC06061Axxx ! 3CC06058Axxx LIU: 3CC06118AAxx ! 3CC05818AAxx and configuration of main and extension IDUs: 2 Mbit/s tributaries: protection	Change the LAU combination board: 3CC07443AAxx ! 3CC06765AAxx and configuration of access IDU: 2 Mbit/s tributaries: 75/120 ohms

UPGRADE	1+0 or 1+1 CONFIGURATIONS	1+1 CONFIGURATIONS WITH MULTIPLEXER PROTECTION	
	MAIN IDU	MAIN/ EXT. IDU	ACCESS IDU
34 Mbit/s # n x 2 Mbit/s (4 < n ≤ 8)	Change boards: LAU: 3CC06061Axxx ! 3CC06059Axxx LIU: 3CC06118AAxx ! 3CC06026AAxx and configuration of main IDU: 2 Mbit/s tributaries: 75/120 ohms	Change boards: LAU: 3CC06061Axxx ! 3CC06059Axxx LIU: 3CC06118AAxx ! 3CC06026AAxx and configuration of main and extension IDUs: 2 Mbit/s tributaries: protection	Change the LAU combination board: 3CC07443AAxx ! 3CC06765AAxx and configuration of access IDU: 2 Mbit/s tributaries: 75/120 ohms
34 Mbit/s # n x 2 Mbit/s (n > 8)	Change boards: LAU: 3CC06061Axxx ! **2** x 3CC06059Axxx LIU: 3CC06118AAxx ! 3CC06026AAxx + 3CC06119AAxx and configuration of main IDU: 2 Mbit/s tributaries: 75/120 ohms	Change boards: LAU: 3CC06061Axxx ! **2** x 3CC06059Axxx LIU: 3CC06118AAxx ! 3CC06026AAxx + 3CC06119AAxx and configuration of main and extension IDUs: 2 Mbit/s tributaries: protection	

7.2.5.2 – Method



- Update the information sheet need to install the station (Appendix 2).
- If the new configuration requires a change of key, change it first (see § 7.3).
- Power down the IDUs, disconnect them from the power source and disconnect the tributary, service channel and inter-unit cables.
- Open the IDUs (§ 7.6).
- Disconnect the plug linking the power supply to the front panel.
- On the front panel, using a 5 mm flat wrench, remove the spacers securing the LAU boards to be replaced.
- Raise and remove the LIU board (held on the lower circuit by two plugs).
- Remove the LAU board (and for the 34 Mbit/s tributary, disconnect the two subclick plugs).
- Remove the LAU board and the corresponding LIU board (linked by ribbon cable).
- For changing from 16 x 2 Mbit/s to 34 Mbit/s, similarly remove the second LAU/LIU board level.
- Install the new LAU and LIU boards after checking the impedance configurations.

  For the 34 Mbit/s version, fit the small daughter board supporting the 1.6/5.6 connectors (3CC07830AAxx), in place of the top I/O connector (9/16) or ACCESS connector (9/16), fit the daughter board at the back of the connector panel and secure it by screwing the spacers to the outside of the connector panel.
- For the 34 Mbit/s version, remember to connect the two subclick plugs to the LAU board.
- Close the IDUs again.
- Reconnect the tributary, service channel, inter unit and power supply cables.
- Power up the IDU.

- Repeat the commissioning operations described in section 5, in particular, declare the new configuration in the following applications:
  - Installation parameters (Tables/Tributary units).
  - Operation parameters (Terminal/Bit rate, Terminal/Inserted tributaries, Terminal/AIS configuration, Terminal/Tributary ports/Time slot mapping).
- Check that there are no alarms.

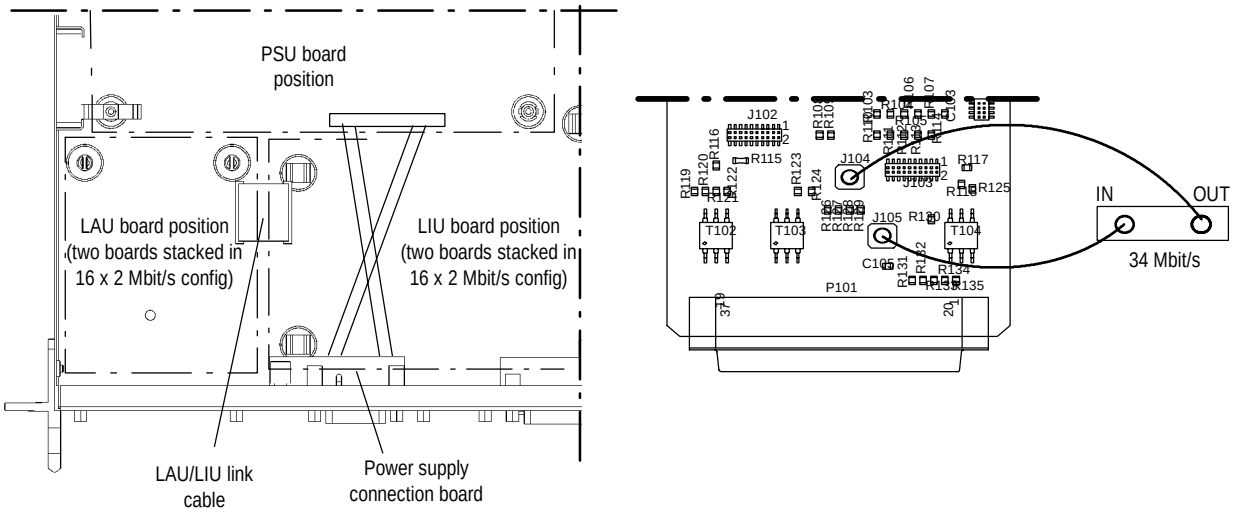


Figure 61 - Indoor unit, positions of LAU and LIU boards and port wiring

7.3 – Changing software key

Not applicable in the Light IDU versions.

When changing configuration (1+0, 1+1, etc) and bit rate, you may need to change the software key if it does not offer the requisite functionality. The procedure for changing the key is as follows:

- Update the information sheet needed to install the station (Appendix 2).
- Open the “Alarms, Status and Controls” application.
- In the main synopsis, zoom in on “Main”, then on “ENVT”.
- Remove the software key from the main IDU and check that the “SOFTWARE KEY MISSING” alarm message appears.
- Fit the new key and check that the “SOFTWARE KEY ALARM” alarm message appears.
- Open the “Operation Parameters” application and resend the configuration (“File/Send”).
- Close the “Operation Parameters” application.
- After resynchronization, zoom in on “ENVT”.
- Wait for the software key alarm to disappear. The alarm will not disappear if the characteristics of the key are incompatible with the configuration, and the message “PROVISIONING REFUSED” will appear to indicate that the key is incompatible with the equipment.

  In this case, change the key and repeat the above procedure from opening the “Operation Parameters” application. If the problem persists, change the IDU (§ 6.7).

7.4 – Changing a NE or IP physical address

7.4.1 – From the station configured as a network element

- Update the information sheet needed to install the station (Appendix 2).
- Open (“Operation parameters/Equipment/Network element”).
- Change the physical address number of the NE (for 946LUX11 or 946LUX12) or IP address (for 946LUX40) (see § 4.5.2).
- Download the new configuration.
- Close the application.
- Wait (about two minutes) for the terminal to be reconfigured.
- Check that there are no alarms.
- Notify the network supervisor of the change of the NE’s physical or IP address.

7.4.2 – From a station with mediation function (master key or local supervision)

- Update the information sheet needed to install the station (Appendix 2).
- Select the remote NE (see § 4.5.2).
- Open “Operation parameters/Equipment/NE physical address” (see § 4.5.2.3).
- Change the physical address number of the remote NE (for 946LUX11 or 946LUX12) or IP address (for 946LUX40) and confirm.
- Download the new configuration.
- Close the application.
- Check the EF alarm on this equipment.
- Declare the new physical or IP address in the “Operation parameters/Equipment/NE physical address” menu on the master station and check that the EF alarm is cleared.

7.5 – Upgrading software (from 946LUXn to 946LUXn+1)

On any change of onboard software in the equipment, consider updating the installation form in Appendix 2 and, if necessary, notify "Alcatel customer service".

If upgrading from the 946LUX10 version of the software to a 946LUX11 or 946LUX12/946LUX40 version, check that the software key used has the number 3CC07619AAAB/AAAC or ABxx. If not, change it.

7.5.1 – Modifying IDU

For the 946LUX40 software, the main IDU's MCU board must be fitted with an 8 Mbyte flash memory and 4 Mbytes of RAM (see § 7.14). If the IDU does not have these memories:

- Open the cover of the main IDU as instructed in § 7.6.
- Install the memories as shown in figure 70, § 7.15.
- Close the main IDU cover, reversing the order of operations on opening.

7.5.2 – Modifying PC software

These operations do not require the craft terminal (PC) to be connected to the equipment.

- PC running WINDOWS only:

  Install in turn (§ 4.1.2):

  - NECTAS
  - RTP
  - 946LUXn (basic and, if appropriate, extension)
  - 946LUXn+1 (basic and, if appropriate, extension)
- PC running WINDOWS, NECTAS, RTP and 946LUXn:
  - Check that the NECTAS version supports 946LUXn+1, if not, update it.
  - Install (§ 4.1.2) the 946LUXn+1 software (basic, and if appropriate, extension).

7.5.3 – Downloading NE software

- Decompress the new file (.cmp) on the PC, then download the software to each NE (.bin) (§ 4.6).
- Wait 2 to 3 minutes for the CT to reopen with the NE running 946LUXn+1.

7.5.4 – Migration from 946LUX1x to 946LUX40 and from 946LUX40 to 946LUX1x

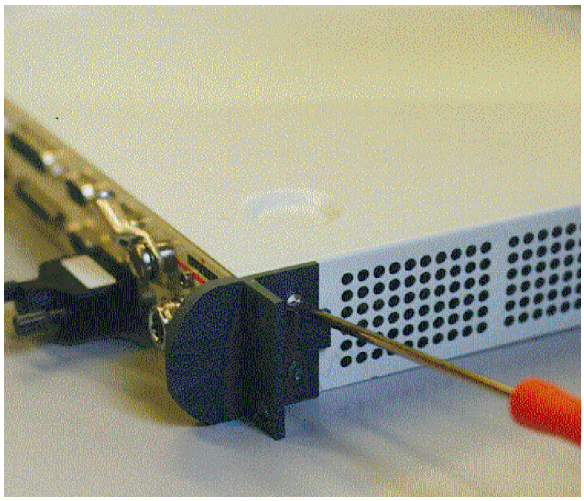
The software migration from 946LUX1x to 946LUX40 and from 946LUX40 to 946LUX1x is explained in the Configuration Manager document which reference is 3CC13299AAAATQBJA.

7.6 – Opening an IDU

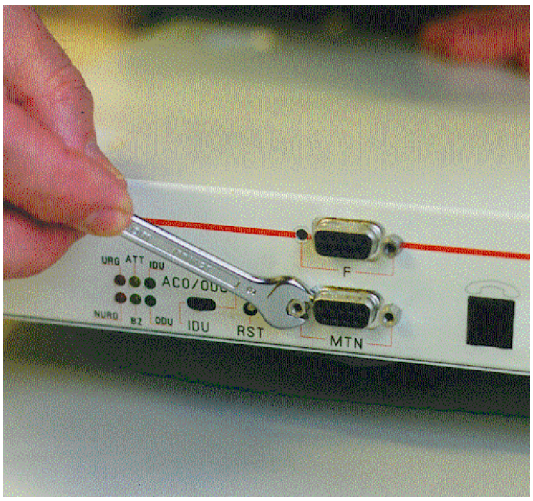
A classic IDU is opened as described in the photos in Figure 62 and the procedure below.

The procedure for opening a **Light IDU** is the same, except the step shown in photo ❷, since this unit does not have a “Display” panel.

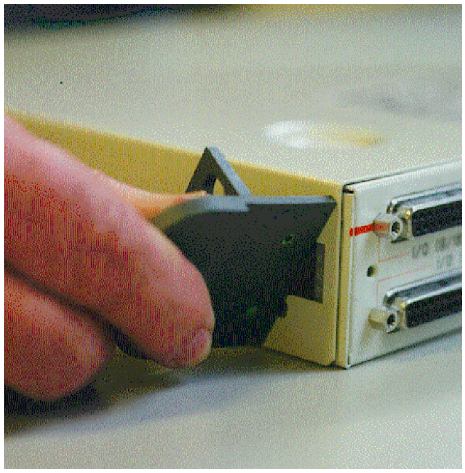
- Isolate the power supply and disconnect all the cables to the IDU.
- Undo the side screws on the unit (screws securing the “fixing brackets” (Photo ❶) and the metal plates concealing the fixing brackets mounting position on the “operation” panel).
- Using a 5 mm flat wrench, undo the connector fixing nuts on the “operation” panel of the IDU (Photo ❷).



❶ Dismantling one of the fixing brackets



❷ Dismantling the connector securing screws on the Operation panel (This phase does not concern the Light IDU version)



❸ Using a fixing bracket to release the cover



❹ Opening the cover

Figure 62 - Opening an IDU

- Slide the cover away from the connector panel towards the operation panel using the “fixing brackets” as a lever to disengage the cover from the connector panel (Photos ❸ and ❹).
- To reassemble, proceed in the reverse order.

7.6.1 – "Indoor" unit configuration

UNIT	CONFIGURATION	ITEM	BOARDS
Light IDU	1+ 0	1 or 1' 2	PSU Light MCU
(Classic) Main IDU	All configurations	1 3 4 5 6 7	PSU MCU LIU 4x2 Mbit/s LIU 8x2 Mbit/s LIU 34 Mbit/s 16x2 Mbit/s extension LIU LAU 4x2 Mbit/s LAU 8x2 Mbit/s LAU 34 Mbit/s 16x2 Mbit/s extension LAU
Extension IDU	1+0 ESC extension	1 8 9	PSU ESC extension SCU ESC daughter boards
	basic 1+1	1 8 9	PSU ESC extension 1+1 SCU ESC daughter boards
	1+1 with multiplexer protection	1 8 9 4 5 6 7	PSU 1+1 protection MUX SCU ESC daughter boards: 4x2 Mbit/s LIU 8x2 Mbit/s LIU 34 Mbit/s LIU 16x2 Mbit/s extension LIU 4x2 Mbit/s LAU 8x2 Mbit/s LAU 34 Mbit/s LAU 16x2 Mbit/s extension LAU
Access IDU	1+1 with MUX protection	10	16x2 Mbit/s LAU combination 34 Mbit/s LAU combination

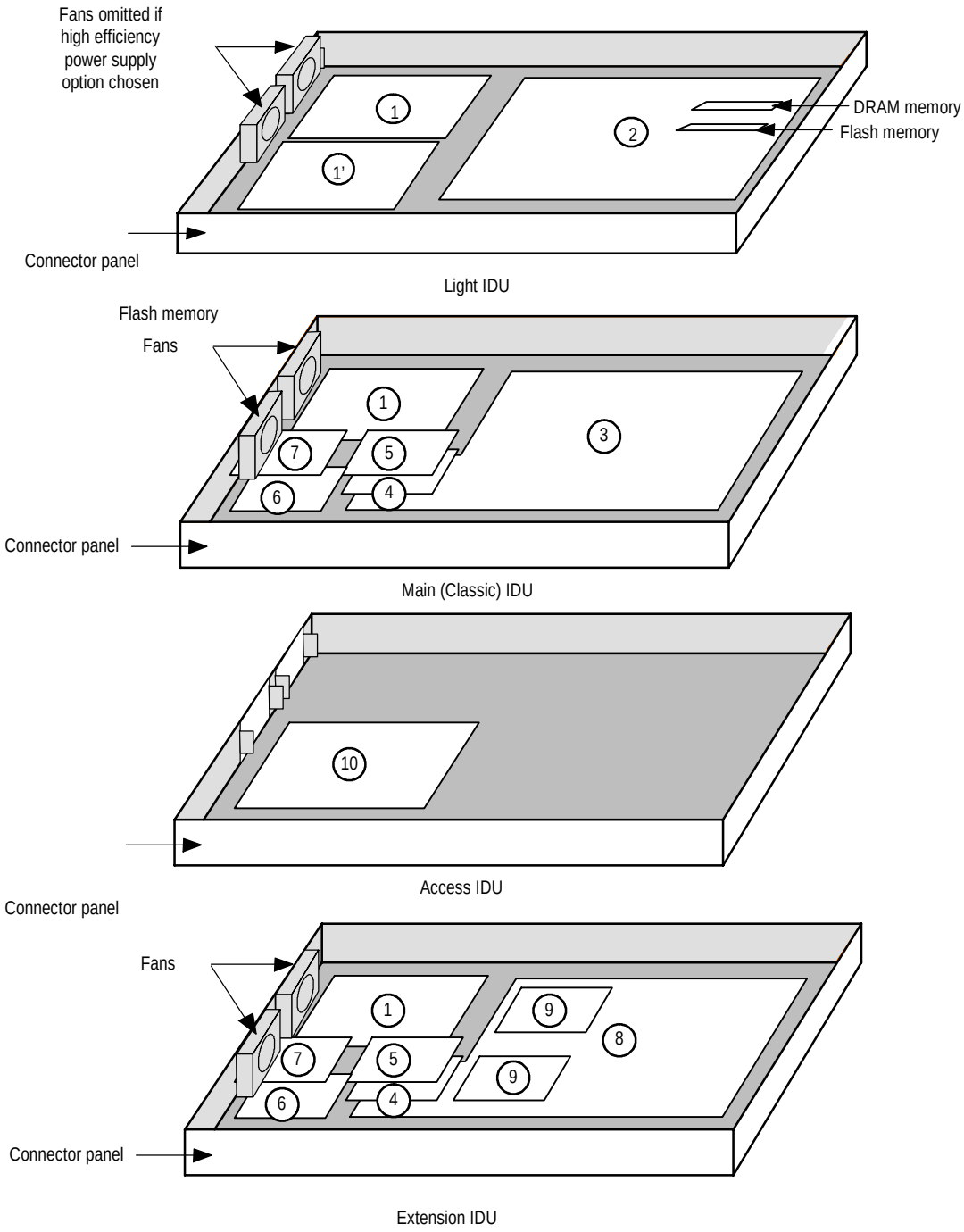


Figure 63 - Layout of boards in the "indoor" units

7.7 – Telephone service channel

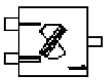
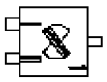
7.7.1 - Deleting, inserting or changing telephone numbers

Not applicable in the Light IDU versions.

- Update the information sheet needed to install the station (Appendix 2).
- Open the “Operation parameters/Terminal/Station number” application.
  - To cancel use of the telephone on the IDU, check the “Audio service channel inhibition” box.
  - To enable use of the telephone on the IDU, make sure that the “Audio service channel inhibition” box is not checked. Otherwise, click it to deactivate it.
  - If the service channel is activated, define the telephone number as described in § 4.5.1.
- Download the new configuration.
- Close the application.
- Check that there are no alarms.

7.7.2 - Configuring the ESC 2

ESC2 can be configured **on the main IDU**, by switches (see figures 64 and 65) on the digital MCU–NUM board (3CC06088AAxx, for an IDU/ODU digital link), or on the analogue MCU board (3CC05714AAxx for an analogue IDU/ODU link), in accordance with the table below:

SWITCH	CHARACTERISTIC		SETTING	
				
S504 (*S104**), S505 (*S105**)	ESC 2 level	audio input	-13 dBm	0 dBm
S506 (*S106**), S507 (*S107**)	ESC 2 level	audio output	0 dBm	+4 dBm

* for an IDU/ODU analogue link

ESC2 is available on the AUDIO 1 and AUDIO 2 connectors.

NOTE: If you want a TTL interface with E&M (TRON/RON) type signalling, close swap SW517 (*SW115**).

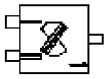
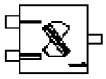
7.8 – Hardware configuration of service channels 3, 4 and 5 (classic and extension IDU)

The 9400 UX can have up to five ESCs (except in a 2 x 2 Mbit/s configuration in which only three are possible); for more details, refer to § 2.3. Only ESC1 (NMS 1 and NMS 2 connectors) cannot be configured manually.

7.8.1 - Configuring ESC 3, 4 and 5 in the main IDU without ESC extension

- Update the information form needed to install the station (Appendix 2).

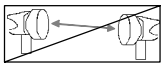
ESC3, ESC4 and ESC5 can be configured **on the main IDU**, by switches (see Figures 64 and 65) located on the digital MCU–NUM board (3CC06088AAxx, for a digital IDU/ODU link), or on the analogue MCU board (3CC05714AAxx for an analogue IDU/ODU link), in accordance with the following table:

SWITCH	CHARACTERISTIC		SETTING	
				
S510 (S110*)	ESC 3 port	Clock inversion	No inversion	Inversion
S511 to S515 *(S111* to S115*)*	ESC 3 port	Format	V11	V28
S508 (*S108**)	ESC 4 port	Clock inversion	No inversion	Inversion
S509 (S*109**)	ESC 5 port	Clock inversion	No inversion	Inversion

* for an analogue IDU/ODU link

S service channels ESC 3, ESC 4 and ESC 5, are available on the ESC EXT connector.

7.8.1.1 – V.11 { V.28 conversion on ESC N_ 3



- Update the information sheet needed to install the station (Appendix 2).
- Power down the main IDU and disconnect it from the power source.
- Open the unit.
- Set switches S511 (*S111**) to S515 (*S115**) on the MCU board to the required position (see § 7.7.2).
- Set switch S510 (*S110**) according to the required clock polarity (see § 7.7.2).
- Close the unit.
- Reconnect the main IDU to the power source and power up again.
- Check that there are no alarms.

* for an analogue IDU/ODU link

POSITIONS OF SWITCHES ON THE MCU–NUM BOARD (DIGITAL): 3CC06088AAxx

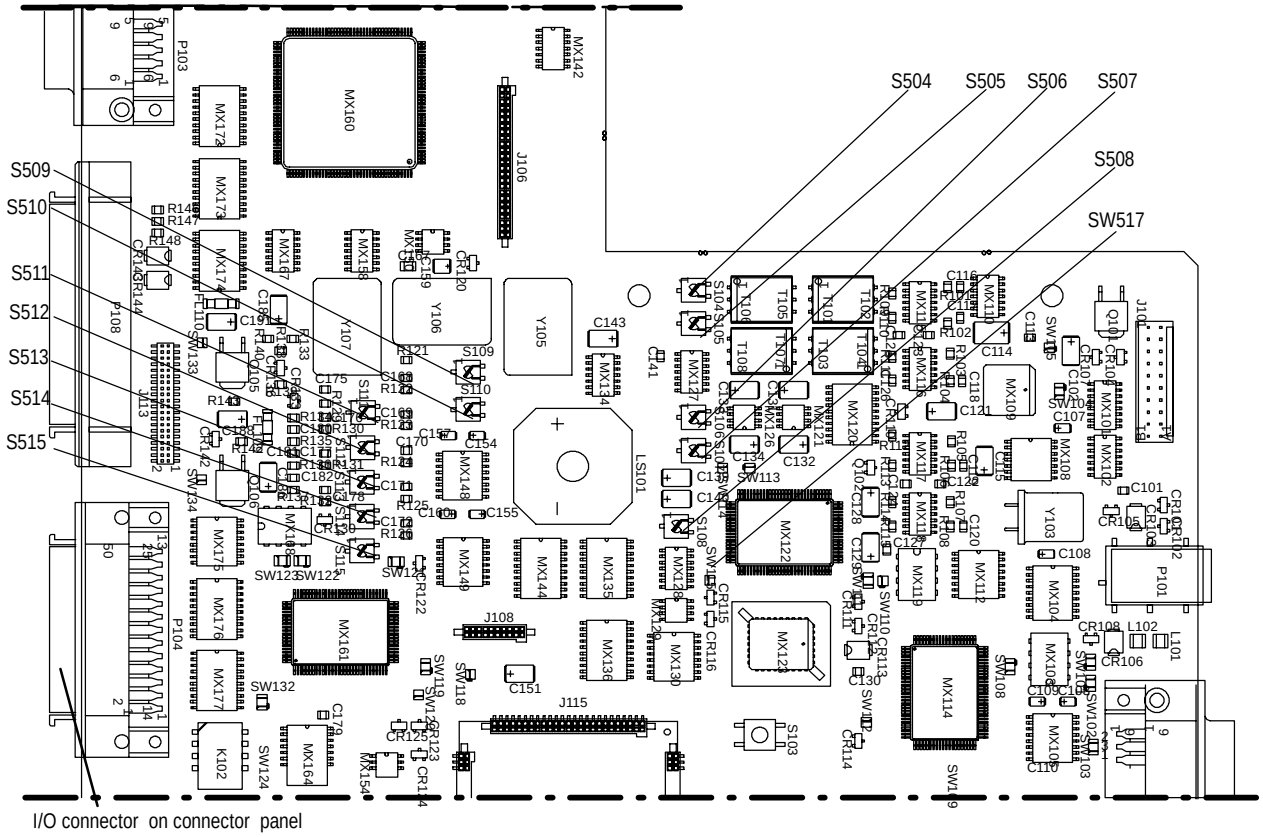


Figure 64 - Locations of switches on the MCU board with digital IDU/ODU link

POSITIONS OF SWITCHES ON THE MCU BOARD (ANALOGUE): 3CC05714AAxx

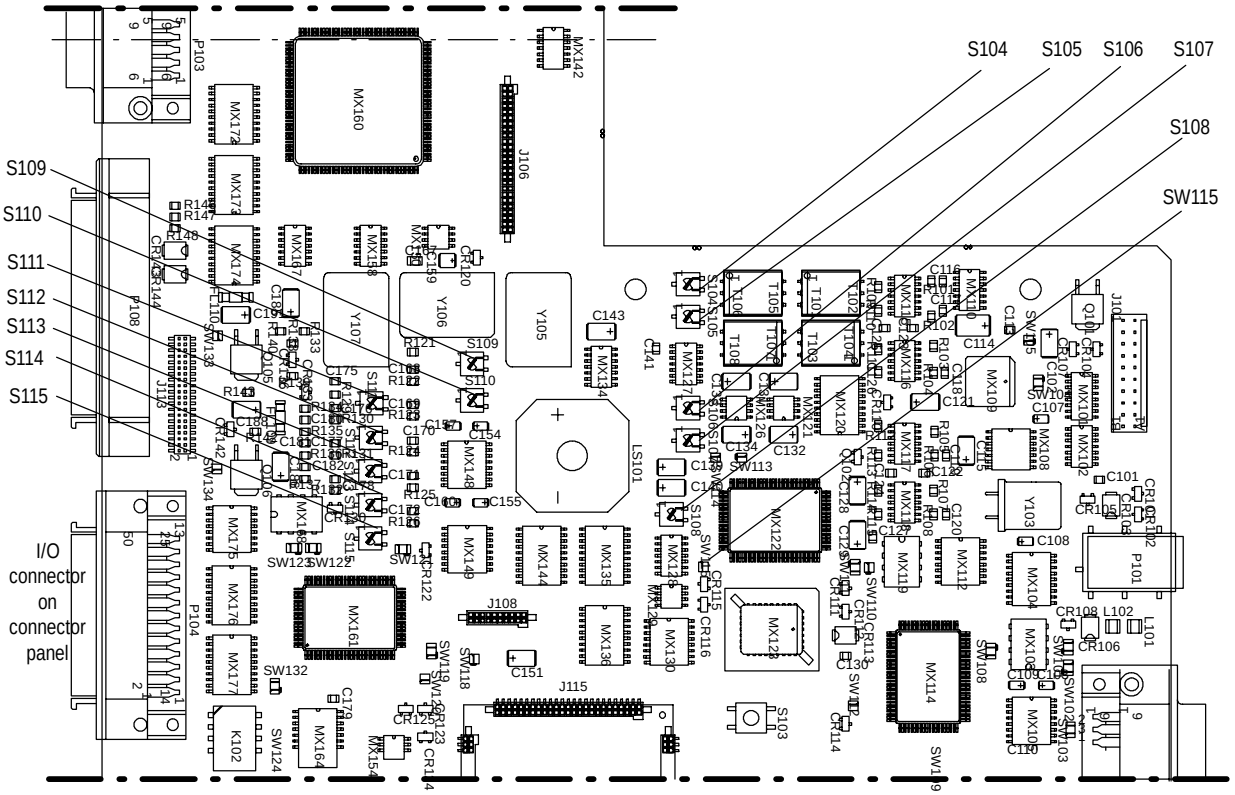


Figure 65 - Locations of switches on the MCU board with analogue IDU/ODU link

7.8.2 – Configuring ESC 3, 4 and 5 in the ESC extension unit

- service channels 3, 4 and 5 can be configured **in the extension IDU**, on the SCU board and the daughter boards that it supports.

7.8.2.1 – Configuring service channels 3 and 4

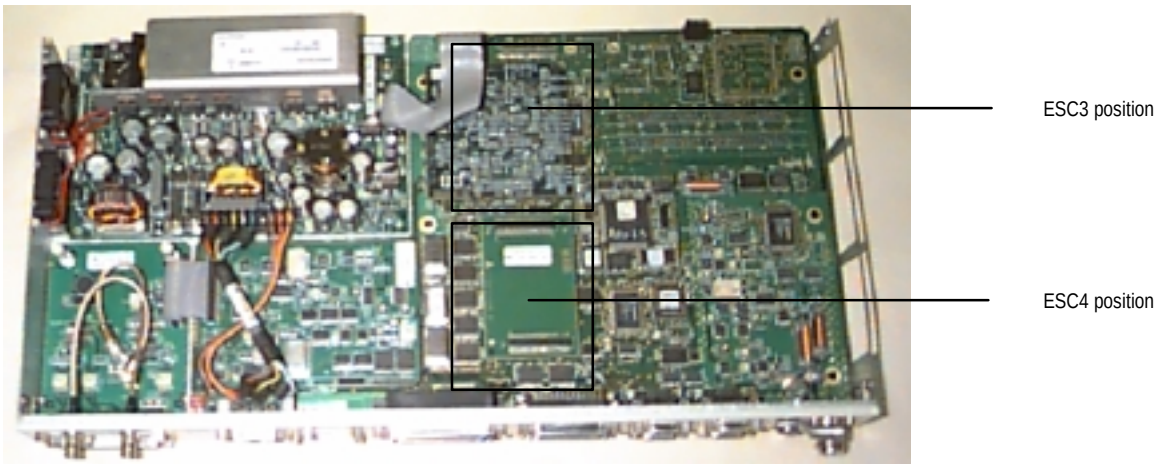


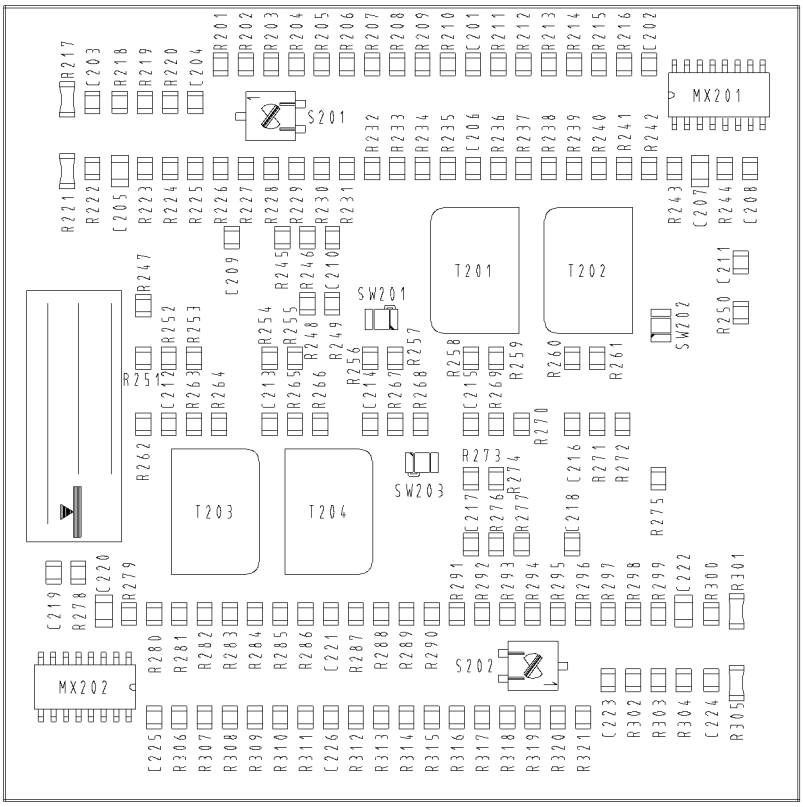
Figure 66 - Positions of service channels 3 and 4 in the ESC extension unit

ESC 3 and ESC 4 can be customized via daughter boards ESC 1 to ESC 5 described below. These boards are installed on the SCU board in the extension unit.

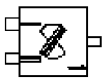
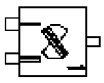
ESC 1 board: 3CC05710AAxx (64 kbit/s G703 point-to-point)

This ESC board is an add-on board which will support a 64 kbit/s ESC with codirectional interface.

Switch positions (customizing ESCs 3 and 4):

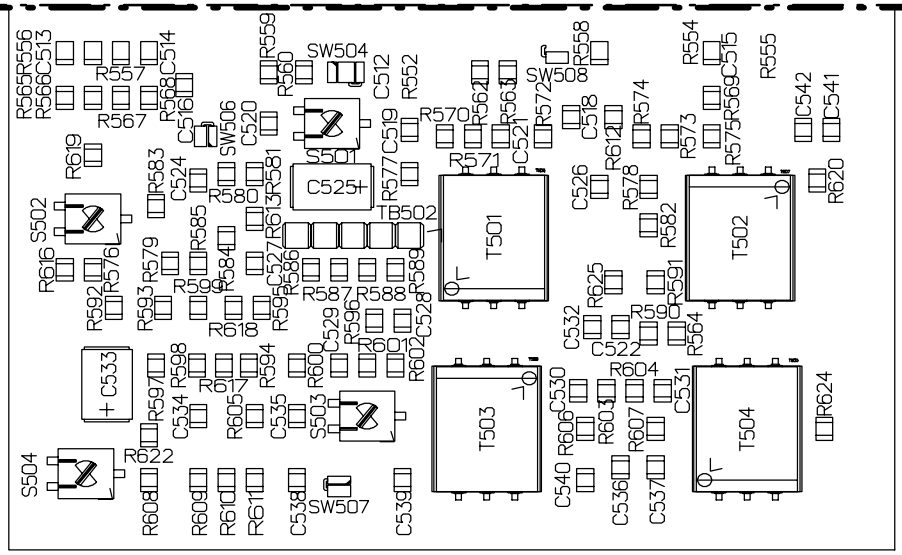


Switch positions on board* 3CC05710AAxx *(Bottom view)

SWITCH	CHARACTERISTIC	SETTING	
			
S201	Synchronization (not active on this equipment)	Channel 2 bit (special)	Channel 2 byte (Normal)
S202	Synchronization	Channel 1 bit (special)	Channel 1 byte (Normal)

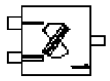
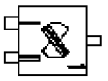
ESC 2 board: 3CC05711AAxx (analogue ESC, point-to-multipoint or point-to-point)

The ESC board 3CC05711AAxx supports a point-to-point or point-to-multipoint type analogue ESC.



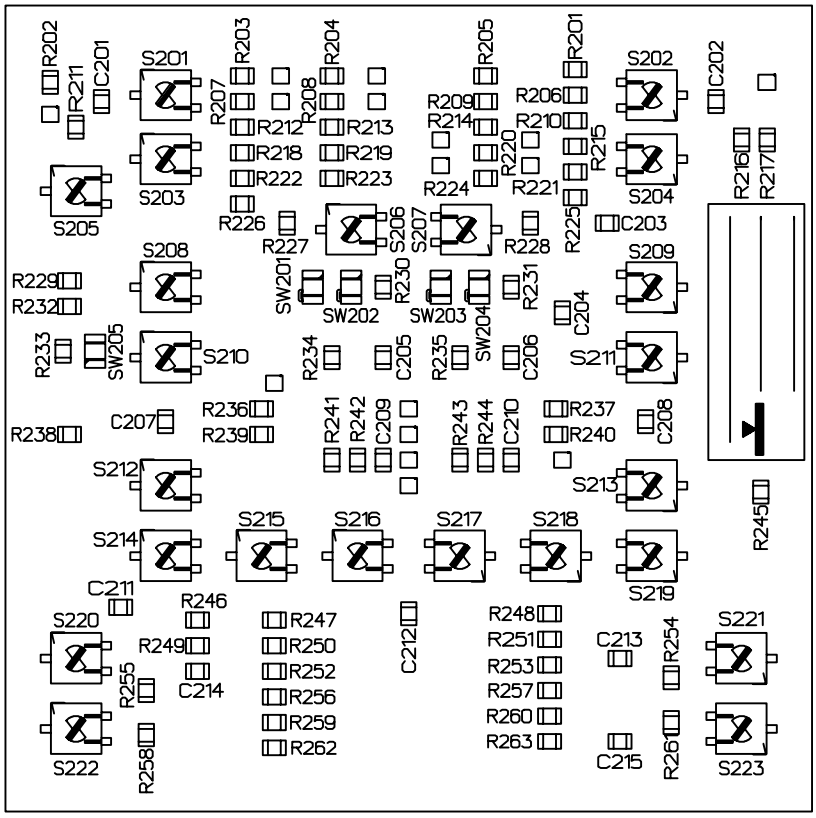
Switch positions on board 3CC05711AAxx (bottom view)

The underside of the board with the settings can be accessed only when removed.

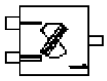
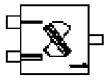
SWITCH	CHARACTERISTIC	SETTING	
			
S501	Output level	+4 dBm	0 dBm
S502	TTL interface with E&M type configuration	disabled	Enabled
S503	Input level	0 dBm	-13 dBm
S504	ESC type	Point-to-multipoint	Point-to-point

ESC 3 board: 3CC05712AAxx(64 kbit/s V11 or V28 point-to-point)

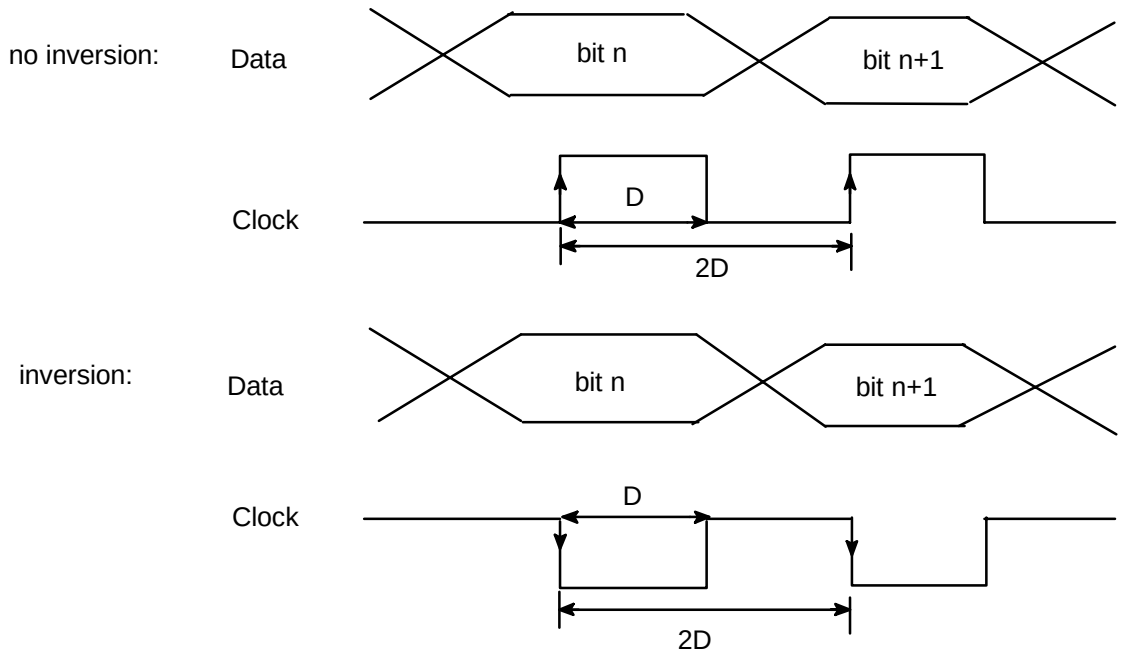
The ESC board **3CC05712AAxx** supports a V11 or V28 compliant ESC operating at 64 kbit/s.



Switch positions on board* 3CC05712AAxx *(bottom view)

SWITCH	CHARACTERISTIC	SETTING	
			
S201, S203, S205, S208, S210, S212, S214, S215, S216, S220, S222	Operator interface type	V11	V28
S206	Transmitted clock inversion	No inversion (*)	Inversion (*)
S207	Received clock inversion	No inversion (*)	Inversion (*)

(*)



CAUTION:

The recommended position is the clock inversion for V11 and no inversion for V28:

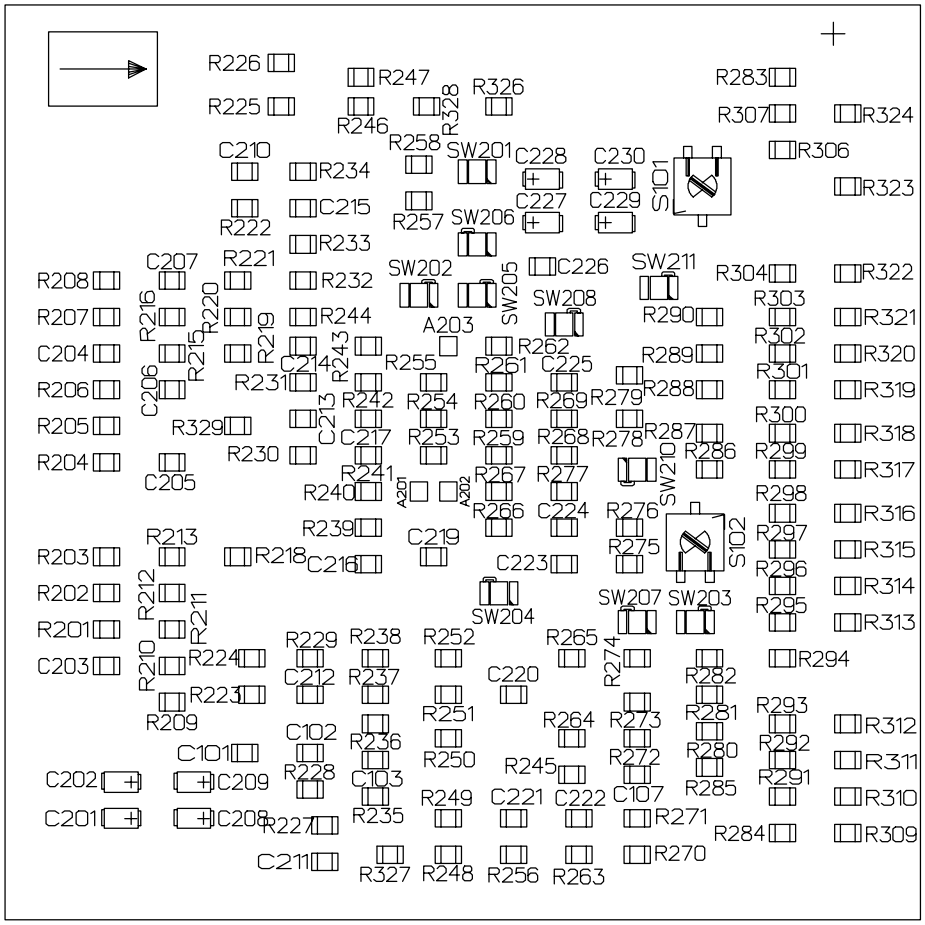
- inversion of S206 in V11, 64 Kbits and no inversion of S207.
- inversion of S207 in V28, 64 Kbits and no inversion of S206.

ESC 4 board: 3CC05713AAxx

The **3CC05713AAxx** ESC board is an optional daughter board that can be installed on the extension IDU SCU board for configurations with engineering service channels 3 and 4.

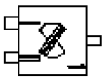
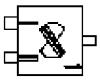
It supports a V11 compliant digital asynchronous engineering service channel:

- a ESC (9.6 kbit/s maximum), point-to-point (PP1) or point-to-multipoint (PM1 and PM2), or
- two engineering service channels (4.8 kbit/s maximum), the first point-to-point (PP1A) or point-to-multipoint (PM1A, PM1B), the second point-to-point (PP2A).



Switch positions on board* 3CC05713AAxx *(Bottom view)

Switch positions

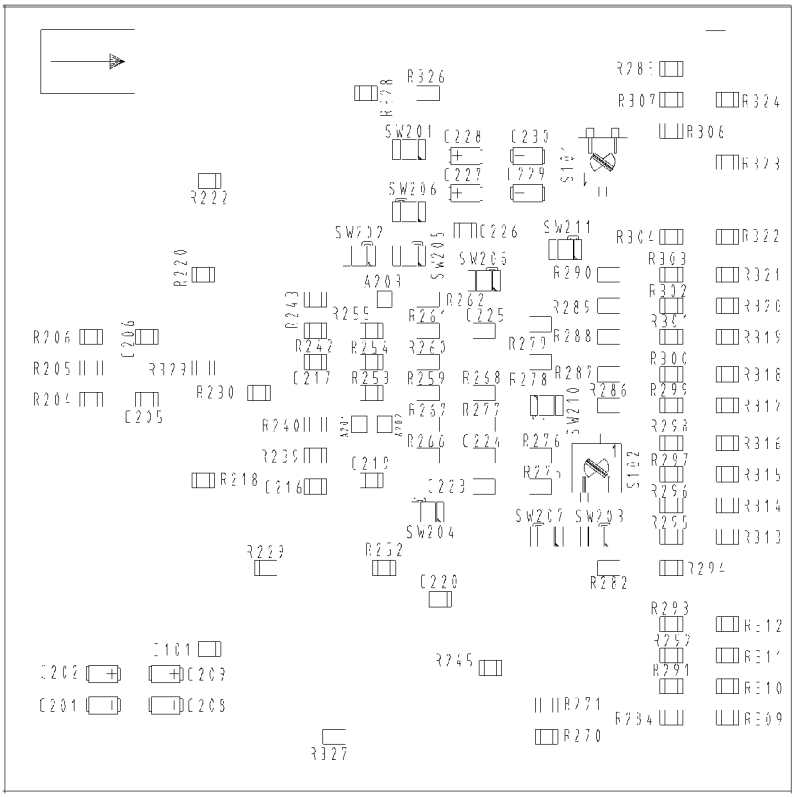
SWITCH	CHARACTERISTIC	SETTING	
			
S101	Link type	Point-to-multipoint	Point-to-point
S102	V11 bit rate	1 x (9600 bit/s max.)	2 x (4800 bit/s max.)

ESC 5 board: 3CC06482AAxx

The engineering service channel board **3CC06482AAxx** is an optional daughter board that is installed on the extension IDU's SCU ESC mother board for configurations with engineering service channels 3 and 4.

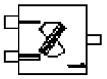
It supports V28 compliant digital asynchronous engineering service channels,

- a ESC (9.6 kbit/s maximum), point-to-point (PP1) or point-to-multipoint (PM1, PM2), or
- two ESC (4.8 kbit/s maximum), the first point-to-point (PP1A) or point-to-multipoint (PM1A, PM1B), the second point-to-point (PP2A).



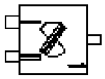
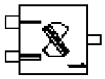
Switch positions on board* 3CC06482AAxx *(Bottom view)

Switch positions

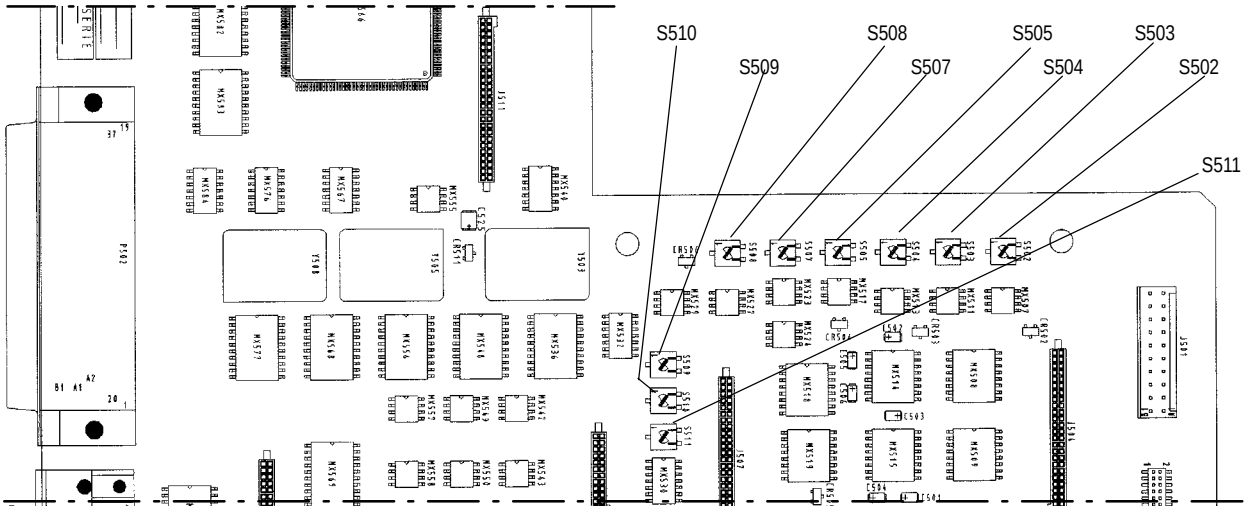
SWITCH	CHARACTERISTIC	SETTING	
			
S101	Link type	Point-to-multipoint	Point-to-point
S102	V28 bit rate	1 x (9600 bit/s max.)	2 x (4800 bit/s max.)

7.8.2.2 – Configuring ESC 5

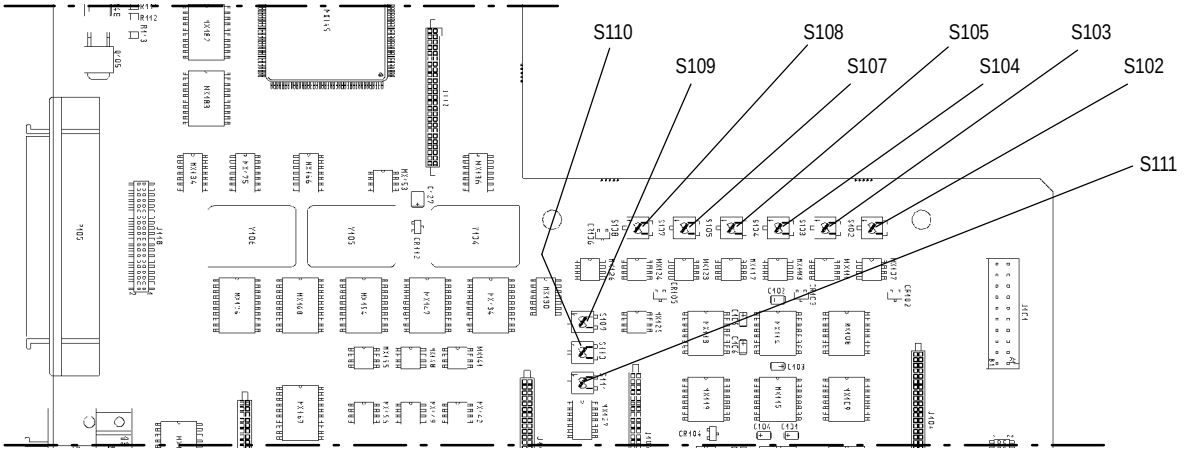
The two possible configurations of ESC 5 (V11 or V28) are obtained by setting switches on the SCU board, of which there are several versions (see page 241).

SWITCH	CHARACTERISTIC	SETTING	
			
S502 to S505 (*S102* to S105**) and S507 to S510 (*S107* to S110**)	ESC 5 format	V11	V28
S511 (*S111**)	Transmit and receive clock inversion on ESC 5 port	Inversion	No inversion

* on SCU with analogue ODU/IDU link



Positions of switches on the SCU board with digital IDU/ODU link



Positions of switches on the SCU board with analogue IDU/ODU link

7.8.2.3 – Method

- Update the information sheet needed to install the station (Appendix 2).
- Power down the extension IDU and disconnect it from the power source.
- Open the unit.
- If the change entails changing the ESC 3 and/or ESC 4 daughter board, change as necessary.
- Configure ESC 3 and/or ESC 4:

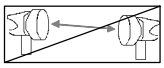
  Configure the replaced board or modify the configuration of the existing board (see § 7.8.2).
- Configure ESC 5:

  Set switches S502 (*S102**) to S505 (*S105**) and S507 (*S107**) to S510 (*S110**) of the SCU board to the V11 or V28 setting according to the required configuration. Set switch S511 (*S111**) according to the required clock polarity (see § 7.8.2).
- Close the unit.
- Reconnect the extension IDU to the power source and power up again.
- If a change has been made to the ESC 3 and/or ESC 4 daughter boards, open the “Installation parameters” application (see § 4.4.1).

  Declare the new boards in “Tables/ESC definition”.
- Download the new configuration.
- Check that there are no alarms.

* for an analogue IDU/ODU link

7.8.3 – Adding an ESC extension IDU



If adding an ESC unit, the positions of ESC 3 and 4 may be fitted with daughter boards.

If the station has back-to-back terminals, transmission of the ESCs from one terminal to the other is performed in baseband and the extension units of each of the terminals must be fitted with daughter boards.

- Update the information sheet needed to install the station (Appendix 2).
- Do not connect the extension IDU to the power source.
- Open the extension IDU to be added and check the configuration of the engineering service channels (ESCs 3 and 4 daughter boards, SCU boards for ESC 5). If the configuration needs to be changed:

  ESC 3 and/or ESC 4 configuration:
  Configure the replaced board or modify the configuration of the existing board (see § 7.8.2).

  ESC 5 configuration:
  Set switches S502 (*S102**) to S505 (*S105**) and S507 (*S107**) to S510 (*S110**) on the SCU board to the V11 or V28 setting according to the required configuration. Set switch S511 (*S111**) according to the required clock polarity (see § 7.8.2).
- Close the unit.
- Install the unit and connect it to the main IDU (see § 3.6).
- Connect the extension IDU to the power source and power up again.
- Open the “Installation parameters” application (see § 4.4.1).
- Declare the new configuration in “Tables/Additional boards” (in a 1+0 configuration), then in “Tables/ESC definition” (in a 1+1 configuration the “Tables/ESC definition” menu appears immediately in the Tables menu).

- Download the new configuration.
- Close the application.
- Check that there are no alarms.

* for an analogue IDU/ODU link

7.9 – Configuring ESC3 and the NMS ports (Light IDU versions)

7.9.1 – Switch positions

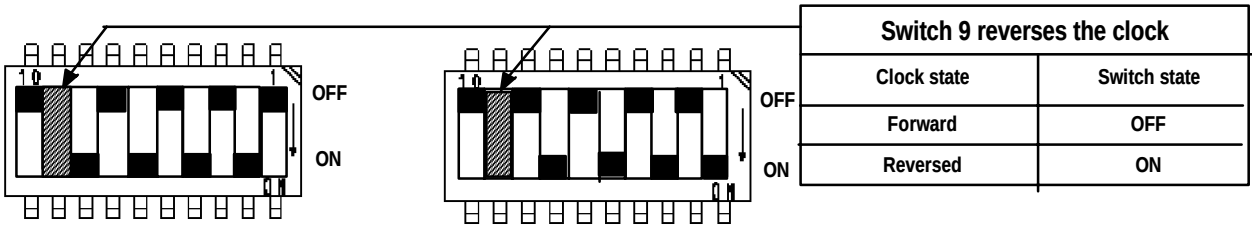
The light 9400 UX has:

- **one engineering service channel, ESC3,**
- **two network management ports, NMS1 and NMS2.**

For more details, refer to § 2.3.

These ports can be set to **V.11 format** (**factory** configuration) or **V.28 format** using the switch modules on the MCU board installed in the IDU (see Figure 58) :

- **S101** configures ESC3,
- **S106** configures the NMS1 port,
- **S107** configures the NMS2 port.

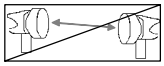


Switch 9 reverses the clock	
Clock state	Switch state
Forward	OFF
Reversed	ON

Figure 67 - Selecting the V.11 or V.28 configuration and clock direction

7.9.2 – Methodology

V.11 { V.28 conversion of ESC3, NMS1 and NMS2



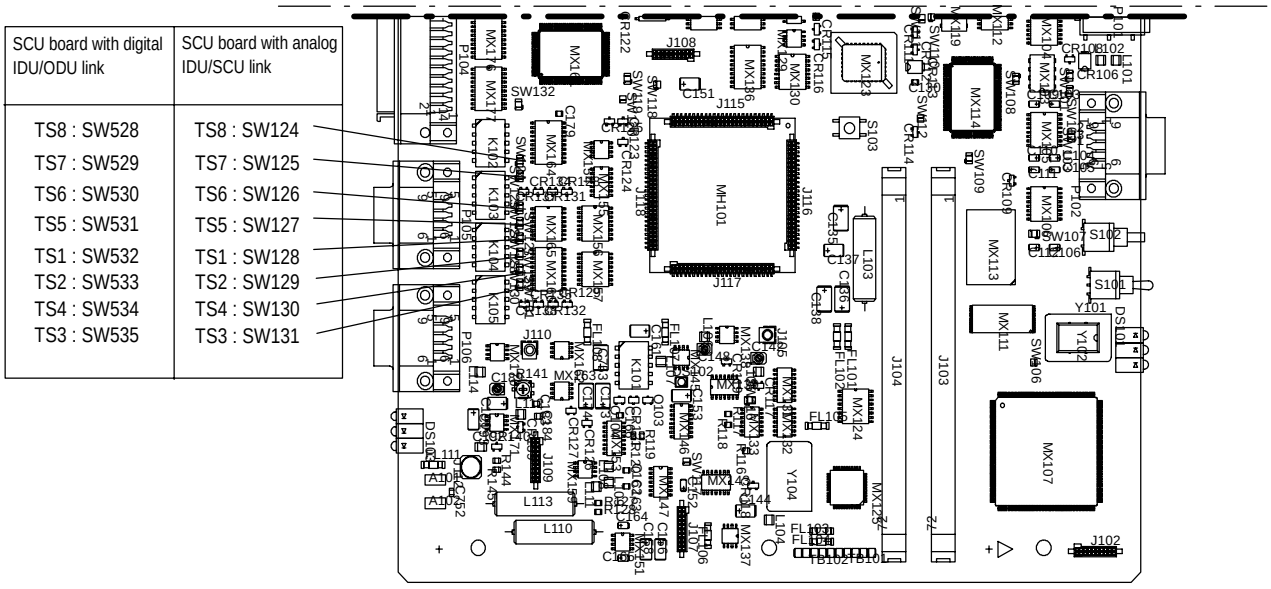
- Update the information sheet needed to install the station (Appendix 2).
- Power down the IDU and disconnect it from the power source.
- Open the unit (see § 7.6).
- Set switch modules S101, S106 and S107 on the MCU board, with switches 1 to 8 set as described in § 7.9.1.
- To alter the polarity of the clock, adjust switch 10 on the same switch modules (see § 7.9.1)
- Close the unit.
- Reconnect the IDU to the power source and power up again.
- Check that there are no alarms.

7.10 – Configuring remote indication loops

7.10.1 - Classic (main) IDU remote indication

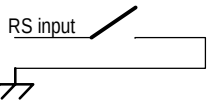
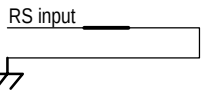
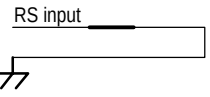
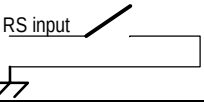
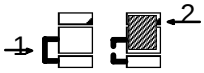
7.10.1.1 - Configuration

The equipment has eight remote indication loops configured as normally open (see § 2.4). These loops can be configured as normally closed by altering the swap configuration on the MCU board according to the drawing below.



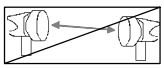
Positions of swaps on the MCU board

To obtain normally closed remote indication loops, you must alter the positions of the soldered links on swaps **SW 124 to SW 127** and **SW 129 to SW 131** as described below.

Loop type	Unactivated	Alarm condition	Swap position
Normally open			Prewired swap 
Normally closed			Change swap position : 1) cut strap 2) insert new strap 

7.10.1.2 - Method

Changing the configuration of remote indication loops 1 to 8 entails moving soldered links on the MCU board.



- Update the information sheet needed to install the station (Appendix 2).
- Power down the main IDU and disconnect it from the power source.
- Disconnect all tributary, service channel and inter-unit cables.
- Open the main IDU (see § 7.6).
- Alter the swap positions on the MCU board.

- Close the unit.
- Connect the new remote indications.
- Reconnect the main IDU to power source and power up again.
- Declare the names of the new RIs via the Operation parameters/Alarms/Housekeeping alarms labels menu.
- Check the operation of the remote indication loops using the “Alarms, states and remote controls” application (Zoom on MAIN/ENVT).
- Check that there are no alarms.

7.10.2 - Light IDU versions remote indication

7.10.2.1 - Configuration

The equipment has eight remote indication loops set as normally open loops (see § 2.4).

These loops can be set to normally closed operation by adding 0 ohm resistors to the MCU board as shown in the diagram below.

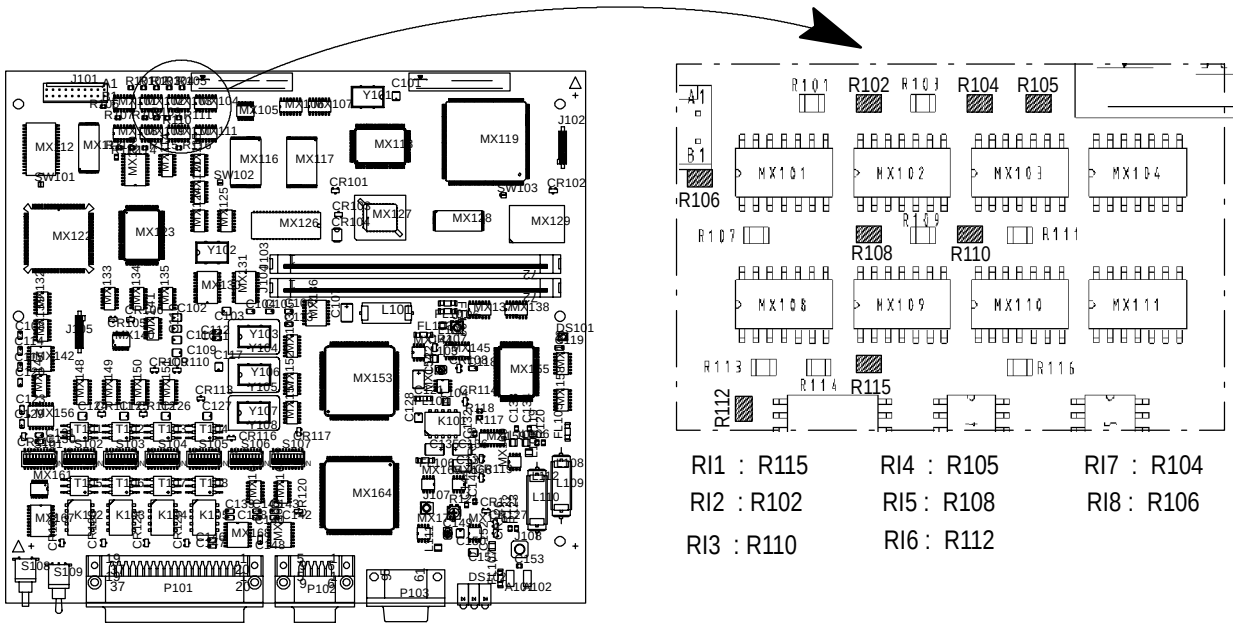
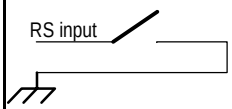
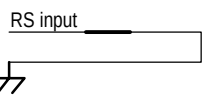
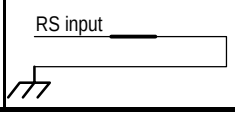
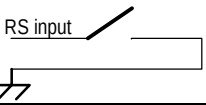
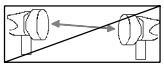


Figure 68 - Locations of housekeeping loop resistors on the MCU board

Loop type	Unactivated	Alarm condition
Normally open		
Normally closed		

7.10.2.2 – Method

To change the configuration of the remote indication loops 1 to 8 to normally closed loops you must fit 0 ohm SMC resistors to the MCU board.

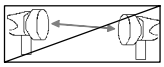


- Update the information sheet needed to install the station (Appendix 2).
- Power down the IDU and disconnect it from the power source.
- Disconnect all tributary, service channel and inter–unit cables.
- Open the IDU (see § 7.6).
- Solder zero value SMC resistors to the MCU board, in the positions shown in Figure 68.
- Close the unit.
- Connect the new remote indication loops directly to the **ESC3** / **ALARM I/O** connector or via a cross connect panel.
- Reconnect the IDU to the power source and power up again.
- Declare the names of the new RI loops via the “Operation parameters/alarms/housekeeping alarms labels” menu.
- Check out operation of the loops via the “Alarms, states and remote controls” application (Zoom in on PRINCIPAL / ENVT).
- Check that there are no alarms.

7.11 – Changing redundancy configurations

7.11.1 – Changing from 1+0 to 1+1 without ESC extension or MUX protection

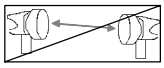
Not applicable in the Light IDU versions.



- Update the information sheet needed to install the station (Appendix 2).
- Install the extension IDU.
- Connect the tributary, engineering service channel and inter-IDU cables.
- Connect the extension IDU to the power source.
- If necessary, change the software key on the main IDU (1+1 max).
- Install the new ODU configuration, referring to section 3: ODU, HSB coupler, antenna, extension IDU link cable.
- Recommission the station and, if necessary, the link (section 5).
- When changing the configuration in HSB, the operator must program the frequencies of the channel x with the same values as for the channel 1.

7.11.2 - Upgrading from 1+0 to 1+1 without multiplexer protection

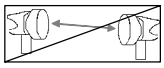
Not applicable in the Light IDU versions.



- Update the information sheet needed to install the station (Appendix 2).
- Power down the main IDU.
- If necessary, change the software key on the main IDU (1+1 max).
- Install the extension IDU (check the ESC configuration, see § 7.8.2).
- Connect the tributary, service channel and inter-unit cables.
- Connect the extension IDU to the power source.
- Install the new ODU configuration referring to section 3: ODU, HSB coupler, antenna, link cable with extension IDU.
- Recommission the station and, if necessary, the link (section 5).
- When changing the configuration in HSB, the operator must program the frequencies of the channel x with the same values as for the channel 1.

7.11.3 - Upgrading from 1+0 to 1+1 with multiplexer protection and access idu

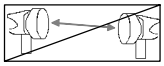
Not applicable in the Light IDU versions.



- Update the information sheet needed to install the station (Appendix 2).
- If necessary, change the software key on the main IDU (1+1 max).
- Power down the main IDU. Disconnect it from the power source.
- Disconnect the tributary, service channel and inter-unit cables.
- Open the main IDU, and the access IDU.
- If there is one, open the extension IDU, remove the customizing boards and install them in the new extension IDU.
- Check that the number of physical ports on the main and extension IDUs are the same.
- On the LAU boards of the extension IDU, check that the tributary impedance selection straps are set to “Protection”.
- On the LAU boards of the main IDU, set the tributary impedance selection straps to “Protection”.
- On the LAU combination board of the access IDU, check the positions of the impedance selection straps.
- Close the units.
- Reconnect the main and extension IDUs to the power source.
- Install the new ODU configuration, referring to section 3: ODU, HSB coupler, antenna, extension IDU link cable.
- Reconnect the tributary, service channel and inter-unit cables.
- Recommission (section 5).
- When changing the configuration in HSB, the operator must program the frequencies of the channel x with the same values as for the channel 1.

7.11.4 – Adding MUX protection with access IDU to a 1+1 configuration

Not applicable in the Light IDU versions.



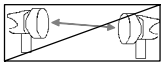
This change of configuration entails replacing the extension IDU with a version supporting multiplexer protection and the addition of an access IDU.

- Update the information sheet needed to install the station (Appendix 2).
- Power down the main and extension IDUs. Disconnect them from the power source.
- Disconnect all tributary, service channel and inter-unit cables.
- Open the main IDU and the new extension IDU.
- On the LAU boards of the main IDU, set the tributary impedance selection straps to “Protection”.
- On the LAU boards of the extension IDU, check that the tributary impedance selection straps are set to “Protection”.
- Close the units again.
- Check that the number of physical ports on the main and extension IDUs are the same.
- Open the access IDU.
- Check, on the LAU combination board, the positions of the impedance selection straps.
- Close the unit again.
- Install the new units (extension and access IDU).
- Reconnect all tributary, service channel and inter-unit cables.
- Reconnect the main and extension IDUs to the power source.
- Install the new ODU configuration, referring to section 3: ODU, HSB coupler, antenna, extension IDU link cable.
- Recommission (section 5).

7.11.5 – Upgrading from 1+0 to 1+1 with multiplexer protection without access idu



This procedure is only avalaible if the IDU type is ❷ (see § 7.2.2.2).

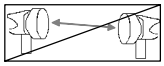


- Update the information sheet needed to install the station (Appendix 2).
- If necessary, change the software key on the main IDU (1+1 max).
- Power down the main IDU. Disconnect it from the power source.
- Disconnect the tributary, service channel and inter-unit cables
- Check that the IDU type is ❷ (see § 7.2.2.2).
- Open the main IDU, and the access IDU.
- If there is one, open the VSD extension IDU, remove the customizing boards and install them in the new extension IDU.
- Check that the number of physical ports on the main and extension IDUs are the same.
- On the LAU boards of the extension IDU, set the straps according to § 7.2.2.2.
- On the LAU boards of the main IDU, set the straps according to § 7.2.2.2.
- Close the units.

- Connect the cable(s) 3CC13659AAxx (see § 3.7.2).
- Reconnect the main and extension IDUs to the power source.
- Install the new ODU configuration, referring to section 3: ODU, HSB coupler, antenna, extension IDU link cable.
- Reconnect the tributary, service channel and inter–unit cables.
- Recommission (section 5).
- When changing the configuration in HSB, the operator must program the frequencies of the channel x with the same values as for the channel 1.

7.11.6 - Adding MUX protection without access IDU to a 1+1 configuration

This procedure is only avalaible if the IDU type is ❷ (see § 7.2.2.2).



This change of configuration entails replacing the extension IDU with a version supporting multiplexer protection.

- Update the information sheet needed to install the station (Appendix 2).
- Power down the main and extension IDUs. Disconnect them from the power source.
- Disconnect all tributary, service channel and inter-unit cables.
- Check that the IDU type is ❷ (see § 7.2.2.2).
- Check that the number of physical ports on the main and extension IDUs are the same.
- Open the main IDU and the new extension IDU.
- On the LAU boards of the extension IDU, set the straps according to § 7.2.2.2.
- On the LAU boards of the main IDU, set the straps according to § 7.2.2.2.
- Close the units.
- Install the new extension IDU.
- Connect the cable(s) 3CC13659AAxx (see § 3.7.2).
- Reconnect all tributary, service channel and inter-unit cables.
- Reconnect the main and extension IDUs to the power source.
- Install the new ODU configuration, referring to section 3: ODU, HSB coupler, antenna, extension IDU link cable.
- Recommission (section 5).

7.12 – Changing power level

7.12.1 – All frequencies bands except 9418UXR201 and A9418UXR202

This operation is a software option.

In the classic IDU version, the main unit must in addition be fitted with a key providing access to this functionality ("TPC:Y" labelling on the key).

- Update the information sheet need to install the station (Appendix 2).
- Modify the output power via the "Operation parameters/Terminal/Radio configuration/Output power" menu (see § 4.5.1).
- Download the new configuration.
- Close the application.
- Check that there are no alarms.

7.12.2 – Frequencies bands without RTPC

You may need to open the ODU to insert an attenuator to adjust the output power using ***integrated*** attenuator kits, the references of which are:

Value	3 dB	6 dB	10 dB
13 GHz to 18 GHz	3CC08543AAxx	3CC08543ABxx	3CC08543ACxx

THIS OPERATION MUST BE CARRIED OUT IN A DRY, DUST FREE LOCATION

PROCEDURE FOR INSTALLING ATTENUATORS :

- Undo the four M4 screws at the corners of the ODU (3 mm Allen key)
  - S The four nuts are unsecured and can easily be lost!
- Remove the cover.
- Depending on version, "carefully" disconnect the top board.

  13 GHz with semi-rigid coaxial cable:
  - To dismantle: undo the two ends of coaxial link W06 between the filter and the RF amplifier (use an 8 mm torque wrench).
  - To reassemble: screw the attenuator onto the RF amplifier plug and the coaxial coupling onto the filter plug.

    Replace the coaxial link between the filter and the RF amplifier.

    Tighten the attenuator, coaxial coupling and the two ends of the coaxial link using the 8 mm torque wrench.

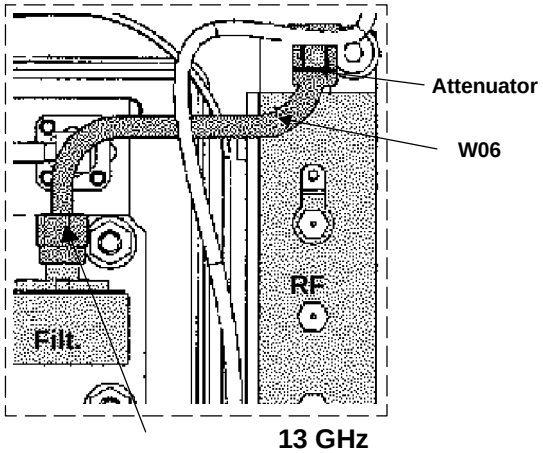
  13 GHz with flexible semi-rigid coaxial cable:
  - Procedure as above, without the use of the coaxial coupling.

  15 and 18 GHz with semi-rigid coaxial cable:
  - To dismantle, undo the plug at the filter end of link W06 between the filter and the RF amplifier (TX output) using an 8 mm torque wrench.
  - To reassemble, screw the attenuator onto the filter plug. Tighten with an 8 mm torque wrench.
  - Replace coaxial link W06 between the filter and the RF amplifier (tighten with an 8 mm torque wrench).
- Replace the top board if applicable (depending on version).

- Press alongside the connectors to lock it in position.

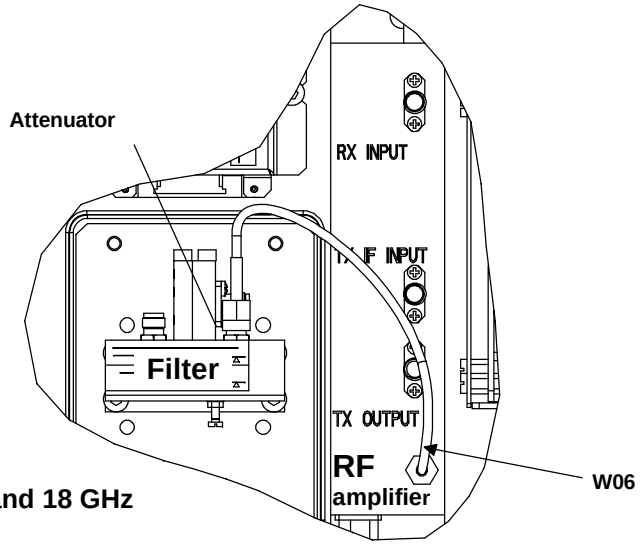
- With a small screwdriver, remove the unit's ring seal from its groove.
  - Do not scratch the casting!
- **Fit a new seal (caution: handle with care because it is fragile!)**
  - **Imperative:** Position the seal's vulcanized coupling in a corner.
- Depending on version, if there is a top board present, check for presence of the CHO-TERM foam pads on the bottom of the cover.
- Replace the cover and check the positioning of the front panel polarizer.
- Tighten the four M4 screws with their nuts using the torque wrench with a socket cap tip to a torque of 1Nm + 0.2/0.
- Affix to the unit (on the handle side), the label supplied with the kit, marking on it the internal attenuator's value.

POSITIONING THE ATTENUATOR IN THE ODU



13 GHz

Position of coaxial coupling. To be fitted in 13 GHz ODU with semi-rigid coaxial cable.



15 and 18 GHz

Depending on frequency band, cable W06 may be connected to the other filter plug.

7.13 – Changing 1+1 HSB coupler polarization

This operation must be carried out before installing the coupler on the pole mounting. **This applies only to configurations with integrated antenna.**

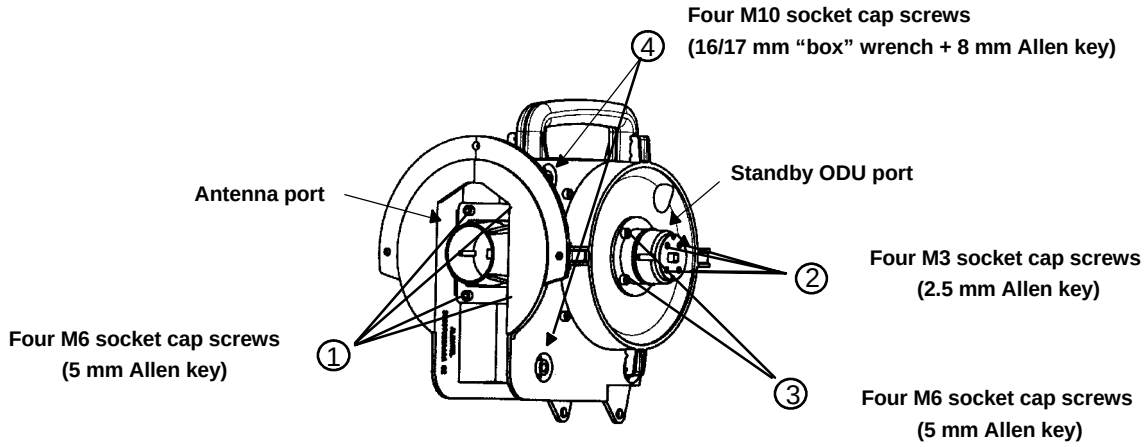
THIS OPERATION MUST BE CARRIED OUT IN A DRY AND DUST-FREE LOCATION.

IMPORTANT NOTE: NEVER REMOVE THE ROUND YELLOW PADS THAT ARE USED TO SEAL THE SYSTEM.

- Update the information sheet needed to install the station (Appendix 2).

Dismantling

- 1) Place the coupler on the table, with the microwave “nose” (which mates with that of the antenna) uppermost, and undo the four M6 screws using an M5 Allen key, then remove the nose; you can now see the waveguide to be replaced.
- 2) On the side nose corresponding to the standby ODU and marked “COUPLED ACCESS” on the casting, insert the 2.5 mm Allen key (M3 screw) and undo the four screws securing the flexible waveguide.
- 3) Using a 5 mm Allen key (M6 screw), undo the four screws securing the side-mounted “COUPLED ACCESS” nose and remove the nose.
- 4) Using a 16/17 mm “box” wrench and an 8 mm Allen key, dismantle: the four M10 socket cap screws + four nuts + eight flat washers + four crinkle washers, holding the two shells of the HSB coupler together.
- 5) Open the HSB casing and, with a 2.5 mm Allen key, remove the straight waveguide on the antenna side to replace it with the twisted waveguide. Pay particular attention to fitting the waveguide gasket on the coupler.

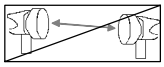


Reassembly

- Insert the four M10 socket cap screws ④ in their positions,
- Fit the side-mounted nose and the four M3 screws ② in their positions,
- Fit the four M6 screws ③ in their positions on the side-mounted nose,
- Tighten the M3 screws and then the M6 screws,
- Fit the front-mounted nose and the four M6 screws ①, and tighten,
- Tighten the four M10 screws ④,
- Replace the HSB coupler, then the ODUs as described in the section on assembling a 1+1 configuration ODU with HSB coupler and integrated antenna.
- On the unit’s label, indicate the antenna waveguide’s direction of polarization: **H** (horizontal) with the twisted guide, **V** (vertical) for the straight guide.

Twisted waveguide reference (flextwist)	13 GHz : **3CC07725AAxx**	15 GHz : **3CC07998AAxx**
	18 GHz : **3CC11263AAxx**	38 GHz : **3CC08024AAxx**
	23/25 GHz : **3CC11264AAxx**	

7.14 – Changing SIMM memories board

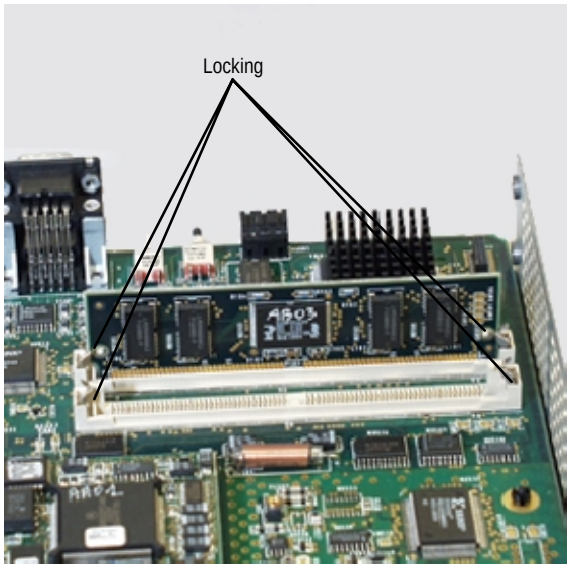


Note:

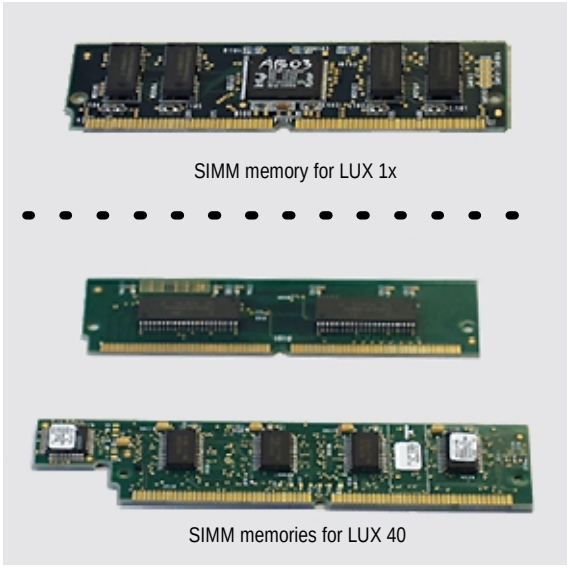
This procedure enables to evolve from the LUX11/ LUX 12/ LUX30 version to the LUX 40 version.

- Power down the main IDU and disconnect it from the power source.
- Open the unit (§ 7.6).
- Open the locking and extract the SIMM board (if any) (Photo ❶).

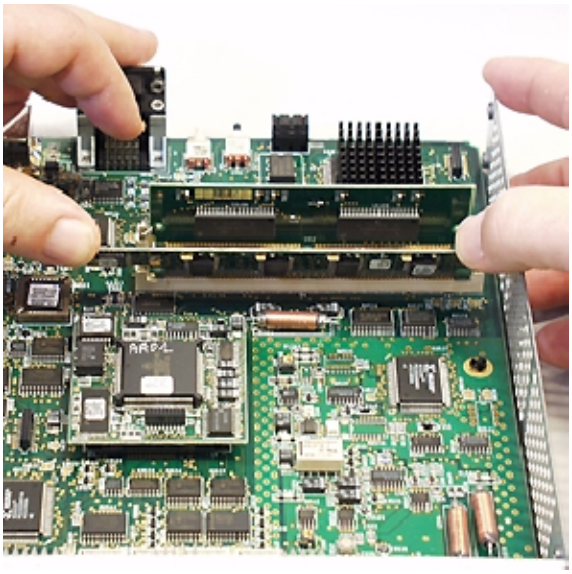
  Note: Not applicable for the light IDU. The IDU Light LUX 12 has no SIMM board.
- Insert the two SIMM boards LUX 40 one after the other (Photo ❸).
- Push the boards backward to lock them.
- Close the unit.
- Connect the unit to the power source.
- Power up the unit again.
- Check that there are no alarms.



❶ Classic IDU with SIMM memory LUX 1x



❷ SIMM memories



❸ Insertion of the SIMM boards



❹ IDU with SIMM memories LUX 40

Figure 69 - Changing SIMM memories

7.15 - Replacing consumable items

7.15.1 - MCU board battery

The MCU board of the main IDU (classic or light) has a battery to safeguard the memory which stores supervision passwords, the last 300 events detected (remote inventory), and so on.

This battery must be replaced every 7 to 10 years if, after powering down, the remote inventory information is lost when you power up again.

CAUTION

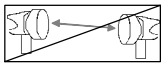
There is a risk of explosion if the incorrect battery is used.

Only replace with a battery of the same type:

M4T28-BR12SH1, Supplier: STMicroelectronics

Observe all applicable national standards regarding the disposal of used batteries.

Do not burn and do not recharge.



- Power down the main IDU and disconnect it from the power source.
- Open the unit (§ 7.6).
- Change the battery (see Figure 70 or 71 depending on the type of IDU). The battery simply plugs into the printed circuit board.
- Close the unit.
- Connect the unit to the power source.
- Power up the unit.
- Wait for the ATT, URG and NURG LEDs to start flashing simultaneously on the main IDU and reload the configuration from the ODU by pushing the “IDU-ACO/ODU” switch on the IDU connector panel to the “ACO/ODU” setting. After two minutes at most, the LEDs should stop flashing simultaneously.
- Check out operation of the equipment.

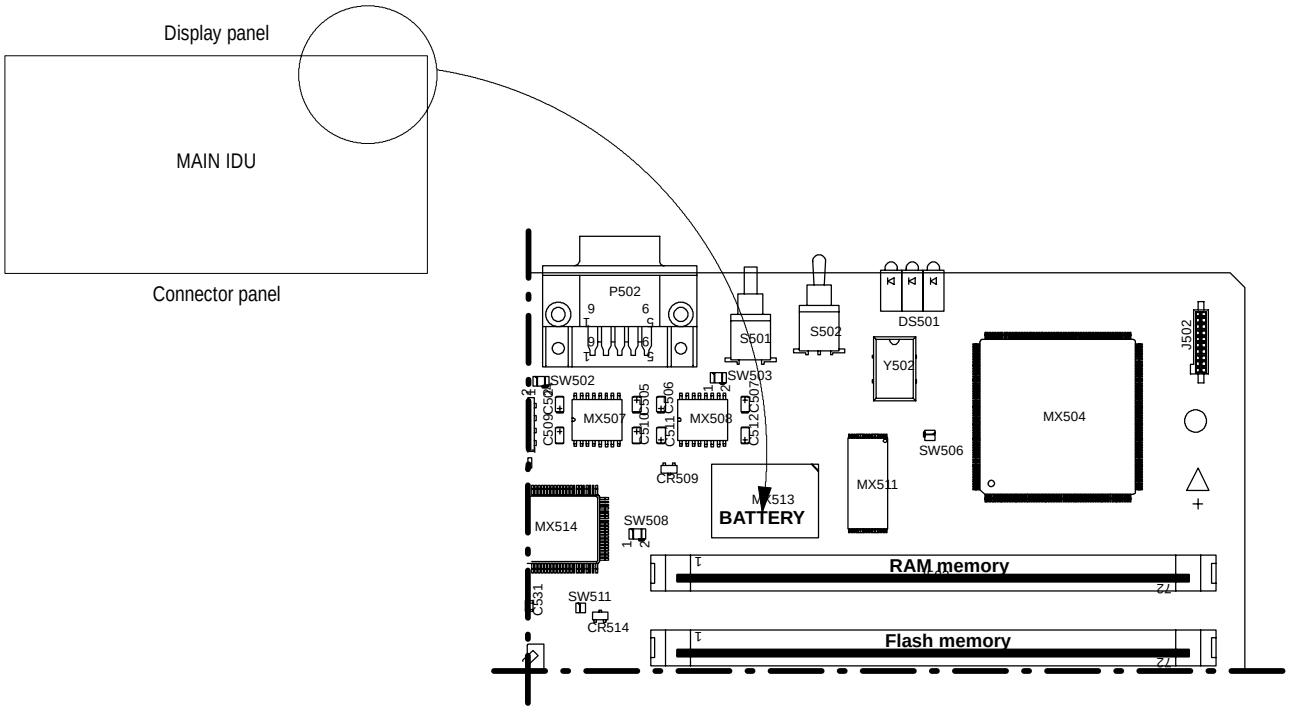


Figure 70 - Position of the battery on the MCU board (classic IDU)

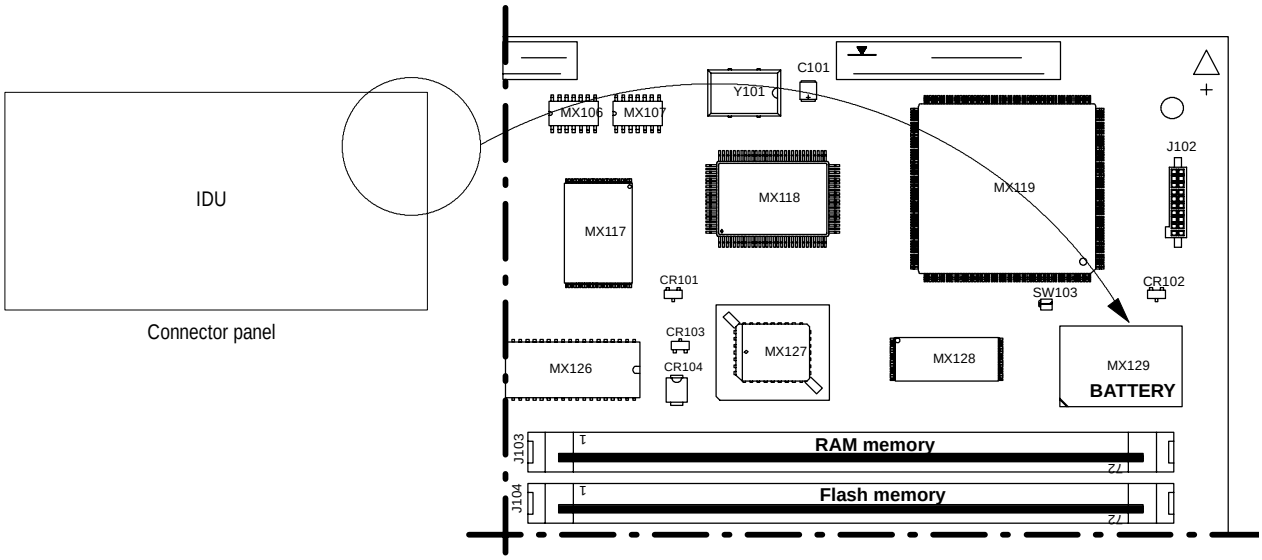
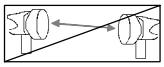


Figure 71 - Position of the battery on the light MCU board (Light IDU)

7.15.2 – Changing fans



Note: Both fans must be changed at the same time.

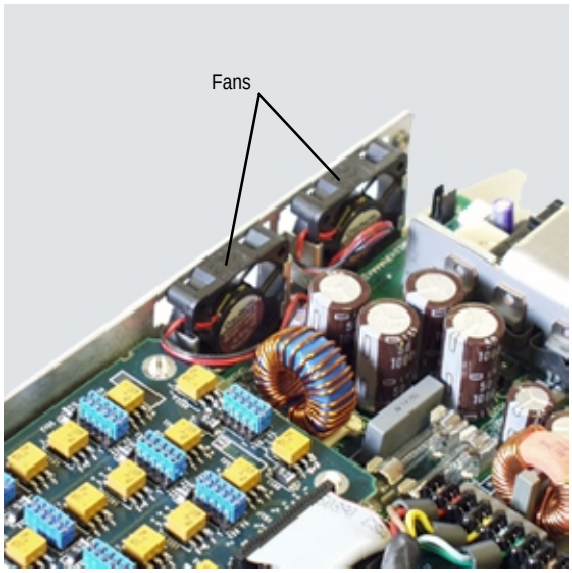
Note the voltage indicated by the used fans (5 or 12 V)

For the 12 Volt fans, use the fan replacement kit 9400UXZ200 (3CC11761AAxx).

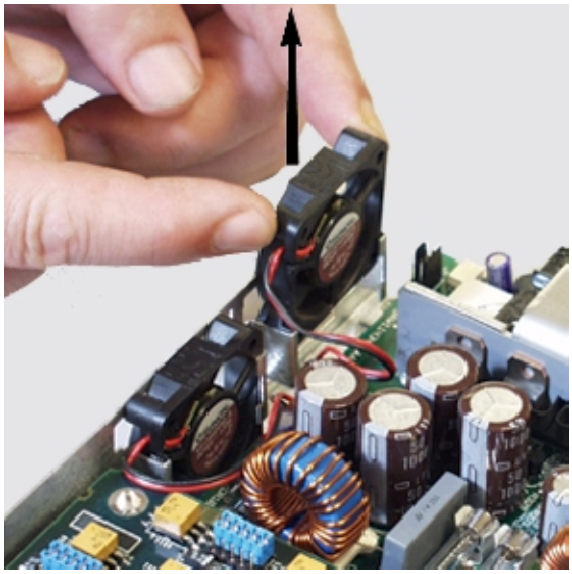
For the 5 Volt fans, use the fan replacement kit 9400UXZ201 (3CC12847AAxx).

- Power down the main IDU and disconnect it from the power source.

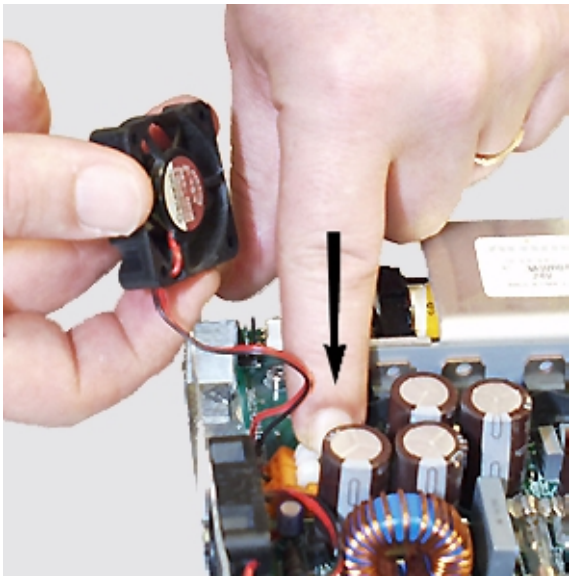
- Open the unit (§ 7.6).
- Identify the orientation of the fan and extract the fan off its recess (Photo ❷).
- Note the polarities and then disconnect the two fan power supply wires (Photo ❸).
- Connect the power supply wires of the replacement fan (the red wire must be connected to the positive terminal, marked on the PCB (Photo ❹)).
- Insert the new fan in its recess, the same way round as the one removed previously, and with the power supply wires run along the same path.
- Close the unit.
- Connect the unit to the power source.
- Power up the unit again.
- Check that the fans extract air (otherwise, reverse their orientation).
- Check that there are no alarms.



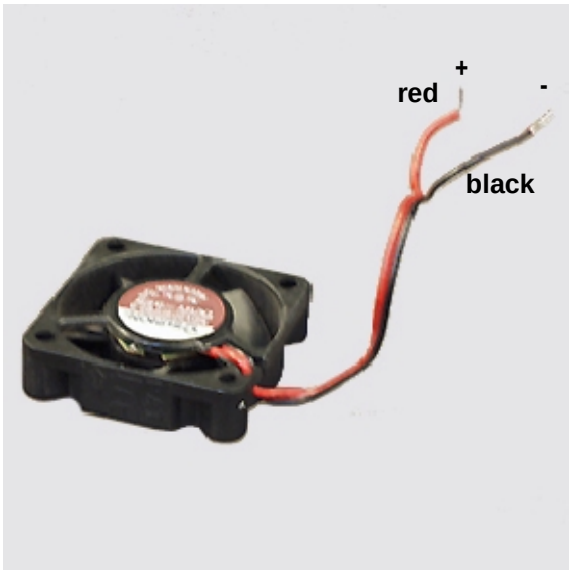
❶ Position of the fans



❷ Extraction of the fan



❸ Disconnection of the power supply wires

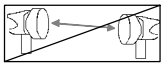


❹ Fan polarity

Figure 72 - Changing fans

7.15.3 – Changing a fuse of an IDU

The power supply units of the main, extension and Light IDUs versions are protected by two fuses which snap into their clip mountings. The procedure below describes how to change these fuses.



- Power down the IDU and disconnect it from the power source.
- Open the unit, for IDU, see § 7.6.
- Remove the suspect fuse and check it with an ohmmeter (if out of service: infinite impedance).
- Insert a good fuse.
- Close the unit.
- Connect the unit to the power source.
- Power up the unit again.
- Check that there are no alarms.



Always replace fuses with fuses of the same type:

§ For the classic IDU (31 x 6 cartridge):
F10H 250 V (10 A fast blow fuse, high breaking capacity).

§ For the Light IDU (20 x 5 mm cartridge):
With Alcatel power supply: **T3,15H250V** (3.15 A slow blow fuse, high breaking capacity),
With ARTESYN power supply: **F3,15H250V** (3.15 A fast blow fuse, high breaking capacity).

If the fuses blow again on power up:

- in an installation phase, check the polarity of the external voltage and/or that there is no short circuit in the power connector,
- if the fault occurs in operation, change the IDU.

PAGE INTENTIONALLY LEFT BLANK

Appendix 1 - Human exposure to electromagnetic fields

A.1.1 - Introduction

This document explains the precautions that are taken to make sure the electromagnetic waves transmitted by Alcatel 9400 family of equipment for Low and Medium Capacity Fixed Radio links are harmless for the public.

It provides an evaluation of a safety perimeter based on calculations derived from the ICNIRP Guidelines and the means (i.e. installation rules) by which this safety perimeter has to be implemented, where necessary.

It applies to the protection of the public from electromagnetic field radiated by the transmit antenna of the equipment. It does not apply to the protection of the workers.

The calculations, based on far-field telecommunications equations, provide, in case this hypothesis is not validated by the result, an over-estimated value for the safety limit in the boresight direction of the antenna.

In this latter case, more realistic evaluations could be derived from near-field calculations, provided that an adequate model of the antenna is used.

Where several radio equipment are installed on the same site, the contributions from each of them have to be taken into account for the definition of the global safety perimeter. Each radio site is then a particular case and the safety perimeter is no more depending on the characteristics of a single equipment only.

A.1.2 - Standards and regulations

A.1.2.1 - ICNIRP guidelines

The International Commission on Non-Ionizing Radiation Protection (ICNIRP) is formally recognized by the World Health Organization (WHO) to draft health based exposure guidelines. This commission has updated the guidelines on the thresholds of exposure to electromagnetic fields for frequencies from 0 to 300 GHz ["Guidelines for Limiting Exposure to Time Varying Electric, Magnetic and Electromagnetic Fields (up 300 GHz)", Health Physics, Vol. 74, n°4, pp. 499–522, April 98]. This document defines the basic restrictions on the population's level of exposure to electromagnetic fields and from that it derives reference levels for more straightforward application in the lowest frequency ranges.

A.1.2.2 - European regulation

The R&TTE Directive [Directive1999/5/EC of the European Parliament and of the Council of 9 March 1999 on radio equipment and telecommunications terminal equipment and the mutual recognition of their conformity] is effective since April 2000. The objective of this Directive is to define the rules for allowing CE marking of these equipment so that they can be placed on the market. These rules rely on harmonised standards. The article 3.1.a of the Directive states that the following essential requirements are applicable to all apparatus: "The protection of the health and safety of the user and any other person, including the objectives with respect to safety requirements contained in Directive 73/23/EEC, but with no voltage limits". The Directive 73/23/EEC of February 1973 defines the harmonisation of the laws of Member States relating to electrical equipment designed for use within certain voltage limits (from 100 V to 100 kV).

The European Commission has also published the Council recommendation 1999/519/EC of July 12 on the limitation of the exposure of the general public to electromagnetic fields (0 to 300 GHz). The limits defined in this recommendation are based on the ICNIRP guidelines of April 1998 for the general public

A.1.3 – Evaluation of safety perimeters

As far as ALCATEL 9400 infrastructure is concerned, the level of human exposure to electromagnetic waves is characterized by the power density, which represents the basic restriction at those frequencies.

The following formula gives the power density assuming far field propagation of RF fields in free space. Then, the power density at a distance D in the main lobe of an antenna of gain G with an input power Pf is:

$$DP = \frac{PfG}{4\pi D^2} \quad (1)$$

Far field propagation assumption is valid at a distance equal to or higher than D_{ff} from the antenna such that $D_{ff} > 2D^2/\lambda$, where D is the largest dimension of the antenna and λ the wave length.

Considering antennas with high gain, the safety perimeter is first evaluated in the direction of the main lobe. The distance D_L where exposure level may reach the exposure limit DPL is provided by formula (2).

$$D_L(m) = \sqrt{\frac{PfG(W)}{4\pi DPL(W/m2)}} \quad (2)$$

Formula (2) overestimates the safety distance if it is lower than the far field distance D_{ff}.

A.1.4 – Safety perimeters

Based on the maximum power of the equipment, the following table gives the safety distances for the general public. Theses distances are valid for 1+0 and 1+1 configurations.

If the transmitted power changes (see § 7.12), the safety distance can be calculated using the formula given in Appendix 1.3 above.

Equipment	Frequency range (GHz)	Antenna gain (dBi)	Maximum power (dBm)	Safety Distance in front of the antenna (meters)
9413 UX	12,75–13,25	35,3	25	2,9
		38,3	25	4,1
		41,5	25	6
		45,1	25	9
9413 UX 16 QAM	12,75–13,25	35,3	21	1,8
		38,3	21	2,6
		41,5	21	3,8
		45,1	21	5,7
9415 UX	14,4–15,35	31,1	21	1,1
		36,5	21	2,1
		40	21	3,2
		42,5	21	4,2
		46	21	6,3

Equipment	Frequency range (GHz)	Antenna gain (dBi)	Maximum power (dBm)	Safety Distance in front of the antenna (meters)
9415 UX RTPC	14,4–15,35	31,1	24	1,6
		36,5	24	3
		40	24	4,5
		42,5	24	6
		46	24	8,9
9418 UX	17,7–19,7	33,3	16,5	0,9
		38,6	16,5	1,6
		42,1	16,5	2,4
		44,6	16,5	3,2
		47,8	16,5	4,6
9418 UX	17,7–19,7	33,3	24,5	2,2
		38,6	24,5	4
		42,1	24,5	6
		44,6	24,5	8
		47,8	24,5	11,6
9423 UX 4QAM	21,2–23,6	34,9	19	1,4
		40,1	19	2,5
		43,6	19	3,8
		46,1	19	5,1
		49,4	19	7,4
9423 UX 16QAM	21,2–23,6	34,9	17	1,1
		40,1	17	2
		43,6	17	3
		46,1	17	4
		49,4	17	5,9
9425 UX 4QAM	24,5–26,5	36	18	1,4
		41,1	18	2,5
9425 UX 16QAM	24,5–26,5	36	16	1,1
		41,1	16	2
9438 UX 4QAM	37–39,5	39,6	16	1,7
		44,5	16	3
9438 UX 16QAM	37–39,5	39,6	14	1,4
		44,5	14	2,4

A.1.5 - Implementation of protection rules

In order to comply with the previously mentioned distance constraints, a safety perimeter must be materialized along with the proper regulatory signaling and warning,

unless the following mounting rules could be implemented for the antennas or outdoor equipment with embedded antennas; they must be placed:

- At the edge of the roof in a position such as nobody can sit or stand in front of the antenna within the applicable distance mentioned above.
  OR
- Against an outside wall or under a roof, with no obstacle within 2 meters in the direction of the signal transmission, taking into account the antenna aperture.If the antenna is placed on a balcony, it should be placed in a position such as nobody can pass in front of it. This means that: either the equipment is fixed on the outside wall, not on the wall between the balcony and the building, and pointed toward the outside with no obstacle within 2 meters; or the access to the balcony must be closed and proper marking used.

  OR

If none of these mountings is possible, then a safety perimeter must be materialized along with the proper regulatory signaling and warning.

These rules apply to the general public and therefore do not apply in case the equipment is installed in professional premises of an Operator with no access to the public, provided that the access to the so defined area around the equipment be restricted to workers having received an adequate information on EMF risks.

These constraints are based on a theoretical general worst case study.

A.1.6 - Conclusion

Alcatel 9400 transmit non-ionizing electromagnetic waves at a very low power, not exceeding 1 Watt at lower frequencies.

Our installation rules make sure that it is impossible for anybody to sit or stand in front of the antenna within these distances either by the physical position of the antenna or with a safety perimeter.

These installation rules are followed by our installation teams and any of our sub-contractors and are they part of the instructions delivered along the equipment for installation by our customers or by third parties.

Appendix 2 – Installation information

Station

Station name				**Name of terminal in station ***				
	Main	Extension	Access	**Name of remote terminal ***				
IDU N_				**Remote terminal site**				
ODU N_				**Remote terminal azimuth**				
Software version				**Power supply**	24 V		48 V	

***** *CAUTION ! No more than eight characters allowed by MS-DOS*

Hardware configuration	1+0 without ESC ext	1+0 with ESC ext	1+1 HSB 1 antenna	1+1 HSB 2 antennas	1+1 Frequency div.

Software key	Master/Slave	RCT	Max config.	Max capacity	TPC	Frequency range

Tributaries

Terminal bit rate *(Limited by the interface boards and the software key installed)*	2 x 2 Mbit/s	4 x 2 Mbit/s	8 x 2 Mbit/s	16 x 2 Mbit/s	1 x 34 Mbit/s
Tributary board type		4 inputs 2 Mbit/s	8 inputs 2 Mbit/s	16 inputs 2 Mbit/s	34 Mbit/s + 2 Mbit/s aux.

Tributary	**Name**	**Impedance**	**BNC-1.6/5.6-Terminal block**	**Tributary**	**Name**	**Impedance**	**BNC-1.6/5.6-Terminal block**
N_ 1				N_ 2			
N_ 3				N_ 4			
N_ 5				N_ 6			
N_ 7				N_ 8			
N_ 9				N_ 10			
N_ 11				N_ 12			
N_ 13				N_ 14			
N_ 15				N_ 16			
2 Mbit/s aux.							

Impedance: 75 or 120 Ohms for 2 Mbit/s channels only.

Tributary cross connect	If no particular matrix is supplied by the network administrator, use the standard configuration.

Radio configuration

Polarisation (H/V)		Normal channel	Standby channel	
Transmit frequency ($_{FTx}$)				*Values supplied by network administrator*
Receive frequency ($_{FRx}$)				*$\|F_{Tx} - F_{Rx}\|$ = Duplex separation*
Duplex separation	MHz			
Link identification code		Transmit	Receive	*Value between zero (default) and 31, inserted in transmission and awaited in reception, supplied by the network administrator*
Amplifier (15 and 18 GHz bands only)		Standard power	High power	
Output attenuator (13, 15 and 18 GHz bands only)		0 dB	3 dB	6 dB / 10 dB
Output power control (13, 23, 25 and 38 GHz bands only**)**		**Normal**	**Standby**	*Values supplied by network administrator*
		dBm	dBm	

Local loop (Option)	Absent	Present	Absent	Present		

Engineering service channels (ESC)

Supervision	Superv. bit rate	NE physical address	*Bit rate: 9600 bit/s for an 9400XX network* *Address: 01 by default* *Values supplied by network administrator*	Inter - terminal cable*
SCC/NMS switching matrix	*Apply standard configuration unless a specific matrix is supplied by the network administrator*			
Station telephone	Nº Input level Output level	 dBm dBm	*Caution! No more than one zero in the number. Value supplied by network administrator.*	
ESC extension IDU	Absent	Present	*Optional in 1+0 configurations* *Standard in 1+1 configurations*	
Definition of ESCs				
S ESC N_ 3	V11 / V28	ESC-1, 2, 3, 4 or 5		
S ESC N_ 4 (Bit rate > 2 x 2 Mbit/s)	64 kbit/s V11	ESC-1, 2, 3, 4 or 5		
S ESC N_ 5 (Bit rate > 2 x 2 Mbit/s)	64 kbit/s V11	64 kbit/s V11 or V28		

* *Check the boxes for which the inter-terminal link cables need to be installed.*

Performance thresholds (triggering alarms)

Early switching request	2 x 2 Mbit/s	4 x 2 Mbit/s	8 x 2 Mbit/s	16 x 2 Mbit/s	1 x 34 Mbit/s
S Minimum values	-90 dBm	-87 dBm	-84 dBm	-81 dBm	-81 dBm
S Terminal values					
Bit error ratio	Early switching request	Switching request	Priority switching request		
S Average values	$1E^{-08}$	$1E^{-06}$	$1E^{-03}$		
S Terminal values					
Propagation thresholds	Nominal power (dBm)	Received power threshold (dB)	*Values specified by network administrator.* *The sum of the two values must not exceed the early switching request alarm value.* *Default values: -35 dBm; -35 dB*		

Maintenance thresholds

Threshold	Section ES	Section US	PSAC	PSAD	Hop ES	Hop US	PSRC
Value							

Telesignalling (TS) and remote controls (TC)

Install the loops required and name them in the software (NO: Normally open/NC: Normally closed).

Loop	Name	NO/NC	Loop	Name	NO/NC
TS N_ 1			TS N_ 2		
TS N_ 3			TS N_ 4		
TS N_ 5			TS N_ 6		
TS N_ 7			TS N_ 8		
TC N_ 1					

Specify the default alarm, equipment and environment alarms (UG, NUG, Disabled, Status).

G784 thresholds

Required if the extended 946LUX11 or 946LUX12 option is installed, or if the standard 946LUX40 is installed.

Threshold	RLTS-1	RLTS21	RLTS-3	RLTS-4	RLTMMax	RLTMMin	
Value							
Threshold	PSAC	PSAD	PSRC	PSRSAD	BBE	ES	SES
Value							

Appendix 3 – Set of consumables

This set is defined for a link, its reference is: **9400UXT002**

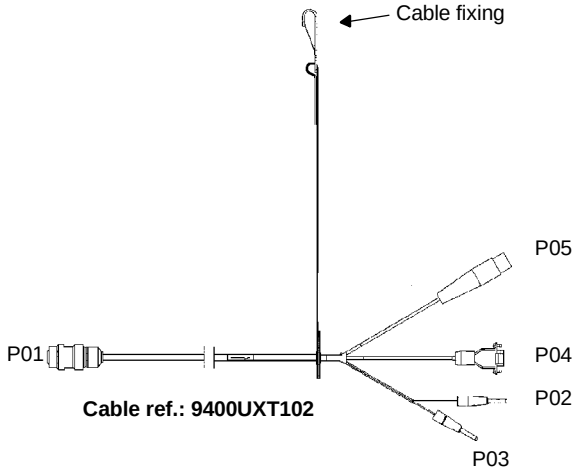
ITEM NUMBER	DESCRIPTION	QUANTITY
1AC000570014	Black heat shrinkable sheathing, 3.2 to 1.6 mm	1.2 m
1AC000570017	Black heat shrinkable sheathing, 4.8 to 2.4 mm	1.2 m
1AC000570010	Black heat shrinkable sheathing, 9.5 to 4.8 mm	1.2 m
1AC001250004	Black heat shrinkable sheathing, 19 to 9.5 mm	1.2 m
1AD012980002	Black Colson cable tie: 350 mm	200
1AD012980003	Black Colson cable tie: 173 mm	100
77085356	Plain cable tie: 95 mm	50
77065004	“Scotchfil” mastic tape, 38 mm wide, 1.5 m long	1
77072089	Black marker pen	1
77970019	Cloth	1 kg
77970278	Desoldering braid	1
77974586	7/10 mm diameter solder wire	100 gr
77062820	White silicon cartridge	1
1AD020450002	“ty-rap” cable tie, 46 M-8	10
1AD020460001	“ty-rap” cable tie, 51 M translucent	20
77081203	Autoamalga. adhesive tape, 19 mm wide, 10 m long	1
77091318	“ty-rap” cable tie, 244 M	20

Appendix 4 – Service kit

A.4.1 – Service kit 9400UXT102 for ODU 94xxUXR201 to 94xxUXR203

The service kit is used when pointing the antennas of each station. It comprises:

- a cable for connecting to the "MAINT" connector on the ODU**, ref. : 9400UXT102, 3CC07972AAxx**
- a headset/microphone assembly for links with the indoor unit and the remote station for using the telephone service channel ESC 2, not available for use with the Light IDU versions, ref.: **1AF00375ABAA**.

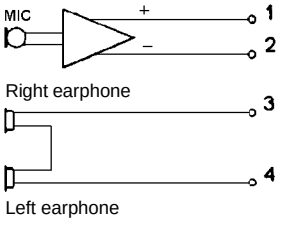


Cable ref.: 9400UXT102

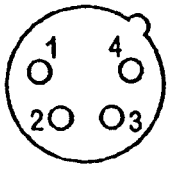
AUDIO PINOUT P05

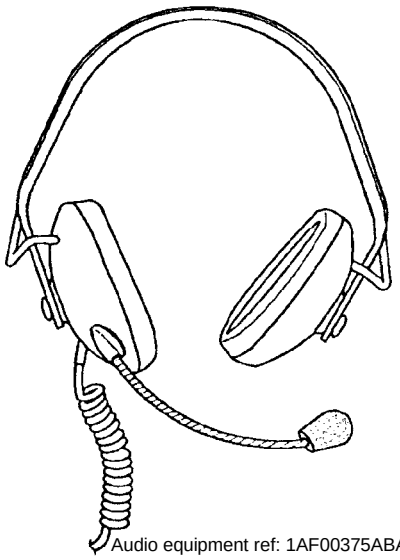
1 – Micro P
2 – Micro N
3 – Speaker P
4 – Speaker N (ground)

- Connector P01 : ODU link, "Maintenance" socket.
- Connector P02 / P03 : Multimeter for measuring AGC voltage.
- Connector P04 : Connector for portable PC, reserved for Alcatel use.
- Connector P05 : Connector for headset (cannot be used with Light IDU versions).



Male connector – External view





Audio equipment ref: 1AF00375ABAA

A.4.2 – Short service kit 9400UXT113 for A9400UX flat ODU

The short service kit 9400UXT113 is used when pointing the antennas of each station. It comprises a cable for connecting to the “MAINT” connector on the ODU, **ref.: 3CC13477AAAA**.

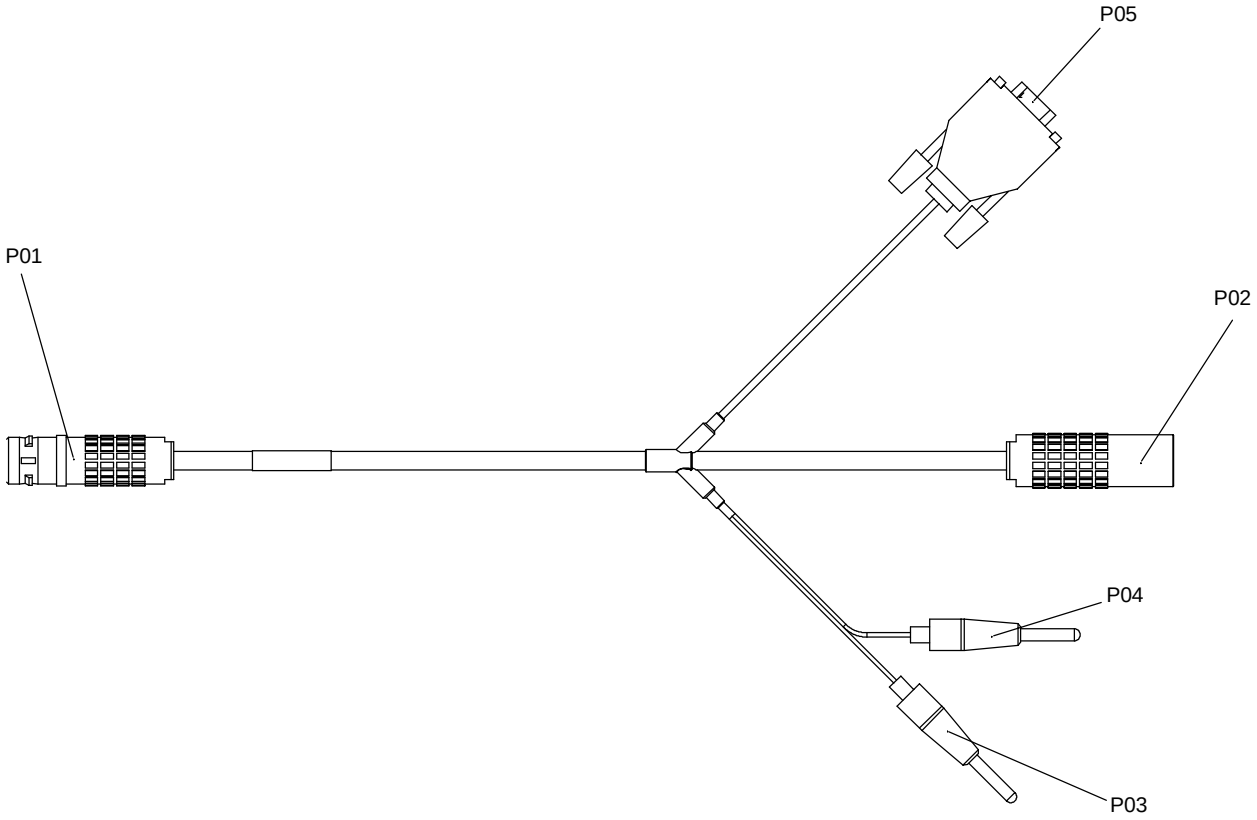


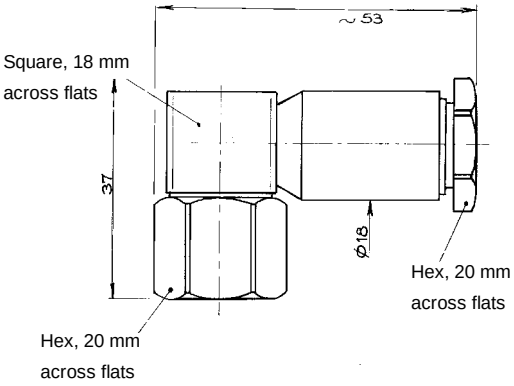
Figure 73 - Cable 3CC13477AAAA

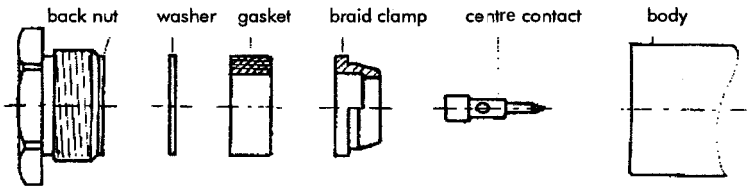
- Connector P01: ODU link, “Maintenance” socket.
- Connector P02: connection of the portable service kit 3DB00511AAAA, if the user wants to use a headset through the service kit, for example.
- Connector P03 / P04:multimeter for measuring AGC voltage.
- Connector P05: connector for portable PC. To be used for maintenance purpose or necessity to upgrade the ODU embedded software.

Appendix 5 – Assembling “N” type coaxial connectors

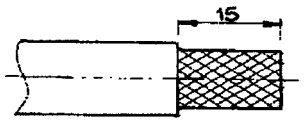
A.5.1 – Type 1 fitting on cable 1AC001100022

Connector BECLER
BK / 224

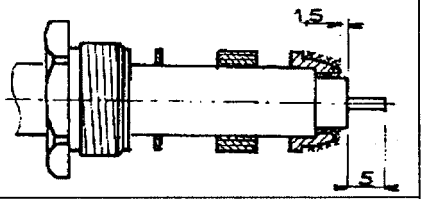




1) Strip 15 mm of outer insulation from the cable and comb out the braid.

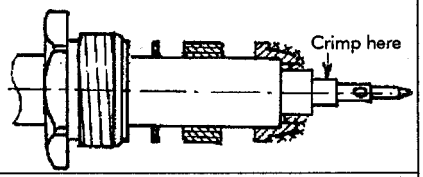


2) Fit onto the cable: the nut, washer, flat gasket and braid clamp, then fold back the braid over the braid clamp and cut away the excess.
Strip the core of the cable to the dimensions opposite.



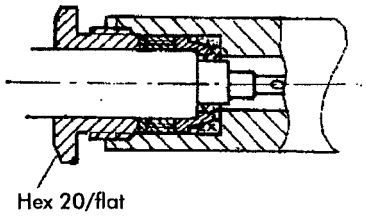
3) Fit the centre contact fully home on the core of the cable. Crimp with DANIELS M22520/5-01 tool fitted with jaws Y215P (hex: .128 across flats).
If not possible, solder.

CAUTION: DO NOT MELT THE CABLE INSULATION.



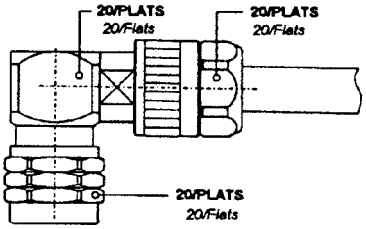
4) Abut the gasket and washer against the braid clamp then offer up the end of the cable into the body of the connector making sure that the contacts mate correctly with each other.
Screw the nut into the body of the connector (torque: 500 N/cm).

The connector is now ready for use.

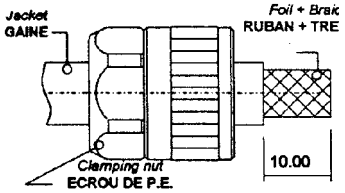


A.5.2 – Type 2 mounting on cable 1AC001100022

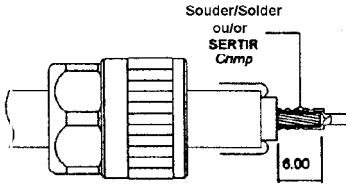
Fiche/*Connector* DELTA OHM
08 250 173



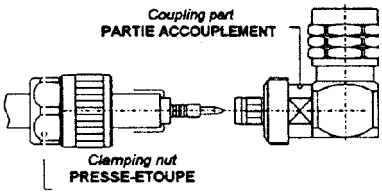
Dénuder la gaine sur 10 mm, mettre l'écrou sur la gaine.
Strip cable (10 mm), put the clamping nut on the cable.



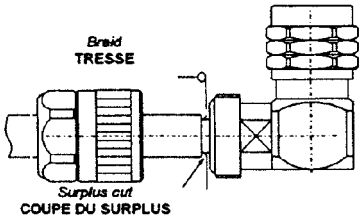
Rabattre les 10 mm de tresse sur la gaine.
Fold back the 10 mm of braid over the outer insulation.
Couper le diélectrique et le ruban sur 6 mm.
Cut dielectric & foil over 6 mm.
Souder ou sertir le contact central.
Solder or crimp the contact on the inner conductor.



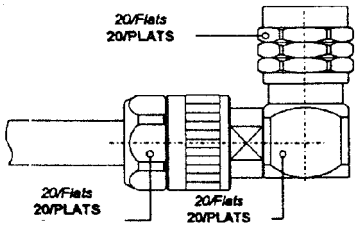
Monter la partie accouplement entre le ruban et la tresse du câble.
Fit the coupling part between the foil and the braid of the cable.



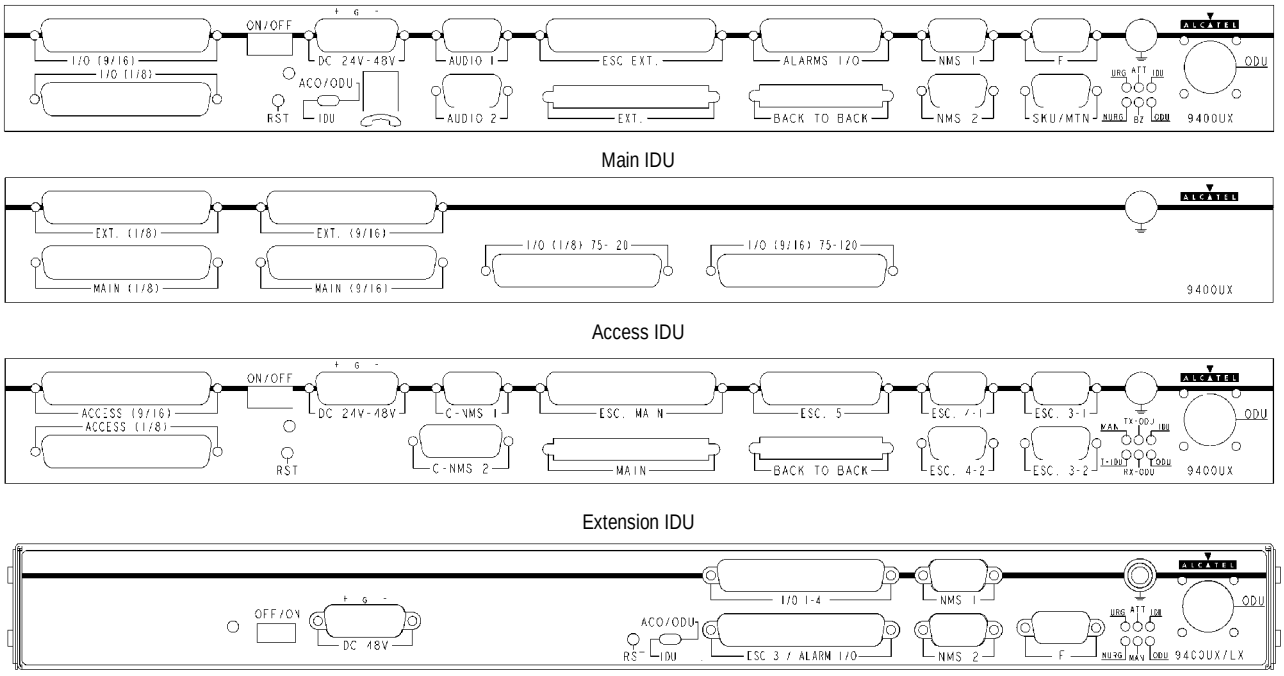
Rabattre la tresse vers l'avant du connecteur et couper le surplus.
Fold the braid towards the front of the connector and cut away any surplus flush with the connector.



Assembler la partie presse étoupe et accouplement, serrer avec une clé plate 20 mm (couple maxi 35 N/m).
Mount the gland and coupling part, tighten with a 20 mm open ended wrench (Max torque 35 N/m).



Appendix 6 – Pin out of user ports



Light IDU

Figure 74 - IDU connector panels

A.6.1 – Tributaries

A.6.1.1 – 2 Mbit/s tributaries

A.6.1.1.1 – 19”IDU connectors (Main, light, access)

Connector		**“I/O 1 - 4” Light IDU “I/O (1/8)” main IDU “I/O (1/8)” access IDU**								**“I/O (9/16)” main IDU “I/O (9/16)” access IDU**							
2 Mbit/s tributary Nº		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
Pin	In+	2	3	4	5	6	7	8	9	2	3	4	5	6	7	8	9
	In-	21	22	23	24	25	26	27	28	21	22	23	24	25	26	27	28
	Out +	12	13	14	15	16	17	18	19	12	13	14	15	16	17	18	19
	Out -	30	31	32	33	34	35	36	37	30	31	32	33	34	35	36	37
Light IDU MCU board		Light IDU MCU board															
4 x 2 Mbit/s boards		LAU: 3CC06058Axxx LIU: 3CC05818Axxx															
8 x 2 Mbit/s boards		LAU: 3CC06059Axxx LIU: 3CC06026Axxx															
16 x 2 Mbit/s boards		LAU: 3CC06059Axxx LIU: 3CC06026Axxx								LAU: 3CC06059AAxx LIU: 3CC06119Axxx							
Access board		LAU: 3CC06765AAxx (requires the same configurations in main and extension IDUs)															

Ground pins: **10** on I/O (1/8) and I/O (9/16) of the **main IDU** and **access IDU,**
6, **7, 8, 9, 10, 11, 16, 17, 18, 19, 20, 25, 26, 27, 28, 29, 34, 35, 36, 37** on I/O 1 - 4 **of the Light IDU**

In 75 ohms version, “In+”, “Out+” are the live input and output pins and “In-”, “Out-” are the ground pins for the same signals.

Depending on the boards installed, some connector pins may not be available (areas shaded in the table).

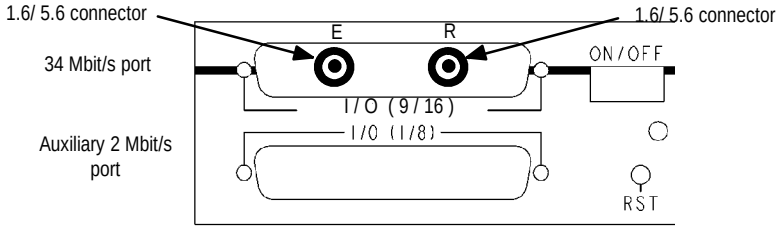
For the automatic configuration, the pin 20 must be grounded at the level of the 75 ohms cable and not connected in 120 ohms otherwise the switching has to be forced (see chapter 7.2.2.4).

A.6.1.2 – 34 Mbit/s tributaries

The 34 Mbit/s port is via 1.6/5.6 connectors:

- "**I/O (9/16)**" on the main IDU in a configuration without MUX protection,
- "**I/O (9/16)**" on the access IDU in a configuration with MUX protection.

If the connector system required for the 34 Mbit/s system is BNC, use a BNC/ 1.6/5.6 adapter kit ref: 3CC08249AAxx (option).



The auxiliary 2 Mbit/s port uses female 37-way Sub-D connectors:

- "**I/O (1/8)**" on the main IDU,
- "**I/O (1/8)**" on the access IDU.

SIGNAL	PIN				**LAU**: 3CC06061AAxx **LIU:** 3CC06118AAxx
	In+	In-	Out+	Out-	
Auxiliary 2 Mbit/s stream	9	28	19	37	X

Pin 10: ground; in 75 ohms configurations, "In+" and "Out+" are the live input and output pins and "In-" and "Out-" are the ground pins for the same signals.

A.6.2 – Engineering service channels

A.6.2.1 – Supervision bus interface

Concerns the "**NMS1**" and "**NMS2**" female 9-way Sub-D connectors on the **main or Light IDU.**

Pin	Signal	Pin	Signal
1	RS 485 data in (-)	6	RS 485 data in (+)
2	RS 485 clock in (-)	7	RS 485 clock in (+)
3	Ground		
4	RS 485 data out (+)	8	RS 485 data out (-)
5	RS 485 clock out (+)	9	RS 485 clock out (-)

A.6.2.2 - Telephone ESC with selective calling

Concerns the “**AUDIO 1**” and “**AUDIO 2**” female 9-way Sub-D connectors on the **main IDU**.

Pin	Signal	Pin	Signal
1	Audio signal from user (+)	6	Audio signal from user (-)
2		7	M-wire signal
3	0 V		
4	Audio signal to user (-)	8	Audio signal to user (+)
5	E-wire signal	9	+5 V protected

A.6.2.3 – Engineering service channels ESC 3, ESC 4 and ESC 5

A.6.2.3.1 – Service channel ESC3 on the Light IDU versions: “ESC3 / ALARM I/O” connector

Female 37-way connector	Out+ pin	Out– or 0 V pin		Pin
64 kbit/s transmitted data input	4	23	Unprotected +5 V	11
64 kHz transmit clock input	5	24	Ground	12 – 30
64 kbit/s received data output	1	20	Reserved for ALCATEL use	3 – 6 – 22 – 25
64 kHz receive clock output	2	21		

A.6.2.3.2 – Without extension unit: “ESC EXT” connector on the main IDU

	ESC3		ESC4		ESC5	
	Out+ pin	Out– or 0 V pin	Out+ pin	Out– or 0 V pin	Out+ pin	Out– or 0 V pin
64 kbit/s transmitted data input	1	20	7	26	13	32
64 kHz transmit clock input	2	21	8	27	14	33
8 kHz transmit byte sync input	3	22	9	28		
64 kbit/s received data output	4	23	10	29	15	34
64 kHz receive clock output	5	24	11	30	16	35
8 kHz receive byte sync output	6	25	12	31		
ESC processing signal (Reserved for Alcatel use)	17		36			
0 volt (ground)	18					
Squelch signal for ESCs (Reserved for Alcatel use)	19					
(Reserved for Alcatel use)	37					

A.6.2.3.3 – ”ESC MAIN” connector on extension IDU

	ESC3		ESC4		ESC5	
	Out+ pin	Out– or 0 V pin	Out+ pin	Out– or 0 V pin	Out+ pin	Out– or 0 V pin
64 kbit/s transmitted data input	4	23	10	29	15	34
64 kHz transmit clock input	5	24	11	30	16	35
8 kHz transmit byte sync input	6	25	12	31		
64 kbit/s received data output	1	20	7	26	13	32
64 kHz receive clock output	2	21	8	27	14	33
8 kHz receive byte sync output	3	22	9	28		
ESC processing signal (Reserved for Alcatel use)	17		36			
0 volt (ground)	18					
Squelch signal for ESCs (Reserved for Alcatel use)	19					
DO NOT CONNECT (Reserved for Alcatel use)	37					

A.6.2.3.4 – With extension unit

The table below concerns the following connectors:

Female 9-way Sub-D, "**ESC. 3-1**", "**ESC. 3-2**", "**ESC. 4-1**", "**ESC. 4-2**" on the extension IDU.

ESC. 3-1 and ESC. 4-1 connectors			ESC. 3-2 and ESC. 4-2 connectors		
Pin	**G703 64 kbit/s signal**	**Analogue ESC signal**	**Pin**	**G703 64 kbit/s signal**	**Analogue ESC signal**
8	Received signal (+)	Audio 1 (+) OUT	8	N.C.	Audio 2 (+) OUT
4	Received signal (-)	Audio 1 (-) OUT	4	N.C.	Audio 2 (-) OUT
1	Transmitted signal (+)	Audio 1 (+) IN	1	N.C.	Audio 2 (+) IN
6	Transmitted signal (-)	Audio 1 (-) IN	6	N.C.	Audio 2 (-) IN
9	N.C.	N.C.	9	N.C.	N.C.
5	N.C.	E signal 1	5	N.C.	E signal 2
2	N.C.	N.C.	2	N.C.	N.C.
7	N.C.	M signal 1	7	N.C.	M signal 2
3	Ground	Ground	3	Ground	Ground

N.C. = Not connected.

ESC. 3-1 and ESC. 4-1 connectors					
Pin	**V11 or V28 * signal**				
	64 kbit/s	**9600 bit/s max. P to MP**	**9600 bit/s max. P to P**	**2x4800 bit/s max P to MP + P to P**	**2x4800 bit/s max. P to P**
8	Received data (+)	Receiv. PM1 signal (+)	Receiv. PP1 signal (+)	Receiv. PM1A signal (+)	Receiv. PP1A signal (+)
4	Received data (-)	Receiv. PM1 signal (-)	Receiv. PP1 signal (-)	Receiv. PM1A signal (-)	Receiv. PP1A signal (-)
1	Transmitted data (+)	Transm. PM1 signal (+)	Transm. PP1 signal (+)	Transm. PM1A signal (+)	Transm. PP1A signal (+)
6	Transmitted data (-)	Transm. PM1 signal (-)	Transm. PP1 signal (-)	Transm. PM1A signal (-)	Transm. PP1A signal (-)
9	Transmit clock (+)	NC	NC	Receiv. PP1A signal (+)	Receiv. PP2A signal (+)
5	Transmit clock (-)	NC	NC	Receiv. PP1A signal (-)	Receiv. PP2A signal (-)
2	Received clock (+)	NC	NC	Transm. PP1A signal (+)	Transm. PP2A signal (+)
7	Received clock (-)	NC	NC	Transm. PP1A signal (-)	Transm. PP2A signal (-)
3	Ground	Ground	Ground	Ground	Ground

N.C. = Not Connected ; PM = Point-to-Multipoint ; PP = Point-to-point ; ***** In V28 mode, the (-) signals are commoned to ground.

ESC. 3-2 and ESC. 4-2 connectors					
Pin	V11 or V28 * signal				
	64 kbit/s	9600 bit/s max P to MP	9600 bit/s max P to P	2x4800 bit/s max P to MP	2x4800 bit/s max. P to P
8	NC	Received PM2 signal (+)	NC	Received PM1B signal (+)	NC
4	NC	Received PM2 signal (-)	NC	Received PM1B signal (-)	NC
1	NC	Transmitted PM2 signal (+)	NC	Transmitted PM1B signal (+)	NC
6	NC	Transmitted PM2 signal (-)	NC	Transmitted PM1B signal (-)	NC
9	NC	NC	NC	NC	NC
5	NC	NC	NC	NC	NC
2	NC	NC	NC	NC	NC
7	NC	NC	NC	NC	NC
3	Ground	Ground	Ground	Ground	Ground

The table below concerns the "**ESC. 5**" female 25-way Sub-D connector on the extension IDU.

Signal	ESC 5 connector pin				Signal	Pin
	In+	In-	Out+	Out-		
DO NOT CONNECT Reserved for Alcatel use	3		2		Reserved for Alcatel use	1, 4, 8, 14, 15, 20
ESC 5: Clock	10	22	12	24	0 volt	6, 7, 9, 16, 17, 18, 19, 21
ESC 5: Data	11	23	13	25	Not connected	2, 3, 5

"In+" and "Out+" are the live input (received signal) and output (transmitted signal) pins and "In–" and "Out–" are the ground pins for the same signals.

A.6.3 – Alarms and TS/TC

The alarm and remote control relays are de-energized in the absence of an alarm.

The remote indication loops (TS) between electrical ground and TS input, can be configured as normally open or normally closed (in the absence of an alarm), depending on the positioning of the soldered links ("swaps") on the MCU board, as indicated in section 7.7.2.

A.6.3.1 - "ESC3 / ALARM I/O" connector on the Light IDU versions

Female 37-way connector

Signal	N/C contact	N/O contact	Common contact	TS signals	
	Pin	Pin	Pin		
Urgent alarm	14	15	32	TS1 input	7
Non urgent alarm	33	34	16	TS2 input	26
Alarm "Attended" state	17	18	35	TS3 input	8
"Housekeeping" remote control	36	37	19	TS4 input	27
Unprotected +5 V	11			TS5 input	9
Ground	12 - 30			TS6 input	28
Reserved for ALCATEL	3 - 6 - 22 - 25			TS7 input	10
				TS8 input	29

A.6.3.2 - "ALARMS I/O" connector on the main IDU

Concerns a female 25-pin Sub-D connector.

Signal	N/C contact	N/O contact	Common contact	TS signals			
	Pin	Pin	Pin		Pin		Pin
Urgent alarm	1	2	14	TS1 input	7	TS5 input	9
Non urgent alarm	15	16	3	TS2 input	20	TS6 input	22
Alarm "Attended" state	4	5	17	TS3 input	8	TS7 input	10
"Housekeeping" remote control	18	19	6	TS4 input	21	TS8 input	23
0 V	25			Reserved for ALCATEL		11 - 12 - 13 - 24	

A.6.4 – "F" connector

A.6.4.1 - IDU "F" connector

The table below concerns the female 9-way sub-D connector used for supervisory PC connection.

Pin	ASSIGNMENT	Pin	ASSIGNMENT
1	Data Carrier Detect	6	Data Set Ready
2	Receive data - level V28	7	Request to send
3	Transmit data - level V28		
4	Data Terminal Ready	8	Clear to send
5	GND	9	Not connected

Appendix 7 – List of equipment alarms

N_	Alarm, status or remote control message	Procedure for tracing the failure or fault
56	NMS1 PORT NMS1 interface disconnected.	Cable on NMS1 port not connected or defective.
57	NMS2 PORT NMS2 interface disconnected.	Cable on NMS2 port not connected or defective.
55	RADIO PORT Radio interface disconnected.	Radio link cut off. To determine the cause of the fault, examine the other equipment alarms, starting from the highest alarm grouping level.
35	TRIB.i RC or AUX 2MB RC: DIST. LOOP Distant loop(back).	Remote control: Remote baseband loopback configuration (ODU port) on tributary n_i or auxiliary 2 Mbit/s bit stream.
32	TRIB.i or AUX 2MB: UNEXPECTED Unexpected tributary input bit rate.	URGent alarm: Incompatibility between bit rate or code at the tributary input or auxiliary 2 Mbit/s bit stream input and that allowed by the LIU board or boards. Check the installation and operational configurations of the LIU boards. **Caution!** Never loop the ports of the unused tributaries at the ends of a link, simply disable them through the software.
34	TRIB.i RC or AUX 2MB RC: SEQ. INS. Pseudo-random sequence insertion.	Remote control: pseudo-random bit pattern inserted on tributary n_i or auxiliary 2 Mbit/s bit stream (Caution! The Wandel & Goltermann PF4 analyzer cannot be used to measure BER on this sequence).
31	TRIB.i or AUX 2MB: SIGNAL LOSS Loss of signal detected at tributary level.	Urgent alarm: no input signal on tributary n_i or auxiliary 2 Mbit/s bit stream. Check the tributary wiring and the presence of the signal.
36	TRIB.i RC or AUX 2MB RC: BER MEAS. Error ratio measure.	Remote control: measurement of bit error ratio on tributary n_i or auxiliary 2 Mbit/s bit stream.
33	TRIB.i or AUX 2MB: TX AIS Generating an AIS at transmission.	Urgent alarm: Insertion on transmission of an AIS on tributary n_i or on the auxiliary 2 Mbit/s bit stream. No input signal? Check for presence of the signal and the wiring.
37	TRIB.i or AUX 2MB: RX AIS Generating an AIS at reception.	Urgent alarm: Generation of an AIS in reception on tributary n_i or auxiliary 2 Mbit/s bit stream. Check the configuration and the presence of the bit stream sent from the far end.
50	SOFTWARE KEY ALARM Software key alarm or (key) modified.	Urgent alarm (alarm generated on a change of key, when the new key does not have the same characteristics as the previous one. Confirm the new key by opening the “Operation parameters” application and downloading the configuration (File/Send). The key alarm should disappear.
15 29	GENERAL ALARM General alarm.	Urgent alarm: Probably a power supply problem or an equipment problem normally requiring a change of equipment. If the alarm persists, replace the IDU (§ 6.7). Check the configuration, particularly the presence or absence of the RF loop configuration. If the ODU has failed, change it (§ 6.8).

N_	Alarm, status or remote control message	Procedure for tracing the failure or fault
38	MUX_DEMUX GEN. ALARM General alarm. MUX_DEMUX NORMAL / STANDBY ALARM Active alarm.	Urgent alarm: Failure of the SCU board in the extension IDU; replace the extension IDU. Urgent alarm: Failure of the MCU board in the main IDU; replace the main IDU.
12	AMPLIFIER Amplifier mode.	Remote control: Cut off transmitted power from the channel 1 or X amplifier. **Remote station:** Automatic disabling after five minutes
19	BB LOOP Local loop(back).	Remote control: local baseband loop at the input of ODU 1 or ODU X. **Set up on remote station:** Remote control rejected
18	RF LOOP Local loop(back).	Remote control: RF local loopback on channel 1 or channel X. **Set up on remote station:** Remote control rejected
43	LOCAL LOOP Local loop(back).	Remote control: local baseband loopback on main or extension IDU. **Set up on remote station:** Remote control rejected
1	CABLE Active alarm.	Cable problem: disconnected, not fastened correctly, cut, short circuited or bad contact in connector. If the cable is short circuited, the IDU power supply is probably generating an alarm condition and crackling may be heard. Check that the power supply alarm disappears when you disconnect the cable. Repair the cable, check its connectors, change the defective components or the cable.
49	SOFTWARE KEY MISSING Software key missing.	Urgent alarm: This alarm prevents any change of configuration or reconfiguration on the equipment. Check that the software key is correctly placed on the SKU/MTN or MTN connector.
51	PROVISIONING REFUSED Configuration rejected.	Urgent alarm: The new software key is incompatible with the equipment. Change the key, or reset the equipment (available only in the 946LUX11 software version; in the 946LUX12 version, a CR alarm replaces it on the right in the summary line)
52	TERM. SHUTDOWN IN 24, 21, ..., 3H Time before terminal shutdown.	Urgent alarm: Time remaining in multiples of three hours before the link is to be cut off because of a key alarm. Fit the right software key.
26	LOW BER Low Binary Error Rate or quality switching request.	Urgent alarm: The BER on normal channel 1 or standby channel X is greater than the value defined in “Operation parameters/thresholds/switching thresholds” for the SR threshold (see § 6.6.2)
25	EARLY WARNING Early warning switching request (on received field or bit error ratio).	Urgent alarm: The BER on normal channel 1 or standby channel X is greater than the value defined in “Operation parameters/thresholds/switching thresholds” for the ESR threshold (see § 6.6.2)
47	HOUSEKEEPING 1 to 8 Active alarm.	Non-urgent alarm on station housekeeping dry loops: depends on how the loops are assigned (see § 7.10).

N_	Alarm, status or remote control message	Procedure for tracing the failure or fault
16 28 30 54	CONFIGURATION ERROR Configuration error.	Inconsistency between the hardware composition of the equipment and its declaration on defining the configuration; for example: – wrong transmit or receive frequency settings (ODU). – wrong bit rates (MCU and/or LAU / LIU). – wrong ESC customizing daughter boards. Make the necessary changes and, if necessary, change the IDU (see § 6.7) or ODU (see § 6.8).
53	EXTENSION MISSING Unit missing or dialogue problem.	Urgent alarm: no extension unit or no dialogue with MCU. Check that the link wiring is correctly connected or that it is not damaged; if not, replace the extension IDU.
41	RX 1 HIGH BER RX X HIGH BER High Binary Error rate or priority switching request.	Urgent alarm: priority switching request on high bit error ratio or loss of frame synchronization in reception on channel 1 or standby channel X.
44	LINK IDENTITY CODE Link identity code mismatch.	Urgent alarm: wrong frame identification code received on normal channel 1 or standby channel X. Check the consistency of configuration between transmitted and received codes. If consistent, check the received field level (disturbing signal received?).
20	FEC INHIBITION Forward Error Correction inhibition.	Remote control: disabling of the error correcting coder.
22	EQUALIZER INHIBITION Receive equalizer inhibition.	Remote control: disable receive equalizer on normal channel 1 or standby channel X.
58	MANAGER ISOLATED Manager isolated	The manager is not polling the NEs (telephone the manager for confirmation).
9	DELAYED MAINTENANCE Delayed (deferred) maintenance alarm.	Urgent alarm: see definition (§ 4.5.3); use the History log programs to identify the alarms. Having located and identified the fault, repair.
8	UNDELAYED MAINTENANCE Undelayed (prompt) maintenance alarm.	Urgent alarm: See definition (§ 4.5.3); use the history log programs to identify the alarms. Having located and identified the fault, replace the failed component or unit.
4	COMMON LOSS Common mode alarm.	In HSB mode, indicates switchover on alarm if the switchover on alarm option has been enabled (§ 4.5.1). **CAUTION!** this alarm can be cancelled only by switching the equipment off then on again. Nota: This alarm is inhibited.
5	MANUAL MODE Forced switching mode at transmission.	Indicates a switchover forced by transmit mode remote control on channel 1 or standby channel X. **On the remote station:** Automatic disabling after five minutes
6	SWITCHING MODE (Forced) switching mode manual.	Multiplexing/demultiplexing equipment selected manually by remote control (MCU of main IDU or SCU of extension IDU). **On the remote station:** Automatic disabling after five minutes
45	FORCED SWITCHING MODE (Forced) switching mode manual.	Remote control: receive channel selected by MCU by manual override. **On the remote station:** Automatic disabling after five minutes

N_	Alarm, status or remote control message	Procedure for tracing the failure or fault
14	UNIT MISSING Unit missing or no dialogue.	Urgent alarm: no outdoor unit or dialogue with microcontroller. Check that the connecting coaxial cable is correctly connected and not damaged; if not, change the ODU (§ 6.8).
46	EXTENSION UNIT General alarm.	Urgent alarm: general extension IDU alarm. The extension IDU or the SCU board has failed; replace the extension IDU.
39	MAIN UNIT General alarm.	Urgent alarm: general main IDU failure alarm; replace the IDU.
40	SUPERVISION UNIT General alarm.	Urgent alarm: MCU board failed in main IDU; replace the IDU.
42	RX 1 LOSS OF FRAME RX X LOSS OF FRAME Loss of frame (sync alarm).	Urgent alarm: Loss of receive mode frame synchronization on channel 1 or standby channel X caused by a receive mode problem resulting in transmission of an AIS. Check the received field level and the frequency configuration of the RF channel.
21	CARRIER UNLOCKED Carrier unlocked or demodulator loss of signal.	Urgent alarm: No signal at normal channel 1 or standby channel X input. Check the frequency of the transmitter on normal channel 1 or standby channel X and the bit rates of the equipment on both stations.
23	PROPAGATION Propagation alarm.	Urgent alarm: The received field is below the threshold defined in “Operation parameters/Thresholds/Propagation”) (see § 6.6.2).
13	OUTPUT POWER Output power (level) alarm.	Urgent alarm: Drop in transmit amplifier output power on channel 1 or standby channel X greater than 5 dB ; check that there are no remote controls to cut off output power; if there are not, replace the ODU (see § 6.8).
24	RECEIVE POWER Receive power alarm.	Urgent received field alarm on normal channel 1 or standby channel X: check the receive frequency of the equipment, continuity of the receive subsystem and then of the remote transmit subsystem. If faulty, change the defective modules.
7	SCU SWITCHING MCU SWITCHING	Indicates multiplexing/demultiplexing equipment not used.
10	SYNCHRONIZING Block synchronization alarm.	Urgent alarm: Bit rates not supported by modulator. Check the configuration of the installation parameters (§ 4.4.1) and the operational parameters (§ 4.5.1). Check the number of 2 or 34 Mbit/s inputs against the installation sheet. If everything checks out OK, change the ODU (§ 6.8).
–	SYNTHESIS Synthesis alarm.	Find your screen in the following screens and refer to the section on “zooming” in on a channel. **Exception :** If the alarm is only in “Main”, zoom in on this item and refer to § 6.6.1.
11	IF SYNTHETIZER Tx synthesizer alarm.	Urgent alarm: transmit IF synthesizer misaligned. On installation, check the transmit frequency, the duplex separation and the compatibility of the frequency with the ODU’s RF band. Restart the transceiver. In the event of failure, replace the ODU.
17	RF SYNTHETIZER RF synthesizer alarm.	Urgent alarm: incompatibility between the transmit and receive frequencies requested and those supported by the RF synthesized LO of channel 1 or X; check the transmit and receive frequencies and the duplex separation. If these are OK, change the ODU.

N_	Alarm, status or remote control message	Procedure for tracing the failure or fault
48	ENVIRONMENT CTRL 1 Remote control activated.	Environment remote control n_1 active.
3	TX X TX 1 Transmitter switched indication (status).	Indicates the transmitter currently operating in HSB mode (switched by remote control: transmitter X or 1 in manual override condition).
2	AIR DISPLACEMENT UNIT Air displacement unit (IDU fan) alarm.	Non-urgent alarm: For the IDU 3CC.....AXAA, one or two fans in the IDU are blocked. Change the fans or the IDU. For the IDU 3CC.....AXAB and 3CC.....AXAC, this means that a too high temperature was reached in the IDU. The two fans must be blocked. Change the fans or the IDU.
27	POWER SUPPLY ALARM	Urgent alarm: At least one of the secondary power supply voltages on ODU 1 or standby X is faulty. Replace the ODU (see § 6.8).

Appendix 8 – List of remote controls

The remote controls available in the “Alarms, States and Remote controls” application are listed in the table below. Depending on the configuration, some may not be available to the operator.

Since the transmission of some remote controls to a **remote station** may result in disconnection of the link entailing a call out to that station:

- the transmission of certain remote controls to a remote station is rejected by the software,
- other remote controls are disabled automatically after five minutes.

These limitations apply only to the 946LUX11 (from revision index N_4) and 946LUX12 (all releases) versions. The remote controls concerned are indicated in the second column of the table below.

REMOTE CONTROL LABEL	DISABLES/ENABLES
AMPLIFIER [Start] : Power Off	Power amplifier for each ODU. **On remote station**: Automatic disabling after five minutes
AMPLIFIER [End] : Power On	
EQUALIZER [Start] : inhibition (1)	Equalizer for each ODU.
EQUALIZER [End] : validation (1)	
FEC [Start] : inhibition (1)	Error correcting encoder for each ODU.
FEC [End] : validation (1)	
BB OUTDOOR [Start] : Local loop validated	Local baseband loop at the ODU input, for each ODU **On remote station**: Remote control rejected.
BB OUTDOOR [End] : Local loop inhibited	
RF [Start] : Local loop validated	Local RF loop for each ODU **On remote station**: Remote control rejected.
RF [End] : Local loop inhibited	
MUX/DEMUX [Start] : normal lockout validated	“Normal” multiplexer/demultiplexer. **On remote station**: Automatic disabling after five minutes.
MUX/DEMUX [End] : normal lockout inhibited	
MUX/DEMUX [Start] : standby switching validated	“Standby” multiplexer/demultiplexer for a 1+1 configuration with MUX protection. **On remote station**: Automatic disabling after five minutes.
MUX/DEMUX [End] : standby switching inhibited	
MCU>TRIB. x [Start] : pseudo rand. seq. ins. valid.	Insertion of a pseudo-random bit pattern on tributary N_ x (for tests) on an active tributary.
MCU>TRIB. x [End] : pseudo rand. seq. ins. inhib.	
MCU>TRIB. x [Start] : BER measure validated	Measurement of BER on tributary N_ x (via the above pseudo-random bit pattern) on an active tributary.
MCU>TRIB. x [End] : BER measure inhibited	
MCU>TRIB. x [Start] : distant loop validated	Remote loopback on tributary N_ x.
MCU>TRIB. x [End] : distant loop inhibited	

(1) Unavailable for16QAM.

REMOTE CONTROL LABEL	DISABLES/ENABLES
MCU>RX 1 [Start] : manual lockout validated	Forced manual switchover to main receiver (in 1+1 configuration). **On remote station**: Automatic disabling after five minutes.
MCU>RX 1 [End] : manual lockout inhibited	
MCU>RX X [Start] : manual switch. validated	Forced manual switchover to standby receiver (in 1+1 configuration). **On remote station**: Automatic disabling after five minutes.
MCU>RX X [End] : manual switch. inhibited	
MCU>AUX. [Start] : pseudo rand. seq. ins. valid.	Insertion of a pseudo-random bit pattern on the 2 Mbit/s auxiliary bit stream (34 Mbit/s rate).
MCU>AUX. [End] : pseudo rand. seq. ins. inhib.	
MCU>AUX. [Start] : BER measure validated	Measurement of BER on the 2 Mbit/s auxiliary bit stream (using the above pseudo-random bit pattern).
MCU>AUX. [End] : BER measure inhibited	
MCU>AUX. [Start] : distant loop validated	Remote loopback on 2 Mbit/s auxiliary bit stream (34 Mbit/s rate).
MCU>AUX. [End] : distant loop inhibited	
TX 1 [Start] : manual lockout validated	Manual override on transmit channel 1 (in 1+1 configuration). **On remote station**: Automatic disabling after five minutes.
TX 1 [End] : manual lockout inhibited	
TX X [Start] : manual lockout validated	Manual override on transmit channel X (in 1+1 configuration). **On remote station**: Automatic disabling after five minutes.
TX X [End] : manual lockout inhibited	
CONTROL 1 [Start] : Environment ctrl 1 validated	Environment remote control.
CONTROL 1 [End] : Environment ctrl 1 inhibited	
RESTART EQT : Restart equipment	Equipment restart.
MAINTENANCE ALARMS : reset	Re-initialization of the maintenance alarm counters and cancellation of the URG(ent).

Appendix 9 – List of RTP application alarms

Alarm mnemonic	MEANING
ALGNT	Alignment alarm
ARAIS	Generation of auxiliary receive AIS
ATAIS	Generation of auxiliary transmit AIS
ATIBR	Auxiliary tributary input bit rate error
ATISL	Auxiliary tributary input signal loss
CABAL	Cable alarm
CAR	Carrier misaligned
CARD	Diversity carrier misaligned
CLAL	Common loss alarm (or common mode alarm)
CRAL	Configuration rejected alarm
DPWSU	Diversity power supply alarm
DWSIP	Power supply alarm
ESCAL	Engineering service channel alarm
GAIS	General transmit AIS
IDUAL	IDU alarm
LICAL	Loss of channel identification code alarm
LSOW	LSU (“logic switching unit”) or order wire (LX n+1) alarm
MUXAL	MUX / DEMUX alarm
NREQT	No response from equipment
NSTAT	No response from station
ODUAL	ODU alarm (“outdoor unit”)
OK	No alarm
PWSUP	Power supply alarm
RDDC	Diversity receiver switching request alarm
RDDCA	Diversity receiver early switching request alarm
RDDCP	Diversity receiver priority switching request alarm
RDMVT	Loss of frame alignment diversity receiver alarm (94..LX ETSI)
RDRPW	Diversity received power alarm

Alarm mnemonic	MEANING
RDSYN	Diversity receive synthesizer alarm
RXAIS	Receive AIS generation
RXAL	Receiver alarm
RXDAL	Receiver diversity alarm (94..LX ETSI)
RXDC	Receiver switching request alarm
RXDCA	Receiver early switching request alarm
RXDCP	Receiver priority switching request alarm
RXMVT	Receiver frame alignment loss alarm
RXRPW	Received power alarm
RXSYN	Receive synthesizer alarm
SKAL	Software key alarm
STMSL	STM0 (96.) synchronization loss
TIAL	Tributary interface alarm (or loss of STM0 sync) TIU LX =TIAL input signal fail
TIBR	Tributary input bit rate error
TISL	Tributary input signal loss
TXAIS	Transmit AIS generation
TXAL	Transmitter alarm
TXBSY	Block synchronization alarm
TXPWA	Transmitted power alarm
TXSYN	Transmit synthesizer alarm

Appendix 10 – Creating user profiles

The profiles are stored in a text file named 946luxxx.prf. This can be edited using any text editor such as NOTEPAD (TM) in WINDOWS (TM).

Figure 75 shows a typical profile file.

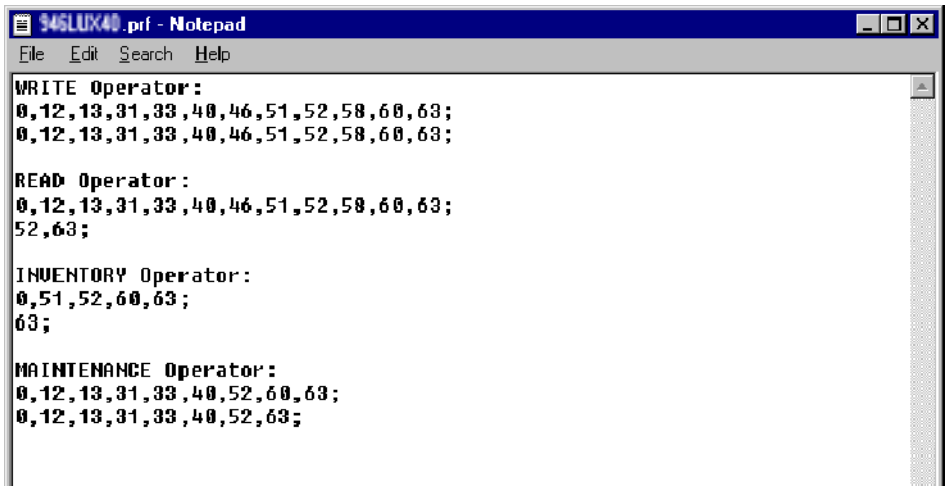


Figure 75 - Typical profile file

Each profile comprises:

- a line identifying the profile, ending with ":"
- a first line of codes, separated by "," characters. These codes each define an application to which the operator with the profile can obtain read mode access. The line ends with the ";" character.
- a second line of codes, separated by "," characters. These codes each define an application to which the operator with the profile can obtain write mode access. The line ends with the ";" character.

The codes are listed in the table below.

CODE	APPLICATION	CODE	APPLICATION
0	Alarms, states and remote controls	12	Performance monitoring (G821)
13	Performance monitoring (G784)	31	Radio transmission parameters
33	Analogue measurements	40	Maintenance memory
46	Software downloading	51	Remote inventory
52	Administrative functions	58	Installation parameters
60	Operation parameters	63	Transparent application for the user (code to be inserted mandatorily in each profile).

Appendix 11 – NECTAS configuration

A.11.1 – Nectas 3.79

The WIN.INI file contains information used to, for example:

- enable the date and time to be updated in an unsupervised network,
- adjust the equipment polling period for refreshing RTP application measurements.

This file, located in the Windows directory, can be edited using any text editor such as WINDOWS(TM) NOTEPAD(TM) . The information to be modified can be found under the [PEX] heading in the file (Figure 76).

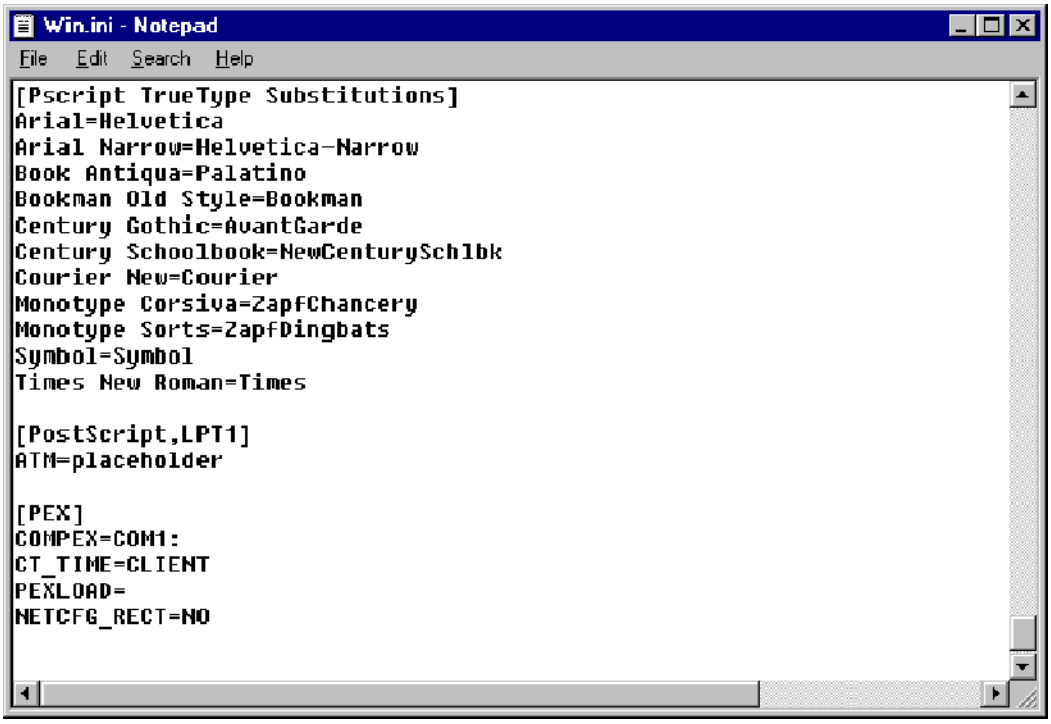


Figure 76 - Typical WIN.INI file

A.11.1.1 – Updating the date and time on an unsupervised network

- With the PC not connected to the equipment:
  - check that the PC's date and time are correct. If not, correct them (it is these values that will be taken into account by the network)
  - edit the WIN.INI file
  - in the [PEX] section, replace the line CT_TIME=CLIENT with the line CT_TIME=MANAGER (or create this line if it does not exist).
  - save the file.
- Connect the PC to the network's "Master" station.
- Run the "Administrative functions/Operator/Date" application and click "OK". the network will then register the date and time on the PC.
- Exit the application.
- Restore the WIN.INI file to its initial state.
- For a network with supervision, see § A.11.1.2.

A.11.1.2 - Time synchronization

When a CT is synchronized by the network, it is set to GMT time. To be able to analyze the results collected by Nectas, it is important to set the PC to local time. This can be done when installing the system software, or manually:

The **SET TZ** function uses the current setting of the TZ environment variable to assign the values of three global variables: **"summer time, time zone and nomtz".**

These variables are used by the PC to make adjustments based on GMT to local time, and by the time computed by the system.

They are set in the PC's autoexec.bat file.

To be interpreted correctly, the TZ variable must not include spaces between the characters.

If the TZ variable is not set, it defaults to PST8PDT, or US west coast summer time

(California-Pacific zone).

The syntax to be used to activate the environment TZ variable is:

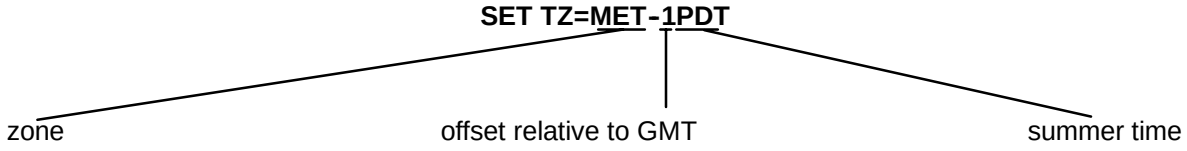
SET TZ=tzn[+/-]**hh[[**:**mm**][[:**ss]]][dzn**]

tzn must be a time zone name on three letters, such as PST, followed by an optional number sign, +/- hh, indicating the time difference between GMT and local time.

To specify the local time exactly, hours may be followed by minutes, :**mm**; seconds, :**ss**; and three summer time zone letters, **dzn**, such as PDT. Separate hours, minutes and seconds by **":".**

If the summer time is not stored, which is the case in some countries, activate TZ with no value for dzn.

Example:**SET TZ=MET-1**



On the "**summer-winter"** time change, initiate an update.

If the installer wants to change the network's time synchronization mode manually, this must be done in the WIN.INI file, under the [PEX] heading, on the CT_TIME line.

A.11.1.3 - Editing the RTP polling period (POLRTP variable)

In the event of network problems, the polling period for displaying the results of RTP application measurements can be increased as follows:

- Edit the WIN.INI file.
- In the [PEX] section, create a line that reads POLRTP=xx, where xx is the value required (in seconds) for the polling interval. The value will not be recognized unless it is greater than the value returned by the NE (one second in the case of a 9400 UX). More than a value of around ten seconds is not recommended. The screen in Figure 77 shows an example in which the polling interval is set to five seconds.
- Save the file.

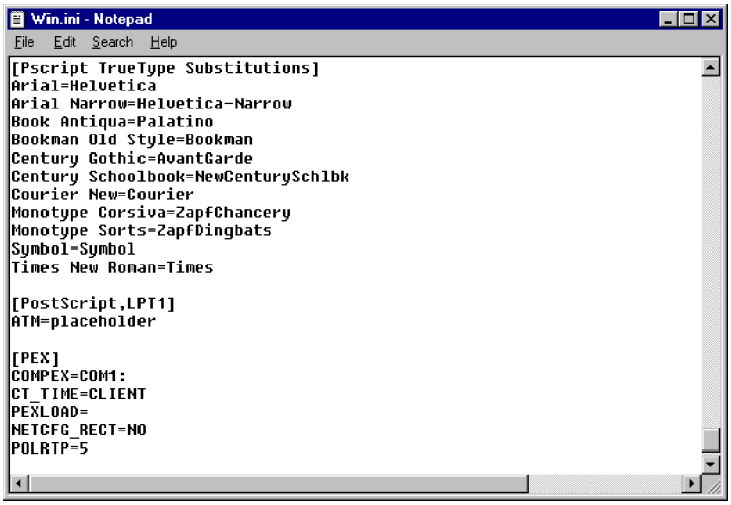
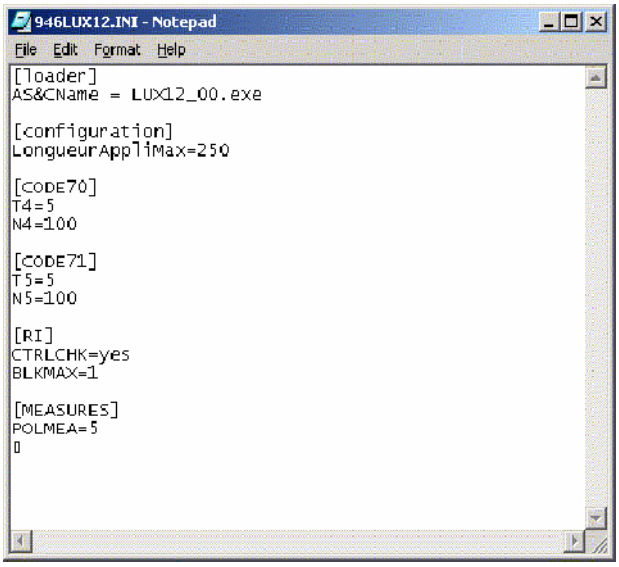


Figure 77 - WIN.INI file giving an RTP polling interval of five seconds

A.11.1.4 – Modification of the polling pediod of analogue measurements (POLMEA variable)



The 946LUXxx.INI file includes a POLMEA variable for editing the polling period of the equipment for refreshing analogue measurements.

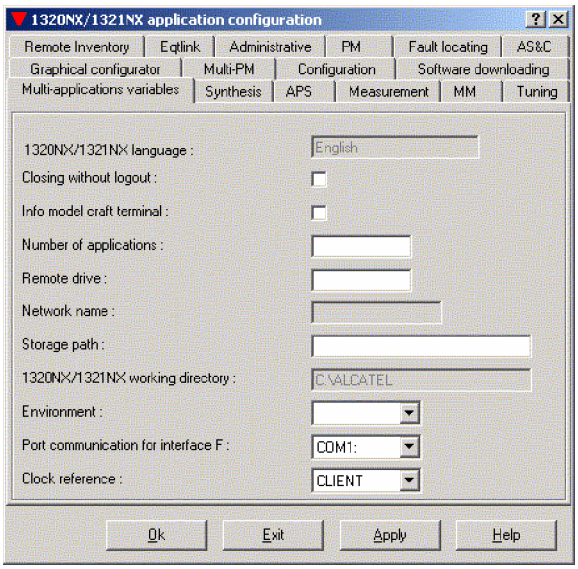
This file, located in the “c:\Alcatel\946LUX1x” directory, can be edited by any text editor such as WINDOWS (TM) NOTEPAD (TM). The information to be edited is in the [MEASURES] section of the file. Edits are made in the same way as for the POLRTP variable in the WIN.INI file.

In the event of network problems, the polling interval for displaying the results of analogue measurements can be increased as follows:

- Edit the 946LUXxx.INI file.
- In the [MEASURES] section, alter the xx value on the POLMEA=xx line, where xx is the value required (in seconds) for the polling interval. Reducing the default setting of five seconds is not recommended because of the risk of overloading the supervision bus. A value of between 30 and 60 seconds is recommended.
- Save the file.

A.11.2 – Nectas 4.1x

With NECTAS 4.1x the system provides a PEX setting tool, to launch this program from the Windows “Start” menu, launch the Program/ALCATEL/Setting tools (”cont1320.exe” file) application, the application configuration Windows appears.



A.11.2.1 - Time and date updating of a network without supervision

- The PC is not connected to the equipment:
  - Check that the date and the time of the PC are correct. Otherwise, correct them (these values will be taken into account by the network).
  - Open the cont1320.exe application
  - In the section “multi-applications variables”, change the “Clock reference” in “MANAGER”
  - Click on “OK”
- Connect the PC to the “Master” station of the network.
- Launch the “Administrative functions/Operator/Date” application and click on “OK”. The network takes into account the date and the time of the PC.
- Quit the application.
- Set back the “Clock reference” in “CLIENT”.

A.11.2.2 - Editing the RTP polling period (POLRTP variable)

In the event of network problems, the polling period for displaying the results of RTP application measurements can be increased as follows:

- Edit the WIN.INI file.
- In the [PEX] section, create a line that reads POLRTP=xx, where xx is the value required (in seconds) for the polling interval. The value will not be recognized unless it is greater than the value returned by the NE (one second in the case of a 9400 UX). More than a value of around ten seconds is not recommended. The screen in Figure 78 shows an example in which the polling interval is set to five seconds.
- Save the file.

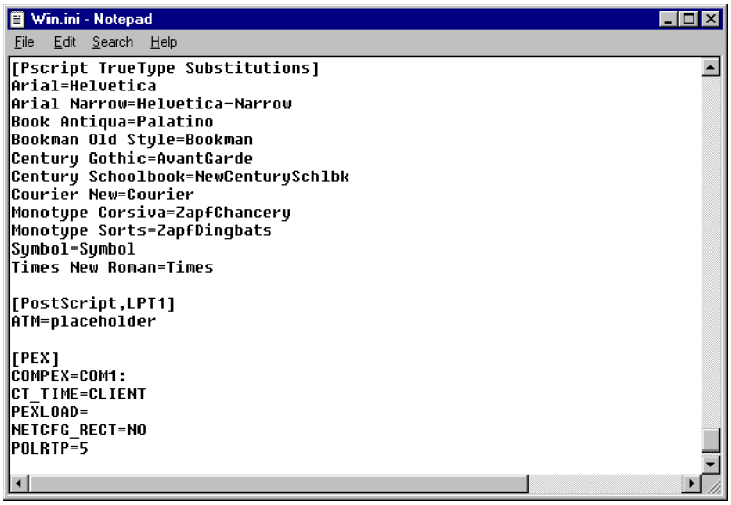
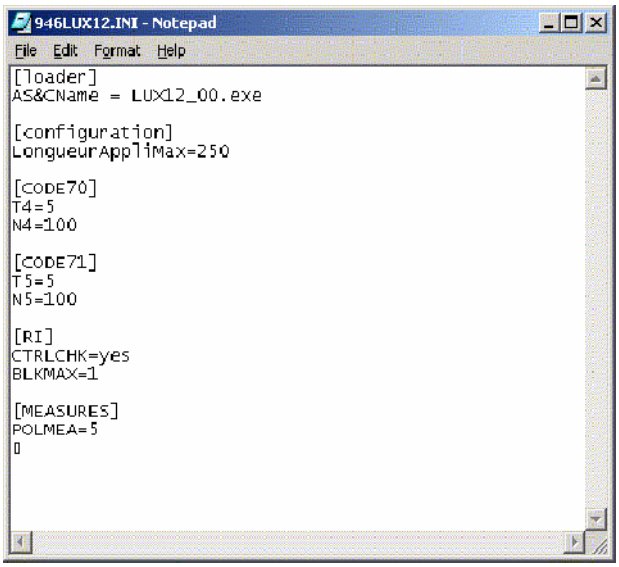


Figure 78 - WIN.INI file giving an RTP polling interval of five seconds

A.11.2.3 - Modification of the polling pediod of analogue measurements (POLMEA variable)



- Edit the 946LUXxx.INI file located in “C:\Alcatel\946LUXxx\946LUXxx.INI” directory with a text editor such as NOTEPAD (TM) of WINDOWS (TM).
- Delete the following section:
  “
  [MEASURES]
  POLMEA=5
  “
- Save the modification.
- Launch the cont1320.exe.
- In the “measure” section, modify the parameter: “polling time interval in seconds”
- Click on “OK”.

Appendix 12 – List of models and commercial items

A.12.1 – Hardware

Outdoor equipment 94xxUX	Mnemonic	Model number	Remote Inventory numbers
Outdoor unit 13 GHz 4QAM without RTPC	9413UXR201	3CC08879AAxx	3CC08879xxxx
Outdoor unit 13 GHz 4QAM with RTPC	9413UXR202	3CC12690AAxx	3CC11737xxxx
Outdoor unit 13 GHz 16QAM	9413UXR203	3CC12683AAxx	3CC11736xxxx
Outdoor unit 15 GHz standard power	9415UXR201	3CC08880AAxx	3CC08880xxxx
Outdoor unit 15 GHz high power	9415UXR202	3CC08978AAxx	3CC08978xxxx
Outdoor unit 15 GHz 4QAM with RTPC	9415UXR203	3CC12979AAxx	3CC12978xxxx
Outdoor unit 18 GHz standard power	9418UXR201	3CC08883AAxx	3CC08883xxxx
Outdoor unit 18 GHz high power	9418UXR202	3CC08979AAxx	3CC08979xxxx
Outdoor unit 23 GHz 4QAM	9423UXR201	3CC08888AAxx	3CC08888xxxx
Outdoor unit 23 GHz 16QAM	9423UXR202	3CC13031AAxx	3CC12755xxxx or 3CC12756xxxx
Outdoor unit 25 GHz 4QAM	9425UXR201	3CC08889AAxx	3CC08889xxxx
Outdoor unit 25 GHz 16QAM	9425UXR202	3CC12684AAxx	3CC11743xxxx
Outdoor unit 38 GHz 4QAM	9438UXR201	3CC08890AAxx	3CC08890xxxx
Outdoor unit 38 GHz 16QAM	9438UXR202	3CC13032AAxx	3CC12797xxxx
HSB Coupler 13 GHz	9413UXR111	3CC07985AAxx	3CC07985AAxx
HSB Coupler 15 GHz	9415UXR111	3CC07999AAxx	3CC07999AAxx
HSB Coupler 18 GHz	9418UXR111	3CC08134AAxx	3CC08134AAxx
HSB Coupler 23/25 GHz	942xUXR111	3CC07987AAxx	3CC07987AAxx
HSB Coupler 38 GHz	943xUXR111	3CC08000AAxx	3CC08000AAxx

Outdoor equipment 94xxUX flat ODU 4QAM/16QAM RTPC	Model number	Inventory number
Outdoor unit 9413 fs=266MHz 1–1’	3DB05524AAAA	3CC13688AAAA
Outdoor unit 9413 fs=266MHz 2–2’		3CC13689AAAA
Outdoor unit 9413 fs=266MHz 1’–1		3CC13690AAAA
Outdoor unit 9413 fs=266MHz 2’–2		3CC13691AAAA
Outdoor unit 9415 fs=315MHz 1–1’		3CC13692AAAA
Outdoor unit 9415 fs=315MHz 2–2’		3CC13693AAAA
Outdoor unit 9415 fs=315MHz 1’–1		3CC13694AAAA
Outdoor unit 9415 fs=315MHz 2’–2		3CC13695AAAA
Outdoor unit 9415 fs=420MHz 1–1’	3DB05526AAAA	3CC13696AAAA
Outdoor unit 9415 fs=420MHz 2–2’		3CC13697AAAA
Outdoor unit 9415 fs=420MHz 1’–1		3CC13698AAAA
Outdoor unit 9415 fs=420MHz 2’–2		3CC13699AAAA

Outdoor equipment 94xxUX flat ODU 4QAM/16QAM RTPC	Model number	Inventory number
Outdoor unit 9415 fs=490MHz 1–1’	3DB05525AAAA	3CC13704AAAA
Outdoor unit 9415 fs=490MHz 2–2’		3CC13705AAAA
Outdoor unit 9415 fs=490MHz 1’–1		3CC13706AAAA
Outdoor unit 9415 fs=490MHz 2’–2		3CC13707AAAA
Outdoor unit 9415 fs=728MHz 1	3DB05527AAAA	3CC13712AAAA
Outdoor unit 9415 fs=728MHz 2		3CC13713AAAA
Outdoor unit 9418 fs=1010MHz 1–1’	3DB05227AAAA	3CC12991AAAA
Outdoor unit 9418 fs=1010MHz 2–2’		3CC12992AAAA
Outdoor unit 9418 fs=1010MHz 1’–1		3CC12993AAAA
Outdoor unit 9418 fs=1010MHz 2’–2		3CC12994AAAA
Outdoor unit 9418 fs=340MHz 1–1’	3DB05228AAAA	3CC12999AAAA
Outdoor unit 9418 fs=340MHz 2–2’		3CC13000AAAA
Outdoor unit 9418 fs=340MHz 1’–1		3CC13001AAAA
Outdoor unit 9418 fs=340MHz 2’–2		3CC13002AAAA
Outdoor unit 9418 fs=1008MHz 1–1’	3DB05530AAAA	3CC13468AAAA
Outdoor unit 9418 fs=1008MHz 2–2’		3CC13469AAAA
Outdoor unit 9418 fs=1008MHz 1’–1		3CC13470AAAA
Outdoor unit 9418 fs=1008MHz 2’–2		3CC13471AAAA
Outdoor unit 9418 fs=1560MHz 1	3DB05229AAAA	3CC12995AAAA
Outdoor unit 9418 fs=1560MHz 2		3CC12996AAAA
Outdoor unit 9423 fs=1008MHz 1–1’	3DB05532AAAA	3CC13718AAAA
Outdoor unit 9423 fs=1008MHz 2–2’		3CC13719AAAA
Outdoor unit 9423 fs=1008MHz 1’–1		3CC13720AAAA
Outdoor unit 9423 fs=1008MHz 2’–2		3CC13721AAAA
Outdoor unit 9423 fs=1232MHz 1–1’	3DB05531AAAA	3CC13722AAAA
Outdoor unit 9423 fs=1232MHz 2–2’		3CC13723AAAA
Outdoor unit 9423 fs=1232MHz 1’–1		3CC13724AAAA
Outdoor unit 9423 fs=1232MHz 2’–2		3CC13725AAAA
Outdoor unit 9425 fs=1008MHz 1–1’	3DB05534AAAA	3CC13734AAAA
Outdoor unit 9425 fs=1008MHz 2–2’		3CC13735AAAA
Outdoor unit 9425 fs=1008MHz 1’–1		3CC13736AAAA
Outdoor unit 9425 fs=1008MHz 2’–2		3CC13737AAAA

Outdoor equipment 94xxUX flat ODU 4QAM/16QAM RTPC	Model number	Inventory number
Outdoor unit 9438 fs=1260MHz 1–1’	3DB05225AAAA	3CC12987AAAA
Outdoor unit 9438 fs=1260MHz 2–2’		3CC12988AAAA
Outdoor unit 9438 fs=1260MHz 1’–1		3CC12989AAAA
Outdoor unit 9438 fs=1260MHz 2’–2		3CC12990AAAA

Indoor equipment	Mnemonic	Model number	Remote Inventory numbers
Light Indoor unit – standalone 19”	9400UXB260	3CC09426AAxx	
Light Indoor unit – standalone 19” low–consumption	9400UXB270	3CC09426ACxx	
Light IDU for BTS (Plug–in IDU)	9400UXB261	3CC11141AAxx	
Main Indoor Unit 4x2 Mbit/s	9400UXB201	3CC08973Axxx	
Main Indoor Unit 8x2 Mbit/s	9400UXB202	3CC08972Axxx	
Main Indoor Unit 16x2 Mbit/s	9400UXB203	3CC08971Axxx	
Main Indoor Unit 34+2 Mbit/s	9400UXB204	3CC08920Axxx	
Extension Indoor Unit 1+0	9400UXB211	3CC08897Axxx	
Extension Indoor Unit 1+1 basic	9400UXB221	3CC08898Axxx	
Extension Indoor Unit 1+1 mux prot. 4x2 Mbit/s	9400UXB231	3CC08976Axxx	
Extension Indoor Unit 1+1 mux prot. 8x2 Mbit/s	9400UXB232	3CC08975AAxx	
Extension Indoor Unit 1+1 mux prot. 16x2 Mbit/s	9400UXB233	3CC08974AAxx	
Extension Indoor Unit 1+1 mux prot. 34+2 Mbit/s	9400UXB234	3CC08908AAxx	
Access Indoor Unit nx2 Mbit/s	9400UXB141	3CC06751ABxx	
Access Indoor Unit 34+2 Mbit/s	9400UXB144	3CC08977AAxx	
Software Key Unit	9400XXB007	3CC07619ABxx	

A.12.2 – Software

Craft Terminal application software	Software identification	Minimum Item Change Status (ICS)
NECTAS 3.79	3CC08601AAxx	19
NECTAS 4.1	3AL78606AAxx	02
Craft Terminal 946LUX 11	3CC07361AAxx	05
Craft Terminal 946LUX 11 extension	3CC07361ABxx	02
Craft Terminal 946LUX 12	3CC08737AAxx	10
Craft Terminal 946LUX 12 extension	3CC08737ABxx	02

Craft Terminal application software	Software identification	Minimum Item Change Status (ICS)
Craft Terminal 946LUX 40	3CC11116AAxx	07
RTP	3CC04458ACxx	04

Downloadable network element software – IDU (360)	Software identification	Minimum Item Change Status (ICS)
NE-SW-IDU 360 LUX 11	3CC07767ABxx	06
NE-SW-IDU 360 LUX 12	3CC07767ACxx	11
NE-SW-IDU 360 LUX 40	3CC07767ANxx	07

Embedded software – IDU (302)	Software identification	Minimum Item Change Status (ICS)
CLASSIC IDU (MCU)	3CC08538AAxx	06
LIGHT IDU (MCU)	3CC08755AAxx	04
EXTENSION IDU (SCU)	3CC08540AAxx	08

Embedded software – ODU	Software identification	Minimum Item Change Status (ICS)
ODU 9400UXR201 to 9400UXR203	3CC06452AJxx	04
ODU (13 GHz)	3CC13007AAxx	01

Set of Diskettes – IDU	Diskette identification	Minimum Item Change Status (ICS)
NE-SW-IDU 360 LUX 12	3CC12848AAxx	02
NE-SW-IDU 360 LUX 40	3CC12849AAxx	01

Set of Diskettes – Craft Terminal	Diskette identification	Minimum Item Change Status (ICS)
Craft Terminal 946LUX 12	3CC08734AAxx	08
Craft Terminal 946LUX 12 EXT	3CC08734ABxx	02
Craft Terminal 946LUX 40	3CC11117AAxx	04

Set of CD-ROM – Craft Terminal	CD-ROM identification	Minimum Item Change Status (ICS)
Craft Terminal 946LUX 12	3CC13241ABxx	01
Craft Terminal 946LUX 12 EXT	3CC13243AAxx	01
Craft Terminal 946LUX 40	3CC13242ABxx	01
RTP	3CC13244AAxx	01
NECTAS 4.1x	3CC78606AAxx	01

Software	Mnemonic	Software Indentification
NECTAS Equipment Craft Terminal software (1320NX)	9000XXN002	3AL78606AAxx 3CC04457AAxx ou 3CC13244AAxx 3CC08597AAxx
Laptop Craft Terminal	9000XXN001	3CC13086ACAA
9400LX/UX RQ2 NE Craft Terminal application software	9400XXN007	3CC08734AAxx ou 3CC13241AAxx
9400LX/UX extended RQ2 NE Craft Terminal application software	9400XXN008	3CC08734ABxx ou 3CC13243AAxx
9400LX/UX SNMP NE Craft Terminal application software	9400XXN009	3CC11117AAxx ou 3CC13242AAxx
Equipment basic software	9400XXN001	
Additional software package for embedded mediation function	9400XXN101	
Additional software package for RCT management	9400XXN102	

A.12.3 – Commercial items

This table maps the various items in the installation manual to the commercial references by which they are sold. The composition of each item is given by way of indication; the definitive list is generated according to the equipment configuration.

Note: The last two letters are meaningless for item definitions.

OUTDOOR PARTS			
Installations	**Commercial code**	**Article**	**Comments**
Outdoor adjustable pole mounting 1+0 (integrated antenna)	**9400UXI102**	3CC10752AAxx	
Outdoor adjustable pole mounting 1+1 (integrated antenna)	**9400UXI101**	3CC06071AAxx	
Wind strengthening	**9400UXI103**	3CC11453AAxx	Optional
Outdoor fix pole mounting (non integrated antenna) 13 GHz	**9413UXI102**	77096359 1AB128510002	Adapter nose Support
	9413UXI103	3CC05751ACxx	Flextwist 13 GHz 60 cm
Outdoor fix pole mounting (non integrated antenna) 15 GHz	**9415UXI102**	1AB146090001 1AB128510002	Adapter nose Support
	9415UXI103	3CC05750ACxx	Flextwist 15 GHz 60 cm
Outdoor fix pole mounting (non integrated antenna) 18 GHz	**9418UXI102**	1AB146090002 1AB128510002	Adapter nose Support
	9418UXI103	3CC05749ACxx	Flextwist 18GHz 60 cm
Outdoor fix pole mounting (non integrated antenna) 23 / 25 GHz	**942xUXI102**	77096356 1AB128510002	Adapter nose (23/25 GHz) Support
	9423UXI103	3CC05749ACxx	Flextwist 23 /25GHz 60 cm
Integrated antenna 13 GHz – 0.6m	**9413UXI002**	3CC06215AAAB	
Integrated antenna 15 GHz – 0.3m	**9415UXI001**	3CC10952AAxx	
Integrated antenna 15 GHz – 0.6m	**9415UXI002**	3CC06321AAAB	
Integrated antenna 18 GHz – 0.3m	**9418UXI001**	3CC10953AAxx	
Integrated antenna 18 GHz – 0.6m	**9418UXI002**	3CC06628AAAB	
Integrated antenna 23 GHz – 0.3m	**9423UXI001**	3CC06217AAAB	
Integrated antenna 23 GHz – 0.6m	**9423UXI002**	3CC06216AAAB	
Integrated antenna 25 GHz – 0.3m	**9425UXI001**	3CC06629AAAB	
Integrated antenna 25 GHz – 0.6m	**9425UXI002**	3CC06630AAAB	
Integrated antenna 38 GHz – 0.3m	**9438UXI001**	3CC06631AAAB	
Integrated antenna 38 GHz – 0.6m	**9438UXI002**	3CC06632AAAB	
Fix attenuator 13/15/18 GHz	**9413UXR120**	3CC08543AAxx 3CC08543ABxx 3CC08543ACxx	3 dB 6 dB 10 dB
H polarization kit for HSB coupler, 13 GHz	**9413UXR112**	3CC07725AAxx	
H polarization kit for HSB coupler, 15 GHz	**9415UXR112**	3CC07998AAxx	
H polarization kit for HSB coupler, 18 GHz	**9418UXR112**	3CC11263AAxx	
H polarization kit for HSB coupler, 23 & 25 GHz	**942xUXR112**	3CC11264AAxx	

Installations	Commercial code	Article	Comments
H polarization kit for HSB coupler, 38 GHz	**9438UXR112**	3CC08024AAxx	
Pipe	**SUPOUTD001**	3CC04658AExx	
Self supported mast	**SUPMAST001**	3CC05148AAxx	

INDOOR PARTS			
Installations	**Commercial code**	**Article**	**Comments**
Fan replacement kit	**9400UXZ200**	3CC11761AAxx	Contains 2 fans (12 V)
Fan replacement kit	**9400UXZ201**	3CC12847AAxx	Contains 2 fans (5 V)
19” supporting rack	**9400XXI300**	77095106	
Indoor wall mounting 3U	**9400UXI301**	3CC09614AAxx 3CC07966AAxx 3CC07957AAxx	Includes a set of connectors and supplies, an alternative to the 19» support rack
Indoor wall mounting 9U	**9400UXI302**	1AD029510001	Alternative to the 19” support rack
Indoor desktop mounting kit (standalone solution)	**9400UXI303**	3CC08295AAxx	Includes a set of connectors and supplies, an alternative to the 19” support rack
19” DC distributor board (6 accesses)	**9400XXC511**	77096294 3CC08165	48 V
19” DC distributor board (6 accesses)	**9400XXC511**	77096294 3CC08211AAxx 3CC08209AAxx 3CC08212AAxx	24 V
Wall mounting DC distributor board (6 accesses)	**9400XXC512**	77096248 3CC08165AAxx 3CC08211AAxx	48 V
Wall mounting DC distributor board (6 accesses)	**9400XXC512**	77096248 3CC08209AAxx 3CC08212AAxx	24 V
Consumable set	**9400UXT002**	3CC06503AAxx	
Service kit	**9400UXT102**	3CC07972AAxx 1AF00375ABxx	Telephone headset cable
Light service kit	**9400UXT112**	3CC07972AAxx	Cable
Light service kit for 9400UX flat ODU	**9400UXT113**	3CC13477AAxx	Cable
Station tool kit	**9400UXT103**	3CC08409AAxx	One for each station
Telephone handset	**9400XXB000**	3CC07946AAxx	
Cable tray – Indoor – Cablofil 6 m	**INFRA00003**	3CC07580AAxx	
Cable tray – Indoor – PVC 4 m	**INFRA00004**	3CC06511AAxx	
Cable tray – Outdoor – CES 12 m	**INFRA00005**	3CC06512AAxx	
Cable tray – Outdoor – CAPRI 12 m	**INFRA00006**	3CC06759AAxx	
Battery		M4T28-BR12-SH1 1AB084760003	

CABLING			
Installations	**Commercial code**	**Article**	**Comments**
Additional cable for 1+1 configuration	**9400UXC130**	3CC07157ABxx	note : the cable 3CC07160ABxx is attached to the commercial reference of the extension IDU
Back to back cables	**9400UXC333**	3CC08729AAxx	Telephone service channel
	9400UXC332	3CC07711AAxx	Supervision
Indoor/Outdoor cable (length < 300 meters)	**9400UXI202**	1AC001100022	Per 50 m
Indoor/Outdoor cable connector set (2 pieces)	**9400UXI203**	1AB095530021	
Indoor/Outdoor cable grounding kit	**9400UXI204**	1AB128500002	
Power supply connection kit	**9400XXC501**	3CC08165AAxx	48 V (per 20m)
		3CC08209AAxx	24 V (per 20m)
Grounding connection kit (IDU + ODU)	**9400XXC502**	3CC08166AAxx	Per 20 m
120 Ohms cable 16 pairs	**9400XXI504**	3CC08809AAxx	Per 15 m
One terminal block 10 pairs	**9400XXI505**	3CC07921AAxx 3CC05527AAxx	
75 Ohms connection kit (2 tributaries)	**9400XXI506**	3CC07917AAxx	Per 15 m
Alarm cable 1 pair	**9400XXI508**	3CC08811AAxx	Per 15 m
Alarm cable 15 pairs	**9400XXI509**	3CC08817AAxx	Per 15 m
Distributor panel 4 x 2 Mbit/s	**9400XXI404**	3CC08061AAxx and 3CC07885AAxx or 3CC07885ADxx	75 Ω / 1.6/5.6. Use adapters if BNC ports are requested
	9400XXI404 + 1 x 9400XXI405	3CC08061ABxx and 3CC07759AAxx or 3CC07759ADxx	75 Ω / BNC
Distributor panel 8 x 2 Mbit/s	**9400XXI408**	3CC08061AAxx and 3CC07885ABxx or 3CC07885AExx	75 Ω / 1,6/5,6
	9400XXI408 + 2 x 9400XXI405	3CC08061ABxx and 3CC07759ABxx or 3CC07759AExx	75 Ω / BNC
	9400XXI408	3CC07810AAxx 3CC07658AAxx	120 Ω – EMC
	9400XXI408	3CC08062AAxx 3CC07658AAxx	120 Ω – non EMC
Distributor panel 16 x 2 Mbit/s	**9400XXI416**	3CC08061AAxx and 2x3CC07885ABxx or 2x3CC07885AExx	75 Ω / 1.6/5.6
	9400XXI416 + 4 x 9400XXI405	3CC08061ABxx and 2x3CC07759ABxx or 2x3CC07759AExx	75 Ω / BNC
	9400XXI416	3CC07810AAxx 2x3CC07658AAxx	120 Ω – EMC
	9400XXI416	3CC08062AAxx 2x3CC07658AAxx	120 Ω – non EMC
BNC adapter	**9400XXI405**	3CC08249AAxx	To be ordered if the 34 Mbit/s tributary is used with BNC connector

Appendix 13 – List of abbreviations

MNEMONIC	ENGLISH	FRENCH
ACO	Alarm cut-off	Coupure d'alarme
ADC	Analog/Digital Converter	Convertisseur Analogique/Numérique (CAN)
AGC	Automatic Gain Control	Commande Automatique de Gain (CAG)
AIS	Alarm Indication Signal	Signal d'Indication d'Alarme (SIA)
BBE	Number of errored blocks that are not part of an SES	Nombre de blocs erronés ne faisant pas partie d'une SES
BTS	Base Transceiver Station	Station de Base
CIA_NUM	Digital Cable Interface Adapter	Carte Interface de Câble numérique
CT	Craft Terminal	Poste d'EXploitation (PEX)
DTMF	Dual-Tone Multi Frequency	Multifréquence en code 2
ECT	Equipment Craft Terminal	Poste d'Exploitation Equipement (PEE)
EMC	ElectroMagnetic Compatibility	Compatibilité ElectroMagnétique (CEM)
ES	Errored seconds	Secondes Avec Erreurs (SAE)
ESC	Engineering Service Channel	Voie De Service (VDS)
ESD	ElectroStatic Discharge	Décharges ElectroStatiques (DES)
ESR	Early Switching Request	Demande de Commutation Anticipée (DCA)
ETSI	European Telecommunications Standards Institute	Institut de Standardisation des Télécommunications Européennes
FD	Frequency Diversity	Diversité de Fréquence
FEC	Forward Error Correction	Code Correcteur d'Erreurs (CCE)
HDB3	High Density Binary 3 code (3rd order)	Code Haute Densité Binaire d'ordre 3
HDLC	High-level Data Link Communication	Liaison de données à haut niveau
HSB	Hot StandBy	Veille active
IDU	InDoor Unit	Coffret intérieur
IF	Intermediate Frequency	Fréquence Intermédiaire (FI)
I^2C	Inter Integrated Circuit	Inter Circuits Intégrés
ITU	International Telecommunication Union	Union Internationale des Télécommunications
LAU	Line Access Unit	Carte d'accès affluent
LED	Light Emitting Diode	Diode électroluminescente
LIU	Line Interface Unit	Carte d'interface affluent
LO	Local Oscillator	Oscillateur Local (OL)
MCU_NUM	Multiplexing and Control Unit with digital cable interface	Carte de Multiplexage et de Commande avec interface de câble numérique
MDU	Modulation/Demodulation Unit	Carte Modulation/Démodulation
NE	Network Element	Elément de réseau (ER)

MNEMONIC	ENGLISH	FRENCH
NMC	Network Management Center	Centre d'exploitation
NMS	Network Management System	Système de Gestion de Réseau
NTP	Network Time Protocol	Protocole de mise à l'heure du réseau
OCT	Office Craft Terminal	Poste d'Exploitation Local (PEL)
ODU	OutDoor Unit	Coffret extérieur
OS	Operation System	Station Centrale (SC)
PCM	Pulse Code Modulation	Modulation par Impulsions et Codage (MIC)
PSAC	Protection Switch Actual Count Normal channel: Number of switchovers from active channel to standby channel Standby channel: Number of switchovers from any active channel to the standby channel	Canal Normal : Nombre de commutations du canal actif vers le canal secours Canal Secours : Nombre de commutations depuis n'importe quel canal actif vers le canal secours
PSAD	Protection Switch Actual Duration Normal channel: Number of one-second periods during which the active channel has been backed up Standby channel: Number of one-second periods during which any active channel has been backed up	Canal Normal : Nombre de périodes de durée une seconde pendant lesquelles le canal actif est secouru Canal Secours : Nombre de périodes de durée une seconde pendant lesquelles n'importe quel canal actif est secouru
PSRC	Protection Switch Request Count Normal channel: Number of automatic switching requests observed on the normal channel Standby channel: Number of automatic switching requests observed on the normal channel	Canal Normal : Nombre de demandes de commutation automatique observé sur le canal normal Canal Secours : Nombre de demandes de commutation automatique observé sur le canal normal
PSRSAD	Protection Switch Request Service Affecting Duration Normal channel: Number of one-second periods during which it has been impossible to meet a channel switching request. Standby channel: Number of one-second periods during which the standby channel has received automatic switching requests and has been unable to satisfy them because it is already backing up another channel (meaningless for UX systems).	Canal normal : Nombre de périodes de durée une seconde pendant lesquelles une demande de commutation du canal n'a pu être satisfaite. Canal secours : Nombre de périodes de durée une seconde pendant lesquelles le canal secours a reçu des demandes de commutation automatique et n'a pu les satisfaire car il assurait déjà le secours d'un autre canal (Non significatif en UX)
PSU	Power Supply Unit	Carte Alimentation
RCT	Remote Craft Terminal	Poste d'Exploitation Distant (PED)
RF	Radio Frequency	Fréquence Radio
RLTMMax	Received Level Tide Mark Max Maximum estimated received power value during the observation period.	Valeur maximum estimée de la puissance reçue pendant la période d'observation.

MNEMONIC	ENGLISH	FRENCH
RLTMMin	Received Level Tide Mark Min Minimum estimated received power value during the observation period.	Valeur minimum estimée de la puissance reçue pendant la période d'observation.
RLTS-1	Received Level Threshold Second -1 Number of one-second periods during which the received power has been more than 10 dB above nominal power.	Nombre de périodes de durée 1 seconde pendant lesquelles la puissance reçue a dépassé de plus de 10 dB la puissance nominale.
RLTS-2	Received Level Threshold Second -2 Number of one-second periods during which received power has exceeded the propagation threshold.	Nombre de périodes de durée 1 seconde pendant lesquelles la puissance reçue a dépassé le seuil de propagation.
RLTS-3	Received Level Threshold Second -3 Number of one-second periods during which received power has exceeded the early switching request threshold.	Nombre de périodes de durée 1 seconde pendant lesquelles la puissance reçue a dépassé le seuil de demande de commutation anticipée.
RLTS-4	Received Level Threshold Second -4 Number of one-second periods during which received power has exceeded the received power threshold.	Nombre de périodes de durée 1 seconde pendant lesquelles la puissance reçue a dépassé le seuil de puissance reçue.
RTPC	Remote Transmit Power Control	Régulation de Puissance d'Emission à Distance
RX	Receiver	Récepteur
SAE	Errored Seconds (ES)	Secondes Avec Erreurs
SCU_NUM	Service Channel Unit with digital cable interface	Carte Voie de Service avec interface de câble numérique
SD	Space Diversity	Diversité d'espace
SES	Severely Errored Seconds	Secondes Gravement Erronées (SGE)
SI	Unavailable seconds	Secondes Indisponibles
SIA	Alarm Indication Signal (AIS)	Signal d'Indication d'Alarme
SNMP	Simple Network Management Protocol resulting from TCP/IP architecture used on Ethernet type networks	Protocole de Gestion de Réseau issu de l'architecture TCP/IP utilisé sur les réseaux de type Ethernet
TS/TC	Remote Signalling/Remote Control	TéléSignalisation/TéléCommande
TX	Transmitter	Emetteur
UF	Manufacturing unit	Unité de fabrication
VCXO	Voltage Controlled Crystal Oscillator	Oscillateur à quartz contrôlé en tension
VDS	Engineering Service Channel (ESC)	Voie De Service

INDEX

A

Administrative functions, application(s), 115

Alarm & Remote control, wiring, 98

Alarms
  description, 42
  equipment, 285

Alarms, Status and Controls
  application(s), 146
  opening screen, 196

Analogue measurements, application(s), 177

Analysing, units and modules, 203
  ENVT, 208
  ESC 3 and 4, 208
  LIU, 207
  main and standby ODU, 204
  main, extension, Light IDU, Plug−in, 203
  MCU and SCU (1+1), 210
  PSU, 207
  SCU, 213

Antenna
  coarse alignement, 79
  polarization, 74

Application(s)
  9400UX, 110
    global synthesis line, 112
    list, 114
    menu bar, 110
    opening a session, 113
  Administrative functions, 115
  Alarms, Status and Controls, 146
    opening screen, 196
  Analogue measurements, 177
  G784 Performance Monitoring, 167
  Installation parameters, 120
  Maintenance Memory, 164
  Operation parameters, 124
  Performance Monitoring (G821), 155
  Radio Transmission Parameters, 158
  Remote inventory, 161
  Software downloading, 143

Assembling, ”N” type coaxial connectors, 276

B

Between a terminal’s units, wiring, 99

Between terminals of a station, wiring, 101

Bit rate
  changing by software, 229
  changing via LAU/LIU boards, 230

C

Capacity, configuration, 29

Changing
  1+1 coupler polarisation, 259
  bit rate by software, 229
  bit rate via LAU/LIU boards, 230
  configuration, 217
  fans, 263
  frequency, 218
  fuse, 265
  IDU, 214
  MCU board battery, 262
  NE physical address, 233
  ODU, 216
  power level, 257
  redundancy configurations, 253
  SIMM memories, 260
  software key, 232
  tributaries, 220
  tributary impedance, 228

Coarse alignment, antenna, 79

Commissioning
  end, 191
  equipment, 181
    checking, 187, 188, 189, 190
  order, 181
  reversing the order, 191
  station A
    (phase 1), 182
    (phase 2), 188
  station B, 186

Configuration, 22, 23
  1+0, 30
  1+1, 34
  capacity, 29
  changing, 217
  hardware, 29
  indoor unit, 236
  multiplexer/demultiplexer protection, 36
  NECTAS, 295
  remote indication loops, 251
  service channels, 238
  service channels 3, 250
  software, 103

Configuration with non−integrated antenna, installing, 87

Consumables, set, 273

Controls, indications and connectors, IDU, 45
  Access version, 49
  Classic version, 45
  Extension version, 48
  Light 19” version, 47

D

Delivery, equipment, 61

Description
  alarms, 42
  equipment, 21
  IDU/ODU cable, 33
  loopback, 42
  service channels, 38
  telesignalling and remote controls, 42

E

Human exposure, Electromagnetic fields, 267

End, commissioning, 191

Engineering service channel, wiring, 98

Equipment
  alarms, 285
  commissioning, 181
    checking, 187, 188, 189, 190
  delivery, 61
  description, 21
  features, 24, 27
  grounding, 93
  installing, 68
  labels, 62
  maintenance, 193
  management, 44
  operation, 193
  remote controls, 290
  RTP alarms, 292
  servicing, 193
  wiring, 92

F

Fans, changing, 263

Features, equipment, 24, 27

Frequency, changing, 218

Fuse, changing, 265

G

G784 Performance Monitoring, application(s), 167

Grounding, equipment, 93

I

IDU
  changing, 214
  controls, indications and connectors, 45
    Access version, 49
    Classic version, 45
    Extension version, 48
    Light 19” version, 47
  opening, 235

IDU/ODU, wiring, 95

IDU/ODU cable, description, 33

Indoor unit, configuration, 236

Installation
  information, 271
  tools, 68

Installation parameters, application(s), 120

Installing
  equipment, 68
  laborack, 70
  Light IDU 19” version, 69
  software, 103
  wall mounting, 70

Item, list, 300

L

Labels, equipment, 62

Laborack, installing, 70

Light IDU 19” version, installing, 69

List
  items, 300
  models, 300

Local supervision, 178

Loopback, description, 42

M

Maintenance
  corrective, 193
  equipment, 193
  preventive, 193

Maintenance Memory, application(s), 164

Management, equipment, 44

Manual
  structure, 15
  using, 15

MCU board battery, changing, 262

Model, list, 300

N

NE physical address, changing, 233

NECTAS, configuration, 295

Network, supervision, 193

O

ODU
  changing, 216
  installing, 73

ODU 9400UX flat ODU, 83
  installing, 90

ODU 94xxUXR201, 94xxUXR202, 94xxUXR203,
  installing, 90

Installing, 83
  configuration with non−integrated antenna, 87
  ODU, 73
  ODU 9400UX flat ODU, 90
  ODU 94xxUXR201, 94xxUXR202, 94xxUXR203,
    90
  pole mounting
    9400UXI101, 81
    9400UXI102, 75
      on the pipe, 78, 84

Opening, IDU, 235

Operation, 30
  equipment, 193
  software, 103

Operation parameters, application(s), 124

Order
  commissioning, 181
  reversing, 191

P

Performance Monitoring (G821), application(s), 155

Pin out, user ports, 278

Pointing the antenna
  pole mounting 1+0, 9400UXI102, 187
  pole mounting 1+1, 9400UXI101, 186

Polarization, antenna, 74

Pole mounting
  9400UXI101, installing, 81
  9400UXI102, installing, 75
    on the pipe, 78, 84

Pole mounting 1+0 (9400UXI101), pointing the
  antenna, 186

Pole mounting 1+0 (9400UXI102), pointing the
  antenna, 187

Profiles, user, 294

R

Radio Transmission Parameters, application(s), 158

Remote controls, equipment, 290

Remote inventory, application(s), 161

RTP alarms, equipment, 292

S

Safety, instructions, 16

Service channel, telephone, 238

Service channels, description, 38

Service kit, 274

Servicing, equipment, 193

Software
  configuration, 103
  installing, 103
  operation, 103

Software key, changing, 232

Station A, commissioning
  (phase 1), 182
  (phase 2), 188

Station B, commissioning, 186

Supervision, network, 193

Symbols, 16, 17

T

Technical characteristics, 51

Telesignalling and remote controls, description, 42

Tools, installation, 68

Tributaries
  changing, 220
  wiring, 96

Tributary impedance, changing, 228

U

Upgrading, software, 234

User, profiles, 294

W

Wall mounting, installing, 70

Wiring
alarm & remote control, 98
between a terminal's units, 99
between terminals of a station, 101
engineering service channel, 98
equipment, 92
IDU/ODU, 95
tributaries, 96

FIN DU DOCUMENT